Wireless InSite®

3D Wireless Communication and Propagation Prediction Software

REFERENCE MANUAL



REMCOM®

Electromagnetic Simulation Software

Wireless InSite® Reference Manual

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Chapter 1

Introduction to Wireless InSite®

In this chapter, you will learn...

- → the main purpose of and flow of operations within Wireless InSite
- → common formats, keywords and symbols used in this manual

1.1 Wireless InSite Summary

Wireless InSite is an electromagnetic simulation tool for predicting the effects of buildings and terrain on the propagation of electromagnetic waves. It predicts how the locations of the transmitters and receivers within an urban area affect signal strength. Wireless InSite models the physical characteristics of irregular terrain and urban building geometries, performs the electromagnetic simulations, and then evaluates the signal propagation characteristics.

The virtual building and terrain environment is either constructed using Wireless InSite's editing tools or imported from a number of popular file formats, such as `DXF`, `shapefile`, `DTED` and `USGS`.

Transmitter and receiver locations can be specified using Wireless InSite's powerful site-defining tools, or imported from an external data file. Separate simulations for portions of the overall area may be specified by defining study areas.

The simulations are made by shooting rays from the transmitters and propagating them through the defined geometry. These rays interact with geometries and make their way to receiver locations. Ray interactions include *reflections* from geometry faces, *diffractions* around geometry edges, and *transmissions* through geometry faces. Wireless InSite's ray-based solvers use the Uniform Theory of Diffraction (UTD) to evaluate a ray path's electric field. UTD provides accurate results when the scenario geometry is large compared to the wavelength of the propagating wave. For typical applications, the UTD-based models provide accurate predictions from approximately 100 MHz to approximately 100 GHz. The X3D propagation model includes atmospheric absorption, extending the validity of its wave propagation simulations up to millimeter wave frequencies.

At each receiver location, contributions from arriving ray paths are combined and evaluated to determine predicted quantities such as electric and magnetic field strength, received power, interference measures, path loss, delay spread, direction of arrival, impulse response, electric field vs. time, electric field vs. frequency, and power delay profile.

Wireless InSite presents results in a number of ways. It provides visual representation of some results, such as transmitter coverage areas and power distributions, placing these visually within the modeled environment. Wireless InSite is also capable of playing movies of time-domain E-field and H-field evolution. For other types of data, Wireless InSite provides an advanced plotting system. Overlays of data allow quick comparison to imported measurements, or even previous Wireless InSite simulations.

1.2 Remcom Support

Remcom's support team provides thorough and professional engineering advice. Contact `support@remcom.com` and an application engineer will respond promptly. Remcom's development team of EM and software engineers will provide detailed expert explanations as needed.

1.3 How to Read This Manual

After installing Wireless InSite, reading the overview (Chapter 2), and working through the tutorials in the Wireless InSite User's Guide, new users of Wireless InSite will find the organization of the software intuitive and will quickly be able to set up projects, run simulations, and view the propagation predictions.

In this manual, the reader will find the text formatted according to the following:

- (icon)WIRELESS INSITE WINDOWS, such as MAIN WINDOW, GEOMETRY VIEW, etc.
- (icon)MAIN WINDOW ELEMENTS, such as ANTENNAS, TRANSMITTERS, etc.
- (icon) *Types* of Main Window Elements, such as *Route* Transmitters, *Horn* Antennas, etc.
- *Menu Options*
- BUTTONS
- KEYBOARD INSTRUCTIONS
- `File Names`

A succession of menu selections will be separated by the → symbol. For example, *File*→*Open*→ *Geometry*→*Object* would be the succession of menu choices to open a file containing object data.

The instruction to "click" will always mean to click with the left mouse button and release. A few operations require the left mouse button to be held down, and the instruction in these cases will be to "click and hold". When it is necessary to click with the right mouse button, the instruction will be to "right-click". This action in most cases will generate what is referred to as a "context menu" throughout the manual.

All values in this manual will be in the following units:

- angles in degrees
- lengths in meters

- E-field magnitude in V/m
- frequency in MHz
- power in dBm
- path loss in dB
- time in seconds

We have also incorporated several key icons in the manual to help organize the material and make it as easy to read as possible.

→ At the beginning of each chapter, there is a summarized list of topics you will find within. This symbol marks each topic.

▶ Wherever there is a reference to another section with additional information on the topic being discussed, this symbol will lead you there.

✓ This symbol marks a "smart tip", which offers a helpful way to think about a topic or complete a task.

There are times when the user must be careful to remember something or especially cautious. This symbol will warn you to pay attention!

Installation Directory

The installation directory is specified in this manual using a placeholder name. This is due to the fact that this location can be changed as a part of the installation process and can vary depending on the version of the operating system you are using.

`install_location\` refers to the location that the installer will place Wireless InSite. By default this will be c:\Program Files\Remcom\Wireless InSite 3.4.4 .

Chapter 2

Overview

In this chapter, you will learn...

- → the main windows and menus available in Wireless InSite
- → how to set up a project, run simulations, and view results
- → how to adjust viewing and simulation preferences

In order to begin using Wireless InSite, it is important to understand the organization of the graphical user interface (GUI) and the elements that comprise an Wireless InSite project. This chapter begins with a description of the primary Wireless InSite windows. Following this is a description of the context menus within Wireless InSite and instructions on accessing these menus. The chapter concludes with an overview of the menus and procedures used in setting up an Wireless InSite project, running simulations and viewing results.

2.1 Wireless InSite Windows

The primary Wireless InSite windows are the MAIN WINDOW, the PROJECT TREE, the GEOMETRY VIEW, the SIMULATION LOG, and the MOVIE PLAYER.

2.1.1 Main Window

The MAIN WINDOW, shown in Figure 2.1, is titled “Wireless InSite” when you first start the application. Once a project is loaded, the title changes to the name of the project. The Main Window is the parent window of the application.

- Closing the MAIN WINDOW will close down the application. All other windows can be closed independently of one another.

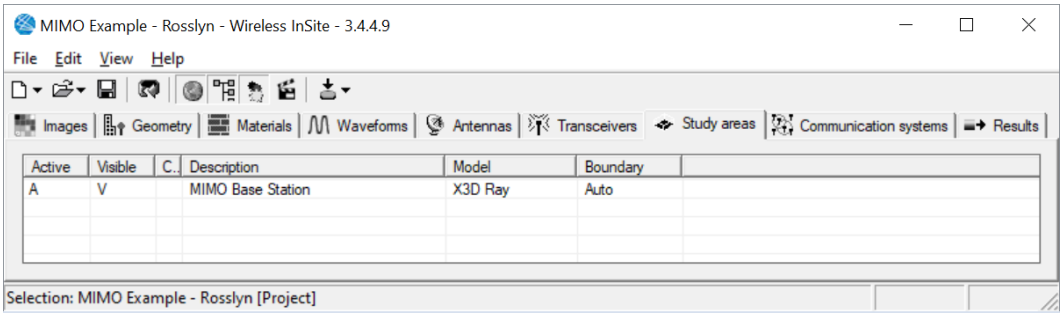


Figure 2.1: The Main Window

Once a project is loaded, the row of tabs below the toolbar allow the user to quickly view important project data. Much of this data is displayed in tabulated form. Other tabs use a hierarchal organization scheme. Right-clicking on any entry on a table opens a context menu. The complete editing options and properties for the selected entry are accessible from this menu.

Depending on the size of the MAIN WINDOW, some tabs may not be visible. When this is the case, you will notice two arrows ( ) to the right or left. These arrows can be used to scroll to the right or left along the row of tabs. It is usually possible to resize the window so that all the tabs are visible. All of Wireless InSite's primary windows can be resized and the new sizes will be saved and used at restart.

Each tab controls an important element of an Wireless InSite project, as described in the following sections.

IMAGES

This tab is a list of all the `*.tiff` images included in the project.

- See Chapter 9 for more information on images.

GEOMETRY

A GEOMETRY comprises all of the building or terrain data. Each geometry is subdivided into structure-groups, structures, substructures and faces. Geometries also contain data on the material properties of each face. The set of material properties are referred to collectively as "material types", and the properties and editing options for these are best accessed from the PROJECT TREE.

- See Chapter 3 for more information on geometry.

MATERIALS

The electromagnetic interactions of each face are determined by the properties of the MATERIALS assigned to the face. The display properties, such as the color and thickness, are also part of the material definition. The *Material filter* is used to filter the display of materials in the project's material table and the material database table. The filter is not case sensitive. Exact expressions can be entered in quotation

marks, and exclusion terms preceded by a minus sign (-) to further refine the filtering. For example, the filter text “dielectric” will show only dielectric materials, while the addition of the term “-concrete” will eliminate those dielectric materials which include concrete in their description.

- ▶ See Chapter 10 for more information on materials.

WAVEFORMS

WAVEFORMS describe the signal radiated from transmitter ANTENNAS and act as a kind of bandpass filter at the RECEIVER. Wireless InSite contains several built-in waveform definitions, the shape of which can be modified by providing carrier frequency and signal duration. All active and inactive waveforms associated with the Wireless InSite project are listed under this tab. Waveforms are assigned to both TRANSMITTERS and RECEIVERS.

- ✓ The *Waveform filter* field follows the same format as the filter in the MATERIALS tab.
- ▶ See Chapter 11 for more information on waveforms.

ANTENNAS

To perform propagation simulations using Wireless InSite, the model requires both TRANSMITTERS and RECEIVERS, each with an associated WAVEFORM and ANTENNA. When an antenna is added to a project and its parameters are set using the ANTENNA PROPERTIES Window, it can be used in multiple instances by associating it with any number of transmitters and/or receivers. The location and position of the antenna will be set by the location and position of the associated transmitter or receiver for each new instance of the antenna. Any number of antennas can be added to the project to simulate real-world scenarios or to test the effects different antennas have on propagation performance.

- ✓ The *Antenna filter* field follows the same format as the filter in the MATERIALS tab.

The antennas used in a simulation can be generated from the following:

- Built-in models for various common antennas including *Dipoles*, *Monopoles*, *Helical*, pyramidal *Horn*, *Rectangular Aperture*, *Rectangular Patch*, and *Parabolic Reflector*
- Synthesized patterns defined by several beam parameters
- Imported patterns from several antenna pattern data standards

- ▶ See Chapter 12 for more information on antennas.

TRANSMITTERS / RECEIVERS

TRANSMITTER RECEIVER and TRANSCEIVER locations and properties are defined in Wireless InSite by means of transmitter, receiver or transceiver sets, which contain one or more location point, an ANTENNA type and orientation, WAVEFORM and other parameters, depending on what type of set it is.

- ▶ See Chapter 13 for more information on transmitters.

STUDY AREAS

This tab lists all STUDY AREAS in the project. Study areas serve several purposes.

- They are used to select a region within the project that limits the space in which the simulation will find ray-paths. Only geometry within the study areas boundary will interact with the ray-casting, so the primary purpose of manually editing the study area boundary is to significantly decrease the runtime of the simulation for scenes with high face counts. Results will only be generated for transmitter and receiver points that lie within the region.
- As an organizational tool, they make it possible to keep predictions made with different parameters separate from each other. The user can create as many study areas as desired.

✓ Use *Duplicate* in the study area context menu to quickly create modified versions of a model.

▶ See Chapter 15 for more information on study areas.

COMMUNICATION SYSTEMS

The user can identify TRANSMITTERS and RECEIVERS to include in a new COMMUNICATION SYSTEM. This system is used to analyze the performance and observe the effect of changing system parameters on the quality of the communications environment.

▶ See Chapter 14 for more information on communication systems.

RESULTS

With this tab the user can access a hierarchical tree view of all currently available RESULTS, including graphs. The results are organized by STUDY AREA.

▶ See Chapter 21 for more information on results.

2.1.2 Geometry View

The GEOMETRY VIEW displays all the currently loaded data. After the simulations are complete, it is also possible to display most of Wireless InSite's propagation predictions, as shown in Figure 2.2.

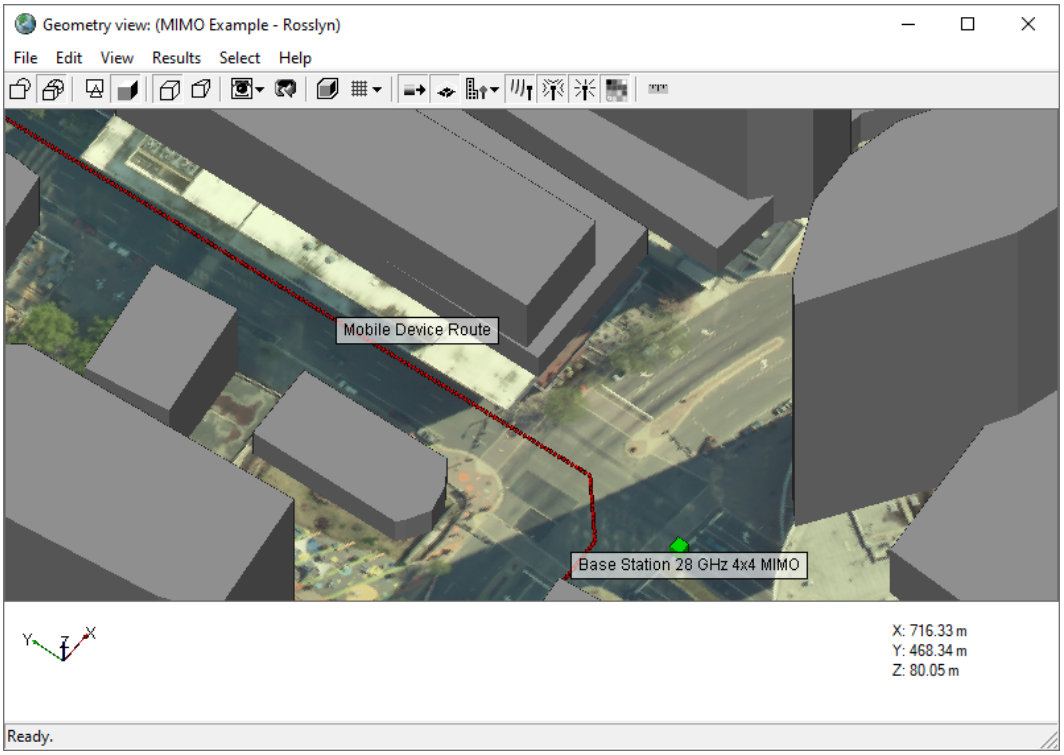


Figure 2.2: The Geometry View

- See Section 22.1 for more information on viewing RESULTS in the Geometry View.

The Menu Bar

The following menus can be found on GEOMETRY VIEW menu bar:

- *File* - The options listed here are identical to those found in the MAIN WINDOW's File menu. In addition to standard Windows menu items, the following options are available:
  - *Explore Project Directory* - Opens a separate explorer window, allowing the user to navigate through the project directory.
  - *Project Statistics* - Opens a window with statistics about the current project.
  - *Results properties* - Opens a window displaying result properties of the current project.
    - A detailed description of the PROJECT RESULT PROPERTIES window can be found in Section 21.4.
  - *Project Properties* - Opens a window displaying properties of the current project. Users can add a description and/or *Set global origin*, which defines the longitude and latitude of the Cartesian origin (0, 0).

- *Edit* - Selected objects in the GEOMETRY VIEW can be removed from the project. The APPLICATION PREFERENCES Window is also accessible under this menu.
  - ▶ See Section 2.7 for more on the APPLICATION PREFERENCES Window.
- *View* - Many of view controls are toggle switches, with a check mark indicating an active status. Several of these options can also be accessed from the toolbar of the GEOMETRY VIEW.
  - *Import* - Reads in settings for a view of the project from a file.
  - *Save* - Saves the current view to the project.
  - *Save as TIFF* - Exports the GEOMETRY VIEW to a `*.tiff` file.
  - *Save as JPEG* - Exports the GEOMETRY VIEW to a `*.jpeg` file.
  - *Orthographic / Perspective*, *2D / 3D*, *Wireframe / Solid-body* - See the **Viewing Modes** section below.
  - *Hi-lites* - See the **Toggle Buttons** section below.
  - *Face normals* - Toggles the display of vectors normal to the faces which compose the GEOMETRY. The green arrow is the outward pointing normal and the red arrow is the inward pointing normal.
  - *Descriptions* - Toggles the display of the short descriptions for all visible TRANSMITTER and RECEIVER sets in the project.
  - *Description backgrounds* - Toggles the display of a box around the *description* making it easier to read it.
  - *Grid* - See the **Toggle Buttons** section below.
  - *Legend* - Displays a list of the materials used by GEOMETRY in the project and the current grid spacing on the right hand side of the GEOMETRY VIEW. Users can access the properties for both by clicking on the square of color next to each entry.
  - *Reset*/*Refresh* - See the **Reset**/**Refresh** section below.
  - *Z-Scaling* - Changes the scale of the vertical axis in the drawing area. This only affects the view and not the Cartesian coordinates of the geometry.
  - *Study areas, Cities, Terrain, Foliage, Floor plans, Objects, Receivers, Transceivers, Transmitters, and Images* - Toggles the display of each of these objects.
- *Results*
  - *View* - Toggles the display of RESULTS results in the drawing area.
  - *Clear all results (no render)* - Eliminates the display of RESULTS.
  - *Flush results (unload)* - Clears RESULTS from memory.
  - *Reload results* - Reloads RESULTS from their data source and refreshes the display to reflect changes.
- *Select* - In order to select an object, it is first necessary to open the *Select* menu and specify its type. Click on the object to select it, and right-click in the GEOMETRY VIEW to access its context menu. To select multiple objects, press and hold CTRL and click on each object. To remove an

object from the group of selected items, press and hold SHIFT and click on the object. To deselect all currently selected objects, click on *Select*→*Clear*. This also clears the selection type.

The *Select* menu contains the following options:

- *Clear* - Clears all current selections and also clears the current selection type.
- *Geometry, Structure-group, Structure, Sub-structure, and Face* - Allows selection of geometry at each of these grouping levels.
- *Tx/Rx Set* - Allows the selection of an entire TRANSMITTER RECEIVER or TRANSCEIVER set.
- *Tx/Rx Point* - Allows the selection of an individual TRANSMITTER RECEIVER or TRANSCEIVER point.
- *Tx/Rx Point Pair* - Allows the selection of a pair of TRANSMITTER RECEIVER or TRANSCEIVER points. This is used primarily when viewing propagation paths between a transmitter and receiver or properties of the path. When selecting a new pair, it is only necessary to click on what is changing. For example, to change the receiver while keeping the transmitter, click only on the new receiver point. It can also be used to align boresights of two antennas.
- *Results* - Allows selection of a result. If the scale bar for the selected result is active it will display a value indicator.

✓ It may be difficult to select an object in WIREFRAME mode. Try switching to SOLID BODY mode and try again.

✓ It may be difficult to select buildings, TRANSMITTERS or RECEIVERS which are on top of terrain. If this happens, try turning off the display of the *Terrain* (*View*→*Terrain*) and selecting the object again. It may also be necessary to turn off the *City* (*View*→*Cities*) when selecting a transmitter or receiver which is adjacent to it.

✓ If you lose track of the last item that was selected you can center it in the GEOMETRY VIEW by selecting *Current Selection* from the *Reset*/*Refresh* menu. See the **Reset/Refresh** section below for this option.

Viewing Modes

Several viewing modes are available:

- WIREFRAME, in 2D or 3D
- SOLID BODY, in 2D or 3D
- ORTHOGRAPHIC, in 3D
- PERSPECTIVE, in 3D

The user has full control over zooming, rotating and panning in all viewing modes. When WIREFRAMEis active selected items will still appear in SOLID BODY. This allows the users to see more easily identify and view the selected item while still being able to visualize its surroundings.

Reset/Refresh

RESET changes the perspective of the GEOMETRY VIEW to the option selected. The default perspective is to look straight down on the project, filling the entire window, with the X-axis pointing to the right and the Y-axis pointing north. REFRESH forces a redraw of the GEOMETRY VIEW.

If *Current Selection* is selected from the menu the GEOMETRY VIEW will snap to the center of the bounding box of the selected item while maintaining the current rotation and zoom level.

Toggle Buttons

The toggle buttons control the following:

- HIGHLIGHTS - Toggles a wireframe around each face. This is only available in the SOLID BODY viewing mode.
- GRID - Accesses display settings of a reference grid in the XY plane. From the GRID PROPERTIES Window, seen in Figure 2.3, it is possible to toggle the grid display and change the spacing in degrees or DMS.

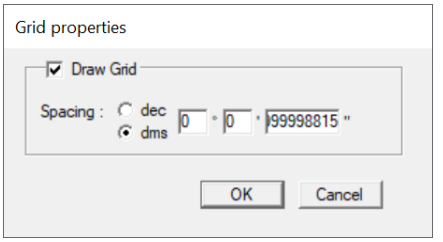


Figure 2.3: The Grid Properties Window

- DRAWING RESULTS - Toggles the display of RESULTS data in the drawing area.
- STUDY AREA - Toggles the display of the STUDY AREA boundary.
- GEOMETRY - Toggles the display of all GEOMETRY.
- TRANSMITTERS - Toggles the display of the TRANSMITTERS.
- RECEIVERS - Toggles the display of the RECEIVERS.
- IMAGES - Toggles the display of the IMAGES overlaid on the project geometry.

✓ The GEOMETRY button has an option to display *Selected Geometry Only*. This hides all other GEOMETRY in order to make it easier to view the selected piece.

The Ruler

The RULER feature is used for measuring distances. Once the ruler is activated, clicking the mouse will place reference points in the GEOMETRY VIEW describing a route. Each segment of the route will have its distance displayed in the window. The status bar at the bottom of the window will provide more details about the route including the total distance of all segments and the bearing of the segment from the last

point and the current mouse position. To clear the ruler, click the icon to clear the current route of points and start over, or right-click to deactivate it until it is needed again.

2.1.3 Project Tree

Wireless InSite's PROJECT TREE provides a convenient means to navigate within the input and results of a project. If the PROJECT TREE is not visible, it can be toggled by clicking on *View→Project tree* from the menu bar of the MAIN WINDOW.

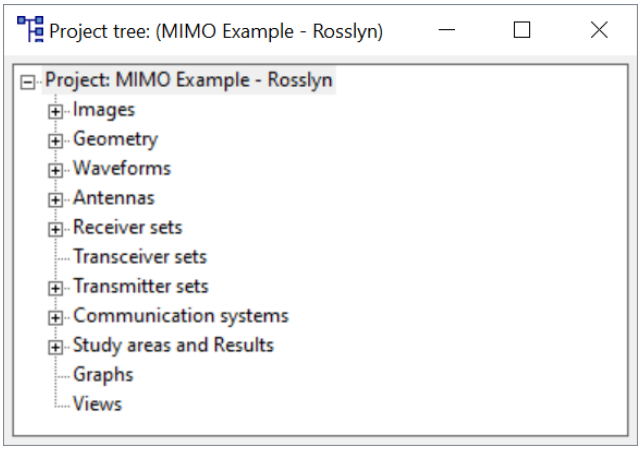


Figure 2.4: The Project Tree Window

Each level in the tree's hierarchy can be expanded to view the underlying levels, as shown in Figure 2.4. Absence of the expand sign indicates that the category is empty. The properties and editing options for most items are available in their context menus. The PROJECT TREE is especially useful for viewing and plotting RESULTS.

- *Antennas* - Lists all ANTENNAS in the project.
- *Geometry* - Includes *Cities*, *Terrain*, *Floor plans*, *Foliage*, and *Objects*. It expands to list the geometry's MATERIALS and structure groups. The structure groups can contain multiple structures, which can contain multiple sub-structures, etc. Typically all object faces are in the same substructure.
- *Images* - Lists all of the IMAGES in the project.
- *Graphs* - Lists all of the graphs in the project.
- *Receiver sets* - Lists all of the RECEIVER sets in the project.
- *Study Areas and Results* - Lists all of the STUDY AREAS in the project. Each study area can be expanded to display the available RESULTS data.
- *Transmitter sets* - Lists all of the TRANSMITTER sets in the project.
- *Views* - Contains the views saved from the GEOMETRY VIEW. Selecting a view under this tab will return the orientation of the Geometry View to the saved settings. It is possible to modify, rename or delete a view from the context menu.

- For more information on saving views in the GEOMETRY VIEW, see the entry under the *View* menu of Section 2.1.2.

- *Waveforms* - Lists all of the WAVEFORMS in the project.

2.1.4 Simulation Log

The SIMULATION LOG, as seen in Figure 2.5, is available by selecting *View→Simulation log* in the MAIN WINDOW. This window records all information generated while performing simulations, including the time and date the simulation starts and finishes. The information displayed in the SIMULATION LOG can be cleared or saved by right-clicking in the window and selecting *Close simulation log* or *Save log to text file* from the context menu.

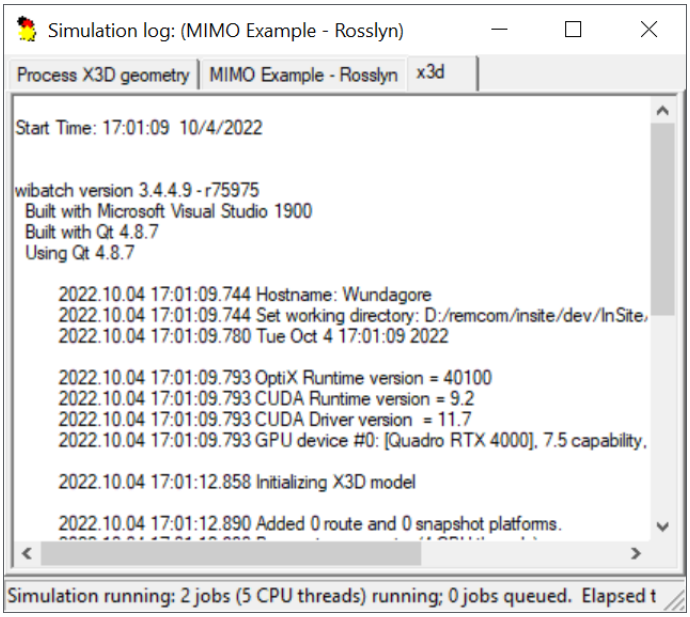


Figure 2.5: The Simulation Log Window

2.1.5 Movie Player

The MOVIE PLAYER provides a complete interface for selecting RESULTS, choosing visualization options, and controlling playback of animated E-fields and propagation path segments. The Movie Player is accessible by selecting *View→Movie Player*. While the Movie Player is open, all other windows are disabled except for the GEOMETRY VIEW, which permits changes to the viewing perspective (2D, 3D, WIREFRAME, SOLID BODY, etc.) while the movie plays.

- See Section 22.3 for more on the MOVIE PLAYER.

2.2 Context menus

Context menus are an important part of the Wireless InSite GUI and contain entries specific to the selected entity. Right-clicking on an entry in one of the MAIN WINDOW tables or on an entry in the PROJECT TREE will open a context menu for that entry. Context menus can also be opened by selecting and right-clicking on an object in the GEOMETRY VIEW. Some of the options on the context menus are also available from the static menu bars.

- See Section 2.1.2 for more on how to select an item in the GEOMETRY VIEW.
- Some context menu options may be grayed out if multiple objects have been selected.

2.3 Setting Up a Project

2.3.1 Creating a New Project

MAIN WINDOW→*File*→*New*→*Project*

Enter a short description and notes (optional) for the new project and press OK. The option to create a new project is also available from the context menu.

2.3.2 Opening an Existing Project

MAIN WINDOW→*File*→*Open*→*Project*

Use this option to locate and load a previously created project `*.setup` file and open it. The five most recently opened projects can be more easily accessed through the menu under MAIN WINDOW→*File*→*Recent projects*.

When opening project files that were created previous to 2.7 you may need to reestablish links from the `*.setup` file to other supporting files within the project directory such as geometry, transmitter and receiver files. If the current version does not support information in any of the files within the project, a warning message will appear detailing what information is not recognized. This will happen either because old information in the file is no longer valid, or you are opening a project file that was created with a newer version of Wireless InSite that has newly added geometries with new values associated with them. Some elements are given defaults if there is a problem reading their type. These are:

Antenna - If it is referred to by a transceiver set the default ANTENNA configured in the application preferences will be used.

Study Area Model - Defaults to X3D.

Waveform - If it is referred to by a transceiver set the default WAVEFORM configured in the application preferences will be used.

2.3.3 Loading a City File

MAIN WINDOW→*File*→*Open*→*Geometry*→*City*

Once a project has been created, it is possible to load *City* geometries from a data file. After navigating to the path above, locate the `*.city` file and open it. City geometries may also be imported into Wireless InSite from AutoCAD `*.dxf` files.

- See Section 4.2 for more on `*.dxf` conversion.

2.3.4 Loading a Terrain File

MAIN WINDOW→*File*→*Open*→*Geometry*→*GDAL Terrain*

Once a project has been created, it is possible to load *Terrain* geometries from a data file. After navigating to the path above, locate the `*.ter` file and open it. Terrain geometries may also be imported into Wireless InSite from other formats, such as `*.DTED` and `*.USGS`.

- See Section 5.3 for more on importing terrain geometries.

2.3.5 Creating a New Material

MAIN WINDOW→*File*→*New*→*Material*

Select the type of MATERIAL to create, and define its parameters in the MATERIAL PROPERTIES Window.

2.3.6 Creating a New Waveform

MAIN WINDOW→*File*→*New*→*Sinusoid Waveform*

When *Sinusoid Waveform* is selected the WAVEFORM PROPERTIES Window will open for you to define the new waveform's parameters.

If *Expose Calcprop options* is selected in the application preferences, multiple types of waveform can be created. The *Sinusoid Waveform* option will be replaced with a *Waveform* submenu containing available waveform types. Select the desired waveform type to open the WAVEFORM PROPERTIES Window.

2.3.7 Creating a New Antenna

MAIN WINDOW→*File*→*New*→*Antenna*

Under the *Antenna* submenu is a list of antenna types. Selecting the desired antenna type will open the ANTENNA PROPERTIES Window where you can enter additional antenna characteristics.

2.3.8 Creating New Transmitter Sets

MAIN WINDOW→*File*→*New*→*Transmitter set*

When creating a new TRANSMITTER set, the GEOMETRY VIEW will come to the foreground. The locations of the new set are defined by clicking the mouse on the drawing area. A right-click is used to exit

this phase of the procedure. The TRANSMITTER/RECEIVER PROPERTIES Window will open to complete the definition of the new set.

One method of creating new TRANSMITTER locations is to create a new set of control points along a *Route*. The segments defined by the distance between each of these points represent the route along which equally spaced transmitter points will be placed. The spacing between points is specified using the properties window, which appears when you right-click after creating the control points. Points can also be placed within the bounds of such geometries as an *Arc*, an *XY Grid*, and a *Cylinder*.

If there are no ANTENNAS and/or WAVEFORMS in the project, a default will be added to the project upon creation of the first transmitter or receiver set. Their properties are accessible from the sets' properties window by clicking the ellipsis button next to the description.

2.3.9 Creating New Receiver Sets

MAIN WINDOW→*File*→*New*→*Receiver sets*

See instructions for creating new transmitter sets above.

2.3.10 Importing Geometry from a CAD File

MAIN WINDOW→*File*→*Import*→*Geometry* →*DAE* <or> *KMZ* <or> *DXF* <or> *GDAL Terrain* <or> *SHP* <or> *STL* <or> *SAT*

GEOMETRY may be imported from CAD files in `*.dae`, `*.dxf`, `*.kmz`, `*.shp`, `*.stl`, and `*.sat` formats as well as many terrain formats supported by the GDAL library..

It is also possible to import multiple geometries of the same file type at once by selecting to import a directory rather than a single file at the time of import. This action will import all the files of the desired type contained within the specified directory. However, due to wide variances in formatting within types of CAD files, it is recommended that users verify that the files to be imported are consistently formatted, as uniform import settings are applied to all files selected for multi-import.

2.3.11 Importing Transceiver Sets from another Project

MAIN WINDOW→*File*→*Import*→*Transceivers*

All TRANSMITTER RECEIVER and TRANSCEIVER set data in a project is saved to a file *ProjectName*`.txrx`. Once a project is created or loaded, you can import transmitter, receiver or transceiver sets from other projects into the new project.

2.3.12 Creating New Study Areas

MAIN WINDOW→*File*→*New*→*Study area*

Wireless InSite STUDY AREAS define the propagation model and its input parameters to be used during simulation, the geometric area of the simulation, and the requested results to be generated during the simulation.

Wireless InSite STUDY AREAS make it possible to specify subsections of GEOMETRY data to be used in a simulation. GEOMETRY, RECEIVERS and TRANSMITTERS outside of the study area are not included in the computations. This allows Wireless InSite to load a large set of geometries but restrict the simulations to a smaller subset, saving considerable simulation time.

The user has the option to automatically *Fit to geometry*, including the Tx/Rx locations, or to *Specify location and size* of the STUDY AREA manually. The latter option allows the user to click on points defining the perimeter of the area in the horizontal plane. Right-clicking will connect the last point to the first. The only restriction on the boundary is that sides are not allowed to cross. Finally, the user specifies the lower and upper height of the study area boundary.

Users can further edit the study area boundary, as well as the type of propagation model and the associated parameters, in the STUDY AREA PROPERTIES Window.

- See Section 15.2 for an explanation of the Study Area Properties Window.

2.3.13 Changing Transmitter and Receiver Set Properties

Select the set using the MAIN WINDOW, GEOMETRY VIEW or PROJECT TREE and then right-click and select *Properties*.

2.3.14 Changing Material Types

To change the properties of an existing MATERIAL, first expand the PROJECT TREE down to the entry for that material. The levels to expand are *Geometry*→*City*→*Materials*. Right-click on the material type to be changed and select *Properties* from the context menu.

To change the MATERIAL assigned to a particular structure, first select the structure in the GEOMETRY VIEW or the PROJECT TREE, and then right-click and select *Change material*. Alternatively, one material can be substituted for another in a GEOMETRY through the REPLACE command. To do so, select a material which is currently in use, right-click, and choose *Replace*. In the window that appears, select a replacement material that is already in the project, or create a new material. All faces in the geometry which were assigned the selected material will be changed to the new material.

- ✓ The *Replace* command is also available for ANTENNAS and WAVEFORMS, and allows their assignment to TRANSMITTER and RECEIVER sets to be changed quickly.

2.4 Running Simulations

After project parameters are completely defined, it is time to generate results. The *Run* menu is accessible from the MAIN WINDOW by clicking the RUN button, as seen in Figure 2.6. Choose *Run Simulation* from this menu to begin a new simulation, which will produce the results that are selected in each active STUDY AREA's *Result Requests* checklist.

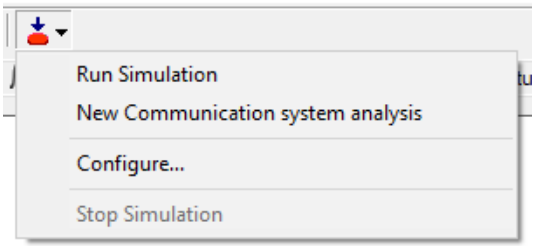


Figure 2.6: The Run Menu

The following options are available in the *Run* menu:

- *Run Simulation* - Run simulations for all active study areas. This will overwrite results that already exist.
- *Run Communication system analysis* - Run after adding or modifying a communication system.
- *Configure...* - Brings up the APPLICATION PREFERENCES SIMULATION tab which contains settings for how to run a simulation for active STUDY AREAS.
- *Stop Simulation* - Stops the currently executing simulation.

Simulation Modes Compatible with Legacy Models

When *Expose Calcprop options* has been enabled in the MODELS tab of theAPPLICATION PREFERENCES, additional RUN menu choices are available for use with the FULL 3D, URBAN CANYON, or VERTICAL PLANE legacy models. These models reuse a previous simulation's path database to generate new results when changes to the project are limited to changing materials, waveforms, or antennas, or adding new transmitter or receiver sets. The following options are available:

- *Rerun Simulation With*
  - *Modified materials* - Run after changing a geometry's material properties.
  - *Modified waveforms* - Run after changing center frequency or bandwidth of waveforms.
  - *Modified antennas* - Run after changing any antenna-related parameters.
  - *Additional transmitter sets* - Run after adding transmitter sets or making existing sets active.
  - *Additional receiver sets* - Run after adding receiver sets or making existing sets active.
  - *Additional result requests* - Run after adding requests for additional results.
- *Prepare cluster scripts...* - Brings up the CLUSTER SCRIPT GENERATOR which is described in detail in section 25.2.
- *Consolidate results from cluster* - When a simulation has been broken up and run on a cluster this command must be invoked to create the final result files and make them available in Wireless InSite.

It is important to note that the accuracy of results generated via one of the rerun options in the *Run* menu depends on the user being careful about what changes have been made to the project.

✓ See Section 2.7.3 for how to enable legacy model capabilities.

It is not possible to delete RECEIVER or TRANSMITTER sets selectively from the database. Deleting a set requires deleting the entire existing path database and requesting a new Wireless InSite simulation.

Changing the number of points or the spacing between points in a RECEIVER and TRANSMITTER set requires a new Wireless InSite simulation.

If *Stop Simulation* is chosen, the path database can get corrupted. The next run must be new to ensure the validity of the generated results.

2.5 Viewing Results

2.5.1 Selecting Result Data

Once the simulation finishes, all *Requested Results* are available for analysis. Results can be selected under RESULTS in the MAIN WINDOW or by using the PROJECT TREE. The results are presented in a tree view, which can be expanded to show the different types of results generated by all completed simulations. The highest levels in the RESULTS tree are the STUDY AREAS. Each study area which is not empty can be expanded to show the different types of results.

Wireless InSite's results are generated in two main result types: "point-to-multipoint" and "point-to-point". For "point-to-multipoint" data, "point" refers to a single TRANSMITTER point and "multipoint" refers to all the points on a RECEIVER set. For "point-to-point" data, "point" refers to a single TRANSMITTER point and single RECEIVER point. When results of a particular type exists, sub-items appear below it. The results are organized according to transmitter set, point number, and receiver set. For example, there might be an entry under *Received Power* such as "(short description for transmitter set 3), point #1→ (short description for receiver set 2)" for a result file that contains the values of that result type involving the first point in transmitter set 3 and all points in receiver set 2.

2.5.2 Creating Plots

To plot a result, locate the entry in the RESULTS tree for the result of interest and select *Plot* from the context menu, as seen in Figure 2.7. This will open the CHOOSE PLOT window which is used to specify properties of the plot desired when viewing the result on a graph. In the case of a receiver set, for example, the user has the option to plot the result as a function of distance or receiver number. If other graphs have been created, the user will be given the option of creating a new graph or adding the plot to an existing graph.

▶ See Section 22.2 for more on plotting results.

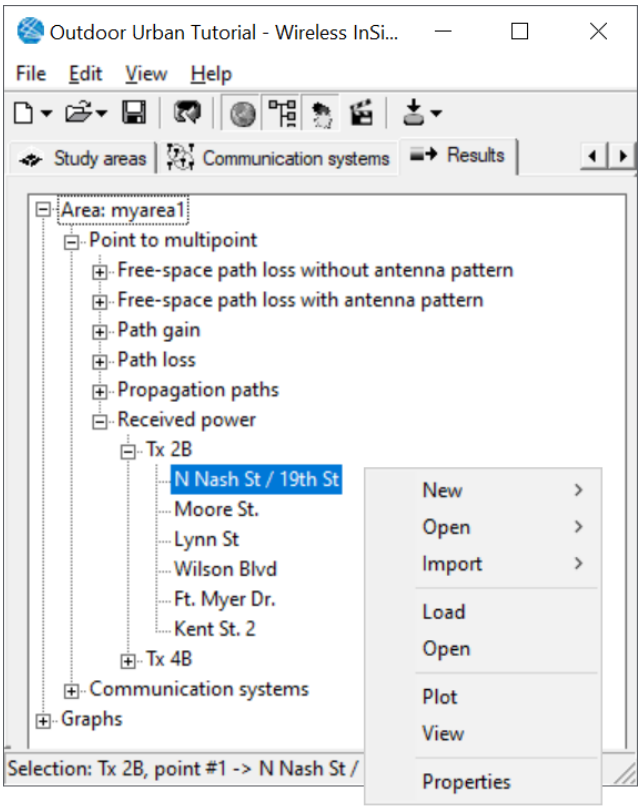


Figure 2.7: The context menu used to plot a result

2.5.3 Displaying Results in the Geometry View

To add a color-coded display of RESULTS data to the GEOMETRY VIEW, select *View* from the context menu. A color-coded representation of the power will appear. The scale bar in the lower-left corner of the GEOMETRY VIEW shows the numerical values of the colors. To change the scale on the color bar:

1. Right-click anywhere on color bar.
2. Select *Scale limits→Manual* scaling.
3. Set *range* and click OK.

If more than one type of result is displayed, the color bar mode can be changed by right-clicking on the color bar. Select *Scale mode* and then choose the result type.

▶ See Section 22.1 for more information on viewing RESULTS in the Geometry View.

2.5.4 Viewing Propagation Paths

The strongest propagation paths between a TRANSMITTER and RECEIVER can be displayed in the GEOMETRY VIEW. First, choose *Load* from the context menu for the particular receiver set of interest.

This is done in the RESULTS tree under the Propagation paths entry, which has one sub-item for each result that represents a particular Tx/Rx set combination. Once the data is loaded, expand the RESULTS tree to view the individual receiver points in the set. To display the propagation paths to one of these points, right-click on the point, and choose *View* from the context menu.

To clear RESULTS from the view, toggle the view action off by right-clicking on the viewed item. You will see a check mark next to *View* that can be turned off. You can clear all the viewed results by clicking *Results→Clear all results* on the GEOMETRY VIEW menu bar.

▶ See Section 22.1 for more information on viewing propagation paths.

2.6 Coordinate Systems

There are three underlying coordinate systems that are used in Wireless InSite: *Cartesian*, *Longitude/Latitude* and *UTM*. Each element that is rendered in the GEOMETRY VIEW has a local origin whose coordinate system can be adjusted manually in the elements' properties window. This defines the coordinate system in which the file for that object is written.

✓ To manually edit the vertices of an element, save the file in the same coordinate system as the object measurements, and then manually edit the resulting file.

The chosen coordinate system also affects how the editor displays/reads data. Following are detailed explanations of the requirements for each system:

- *Cartesian* - The vertices of the element are written to the file in Cartesian coodinates. They are based on the user-entered values for the latitude and longitude of a reference point, which Wireless InSite uses to properly place the element on the earth.
- *Longitude/Latitude* - The vertices of the element are written in latitude and longitude coordinates.
- *UTM* - The vertices of the element are written in UTM coordinates. In order to properly place the element on the earth, the zone number (1 to 60) and ellipsoid must be specified so that Wireless InSite can properly conform the points to the curvature of the earth.

Regardless of the coordinate system, the user can specify whether the elevation (Z-coordinate) is relative to *Sea level* or to the *Terrain*.

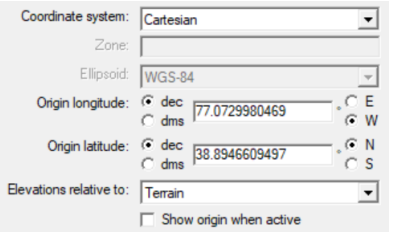


Figure 2.8: The coordinate system properties

2.7 Application Preferences

The overall appearance of Wireless InSite can be customized in several ways. To customize these aspects, select *Edit→Application Preferences* in the MAIN WINDOW, which will bring up the window shown in Figure 2.14.

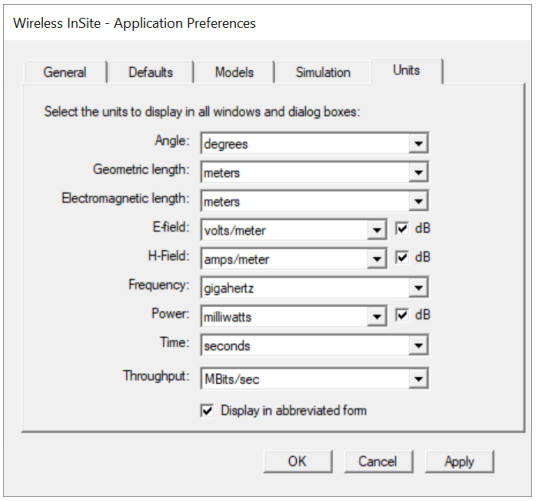


Figure 2.9: The Application Preferences Window

2.7.1 General

This section contains general settings that control how data in Wireless InSite is presented.

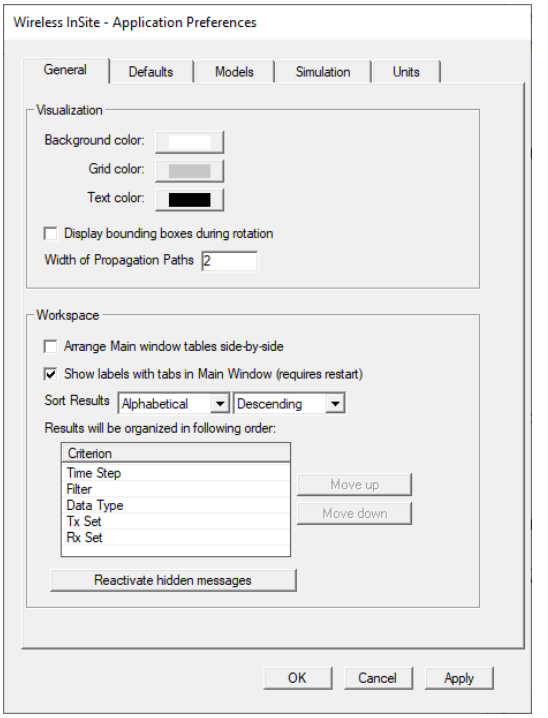


Figure 2.10: The General tab of the Application Preferences Window

- *Visualization*
  - *Background color* - This option will change the background color of all viewing windows. Any text in the viewing windows will be shown in a color that provides the best contrast to the newly selected color.
  - *Grid color* - This option will change the color of the grid lines in all editor windows.
  - *Text color* - This option controls the color of text within all of the viewing windows. It also controls the text of the Tx/Rx set descriptions that appear in the GEOMETRY VIEW.
  - *Display bounding boxes during rotation* - When selected, the GEOMETRY VIEW will render the geometry while the user is performing any movement commands. Instead, a box will be rendered for every structure to indicate its maximum boundary and position. This option should be selected for large and/or complicated geometries to improve rendering performance.
  - *Width of Propagation Paths* - Controls the width of propagation paths when they are rendered in the GEOMETRY VIEW.
- *Workspace*
  - *Arrange Main window tables side-by-side* - Some of the tabs in the MAIN WINDOW contain more than one table, such as the database tables for antennas, materials and waveforms. By setting these tables to be side-by-side as opposed above/below each other, the user can show more entries (with less detail immediately visible).

- *Show labels with tabs in Main Window* - This option is used when the user wants to reduce the width of the MAIN WINDOW and keep all tabs easily accessible.
- *Sort Results* - All of the possible sorting criterion for the PROJECT TREE are presented in the table, as shown in Figure 2.10. In addition, each level of the hierarchy can have its items sorted in *Ascending* or *Descending* order based on their short description (*Alphabetical*) or the order of the objects as they appear in their respective tabs in the MAIN WINDOW (*Project order*). The table controls the order of the grouping levels used in the RESULTS tree or under a STUDY AREA in the PROJECT TREE.
- *Reactivate hidden messages* - Various message boxes appear throughout the application and contain helpful information about the particular action being performed. Some of these messages can be hidden once the user becomes familiar with the application in order to allow work to be performed without interruption. This button will reactivate all of these messages so that they will appear the next time the action they are related to is performed.

2.7.2 Defaults

This tab, seen in Figure 2.11, contains the following preferences:

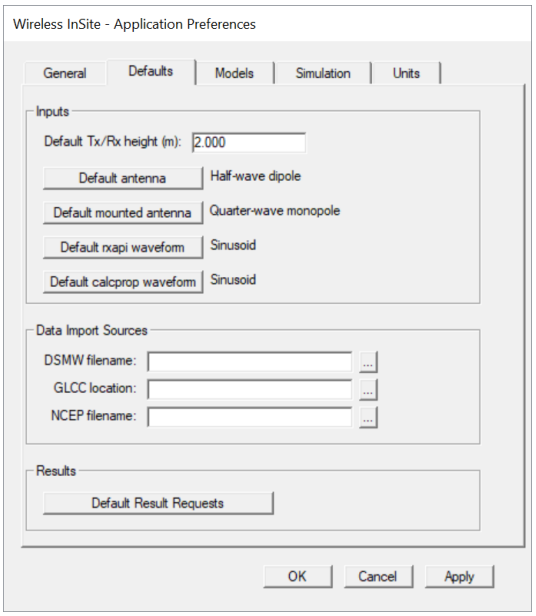


Figure 2.11: The Defaults tab of the Application Preferences Window

- *Inputs*
  - *Default Tx/Rx height* - New transmitter or receiver points will be initially set to this height.
  - *Default antenna and waveform* - If an antenna or waveform is required to complete an action and none currently exist in the project, these preconfigured options are automatically added and used for the current operation. When this happens, the user is informed by a pop-up

message. A default antenna is specified for both surface mounted and regular transmitter/receiver sets separately.

- *Data Import Sources*
  - *DSMW filename* - This is the location of the DSMW GIS file. This file must be in ERDAS 7.5 raster format. The default location in the installation directory is specified.
  - *GLCC location* - This is the location of the customized GLCC legend files that are required for the GLCC foliage import operation. The default location of the files in the installation directory is specified.
  - *NCEP filename* - This is the location of the NCEP soil moisture database file. Its default location in the installation directory is specified.
- *Results*
  - *Default result requests* - This opens the REQUESTED RESULT CATEGORIES list. This the default set of requests to use when a creating a new STUDY AREAṪo modify the requests for specific STUDY AREAS see Section 21.5.

2.7.3 Models

This section has a group for activating experimental features that are described in more detail in Appendix B. There are also options to *Expose Calcprop options* that are only supported by the legacy models FULL 3D, VERTICAL PLANE, and URBAN CANYON and to *Use Calcprop engine to calculate antenna patterns* which switches between using legacy models and the *X3D* model to calculate the pattern that is displayed for analytic antennas.

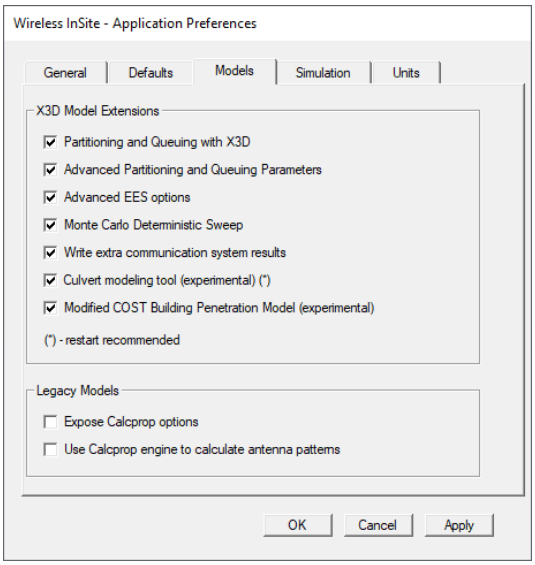


Figure 2.12: The Models tab of the Application Preferences Window

2.7.4 Simulation

Wireless InSite has the ability to make use of multi-core processors and can run multiple instances of the X3D, FULL 3D, URBAN CANYON, and VERTICAL PLANE propagation models in parallel for projects that contain multiple TRANSMITTER points or RECEIVER sets. When a new run is initiated by selecting RUN→*New* from the menu, the user interface will initiate multiple simulations based on the settings listed under the SIMULATION tab, as seen in Figure 2.13.

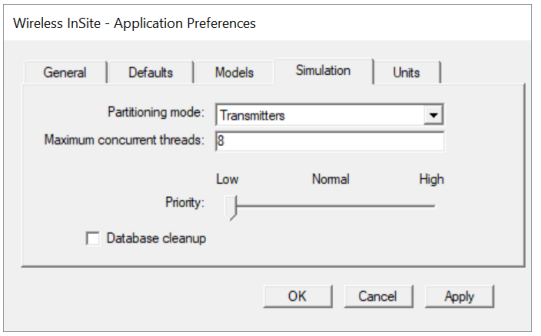


Figure 2.13: The Simulation tab of the Application Preferences Window

- *Multi-thread mode* - This controls if the simulation is distributed by TRANSMITTER points or RECEIVER sets. For majority of projects with multiple TRANSMITTER points, using the TRANSMITTER option will result in the greatest decrease in simulation time. The RECEIVER set option will only result in faster run times when the time to evaluate the result at the RECEIVER locations greatly exceeds the ray tracing portion of the simulation.
- *Maximum concurrent threads* - This controls how many process threads can be used at the same time during simulations. This limit should not be set higher than the total number of logical processor cores that are on the PC running the simulation, as there is no advantage to be gained beyond this point.
- *Priority* - This sets the priority of the processes. It allows the user to prevent simulations from interfering with other processes that have more immediate priority or allows the simulation to use more resources to complete faster.
- *Database cleanup* - Remove the `*.sqllite` database file after a simulation completes. This file can be very large and is not required by the UI to browse results.

2.7.5 Units

Under this tab, the user can change the units of measure for all data displayed in Wireless InSite. By checking the *Display in abbreviated form* option, units will be shown as an abbreviation (e.g. “m” instead of “meters”). All aspects of the application that involve the display of quantitative information will immediately update to reflect any changes made, including graphs that are currently open.

✓ The *Electromagnetic length* preference controls the units used to describe ANTENNA dimensions and MATERIAL thicknesses, and is separate from geometric lengths.

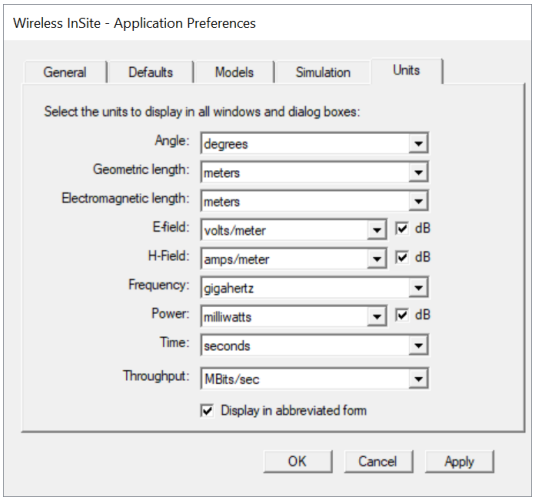


Figure 2.14: The Units tab of the Application Preferences Window

2.8 Images

Overlay of the imported image files in the GEOMETRY VIEW can be performed by selecting *File→ Open→Image*. The position of the image and the distance per pixel of the image is controlled in the IMAGE PROPERTIES Window. The default settings will result in the image fit to the GEOMETRY as well as the TRANSMITTER and RECEIVER sets in the project. Cropping the image is also possible by checking the *Crop image* option when the image is loaded. The IMAGES tab in the MAIN WINDOW lists all images that have been added to the project.

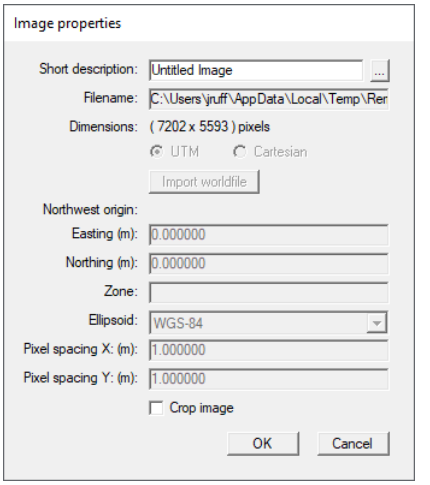


Figure 2.15: Adding an image to the project

2.9 Simulation Engine

The simulation engine for Wireless InSite is called `calcprop`. It is located beneath the bin directory of the Wireless InSite installation path on Linux, or in `install_location\\calc*` by default in Windows. The simulation engine may be run from the command line.

- See Section 25.1 for more on simulation engine command line options.

Chapter 3

Geometry

In this chapter, you will learn...

- → how geometries are structured at the most basic level
- → the differences among geometry types
- → how to open a geometry from COLLADA or KMZ files
- → how to export existing geometry to COLLADA or KMZ files
- → the properties and operations of geometries

There are five main GEOMETRY types used in Wireless InSite to define problem geometry: cities (`*.city`), Floor plans (`*.flp`), foliage (`*.veg`), objects (`*.obj`), and terrain (`*.ter`). This chapter describes the general composition, properties, operations and file structure of geometries.

3.1 Hierarchical Organization of Geometry Data

GEOMETRY are essentially composed of planar polygons with three or more vertices, also known as "faces." The coordinates of the vertices are specified with respect to the same coordinate axes. The vertices on each face are defined in counter-clockwise order, with the outward normal given by the right-hand rule.

The FULL 3D, VERTICAL PLANE, and URBAN CANYON propagation models make a distinction between "one-sided" and "double-sided" faces. When faces are "one-sided," rays will only interact with the face when incident on the side with the outward normal, and they will not "see" the face from the "back" side. This means that in order to model a knife edge or a free-standing wall using these models, it is necessary to either place two facets back-to-back or specify that the facet is "double-sided." Each face is also assigned a MATERIAL. X3D lets rays hit either side of a face, ignoring the single-sided versus double-sided property of a face.

The geometric data is grouped in a hierarchy. Its levels, from lowest to highest, are described below.

- **Face** - An individual planar surface described by a set of coplanar points that form its perimeter.
- **SubStructure** - A collection of faces; each must share an edge with at least one other face. The faces of *Terrain* geometry are usually all contained in one sub-structure.
- **Structure** - A collection of sub-structures; each must overlap with at least one other. If there is only one sub-structure in the structure, then no overlap with another sub-structure is necessary.
- **Structure-Group** - A collection of structures. This is typically used to group structures into city blocks. However, an entire city could be loaded into one structure-group if desired. Typically, the DXF converter will load all the structures into one structure-group.
- **Geometry** - A collection of structure-groups, which together form a complex model.

Each level can contain multiple items of the level below it. For example, a sub-structure can have more than one face, a structure more than one sub-structure, etc. Figure 3.1 illustrates the hierarchal organization of geometrical data comprising each GEOMETRY.

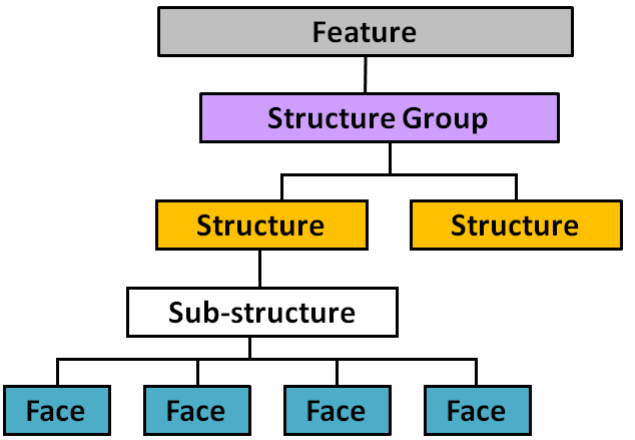


Figure 3.1: Typical organizational hierarchy of geometry

3.2 Geometry Types

The GEOMETRY available and descriptions of their specific properties are listed below.

- *City* - Represents the layout of buildings and structures that make up a city. The CITY EDITOR provides ways to create and delete building structures, set building heights and change the footprint of a building. In the X3D and FULL 3D propagation models the buildings can have any shape, but in the URBAN CANYON model the buildings must consist of rectangular vertical walls and a horizontal roof.

  ▶ See Chapter 4 for more information on *City* geometry.

- *Terrain* - Represents the landmass underneath all of the other geometries that are in a project. The editor is geared towards creating new facets and setting heights of vertices to create the terrain

profile. In most cases a new terrain will be imported from a specific file format, such as `*.DTED` or `*.DEM`, and not need to be edited.

▶ See Chapter 5 for more information on *Terrain* geometry.

- *Foliage* - Represents an area of vegetation on the terrain. Foliage areas are represented by a polygonal shape only. When creating new foliage, it is important to pick a material that has the characteristics of the vegetation contained in it before creating the representation of its area. After creating foliage, you can change the material it uses by selecting *Choose Material* in its context menu under the *Geometry* tab.

  ▶ See Chapter 6 for more information on *Foliage* geometry.

- *Floor plan* - Represents the layout of the room and hallways that make up a floor in a building. The editor allows the user to add ceiling, floors and walls and provides a simple interface for creating doors and windows.

  ▶ See Chapter 7 for more information on *Floor plan* geometry.

- *Object* - Represents items such as office equipment, vehicles, signposts, etc. The editor enables you to create simple polygonal shapes, but in most cases objects will be imported from one of the CAD formats that are supported.

3.3 Importing COLLADA Files

Geometry can be imported from COLLADA files which can also be embedded in KMZ files in order to provide geographic reference information so that the Geometry will be correctly located in the GEOMETRY VIEW.

To import a COLLADA `(*.dae)` or KMZ `(*.kmz)` file in an existing project, select *File→Import→ Geometry* and select the type of file to be imported. Wireless InSite will prompt you for the type of geometry that the file will be. The option to *Import by folder* is also present, allowing multiple files of the same type to be imported by specifying a directory to import rather than a single file.

A KMZ file will contain the following files when uncompressed:

1. `doc.kml` - Provides the information to position the asset within the scene.
2. `models/<kmz_filename>.dae` - The COLLADA asset.
3. `models/<kmz_filename>.shadow` - A version of the asset that contains information about the materials used by the Geometry that is required by the simulation models

3.4 Exporting COLLADA Files

Geometries can be exported as either individual COLLADA files or embedded into KMZ files with the corresponding geographic reference information included. Geometries can be exported on an individual basis, or all geometry in a given project can be exported at once.

To export an individual COLLADA `(*.dae)` or KMZ `(*.kmz)` file in an existing project, select the desired geometry on the GEOMETRY tab in the MAIN WINDOWand from the right-click context menu select

either *Export→COLLADA* or *Export→KMZ*. The new export will be placed into the project folder, and will be given the original geometry's file name automatically.

To export COLLADA (`*.dae`) or KMZ (`*.kmz`) files for an entire existing project, select *File→Export*, select the type of export desired, and then select which set of geometry should be exported:

- *Active Geometry* - Exports all currently active geometry in the project.
- *All Geometry* - Exports all geometry in the project.

3.5 Geometry Properties

The GEOMETRY PROPERTIES Window, seen in Figure 3.2, displays the hierarchy of structure types and faces represented by the tag structure of the file. To access this window, right-click on the geometry under the GEOMETRY tab in the MAIN WINDOW, and select *Properties*.

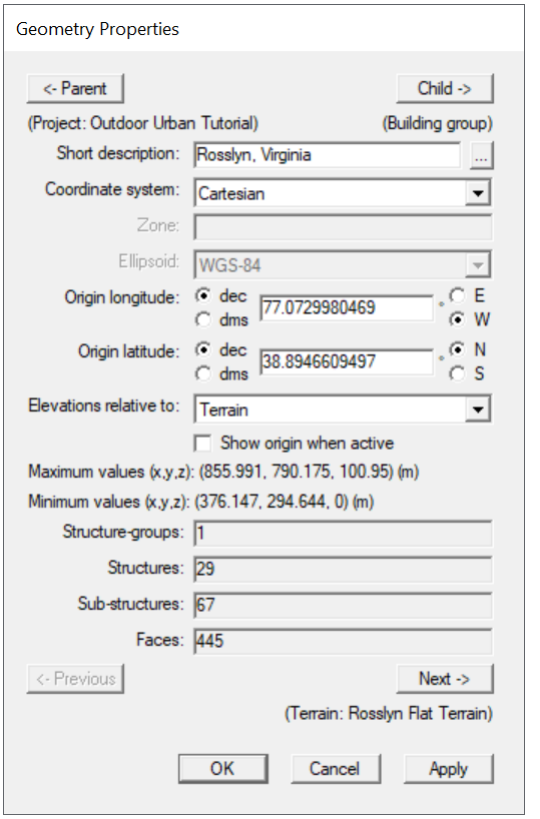


Figure 3.2: Properties window for a geometry

Information used to generate the project statistics, as well as the short description and notes, are available from this window. The lower half of the window displays information about the region that the geometry occupies and the number of sub-level structures. By using the buttons PARENT and CHILD, it is possible to browse the GEOMETRY PROPERTIES and PROJECT PROPERTIES which contain the geometry. The

PREVIOUS and NEXT buttons allow you to cycle through the geometries in the project, and when you are at a lower level you can cycle among the elements at that level as well. For example, clicking the CHILD button in this case shows the properties for the first structure group, and clicking on the NEXT button would begin cycling through the other structure groups within the geometry.

The GEOMETRY PROPERTIES window for a terrain contains a *Do sparse rendering* check box that allows the user to choose to render the terrain at a lower resolution. This only modifies the on-screen rendering of the terrain to improve performance and does not affect the level of detail that will be used for simulations. A manual value for the resolution can be entered in the *Sparse rendering distance* field. When importing or opening a terrain with a high face count, Wireless InSite will produce a warning of potential performance issues and suggest to use sparse rendering with an automatically generated resolution value. Note that the *Color by height* function is disabled while using sparse rendering.

3.6 Editing Vertex Coordinates

The FACE PROPERTIES Window allows exact specification of vertex coordinates to precisely position and size a face. To access this window, select a face, right-click and select *Properties*. To edit the coordinates, click on the VIEW VERTICES button and edit the values in the VIEW/EDIT VERTICES Window, as seen in Figure 3.3. Double-clicking any entry will enable editing.

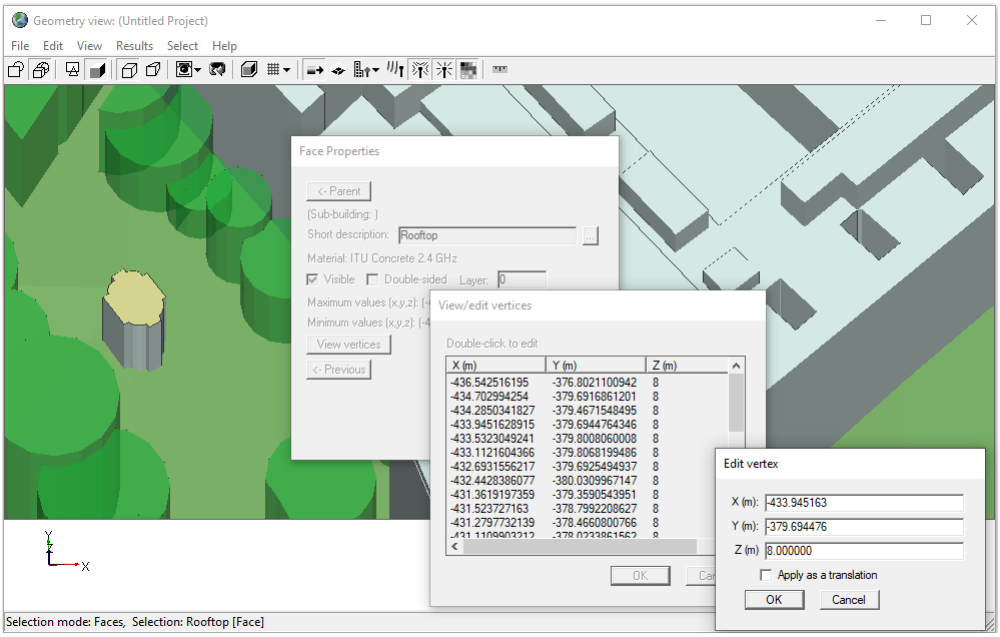


Figure 3.3: Modifying a face by manually editing its vertices

3.7 Geometry Operations

A number of operations are available to position and modify the properties and appearance of each GEOMETRY. To perform an operation, select a geometry by navigating to *Select→Geometry* from the GEOMETRY VIEW. Click on the desired geometry, or select multiple geometries by holding the CTRL key while clicking on each one.

✓ If finer control is needed for any of the operations, then groups of faces can be selected at any of the intermediate levels of the geometry hierarchy.

▶ See Sections 2.1.2 and 2.1.3 for more information on selecting items in the GEOMETRY VIEW or the PROJECT TREE.

The geometry context menu includes the following operations.

- *Graph profile* - Creates a graph of the terrain profile. After selecting the option, left-click and drag across the GEOMETRY VIEW to specify the start and end points of the X-axis of the plot.
- *Render smooth* - Displays the terrain as a smooth surface.
- *Remove empty levels* - Cleans up the geomtry's organization to only sections that have faces.
- *Remove unused materials* - Removes any materials from the geometry that are no longer associated with any of its faces.
- *Visibility* - Sets the selected items as visible or invisible (transparent). In either case, the items will still be included in simulations.
- *Sided* - Specifies whether a face is single- or double-sided.
- *Color by height* - Assigns a color to the faces of a geometry according to height. Select this option again to turn it off.
- *Change height* - Provides two alternatives for changing the heights of all the selected structures. The first is to assign the same height to all selected structures. The second is to multiply the current height by a scaling factor.
- *Change height* (*Foliage* only) - This allows the user to specify the top and bottom heights of the selected canopies.
- *Change material* - Changes the material assigned to all selected items. This action will bring up the CHOOSE MATERIAL window as shown in figure 3.4. This allows specification of a material to assign to the selected geometry from one of the following sources.

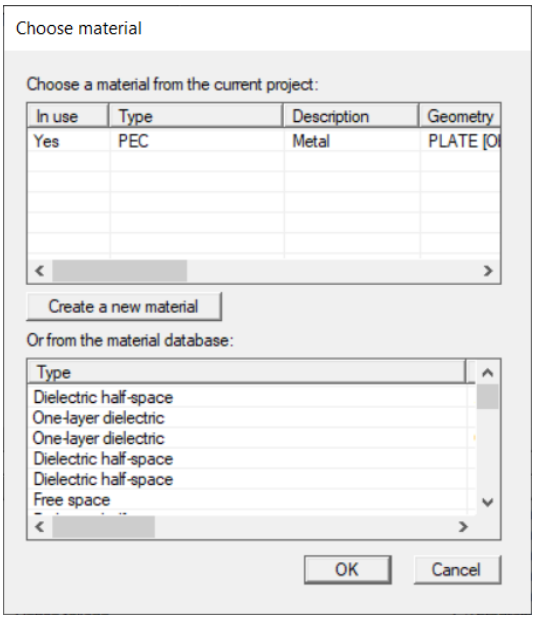


Figure 3.4: Selecting a material to apply to selected geometry.

  - *From the current project* - Select from materials that are currently owned by geometry in the project. Each geometry has a separate list of materials for its faces so if you select a material that does not have an equivalent one in the geometry of the selected face(s) a duplicate is created for them to use in that context.
  - *Create a new material* - This is the same as selecting *File→New→Material* in the MAIN WINDOW menu, but makes it available for immediately use in this operation.
  - *From the material database* - See Section 10.4 for information about how to populate and configure materials for reuse in the database.

  An engineered surface material can only be assigned to a single face using the *Create a new material* option. This is done so that the origin of the material can be initialized with a value that would make it flush and centered on the face.

  At the face level this action will be under a *Material* sub-menu that includes options for creating and assigning an EES material file if one is not available as described in section 10.7.

  ▶ See Chapter 10 for more information about assigning material properties.

- *Translate in* GEOMETRY VIEW - Uses the GEOMETRY VIEW to do a visual translation in the x-y plane. When this is activated the part of the GEOMETRY that will act as the anchor point for the operation must be selected. The GEOMETRY can then be moved around in the GEOMETRY VIEW until the desired location is achieved. Clicking at this point will translate the GEOMETRY to that position. In figure 3.5 the southeast corner of a building was selected as the anchor point and then used to translate it to the position indicated by its outline at the new location.

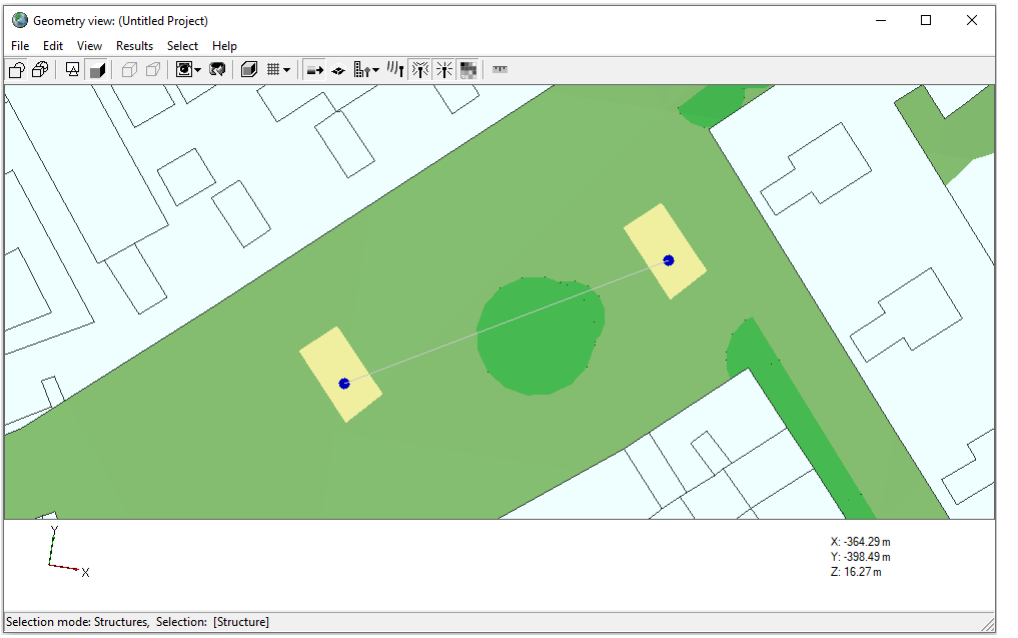


Figure 3.5: Visually translating a building in the Geometry View.

- *Translate* - Moves any item by selecting *Translate→Relative to local origin*. This will bring up a window in which the Cartesian components of the translation vector can be specified. The other option in the translation menu is *Coordinate reduction*. Selecting this option will automatically translate the local origin for the geometry to its center.
- *Scale by* - Increases or decreases the size of the selected item's along each principal axis by the specified factors. Separate factors can be entered for each Cartesian axis. When the scaling is complete, the selected items will be positioned so that the position of the object's control point is unchanged in the global frame.
- *Rotate* - Brings up a window for selecting the axis and rotation angle (in degrees). A positive angle gives a counter-clockwise rotation, and a negative angle a clockwise rotation. The axis of rotation is defined by selecting two points (point, point), a point and vector (point, vector), or the principle axes of the global reference frame (X, Y, or Z).
- *Pre-processing* - Displays all of the operations that can be performed on the geometry to make it easier to interact with the simulation or to make it run more efficiently.
  - *Add indices to description* - (All geometry) - Adds a number to the beginning of each item's short description indicating its position within the next higher grouping level.
  - *Join coplanar faces* - (*City*, *Terrain*, or *Object* only) - Joins all faces that share a common edge and lie in the same plane.
  - *Validate faces* - (*City*, *Terrain*, or *Object* only) - Displays the faces in the geometry that prevent it from creating a proper normal. The failure conditions for a face check are:
    1. The vertices do not describe an area.

2. Two or more sides intersect.
3. A side is described by two or more colinear points.
4. All vertices do not lie in the same plane.
5. The face has less than three vertices.
6. The face describes an acceptably large area, but is too narrow.

Such faces are ignored by the simulation and should be edited to correct the problem or deleted before running a simulation.

✓ Generally, deleting the faces that fail validation for the last two conditions will not affect the results.

– *Group sub-structures* - (*City* only) - Combines all of the sub-structures within the group so that all sub-structures that overlap and only sub-structures that are part of a chain of overlapped sub-structures will be contained in the same structure when finished.

– *Import foliage coverage from GLCC database* - (*Terrain* only) - Opens the IMPORT CONTROLLER Window for geo-referenced foliage import.

▶ The GLCC database is described in Section 6.2.

– *Extract soil information from DSMW database* - (*Terrain* only) - Allows the user to change the terrain material to a special composition from the Digital Soil Map of the World database.

▶ The DSMW database is described in Section 5.7.

– *Simplify* - (*City* only) - Opens the BUILDING SIMPLIFICATION Window.

▶ The BUILDING SIMPLIFICATION Window is described in Section 4.6.

– Available only when there is a *Terrain* in the project:

∗ *Crop terrain to footprint* - (*City* only) - Removes all faces of the terrain that do not lie within the region of the city.

∗ *Extrude to terrain* - (*City* only) - Solves a common problem when running a simulation. The issue is that the building walls may not be in contact with the terrain. This will introduce errors since rays may propagate under these 'floating' buildings. When selected, the floors of all sub-structures will be lowered to the local terrain elevation so that all walls of the sub-structure touch the terrain. The rooftops will remain at the same elevation as they were at the beginning of the operation.

∗ *Lower to terrain* - (*City* or *Foliage* only) - This will perform a vertical translation of all structures so that their bottoms are flush with the terrain surface.

∗ *Conform to terrain* - (*Foliage* only) - This will shape the foliage to follow the contours of the terrain. This will make a best guess about the desired height of the foliage structures. For more control over this use the *Change height* option.

– *Simplify Geometry* - (*Object* only) - Provides a method for removing excessive details from objects that can result in slow simulation times and that do not affect the results. For example, an imported model of a vehicle may include details of the tire treads, but at a low enough frequency this appears as a flat surface and should be removed.

– *Triangulate Geometry* - (*Object* only) - Splits all of the faces in the geometry into triangles. If a geometry has faces with more than three vertices, the simplification process will not work correctly. This is provided as a separate operation as it can be computationally expensive for large geometries and should be avoided when possible.

3.8 Face Operations

Similar to the operations available for modifying GEOMETRY, there exists a set of operations to edit the position, appearance, and properties of faces. While most of the operations for a selected level of a geometry (i.e. structure group, structure, or substructure) work on the faces contained within that level, the additional operations listed below are specific to faces and are not available at any other level in the geometry hierarchy. The operations will be applied to selected faces only.

▶ See Section 2.1.2 for a description of how to select objects.

- *Cover with (Transmitter or Receiver Points)* - Applies a uniform grid of transmitters or receivers to the surface of a face.
- *Place Points (Transmitters or Receivers)* - Opens the selected face in an editor window where transmitter or receiver points can be placed at specific locations.

  ▶ See Chapter 13 for more information about the Points-on-Face Tx/Rx set.

- *Sided (Single or Double)* - Determines if the face will be visible only from the front (single) or both sides (double).
- *Change material* - Changes the material that composes the face.
- *Reverse normals* - Reverses the direction of the face normal, effectively flipping the face in its current position.
- *Translate, Scale by, and Rotate* - Performs normal geometric operations.

Chapter 4

Cities

In this chapter, you will learn...

- → how to create and edit buildings
- → how to import DXF files, ESRI files, and Raster data
- → how to pre-process and simplify buildings and cities

Wireless InSite requires full three-dimensional building data for modeling propagation in urban environments. A growing number of sources now provide accurate data for a large number of cities. This data is available in a number of formats, including AutoCAD's DXF and ESRI Shapefiles. Wireless InSite is able to read and convert these files to its *.city file format. Once the file is read, material properties can quickly be assigned to each building. Wireless InSite can also fit buildings to the underlying terrain or shift buildings horizontally to align with the terrain. Building data can be in latitude/longitude, UTM, or Cartesian coordinates.

The excessive detail sometimes included in imported city files can considerably increase computation time. In some cases it can even produce less accurate results. Wireless InSite contains an automated building simplifier which re-formats *.city files to enhance the performance of the propagation models.

Users can make manual changes to urban geometry with Wireless InSite's CITY EDITOR. This tool enables users to create new cities, add buildings to cities, or modify existing buildings by defining their horizontal footprint and their height.

4.1 City Editor

The Wireless InSite CITY EDITOR, as seen in Figure 4.1, can be used to create urban geometry and to edit existing buildings. Each building is created by defining the footprint in the horizontal plane and then entering the height of the building. The side walls are all vertical rectangular faces, and the roof is a flat, horizontal polygon. No floor is created for the building. The materials for the walls and roofs of the buildings that are added to the *City* can be set separately.

By default, both building walls and roofs are set to concrete.

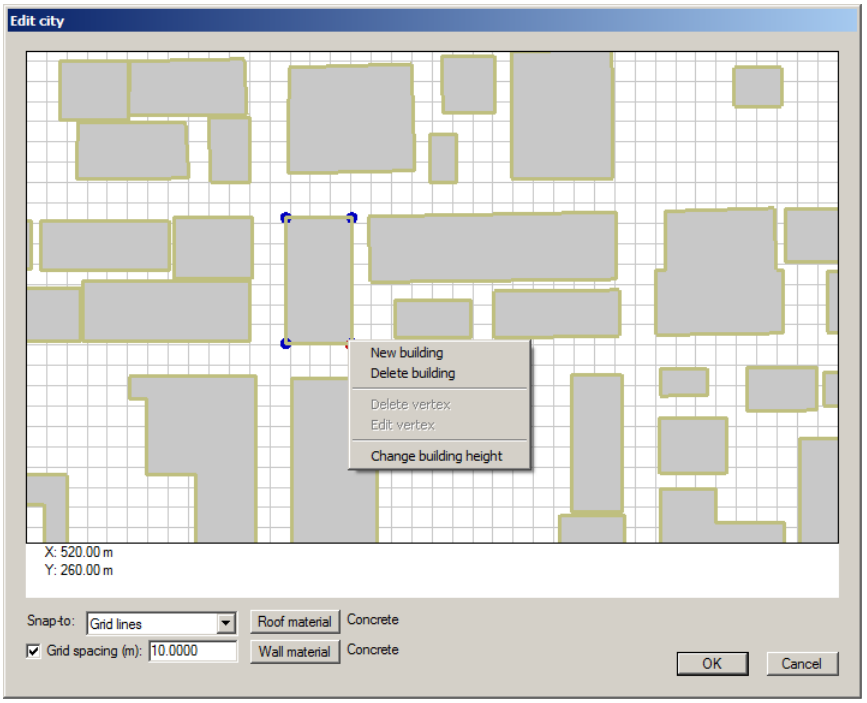


Figure 4.1: The City Editor

Clicking on the button for each material allows the user to either view the current materials properties, or to replace the material with:

- A new material of a different type
- A material in the material database
- A material that is in another non-organic geometry within the project.
  - ✓ If this option is used, then the geometry currently being edited will make a copy of that material for its own use

Creating a New Urban Geometry

To create a new urban geometry:

1. Click on *File→New→Geometry→City*
2. Select *New building* from the editors context menu.
3. Rotate, zoom or pan the view whenever necessary. The EDIT CITY Window may also be resized or maximized if needed.
4. Define the footprint by clicking on the vertex locations, then right-click after the last point.

✓ There is no need to close the footprint by clicking on the starting vertex.

5. Enter the *Height*, as seen in Figure 4.2.

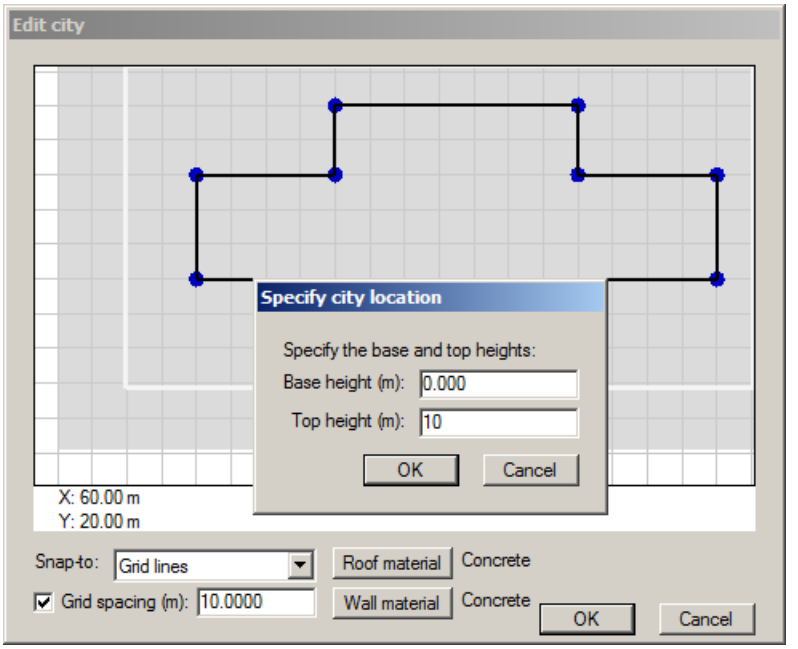


Figure 4.2: Creating a new building in the City Editor

6. Repeat Steps 2-5 to add another building.
7. Click OK to close the EDIT CITY Window.

When drawing the footprint, it may be useful to turn on the grid by clicking the check box in the lower left. The grid spacing can be set to any desired interval. If the vertices of the footprint fall exactly on the grid it may also be useful to set the *Snap-to* mode at the lower left to "Gridlines."

Editing a Building Footprint

To edit the footprint of an existing building substructure:

1. Set the selection mode in the GEOMETRY VIEW to *Geometry* (*Select→Geometry*).
2. Select the substructure and choose *Edit* from the context menu to open the EDIT CITY Window.
3. Rotate, zoom or pan the view if necessary.
4. In the EDIT CITY Window, click on the desired building substructure.
5. **To graphically move a vertex**: set the *Snap-to* mode to "Off" and then click and hold on the vertex and drag it to the new position.
6. **To move a vertex to an exact coordinate location**: right-click on the vertex and select *Edit vertex*. The X, Y, and Z coordinates of the point can be entered in EDIT VERTEX Window.

7. **To delete a vertex**: set the *Snap-to* mode to "Vertices," right-click on the point and select *Delete vertex*.
8. **To add a point**: click on the line between two existing points.
9. Repeat Steps 5-8 to change other substructures on the selected geometry.
10. Click OK to close the EDIT CITY Window.

4.2 Importing DXF Files

DXF files can be created in many different ways. Unfortunately, not all of these are compatible with Wireless InSite. The Wireless InSite DXF converter can only convert the following DXF objects:

- Polylines
- Polyface meshes
- 3D faces

- Roofs should be placed on the tops of all buildings to ensure accurate simulations.
- Buildings should be created using AutoCAD "grip points" to specify points used for face edges. This will force the spatial positions of the face points to be identical. This practice is especially important when the point is an endpoint of an edge that is common to two faces. If the face edges are not exactly coincident, the DXF converter cannot identify the faces as touching.
- Architectural symbols (e.g. door swings, graduated scales, lettering, window and stair symbols) should be removed prior to import.

To import a DXF file into an existing project, select *File→Import→Geometry →DXF*. The IMPORT FEATURE Window will appear. Choose *City* and click OK. The IMPORT DXF DATA Window will appear, as seen in Figure 4.3. Ensure that the *Import type* is set to "Single file", and select the desired DXF file as the *Import Source*.

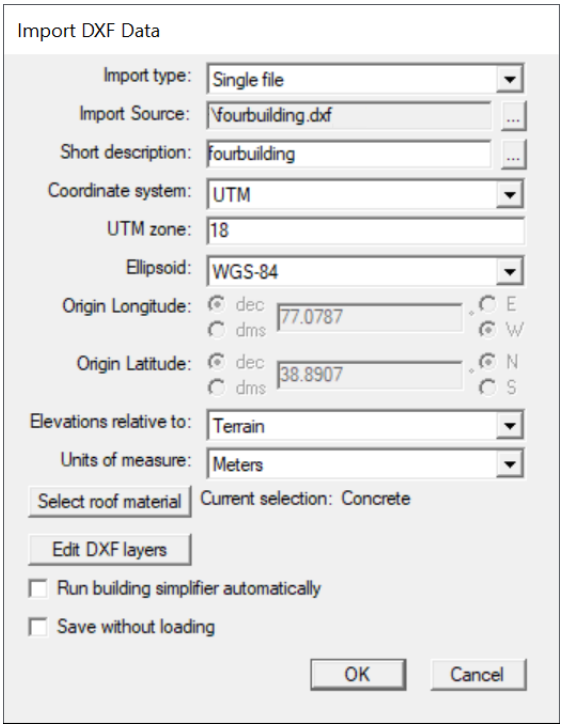


Figure 4.3: Import DXF Data Window

The fields in this window are described below.

- *Short description* - Identifies the imported city.
- *Coordinate system* - Defines the coordinate system of the DXF for proper importing, scaling, and placement of the *City* into the project. After selecting the coordinate system from the drop-down menu, the *Origin* of the file can be entered in the appropriate fields. For Cartesian coordinates, the *Origin longitude* and *Origin latitude* are needed. For UTM, the *UTM zone* and *Ellipsoid* are needed. No additional information is required if the file's coordinate system is Latitude/longitude.
- *Elevations relative to* - Sets building heights relative to terrain or sea level.
- *Select roof material* - Specifies the roof material for all buildings in the imported city.
- *Edit DXF layers* - Opens a window with a list of the layers in the DXF files. Wireless InSite will only import layers identified as *Active*. Double-clicking on a layer opens a window where layers may be set to inactive by deselecting the *Active* checkbox. The MATERIAL assigned to the layer may also be changed by clicking on SELECT MATERIAL and choosing a new layer from the material database.
- *Run building simplifier automatically* - This option automatically performs the simplification necessary for propagation simulations. When unchecked, the BUILDING SIMPLIFIER Window will appear after import to enable more control of the simplification process. To import the DXF file without performing any simplification, click *Cancel* in the BUILDING SIMPLIFIER Window.
- *Save without loading* - When checked, the DXF file is converted into a `*.city` file but is not loaded

into the project. When unchecked, the *City* is added to the project, but the `*.city` file is not saved. Save the file by selecting *File→Save Project*.

- *Double Sided Faces* (available for floor plan and object import). When checked, the faces of the imported geometry are created as double-sided. When unchecked, the faces are created as one-sided.

When everything is set up correctly, click OK . The BUILDING SIMPLIFICATION Window will appear. If none of the functionality listed here is desired, then click CANCEL and the DXF file will be converted.

▶ The operation of the Building Simplification Window is given in greater detail in Section 4.2.

DXF Import Example

Two example DXF files are included with the Wireless InSite distribution. During installation these files are copied to the Remcom program group located at `install_location\data\DXF`. The file names are:

`fourbuilding.dxf` - Written in Cartesian file format. It is a simple geometry of four rectangular buildings that allows users to try out Wireless InSite on an uncomplicated data set. The longitude and latitude origins are 0, 0.

`dupont_circle.dxf` - Written in UTM file format. This is a description of a complex urban area centered around Dupont Circle in Washington, DC. The UTM zone is "17S" and the ellipsoid is "Clarke 1866".

To import `fourbuilding.dxf`:

1. Create a new project in Wireless InSite.
2. Go to the GEOMETRY VIEW and select *File→Import→Geometry→DXF*.
3. Choose the format of the DXF file and click OK.
4. Choose the *City* format from the drop-down list and click OK.
5. Browse to the file on the hard drive and open it.
6. Change the *Coordinate system* to Cartesian.
7. Click OK. The BUILDING SIMPLIFICATION Window will come up.
8. Click CANCEL, as only non-simplifying pre-processing is needed for this *City*. The converted four buildings should appear in the GEOMETRY VIEW.
9. Select the *City* in the GEOMETRY tab in the MAIN WINDOW window and select the following operations from the context menu:
   - *Pre-processing→Add indices to descriptions*
   - *Pre-processing→Group sub-structures*
   - *Correct Normals*

An example of the `fourbuilding.dxf` geometry, imported and changed to the Wireless InSite `*.city` file format, can be found in `install_location\data\city\fourbuilding.city`. This can be loaded into Wireless InSite and compared with the results of the example procedure.

To import dupont_circle.dxf:

1. Create a new project in Wireless InSite.
2. In the MAIN WINDOW, select *File→Import→Geometry→DXF*.
3. Choose the format of the DXF file and click OK.
4. Choose the *City* format from the drop-down list and click OK.
5. Browse to the file on the hard drive and open it.
6. Set the *Coordinate system* to UTM.
7. Fill in the *UTM zone* to "17S".
8. Make sure that the *Ellipsoid* is set to "Clarke 1866".
9. Set the *Relative to field* to be Sea level.
10. Click SELECT ROOF MATERIAL and set it to *Wood*.
11. Click EDIT DXF LAYER.
12. Double click on the active layer labeled 'building'. A popup window appears.
13. Click SELECT MATERIAL.
14. Set the building's material to *Brick* by selecting it from the material database.
15. Click OK until the IMPORT DXF DATA Window is active. Clicking OK again will cause the BUILDING SIMPLIFICATION Window to appear.
16. For now, click CANCEL as we need to perform more pre-processing before actually simplifying the geometry. Wireless InSite will begin converting the file to *.city format. The converted DuPont Circle buildings should appear in the GEOMETRY VIEW.
17. Select the *City* in the GEOMETRY tab in the MAIN WINDOW window and select the following operations from the context menu:
    - *Pre-processing→Add indices to descriptions*
    - *Pre-processing→Group sub-structures*
    - *Correct Normals*
    - *Translate→Coordinate reduction*
18. Select *Properties* from the cities' context menu.
19. Change the *Coordinate system* to Cartesian in the Properties Window.
20. Click OK.

This last change did not affect the physical appearance of the *City*. However, it did affect the way Wireless InSite stores the *.city file. This demonstrates the capability of Wireless InSite to change to the coordinate system for existing *.city files. It is vital, therefore, that the DXF converter receives the correct coordinate system when performing the original conversion from DXF to Wireless InSite *.city format. In order to check your results, see an example of dupont_circle after it is imported, located in install_location\data\city\dupont_circle_raw.city.

Note that the coordinate system display at the bottom of the GEOMETRY VIEW is not determined by the coordinate system used for the specific `*.city` file. The display may be changed to a convenient coordinate system by selecting from the available coordinate system choices in its context menu. This independence is a useful feature of Wireless InSite. For example, the user may import one part of a *City* in UTM coordinates, another in Latitude/Longitude, specify Tx/Rx coordinates in Cartesian, and Wireless InSite will simultaneously display all of these in whichever coordinate system is chosen.

This `*.city` file contains large amounts of building detail. This detail forces Wireless InSite to calculate many rays that do not affect the propagation predictions, thus slowing the simulations. To make the *City* description more suitable, we will use the Wireless InSite building simplifier.

- This is described in more detail in Section 4.6.

To run the simplifier on DuPont Circle:

1. Select the *City* and choose *Pre-processing→Simplify* from its context menu.
2. Set the *Complexity* slider to 90% in the BUILDING SIMPLIFICATION Window.
3. Retain sub-structures based on *Area*.
4. Adjust the *Building Retention* slider so that the cutoff value is approximately 100 m^2.
5. Press ACCEPT on the BUILDING SIMPLIFICATION Window. The buildings process. The modified city should appear in the GEOMETRY VIEW in a few seconds.

You should be able to see that the buildings are simplified but still recognizable. To check your results against an example of dupont_circle, access `install_location\data\city\dupont_circle_processed.city`.

4.3 Importing ESRI Shapefiles

To import an ESRI Shapefile containing building footprint and height information into the current project, select *File→Import→Geometry→SHP*. Only Polygon and PolygonZ shape entities are supported by Wireless InSite.

- ✓ The type of shape entities found in the file is displayed in brackets after the filename.

Some Shapefiles are accompanied by a `*.prj` file containing coordinate projection information. While the `*.prj` file projection information is supported by Wireless InSite to properly geo-locate the imported geometry, it is not required. If a `*.prj` file is present when specifying a Shapefile, its contents will be read to automatically specify some of the import settings. If a `*.prj` file is not present when specifying a Shapefile, a message will be displayed with a reminder that some import settings may not be automatically set without one. All Shapefiles must be accompanied in the same location by corresponding `*.dbf` and `*.shx` files containing attribute and index information respectively. If either `*.dbf` or `*.shx` files are not present when specifying a Shapefile, it cannot be imported and the import process will be stopped. The IMPORT SHAPEFILE Window, seen in Figure 4.4, will attempt to select the coordinate system automatically when a `*.prj` file is present.

As with other geometry file types, multiple Shapefiles can be imported at once by selecting “Directory” from the *Import type* drop-down on the IMPORT SHAPEFILE window.

Attempting to import multiple Shapefiles at once with varying content formatting (e.g. different geo-referencing methods, inconsistent attributes) can lead to improperly imported models. It is recommended that users verify consistent file formatting when importing multiple geometry files at once.

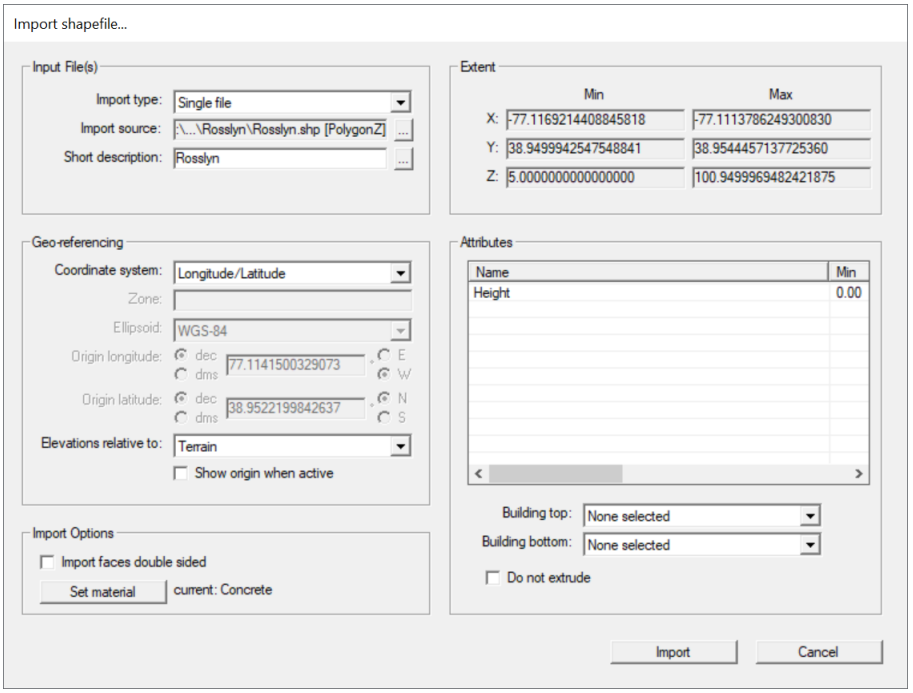


Figure 4.4: The Import Shapefile Window

After specifying one or more Shapefiles, the IMPORT SHAPEFILE Window will attempt to read the selected files and automatically set several import options, including filling in the extent and attribute information fields. The extent information fields contain the absolute minimum and maximum span values present in the Shapefile. If the Shapefile elements are in the PolygonZ format, the minimum and maximum Z-values will be determined and displayed with the X and Y extent values. When importing multiple Shapefiles, the extent information will be the resultant minimum and maximum of all Shapefiles selected for import. extent information can be used to help determine what coordinate reference system should be used to import the selected Shapefiles. Attribute information will be read from the given Shapefiles and used to populate the attributes table. The attributes table will display each attribute's name, minimum, and maximum values with the minimum and maximum values having been determined by analyzing the contents of the specified Shapefiles. The information provided in the attributes table can help to determine what attributes are most suitable for specifying the building top and bottom.

✓ Wireless InSite will only display numeric attributes that can be used to set the building top or bottom. All non-numeric attributes contained in the Shapefile will not be displayed for the attributes table, top, or bottom options.

Wireless InSite also reads information from the selected files to determine the appropriate coordinate system of the Shapefiles for proper importing, scaling, and placement of the geometry into the project. This process will set the type of coordinate system and the *UTM Zone* and *Ellipsoid* for UTM coordinate

systems, or the *Origin Longitude* and *Origin Latitude* for Cartesian coordinate systems. In cases of Latitude/Longitude coordinate systems, no additional options need to be set. While Wireless InSite will determine these import settings automatically for you, note that they can be changed manually if your specific import requires different import settings.

Building height data is needed in order to model urban propagation, and as such the user must identify which, if any, of the Shapefile attributes represent building heights. The *Building top* and *Building bottom* options permit the user to select from all the available numerical attributes found in the current Shapefile, with the attributes table giving the minimum and maximum values for each available attribute. If building height information is contained in the Z-values for each shape entity, or if no relevant height information is contained in the attributes, it is typical to leave the options set to "None selected". In addition, the user must specify whether those elevations are relative to terrain or sea level by appropriately setting the *Elevations relative to* drop down. If desired, the user may choose not to extrude height for the structures in the shapefile(s) by checking the *Do not extrude* box. This will import shapefiles as two-dimensional footprints, which can be useful for tasks such as adding floors and ceilings to a building that has already been imported with height. Finally, the user must assign MATERIALS to the imported buildings. A default material is always provided for import, though this can be changed before the import takes place.

4.4 Importing Raster Data

In contrast to DXF and Shapefiles, which contain facets (vector data), raster data files contain grids of height values. The raster converter extracts building facets from these files for use in Wireless InSite's ray tracing propagation models. Raster data is commonly obtained by aerial or satellite photography and Light Detection And Ranging (LIDAR). It contains significant information and can be used to obtain data on manmade structures, particularly in urban environments. The raster converter can be accessed through *File→Import→Data→Raster*. The raster data will be displayed on the right side of the window, as seen in Figure 4.5, colored by height, where purple is the lowest elevation and red is the highest elevation.

✓ Wireless InSite currently supports the ARC ASCIIGRID file format and Portable Gray Map (*.pgm) raster images.

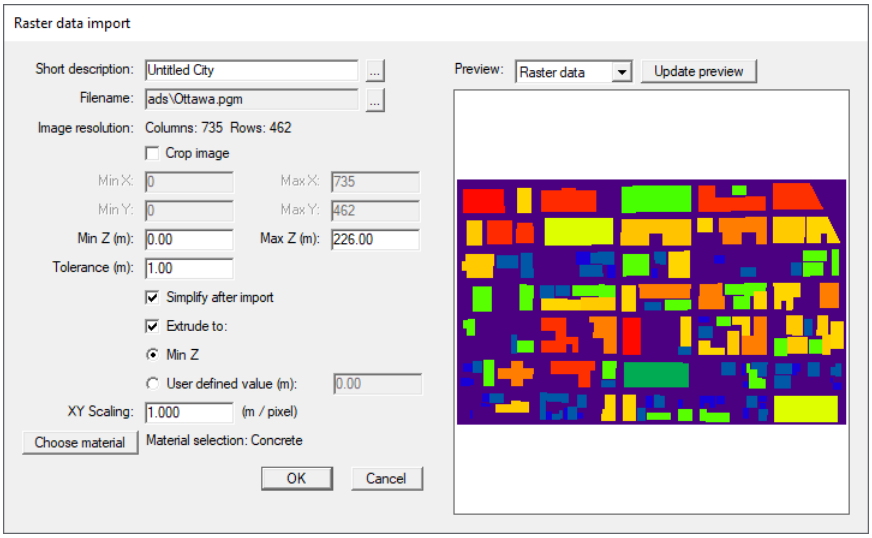


Figure 4.5: Previewing raster data. The color by height option is always activated.

Following is a description of the RASTER DATA IMPORT Window:

- If some data is not useful for the simulation, it can be cropped in the X, Y and Z directions in the respective *Min* and *Max* boxes. It is often necessary to specify a minimum Z-value so that the buildings are separated from the *Terrain*.
- The *Tolerance* is used when creating building footprints from the data. Data points within the specified tolerance of each other will be grouped together to form the building footprint.
- *Simplify after import* is used to remove jagged edges caused by the discrete nature of the raster data files.
- The *Extrude to* field specifies how far the building footprints should be extruded: to a *Min Z* value or a *User-defined value*.
- The *XY Scaling* is used to properly scale the *City* based on the spatial resolution of the raster data, in meters per pixel.

To preview the imported *City* choose the "Vector data" option from the *Preview* menu. Once the correct parameters are entered, click OK to add the city to the project.

▶ The Ottawa example .pgm is available in the `install_location\data\Raster` directory.

The UPDATE PREVIEW button is not required when switching between previewing raster and vector data (only after updating import parameters).

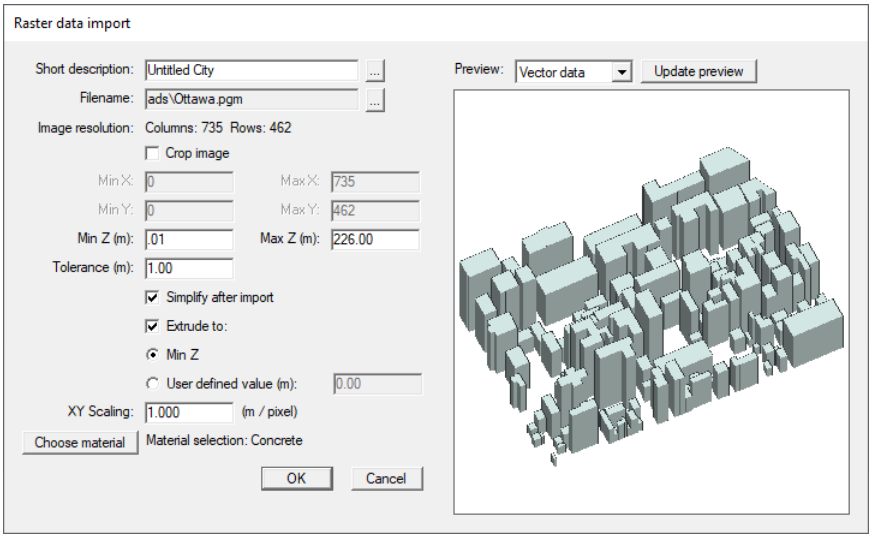


Figure 4.6: Importing the Ottawa City using the Raster Data Import Window

4.5 Building Pre-processing

After creating or importing a *City* into Wireless InSite, there are several options available for pre-processing the geometry before using it to perform simulations. These options are available by selecting a piece of the geometry in the GEOMETRY VIEW and choosing *Pre-processing* from its context menu.

- The *Pre-processing* menu options are described in Section 3.7.

4.6 Building Simplification

Wireless InSite enables the user to modify building geometry as needed to correct shortcomings in the imported `*.city` file, as well as reduces the number of faces that are needed to describe the buildings. These adjustments allow for more accurate and efficient simulations. This is necessary because building geometry descriptions obtained from different sources are not consistent in organization. For example, many individual facets may be used to describe one flat building face. The Wireless InSite ray models will be more efficient if these individual facets are combined into a single surface.

Additionally, building faces in some imported *Cities* may not be properly connected to form buildings. The simplifier will look for sub-structures that overlap and combine them so that the ray models can quickly determine building intersections. Thus ensuring the rays will not propagate through buildings.

Each building face has a normal (perpendicular) direction specified. Wireless InSite uses these normals to determine if a point in space is inside of a specific building. These normals are often not consistent in `DXF` files. The simplifier forms a consistent set of normal vectors for the faces of each building.

Other imported cities may include small geometries that do not impact the propagation prediction. Intuitively it may seem that providing greater detail in the building description will improve accuracy.

However, the high frequency methods used in Wireless InSite assume that geometries are large compared to the wavelength, and including small geometries will actually reduce accuracy. The simplifier may be used to remove these small geometries.

Building descriptions may also include small indentations in the building surface, perhaps due to ornamentation. These indentations may be so small that they do not to affect the propagation results, but rays will bounce back and forth in these building regions. The simplifier will remove such ornamentation, decreasing simulation time while preserving prediction accuracy.

The Building Simplification Window

The BUILDING SIMPLIFICATION Window, seen in Figure 4.7, provides an easy-to-use interface for controlling how a *City* is simplified. On the left side of the window is a view of the geometry. When the PREVIEW button is clicked, the city will be simplified using the current settings. Results are shown in the viewing window. After simplification runs, the footprint of each buildings original shape will be visible around the edges of any buildings that were modified.

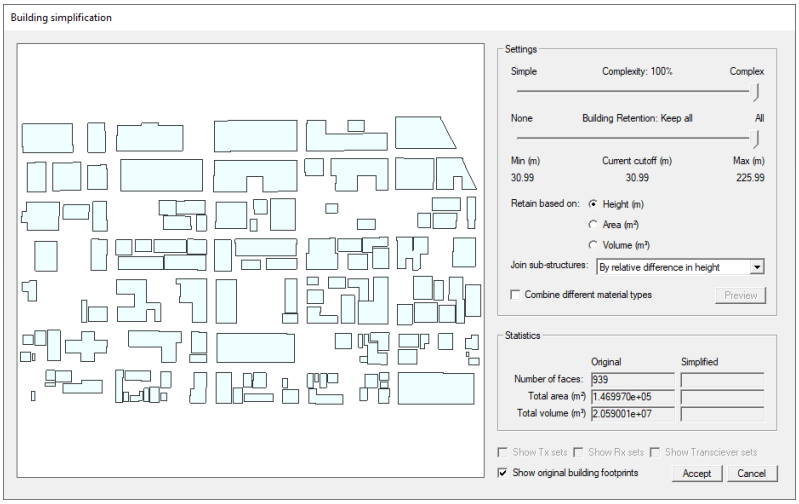


Figure 4.7: The Building Simplification Window

The settings on the right side of the window are described below.

Settings control the simplification process.

Complexity - Controls the tolerance that determines if parts of a sub-structure are kept in the simplified version. The lower the slider is set, the fewer faces will be generated in the resulting geometry.

Building Retention - Controls the tolerance that determines if a building in the unsimplified version of the city is retained in the simplified version.

Retain based on - Allows the user to determine if sub-structures should be retained based on their current *Height*, *Area* or *Volume*.

Join sub-structures - Allows the user to control under what conditions joining will be performed. The default setting, *By relative difference in height*, uses the *Complexity* slider to determine if sub-structures of dissimilar roof height should be combined. As the *Complexity* setting is lowered, greater differences in roof height can be joined.

✓ Joining is the process by which the footprints of several overlapping sub-structures of different heights are combined into a common structure with a single roof.

Combine different material types - Combines sub-structures that are composed of different materials. The material of the larger sub-structure, as determined by the *Retain based on* setting, will be used for the resulting structure.

PREVIEW - Runs the simplifier on the city using the current settings and displays the results in the preview window.

Statistics show the difference between the original city and the most recently created preview.

Below the statistics window are options to *Show Tx sets* and *Show Rx sets* in the project. These are useful for showing whether a simplified version of the *City* will cause points in the sets to be hidden inside a building.

The ACCEPT button simplifies the *City* based on the current settings and immediately replaces the original city with the result. Clicking CANCEL will close the BUILDING SIMPLIFICATION Window without replacing the original city with a simplified version.

Chapter 5

Terrain

In this chapter, you will learn...

- → how to create or import a new terrain into Wireless InSite
- → how to optimize the geometry and appearance of the terrain
- → the unique features of importing from the data file formats which Wireless InSite supports

To obtain realistic results, urban and rural environments in Wireless InSite must include a *Terrain*. In most cases, a terrain will be imported from one of several supported data file formats. However, users can manually create new terrains via the included editing tools.

This chapter explains what operations users can perform on *Terrains*, how to import terrain information from several different sources, and how to use the editors to modify any terrain geometry.

5.1 Creating a New Terrain

To manually create a new *Terrain*, select *File*→*New*→*Geometry*→*Terrain*. This will bring up the CREATE NEW TERRAIN Window, shown in Figure 5.1.

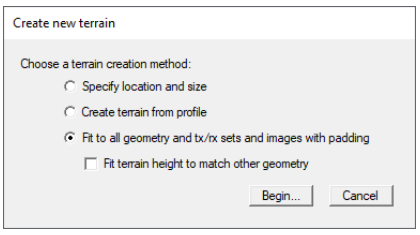


Figure 5.1: Creating a new terrain

If users decide to specify the location and size of their *Terrain*, they will be presented with the NEW TERRAIN Window, as seen in Figure 5.2. This window enables users to enter their exact specifications.

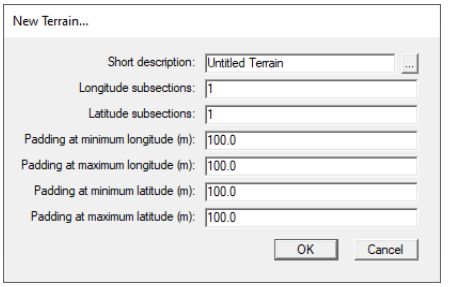


Figure 5.2: Specifying the terrain size and location

If users instead decide to *Create from terrain profile*, they will be able to create a terrain that is a set of faces along a linear path. Selecting the *Fit to all geometry and tx/rx sets and images with padding* button will create a *Terrain* so that all of the elements currently in the project will be over the terrain. Checking the *Fit terrain height to match other geometry* checkbox will cause the terrain that is created to conform to the heights of the other elements in the project.

After clicking OK, the CREATE NEW TERRAIN PROFILE Window will appear, as seen in Figure 5.3. By default, the *Width* of the terrain will be 1/10th the length of the *Total length*. For extremely long profiles, this can be overridden with a better value.

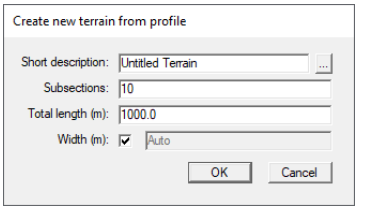


Figure 5.3: Specifying a terrain to be created from a profile

Wireless InSite creates a new terrain from the entered data and displays it in an editor window for users to make modifications, such as adding vertices or adjusting height values to create a non-planar terrain.

Once all the information is collected by Wireless InSite, it creates a new terrain and displays it in an editor window for users to make modifications, such as adding vertices and adjusting height values to create a non-planar terrain.

5.2 Terrain Editor

After users create a *Terrain*, the editor window automatically appears. This window is also accessible by clicking *Edit* from the terrain's context menu (under the GEOMETRY tab). The style of editor that appears depends on whether or not the user selected *Create from profile* in the CREATE NEW TERRAIN window. If so, then a special editor for editing profile information will appear.

Editing a Terrain Surface

A picture of the EDIT NEW TERRAIN Window, for a non-profile based *Terrain*, is shown in Figure 5.4. The example presented here is how a manually-created new *Terrain* will appear if it is created with five latitudinal and five longitudinal subsections. Users can add more vertices by clicking anywhere in the editor window inside the current terrains' area. If the mouse is lying on a line between vertices, then Wireless InSite will split the two facets that the line borders into two. Otherwise, the new vertex will split the facet into three separate facets.

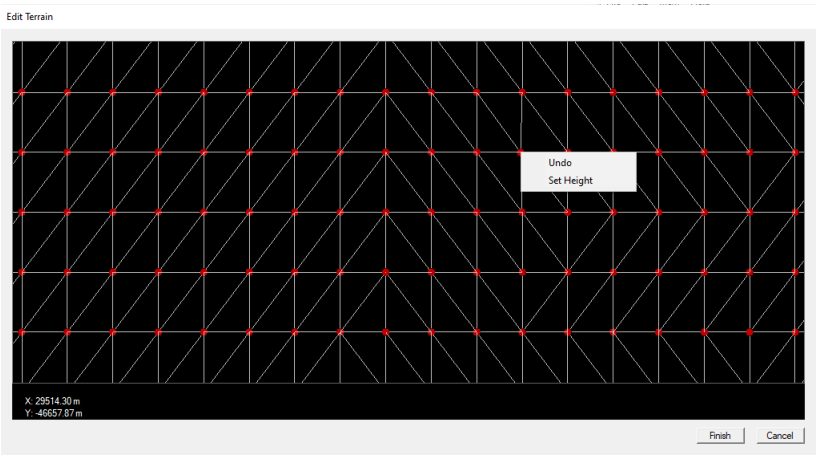


Figure 5.4: The area terrain editor

All vertices can be manipulated to change the appearance of the terrain. By right-clicking on a vertex, you can change its height in the white rectangle that appears. This is also shown in Figure 5.4. The EDIT NEW TERRAIN Window also enables users to undo the last action performed on a vertex including creation, setting height, and moving points.

Editing a Terrain Profile

When a *Terrain* is generated from a profile, Wireless InSite displays the EDIT TERRAIN PROFILE Window, shown in Figure 5.5. This window displays the terrain as a set of points in the plane of the profile. The editor has a fieldHeight scaling field that allows the terrain to be scaled in the Z-direction so that the contours of the terrain are easier to distinguish. The context menu offers options for adding, removing and manually specifying points as well as a way to change the MATERIAL for any of the faces being generated. The material for any face is changed at its left vertex.

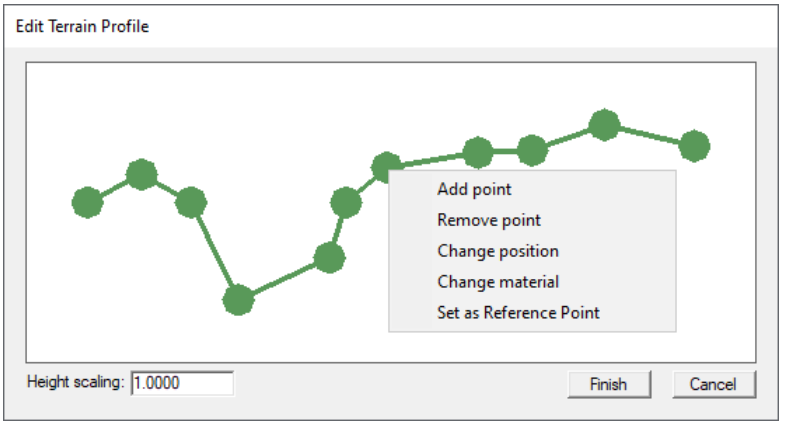


Figure 5.5: The Edit Terrain Profile Window

5.3 Importing a Terrain

Wireless InSite's user interface uses the Geospatial Data Abstraction Library (GDAL) for importing terrain data from raster sources. Currently supported terrain data formats include:

- Arc/Info ASCII Grid: *.txt, *.ascii, *.asc
- Digital Elevation Model(DEM): *.dem
- Japanese DEM: *.dem, *.mem
- USGS SDTS formatted DEM: *.catd.ddf
- Digital Terrain Elevation Data (DTED): *.dt*
- TIFF, BigTIFF, GeoTIFF: *.tif

Selecting "Unsupported formats (*.*)" allows the user to attempt to import any terrain format supported by the GDAL library.

To begin importing a *Terrain* file:

1. Open or create a new project.
2. Select *Import→Geometry →GDAL Terrain* to open the IMPORT TERRAIN Window, as seen in Figure 5.6.

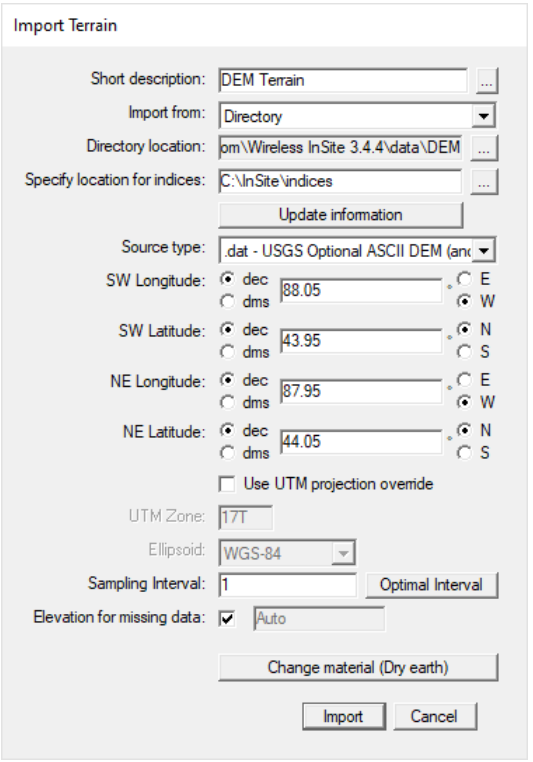


Figure 5.6: The Import Terrain window

3. Fill in the remaining fields, as described below.

- *Short description* - Describes the terrain. This field is optional.
- *Import from* - Select “Single file” or “Directory” to specify whether to import data from a single file or assemble terrain from multiple files contained in the same directory.
  - Selecting to import from a directory enables the following fields:
    - *Specify location for indices*
    - *Source type*
- *File name / Directory location* - Select the terrain file or directory to be imported. The name of this field changes based on whether terrain is being imported from a single file or a directory.
- *Specify location for indices* - Select the location for indices to be created when assembling a terrain from multiple files in a directory.
- *Update information* - Refreshes the IMPORT TERRAIN window based on information specified by the user.

  The UPDATE INFORMATION button does the following:

  - Single-file terrains:

    - Automatically fills *Latitude / Longitude* fields using data from the file.
  - Terrains imported from a directory:
    - Populates the *Source type* drop-down menu with file types found in the selected directory.
  - If using UTM projection, the *Latitude / Longitude* fields are automatically filled using the specified UTM projection information.
  - Automatically detects the default no data value of the terrain files for the *Elevation for missing data* field when in "auto" mode.

- *Source type* - Select the type of file that will be used when importing terrain from a directory. The importer currently supports DTED, DEM, etc. The importer will only work on files that are of the same file type and resolution.
- *Latitude / Longitude* values - Determine the boundaries of the imported terrain. The region is specified as a rectangular area with the southwest and northeast corners that are entered.
- *Use UTM projection override* - This checkbox should be enabled when importing terrain from UTM based files that do not contain projection data. Checking this box enables the *UTM Zone* and *Ellipsoid* fields.
- *UTM Zone* - Specify the UTM Zone that the terrain files lie in.
- *Ellipsoid* - The ellipsoid you want to use when performing the coordinate conversions.
- *Sampling interval* - Reduces the resolution of the imported terrain data. For example, if a file has 10 meter spacing and a sampling interval of 'Every 3rd point' is specified, then the final terrain will have a spacing between samples of 30 meters.
- *Elevation for missing data* - When checked, i.e. in Auto mode, this field will be filled in with the default missing data value specified in the selected file's header when the UPDATE INFORMATION button is clicked. If the *Import from* field is set to "Directory", this value will be extracted based on the file type selected from the *Source type* drop-down menu. If the Auto check box is checked or unchecked with the default value remaining unmodified, the import operation will not create facets connected to samples that have this value. If a value is provided other than the default, then the importer will replace these values and create facets attached to those sample points.
- *Change material* - Allows selection of the terrain material. The default is Wet Earth.

If there are any errors in the IMPORT TERRAIN window, the user will be informed and returned to the window to correct errors or modify the information.

5.4 Terrain Operations

Z-Scaling

It is possible to scale any *Terrain* along the Z-axis. This option can be turned on by selecting *View*→ *Z-scaling*→*On* in the GEOMETRY VIEW. When a scaling value is given, the terrain and all other geometries in the viewing window are scaled along the Z-axis, as shown in Figure 5.7. This feature is

especially useful for terrain viewing because the vertical terrain dimensions are often small relative to the horizontal dimensions. When Z-scaling is applied, none of the height information about the points is modified, as this option only affects the way GEOMETRY are rendered.

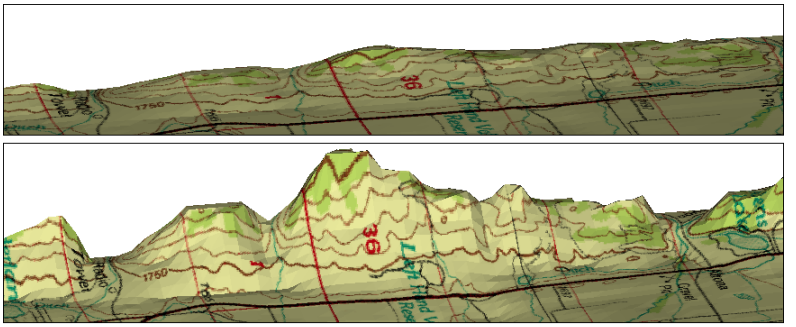


Figure 5.7: An example of Z-scaling (by a factor of 5) on a Colorado terrain

Color by Height

It is possible to color any GEOMETRY by height. This capability is mainly intended for *Terrain*, but may also be useful for buildings. First select the geometry by setting the selection mode to *Geometry* by choosing *Select→Geometry* in the GEOMETRY VIEW. Then click on any facet of the geometry and select *Color by height* from the context menu. Note that *Color by height* is disabled for terrains using sparse rendering. An example of coloring by height is shown in Figure 5.8.

✓ Multiple geometries can be selected by holding the CTRL key.

▶ For more information on color scheme selection, see Section 22.1.

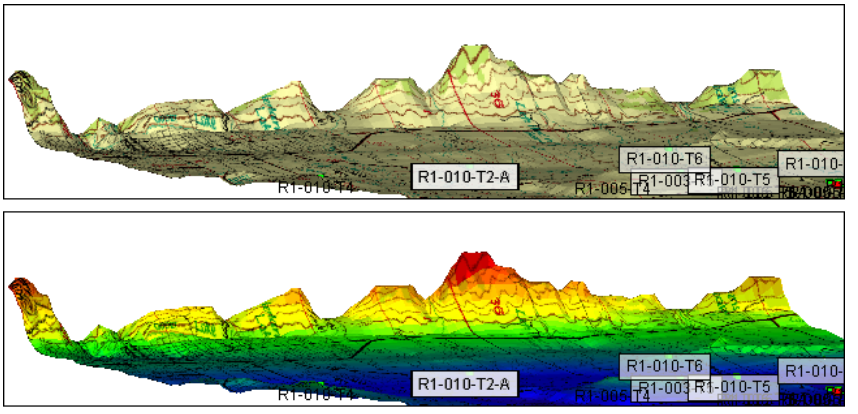


Figure 5.8: An example of coloring by height on a Colorado terrain

5.5 DTED (Digital Terrain Elevation Data)

The National Geospatial-Intelligence Agency (NGA) developed the Digital Terrain Elevation Data (DTED) standard for specifying terrain data as a uniform matrix of terrain elevation values. DTED Level 0 elevation post spacing is 30 arc second (nominally one kilometer), and has been made available (within copyright restrictions) to the public at no charge through the Internet. It allows a gross representation of terrain elevations for general modeling purposes. This data comes in three different levels, all in 1°x1°tiles. The resolutions of the various levels are shown in Table 5.1.

- DTED level 0 data is available from the United States Geological Survey (USGS) as SRTM 1 Arc-Second data via their Earth Explorer site, https://earthexplorer.usgs.gov/.

Using DTED levels 1 and 2 requires government permission.

Table 5.1: Resolution Values of DTED Levels

ZONE	LATITUDE RANGE	MATRIX INTERVAL
DTED level 0		*latitude x longitude*
I	0°- 50° North-South	30” x 30”
II	50°- 70° North-South	30” x 60”
III	70°- 75° North-South	30” x 90”
IV	75°- 80° North-South	30” x 120”
V	80°- 90° North-South	30” x 180”
DTED level 1		*latitude x longitude*
I	0°- 50° North-South	3” x 3”
II	50°- 70° North-South	3” x 6”
III	70°- 75° North-South	3” x 9”
IV	75°- 80° North-South	3” x 12”
V	80°- 90° North-South	3” x 18”
DTED level 2		*latitude x longitude*
I	0°- 50° North-South	1” x 1”
II	50°- 70° North-South	1” x 2”
III	70°- 75° North-South	1” x 3”
IV	75°- 80° North-South	1” x 4”
V	80°- 90° North-South	1” x 6”

Wireless InSite determines which specific DTED files to read and how to piece them together to produce a seamless terrain `*.ter` file. Because `*.ter` files generally consist of numerous faces, which slows visual rendering and manipulation, it is recommended that terrain areas be kept to relatively small areas.

✓ Alternatively, the sampling interval can be increased leading to lower resolution, but increased performance for larger areas.

DTED Example

The installation files of Wireless InSite contain several Level 0 DTED digital elevation maps. These files are located in the `install_location\data\DTED` folder. This directory includes DTED Level 0 data for a region inside 109°W 39°N to 106°W 41°N, which roughly corresponds to one-quarter of Colorado in the northwest part of that state. This is an area of Colorado that includes Glenwood Springs and Rifle. Interstate Route 70 also goes through the region. The overall area is approximately 220 km x 220 km.

To work with the DTED example:

1. Either initialize a new project or load an existing project. For this example, we will select an area within the directory described above.
2. Load DTED data into Wireless InSite by selecting *Import→Geometry →GDAL Terrain* from the current context menu. The IMPORT TERRAIN window displays.
3. Enter "Glenwood and Rifle" as the *Short Description* (the region is roughly between these towns).
4. Select "Directory" from the *Import from* drop-down menu.
5. Select the directory to be imported in the *Directory location* field by navigating to the `install_location\data\DTED` folder and clicking OK.
6. If a location for indices has not already been designated, a message will appear asking you to do so. Click OK.
7. Specify a location of your choosing for indices in the *Specify location for indices* field.
8. Click UPDATE INFORMATION.
9. Select ".dt0 - DTED Elevation Raster" in the *Source type* drop-down menu.
10. Specify that Wireless InSite import the southwest and northeast corners of the rectangular region.

    Here, we know the region of interest in Long/Lat decimal format and that our region of interest is in the western and northern hemispheres.
11. For all longitudes and latitudes entered, select the button indicating *dec*.
12. Type 108.2 for the southwest longitude. Select the *W* button to the right of the edit box.
13. Type 39.2 for the southwest latitude. Select the *N* button to the right of the edit box.
14. Type 107.8 for the northeast longitude. Select the *W* button to the right of the edit box.
15. Type 39.6 for the northeast latitude. Select the *N* button to the right of the edit box.
16. Enter "1" for the *Sampling interval*.
17. Click CHANGE MATERIAL and select *Dry Earth* in the CHOOSE MATERIAL Window, as seen in Figure 5.9.

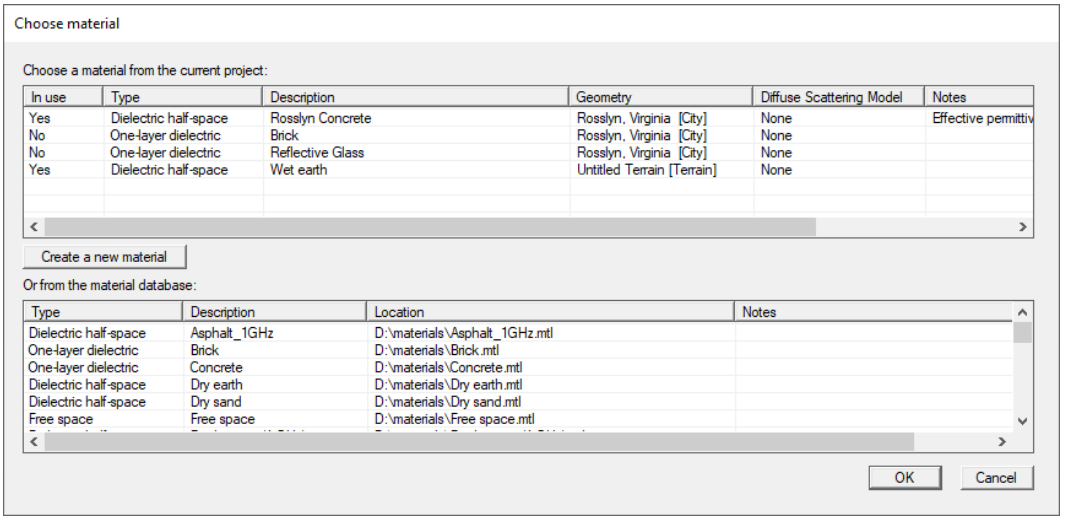


Figure 5.9: The Choose Material Window

18. Click OK. The *Import Terrain* window should now appear.
19. Click IMPORT. Wireless InSite imports the terrain and automatically renders it as seen in Figure 5.10.

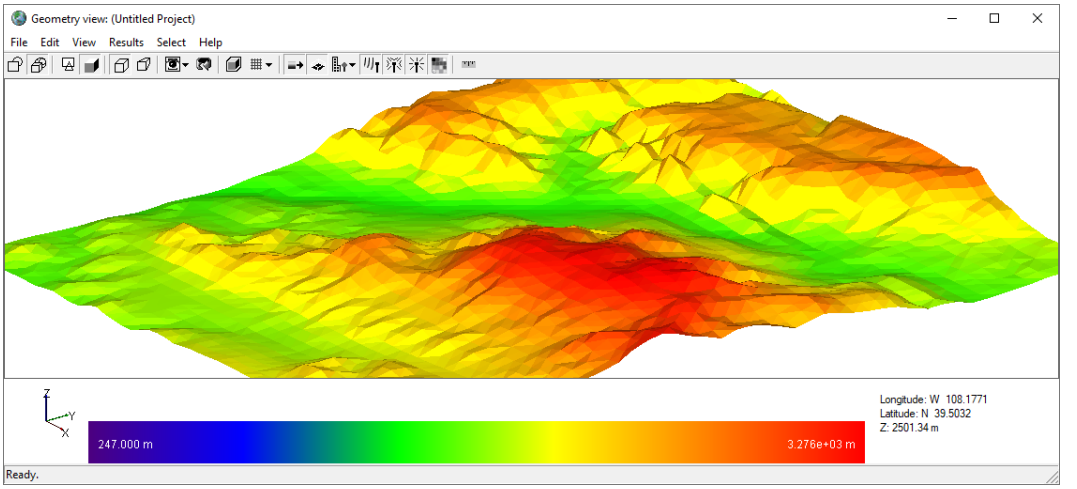


Figure 5.10: Imported DTED terrain example

Users can change the appearance of the *Terrain* by clicking 3D and SOLID BODY, or by navigating to *View→Z-scaling* and using the mouse to change the perspective view. Figure 5.10 has the Z-scaling feature turned on and uses the *Color by height* option for the GEOMETRY. A factor of three was used, multiplying the vertical dimension by three.

5.6 DEM (Digital Elevation Model)

U.S. Geographical Survey (USGS) digital elevation maps (DEMs) provide detailed topographical data widely available to the public either for free or for a modest price.

- DEM files are available for download at https://earthexplorer.usgs.gov. The files are in ASTER Global DEM (GDEM) format.
- ✓ To use older (SDTS) format DEM files in Wireless InSite, users must convert the files to the DEM format with the `sdts2dem.exe` utility available through various academic websites.

DEM Example

The installation files of Wireless InSite contain several USGS DEM files. These files are located in the `install_location\data\DEM` folder.

For this example, we will read in a small portion of central Wisconsin.

1. Create a new project by selecting *File*→*New*→*Project*.
2. Load DEM data into Wireless InSite by selecting *Import*→*Geometry* →*GDAL Terrain* from the current context menu. The *Import Terrain* window displays.
3. Enter “DEM terrain” into the *Short Description* field.
4. Select “Directory” from the *Import from* drop-down menu.
5. Select the directory to be imported in the *Directory location* field by navigating to the `install_location\data\DEM` folder and clicking OK.
6. Specify a location of your choosing for indices in the *Specify location for indices* field.
7. Click UPDATE INFORMATION.
8. Select “.dem” in the *Select type* drop-down menu.
9. For all longitudes and latitudes entered, choose the button indicating *dec*.
10. Type 88.05 for the southwest longitude. Select the *W* button to the right of the edit box.
11. Type 43.95 for the southwest latitude. Select the *N* button to the right of the edit box.
12. Type 87.95 for the northeast longitude. Select the *W* button to the right of the edit box.
13. Type 44.05 for the northeast latitude. Select the *N* button to the right of the edit box.
14. Select “Every Point” as the *Sampling interval*.
15. Click the CHANGE MATERIAL button and select *Dry Earth* in the CHOOSE MATERIAL Window.
16. Click OK. The IMPORT TERRAIN window should now appear, as seen in Figure 5.6.
17. Click the Import button. Wireless InSite automatically renders the terrain, as seen in Figure 5.11.
18. Click IMPORT. Wireless InSite imports the terrain and automatically renders it as seen in Figure 5.11.

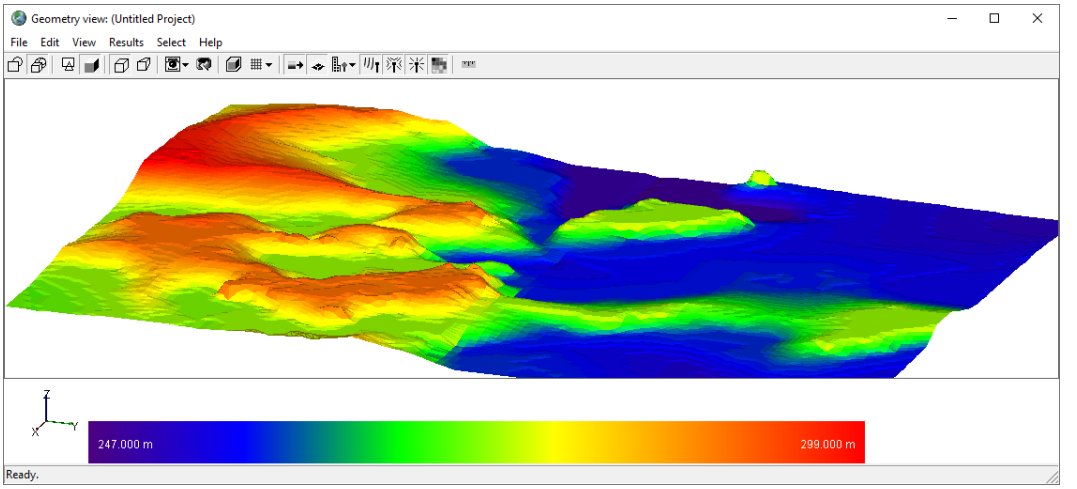


Figure 5.11: Imported DEM terrain example

Users can change the appearance of the *Terrain* by clicking 3D and SOLID BODY, and by navigating to *View*→*Z-scaling*→*On* and setting the *Z-scaling factor* to 10. Additionally, the *Terrain* can be colored by height by selecting the terrain, right-clicking, and choosing *Color by height*. Figure 5.11 shows the terrain with *Z-Scaling* and *Color by height* enabled.

5.7 DSMW (Digital Soil Map of the World) Materials

The Digital Soil Map of the World (DSMW) is a digital version of the FAO/UNESCO Soil Map which contains geo-referenced information about soil materials [1]. In order to create a more accurate simulation for conductivity and permittivity, Wireless InSite uses the information found in the DSMW to change the materials of a *Terrain* to a special soil material. This material uses the compositional makeup of sand, silt and clay as well as texture.

- For more information on the soil terrain material, see Section 10.3.

In addition, moisture content is also extracted from the National Center for Environmental Prediction (NCEP) climate history database. By combining these two sources, Wireless InSite creates a soil material for terrain geometries that provides for a more comprehensive simulation of the terrain's conductivity and permittivity for propagation analysis.

When the DSMW database indicates a water region, a default fresh water dielectric half-space is used. If there are gaps in the DSMW information, the presence of water is assumed and the *Terrain* faces in that area are modified accordingly.

In order for Wireless InSite to import data from these sources, the location of the ERDAS 7.5 raster GIS file and the NCEP climate history file must be specified in the *Application Preferences* under *Defaults*. The files included on the installation CD are used by default, so these settings should only need to be adjusted if newer files become available.

✓ To access this window, select *Edit→Application Preferences* in the MAIN WINDOW, then select the *Defaults* tab.

▶ See Section 2.7 for more on the *Defaults* tab of the APPLICATION PREFERENCES Window.

DSMW Example

To show how to import soil information from the DSMW, we will first import a large terrain surface. In this example, we will use the terrain for Yellowstone National Park with a topographic map overlay. This terrain was imported using DTED Level 0, which has a resolution of 5 km (a lower resolution than typically used). The Level 0 data was selected to show the complexity of the data within the database.

1. Create a new project by selecting *File→New→Project* and call it Yellowstone.
2. Load DTED data into Wireless InSite by selecting *Import→Geometry →GDAL Terrain* from the current context menu. The IMPORT TERRAIN window displays, as seen in Figure 5.12.

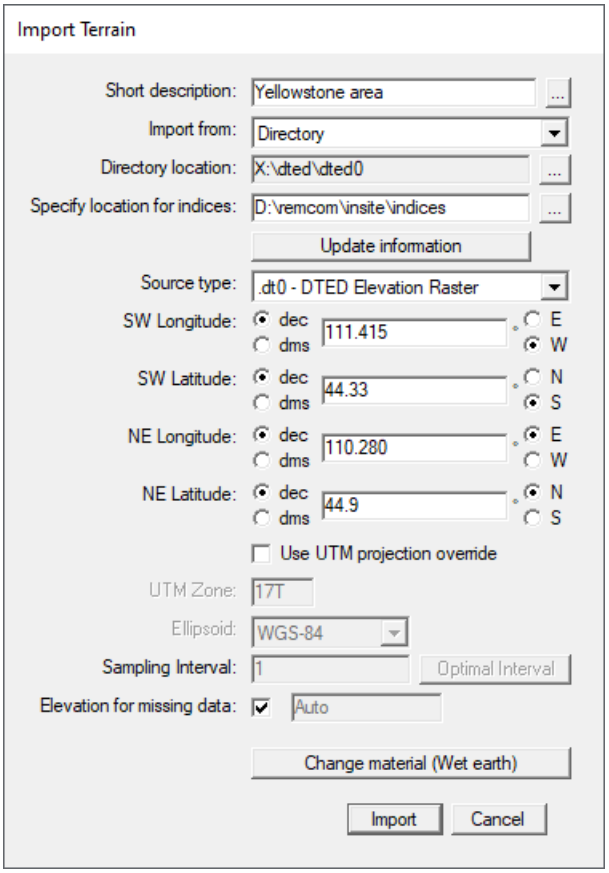


Figure 5.12: Importing the DTED terrain files for this exercise.

3. Enter “Yellowstone area” into the *Short Description* field.

4. Select “Directory” from the *Import from* drop-down menu.
5. Select the directory to be imported in the *Directory location* field by navigating to the `install_location\data\DTED` folder and clicking OK.
6. Specify a location of your choosing for indices in the *Specify location for indices* field.
7. Click UPDATE INFORMATION.
8. Select “.dt0” in the *Source type* drop-down menu.
9. Specify that Wireless InSite import the southwest and northeast corners of the rectangular region.

   Here, we know the region of interest in Long/Lat decimal format and that our region of interest is in the western and northern hemispheres.
10. For all longitudes and latitudes entered, select the button indicating *dec*.
11. Type 111.415 for the southwest longitude. Select the *W* button to the right of the edit box.
12. Type 44.330 for the southwest latitude. Select the *N* button to the right of the edit box.
13. Type 110.280 for the northeast longitude. Select the *W* button to the right of the edit box.
14. Type 44.900 for the northeast latitude. Select the *N* button to the right of the edit box.
15. Select “Every Point” as the *Sampling interval*.
16. Click the IMPORT button.
17. Once the terrain has been imported, select the terrain geometry through the GEOMETRY VIEW by choosing *Select→Geometry* and clicking on the terrain.
18. From the terrain’s context menu, select *Pre-processing→Extract soil information from DSMW database*.
19. You will be asked to specify the month to extract the climate information from the NCEP database, as seen in Figure 5.13. Enter 1 into the *Month* field and click OK.

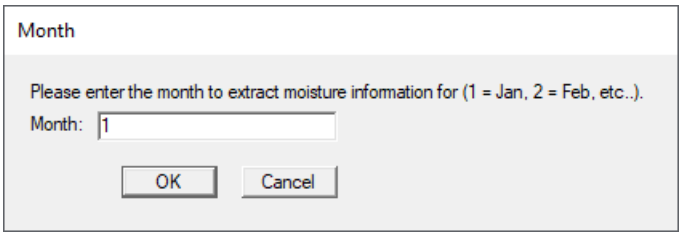


Figure 5.13: Specifying the month for the NCEP climatology lookup

The GEOMETRY VIEW, as seen in Figure 5.14, shows the results of the import operation after the DSMW layer is added. The import operation will assign colors based on the material’s FAO type.

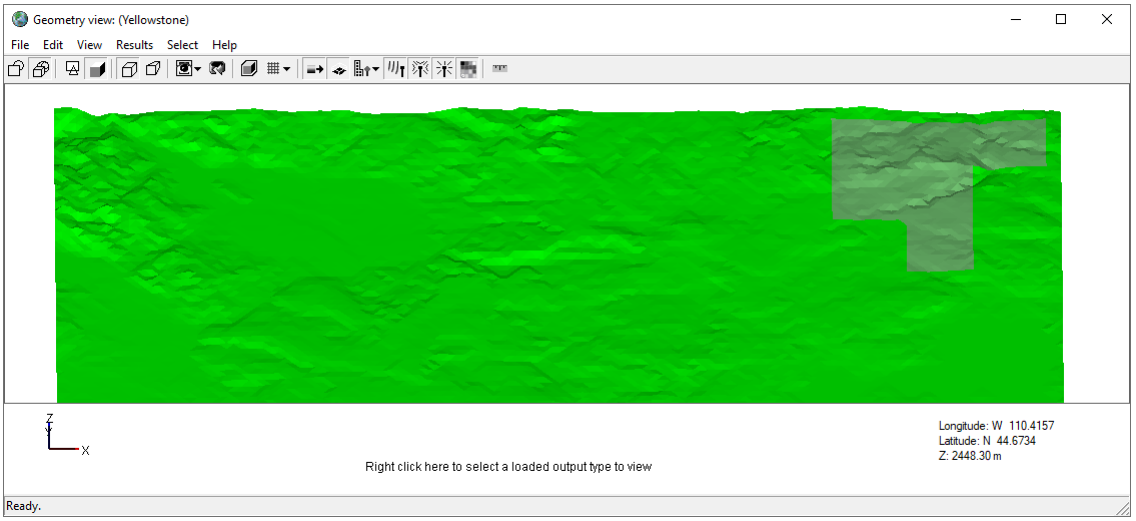


Figure 5.14: The result of importing terrain with the addition of the DSMW information

Once the materials, as shown in Figure 5.15, are imported, users can plot conductivity and permittivity versus frequency as well as normal coefficient plots for the material by right-clicking on the material and selecting *Plot*. Plots for Albic Luvisols are shown for both winter and summer months in Figures 5.16, 5.17 and 5.18 to show the difference that the soil moisture content can produce in the soil material.

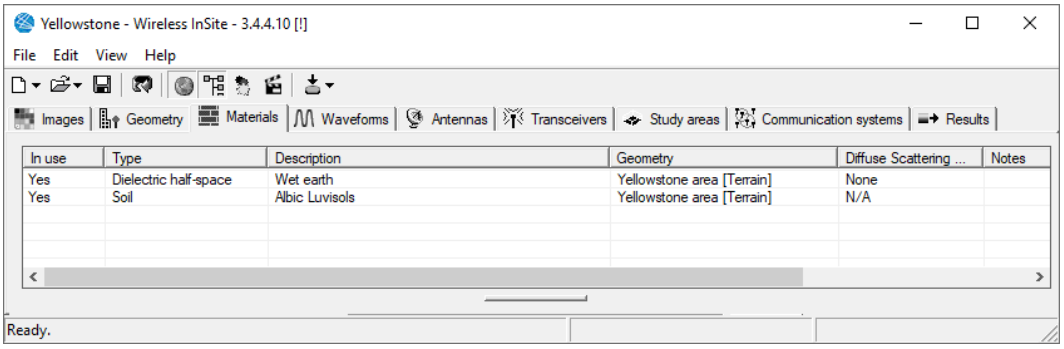


Figure 5.15: The materials produced by the import operation

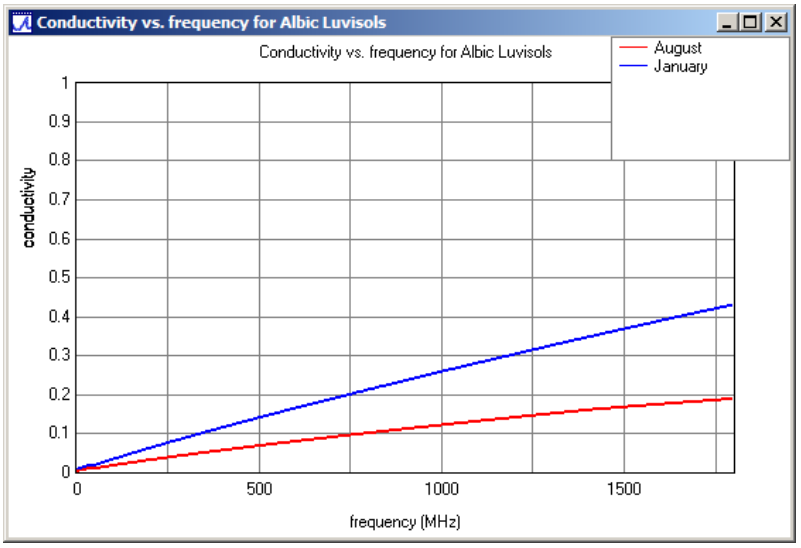


Figure 5.16: Conductivity vs. frequency plot

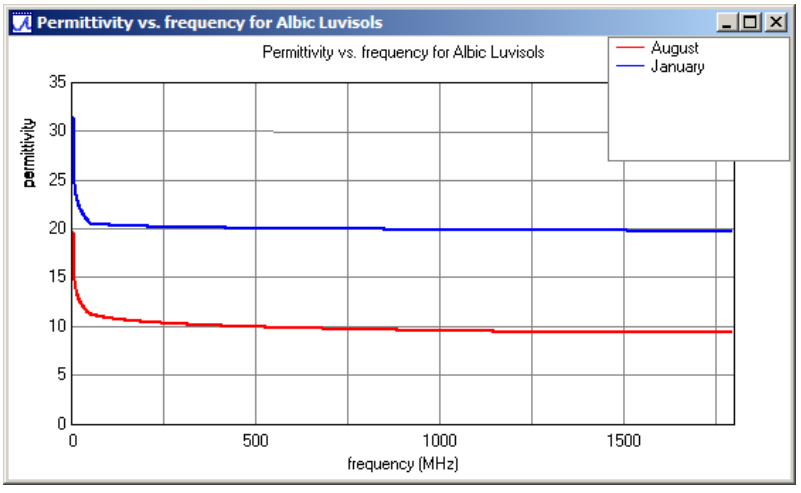


Figure 5.17: Permittivity vs. frequency plot

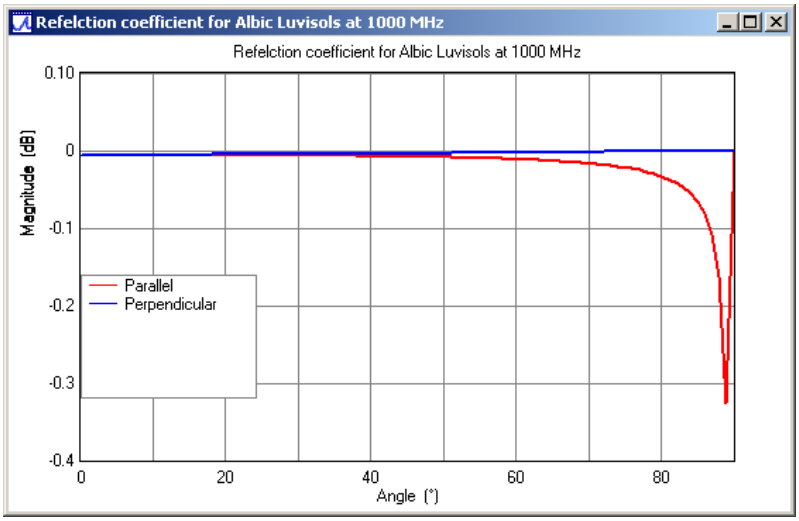


Figure 5.18: Reflection coefficient plot

Chapter 6

Foliage

In this chapter, you will learn...

→ how to create or edit foliage in Wireless InSite

→ how to import foliage information from the GLCC database

To enhance the accuracy of a simulation, Wireless InSite enables the addition of *Foliage* to a propagation environment. Foliage data is available for import from the GLCC database. Wireless InSite is able to read and convert this data to its `*.veg` file format.

Users can make manual changes to foliage with Wireless InSite's FOLIAGE EDITOR. This tool enables users to create new foliage or modify existing foliage. It is also possible to conform foliage to the surface of the *Terrain*.

Note that the X3D and FULL 3D models handle *Foliage* differently: The X3D model uses the foliage model selected in the STUDY AREA. FULL 3D uses the properties defined in the `*.veg` file.

6.1 Creating and Editing Foliage

Foliage Editor

The EDIT FOLIAGE Window is primarily used when creating and editing *Foliage* in Wireless InSite. This window is used to create new groups of foliage and to edit existing groups. This window, as shown in Figure 6.1, is similar to the City Editor in that Wireless InSite creates foliage geometries from footprints. Foliage geometries are completely enclosed, having bottom, top, and side faces. All faces in a foliage area must have the same material.

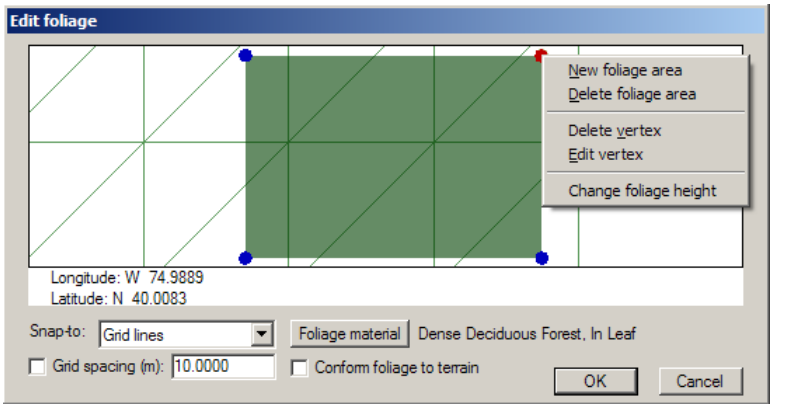


Figure 6.1: The EDIT FOLIAGE Window

The *Conform foliage to terrain* checkbox determines the way the foliage is classified. The types of *Foliage* are:

- **Non-conformal foliage** - Has flat top and bottom faces.
- **Conformal foliage** - Fits the top and bottom faces to the *Terrain* beneath the foliage. This ensures that the height difference between the top and bottom of the foliage remains constant.

Examples of non-conformal and conformal foliage are shown in Figure 6.2.

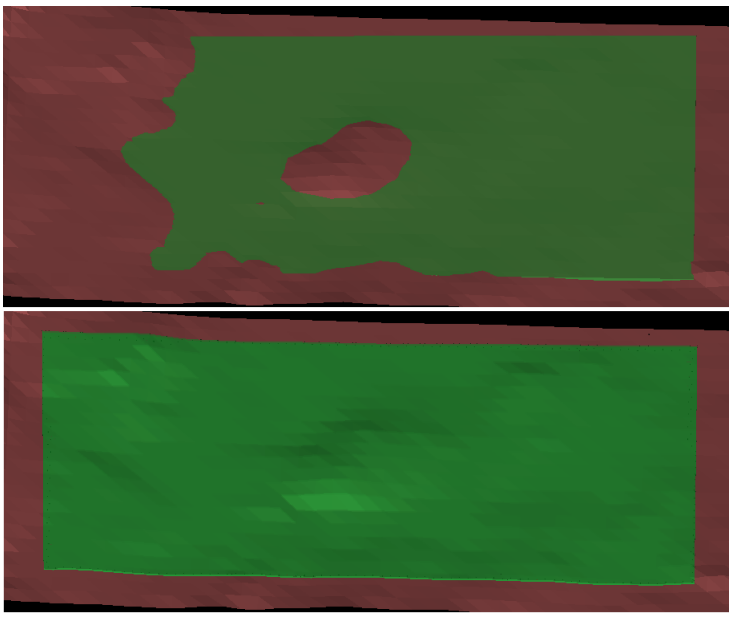


Figure 6.2: Non-conformal foliage (top) and conformal foliage (bottom)

The MATERIAL that is used for areas as they are added can be configured by the user. The default material is *Dense Deciduous Forest*. Clicking on the FOLIAGE MATERIAL button for the material allows

the user to either view the current material *Properties*, or to *Replace* the material with:

- A new material of a different type
- A material in the material database
- A material in another organic geometry within the project. If this option is used, the geometry currently being edited will make a copy of that material for its own use.

After choosing to *Replace* a material, the CHOOSE MATERIAL Window will appear, as seen in Figure 6.3.

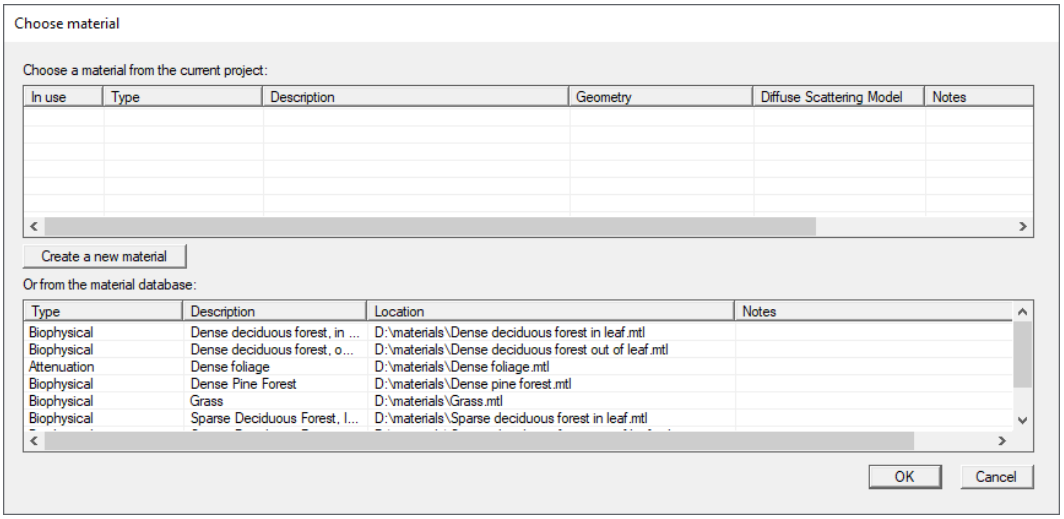


Figure 6.3: The Choose Material Window for foliage

After selecting a material from the database or creating a new material, click OK to begin using that material to create *Foliage*.

- Clicking the FOLIAGE MATERIAL button and selecting *Properties* will bring up the BIOPHYSICAL PROPERTIES Window, which is described in Section 10.6.

Creating New Foliage

To create a new *Foliage* geometry:

1. Click *File*→*New*→*Geometry*→*Foliage*. The EDIT FOLIAGE Window will appear.
2. If you would like to conform the foliage to the *Terrain*, and the project contains terrain, click the *Conform foliage to terrain* checkbox.
3. Click the *Foliage material* button to *Replace* the foliage MATERIAL or view its *Properties*.
4. Right-click in the EDIT FOLIAGE Window and select *New foliage area* from the context menu.
5. Define the footprint of the foliage by clicking the control points of the footprint. When the last point has been added, right click to finish creating foliage.

✓ When drawing the footprint, it may be useful as a guide to turn on the grid as a guide by adjusting the *Grid spacing*. If the vertices of the footprint fall exactly on the grid it may be useful to set the *Snap-to* mode to “Gridlines”.

6. Specify the *Base height* and *Top height* of the foliage, as seen in Figure 6.4.

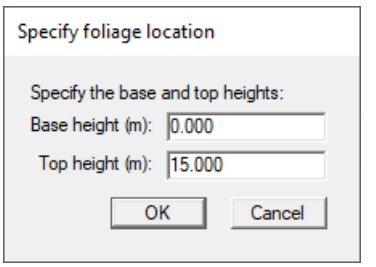


Figure 6.4: Specifying the height of a new foliage

7. Click OK when finished.

Adding Foliage

To add extra *Foliage* to an existing foliage geometry:

1. Set the selection mode in the GEOMETRY VIEW to *Geometry* (*Select→Geometry*).
2. Right-click on the foliage geometry that will be modified.
3. Select Edit from the right-click menu to open the EDIT FOLIAGE Window.
4. Follow Steps 4-7 above for creating new foliage.

Editing the Foliage Footprint

To edit the footprint of existing *Foliage*:

1. Set the selection mode in the GEOMETRY VIEW to *Geometry* (*Select→Geometry*).
2. Right-click on the *Foliage* that will be edited.
3. Select *Edit* from the current context menu of the GEOMETRY VIEW to open the EDIT FOLIAGE Window.
4. Rotate, zoom or pan the view if necessary.
5. In the EDIT FOLIAGE Window, click on the foliage group you want to change.
6. **To move a vertex**: Set the *Snap-to* mode to “Off”, click on the vertex, hold the mouse button down, and drag it to the new position.
7. **To delete a vertex**: Set the *Snap-to* mode to “Vertices”, then right-click on the point and select *Delete vertex*.
8. **To add a point**: Click on the line between two existing points.

9. **To change the height of existing Foliage**: Select *Change foliage height* from its context menu and then enter new *Base height* and *Top height* for the foliage.
10. Click OK to close the EDIT FOLIAGE Window.

6.2 GLCC (Global Land Cover Characteristics) Database

Geo-referenced foliage information can be imported from data in the GLCC database. This database has information on Seasonal Land Cover Regions (SLCR), which has a resolution of 1x1 km. The compressed image files.are installed along with Wireless InSite.

The regions and the required files are shown in Table 6.1.

Table 6.1: Data Files for Seasonal Land Cover Regions

Region	Data File
Africa	afslcr1_2g.img.gz
Australia Pacific	apslcr1_2g.img.gz
Eurasia	easlcr1_2g.img.gz
North America	naslcr1_2g.img.gz
South America	saslcr1_2g.img.gz

6.2.1 Importing Foliage from the GLCC Database

To import foliage from the GLCC database, a project must first contain a geo-referenced terrain, preferably a terrain imported from DTED or DEM files. The boundary of the geo-referenced terrain is used to located foliage for the same region. To import foliage from the GLCC database:

1. Select the *Terrain* in the MAIN WINDOW and choose *Pre-processing→Import foliage coverage from GLCC database* from the context menu.
2. The FOLIAGE CREATION window appears showing all of the foliage areas that were found in the database and the measurements that will be used to determine the material parameters of the *Foliage*. To model foliage properties, Wireless InSite uses the *Biophysical* material type.
   - The *Biophysical* material type is described in detail in Section 10.6.

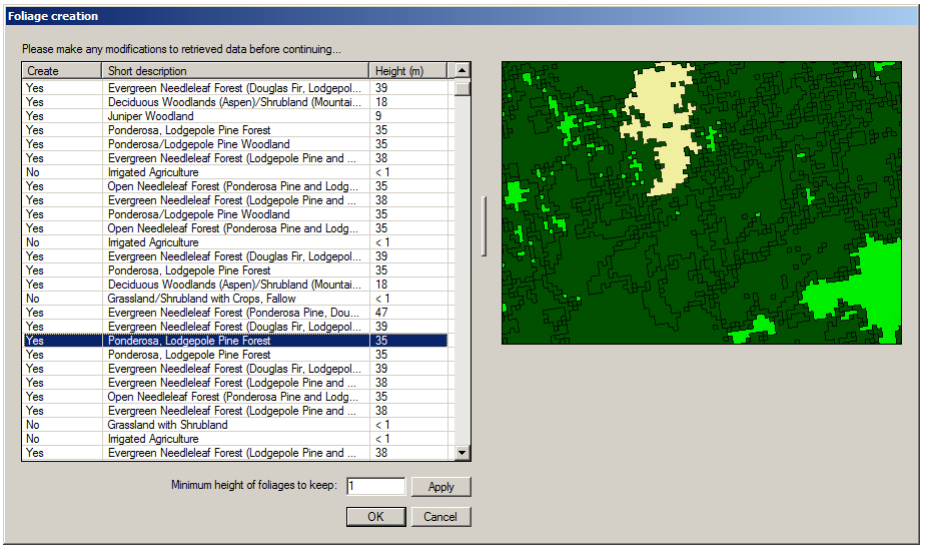


Figure 6.5: Previewing the GLCC data to be imported

Many of the foliage areas found in the database may be relatively short in height, and thus do not have a great effect on the radio wave's propagation. In this case it is convenient to set a minimum cutoff height for accepting foliage so that trees under the cutoff height are ignored by the importer. Doing so will increase the performance of the simulation and the refresh rate of the project display window by decreasing the amount of foliage that Wireless InSite needs to model.

There may be cases where more than one type of foliage occupies the same area. Here, larger and taller trees stand over shorter and smaller vegetation. To see the smaller foliage, make larger foliage invisible by toggling its visibility through its context menu.

The following options are available when editing foliage information for import. It can be found in the context menu of any entry in the table.

- *Visibility* - Indicates whether foliage should be visible in the preview window.

  ✓ Sometimes it is desirable to turn off visibility to see the foliage that lies underneath it.

- *Create* - Indicates if the foliage should be kept in the project during the import process.
- *Edit foliage information* - Displays a window where users can modify the foliage parameters.
- *Edit material information* - Displays a window for editing the properties of the MATERIAL the foliage area is composed of.

  ▶ See Section 10.6 for information about how to set the properties of the biophysical material type.

3. Once all of the modifications have been completed, click OK to complete the import process. All accepted foliage will be added to the GEOMETRY table. All materials that are needed for the selected areas will be created and added to the MATERIALS table.
4. To raise or lower the foliage to the terrain, Select the *Foliage* in the MAIN WINDOW and choose

Pre-processing→Conform to terrain from the context menu.

The conforming process may take a significant amount of time if there are many foliage areas to import, a large number of terrain facets, or if they cover a large portion of the terrain.

Chapter 7

Floor Plans

In this chapter, you will learn...

- → how to create and edit floor plans in Wireless InSite
- → how to import DXF floor plan data

Wireless InSite is capable of modeling signal propagation for virtually any indoor environment. Floor plans may be read into Wireless InSite from CAD files, such as DXF, or they can be created from scratch using the Wireless InSite Floor Plan Editor. This editor allows the user to create a custom indoor environment by specifying wall locations, wall heights, ceilings, floors, windows, and doorways. The material properties of each of these structures can be changed to accurately reflect the real environment. The editor also provides a method for modifying imported and existing floor plans.

7.1 Indoor Floor Plan Editor

Wireless InSite's Floor Plan Editor is used to create an indoor environment with walls, windows, doorways, a floor and a ceiling. It is also used to edit an existing *Floor plan*. Wireless InSite creates each new floor plan as a separate GEOMETRY and saves it to a separate file with the extension `*.flp`.

Materials for the walls, windows, doors, floor and ceiling of the floor plan are set separately. The default values of these materials are shown in Figure 7.1.

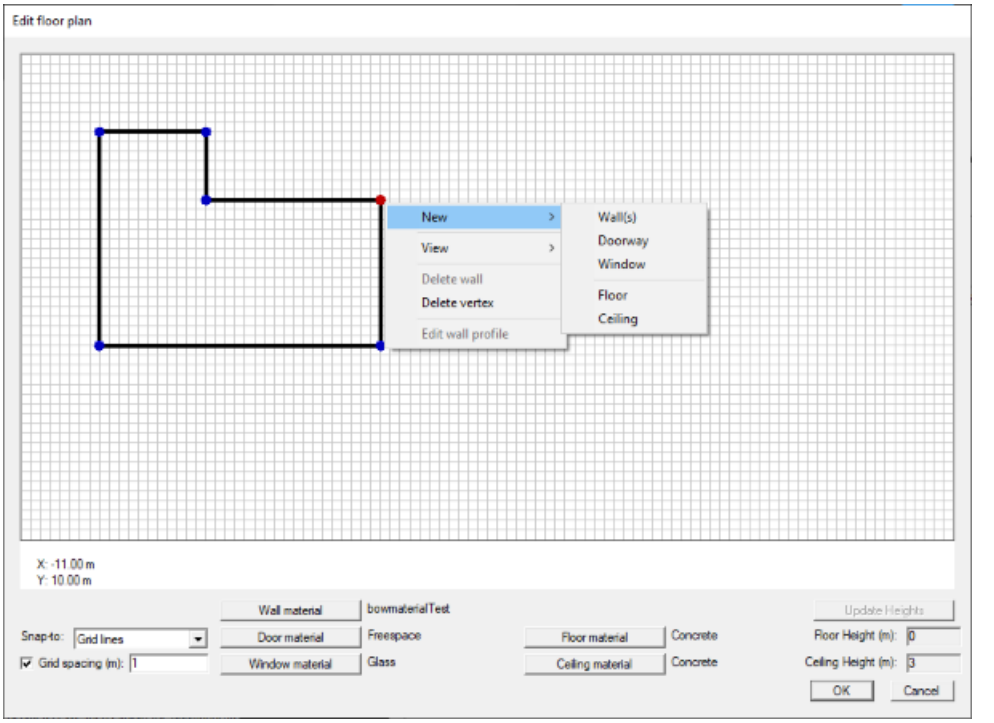


Figure 7.1: The Edit Floor Plan Window

Clicking the button for each material allows the user to either view the current materials properties, or to replace the material with:

- A new material of a different type
- A material in the material database
- A material that is in another non-organic geometry within the project. If this option is used, then the geometry currently being edited will make a copy of that material for its own use.

It is possible to create floor plans to fit inside of an existing building, or with an arbitrary outer perimeter. First we describe the steps for creating a new floor plan with an arbitrary outer perimeter. We then explain the additional steps required to fit the floor plan to an existing building substructure.

Creating a New Floor Plan

To create a new *Floor plan*:

1. Click *File*→*New*→*Geometry*→*Floor plan*.
2. Enter the elevation of the floor and ceiling.

   ✓ For example, if the indoor geometry with 3 m high walls is going to be placed on the third floor of a building, the base height might be set to 6 m and the top height to 9 m. All interior walls will be created with a height equal to the difference between the top and base heights.

3. After the heights are entered, the EDIT FLOOR PLAN Window, as seen in Figure 7.1, will start automatically.
4. Zoom or pan to adjust the view whenever necessary. The editor window may be resized or maximized if needed.
5. Set *Snap-to* mode "Grid lines" (optional).
6. Select *New→Wall(s)* from the editor's context menu.
7. Click on the endpoints of one or more connected wall segments.
8. Repeat Steps 6 and 7 to place additional walls.
9. Select *New→Doorway* from the editor's context menu.
10. Click on the location of the new doorway. At present doors are all initially created with *Free space*, but this can be reset to different material type after exiting from the editor.
11. Select *New→Window* from the editor's context menu.
12. Click on the location of the new window. At present all windows are created with the default *Glass*.
13. To add a floor or ceiling after the outer perimeter has been defined, set *Snap-to* mode to "Vertices" (optional).
14. Select *New→Floor (or Ceiling)* from the editor's context menu.
15. Click on vertices which define the polygon, then right-click after the last point.

Fitting a Floor Plan

To fit a *Floor plan* to an existing building:

1. Select the sub-structure which will contain the floor plan. Set the selection mode to sub-structures by clicking *Select→Sub-structure* in the GEOMETRY VIEW.
2. Right-click on the desired sub-structure and select *Insert new Floorplan*. The INSERT FLOOR PLAN dialog appears as shown below.
3. In this dialog select which parts of the *Floorplan* you want to add and change the material to use for them if the default is not desired. Set the desired *ceiling* and *floor* heights which are relative to the base of the selected sub-structure.
4. Click *OK* and the EDIT FLOOR PLAN dialog will appear.
5. Turn on the view of the floor and/or ceiling by selecting *View→Floor/Ceiling* from the context menu.
6. Continue adding walls, windows and doors as described above.

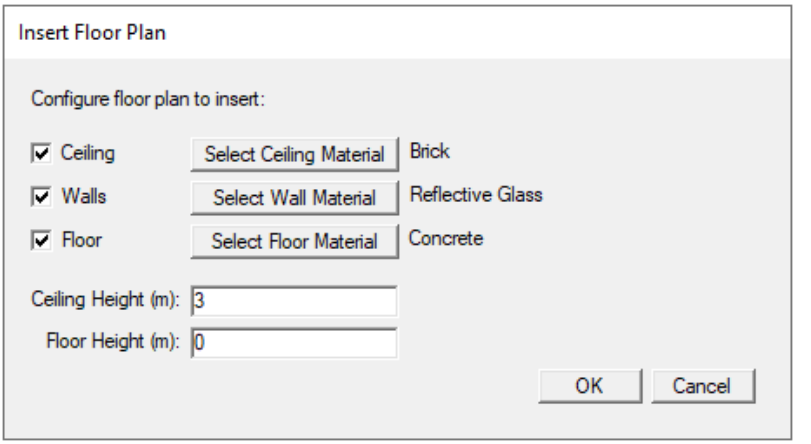


Figure 7.2: The dialog for inserting a floor plan into a substructure.

Updating Legacy Floor Plan Height

When editing a floor plan that was generated in Wireless InSite 3.3 or earlier, it may be necessary to update the floor and ceiling heights when adding new elements. In this case the UPDATE HEIGHTS button will be active, and the floor and ceiling heights displayed below. Note that these values apply to the next drawn item.

If the UPDATE HEIGHTS is active and you want to change the currently displayed heights:

1. Click the UPDATE HEIGHTS button.
2. Input the desired heights for the floor and ceiling. By default they will be filled with the floor plan's minimum and maximum height if they can be calculated.
3. A message box will pop up telling you the values you entered. Click OK to save values.
4. Continue editing the floor plan as usual, noting from now on all items will be created with the new heights.
5. Click OK to save the new height. Clicking CANCEL out of the EDIT FLOOR PLAN Window will result in all unsaved changes being lost including the heights.

7.2 Importing Indoor Geometries

To import floor plan data, navigate to *File→Import→Geometry*, and select the desired format (e.g. *DAE,KMZ,DXF*). Next select "Floor Plan" and click OK. For more information on importing DAE or KMZ files see Section 3.3.

For DXF formats, the IMPORT DXF DATA Window, as seen in Figure 7.3, will appear. See Section 4.2 for more details on DXF file content.

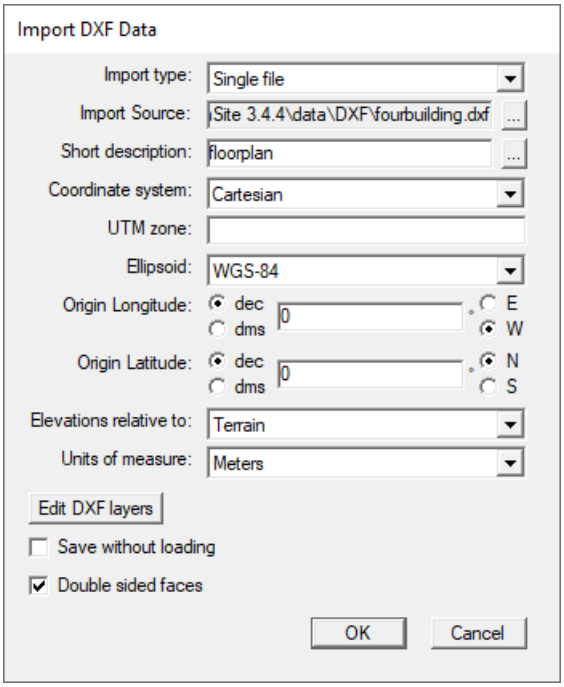


Figure 7.3: The DXF Import Window

The settings in the IMPORT DXF DATA Window are described below.

- *Coordinate System* - Allows the user to choose between *Cartesian* and *UTM* coordinates to match the format of the DXF file. Both coordinate systems allow the user to specify a rotation applied to the *Floor plan* when importing.
  - *Cartesian* coordinates allow the user to place the imported floor plan in the project by entering the appropriate values for the *Origin Longitude* and *Origin Latitude*.
  - *UTM* coordinates allow the user to enter data into the *UTM zone* and *Ellipsoid* fields to place the floor plan.
- *Elevations relative to* - Sets the height of the floor plan relative to "Terrain" or "Sea level".
- *Units of measure* - Enables users to identify the units used in the DXF file so that Wireless InSite can correctly convert the floor plan to meters. If the DXF file's units are not given as an option, the user can select "User-defined" and then may enter the conversion factor in the Meters per unit field.
- *Save without loading* - Converts the DXF file directly to a `*.flp` file without adding it to the project.
- *Double sided faces* - Defines the faces in floor plans as single- or double-sided.

After importing data, you can make use of the editing tools of the GEOMETRY VIEW and the EDIT FLOOR PLAN Window to make further modifications or to delete unwanted objects.

Chapter 8

Objects

In this chapter, you will learn...

- → how to create and edit an *Object* in Wireless InSite
- → how to import an *Object* from DXF

Objects represent generic geometric data, usually in the form of objects such as automobiles in a *City* scene or a table in a *Floor Plan*. They can be created from scratch using the Wireless InSite Object Editor, but are normally used to represent CAD data imported from various formats.

8.1 Object Editor

Wireless InSite's Object Editor is used to layout components in the same manner that the City Editor is used to layout buildings or to edit an existing *Object*.

The current material to use when creating components can be changed during the editing process and is applied to new components as they are created. The default material type is a PEC called *Metal* as shown in Figure 8.1. Clicking the button for the material allows the user to either view the current material properties, or to replace the material with:

- A new material of a different type
- A material in the material database
- A material that is in another non-organic geometry within the project. If this option is used, then the geometry currently being edited will make a copy of that material for its own use.

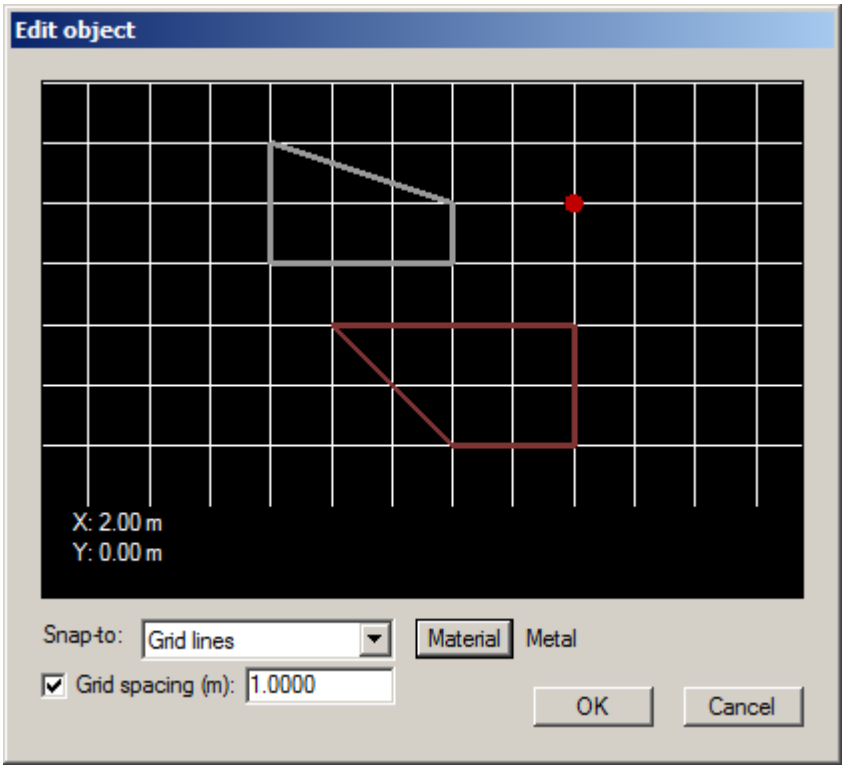


Figure 8.1: The Edit object Window

Creating a New Object

To create a new *Object*:

1. Click on *File*→*New*→*Geometry*→*Object*
2. Select *New component* from the editor's context menu.
3. Zoom or pan the view whenever necessary. The EDIT OBJECT Window may also be resized or maximized if needed.
4. Define the footprint by clicking on the vertex locations, then right-click after the last point.
   ✓ There is no need to close the footprint by clicking on the starting vertex.
5. Enter the *Height*, as seen in Figure 8.2.

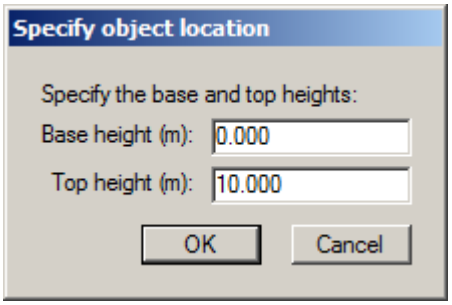


Figure 8.2: Creating a new component in the Object Editor

6. Repeat Steps 2-5 to add another component.
7. Click OK to close the EDIT OBJECT Window.

8.2 Importing DXF Files

Objects can be imported from DXF files in the same manner as *Cities*. This is described in section 4.2.

Chapter 9

Images

In this chapter, you will learn...

→ about the image file formats supported by Wireless InSite

→ how to import and adjust the properties of images

→ where to find TIFF images for use in Wireless InSite

To aid in creating projects and referencing results, Wireless InSite includes the capability to load images into projects. Images can be used to increase accuracy when placing GEOMETRY, TRANSMITTERS and RECEIVERS in a project. Images can contain any type of raster data. Common types of geo-referenced images include scanned maps, aerial photographs and satellite photos.

9.1 Geo-Referenced File Formats

Wireless InSite supports two raster image formats, GeoTIFF and TIFF. GeoTIFF is a TIFF file that contains embedded geo-referencing information. This data enables the image to be placed on the surface of the earth at an exact position. Because of this embedded data, GeoTIFFs cannot be translated to another position on the earth's surface. GeoTIFFs have one of the following file extensions: `*.tif` or `*.tiff`.

▶ The GeoTIFF standard is described in [2].

The second type of supported raster format is the standard TIFF file. A world file must accompany any TIFF file to be geo-referenced. Changing the values in the world file will move the image. The world file consists of UTM coordinates for the upper left-hand corner of the image as well as meter-per-pixel spacing in the X- and Y-directions. When using world files, the user must enter UTM zone and ellipsoid data. In addition to the UTM world file, Wireless InSite provides a Cartesian-world-file-like feature to allow users to specify the image coordinates in the global Cartesian coordinates of the project. World files can be entered manually or entered by the user directly into the OPEN IMAGE Window.

Changes to the global project origin will cause the image to shift if the image coordinates are specified using the Cartesian option.

9.2 Importing and Editing Images

To load an image, click *File→Open→Image*. The IMPORT IMAGE Window will appear, as seen in Figure 9.1. Wireless InSite can display rectangular portions of images, whether it is the whole image or a subset of it. When users select a GeoTIFF, Wireless InSite disables the boxes used to choose the coordinate system because the georeferencing information is embedded in the file. The boxes are enabled for other image types.

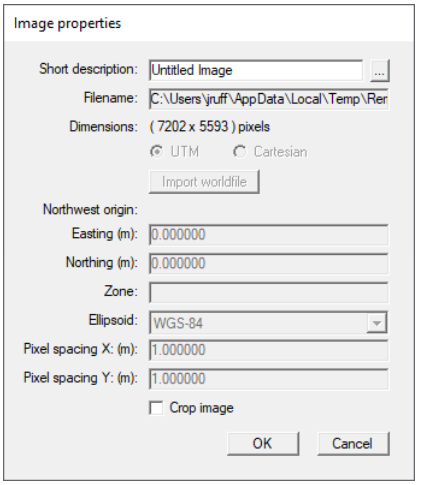


Figure 9.1: Importing an image

The settings in the IMPORT IMAGE Window are described below.

- *UTM* or *Cartesian* - Enables the user to switch between positioning the image in UTM or Cartesian coordinates within the project.
- IMPORT WORLDFILE - For geographic images which do not have embedded positioning information, their corresponding Worldfile is selected here.
- *Northing* - When the *UTM* coordinate system is selected, the northing of an images' top right corner is specified here.
- *Easting* - When the *UTM* coordinate system is selected, the easting of an images' top right corner is specified here.
- *Zone* - When the *UTM* coordinate system is selected, this specifies the zone used for the the position information.
- *Ellipsoid* - When the *UTM* coordinate system is selected, this specifies the ellipsoid used for the position information.
- *Pixel spacing X* - Defines the geographic width of each pixel in the X-direction (along a latitudinal line).
- *Pixel spacing Y* - Defines the geographic width of each pixel in the Y-direction (along a longitudinal line).

✓ Sometimes it may be necessary to modify the image pixel spacing after it has been imported to match the geometry more accurately. This can be done in the IMAGE PROPERTIES window.

Usually the Y-pixel spacing is a negative coordinate because (0, 0) in image coordinates is the upper left-hand corner.

✓ If an image appears inverted after it is imported you can change the pixel spacing from a positive to negative value to reflect it about the X or Y axis. For example, changing *Pixel spacing X* from 5 to -5 will reflect the image about the Y-axis.

- *Crop image* - After clicking OK, this option brings up the window seen in Figure 9.2. The crop box allows a subset of the image to be displayed. Cropping can be specified in UTM, Longitude/Latitude, Cartesian or pixel coordinates. If there is *Terrain* in the project, the FIT TO TERRAIN button will be enabled and will automatically set the coordinates to cover as much terrain as possible.

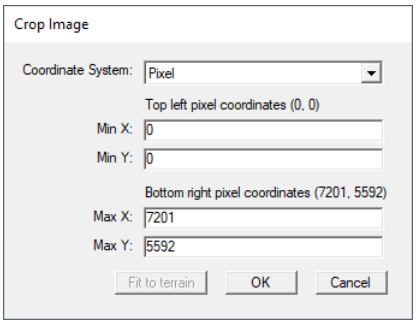


Figure 9.2: Cropping an image at import

When a *Terrain* is present and the GEOMETRY VIEW is in solid mode, the images will overlay the terrain. Otherwise in the Geometry View and in the various editor windows the images will appear flat at the minimum height of the project. Multiple images can be loaded into a project at the same time.

Images are not transparent when superimposed. Therefore, multiple images can be seen at once only if they are not occupying the same region of the terrain. In regions where images overlap, newly loaded images appear above the older images.

The images contained in a project are listed in the IMAGES tab, as seen in Figure 9.3. Changing the *Visible* state will toggle the image on or off.

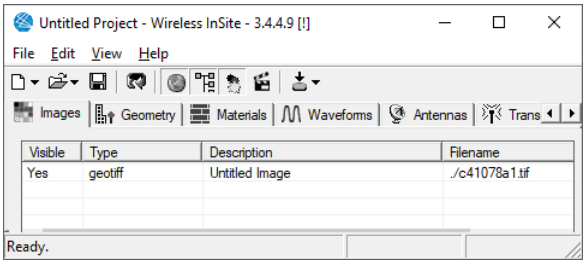


Figure 9.3: The table of images in the Main Window

Selecting *Properties* from the context menu of an image enables users to change the world file values of the image. The IMAGE PROPERTIES Window uses the same interface as the image loading window.

The combination of terrain and images has some interesting characteristics.

- Images blend with the color of the terrain.
- Terrain can be seen through images.
- Color by height can be used on terrain with images.
- Changing the color of the terrain material makes it easier to see the image.

In Figure 9.4, terrain is covered with a DOQ file and colored by height.

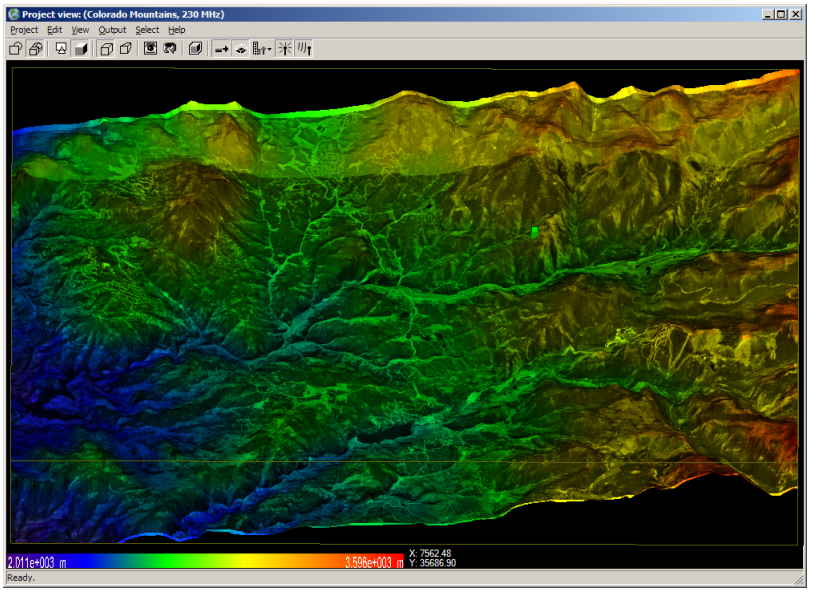


Figure 9.4: A DOQ image applied to a terrain with Color by height activated

When there are visible images in a project, they appear not only in the GEOMETRY VIEW but also in the editor windows. This is useful for moving and adding new GEOMETRY, TRANSMITTERS and RECEIVERS to the project. An example of the floor plan editor with a TIFF used to assist tracing the walls is shown in Figure 9.5.

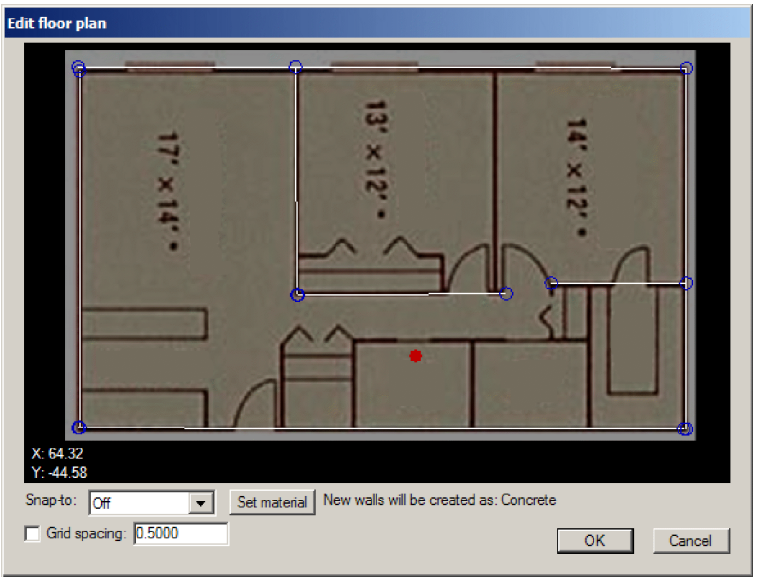


Figure 9.5: Creating a new floor plan using a background image for reference

9.3 Common Supported Files

9.3.1 USGS DRG Files

USGS (United States Geological Survey) DRG (Digital Raster Graphic) files are scanned representations of USGS topographic maps. They are available in GeoTIFF format, which is the only DRG format Wireless InSite supports.

- Further information about USGS DRGs is available at [3].

An example of part of an USGS DRG image is shown in Figure 9.6.



Figure 9.6: USGS DRG overlaid on imported terrain model

9.3.2 USGS DOQ And DOQQ Files

USGS DOQ images, as seen in Figure 9.7, are aerial photographs. Problems such as terrain shadowing and camera angle have been corrected so that these photos have the same accuracy as maps. DOQQ (Digital Orthophoto Quarter Quadrangle) are quarters of the 7.5-minute DOQ files.

- Further information about DOQ files is available in [4].



Figure 9.7: USGS DOQ of Rosslyn, VA area

9.4 Exporting The Geometry View

The contents of the Geometry View can be saved to a `*.tiff` image by selecting *View→Save as TIFF* from the GEOMETRY VIEW menu. In order to use this function, the color depth must be set to 16-bit, 24-bit or 32-bit. *View→Save as JPEG* is also available.

9.5 Data Sources

There are many sites on the Internet where TIFFs covering certain regions can be downloaded for free. Areas not available for free are usually available directly from the USGS, or data providers such as GIS Data Depot. Several sites that provide TIFFs and directories of sites that contain TIFFs include the following:

Table 9.1

More GIS Links:	http://mapdawg.tripod.com/data.htm
USGS DRG/DOQ page:	http://earthexplorer.usgs.gov/
GIS Data Depot:	http://data.geocomm.com/
Terraserver:	http://www.terraserver.com/

Chapter 10

Materials

In this chapter, you will learn...

- → the types of materials available in Wireless InSite, and their properties
- → equations used to define material properties
- → how to plot reflection and transmission coefficients
- → material file formats

In Wireless InSite, all GEOMETRY are ultimately composed of MATERIALS. The properties of a material are collectively referred to in this manual as "material types". The material properties include both the electromagnetic properties of the face as well as the display properties. The reflection and transmission coefficients are determined directly from the material properties, but the diffraction coefficients are determined indirectly through their dependence on the reflection and transmission coefficients. For some material types, the thickness is used to determine the reflection and transmission coefficients, but for others it only affects how the facet is displayed. Color and shininess, on the other hand, only affect how the facet is displayed for all material types. Wireless InSite provides tools which allow the user to easily assign a MATERIAL to individual faces as well as groups of faces. Materials may also be saved to the material database for use in other projects.

- ▶ See Chapter 3 for more on the properties of GEOMETRY, including the options for changing the MATERIAL assigned to a face.
- The reader should be aware that in some ways the term "materials" may be misleading, since it usually describes the reflection and transmission coefficients rather than the intrinsic properties of the material itself. (In the case of *Foliage*, however, the volumetric properties rather than surface scattering properties are most important. Foliage surface refraction is also important.) Because no single term seems to describe the various ways in which the electromagnetic characteristics are determined, we have decided to use the generic term "materials" rather than a more specific term which may be more accurate in some contexts, but less so in others.

10.1 Creating a Material

A MATERIAL is created by selecting *File→New→Material* and choosing from the list of material types. This will bring up a dialog to select the type of material to create as shown in figure 10.1.

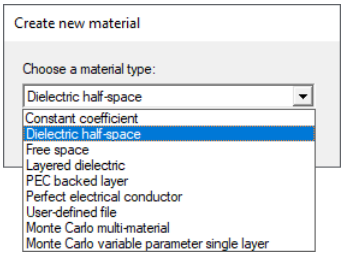


Figure 10.1: Selecting a new material to create.

10.2 Material Properties

Materials used in an Wireless InSite project are listed under the MATERIALS tab in the MAIN WINDOW, as shown in Figure 10.2. Clicking on the bar near the bottom of the window reveals a list of materials in the material database, displayed in a similar format.

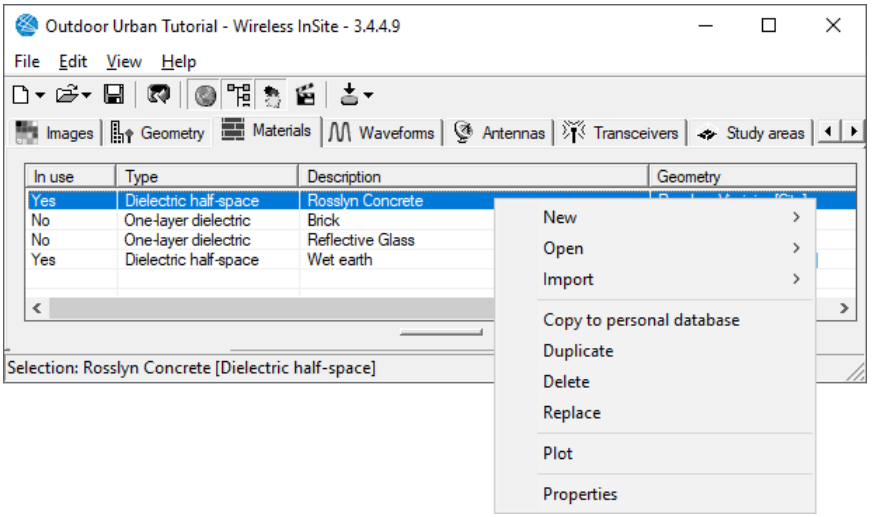


Figure 10.2: The list of materials under the Materials tab

The fourth column shows the GEOMETRY to which the MATERIAL belongs. It is important to understand that materials are saved separately for each geometry. Therefore, the user may create a material called “Metal” for more than one geometry, but each of these “Metals” may have different properties than the others. The material properties for each geometry are saved to the same file which contains the geometrical data of the geometry. In order to add the same material to a number of

geometries, first add the material to the material database, and then add it to each geometry from the database.

- The material database is discussed in Section 10.4.

The context menu for each MATERIAL includes the following options:

- *Copy to personal database* places a copy of the selected material into the material database for use in other Wireless InSite projects. This option is not available for engineered surfaces in order to enforce the workflow defined in section 3.7 for the *Change Material* operation.
- *Duplicate* creates a copy of the selected material. This option is not available for engineered surfaces in order to enforce the workflow defined in section 3.7 for the *Change Material* operation.
- *Delete* removes the selected material from the geometry which contains it.
- *Replace* assigns a different material to all faces in the geometry which are made of the selected material.
- *Plot* displays a 2D plot of material reflection or transmission coefficients versus incident angle or frequency. This option is not yet available for the engineered surface material.
- *Properties* accesses the properties of the selected material.

The properties window for each material type contains all parameters which define the material. For materials that are not file based there are common values that appear in each window, such as thickness and roughness, but most values are unique to a material type. Below is a list of common fields present in the material properties window for these material types.

- *Roughness* - The standard deviation of the surface height relative to the mean height, in meters.
- *Thickness* - The thickness of the material, in meters. Several material types use the thickness to determine the reflection and transmission coefficients, but for other material types the thickness is only used for display purposes. For example, a thickness must be entered for a *Dielectric Half-Space*, even though the reflection and transmission coefficients do not depend on the thickness. It is best in this case to enter a reasonable value, which can be zero, so the wall is displayed in an appropriate manner. Material types with more than one layer have a thickness for each layer, and in these cases the total thickness is used in the display.
- *Permittivity* - The permittivity of the material, specified relative to free space permittivity (ϵ_0 = 8.854x10^{-12}). Several material types allow the user to enter one or more permittivity values. In most cases only the real part of the permittivity is entered, but for *Foliage*, the complex permittivity in the form of $\epsilon_r = \epsilon_r' - j\epsilon_r''$ must be entered.
  - More information about different representations of the dielectric parameters and conversions between them can be found in Section 10.5.
- *Conductivity* - The conductivity of a material is entered in units of S/m. Several material types allow the user to enter one or more conductivity values.
  - More information about different representations of dielectric parameters and conversions between them can be found in Section 10.5.
- *Reflection Coefficient (parallel), etc.* - When using a *Constant Coefficient* Material, values are entered for the reflection and transmission coefficients for the electric field (polarized, parallel, and perpendicular to the plane of incidence).

- *Reflection/Transmission Coefficient Filename* - The names of the files containing the reflection and transmission coefficients for a *User-Defined* Material.
- *DS Enabled* - When checked, this field enables DIFFUSE SCATTERING properties for the material, which can be accessed and modified by clicking the DIFFUSE SCATTERING button. More information on DIFFUSE SCATTERING material properties can be found in Section 20.2.
- *Origin* - Sets the origin for an engineered surface to place it in the global coordinate reference frame.
- *Center On Face* - Resets the origin for an engineered surface so that it is centered on the face that it is assigned to.
- *Rotation* - Sets the rotation for an engineered surface which along with its origin defines its orientation.
- *Definition filename* - The source file for the data defining an engineered surface. More information about the format of these files can be found in Appendix G.
- *Volumetric* - When using a *Layered dielectric* material, this property can be enabled to indicate that two facets bound a single volume. This requires that facets on opposing sides of an object (with facet normal facing in opposite directions to within 1 degree) both have the *Volumetric* property enabled. Rays that pass through the two facets will then be treated as having transmitted through a single material. The *Thickness* property for the opposing facets should match the physical separation distance between them.

10.3 Material Types

Material types are divided into two groups: Foliage and Non-Foliage. Foliage materials may only be used in *Foliage* geometries, while the Non-Foliage materials may be used freely in all other geometries.

10.3.1 Materials for Use with Non-Foliage Geometry

The following Non-Foliage material types are available in Wireless InSite.

1. *Constant Coefficient*

   This material is defined independent of angle and frequency. Separate coefficients are assigned for each polarization. The coefficients are given as the ratio of the reflected or transmitted field to same component of the incident field, that is:

2. *Dielectric Half-Space*

   The reflection coefficients for Dielectric Half-Space are that of a semi-infinite dielectric block. The transmission coefficients are always zero for this material type. The angle- and frequency-dependent reflection coefficients are calculated from the Fresnel plane wave reflection coefficients. Two typical applications for Dielectric Half-Space materials are exterior building walls and *Terrain*. The parameters needed are permittivity, conductivity, roughness, and thickness (used only for display purposes).

- See Section 10.11 for information on Fresnel Coefficients.

3. *Engineered Surface*

   These materials are represented by a grid of surface impedances and phase gradients, allowing the X3D model to incorporate CRC's Ray-Optical Model for electromagnetically engineered surfaces.

   - See Appendix G for information on how to define an *Engineered surface*.

4. *Free Space*

   Faces assigned this type are rendered as transparent. The reflection coefficient is zero and the transmission coefficient is one for both polarizations.

5. *Layered Dielectric*

   This material is defined by plane wave reflection and transmission coefficients for multiple finite thickness dielectric layers. The parameters needed for each dielectric layer are permittivity, conductivity, roughness, and thickness. The field is always assumed to be incident on the first dielectric layer.

6. *Perfect Electrical Conductor (PEC)*

   Faces assigned with this material act as perfect electrical conductors, from which all energy is reflected. The transmission coefficient is zero, and the parameters are roughness and thickness. It should typically be used as an approximation when a good conductor is needed in a simulation and losses aren't important.

7. *PEC Backed Layer*

   This material is defined by plane wave reflection and transmission coefficients for a single finite thickness dielectric layer backed by a perfectly conducting layer. The field is always assumed to be incident on the dielectric layer, not the PEC layer. The transmission coefficient is zero. The parameters required for this material are the permittivity, conductivity, roughness, thickness of the dielectric, and thickness of the PEC layer, with the latter only used for display purposes.

$$R_{\perp} = \pm \left| \frac{E_{\perp}^{r}}{E_{\perp}^{i}} \right| \tag{10.1}$$

   The coefficients must be real numbers, and as such cannot include phase except for an overall negative sign.

8. *User-Defined*

   Wireless InSite can read user-defined material files with reflection and transmission coefficients. The coefficients can vary in angle of incidence, orientation of the incident plane (optional), and frequency (optional). When a Wireless InSite project with one or more user-defined materials is started, Wireless InSite reads and stores the coefficient table(s) from the specified file(s). During the run, when a reflection, transmission, or diffraction interaction involving a user-defined material is evaluated, Wireless InSite determines the incidence plane and incidence angles, interpolates the reflection and/or transmission coefficients from the user's data, and then applies those coefficients to the respective interaction calculations. The material acts in all other respects during the simulation as any other material would. A complete description of the user-defined material file is available in Appendix F.

10.3.2 Materials for Use with Foliage Geometry

The following foliage materials are recognized by the FULL 3D and VERTICAL PLANE propagation models. See 15.3.2 for using foliage with the X3D model.

1. *Attenuation*

   The attenuation in dB/m and the real part of the permittivity are specified for both the vertically and horizontally polarized fields.

2. *Biophysical*

   The effective permittivity is determined from the biophysical characteristics of the tree, such as leaf area, leaf density, branch size, branch density, etc. This material type also takes the permittivity of the leaves and branches into account when determining the effective permittivity of the foliage. There is also an option for adjusting the effective permittivity to agree with a measured attenuation at a particular frequency, polarization and incidence angle.

   ▶ This material is described in detail in Section 10.6.

3. *Complex Permittivity*

   A complex permittivity in the form $|\epsilon_r = \epsilon_r' - j\epsilon_r''|$ must be entered for both vertically and horizontally polarized fields.

4. *Lossy Dielectric*

   The real part of the permittivity and the effective conductivity are specified for both vertically and horizontally polarized fields.

10.4 Material Database

The material database includes a small number of common building, ground, and foliage materials, in addition to a few generic material types. When one of these materials is assigned to one or more faces within a GEOMETRY, the material properties are written to the geometry file and become part of that particular geometry. No link to the entry in the database is retained. Once a material in the database is applied to a geometry, modifications to the properties of the material are only applied to that particular geometry, and the original properties of the material in the database are left unchanged.

▶ The procedures for adding new materials to the database and modifying existing ones can be found in Chapter 24.

A brief description of the materials in the database follows. The symbols ϵ and σ are used to designate the relative permittivity and the conductivity, respectively.

Exterior Building Materials

- **Concrete Wall**: *type* = *Layered Dielectric*
  - Layer 1: $\epsilon = 7$, $\sigma = 0.015$ S/m, $thickness = 0.3$ m
- **Brick Wall**: *type* = *Layered Dielectric*

  - Layer 1: $\epsilon = 4.44$, $\sigma = 0.001$ S/m, $thickness = 0.125$ m

- **Wood**: *type* = *Layered Dielectric*
  - Layer 1: $\epsilon = 5$, $\sigma = 0$ S/m, $thickness = 0.03$ m
- **Plate Glass**: *type* = *Layered Dielectric*
  - Layer 1: $\epsilon = 2.4$, $\sigma = 0$ S/m, $thickness = 0.003$ m
- **Asphalt (1 GHz)**: *type* = *Dielectric Half-Space*
  - Layer 1: $\epsilon = 5.72$, $\sigma = 5e - 4$ S/m

Interior Building Materials

- **Layered drywall**: *type* = *Layered Dielectric*
  - Layers 1 and 3: Drywall $\epsilon = 2.8$, $\sigma = 0.001$ S/m, $thickness = 0.013$ m
  - Layer 2: Air $\epsilon = 1$, $\sigma = 0$ S/m, $thickness = 0.089$ m

Terrain Materials

- **Wet Ground**: *type* = *Dielectric Half-Space*
  - Layer 1: $\epsilon = 20$, $\sigma = 0.02$ S/m
- **Dry Ground**: *type* = *Dielectric Half-Space*
  - Layer 1: $\epsilon = 4$, $\sigma = 0.001$ S/m
- **Dry Sand**: *type* = *Dielectric Half-Space*
  - Layer 1: $\epsilon = 4$, $\sigma = 0.0002$ S/m
- **Fresh Water**: *type* = *Dielectric Half-Space*
  - Layer 1: $\epsilon = 81$, $\sigma = 0.22$ S/m
- **Sea Water**: *type* = *Dielectric Half-Space*
  - Layer 1: $\epsilon = 81$, $\sigma = 20$ S/m

Foliage Materials

- **Dense Deciduous Forest In Leaf**: *type* = *Biophysical*
- **Sparse Deciduous Forest In Leaf**: *type* = *Biophysical*
- **Dense Deciduous Forest Out of Leaf**: *type* = *Biophysical*
- **Sparse Deciduous Forest Out of Leaf**: *type* = *Biophysical*

- **Dense Pine Forest**: *type* = *Biophysical*
- **Sparse Pine Forest**: *type* = *Biophysical*
- **Grass**: *type* = *Biophysical*

Frequency-Sensitive Materials

Materials generally impact propagation differently depending on the frequency of the radio waves interacting with them. Wireless InSite provides several material definitions in its material database based on the ITU (International Telecommunication Union) recommendations for four representative Wi-Fi and 5G millimeter wave frequencies. The building material properties are calculated using the formulas described in [5]; those for soil are based on the plots published in [6]. The *relative permittivity* and *conductivity (S/m)* of the included materials are shown in tables 10.1 and 10.2.

Table 10.1: Properties of Included Wi-Fi Frequency-Sensitive Materials

	2.4 GHz		**5 GHz**	
Material	*Relative Permittivity*	*Conductivity (S/m)*	*Relative Permittivity*	*Conductivity (S/m)*
Concrete	5.31	0.066221437	5.31	0.119959271
Drywall	2.94	0.021552391	2.94	0.036228401
Wood	1.99	0.012011805	1.99	0.026378732
Glass	6.27	0.01221435	6.27	0.02930827
Ceiling board	1.5	0.001384547	1.5	0.003252009
Floorboard	3.66	0.014365084	3.66	0.038735631
Very dry ground	3	0.001362146	3	0.008659557
Medium dry ground	13.74263889	0.145818413	12.77009884	0.482379843
Wet ground	21.1366793	0.468129353	15.75916683	1.215492448

Table 10.2: Properties of Included 5G Frequency-Sensitive Materials

	28 GHz		**60 GHz**	
Material	*Relative Permittivity*	*Conductivity (S/m)*	*Relative Permittivity*	*Conductivity (S/m)*
Concrete	5.31	0.483829146	5.31	0.896666691
Drywall	2.94	0.122593378	2.94	0.210221831
Wood	1.99	0.167171345	1.99	0.3783732
Glass	6.27	0.228667481	6.27	0.567431997
Ceiling board	1.5	0.024132019	1.5	0.058569453
Floorboard	3.66	0.397453894	3.66	1.11333046
Very dry ground	3	0.3	3	0.75
Medium dry ground	5.7	6.5	4.3	14
Wet ground	5.7	9.5	4.3	15

For convenience, we have plotted *relative permittivity* and *conductivity* as a function of frequency for several of the building materials as defined in the two ITU references. Figure 10.3 shows the *relative*

permittivity, and Figure 10.4 shows the *conductivity (S/m)* of the included building materials between 100 MHz and 100 GHz based on the ITU formulas. These can be used to define materials for other frequencies of interest. As shown in the plots, for many of the materials, the ITU references suggest that the *relative permittivity* is fairly stable with frequency, whereas the *conductivity* tends to vary significantly over this frequency range.

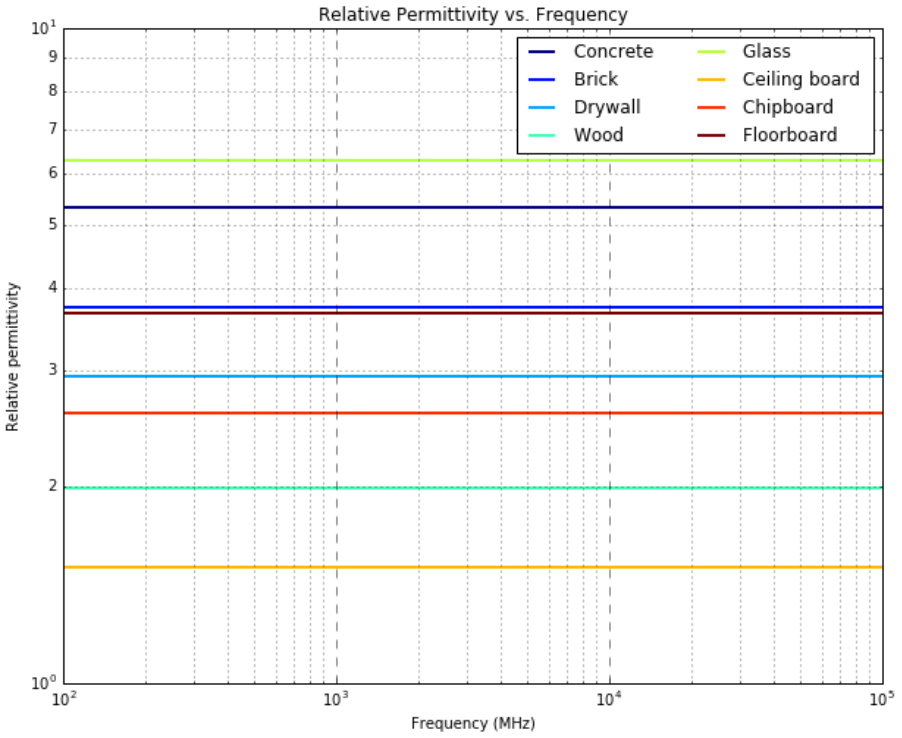


Figure 10.3: Relative permittivity of building materials, based on ITU recommendation.

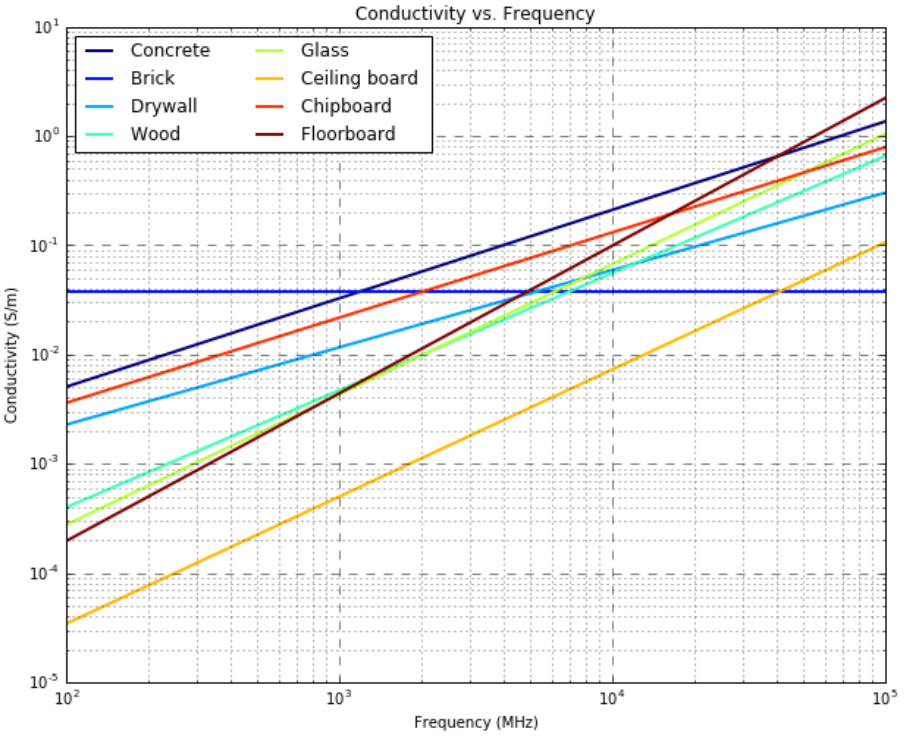


Figure 10.4: Conductivity of building materials, based on ITU recommendation.

Miscellaneous Materials

- **Free Space**: *type* = *Free Space*
  - $R_{\perp} = 0, R_{\parallel} = 0, T_{\perp} = 1, T_{\parallel} = 1$
- **Perfect Absorber**: *type* = *Constant Coefficient*
  - $R_{\perp} = 0, R_{\parallel} = 0, T_{\perp} = 0, T_{\parallel} = 0$

10.5 Dielectric Parameters

There are various ways of representing the constitutive parameters of a lossy dielectric medium. Wireless InSite commonly uses the real part of the materials' permittivity and conductivity. The one exception is the use of the *Complex Permittivity* material for *Foliage*. Complex Permittivity is an option because it is one of the more commonly encountered representations for foliage. At present, all MATERIALS in

Wireless InSite are nonmagnetic, and the permeability for all materials is that of free space ($\mu_0 = 4\pi\ x\ 10^{-7}$ H/m).

Some of the commonly encountered parameters used for lossy dielectric materials, along with formulas for converting between them, are discussed below. In all formulas, the permittivity is designated by ϵ, and the real and imaginary parts by ϵ' and ϵ'', respectively. The real part of the relative permittivity is sometimes referred to as the dielectric constant, although this term is also used for the complex quantity. In many formulas it is the relative permittivity which is one of the parameters, and the absolute permittivity is obtained by multiplying the relative value by the permittivity of free space ($\epsilon_0 = 8.854x10^{-12}$ F/m). To avoid any confusion, an "r" subscript will be placed on all relative values, and an absolute value should be assumed otherwise.

- The notation used here generally follows that described in [7, Ch. 2]. More information on dielectric constitutive parameters is also found in this reference.

It is important to note that many of the formulas given below are only accurate for a single frequency, so if calculations are made at significantly different frequencies, it may be necessary to recompute the dielectric parameters.

Permittivity and Conductivity

The complex permittivity can be written as:

$$\epsilon = \epsilon' - j\frac{\sigma_e}{\omega} \tag{10.2}$$

where ϵ' is the real part of the permittivity and σ_e is the effective conductivity of the material.

Complex Permittivity

The permittivity can also be expressed as:

$$\epsilon = \epsilon' - j\epsilon'' \tag{10.3}$$

where the static conductivity is assumed to be zero ($\sigma_s = 0$).

The effective conductivity is given by:

$$\sigma_e = \epsilon''\omega \tag{10.4}$$

where ω is the angular frequency in radians/s.

Complex Permittivity and Static Conductivity

When both the imaginary part of the permittivity and the static conductivity, σ_s , are given, the complex permittivity can be expressed as:

$$\epsilon = \epsilon' - j\epsilon'' - j\frac{\sigma_s}{\omega} \tag{10.5}$$

where an additional term involving the static conductivity is also present in this representation.

The effective conductivity is then given by:

$$\sigma_e = \epsilon''\omega + \sigma_s \tag{10.6}$$

where ω is the angular frequency in radians/s and σ_s is the static conductivity.

The effective conductivity can also be expressed as:

$$\sigma_e = \sigma_s + \sigma_a \tag{10.7}$$

where

$$\sigma_a = \epsilon''\omega \tag{10.8}$$

and is sometimes called the alternating field conductivity.

The effective complex permittivity can also be written as:

$$\epsilon = \epsilon' - j\epsilon_e'' \tag{10.9}$$

where the imaginary part of the effective permittivity is given by:

$$\epsilon_e'' = \epsilon'' + \frac{\sigma_s}{\omega} \tag{10.10}$$

Electric Loss Tangent

Instead of specifying an imaginary part of the permittivity and/or a conductivity, the loss can be expressed as a tangent of the phase angle between the real and imaginary part of the effective permittivity:

$$tan(\delta_e) = \frac{\sigma_e}{\omega\epsilon'} = \frac{\sigma_s}{\omega\epsilon'} + \frac{\epsilon''}{\epsilon'} \tag{10.11}$$

The effective conductivity is given by:

$$\sigma_e = \omega\epsilon' tan(\delta_e) \tag{10.12}$$

Susceptibility

In terms of the susceptibility, the complex permittivity (omitting the conductivity) is:

$$\epsilon_r = (1 + \chi') - j\chi'' \tag{10.13}$$

where

$$\epsilon_r' = 1 + \chi' \tag{10.14}$$

$$\epsilon_r'' = \chi'' \tag{10.15}$$

The effective conductivity can be found from:

$$\sigma_e = \chi''\epsilon_0\omega + \sigma_s \tag{10.16}$$

Index of Refraction

Instead of the permittivity one may occasionally find values for the index of refraction of a material, usually given as a complex number, $n = n' - jn''$. The index of refraction is related to the permittivity by $n = \sqrt{\epsilon_r}$. It is straightforward to show that the loss tangent angle is given by:

$$\delta = 2tan^{-1}\left(\frac{n''}{n'}\right) \tag{10.17}$$

and the real part of the permittivity is found from:

$$\epsilon_r' = \sqrt{\frac{n'}{sec(\delta)cos\left(\frac{\delta}{2}\right)}} = \sqrt{\frac{n''}{sec(\delta)sin\left(\frac{\delta}{2}\right)}} \tag{10.18}$$

and the effective value for the imaginary part of the permittivity is given by:

$$\epsilon_r'' = \epsilon_r' tan(\delta) \tag{10.19}$$

and the effective conductivity by:

$$\sigma_e = \epsilon''\epsilon_0\omega \tag{10.20}$$

Propagation Constant

The propagation vector of a wave propagating in a dielectric medium is given by:

$$k = nk_0 = \sqrt{\epsilon_r k_0} = k' - jk'' \tag{10.21}$$

where

$$k_0 = \frac{2\pi}{\lambda_0} = \omega/c \tag{10.22}$$

and λ_0 is the wavelength in free space.

Equations 10.18 and 10.19 can be applied to the propagation vector by replacing n' by $\frac{k'}{k_0}$ and n'' by $\frac{k''}{k_0}$.

In some cases, it is useful to write the propagation constant in the form:

$$k = k_0(1 + \kappa' - j\kappa'') \tag{10.23}$$

Solving for the permittivity gives:

$$\epsilon_r' = (1 + \kappa')^2 - \kappa''^2 \tag{10.24}$$

$$\epsilon_r'' = 2\kappa''(1 + \kappa') \tag{10.25}$$

When the real part of the permittivity and the attenuation are known, the imaginary part of the permittivity is given by:

$$\epsilon_r'' = 2\kappa''\sqrt{\epsilon_r' + \kappa''^2} \tag{10.26}$$

When the medium is *Foliage*, it is usually possible to assume that κ',κ'' $<<$ 1, and the equations for the relative permittivity reduce to:

$$\epsilon_r' \approx 1 + 2\kappa' \tag{10.27}$$

$$\epsilon_r'' \approx 2\kappa'' \tag{10.28}$$

Skin Depth

The skin depth is the distance into a material at which the amplitude of the electric field has decayed to $e^{-1} = 0.37$ of its value at the surface of the dielectric for a plane wave normally incident on the surface. The skin depth is easily shown to be equal to $\frac{1}{k''}$ where k'' is the imaginary part of the propagation constant. In general, k'' is given by:

$$k'' = k_0\sqrt{\epsilon_r'^2 + \left(\frac{\sigma_e}{\omega\epsilon_0}\right)^2} sin\left(\frac{\delta_e}{2}\right) \tag{10.29}$$

For a good dielectric $\left(\left(\frac{\sigma}{\omega\epsilon'}\right)^2 << 1\right)$ the skin depth is equal to:

$$\frac{1}{k''} \approx \frac{2}{\sigma}\sqrt{\frac{\epsilon'}{\mu_0}} \tag{10.30}$$

and for a good conductor $\left(\left(\frac{\sigma}{\omega\epsilon'}\right)^2 >> 1\right)$ the skin depth is equal to:

$$\frac{1}{k''} \approx \sqrt{\frac{2}{\omega\mu_0\sigma}} \tag{10.31}$$

10.6 Biophysical Foliage Material

The effective permittivity of a region containing *Foliage* can be approximated by modeling the foliage as a dielectric slab containing randomly oriented leaves and branches which act to scatter and attenuate the propagating field ([8], [9], [10]). The leaves and branches are represented by lossy-dielectric discs and cylinders with orientations described by random distributions, as seen in Figure 10.5. The size and number density of the leaves and branches are described by biophysical parameters, which include leaf radius, leaf thickness, branch radius, and branch length. The model also accounts for the dielectric properties of the leaves and branches.

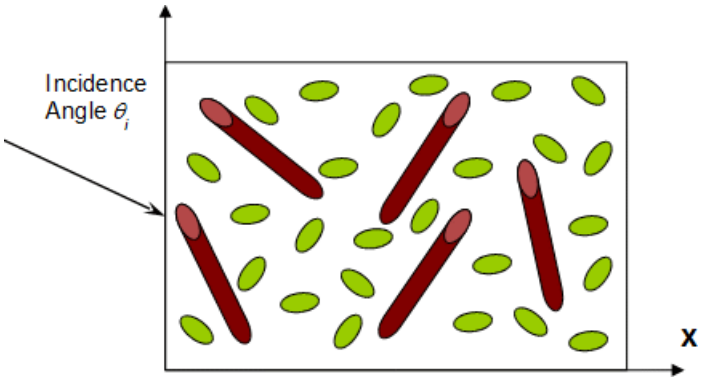


Figure 10.5: Representation of the foliage as a collection of randomly oriented scattering disks and cylinders

By considering the propagation of the mean field through the foliage, a frequency- and polarization-dependent effective permittivity can be calculated from the propagation constant of the mean field through the foliage. The mean field within the foliage can be considered to be equal to the incident field on the foliage, provided the volume fraction of the leaves and branches is small. The biophysical model is valid for frequencies between 200 MHz and 2 GHz.

Figures 10.6 and 10.7 show the typical frequency-dependent behavior of the real and imaginary parts of the effective permittivity generated by this type of model.

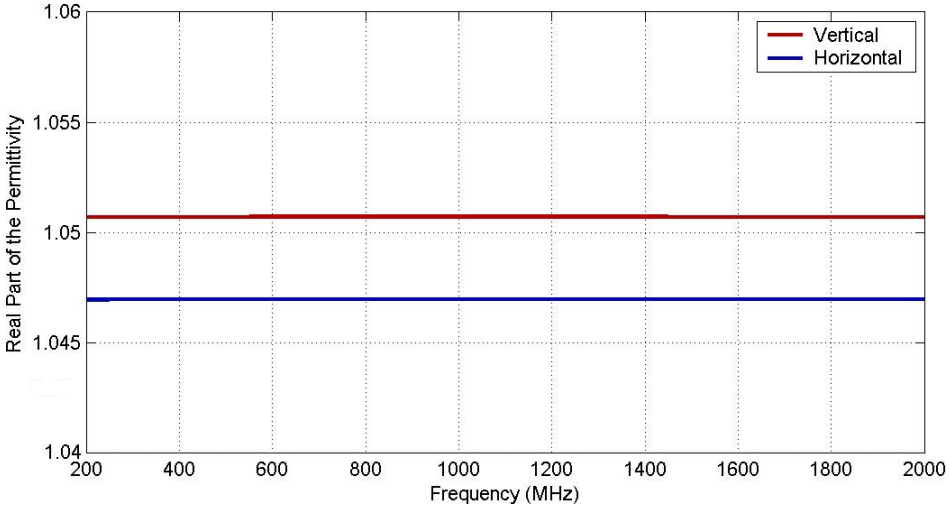


Figure 10.6: Real part of the effective permittivity versus frequency for a given set of foliage parameters

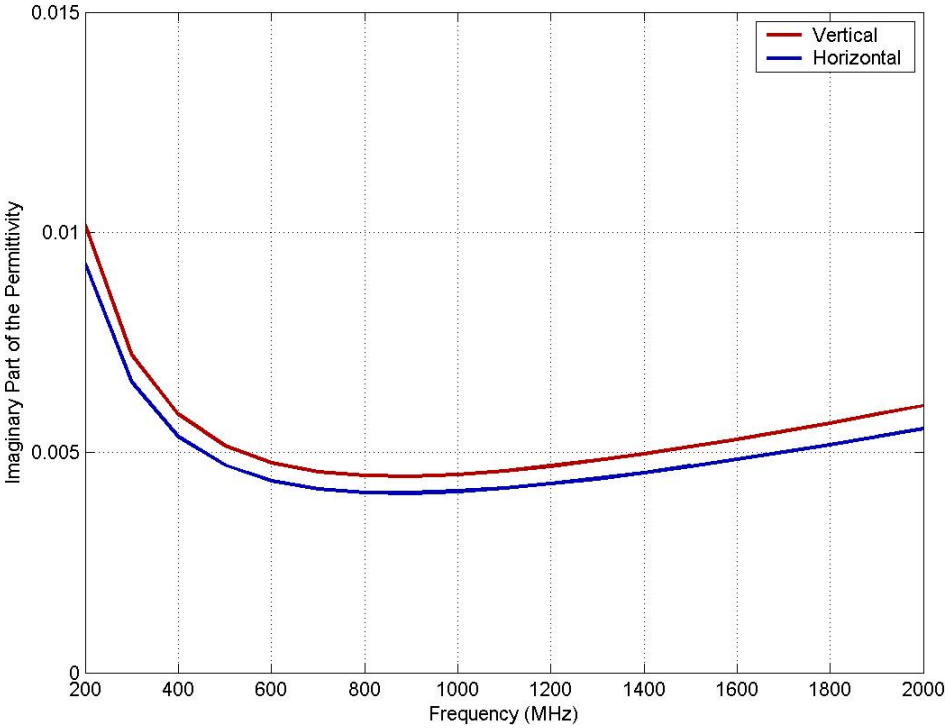


Figure 10.7: Imaginary part of the effective permittivity versus frequency for a given set of foliage parameters

The effective complex permittivity ϵ_{eff} of a foliage is related to the foliage's susceptibility χ_{eff} and propagation constant K_{eff} by the following relation [11, Ch. 7]:

$$\epsilon_{eff} = 1 + \chi_{eff} = \frac{K_{eff}^2}{k_0^2} \tag{10.32}$$

where k_0 is the free space propagation constant, given by [12]:

$$k_0 = \frac{2\pi f}{c} \tag{10.33}$$

where f is the frequency and c is the speed of light.

The propagation constant within a volume containing vegetation can be represented as the sum of the free space propagation constant and the deviations from the free space propagation constants caused by the different types of scatterers (leaves and branches) within the volume [7, Ch. 5]:

$$K_{eff}^p = k_0 sin\theta_i + \sum_t \kappa_t^p \tag{10.34}$$

where

p is the polarization of the mean field

θ_i is the angle of incidence of the field on the foliage

κ is the complex deviation

t is the type of scatterer

Presently, the model only accounts for scattering from leaves and branches, reducing equation above to:

$$K_{eff}^{p} = k_0 sin\theta_i \; + \; \kappa_{leaves}^{p} \; + \; \kappa_{branches}^{p} \tag{10.35}$$

The following equations describe the deviation from the free space propagation constant for leaves and branches for horizontally and vertically polarized fields [13] [14].

$$K_{leaf}^{V} = k_0 \left(\frac{\rho_l}{2}\right) \frac{(\pi r_l^2 t_l)\chi}{sin\theta_i} \left[1 \; - \; \frac{\chi}{2(1+\chi)} \left(I_1 cos^2\theta_i \; + \; 2I_2 sin^2\theta_i\right)\right] \tag{10.36}$$

$$K_{leaf}^{H} = k_0 \left(\frac{\rho_l}{2}\right) \frac{(\pi r_l^2 t_l)\chi}{sin\theta_i} \left[1 \; - \; \frac{\chi}{2(1+\chi)} I_1\right] \tag{10.37}$$

$$K_{branch}^{V} = k_0 \left(\frac{\rho_b}{2}\right) \frac{(\pi r_b^2 l_b)\chi}{sin\theta_i} \left[\frac{\chi}{2(2+\chi)} \left(I_1 cos^2\theta_i \; + \; 2I_2 sin^2\theta_i\right) \; + \; \frac{2}{2+\chi}\right] \tag{10.38}$$

$$K_{branch}^{H} = k_0 \left(\frac{\rho_b}{2}\right) \frac{(\pi r_b^2 l_b)\chi}{sin\theta_i} \left[\frac{\chi}{2(2+\chi)} I_1 \; + \; \frac{2}{2+\chi}\right] \tag{10.39}$$

where

ρ_i is the number of leaves per cubic meter

r_i is the leaf radius

t_i is the leaf thickness

ρ_b is the number of branches per cubic meter

r_b is the branch radius

l_b is the length of the branch

χ is the complex susceptibility of the scatterer type

These values represent the average physical parameters present in the foliage. I_1 and I_2 are factors that account for the randomly distributed orientations of the leaves and branches in the foliage:

$$I_1 = \int_{\theta_1}^{\theta_2} p(\theta) sin^2\theta d\theta \tag{10.40}$$

$$I_2 = \int_{\theta_1}^{\theta_2} p(\theta) cos^2\theta d\theta \tag{10.41}$$

where $p(\theta)$ is a probability function describing the inclination angle θ of the leaves and branches. In this implementation, the orientations are described by a uniform random distribution:

$$p(\theta) = \frac{1}{\theta_2 - \theta_1} \tag{10.42}$$

where θ_1 and θ_2 are both measured with respect to the positive Z-axis.

The electrical properties of the leaves and branches are contained in the permittivity of the individual components. The real and imaginary parts of leaf and branch permittivity can be specified using a constant or frequency-dependent model [13] [15, Ch. 8].

Constant permittivity model

$$\epsilon = \epsilon' - j\epsilon'' \tag{10.43}$$

Frequency-dependent permittivity model

$$\epsilon = \epsilon_\infty + \frac{\epsilon_S - \epsilon_\infty}{1 + j(2\pi f)t_0} - j\frac{\sigma}{\epsilon_0\, 2\pi f} \tag{10.44}$$

where

ϵ' is the real component of the permittivity

ϵ'' is the complex component of the permittivity

f is the frequency in Hz

ϵ_S is the static permittivity

ϵ_∞ is the infinite frequency permittivity, assumed to be 1.0

ϵ_0 is the permittivity of free space

σ is the conductivity

t_0 is the relaxation time in seconds, assumed to be 9.4e-12

Finally, the specific attenuation of the foliage can be calculated from the imaginary part of the effective propagation constant [9]:

$$\alpha_{foliage} = -(20log_{10}e)Im\{K_{eff}\} = -8.686Im\{\kappa^p_{leaves} + \kappa^p_{branches}\} \tag{10.45}$$

Defining a Biophysical Material

The BIOPHYSICAL PROPERTIES Window, as seen in Figure 10.8, contains parameters used to define *Biophysical* material, including *Theta*, the *Data type* and relevant input parameters, the *Permittivity model* and relevant parameters, and *Waveform* characteristics.

✓ When entering values into the BIOPHYSICAL PROPERTIES Window, all lengths are assumed to be in meters, all densities are entered as the number of items per cubic meter, and all angles are given in degrees with respect to the positive Z-axis.

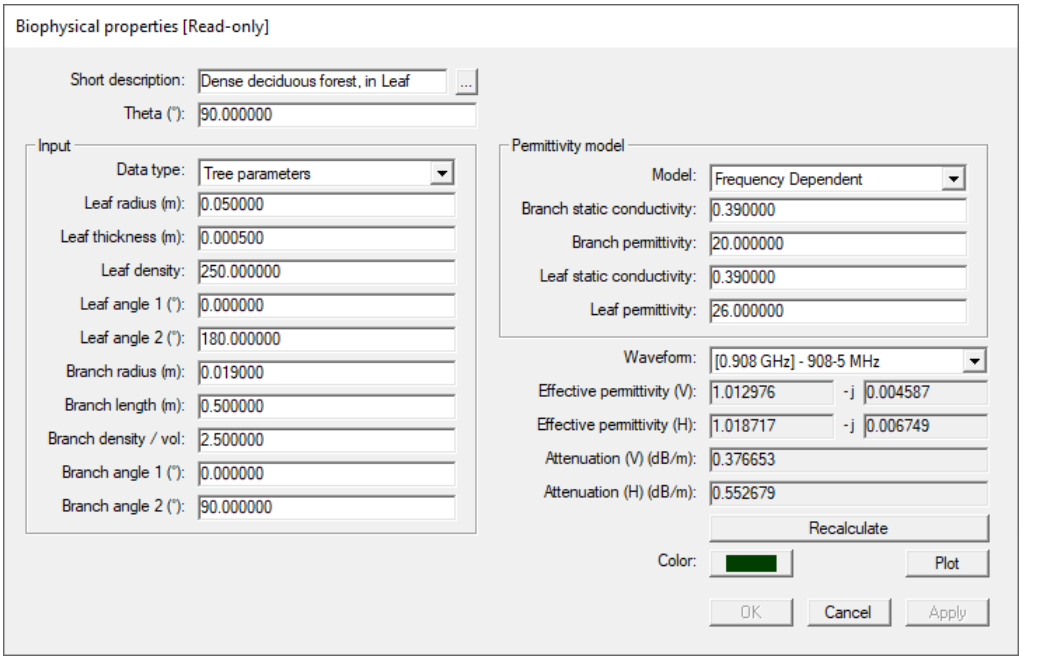


Figure 10.8: The Biophysical Properties Window

The *Theta* field gives the average incidence angle the waveform makes with the *Foliage*. As the incident angle varies, the effective areas of the leaves and branches change, resulting in different attenuation values. Acceptable values for *Theta* range from 0° to 90°.

- ✓ *Theta* is measured with respect to Z-axis. Therefore, setting "Theta = 90°" represents a wave normally incident with the side of the foliage.
- ✓ For TRANSMITTERS that are below the height of the foliage or that are sufficiently far from the foliage, *Theta* should be set to 90°.

The default inputs into the *Biophysical* model are shown for the *Tree parameters* data type, such as leaf size, branch size, number density, and orientation. *Leaf angle 1* and *Leaf angle 2* are used as the limits of a uniform random distribution of leaf inclination angle. Similarly, *Branch angle 1* and *Branch angle 2* give the limits of a uniform random distribution of branch angle in the foliage.

The *Grass parameters* data type receives similar input relative to the blades of grass.

By selecting the *Attenuation* data type, the scattering model can be used in conjunction with measurements to calculate the effective permittivity of foliage at different frequencies. By entering a measured *Attenuation* coefficient, measurement *Frequency*, and incidence angle *Theta*, users can calculate the real and imaginary parts of the propagation constant from the scattering model. The model will then extrapolate the attenuation from the measurement frequency to the frequency of the waveform given in the *Waveform* drop-down menu.

10.7 Engineered Electromagnetic Surface Materials

Wireless InSite's X3D model supports Engineered Electromagnetic Surface (EES) materials to model locally periodic metasurfaces. This capability is based on the Communications Research Centre Canada's (CRC) Ray-optical Engineered Electromagnetic Surface scattering model (EES Scattering Model). Unlike conventional materials, an EES material's properties vary spatially along the surface of the material. For this reason, they are defined by assigning an `*.ees` material specification file to a single face.

- See Section 16.2 for details on the EES Scattering model.
- The number of times the simulation will interact with an EES material is specified as a part of the STUDY AREA properties as described in section 15.3.2.
- The `*.ees` file format is described in Appendix G.

Engineered electromagnetic surface (EES) materials can be generated in Wireless InSite using one of the following methods.

- *Create a new plate* - Available in the main menu under *File* →*New* →*Geometry* →*EES plate*. This will create a new free standing plate object and its associated EES material. The object is centered in the project, and can then be translated and rotated into a final position.
- *Face operations* - The following operations are available in the context menu of a selected face under the *Material* sub-menu.
  - ✓ These operations are useful when defining an EES in relation to existing geometry.
  - *Attach EES Plate* - Create an EES object with desired dimensions, material properties, and offset from selected face. The new object is centered on the face with a specified offset. It is put into its own GEOMETRY so that it can be deactivated independently, making it easier to simulate and compare results that include the EES's effects with those that do not. The associated EES material is generated and saved to the project.
  - *Generate EES material* - This operation generates an EES material with desired properties and assigns it to the selected face. No new geometry is created.
  - *Change material* - Invoke the normal work flow for creating an EES material as described in section 3.7. This action should be used if the EES material file already exists and needs to be assigned to the face.

When any of these actions are invoked the EES material generator window will appear as shown below.

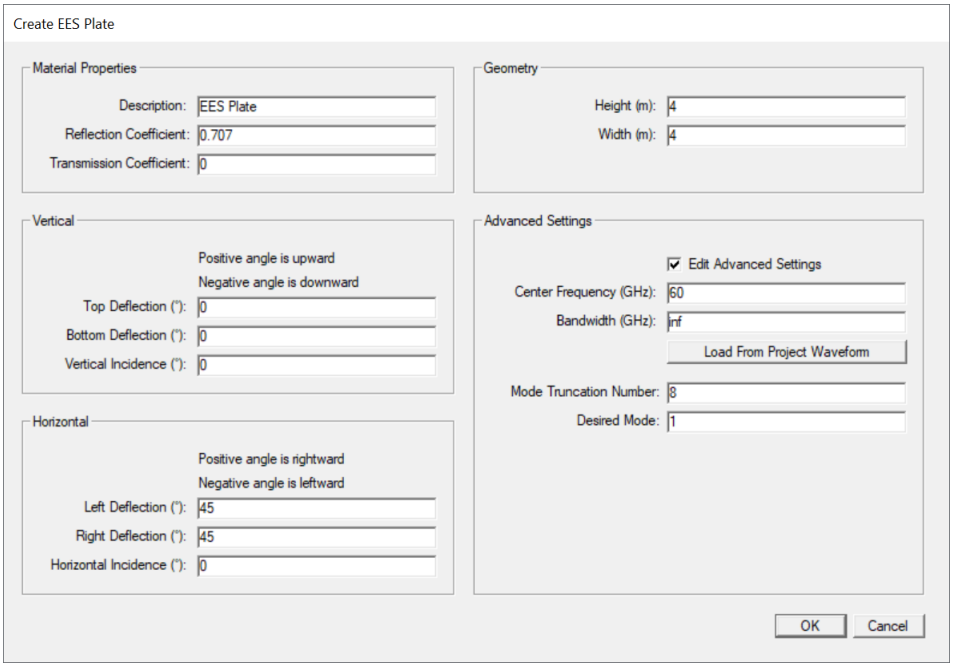


Figure 10.9: Generating an EES material

The inputs to this operation and the effect they have are:

- *Material Properties* - Specifies the *reflection* and *transmission* coefficients which affect how the mode data in the file is generated. The *description* is assigned to the material, the free-standing plate when it is created, and the `*.ees` filename.
- *Vertical* - Specify the *deflection* angles from the face normal for the *top* and *bottom* edges. These angles represent the outward going angle of a ray that hits each edge at the *vertical incidence* angle to the EES plate.
- *Horizontal* - Specify the *deflection* angles from the face normal for the *left* and *right* edges. These angles represent the outward going angle of a ray that hits each edge at the *horizontal incidence* angle to the EES plate.
- *Geometry* - Specify the *height* and *width* of the face that the generated material will cover. This is the size of the plate that is creating when attaching or creating a free-standing plate for the material. When generating a material for the selected face these fields are derived from it and only shown for informational purposes. The *offset* is used to position the plate when attaching one to the selected face.
- *Advanced Settings* - When activated these settings provide more detailed control of how the mode information is created for the EES file.
  - *Edit Advanced Settings* - This activates the ability to modify these parameters from their default settings.
  - *Center Frequency* - When present, this, along with the bandwidth define a frequency range of

operation for the EES material. If the operating frequency in the Wireless InSite simulation falls outside this band, the simulation will not run.

- *Bandwidth*
- *Load From Project Waveform* - Clicking this button opens a window to select a waveform from the current project to use to set the *Center Frequency* and *Bandwidth*.
- *Mode truncation number* - See Appendix G for more details.
- *Desired mode* - See Appendix G for more details.

The incident and deflection angles are defined as illustrated in Figure 10.10. Vector $\hat{k}$ represents the direction of an incident or deflected ray, and $\hat{n}$ represents the face's normal vector. Angle α is the horizontal component of this vector, defined to be positive when incident or deflected from the right side of the normal vector, and negative when incident from the left side. Similarly β is the vertical component of the vector, and is defined to be positive when incident from above the normal vector, and negative when incident or deflected below the normal vector. In the figure, $\hat{k}$ has a positive horizontal angle and a negative vertical angle.

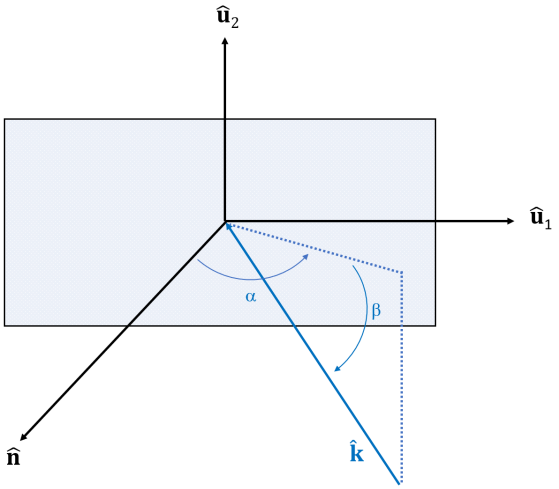


Figure 10.10: Definitions for EES incident angles

When the deflection angles off of opposite edges of the EES plate are identical, the EES material redirects most of the incident rays in the direction of that angle. This type of EES material is referred to as a grating.

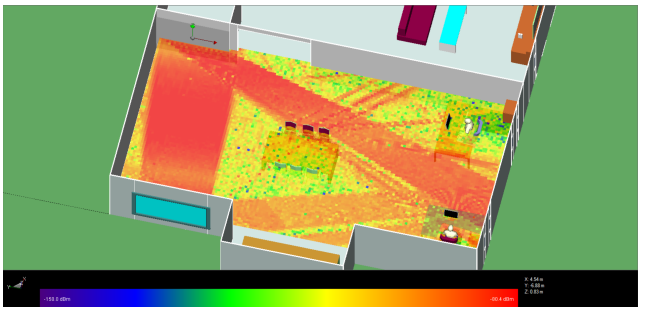


Figure 10.11: Received power reflecting off a grating EES plate

When the deflection angles off of opposite edges of the EES plate are unequal, the EES material redirects energy across that range of angles; this type of EES material is referred to as a diffuser. Because of the ray spreading associated with interactions with a diffuser EES, run times can increase.

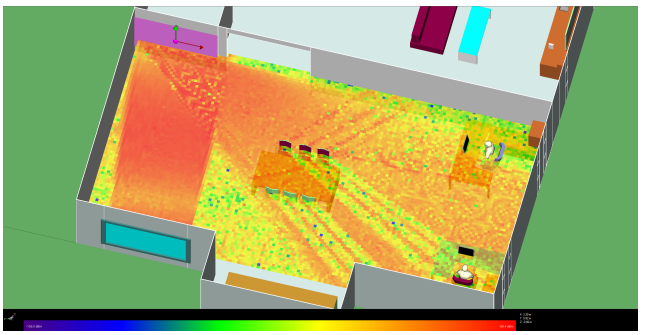


Figure 10.12: Received power reflecting off a diffuser EES plate

- See Section 15.3.2 for how to limit these interactions.

10.8 Plotting Reflection and Transmission Coefficients

There are two methods for accessing the PLOT COEFFICIENTS Window, as seen in Figure 10.13. The first is to right-click on a material in the MATERIALS list, and the other is by clicking on the PLOT button on a MATERIAL PROPERTIES Window.

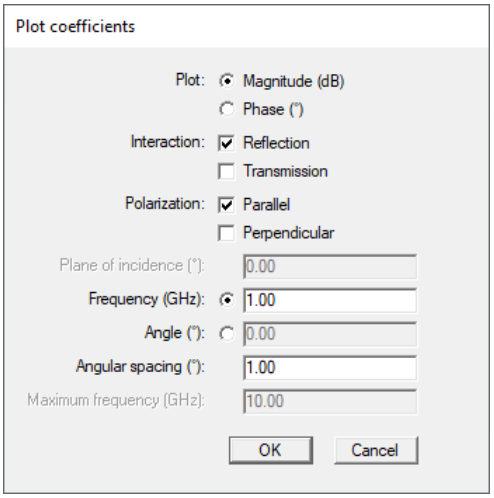


Figure 10.13: Plotting the coefficients of a material

The following options are available when plotting coefficients:

- *Plot* specifies whether the magnitude or the phase is plotted
- *Interaction* specifies whether reflection or transmission coefficient is plotted
- *Polarization* specifies the component
- *Frequency* shows plot vs. angle of incidence at the specified frequency
- *Angle* shows plot vs frequency at the specified angle of incidence
- *Angular/Frequency Spacing* increments in angle or frequency depending on plot type
- *Maximum frequency* is only active for plots as a function of frequency

Once the graph is displayed, it can be modified using the Graph and Plot Properties Windows.

▶ For more on adjusting graph properties, see Chapter 22.

10.9 Plotting Foliage Permittivity and Attenuation

The frequency- and polarization-dependent permittivity and attenuation of *Foliage* described by the biophysical model can be plotted from the MATERIAL PROPERTIES Window. To access the properties window of a biophysical material:

1. Click the MATERIALS tab.
2. Double-click the entry for the *Biophysical* material to open the properties window.
3. Click the PLOT button.
4. In the PLOT BIOPHYSICAL MATERIAL PARAMETERS window, as seen in Figure 10.14, specify the material *Value to Plot* as *Complex permittivity* or *Attenuation*.

5. Choose either the *Real* or the *Imaginary* component for the complex permittivity
6. Select the *Polarization* of the property.

Both polarizations can be plotted on the same graph to view the effects of waveform polarization on signal propagation. By default, the resulting plot is generated from 200 MHz to 2 GHz.

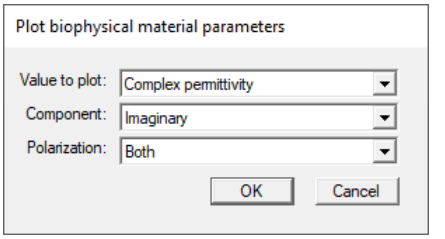


Figure 10.14: Plotting the biophysical material parameters

10.10 Ray-Fixed Coordinate System

Most materials can have different values for the field components parallel and perpendicular to the plane of incidence. It is especially important to understand this coordinate system if *User-Defined* coefficient files are being created. Wireless InSite uses one of two commonly used ray-fixed systems, which differ in the sign of the field component parallel to the plane of incidence.

- The ray-fixed system used by Wireless InSite is defined in Section 16.9, under Electric Field Evaluation.
- This system corresponds to the one used in most UTD literature, but it differs from the system used in many electrical engineering textbooks on electromagnetics, such as [7, Ch. 5].

Using the incorrect coordinate system can lead to a 180°phase error in the parallel component.

10.11 Fresnel Coefficients

The first four material types are based on the Fresnel plane wave transmission and reflection coefficients for a semi-infinite lossy dielectric half space of relative permittivity ϵ_r. Assuming that the incident field is propagating in free space, the Fresnel coefficients are given by the following formulas:

$$R_{\perp} = \frac{cos(\theta_i) - \sqrt{\epsilon_r}cos(\theta_t)}{cos(\theta_i) + \sqrt{\epsilon_r}cos(\theta_t)} \tag{10.46}$$

$$R_{\parallel} = \frac{\sqrt{\epsilon_r}cos(\theta_i) - cos(\theta_t)}{\sqrt{\epsilon_r}cos(\theta_i) + cos(\theta_t)} \tag{10.47}$$

$$T_{\perp} = \frac{2cos(\theta_i)}{cos(\theta_i) + \sqrt{\epsilon_r}cos(\theta_t)} \tag{10.48}$$

$$T_{\|} = \frac{2cos(\theta_i)}{\sqrt{\epsilon_r}cos(\theta_i) + cos(\theta_t)} \tag{10.49}$$

where θ_i is the angle of incidence measured from the face normal, and θ_t is the angle given by Snell's law of refraction:

$$sin(\theta_i) = \sqrt{\epsilon_r}sin(\theta_t) \tag{10.50}$$

10.12 Roughness Correction

To account for the decrease in the reflected energy in the specular direction, the reflection coefficient for a rough surface is determined using [16]:

$$R = R_0 exp\left[-8\left(\frac{\pi(\triangle h)cos\theta_i}{\lambda_0}\right)^2\right] I_0\left[8\left(\frac{\pi(\triangle h)cos\theta_i}{\lambda_0}\right)^2\right] \tag{10.51}$$

where

R_0 is the smooth surface reflection coefficient

θ_i is the angle of incidence

$\triangle h$ is the standard deviation in the surface height about the mean height, and corresponds to the roughness parameter that is entered in the GUI for roughness

λ_0 is the wavelength

I_0 is the modified Bessel function of zero order

Typically, $\triangle h$ should be between 0 and one wavelength. The maximum additional loss in the reflection coefficient is limited to $\triangle h$ equal to one wavelength in the calculation. The same roughness correction is used for both sides of a double-sided face. The transmission coefficients are unaffected by the surface roughness.

10.13 Data Sources for Materials

Building Materials

Reliable measured values for the permittivity and conductivity of common materials over the VHF and UHF bands are just now becoming available. However, some values that are frequently quoted in the literature may have obtained more validity than they deserve simply by repetition. It is also important to be aware that there is, at present, no good model for the frequency dependence of any building material (concrete, brick, glass, wood, etc.), so it is impossible to confidently extrapolate from values measured at one frequency to a significantly higher or lower frequency.

- Some sources for material permittivity and conductivity include [7], [17], [18], [19], [20], [21], [22], [11, Ch. 3], [23], and [24], [25].

While it would be preferable if accurate angle-dependent coefficients could be assigned to every surface, this is unfortunately rarely possible. However, fairly accurate predictions can often still be obtained in urban environments without precise information of the material properties of each building. It is generally agreed that in American and European cities a relative permittivity between 4 and 8 yields good results in the UHF band. The permittivity is sometimes considered to be the sole adjustable parameter through which a ray-based model can be "tuned" to give the best agreement with measurements. If the *Dielectric Half-Space* Material type is used, the conductivity will not usually be important, but if the dielectric layer coefficients are used with a realistic wall thickness, it will be important to enter a reasonably accurate value for the conductivity. For simulations of propagation in an indoor environment where transmissions through walls are almost always important, an accurate conductivity should always be specified.

- Formulas for the reflection and transmission coefficients for a finite thickness dielectric layer containing a regular mesh of reinforcing bar are presented in [17]. These coefficients can be imported into Wireless InSite by using the *User-Defined* Material.

Terrain

More data is available for ground materials than building materials, at least in the United States, although much of this data should also be used with caution. The most widely quoted values for areas in the United States appear to be the permittivity and conductivity values compiled for modeling AM radio propagation [23], [26]. It is not known whether this data is appropriate for modeling propagation in the VHF and UHF bands. However, since the results at VHF and UHF will often not strongly depend on the dielectric parameters when the incidence angles are near grazing, any reasonable values may be satisfactory when the distance separating the antennas is much greater than the heights of the antennas.

Foliage

Accurate data for different types of trees is difficult to acquire. Sources of effective dielectric constitutive parameters for foliage include [11, Ch. 7], [27], [28], [29], [30], [31], [8]. In many cases, the permittivity has been chosen to match a particular model with the measured path loss, rather than being derived from the bio-physical parameters. However, methods for determining the effective constitutive parameters from the bio-physical parameters are described in [31] and [9].

Chapter 11

Waveforms

In this chapter, you will learn...

- → how to define a waveform
- → how to define a frequency sweep
- → each waveform type and its properties

The WAVEFORMS in Wireless InSite allow the user to enter parameters which describe the time and frequency dependence of the signal radiated from the transmitting ANTENNA. The X3D only uses the WAVEFORM's frequency in its simulation, and is therefore limited to using the *Sinusoidal* waveform. For FULL 3D the waveform also acts as a kind of bandpass filter at the RECEIVER, and other pulse shapes are suported.

Once a waveform is defined, it may be assigned to any antenna, transmitter set or receiver set. Waveforms may also be saved to the waveform database for use in other projects.

11.1 Creating a Waveform

A Sinusoidal WAVEFORM can be created by selecting *File→New→Sinusoidal Waveform*. The SINUSOID PROPERTIES window will appear with plots of the waveform versus time and versus frequency, as seen in Figure 11.1. The APPLY button causes the plots to redraw in the properties window and display any changes made to the waveform.

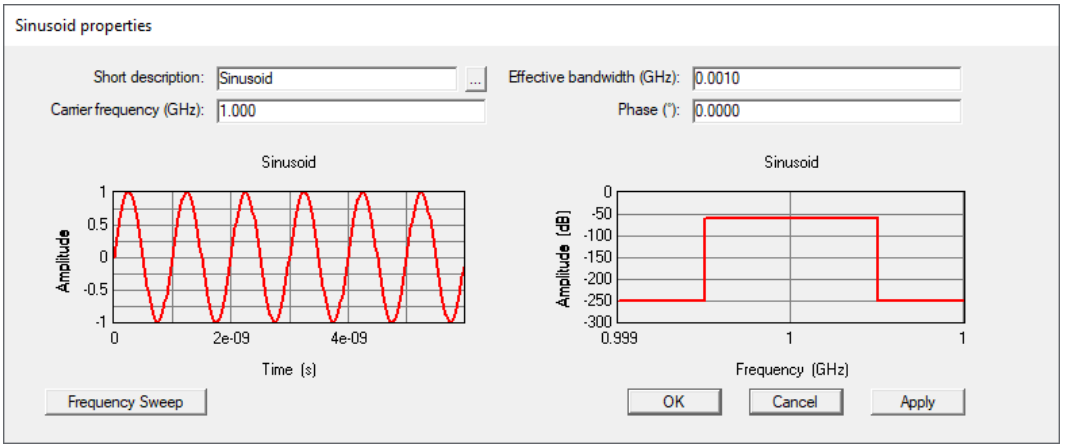


Figure 11.1: The Waveform Properties Window for a Sinusoid Waveform

To interact with the waveforms in a project, click on the WAVEFORMS tab in the MAIN WINDOW, as seen in Figure 11.2.

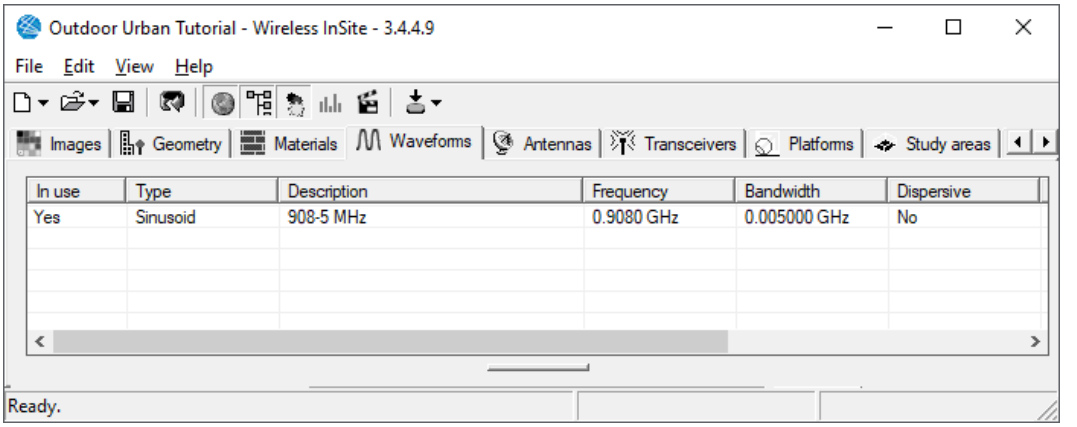


Figure 11.2: The Waveform Table in the Main Window

Right-click on a waveform in this tab to access a context menu with the following options:

- *Copy to personal database* - Places a copy of the selected waveform into the waveform database for use in other Wireless InSite projects.
  - ▶ See Chapter 24 for more information on the waveform database.
- *Duplicate* - Creates a copy of the selected waveform in the current project.
- *Delete* - Removes the selected waveform from the project.
- *Replace* - Assigns a different waveform to all transmitter sets, receiver sets, and antennas in the project which reference the selected waveform.
- *Export time domain envelope* - Writes time-domain samples of the waveform to a file.

- *Plot*
  - *Time domain* - Plots the time domain envelope of the selected waveform.
  - *Frequency* - Plots the selected waveform in the frequency domain.
- *Properties* - Accesses the properties of the selected waveform.

11.2 Frequency Sweeping

The X3D has the ability to simulate multiple frequencies with a single waveform definition. When active, X3D performs its ray trace only once, then calculates the outputs for each frequency, thus saving simulation time. To do define a frequency sweep click on the FREQUENCY SWEEP button in the WAVEFORM PROPERTIES window. This will bring up the SWEEP PARAMETERS FOR CARRIER FREQUENCY window that can be used to configure the sweep with the following options:

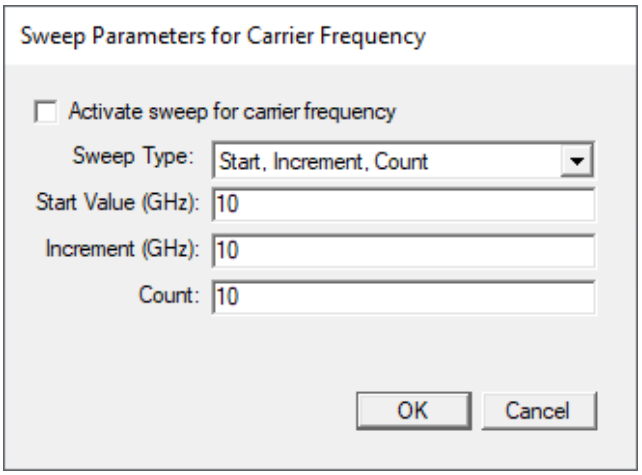


Figure 11.3: Specifying a frequency sweep

- *Activate sweep for carrier frequency* - Activates the sweep for the waveform.
- *Sweep Type* - Indicates how the frequencies of the sweep should be calculated. The options are:
  - *Start, Increment, Count* - This will calculate the frequencies based on a *start* frequency, the value to *increment* it, and a *count* of the number of frequencies to simulate for the WAVEFORM.
  - *Start, End, Count* - This will calculate the frequencies based on a *start* frequency, the final value to *end* the sweep on, and a *count* of the number of frequencies to simulate for the WAVEFORM.

11.3 Waveforms Compatible with Legacy Models

See Section 2.7.3 for how to enable legacy model capabilities. When *Expose Calcprop options* has been enabled in the APPLICATION PREFERENCES, a variety of waveforms supported by FULL 3D, VERTICAL PLANE, and URBAN CANYON can be created by selecting *File→New→Waveform* and selecting one of the following options from a drop-down menu:

- *Blackman* envelope
- *Chirp*
- *Gaussian*
- *Gaussian Derivative*
- *Hamming* envelope
- *Hanning* envelope
- *Raised Cosine*
- *Root Raised Cosine*
- *Sinusoid*
- *Tukey* envelope
- *User-Defined*

▶ A description of the *User-Defined* file format is given in the Section 11.3.10.

The *Carrier frequency* may be specified for all waveforms except *Gaussian Derivative* and *Chirp* pulse, which are not modulated, and the *User-Defined* Waveform. The *Pulse width* specifies the time duration of all waveforms except the *Sinusoid* and *User-Defined* Waveforms.

✓ The bandwidth of a waveform, which is determined by the pulse width, must not exceed twice the carrier frequency.

The *Phase* field specifies the phase shift of all waveforms except *Gaussian Derivative*, *Chirp*, and *User-Defined*. The *Dispersive* checkbox controls how certain result types are generated when simulations are performed. Additionally, some WAVEFORM PROPERTIES Windows contain entries for parameters specific to that waveform type.

▶ See Chapter 21 for more information regarding generating results.

11.3.1 Blackman

The BLACKMAN WAVEFORM PROPERTIES Window is shown in Figure 11.4.

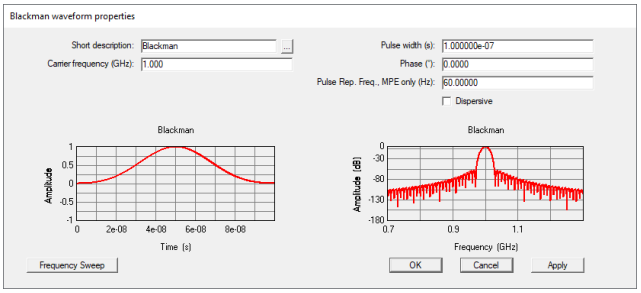


Figure 11.4: The Blackman Waveform Properties Window

The *Blackman* envelope is defined by:

$$0.42 - \frac{1}{2}cos\left(\frac{2\pi t}{\tau}\right) + 0.08cos\left(\frac{4\pi t}{\tau}\right) \tag{11.1}$$

where τ is the pulse width and $0 \leq t \leq \tau$. [32]

11.3.2 Chirp

The CHIRP WAVEFORM PROPERTIES Window is shown in Figure 11.5.

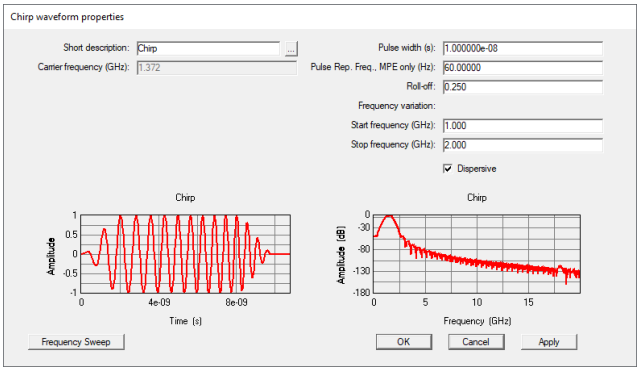


Figure 11.5: The Chirp Waveform Properties Window

The linear *Chirp* pulse is defined by:

$$P_{RC}(t)sin\left(2\pi f_0 + \frac{2\pi(f_1 - f_0)}{2T}t\right)t \tag{11.2}$$

for $0 \leq t \leq T$.

The exponential *Chirp* pulse is defined by:

$$P_{RC}(t)sin\left(2\pi\left(\frac{T}{ln(f_1-f_0)}\left(f_0\left(\frac{f_1}{f_0}\right)^{\frac{t}{T}}-f_0\right)\right)\right) \tag{11.3}$$

for $0 \leq t \leq T$, where T is the pulse width, f_0 is the start frequency, f_1 is the stop frequency, and $P_{RC}(t)$ is defined by:

$$P_{RC}(t) = \begin{cases} \frac{1}{2}\left[1+cos\left(\frac{(1+\beta)\pi}{\beta T}\left(t-\frac{\beta T}{1+\beta}\right)\right)\right] & \text{for } 0 \leq t \leq \frac{\beta T}{1+\beta} \\ 1 & \text{for } \frac{\beta T}{1+\beta} \leq t \leq \frac{T}{1+\beta} \\ \frac{1}{2}\left[1+cos\left(\frac{(1+\beta)\pi}{\beta T}\left(t-\frac{T}{1+\beta}\right)\right)\right] & \text{for } \frac{T}{1+\beta} \leq t \leq T \\ 0 & \text{otherwise} \end{cases} \tag{11.4}$$

where $0 \leq \beta \leq 1$ is the roll-off factor.

Up-chirp waveform is obtained when $f_0 < f_1$, while down-chirp waveform is obtained when $f_0 > f_1$. The Chirp pulse has no specific carrier frequency. Consequently, the Chirp pulse always generates results using the dispersive option. The Chirp pulse properties window displays the frequency with maximum amplitude in the *Carrier frequency* field.

11.3.3 Gaussian

The GAUSSIAN WAVEFORM PROPERTIES Window is shown in Figure 11.6.

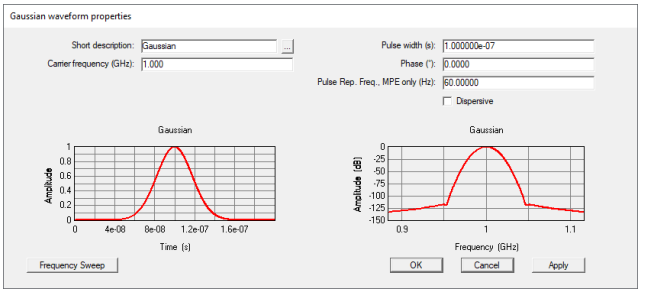


Figure 11.6: The Gaussian Waveform Properties Window

The *Gaussian* pulse is defined by:

$$exp\lfloor -\alpha(t-\tau)^2 \rfloor \tag{11.5}$$

for $0 \leq t \leq 2\tau$, where τ is the pulse width and $\alpha = \frac{16}{\tau^2}$. [15]

The pulse will exist from $t = 0$ until $t = 2\tau$; it is approximated as zero outside this range, with a peak value at $t = \tau$. The value at truncation ($t = 0$ and $t = 2\tau$) is determined by α, and the Gaussian pulse at truncation will have a value $exp(-\alpha\tau^2)$ down from the maximum value. With $\alpha = \frac{16}{\tau^2}$, the value at truncation will be e^{-16}, approximately 140 dB down from peak.

11.3.4 Gaussian Derivative

The GAUSSIAN DERIVATIVE WAVEFORM PROPERTIES Window is shown in Figure 11.7.

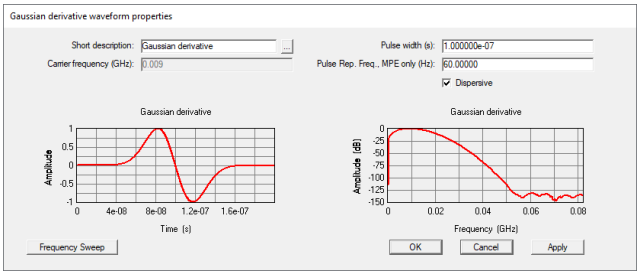


Figure 11.7: The Gaussian Derivative Waveform Properties Window

The *Gaussian Derivative* pulse is defined by:

$$-2\sqrt{\frac{e}{2\alpha}}\alpha(t-\tau)exp[-\alpha(t-\tau)^2] \tag{11.6}$$

for $0 \leq t \leq 2\tau$, where τ is the pulse width and $\alpha = \frac{16}{\tau^2}$.

The pulse will exist from $t = 0$ until $t = 2\tau$; it is approximated as zero outside this range. As a broadband pulse, the *Gaussian Derivative* Waveform has no specific carrier frequency. Consequently, the Gaussian Derivative always generates results using the dispersive option. The GAUSSIAN DERIVATIVE PROPERTIES Window displays the frequency with maximum amplitude in the *Carrier frequency* field.

11.3.5 Hamming

The HAMMING WAVEFORM PROPERTIES Window is shown in Figure 11.8.

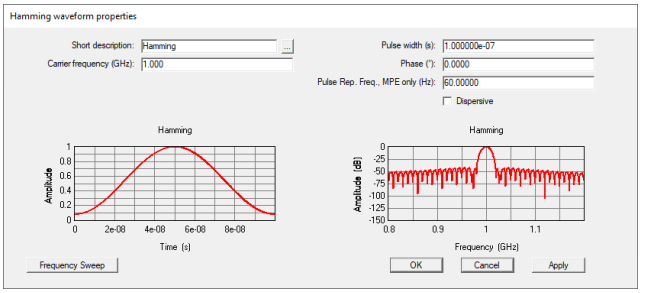


Figure 11.8: The Hamming Waveform Properties Window

The *Hamming* envelope is defined by:

$$0.54 \ - \ 0.46cos\left(\frac{2\pi t}{\tau}\right) \tag{11.7}$$

where τ is the pulse width and $0 \ \leq \ t \ \leq \ \tau$. [32]

11.3.6 Hanning

The HANNING WAVEFORM PROPERTIES Window is shown in Figure 11.9.

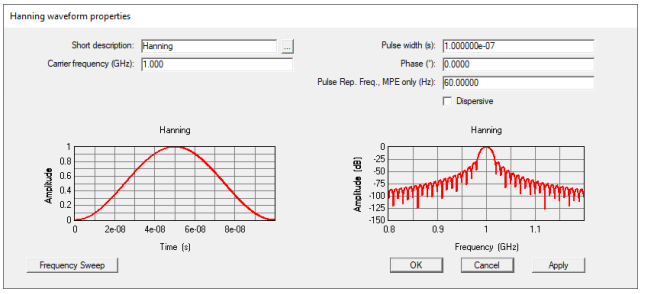


Figure 11.9: The Hanning Waveform Properties Window

The *Hanning* envelope is defined by:

$$0.5 \ - \ 0.5cos\left(\frac{2\pi t}{\tau}\right) \tag{11.8}$$

where τ is the pulse width and $0 \ \leq \ t \ \leq \ \tau$. [32]

11.3.7 Raised Cosine and Root Raised Cosine Pulses

The RAISED COSINE PROPERTIES Window is shown in Figure 11.10. The *Root Raised Cosine* properties window is nearly identical.

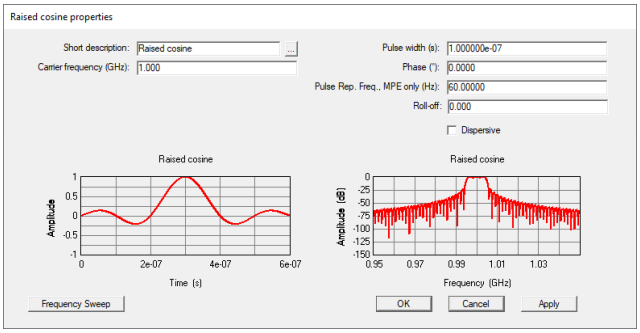


Figure 11.10: The Raised Cosine Properties Window

The *Raised Cosine* pulse [33] is defined by:

$$sinc\left(\frac{\pi t}{\tau}\right)\frac{cos\left(\frac{\pi\beta t}{\tau}\right)}{\frac{1-4\beta^2t^2}{\tau^2}} \tag{11.9}$$

and the *Root Raised Cosine* [34] by:

$$\frac{4\beta cos\left[\frac{(1+\beta)\pi t}{\tau}\right]+sin\left[\frac{(1-\beta)\pi t}{\tau}\right]\left(\frac{4\beta t}{\tau}\right)^{-1}}{\pi\sqrt{\tau}\left[\frac{1-16\beta^2t^2}{\tau^2}\right]} \tag{11.10}$$

where τ is the pulse width and $0 \leq \beta \leq 1$ is the excess bandwidth roll-off factor. In order to ensure a causal system, the Raised Cosine pulse is typically delayed by 3τ and truncated at 6τ.

11.3.8 Sinusoid

Although the *Sinusoid* Waveform continues infinitely in the time-domain, it is modeled by a flat power spectral density over its effective bandwidth $B << f_c$. Thus, the transmitted power of the *Sinusoid* is uniform over B Hz, centered at f_c. Receiver sets use waveform information as a front-end bandpass filter, with uniform response over the band B Hz wide, centered at f_c. The purpose of specifying the effective bandwidth B is to allow calculation of the carrier-to-interferer ratio when transmitted waveforms overlap. An initial phase offset, θ_0 in degrees, is also permitted.

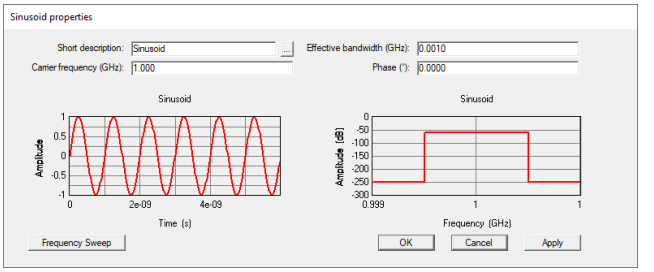


Figure 11.11: Sinusoid Waveform Properties Window

Mathematically, the *Sinusoid* is defined by

$$s(t) = sin(2\pi f_c t + \theta_0) \tag{11.11}$$

And

$$S(f) = \begin{cases} \dfrac{1}{B} & \text{for } f_c - \dfrac{B}{2} < f < f_c + \dfrac{B}{2} \\ 0 & \text{otherwise} \end{cases} \tag{11.12}$$

Lastly, because the *Sinusoid* is not time-limited; it cannot generate results using the dispersive option.

11.3.9 Tukey

The TUKEY WAVEFORM PROPERTIES Window is shown in Figure 11.12.

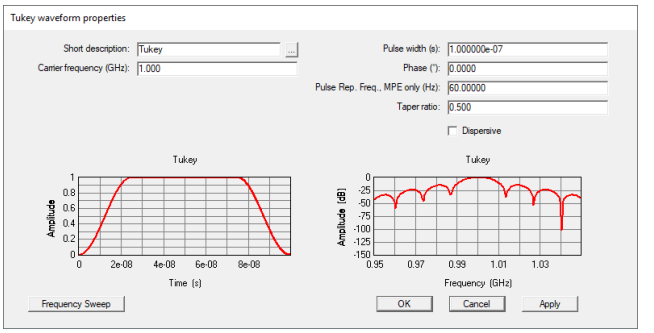


Figure 11.12: The Tukey Waveform Properties Window

The *Tukey* envelope is defined by:

$$\begin{cases} 1 & \text{for } \left|t-\frac{\tau}{2}\right| \leq \alpha\frac{\tau}{2} \\ \frac{1}{2}\left[1+cos\left(\frac{\pi\left(t-\frac{(1+\alpha)\tau}{2}\right)}{\frac{(1-\alpha)\tau}{2}}\right)\right] & \text{for } \alpha\frac{\tau}{2} \leq \left|t-\frac{\tau}{2}\right| \leq \frac{\tau}{2} \\ 0 & \text{otherwise} \end{cases} \quad (11.13)$$

where τ is the pulse width and $0 \leq \alpha \leq 1$ is the constant-to-taper ratio. When $\alpha = 0$, the envelope of the Tukey is that of a cosine, while $\alpha = 1$ is a flat square pulse. All other values of α provide a "tapered" sinusoidal shape which ramps up, remains flat, and ramps down to 0 again. [35]

11.3.10 User-Defined Waveform

The USER-DEFINED WAVEFORM PROPERTIES Window displays the specified filename and the frequency at which the maximum amplitude occurs. No additional modulation is performed on the waveform, and thus the carrier frequency cannot be specified through the user interface. The display of the waveform in both the time- and frequency-domains is provided as a reference by taking the FFT or IFFT, as necessary. All results generated using a *User-Defined* Waveform is performed dispersively.

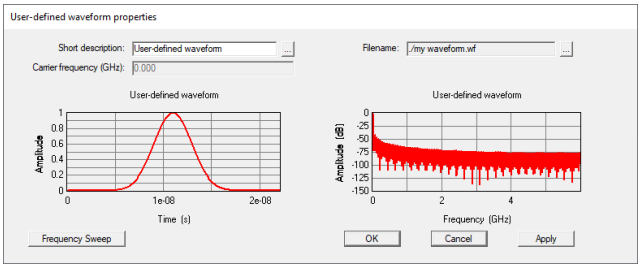


Figure 11.13: User-Defined Waveform Properties Window

When creating a *User-Defined* waveform, keep in mind the following guidelines:

- ✓ Time domain waveforms should start at 0 amplitude at (t=0) and should end at zero amplitude.
- ✓ It may be necessary to ramp up or down the amplitude at the start and end times to avoid high frequency components in the FFT.
- ✓ The minimum sampling rate for time domain waveforms should be the period/10.

Chapter 12

Antennas

In this chapter, you will learn...

- → how to create and define an antenna
- → the types of antennas and patterns available in Wireless InSite
- → how to plot an antenna pattern

Propagation simulations in Wireless InSite require both TRANSMITTERS and RECEIVERS, each with an associated ANTENNA and WAVEFORM. When an antenna is added to a project and its parameters are set using the ANTENNA PROPERTIES Window, it can be used in multiple instances by associating it with any number of transmitters and/or receivers. Each time the antenna is assigned to a transmitter or receiver set, it becomes a new "copy" of the antenna. The location, orientation, and polarization of the antenna are set by the location of the associated transmitter or receiver and the rotation angles about X-, Y-, and Z-axes for each association of the antenna with a transmitter or receiver. The coordinate system used for these rotations is shown in Figure 12.1. These rotations are specified in the TRANSMITTER/RECEIVER PROPERTIES Window.

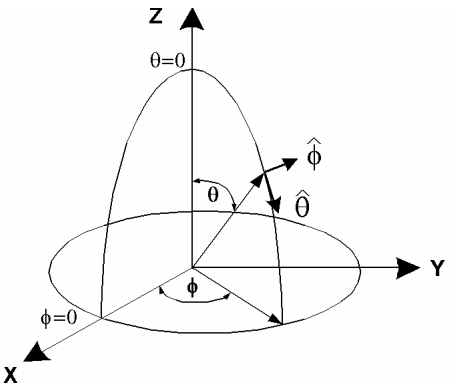


Figure 12.1: The Wireless InSite spherical coordinate system

Any number of different types of antennas can be added to the project to simulate real-world scenarios or to test the effects that different antennas have on propagation performance. Antennas may also be saved to the antenna database for use in other projects.

12.1 Creating an Antenna

To add a new ANTENNA to an existing project, select *File→New→Antenna* and choose from the list of antenna types. If you have added an antenna to the database, it will also be available from the database window by selecting *Add to Project* from its entries context menu.

- See Chapter 24 for more information on the antenna database.

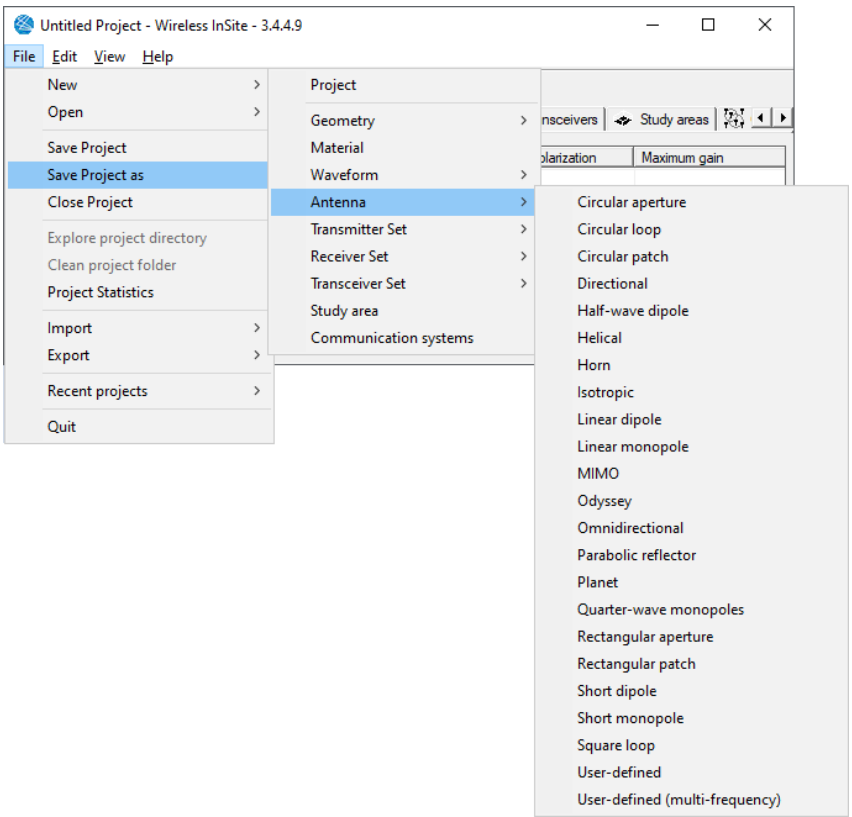


Figure 12.2: Adding a new antenna to the project

After selecting an antenna, the ANTENNA PROPERTIES Window will appear. The properties windows includes parameters specific to that antenna type. To access the antennas after they are created, click on the ANTENNAS tab in the MAIN WINDOW.

12.2 Antenna Properties

All of the ANTENNAS used in a Wireless InSite simulation include a set of common parameters used to define the antenna and some immediate system parameters. They are:

- Gain (maximum gain)
- Waveform
- Transmission line loss

The the ANTENNA PARAMETERS Window, seen in Figure 12.3, allows the user to enter the antenna characteristics. The WAVEFORM associated with the antenna can be selected from a pull-down menu of the waveforms already defined in the project or by clicking the "..." button and defining a new waveform. In the TRANSMITTER/RECEIVER PROPERTIES Window, the antenna waveform selection can be overridden or the antenna's waveform may be selected.

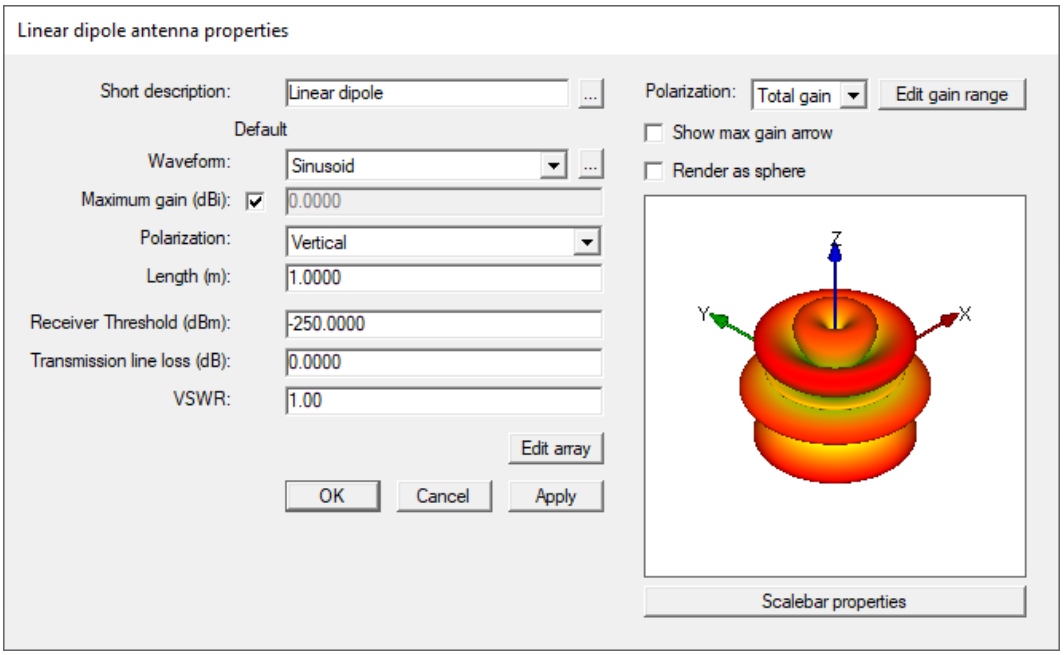


Figure 12.3: Properties for the Linear Dipole Antenna

The remaining shared antenna parameters are summarized below.

- *Maximum gain* - Sets the maximum gain of the ANTENNA to the value listed in the field. It is possible to enter nonphysical gain values for an antenna, so the user is responsible for realistic manual entries. User-defined antennas have additional options for setting the maximum gain that are detailed in section 12.5.1.1.
- *Receiver threshold* - Determines which individual ray paths to ignore when evaluating the power at a RECEIVER point. Ray paths are ignored when their power falls beneath this value.

- *Transmission line loss* - Simulates line losses, return losses, feed losses, or other system losses between the TRANSMITTER or RECEIVER and its ANTENNA. The number entered here is taken as a loss (in dB) and has the effect of reducing signal strength between antenna and Tx/Rx, whether it is entered as a negative or positive number.
- *VSWR* - Defines the Voltage Standing Wave Ratio (VSWR) that must be greater than or equal to 1, with 1 representing a perfect impedance match between the ANTENNA and transmission line resulting in no addition loss. VSWR values greater than 1 will result in an additional mismatch loss during the calculation of received power:

$$Mismatch\ loss(dB) = 10log_{10}\left[1 - \left(\frac{VSWR-1}{VSWR+1}\right)^2\right] \tag{12.1}$$

12.2.1 Maximum Gain

The *Maximum gain* in the ANTENNA PROPERTIES Window allows the automatically calculated gain of an antenna pattern to be overridden with a user specified value. The calculation used to scale the antenna pattern is depended on the propagation model. In general, FULL3D, URBAN CANYON and VERTICAL PLANE will set the maximum antenna component, θ or ϕ, to the *maximum gain*. X3D will scale the pattern so the antenna's maximum total gain is equal to the *Maximum gain*. The two methods are equivalent for purely θ or ϕ polarized antennas, but lead to different results for dual-polarized antennas. For instance, for the same *Maximum gain* setting, circularly-polarized isotropic antennas will have a 3 dBi higher gain in FULL3D simulations than in X3D simulations. Details of the calculations are presented below.

Maximum Gain for FULL3D, URBAN CANYON, & VERTICAL PLANE

With the distance dependence suppressed, the far zone electric field of the antenna can be expressed as:

$$E(\theta,\phi) = f_\theta(\theta,\phi)\hat{e}_\theta + f_\phi(\theta,\phi)\hat{e}_\phi \tag{12.2}$$

where $E(\theta,\phi)$ has been normalized such that the larger of either $|f_\theta(\theta,\phi)|$ or $|f_\phi(\theta,\phi)|$ has a maximum value of 1. The maximum gain (and directivity) is then defined to be:

$$G_{max} = \frac{4\pi}{\left[\int\ |f_\theta(\theta,\phi)|^2 + |f_\phi(\theta,\phi)|^2\ d\Omega\right]} \tag{12.3}$$

and the maximum gain in dBi is $G_{max}[dBi] = 10log_{10}(G_{max})$.

If the *Automatic* box next to this field is unchecked and a value is entered, $G_{max}[dBi]$ will be set to the user specified value.

Once the maximum gain is set, either by the above calculation or to the value entered by the user, the components of the gain pattern in dBi are calculated from:

$$G_\theta(\theta,\phi) = 10log_{10}\left[G_{max}\,|f_\theta(\theta,\phi)|^2\right] = 10log_{10}\left[\,|f_\theta(\theta,\phi)|^2\right] + G_{max}[dBi] \tag{12.4}$$

$$G_\phi(\theta,\phi) = 10log_{10}\left[G_{max}\left|f_\phi(\theta,\phi)\right|^2\right] = 10log_{10}\left[\left|f_\phi(\theta,\phi)\right|^2\right] + G_{max}[dBi] \quad (12.5)$$

Maximum Gain for X3D

For X3D, the pattern is normalized such that the maximum magnitude of the complex vector $E(\theta,\phi)$ in equation 12.2 in the direction of the maximum total gain is 1. The direction where $|E| = 1$ is the direction of the *maximum gain*. The *maximum gain* is also calculated from equation 12.3. Once the maximum gain is set, either by the above calculation or to the value entered by the user, the components of the gain pattern in dBi are calculated from equations 12.4 and 12.5.

12.2.2 Antenna Arrays

ANTENNA arrays can be created from any non-MIMO antenna. Wireless InSite uses the amplitude, phase, and relative location of each element to create a combined antenna pattern that users can assign to a single TRANSMITTER, RECEIVER, or TRANSCEIVER. Antenna arrays do not treat each antenna element as independent. Instead, the combined effect of the array is represented as a single antenna pattern. Antenna arrays can be specified by clicking on EDIT ARRAY in the ANTENNA PROPERTIES window. This will bring up the window shown in Figure 12.4. In this window, begin specifying points by selecting *New array point* from the table's context menu. After selecting an element, the user can edit it by double-clicking on the point. Users can also delete points by choosing *Delete* from the context menu.

Array point properties

Users can specify the following properties of an array point:

- *Amplitude and Phase* - The amplitude represents a scaling factor that is applied to the gain of the element's antenna pattern. The phase represents a relative phase offset between the elements in the array.
- X, Y *and* Z *position* - This is an offset from the origin of the antenna indicating where the point is located within the array.

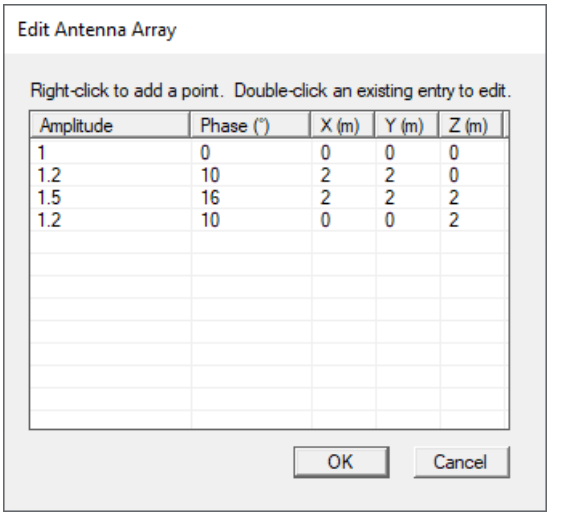


Figure 12.4: The Edit Antenna Array Window, used to create the antenna array in Figure 12.5

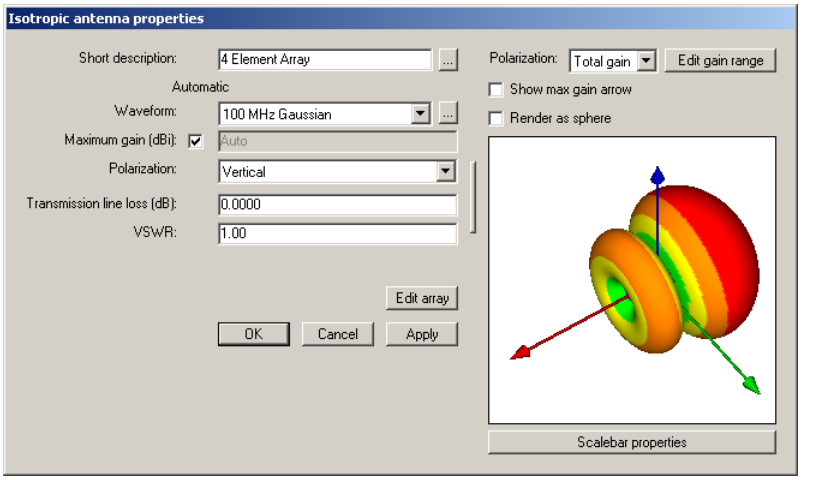


Figure 12.5: The radiation pattern for an array of four isotropic elements

The antenna pattern will not render using the array information unless *Use Calcprop engine to calculate antenna patterns* is activated in the MODELS tab of the APPLICATION PREFERENCES.

12.3 Antenna Types

Wireless InSite has several choices for the ANTENNAS used in a simulation. The antenna patterns can be generated from:

- **Built-in models** (See 12.4)

  - *Short Dipole*
  - *Short Monopole*
  - *Linear Dipole*
  - *Half-Wave Dipole*
  - *Linear Monopole*
  - *Quarter-Wave Monopole*
  - *Axial Mode Helix*
  - *Circular and Square Loops*
  - *Circular and Rectangular Apertures*
  - *Circular and Rectangular Patch*
  - *Horn* - pyramidal, E-plane sectoral, H-plane sectoral
  - *Parabolic Reflector*
  - *Directional*
  - *Omnidirectional*
  - *Isotropic*
- **User-Defined and Imported Patterns** (See 12.5)
  - *Planet*
  - *Odyssey*
  - *UAN User Defined*
- *MIMO* (See 12.6)
- *Multi-frequency User Defined* (See 12.7)

12.4 Built-in Antennas

There are several built-in choices for the ANTENNAS used in a simulation. The ANTENNA PROPERTIES window requests only a few parameters from the user for each type and the patterns are generated automatically for both propagation calculations and plotting.

The models used for these pattern calculations are approximate. In each case, a reference is included for further information on how the pattern was calculated.

Each of the built-in ANTENNAS in Wireless InSite is in one of two groups: **Freestanding** or **Ground-plane mounted**. The following sections describe the antennas in each classification.

12.4.1 Freestanding Antennas

The freestanding ANTENNAS include:

- *Dipoles*
- Pyramidal *Horn*
- *Parabolic Reflector*
- *Circular Loop*
- *Square Loop*
- *Axial Mode Helix*
- *Directional*
- *Omnidirectional*
- *Isotropic*

For *Dipoles*, if vertical polarization is selected, the wire is along the Z-axis with the omnidirectional E-plane pattern in the XY plane. If horizontal polarization is chosen, the wire is along the Y-axis and the omnidirectional E-plane pattern is in the ZX plane. The other types of freestanding antennas, *Horns* and *Parabolic Reflectors*, have a default orientation such that the main beam is X-directed.

- The polarizations for these antennas are described in the next several sections.

Circular Loop and *Square Loop* Antennas are considered freestanding, but do not allow polarization to be defined as an antenna parameter. By default, these antennas lie in the XY plane. The orientation used in the simulation is achieved by rotating the associated TRANSMITTER or RECEIVER.

The remaining freestanding antenna is an *Axial Mode Helix*. The axis of the helix is along the X-axis by default and produces a circularly-polarized main beam along the X-axis.

12.4.1.1 Short Dipole

This antenna is also called the *Infinitesimal Dipole* and is ideal in the mathematical sense. Its length is much less than one wavelength. It generates a field proportional to $\sin(\theta)$. If the antenna *Polarization* is set to "Vertical," its orientation is along the Z-axis and the main beam is omnidirectional in the XY plane. Selecting "Horizontal" polarization will rotate the antenna about the X-axis by 90° so that it will lie along the Y-axis and the main beam is omnidirectional in the ZX plane.

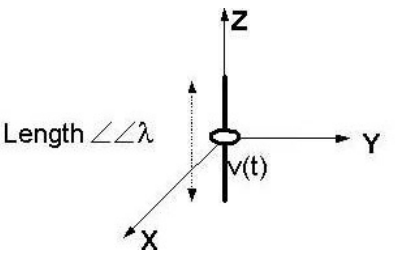


Figure 12.6: Vertically-polarized Short Dipole Antenna

12.4.1.2 Linear Dipole

For the *Linear Dipole*, the user sets the antenna *Length* and *Polarization* in the ANTENNA PROPERTIES Window, seen in Figure 12.7. The wire is considered thin with respect to a wavelength. The excitation is considered to be a truncated sinusoidal current distribution.

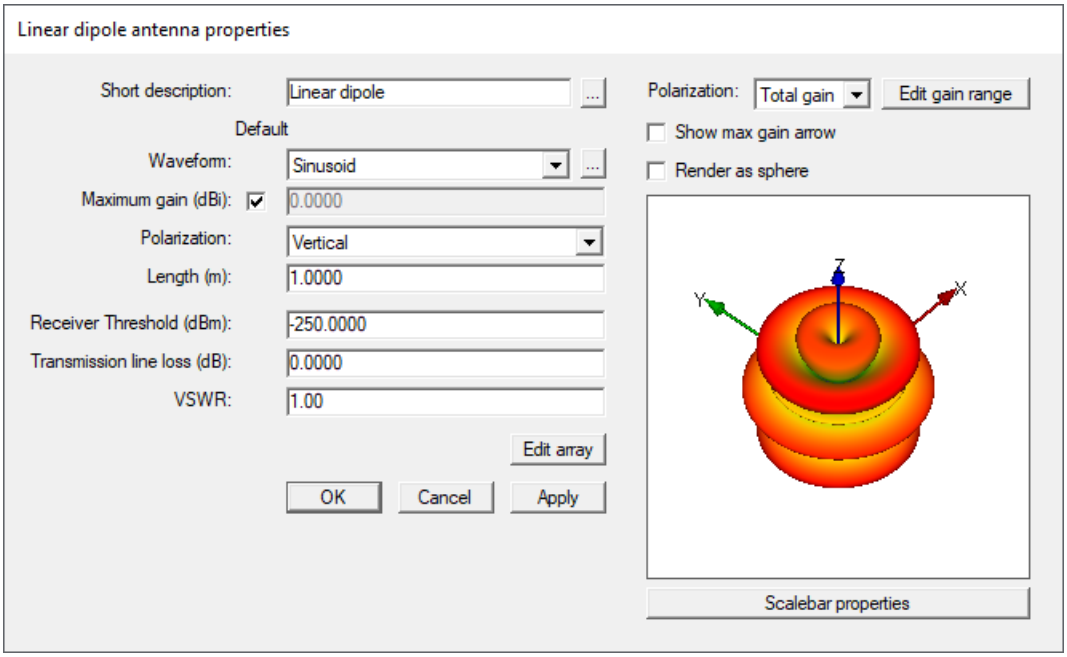


Figure 12.7: Properties window for a Linear Dipole Antenna

If the antenna *Polarization* is set to "Vertical", its orientation is along the Z-axis and the main beam is omnidirectional in the XY plane. Selecting "Horizontal" polarization will rotate the antenna about the X-axis by 90° so the antenna will lie along the Y-axis and the main beam is omnidirectional in the ZX plane.

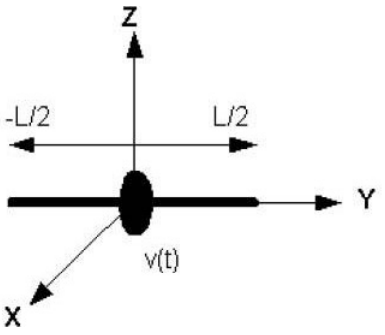


Figure 12.8: Horizontal Linear Dipole Antenna

12.4.1.3 Half-Wave Dipole

The length of the *Half-Wave Dipole*, seen in Figure 12.9, is automatically adjusted to be one-half wavelength at the waveform frequency. The wire is considered thin with respect to a wavelength. The polarization is set in the ANTENNA PROPERTIES Window. If the antenna *Polarization* is set to "Vertical," its orientation is along the Z-axis and the main beam is omnidirectional in the XY plane. Selecting "Horizontal" polarization will rotate the antenna about the X-axis by 90° so the antenna will lie along the Y-axis and the main beam is omni-directional in the XZ plane.

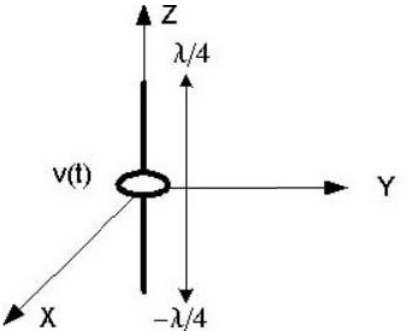


Figure 12.9: Vertical Half-Wave Dipole Antenna

12.4.1.4 Horn

The *Horn* is a freestanding antenna, as shown in Figure 12.10. It is assumed to be made of perfect electrical conductor (PEC). The input parameters, entered through the ANTENNA PROPERTIES Window, are the *Aperture dimensions*, *Feed dimensions*, and the distance from the feed aperture to the horn aperture (*Feed-aperture length*). With these parameters, users can define a pyramidal horn, E-plane sectoral horn, or H-plane sectoral horn based on flare angles of the horn and the polarization. [36, p. 682]

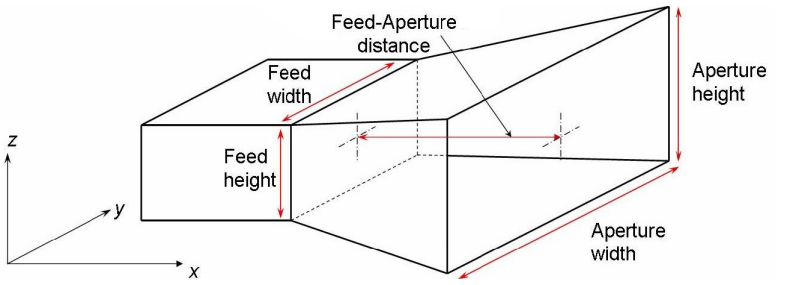


Figure 12.10: Description of a Horn Antenna

The default orientation of the *Horn* aligns the main beam with X direction. The antenna can be oriented in the project by entering the appropriate rotations in the TRANSMITTER/RECEIVER PROPERTIES Window. Pyramidal *Horns* are created when the horn has nonzero flare angles in both the E and H-planes. E-plane and H-plane sectoral horns are created when the *Horn* flares in only one plane. For E-plane sectoral horns, the flare angle of the horn is parallel to the E-Plane or polarization. H-plane sectoral horns

are created when the horn flare is perpendicular to the E-plane. If the antenna *Polarization* is set to "Vertical," the horn E-plane lies in the XZ plane for the default orientation. "Horizontal" polarization aligns the E-plane with the XY plane. Diagrams of an E-plane and H-plane sectoral horn, assuming the E-plane is in the XZ plane, are shown in Figure 12.11.

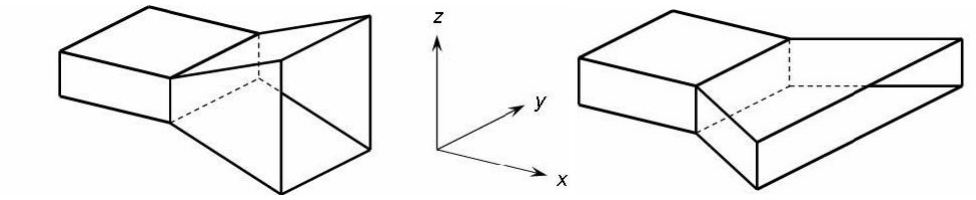


Figure 12.11: Sectoral Horns aligned to the E-plane (left) and the H-plane (right

Figure 12.12 shows a HORN ANTENNA PROPERTIES Window defining a Pyramidal *Horn*. The resulting 2D cut plane pattern produced using the *Plot pattern* function is shown in Figure 12.13.

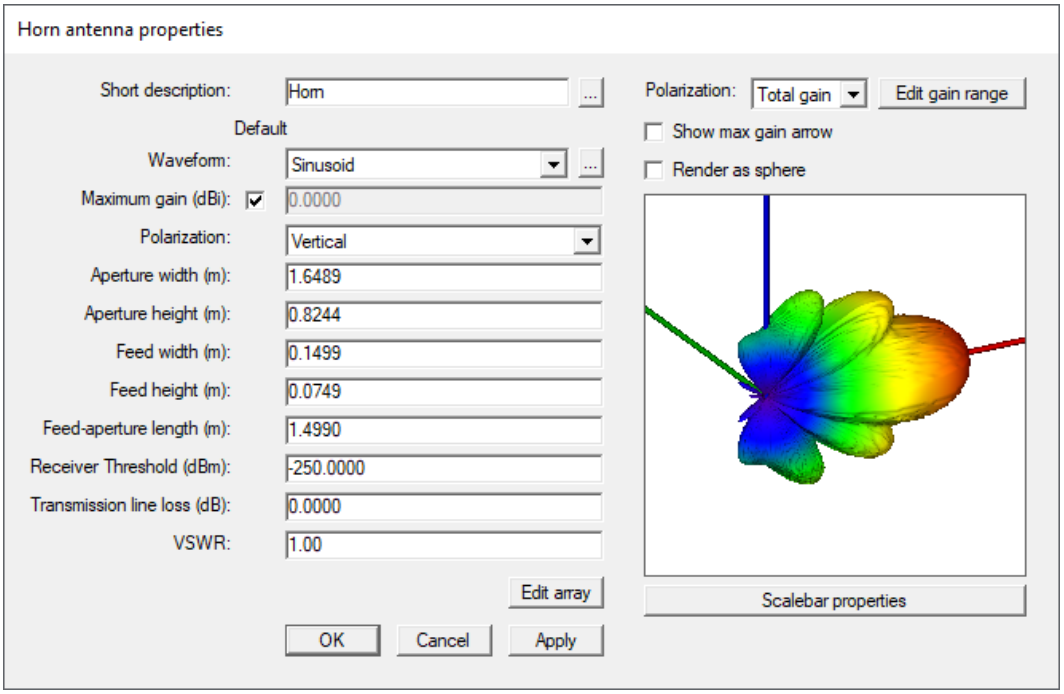


Figure 12.12: Properties for a Horn Antenna configured as a Pyramidal

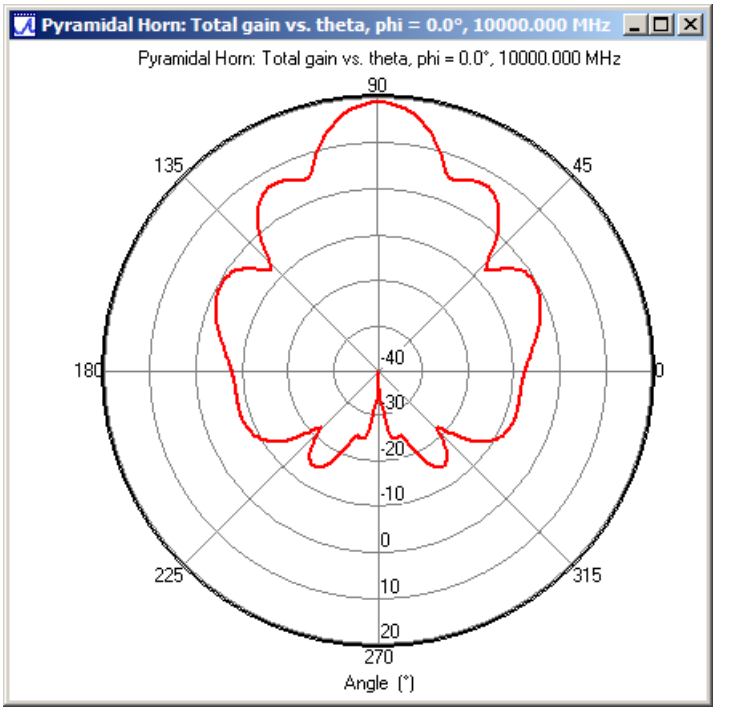


Figure 12.13: Plot of the Horn Antenna pattern

12.4.1.5 Circular Loop

This antenna is a single continuous loop of wire that is thin compared to a wavelength. A uniform, in-phase current is assumed on the loop. The loop radius is set in the ANTENNA PROPERTIES Window. For the default antenna polarization, "Vertical," as seen in Figure 12.14, the loop lies in the XY plane with the normal to the loop being the Z-axis. This produces an omnidirectional field pattern in the XY plane. To operate the antenna in the "Horizontal" polarization, the associated transmitter is rotated to the desired direction in the TRANSMITTER PROPERTIES Window.

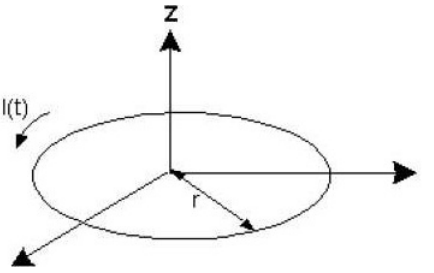


Figure 12.14: Circular Loop Antenna

12.4.1.6 Square Loop

This antenna is a single continuous square loop of wire that is thin compared to a wavelength, as seen in Figure 12.15. A uniform, in-phase current is assumed on the loop. The dimension of the loop is $\frac{\lambda}{4}$ or the perimeter is 1λ at the frequency of the associated waveform. The loop lies in the XY plane and produces an omnidirectional pattern in the YZ plane.

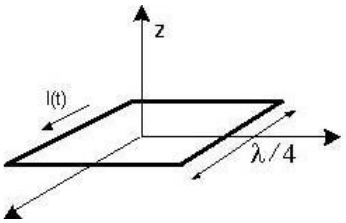


Figure 12.15: One wavelength-long Square Loop Antenna

12.4.1.7 Axial Mode Helix

The *Axial Mode Helix* is a single continuous monofilar circular helix of wire that is thin compared to a wavelength. The antenna is assumed to be mounted on a finite ground plane [37]. The input parameters, entered through the ANTENNA PROPERTIES Window, are the helix *Radius*, *Length* and *Pitch (turns/meter)*. The antenna *Polarization* is "Circular." Both "Left-hand" and "Right-hand" polarizations are available. The default orientation is in the X-direction. The orientation is associated with the transmitter's and is adjusted through the TRANSMITTER PROPERTIES Window.

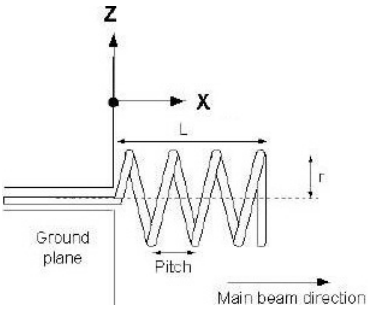


Figure 12.16: Default orientation of the Axial Mode Helix

12.4.1.8 Parabolic Reflector

The *Parabolic Reflector* is a *Circular Aperture* with a particular field distribution and a finite amount of central blockage to simulate the antenna feed. The parabola can have one of three commonly used *Aperture distributions*: "Uniform," "Parabolic," or "Parabolic squared." The main parameters, entered through the ANTENNA PROPERTIES Window, are the parabola *Radius*, the feed or central *Blockage radius*, the *Aperture distribution*, and the antenna *Polarization*. The default antenna orientation points the main

beam in the Z-direction with linear polarization (X-directed). The desired orientation is set by rotating the associated transmitter(s) or receiver(s).

If the antenna *Polarization* is set to “Vertical” (default), the antenna E-plane lies in the XZ plane. Setting the polarization to “Horizontal” will rotate the antenna about the X axis by 90° to produce an E-plane pattern in the XY plane. [38]

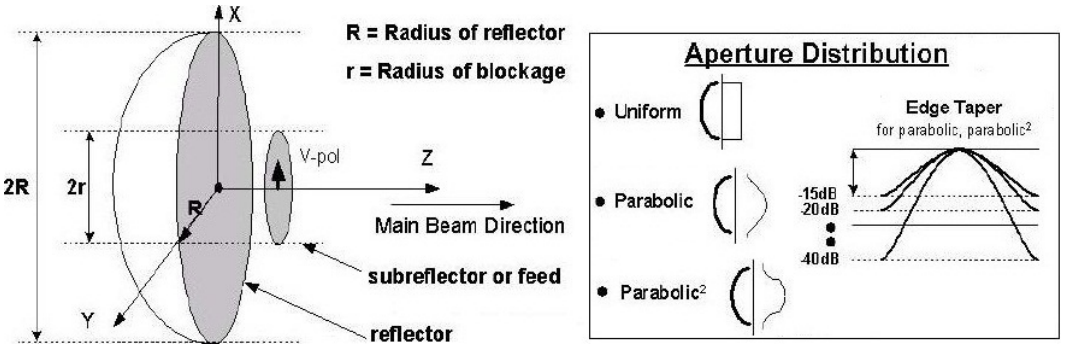


Figure 12.17: Description of a Parabolic Reflector Antenna

Shown below are the PARABOLIC REFLECTOR ANTENNA PROPERTIES Window and the resulting antenna field pattern.

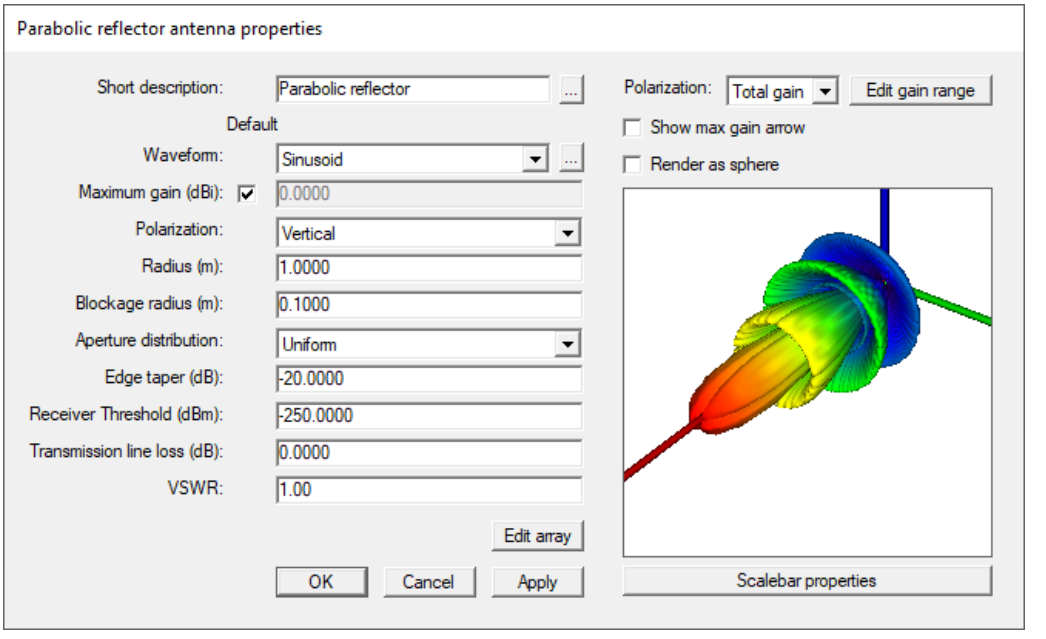


Figure 12.18: Properties of a Parabolic Reflector Antenna and the resulting gain pattern

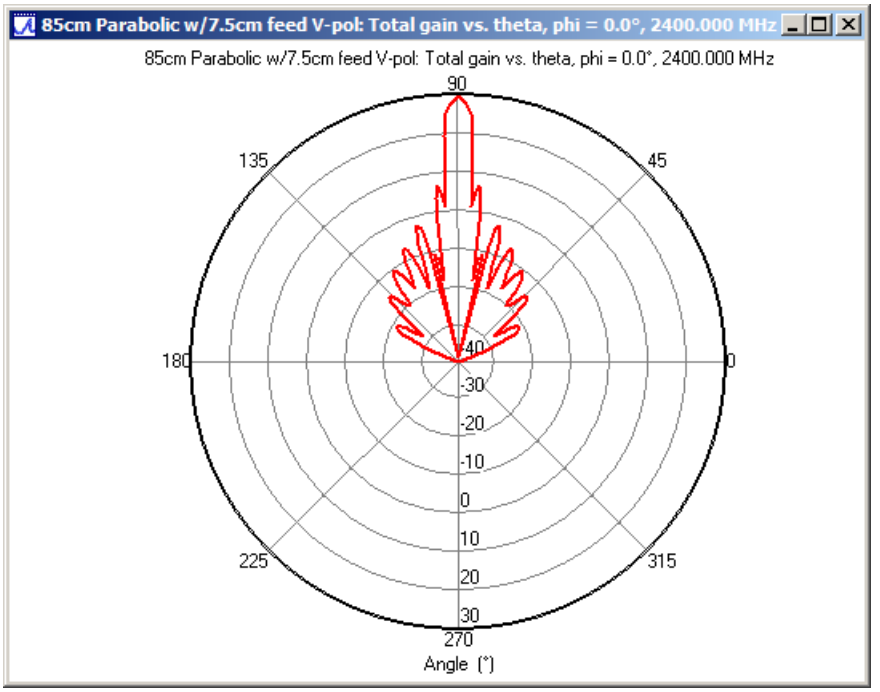


Figure 12.19: Plot of the Parabolic Reflector Antenna pattern

12.4.1.9 Directional

The geometrically defined *Directional* antenna beam is one of the idealized antenna patterns in Wireless InSite. It is not associated with any particular type of physical antenna, rather it is completely defined by the beamwidth parameters. The *Directional* antenna provides a simple means of defining the main beam of a high gain antenna while disregarding the sidelobes. It produces a beam pattern aimed along the X-axis.

This antenna pattern is specified by:

- *E-plane half-power beamwidth*
- *E-plane first null beamwidth*
- *H-plane half-power beamwidth*
- *H-plane first null beamwidth*

For “Circular” polarization, the beamwidth parameters are:

- *Vertical-plane half-power beamwidth*
- *Vertical-plane first null beamwidth*
- *Horizontal-plane half-power beamwidth*
- *Horizontal-plane first null beamwidth*

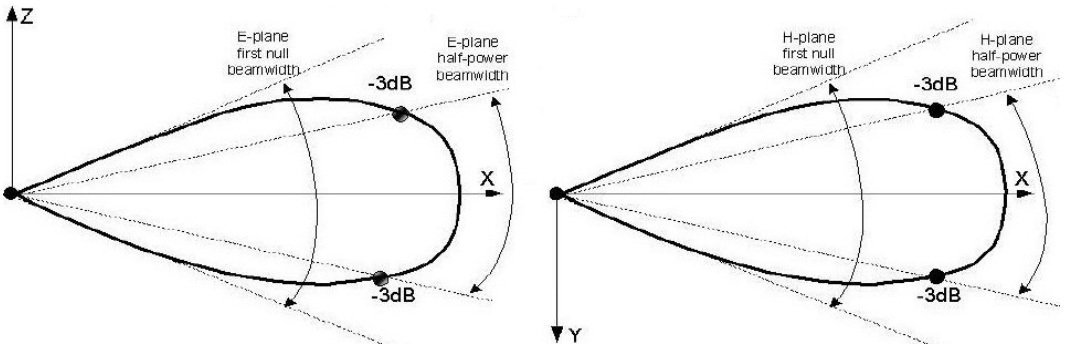


Figure 12.20: The E-plane pattern (left) and H-plane pattern (right) of a generic Directional Antenna

Figure 12.21 shows the DIRECTIONAL ANTENNA PROPERTIES Window.

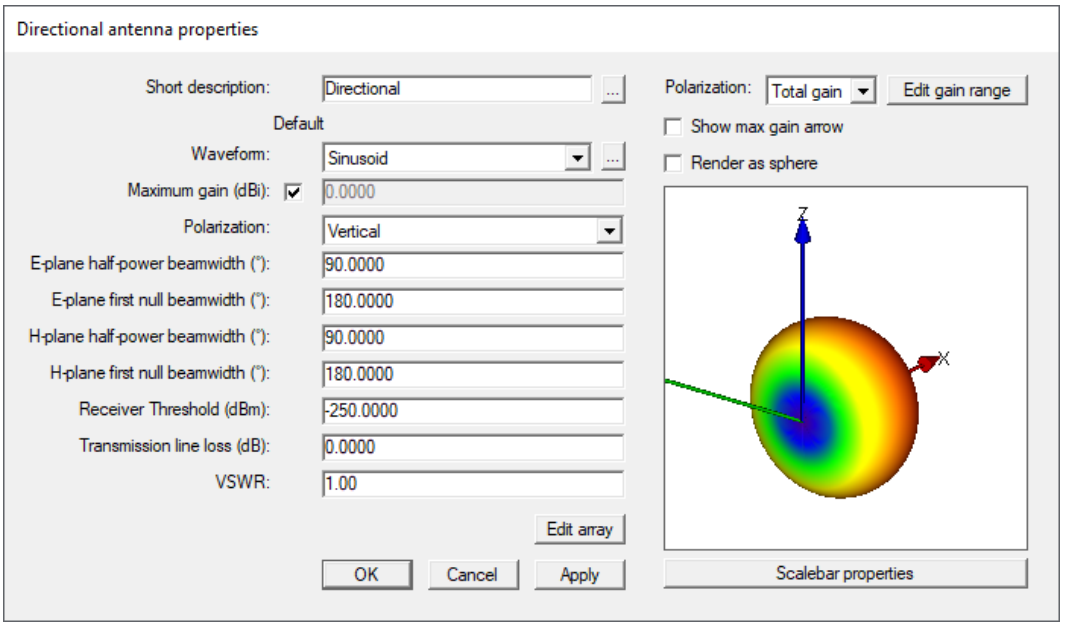


Figure 12.21: Properties of a Directional Antenna

12.4.1.10 Omnidirectional

The geometrically defined *Omnidirectional* antenna pattern is one of the idealized antenna patterns in Wireless InSite. It is not associated with any particular type of physical antenna, rather it is completely defined by the beamwidth parameters. The *Omnidirectional* antenna produces an omni pattern in the azimuthal XY plane with a null on the Z-axis.

For "Vertical" polarization, this antenna pattern is specified by:

- *E-plane half-power beamwidth*

- *E-plane first null beamwidth*

For “Horizontal” polarization, the antenna pattern is specified by:

- *H-plane half-power beamwidth*
- *H-plane first null beamwidth*

For “Circular” polarization, the beamwidth parameters are:

- *Vertical-plane half-power beamwidth*
- *Vertical-plane first null beamwidth*

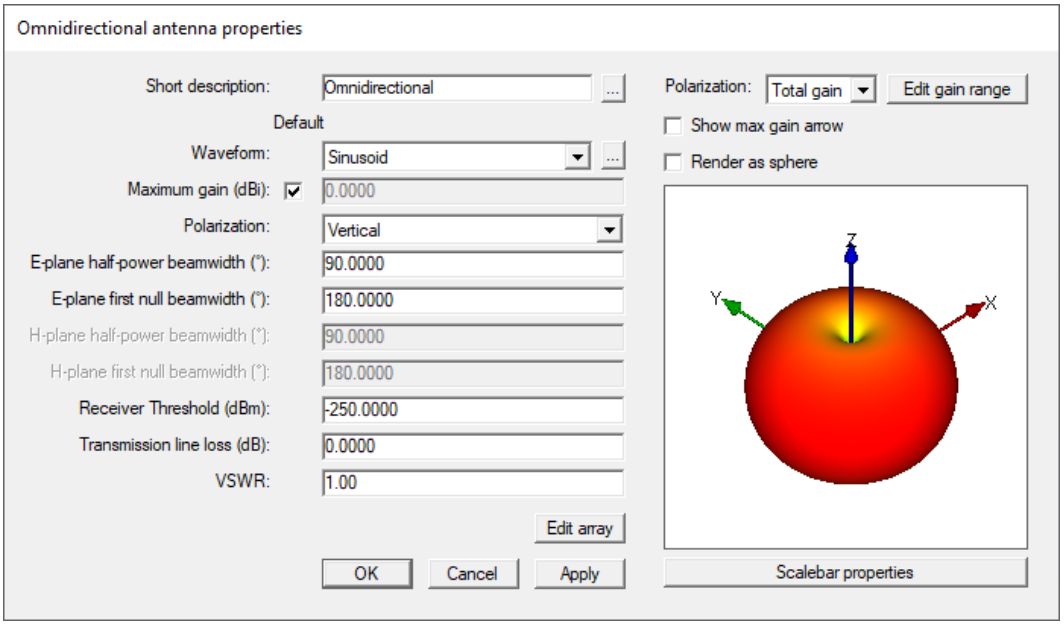


Figure 12.22: Properties of an Omnidirectional Antenna

12.4.1.11 Isotropic

The *Isotropic* pattern is one of the idealized antenna patterns in Wireless InSitenot associated with any particular type of physical antenna. It provides a uniform field strength in one linear polarization or circular polarization. The default polarization is vertical, but a different polarization can be set in the ANTENNA PROPERTIES Window. When using this type of antenna, it is important to keep in mind that away from the horizontal plane, the vertically polarized Isotropic is more accurately described as being theta-polarized, and the horizontal Isotropic is more accurately regarded as phi-polarized.

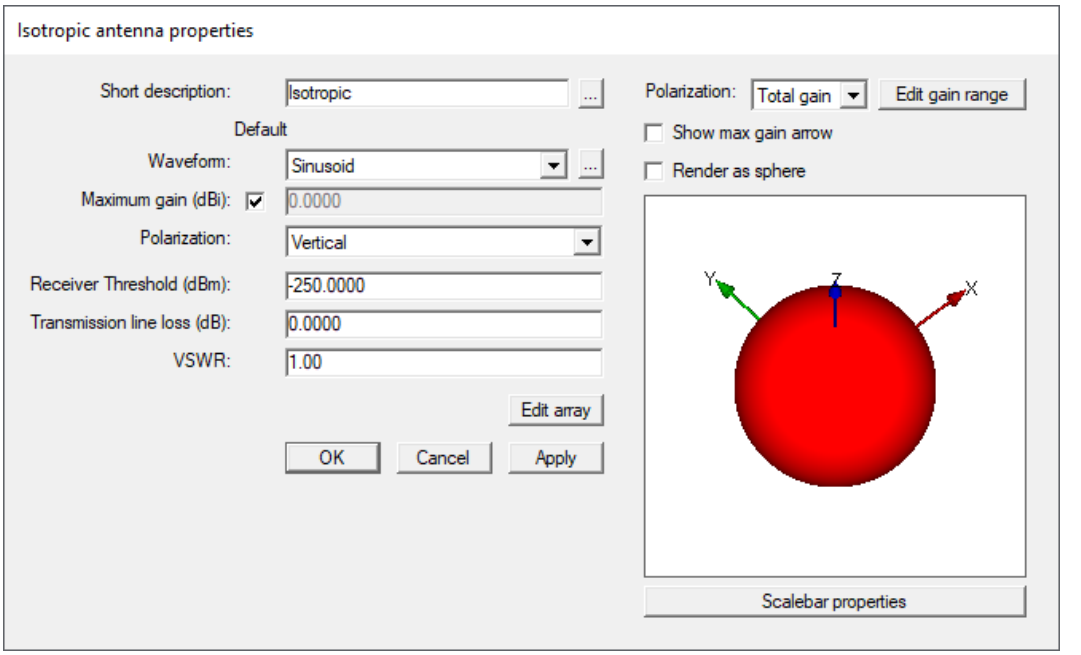


Figure 12.23: Properties of an Isotropic Antenna

12.4.1.12 Orientation and Rotation of Freestanding Antennas

It should be noted that these specifications are for the antenna parameter definitions. The actual antenna orientation and polarization for a Wireless InSite simulation is also determined by rotating the associated TRANSMITTER or RECEIVER control vectors. Figure 12.24 shows the properties window of a TRANSMITTER that is using an ANTENNA. The rotation parameters that control the orientation of the antenna are associated with the Tx/Rx set that is using it, rather than with the antenna. For *Omnidirectional* patterns and patterns "boresighted" along the X-axis, this is no restriction at all. The final orientation of the antenna after these rotations are applied can be viewed in the GEOMETRY VIEW by making the transmitter or receiver's control vectors visible.

- For more information on antenna alignment options see 13.3.

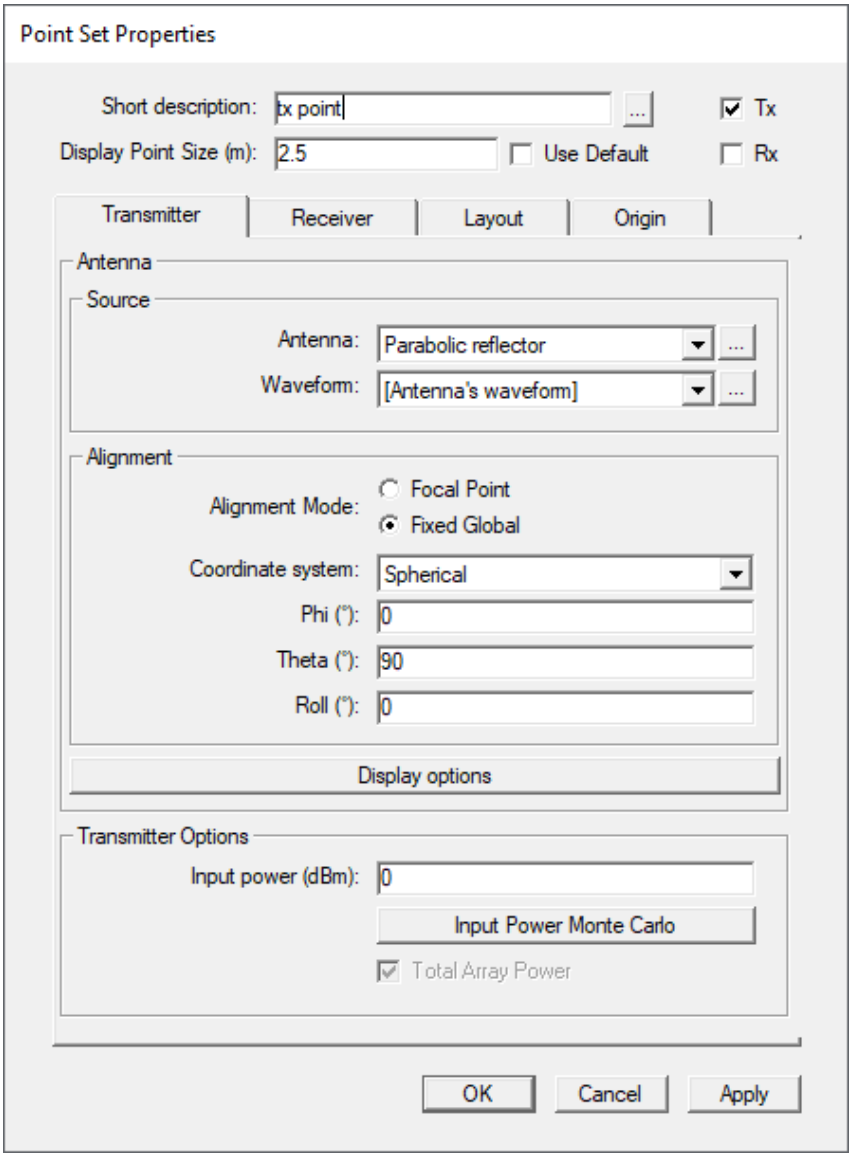


Figure 12.24: The properties window of a transmitter using an antenna

12.4.2 Ground-Plane Mounted Antennas

Aperture, *Monopole*, and *Patch* Antennas, which are mounted on a ground plane or other planar surface, are oriented such that the antenna normal points in the $+Z$ direction by default. Thus for *Aperture* or *Patch* Antennas the pattern is plotted with the main beam in the $+Z$ direction. For *Monopoles* the pattern is plotted with the wire in the $+Z$ direction. *User-Defined* Antennas can also be treated as being mounted on a ground plane.

12.4.2.1 Short Monopole

This antenna is similar to the *Short Dipole*. The length is much less than one wavelength. It generates a field pattern proportional to $\sin(\theta)$ in the upper half-space only. The lower boundary of the space where the *Short Monopole* radiates is defined by the surface the antenna is mounted on. The orientation is parallel to the normal of that surface.

12.4.2.2 Linear Monopole

This antenna is similar to the *Linear Dipole*. The *Length* of the *Linear Monopole* is defined in the ANTENNA PROPERTIES Window. This antenna generates a field pattern proportional to $\sin(\theta)$ in the upper half-space only. The lower boundary of the space where the *Linear Monopole* radiates is defined by the surface the antenna is mounted on. The orientation is parallel to the normal of that surface.

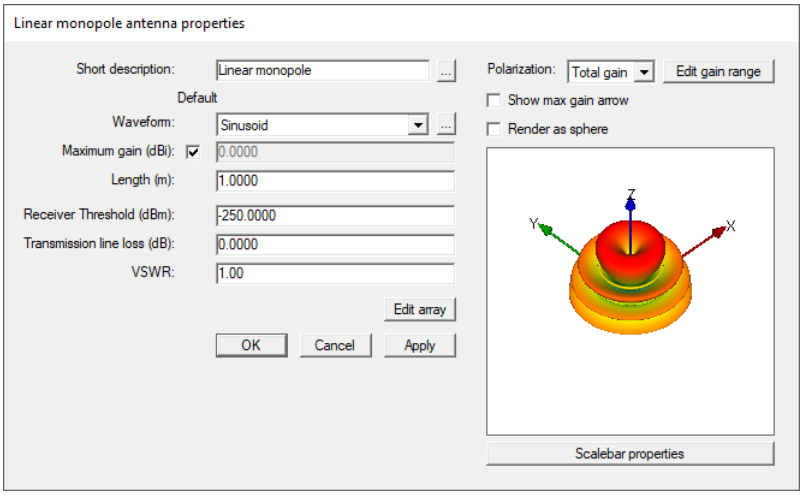


Figure 12.25: Linear Monopole Antenna with antenna pattern display active

12.4.2.3 Quarter-Wave Monopole

The *Quarter-Wave Monopole* is identical to the *Linear Monopole*, with the exception that the length of the monopole is always set to be one quarter of the wavelength of the assigned waveform.

12.4.2.4 Rectangular Aperture

The *Rectangular Aperture* is a rectangular opening in a PEC sheet. The pattern is calculated as if the PEC sheet were infinite in extent. The actual surface used to support the antenna will, of course, be finite. The opening can support one of two possible electric field distributions: “Uniform” or “TE10.” The input parameters, entered through the ANTENNA PROPERTIES Window, are the aperture dimensions (*Side A* and *Side B*) and the *E-field distribution*. The default antenna orientation is such that the main beam is in the Z-direction. The desired orientation is achieved by placing the antenna’s associated TRANSMITTER or RECEIVER on a surface. The antenna aperture will lie in the plane of that surface. The main beam will be parallel to the mounting surface normal.

The antenna’s polarization is linear (Y-directed, E-plane is the YZ plane) by default. The polarization can be adjusted by setting the rotation about the antenna’s Z axis (the normal to the transmitter or receiver’s mounting surface) in the TRANSMITTER/RECEIVER PROPERTIES Window. Once the antenna and associated transmitter or receiver is placed on a surface, the antenna’s Z axis lies parallel to the mounting surface’s normal. Rotations applied in the TRANSMITTER/RECEIVER PROPERTIES Window about the Z axis are about this mounting surface normal. [36, p.584]

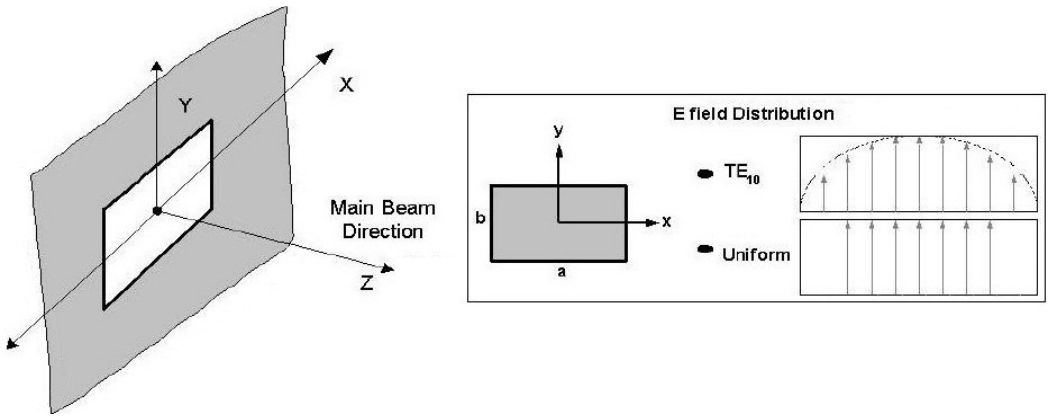


Figure 12.26: Description of the Rectangular Aperture Antenna

12.4.2.5 Circular Aperture

The *Circular Aperture* is a circular opening in an infinite PEC sheet. The opening can support one of two possible electric field distributions: “Uniform” or “TE11.” The input parameters, entered through the ANTENNA PROPERTIES Window, are the aperture *Radius* and the *E-field distribution*. The default antenna orientation is such that the main beam is in the Z-direction. The desired orientation is achieved by placing the antenna’s associated transmitter or receiver on a surface. The antenna aperture will lie in the plane of the surface. The main beam will be parallel to the mounting surface normal. The antenna polarization is linear (Y-directed) by default. The polarization can be adjusted by setting the rotation about the antenna’s Z-axis (the normal to the surface) in the TRANSMITTER/RECEIVER PROPERTIES Window. Once the antenna and associated Tx/Rx are placed on a surface, the antenna’s Z-axis lies parallel to the mounting surface’s normal. Rotations applied in the TRANSMITTER/RECEIVER PROPERTIES Window about the Z-axis are about this mounting surface normal. [36, p. 603]

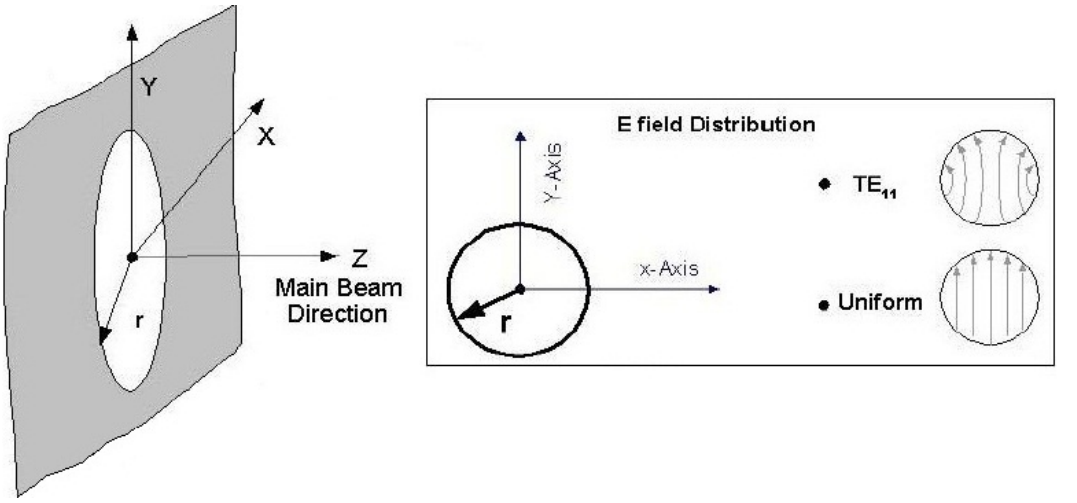


Figure 12.27: Description of the Circular Aperture Antenna

12.4.2.6 Circular Patch

The *Circular Patch* is a circular printed patch of PEC lying on a dielectric slab of a given relative dielectric constant. The horizontal extent of the dielectric substrate is that of the patch surface. The patch excitation and feed point are not modeled and it is assumed the patch is excited near the dominant TM_{11} mode.

The input parameters, entered through the ANTENNA PROPERTIES Window, are the patch *Radius*, the relative *Dielectric constant* of the substrate and the substrate height. The default antenna orientation is such that the main beam is in the Z-direction. The desired orientation is achieved by assigning the antenna to a transmitter or receiver on a surface with the desired orientation. The antenna polarization is linear. [36, p. 752]

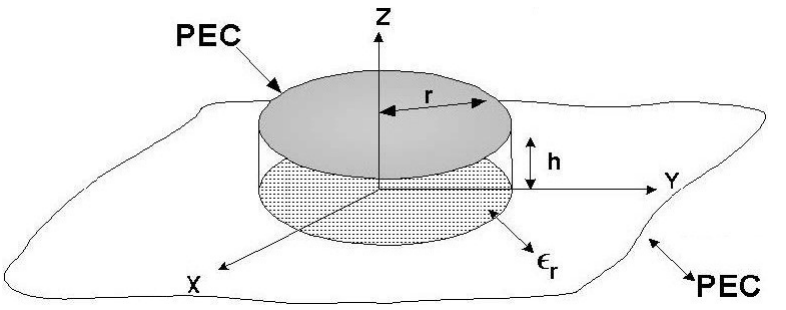


Figure 12.28: Description of the Circular Patch Antenna

12.4.2.7 Rectangular Patch

The *Rectangular Patch* is a rectangular printed patch of PEC lying on a dielectric slab of a given relative dielectric constant. The horizontal extent of the dielectric substrate is that of the patch surface. The patch excitation and feed point are not included in the model.

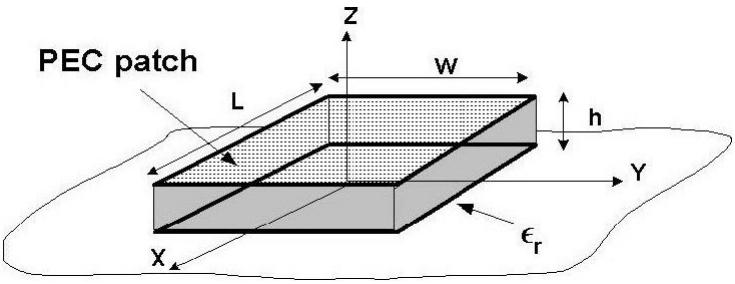


Figure 12.29: Description of the Rectangular Patch Antenna

The input parameters, entered through the ANTENNA PROPERTIES Window, are the patch *Length*, the patch *Width*, and the relative *Dielectric constant* and *Height* of the substrate. The default antenna orientation is such that the main beam is in the Z-direction. The desired orientation is achieved by assigning the antenna to a transmitter or receiver on a surface with the desired orientation. The antenna polarization is linear. The E-plane for this antenna is parallel to the longer of the patch dimensions *Length* and *Width*. For example, if *Length* (X-dimension) $>$ *Width* (Y-dimension), the dominant mode is the TM^z_{010} mode. This will orient the E-plane in the XY plane and is the default orientation. Setting *Length* $<$ *Width* produces the TM^z_{001} mode and the E-plane will be the YZ plane.

The polarization can be adjusted by setting the rotation about the antenna's Z-axis (the normal to the transmitter or receiver's mounting surface) in the TRANSMITTER/RECEIVER PROPERTIES Window. Once the antenna and associated transmitter or receiver are placed on a surface, the antenna's Z-axis lies parallel to the mounting surfaces normal. Rotations applied in the TRANSMITTER/RECEIVER PROPERTIES Window about the Z-axis are about this mounting surface normal. [36, p. 727]

12.4.2.8 Orientation and Rotation of Ground-Plane Mounted Antennas

One way to orient these ANTENNAS for the simulation is to position the *Aperture*, *Monopole*, or *Patch* on a facet. Once the antenna and associated TRANSMITTER or RECEIVER is placed on a surface, the antenna's Z-axis lies parallel to the mounting surface's normal. Rotations applied in the TRANSMITTER/RECEIVER PROPERTIES Window about the Z-axis are about this mounting surface normal. For the *Monopole*, this rotation has no effect since the pattern is symmetric in the XY plane.

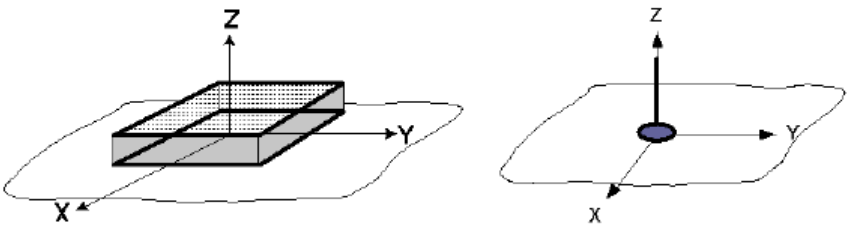


Figure 12.30: A Rectangular Patch in default orientation (left), and a Monopole in default orientation (right)

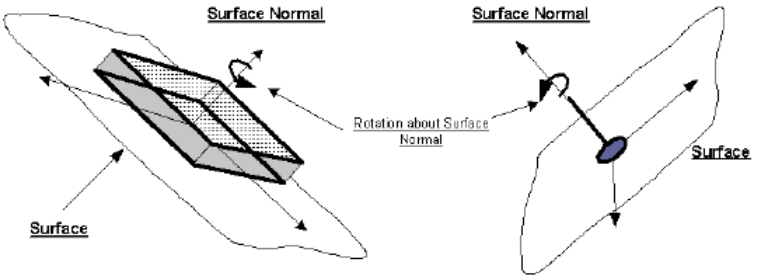


Figure 12.31: A Rectangular Patch after placement on a surface (left), and a Monopole mounted on a surface (right)

This approach to specifying and locating antennas allows great flexibility. Each antenna type is defined independently of how it will be oriented in the simulation. Thus the same *Monopole* may be mounted on several different model surfaces with different orientations, or the same *Horn* may be directed in one way for transmit and in a different direction for receive.

12.5 User-Defined and Imported Antenna Patterns

Wireless InSite also has the capability to import *User-Defined* Antenna patterns. These antennas are considered to be freestanding. Wireless InSite has its own pattern data file format or can import data in several commonly used formats. If the pattern data is only available on the E-plane and H-plane, a full three-dimensional pattern will be generated through interpolation [39] [40] [41].

The pattern data used for import can be obtained from manufacturer websites, or generated from a full-wave electromagnetic simulator such as NEC or XFdtd®.

12.5.1 Wireless InSite User-Defined Format

The Wireless InSite *User-Defined* Antenna format (`*.uan`) starts with a parameter section as shown below.

```
begin_<parameters>
     format free
     phi_min 0
     phi_max 360
```

```
      phi_inc 4
      theta_min 0
      theta_max 180
      theta_inc 3
      complex <or> real
      mag_phase <or> real_imag (only required when complex is specified)
      pattern gain <or> power <or> field
      magnitude dB <or> linear
      maximum_gain 0 (optional parameter, measured in dBi)
      phase degrees <or> radians
      direction degrees <or> radians
      polarization theta_phi <or> phi_theta <or> theta <or> phi
end_<parameters>
```

✓ If additional parameters are included in the file beyond what is described above, Wireless InSite will ignore them. This is particularly the case when the `*.uan` file is created using an external program, such as XFdtd®.

Following the parameters section, the first four lines of data might be:

```
90.000 0.000 -10.0 -25.0 90.0 172.50
90.000 4.000 -11.5 -24.0 92.0 174.50
90.000 8.000 -12.0 -24.0 94.0 176.50
90.000 12.00 -13.5 -20.0 97.0 178.50
```

where the data format is theta, phi, gain (theta component), gain (phi component), phase (theta component), phase (phi component).

❗ The min, max and inc values for theta and phi must be integers.

If phase is not given, the keyword `complex` should either be omitted from the parameter section, or replaced by `real` (which is the default). Further, the mag_phase **or** real_imag keywords, which specify the format of the complex values, are not required in the header. The data in the file would be:

```
90.000 0.000 -10.0 -25.0
90.000 4.000 -11.5 -24.0
90.000 8.000 -12.0 -24.0
90.000 12.00 -13.5 -20.0
```

✓ If the data only specifies a horizontal cut at theta = 90 and a vertical cut at phi = 0/180, then Wireless InSite will automatically interpolate the pattern to produce a full 3D-gain pattern. When doing this, ensure that the theta min and max are specified at 90°.

The Wireless InSite antenna type also allows full 3D far zone radiation patterns calculated by XGtd and XFdtd to be imported into other Wireless InSite projects and used as antenna patterns.

12.5.1.1 Maximum Gain Settings for User-Defined Antennas

Additional options are available for setting the maximum gain of user-defined antenna patterns in the antenna's properties window:

- *From UAN File* - The value of this field is taken from the `Maximum_gain` keyword in the `*.uan` header. If `maximum_gain` is omitted, the maximum total gain of the pattern data is used. Users cannot edit the value in this field.
- *From integration* - Determines the gain by first normalizing the user-defined antenna pattern data and using equation 12.3. Users cannot edit the value in this field.
- *User-specified* - Sets the maximum gain of the ANTENNA to the value listed in the field. It is possible to enter nonphysical gain values for an antenna, so the user is responsible for realistic manual entries.

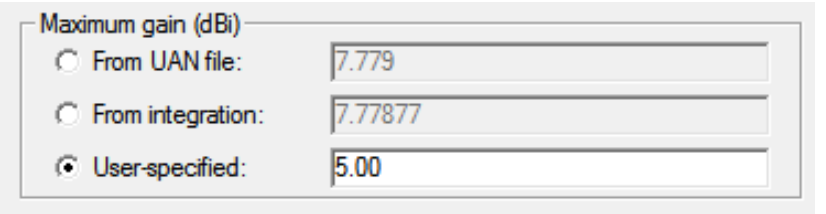


Figure 12.32: Maximum gain settings for user-defined antenna patterns

12.5.2 Odyssey

Aethos *Odyssey* Antenna files contain horizontal and vertical cut-planes which are interpolated to form a full 3D pattern. Supported keywords include:

- `NAME`
- `FREQUENCY`
- `BEAM_WIDTH`
- `DIAMETER`
- `GAIN`
- `GAIN_UNIT`
- `TILT`
- `ANTENNA_TYPE`
- `COMMENTS`
- `POLARIZATION`

Possible values for select keywords are shown below. The gain unit may either be on the same line as the gain value or may be listed on a separate line using the keyword `GAIN_UNIT`. Information found using these keywords is saved in the antenna's *Notes*.

Each cut-plane consists of 360 data points, for angles 1 to 360 degrees. Zero degrees corresponds to north in the horizontal plane and the horizon in the vertical plane. The data columns for angle and gain may be separated by either tabs or spaces. Cut-plane values are given in dBm relative the maximum gain (0 dBm), and must not include the minus sign.

```
NAME BCD-8706-0-25
FREQUENCY 900
BEAM_WIDTH 360
GAIN 8.14 dBi [dBd]
GAIN_UNIT dBi [dBd]
TILT ELECTRICAL
POLARIZATION VERTICAL [HORIZONTAL]
HORIZONTAL 360
1 0.050
2 0.090
.
.
359 0.090
360 0.050
VERTICAL 360
1 0.090
2 0.170
.
.
359 0.090
360 0.000
```

12.5.3 Planet

An MSI *Planet* Antenna file consists of ASCII pattern data for the vertical and horizontal cut-planes. Valid parameters at the beginning of the file are shown below. This information is read and saved in the antenna's *Notes*. antenna gain is given in dBd by default; if the gain is specified in dBi it must be so labeled after the gain value, separated by a space.

Each cut-plane consists of 360 data points, for angles 0 to 359 degrees. Zero degrees corresponds to north in the horizontal plane and the horizon in the vertical plane. Data columns may be separated by either tabs or spaces. Cut-plane values are given in dBm relative the maximum gain (0 dBm), and must not include the minus sign.

By convention, the antenna name is listed first.

```
NAME Andrew Corp
MAKE PCS19HA-11015-2
FREQUENCY 1920
H_WIDTH 115.97
V_WIDTH 4.77
FRONT_TO_BACK 25.32
GAIN 14.50 dBi
```

```
TILT Electrical
HORIZONTAL 360
0 0.000
1 0.261
.
.
358 0.261
359 0.261
VERTICAL 360
0 1.401
1 0.300
.
.
359 3.098
360 0.000
```

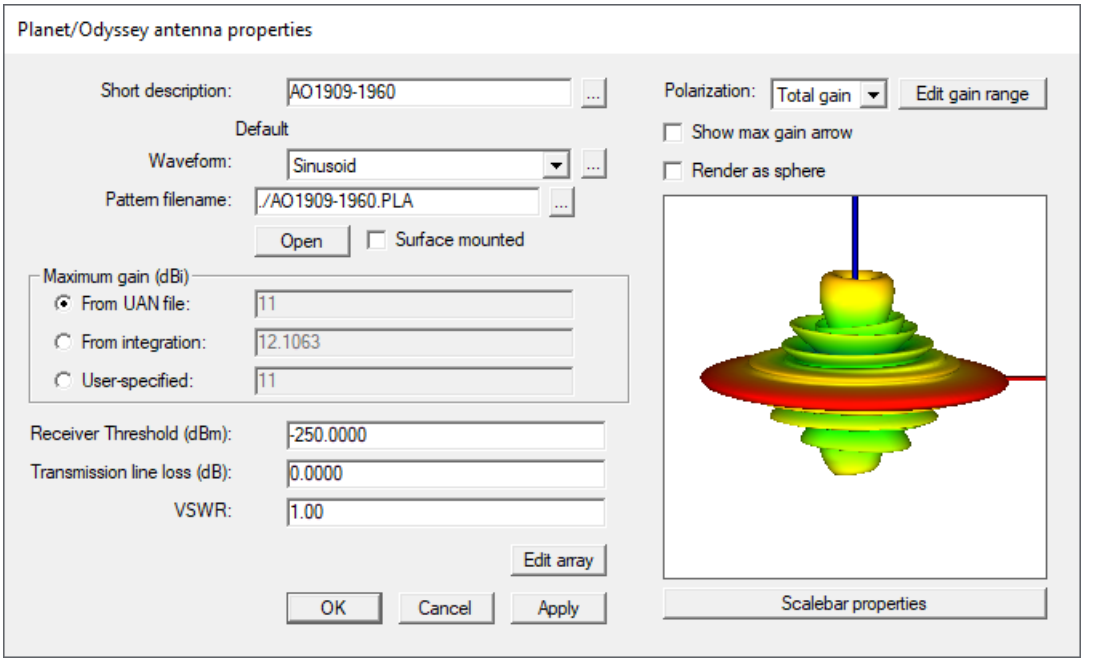


Figure 12.33: Properties of an imported Planet Antenna

12.5.4 Orientation of User-Defined and Imported Antenna Patterns

Just as with the built-in antenna types, the *User-Defined* Antenna patterns may be rotated to aim the antenna in the desired direction. The ANTENNA orientation and polarization for a Wireless InSite simulation is acheived by rotating the associated transmitter or receiver's control vectors. As stated earlier, the *User-Defined* Antenna is considered freestanding. As such, the rotations follow the same paradigms

as described in Section 13.3. The final orientation of the ANTENNA after these rotations are applied can be seen in the Geometry View by making the transmitter or receiver's pattern or control vectors visible.

✓ It is best to generate *User-Defined* patterns such that the direction of maximum gain is along the x-axis, since this will align the main beam with the orientation and control vectors.

12.6 MIMO Antennas

Multiple-input and multiple-output (MIMO) antennas are supported in Wireless InSite as arrays of the previously mentioned antenna types. A MIMO antenna can be composed of all built-in antenna types (12.4), all multi-frequency user-defined antennas (12.7), or single-frequency user-defined antennas (12.5.1), but not a mixture of these types.

12.6.1 MIMO Antenna Properties

The MIMO ANTENNA PROPERTIES window (Figure 12.34) provides all the tools necessary to create and customize *MIMO* arrays.

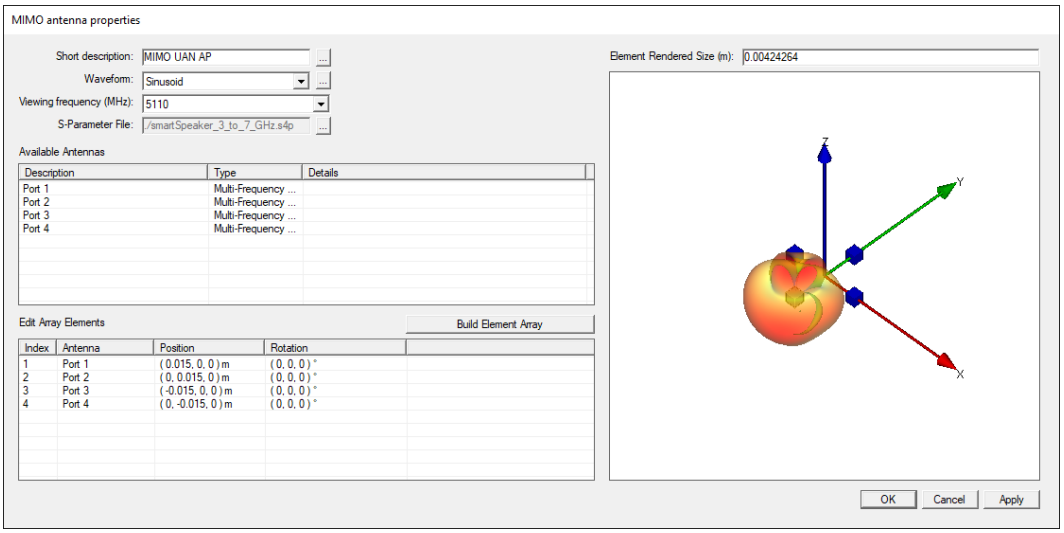


Figure 12.34: The MIMO ANTENNA PROPERTIES window.

The properties of *MIMO* antennas are as follows.

- *Short description* - A word or phrase to identify the antenna.
- *Waveform* - The existing waveform in the project to be used by the *MIMO* array.
- *Viewing Frequency* - Selects the frequency of the waveform to use when determining which antenna pattern to display.
- *S-Parameter File* - (Optional) Specify an S-parameter file to describe impedance matching conditions and mutual coupling between antenna elements. Details about file format can be found

in Appendix J. If file is specified, impedance matching values (e.g. VSWR) used for simulation engine calculations are derived from the S-parameter file assuming mutual coupling is zero. The effect of mutual coupling can be included in the channel matrix output (see 17.1 and Appendix M) and in Communication System analyses (see 14.2.4).

- *Available Antennas* - A list of the sub-antennas to be used by the elements of the *MIMO* array. New entries can be added or copied from the main list in the project via the context menu.
- *Build Element Array* - Button that opens the MIMO ARRAY BUILDER window. Details about this window can be found in 12.6.1.1.
- *Edit Array Elements* - A list of each element currently in the *MIMO* array. The antenna, position, and rotation of each individual element can be modified by selecting "Properties" from its context menu. New elements can also be added through the context menu. The displayed pattern is for the selected element based on the current *Viewing Frequency*. If an element is not currently selected, then the first one is used.
- *Element Rendered Size* - The size (in meters) of the elements shown in the MIMO array viewer.

12.6.1.1 MIMO Array Builder

The MIMO ARRAY BUILDER (Figure 12.35) allows for the rapid creation of two and three-dimensional grids of elements in a *MIMO* array.

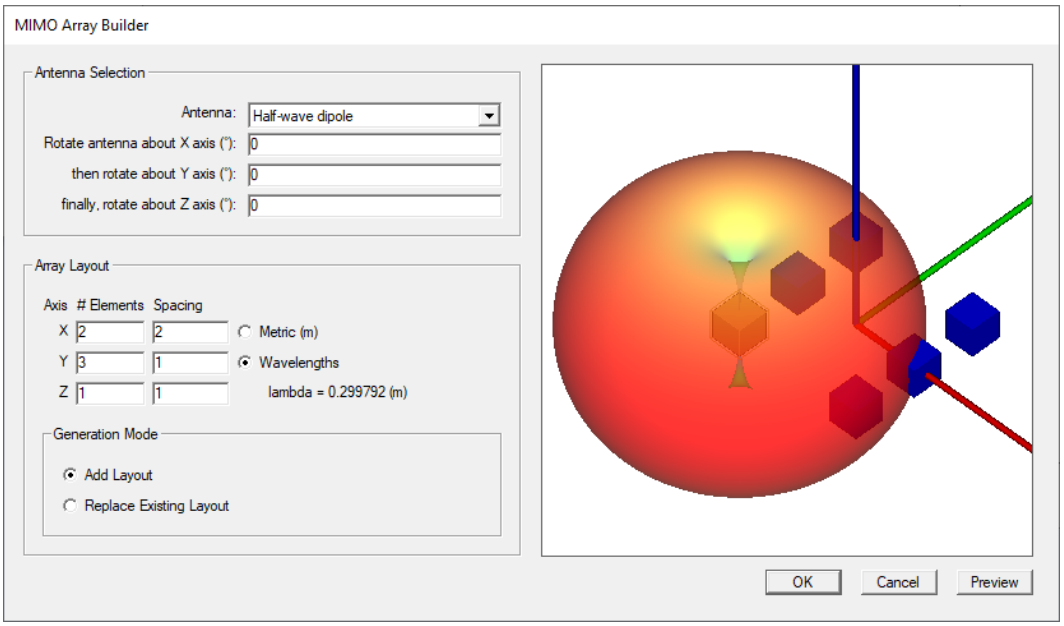


Figure 12.35: The MIMO ARRAY BUILDER window.

The parameters of the MIMO ARRAY BUILDER are listed below.

- *Antenna* - Specifies the antenna to be used by each element added by the MIMO Array Builder.

- *Rotate antenna about X/Y/Z axis* - The antenna's rotation (in degrees) around each axis to be applied to all elements added to the array.
- *# Elements (X/Y/Z)* - The number of elements to be added in each dimension.
- *Spacing (X/Y/Z)* - The spacing between the elements in each dimension. The distance can be set in meters or by wavelengths of the waveform specified in the MIMO ANTENNA PROPERTIES window (12.6.1).
- *Generation Mode* - Choose "Add Layout" to combine the newly-defined grid of elements with any pre-existing elements in the *MIMO* array. Choose "Replace Existing Layout" to have the new elements overwrite any existing elements.

Each individual element created using the MIMO ARRAY BUILDER can be modified via the *Edit Array Elements* table in the MIMO ANTENNA PROPERTIES window described in 12.6.1 (Figure 12.36).

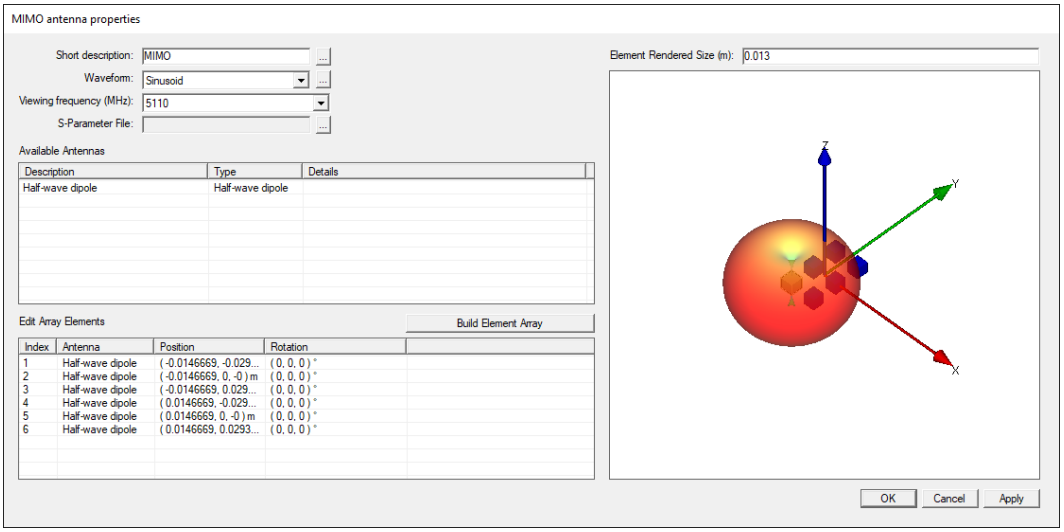


Figure 12.36: An array constructed using the MIMO ARRAY BUILDER and individual elements.

12.7 Multi-Frequency User Defined Antennas

A special type of user-defined antenna is available that is composed of multiple user-defined antennas where each specifies the pattern at a specific frequency. This allows the antenna to have a different shape for each frequency specified as a part of a frequency sweep as detailed in 11.2. When the antenna is used at a frequency that it does not have an exact match for, it will use the pattern that has the frequency closest to the one that is requested.

12.7.1 Multi-Frequency User Defined Antenna Properties

The MULTI-FREQUENCY USER DEFINED ANTENNA PROPERTIES window (Figure 12.37) provides all the tools necessary to specify patterns for various frequencies and to visualize them.

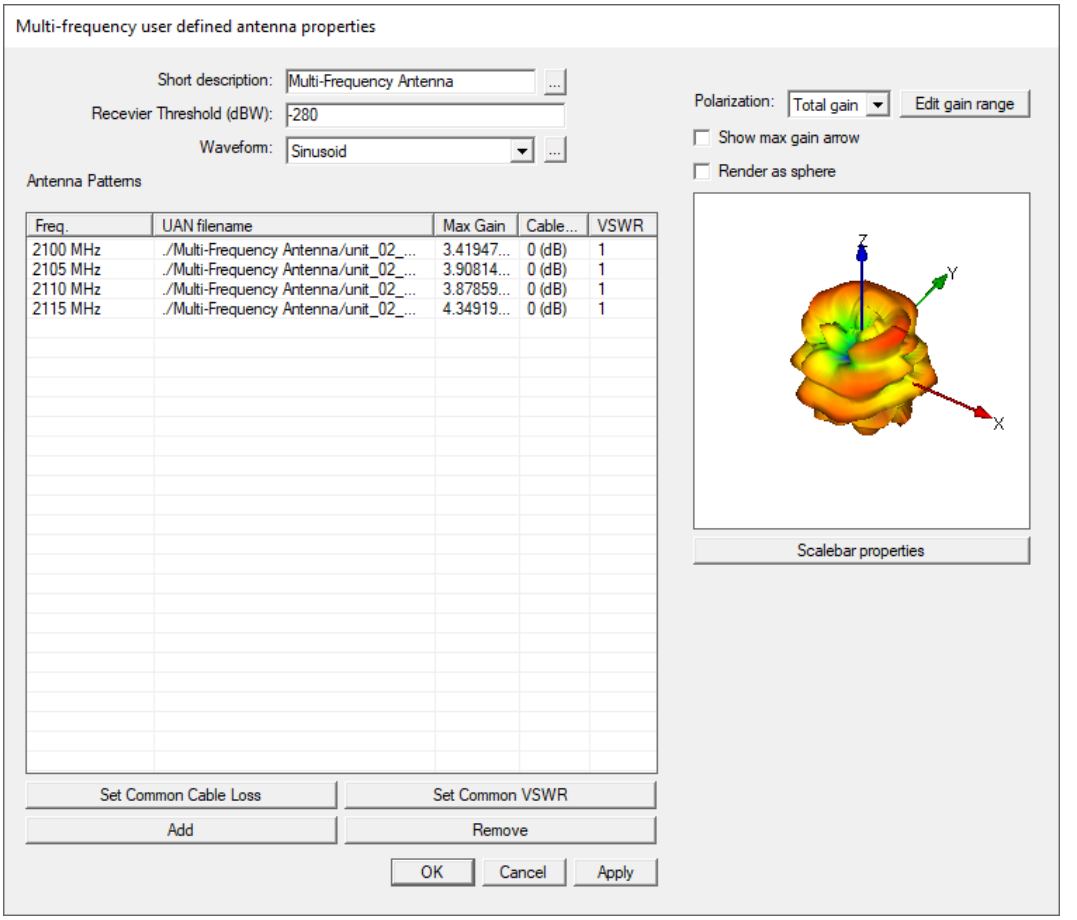


Figure 12.37: The MULTI-FREQUENCY USER DEFINED ANTENNA PROPERTIES window.

The properties of *Multi-frequency user defined* antennas are as follows.

- *Short description* - A word or phrase to identify the antenna.
- *Receiver threshold* - Determines which individual ray paths to ignore when evaluating the power at a RECEIVER point. Ray paths are ignored when their power falls beneath this value.
- *Waveform* - This is the WAVEFORM associated with the antenna that is the default option when editing a transceiver set in the TRANSMITTER/RECEIVER PROPERTIES window where this can be overridden with another waveform in the project. This can be selected from the WAVEFORMS already defined in the project and clicking the "..." button brings up options to add a new WAVEFORM to use for this purpose or modify the one that is selected. This antenna is designed to be used in conjunction with a WAVEFORM that is configured to provide a frequency sweep. See 11.2 for information on how to configure a WAVEFORM for this purpose.
- *Antenna patterns* - This table lists all of the patterns that have been loaded for the indicated frequencies. The columns in the table are as follows and can be edited by double-clicking its entry which brings up the editor window shown in figure 12.38 which has an option to *Open* the *Pattern*

File in addition to editing the other values related to it in the table. Selecting a pattern in the table will display it in the 3D display to the right which is described in more detail in 12.8.2.

- *Frequency* - The frequency at which the pattern should be used when accessing this ANTENNA. This should be the frequency at which it was calculated.
- *UAN filename* - This is the location of the UAN file that the pattern is using relative to the project root folder. This cannot be changed after the pattern is added. To do this *Remove* it and *Add* the other pattern file.
- *Max Gain* - This is the maximum gain that will be used for the pattern. If this is not in the header of the UAN file, then the maximum gain value in the patterns data will be used. This cannot be edited and is only provided for verification purposes.
- *Cable loss* - Simulates line losses, return losses, feed losses, or other system losses between the TRANSMITTER or RECEIVER and its ANTENNA. The number entered here is taken as a loss (in dB) and has the effect of reducing signal strength between the antenna and the Tx/Rx, whether it is entered as a negative or positive number.
- *VSWR* - Defines the Voltage Standing Wave Ratio (VSWR) that must be greater than or equal to 1, with 1 representing a perfect impedance match between the ANTENNA and transmission line resulting in no addition loss. VSWR values greater than 1 will result in an additional mismatch loss during the calculation of received power:

$$Mismatch\ loss(dB) = 10log_{10}\left[1 - \left(\frac{VSWR-1}{VSWR+1}\right)^2\right] \tag{12.6}$$

- *Set Common Cable Loss* - Set the cable loss for all patterns to the same value.
- *Set Common VSWR* - Set the VSWR for all patterns to the same value.
- *Add* - Add a pattern at a new frequency.
- *Remove* - Remove the currently selected pattern in the table.

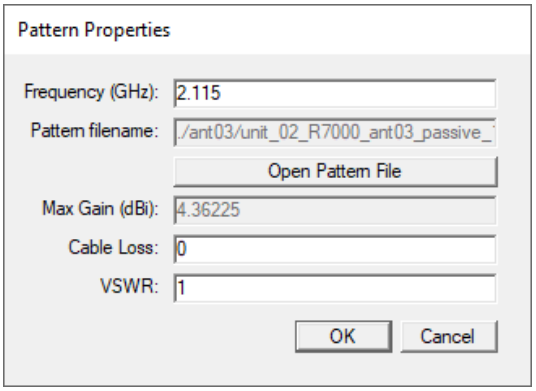


Figure 12.38: The PATTERN properties editor.

12.7.2 Importing from an Antenna Index File

A collection of multi-frequency antennas can be specified using the antenna index file format. This file can be generated by XFdtd®, along with the supporting UAN files, and can specify either a *Multi-frequency user defined* (single-port) antenna or a *MIMO* antenna composed of them depending on the number of ports in the file.

To import an antenna index file:

1. Select *File→Import→Indexed Antenna Set*.
2. Navigate to and select the `XML` file for the desired antenna.

This will create an antenna of the type indicated along with waveforms for each frequency all of which are fully modifiable in the project.

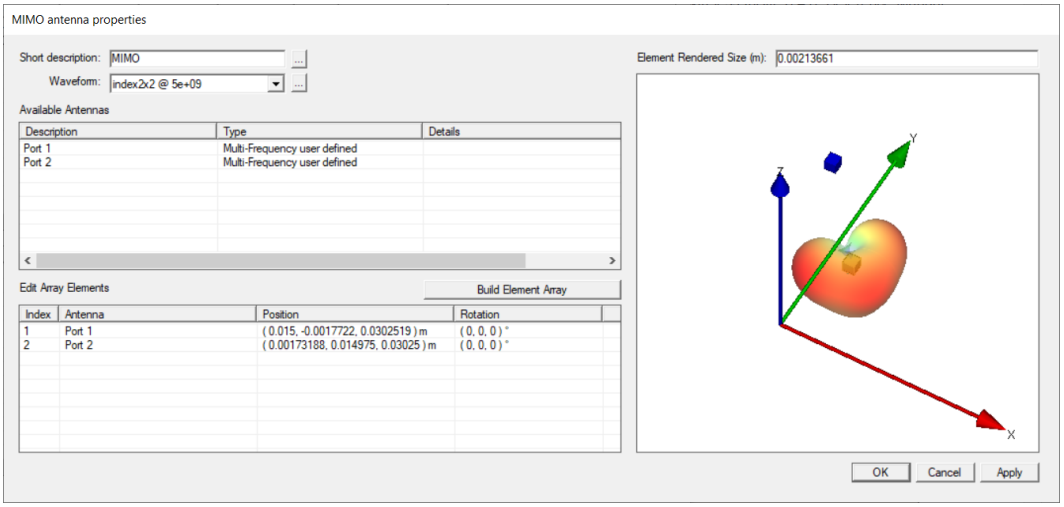


Figure 12.39: An example of a MIMO antenna array imported from an antenna index file generated by XFdtd®.

12.7.3 Antenna Index File Schema

The antenna index file is organized as follows:

- *MIMO_index* - This element contains the information needed to generate the ANTENNA. Even though the tag indicates it is for a *MIMO*, a *Multi-frequency user defined* antenna will be created if there is only one *ActivePort* element.
- *Frequencies* - This begins the list of frequencies that each *Pattern* will reference later in the file. This element should have a *count* attribute indicating the number of *Frequency* elements it contains.
- *Frequency* - There should be an element for each frequency that is referenced by *Pattern* elements. It should have an *index* attribute which is used by the *Pattern* element to indicate which one is assigned to it. The frequency is listed in Hertz contained in the *frequencyHz* attribute.

- *ActivePorts* - Begins the list of *ActivePort* elements. This field should have a *count* attribute indicating the number of *ActivePort* elements it contains.
- *ActivePort* - Encapsulates a collection of *Pattern*s that will be converted into a *multi-frequency user defined* antenna. This element has a the following attributes:
  - *number* - Its index in the list of *ActivePort*s.
  - *farZoneReferenceLocationM* - The Cartesian offset of this port from the antennas center point which is used to position the antenna in the scene using a TRANSMITTER or RECEIVER.
  - *name* - A description of the port which is used as the *Short Description* of the created ANTENNA.
- *FarZonePatterns* - The list of *Pattern*s.
- *Pattern* - The location of the UAN file that this pattern will be loaded from. This element has the following attributes:
  - *frequencyIndex* - This should match the *index* of a *Frequency* in the list of *Frequencies*.
  - *maxGain* - This is the maximum gain to use for the UAN file. If this is missing the value in the UAN header will be used. If that value is not present, then the maximum value from the pattern data will be used.

12.7.4 Antenna Index File Example

This example file will create a *MIMO* antenna with two elements each of which references a *multi-frequency user defined* antenna with patterns for 2.412 GHz and 2.45 GHz. The pattern files, e.g. *p1_f0.uan*, are specified with a path relative to the index files location. The first *MIMO* element is located at (0.01,0.02,0.03) and uses the antenna named *5GC1*, and the second at (0.02,0.01,0.03) and uses the antenna named *5GC2*.

```
<?xml version="1.0" encoding="UTF-8"?>
<MIMO_index>
  <Frequencies count="2">
    <Frequency index="0" frequencyHz="2.412e+09"/>
    <Frequency index="1" frequencyHz="2.45e+09"/>
  </Frequencies>
  <ActivePorts count="2">
    <ActivePort number="1" farZoneReferenceLocationM="0.01,0.02,0.03" name="5GC1">
      <FarZonePatterns>
        <Pattern frequencyIndex="0" maxGain="1.25">p1_f0.uan</Pattern>
        <Pattern frequencyIndex="1" maxGain="1.21">p1_f1.uan</Pattern>
      </FarZonePatterns>
    </ActivePort>
    <ActivePort number="2" farZoneReferenceLocationM="0.02,0.01,0.03" name="5GC2">
      <FarZonePatterns>
        <Pattern frequencyIndex="0" maxGain="5">p2_f0.uan</Pattern>
        <Pattern frequencyIndex="1" maxGain="7.931064">p2_f1.uan</Pattern>
      </FarZonePatterns>
    </ActivePort>
```

```
  </ActivePorts>
</MIMO_index>
```

12.8 Antenna Pattern Plotting

12.8.1 Cut-plane Pattern Plotting

The Wireless InSite GUI can verify antenna characteristics by plotting antenna patterns. In order to plot the pattern of an antenna, the user right-clicks on the antenna and selects *Plot pattern*, as seen in Figure 12.40.

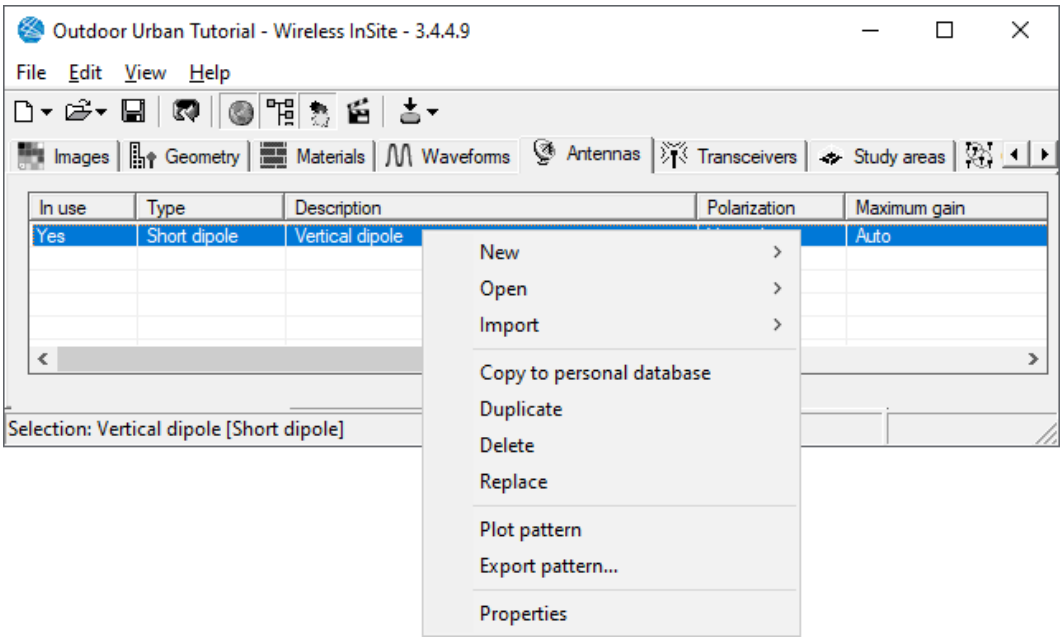


Figure 12.40: The Antenna tab of the Main Window

After right-clicking on the antenna, the user can select the cut-plane using the window shown below. Options include:

- *Complex Component* - Select the magnitude or phase to plot.
- *Polarization* - Gives the option of selecting one of "Total gain," "Theta," "Phi," "LHCP" or "RHCP" polarizations.
- *Mode* - Extracts one of two types of cut planes at a particular angle; those with "Constant Phi" or "Constant Theta."
- *Magnitude format* - Gives the option of selecting between plotting actual gain values and a normalized plot.

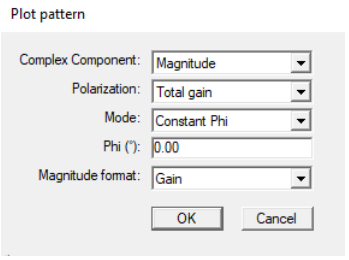


Figure 12.41: Setting the parameters for an antenna pattern plot

When plotting an antenna, it is shown in the default orientation. The actual direction used in the simulation will depend on the orientation of the associated transmitter or receiver.

12.8.2 3D Pattern Display

Wireless InSite also has the capability of showing the full 3D pattern of an ANTENNA in the ANTENNA PROPERTIES Window. The standard Wireless InSite controls for zooming, rotating and panning operate as they do in the GEOMETRY VIEWṪo show the 3D pattern in the ANTENNA PROPERTIES Window, left-click on the vertical button on the right hand side of the window.

Once the antenna pattern appears, there are options to change the manner in which it is displayed. These options only change the display of the pattern and do not have any effect on simulations. The *Component* option changes the component displayed to total gain, theta or phi components or either “LHCP” or “RHCP” polarizations. The *Gain range* sets the value below the maximum gain that will be considered zero or the center of the pattern when it is displayed. This setting changes both the shape of the surface and the colors of the pattern. Red always represents the maximum gain and violet the minimum. By lowering the gain range it becomes easier to see the directionality of the antenna. The pattern can also be shown as a sphere and an arrow in the direction of maximum gain can be activated. The colors that the antenna uses can be changed by clicking on the SCALEBAR PROPERTIES button and the scale bar in the GEOMETRY VIEW shows the range used to display the last antenna that was selected for displaying in this manner. The red, green and blue axes correspond to X, Y and Z respectively, and show the default orientation in the global coordinate system of the antenna pattern.

Chapter 13

Transmitters and Receivers

In this chapter, you will learn...

→ the types and properties of transmitters, receivers, and transceivers

→ how to edit, aim and align them

→ how to change the way they display in Wireless InSite

In Wireless InSite, a TRANSMITTER (Tx) point is the location of a source of input radiation, a RECEIVER (Rx) acts as a field point that collects radiation, and a TRANSCEIVER acts as both a transmitter and receiver. Transmitter and receiver points are grouped into sets where all the points in a given set share the same properties, such as:

- Coordinate system
- Relative elevation
- Antenna
- Waveform
- Antenna rotations
- Input power (transmitters only)

▶ Additionally, distant transmitters can be modeled using *Plane Wave* transmitter sets, which are discussed separately in Section 13.1.13.

Sets, as seen in Figure 13.1, provide a way to create a large amount of points quickly. They are arranged in patterns that provide the type of feedback that would be most useful during the analysis process. Once a simulation is complete, the results are arranged based on how these sets are layed out. Line plots are generated which follow the arrangement of the points in a set.

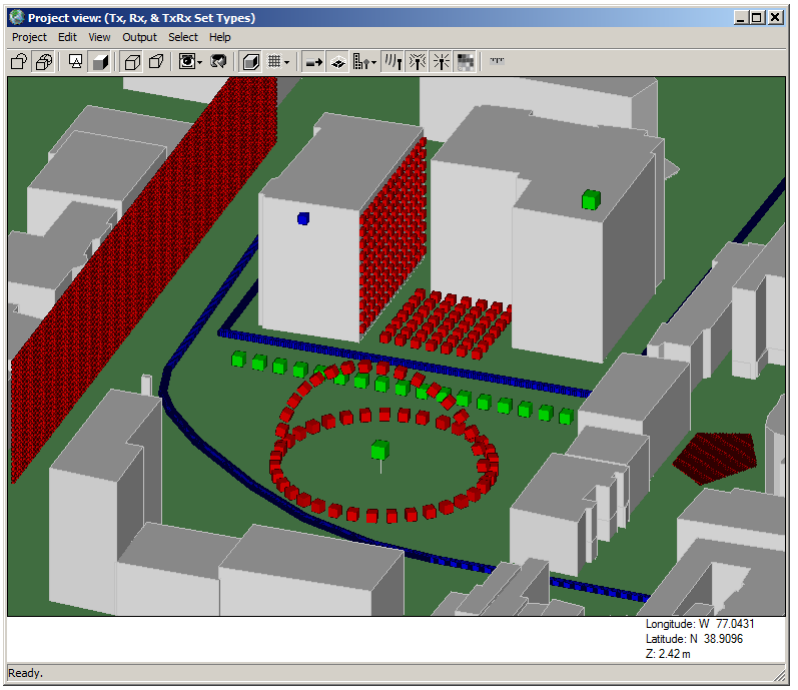


Figure 13.1: A project showing different types of transmitter (green), receiver sets (red), and transceiver (blue) sets

The remainder of this chapter describes the types of Tx/Rx layouts that are possible, Tx/Rx properties, advanced operations for aiming antennas in Tx/Rx set, and setting bounding boxes around sets.

All examples and references in this chapter relate to receivers, but the user should be aware that transmitters and transceivers function identically unless otherwise noted.

13.1 Types of Transmitters and Receivers

13.1.1 Points

The simplest set of transmitter and receiver locations can be defined using *Points*, where each point can be located independently. To create a receiver set of Points, select *File*→*New*→*Receiver Set*→*Points*. The GEOMETRY VIEW will change to 2D WIREFRAME mode, allowing the user to click on as many points as desired. Figure 13.2 shows a point, represented by a small square, which has been placed in the center of a city. To complete the set, right-click and verify the values in the POINT SET PROPERTIES window.

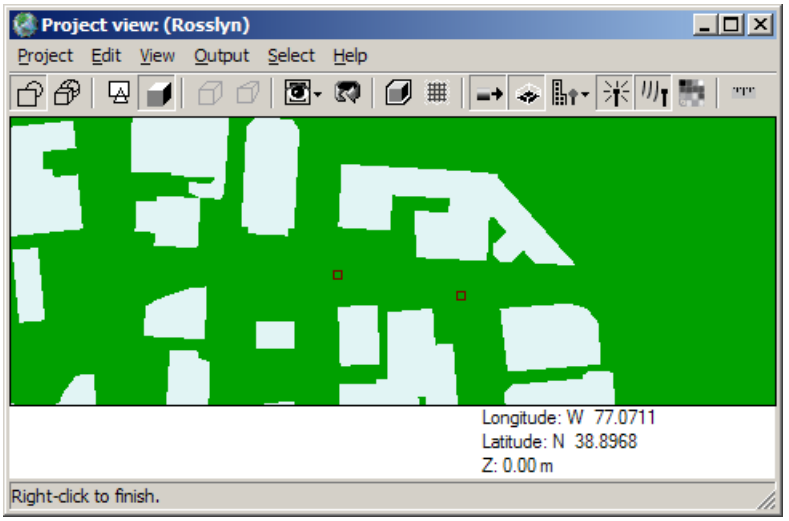


Figure 13.2: Creating a Tx/Rx Point set

13.1.2 Routes

Routes are composed of evenly spaced points along a connected series of line segments. They are typically used to place points along a street in an urban environment. To create a RECEIVER Route, select *File→New→Receiver Set→Route*. Each point that is clicked in the GEOMETRY VIEW defines the beginning or end of a line segment to be included in the route. The route is completed when the user right-clicks.

The heights of the receiver points along the length of the route are defined by the heights of the points which were clicked when the set was created. While the default height of the control points is two meters, these heights can be edited in the LAYOUT tab in the ROUTE SET PROPERTIES window that appears after the set is created. The height of receivers between control points is determined by a linear interpolation of the heights of the closest control points.

Figure 13.3 shows the GEOMETRY VIEW in the middle of the creation of a *Route*. The values displayed between the control points indicate the length of the line segments. In the LAYOUT tab of the ROUTE SET PROPERTIES window the number of points along a particular route can be adjusted by changing the value in the *Spacing* field.

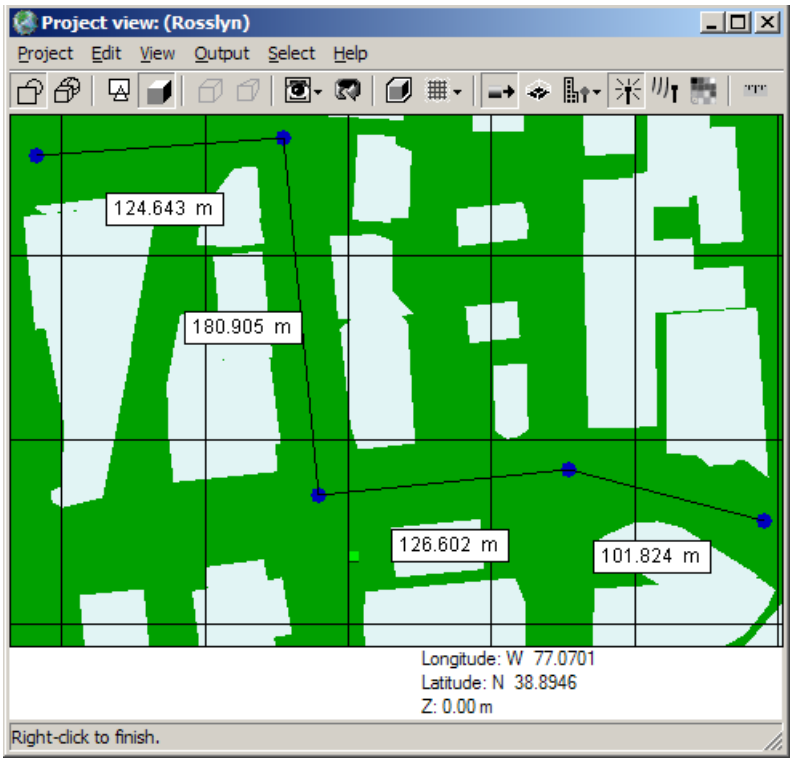


Figure 13.3: The control points in blue define a route of Tx/Rx points

13.1.3 Trajectories

Trajectory sets are a special type of route that can be used to model moving TRANSMITTER or RECEIVER points. For example, a Trajectory can be used to define the path an aircraft or ground-based vehicle takes through the project geometry and the velocity at which it is moving. Trajectories differ from other sets in two important ways:

- Antenna orientation is not uniform throughout the set, but rather differs from point to point based on the location of the next point, radius of curvature, and velocity.
- Received power, path loss and path gain RESULTS for Trajectory sets can be plotted versus time.

Trajectory sets, like other sets, are created by specifying control points that define its location in the project. A spline is fit to the control points and then individual transmitter or receiver points are located along the spline according to the spacing provided by the user in the set's property window. Figure 13.4 shows the editor window with a series of control points (in blue).

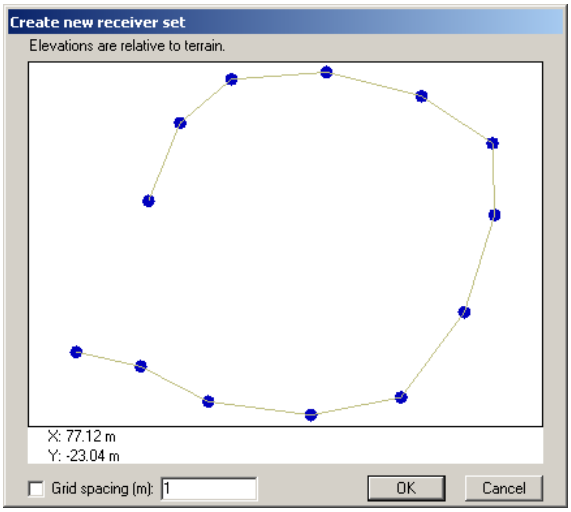


Figure 13.4: Editing window for placing Trajectory control points

After the control points are placed, the user can enter the desired *Spacing* and assign a *Velocity* to the Trajectory set in the LAYOUT tab of the TRAJECTORY SET PROPERTIES window. The *Velocity* for a Trajectory set is assumed to be constant along the entire Trajectory and must be greater than zero. The *Start time* can also be specified for the Trajectory in this tab which acts as an offset for plotting results versus time.

When placing the control points for this type of set, the final arrangement of points is very sensitive to the curvature between the control points. If the roll angles for the created points have any discontinuities, this can be corrected by making small adjustments to the control points that make curves more gradual.

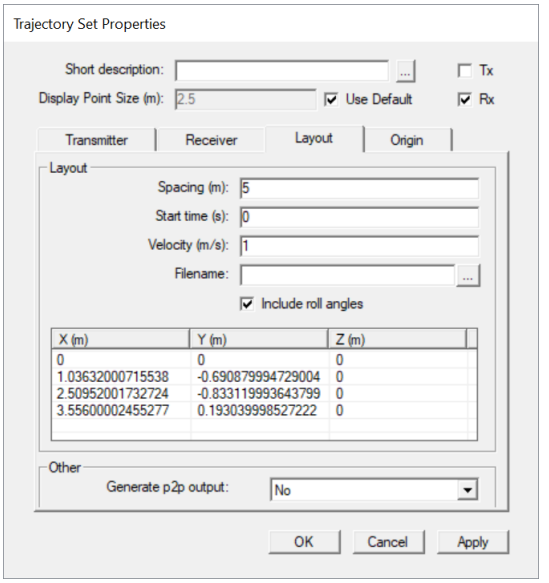


Figure 13.5: Tx/Rx layout properties window for a trajectory set

After definition of the set is complete, the interface calculates the point locations along the spline and displays the set in the GEOMETRY VIEW, as seen in Figure 13.6.

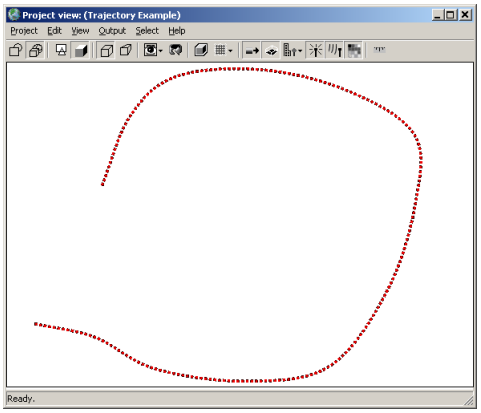


Figure 13.6: Trajectory set defined by the control point shown in Figure 13.4

Figure 13.7 shows the orientation of one of the receiver points along the *Trajectory* route. Antenna patterns are oriented by first aligning their X-axis to the next point in the *Trajectory* set. Roll angle calculations are then applied about this direction.

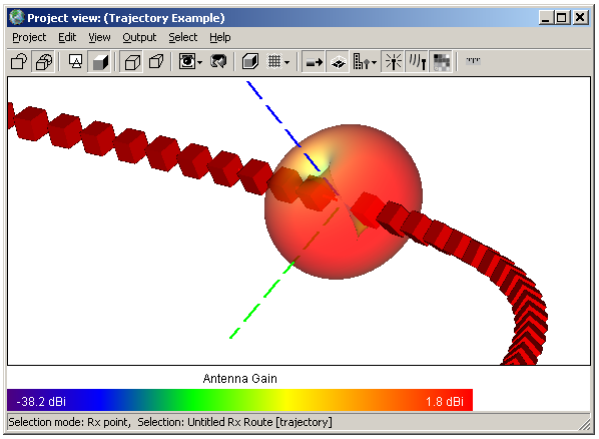


Figure 13.7: Orientation of antenna pattern along a Receiver Trajectory

The roll angles are calculated from a force balance between the aircraft's lift and centrifugal force:

$$R = \frac{V^2}{gtan\phi} \tag{13.1}$$

where

V is the velocity

g is the acceleration due to gravity

ϕ is the roll angle

The calculation of ϕ is only valid for aircraft making banking turns at a constant velocity and constant elevation. This calculation should be disabled for all other cases. More complicated Trajectory orientations can be specified with the user-defined Trajectory set.

Projects may only contain transmitter *Trajectory* sets or receiver *Trajectory* sets, but not both. Transmitter Trajectories in particular can result in long run times and a large number of result files. Users should consider if they can make use of the principle of reciprocity and replace the Transmitter Trajectory with a Receiver Trajectory in order to reduce the number of transmitter points in the project.

13.1.3.1 User-Defined Trajectories

Trajectory sets can also be imported from a `*.traj` file by selecting *File→Import→Transceivers→TRAJ*. The *Files of type* setting in the bottom of the window should be changed to **Tx/Rx trajectory files (*.traj)**. Defining *Trajectories* using this file provides more control over the time, location, and orientation of the ANTENNA at each point in a Trajectory set. A constant time increment is required, but the actual locations of the points can vary, allowing users to create trajectories with a varying velocity. Further, complete control over the antenna orientation at each point is accessible, allowing users to specify more complicated trajectories than what is possible with the built-in Trajectory set. Each user-defined Trajectory

file requires a header describing the data contain in the file. The components of the header are described below.

```
format rx <or> tx
minimum_time [float]
maximum_time [float]
time_increment [float]
time_units seconds
angle_units degrees
cartesian
longitude [double]
latitude [double]
sealevel <or> terrain
spline
has_rotations
[time] [x] [y] [z] [roll] [pitch] [yaw]
```

where:

format - Must be `tx` for a transmitter set and `rx` for a receiver set.

minimum_time - Minimum time listed in the file

maximum_time - Maximum time listed in the file

time_increment - Time spacing used in the file, must be constant

time_units - Must be seconds

angle_units - Must be degrees

cartesian - Cartesian indicates that point locations are metric offsets from the sets origin; otherwise they are assumed to be geographic coordinates.

longitude - Geographic longitude of the set's origin

latitude - Geographic latitude of the set's origin

sealevel or *terrain* - Indicates if height values are given relative to terrain or sea-level.

spline - If this is set the points are interpreted as a set of waypoints from which the rest of the points positions and roll/pitch/yaw are calculated.

spacing - If *spacing* is set this controls the spacing between the calculated points.

[time] - The temporal locations of each point. This field should only be present if *spline* is not specified. The header must be present in the file or the points will not be read.

13.1.4 XY Grid

The *XY Grid* allows a large area to be easily covered with evenly spaced points. To create an *XY Grid*, select *File→New→Receiver Set→XY Grid*. Click and hold the left mouse button down while moving the mouse in the GEOMETRY VIEW to outline the area to be filled with points. Release the left mouse button when the area has been drawn. Figure 13.8 shows an XY Grid drawn in the Geometry View.

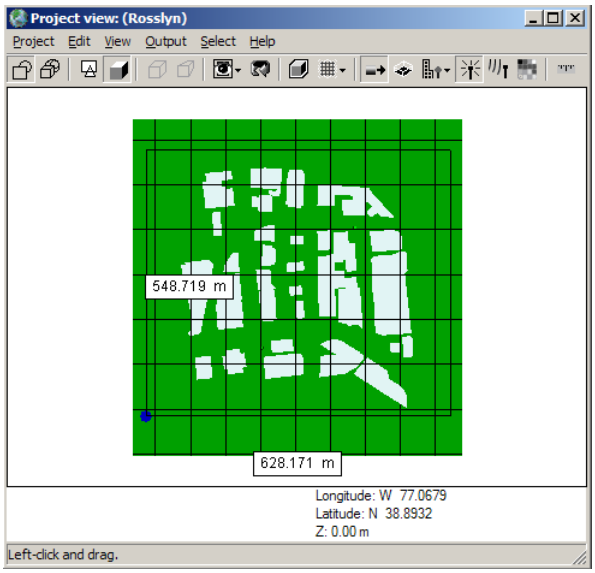


Figure 13.8: Defining the outline of an XY Grid of Tx/Rx points

The LAYOUT tab of the GRID SET PROPERTIES window can be used to specify the *Spacing* between points in the grid and the lengths of the *Length*s of its edges in the x and y directions which are determined by the *Rotation* of the set.

13.1.5 Arc

A horizontal arc of receiving points may be placed using type *Arc*, which requires the user to specify the center position and radius, in meters, of a circle. To create a receiver set of type *Arc*, select *File*→ *New*→*Receiver Set*→*Arc*. In the GEOMETRY VIEW, click and hold the left mouse button to define the center point, then move the mouse to the desired radius and release the left mouse button. The radial distance is shown in meters as the mouse is moved. Figure 13.9 shows an Arc defined in the Geometry View.

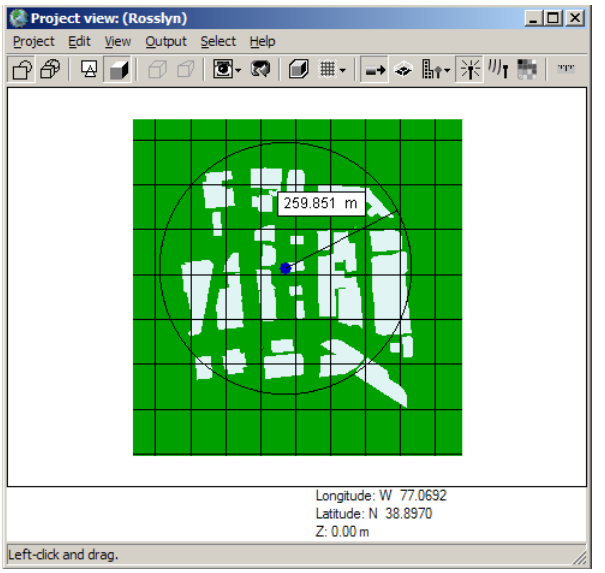


Figure 13.9: Defining the outline of an Arc of Tx/Rx points

By default the *Arc* is a complete circle, however the user may input start and stop angles through which receivers are placed on the circle. In addition, the user may choose to order the points in a clockwise or counter-clockwise direction. The height of the center point determines the height of all RECEIVERS in the Arc. These fields along with the *Radius* can be modified in the LAYOUT tab of the ARC SET PROPERTIES window.

13.1.6 Vertical Arc

Similar to a receiver set of type *Arc*, a *Vertical Arc* allows points to be placed around an arc which extends above and below the XY plane. To create a Vertical Arc, select *File→New→Receiver Set→Vertical arc*. In the GEOMETRY VIEW, click and hold the left mouse button to define the center of the *Vertical Arc* and move the mouse to the desired radial distance. Note that the orientation of the Vertical Arc is shown as a double-sided line segment and is controlled by the mouse placement. Release the left mouse button to complete the creation of the *Vertical Arc*, as seen in Figure 13.10.

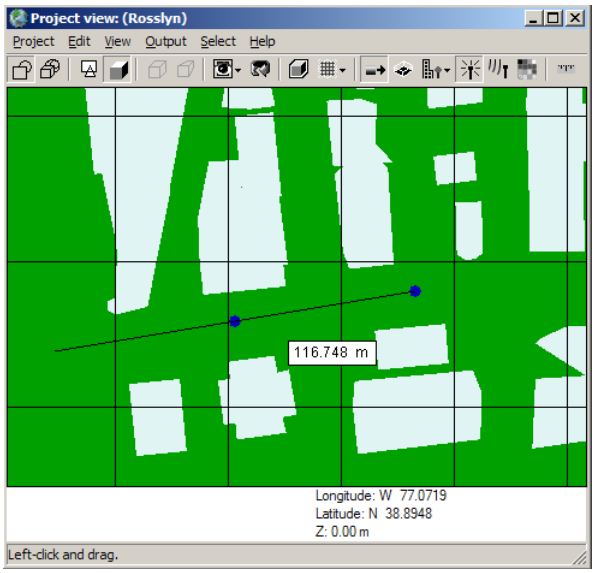


Figure 13.10: Describing the outline of a Vertical Arc of Tx/Rx points

In the LAYOUT tab of the VERTICAL ARC SET PROPERTIES window, the angle of the *Vertical Arc* out of the XZ plane can be modified through the *Rotation* field.

✓ The 0-degree reference point for the VERTICAL ARC lies directly above the arc's center.

13.1.7 Cylinder

A vertical *Cylinder* of receiver points can be created by selecting *File→New→Receiver Set→Cylinder* and following the same procedure for that of an *Arc*. When the left mouse button is released to complete creation of the *Cylinder*, the SPECIFY HEIGHTS Window appears, which allows the user to define the base and top heights of the central axis of the *Cylinder*.

13.1.8 Sphere

You can create a three-dimensional *Sphere* of receiver points by selecting *File→New→Receiver Set→Sphere*, clicking a center point, and dragging the mouse to the desired radius. When the left mouse button is released to complete the *Sphere*, the SPHERE SET PROPERTIES window appears where additional parameters for creating partial *Sphere*s are located.

13.1.9 Polygon

A *Polygon* set is defined by a closed planar region consisting of three or more points. To create a Polygon set, select *File→New→Receiver Set→Polygon*. The GEOMETRY VIEW will change to 2D WIREFRAME mode, and each point selected here defines a vertex on the *Polygon*, as seen in Figure 13.11. The *Polygon* is completed when the user right-clicks.

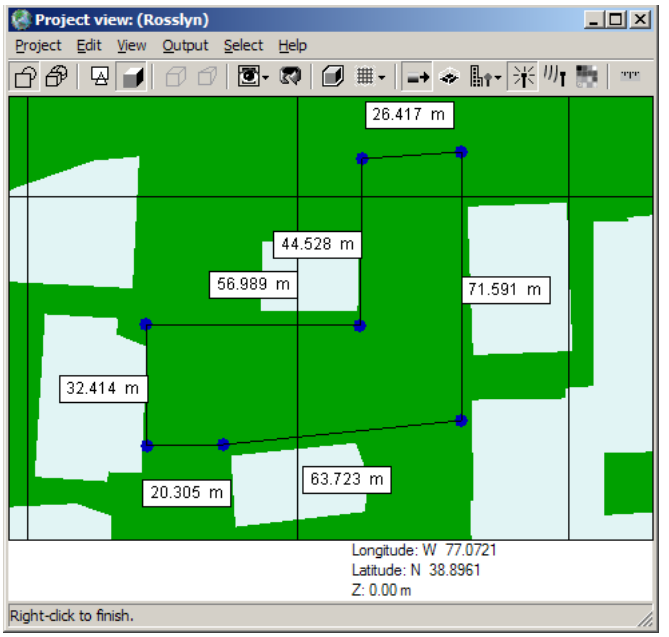


Figure 13.11: Creating a Polygon set

After the editing is complete, the *Polygon* can be rotated out of the XY-plane. However, once a set has been rotated out of the XY-plane it can no longer be modified using the graphical editor. The coordinates of the vertices can be edited manually in the LAYOUT tab of the POLYGON SET PROPERTIES window. You can also add and delete vertices and change the order of the points in this window.

The *Offset* is used to adjust the placement of the points slightly above or below the selected face. Positive offset values move the points the specified distance in the direction of the face's normal vector; negative values move the points opposite to the normal vector.

✓ This parameter is especially useful, and necessary, when the selected face is set double-sided and its material thickness is large. In this case, the offset is used to ensure that the points are visible and not contained within the thick face. A warning message will appear if an adjustment is necessary for this reason.

13.1.10 Surface Illumination

A *Surface Illumination* set provides a way to cover a selection of geometry with points and treat them as a single set as opposed to covering individual faces with unique *Polygon* sets. To create a *Surface Illumination* set assemble a selection of geometry to cover and select *Cover with points* from the context menu. Multiple pieces of geometry can be selected and each piece can be of any grouping level (Geometry, Structure-group, Structure, Sub-structure, or Face), but only faces in a *Floor plan*, *City*, or *Object* will be covered.

▶ See section 2.1.2 for information about how to perform multiple selection in the GEOMETRY VIEW.

If you move the face the surface was based on within the scene after the set is created the set will not move with it. If you want to create points on a face that move with the face, see Section 13.1.12.

If any of the faces were selected as a part of a grouping level, then this action will bring up one of the windows shown in figure 13.12. This is used to indicate which faces should be included in the illumination based on their categorization. Any faces in a selection that were selected at the Face level are automatically covered.

The window on the left in figure 13.12 is for the general case when *any* of the faces are in a *City* or *Object* and puts each face into one of the following categories that can be marked for inclusion when creating the set.

- *Rooftops* - Any faces having a normal that is within 45 degrees of up.
- *Exterior Walls* - Any face that does not qualify for *Rooftops*.
- *Interior Floor Coverages* - This will add virtual polygons that match the footprint of each selected sub-structure. Multiple coverages will be inserted starting at the floor and then one at each *Floor Interval* until the roof is reached.

When *all* of the selected faces are contained in a *Floor plan* then the window shown on the right of figure 13.12 appears which provides the following face categories.

- *Outside Surfaces*
  - *Roof* - The top side of faces that have a normal that is within 45 degrees of up and have an edge that lie over the floor plan's footprint.
  - *Walls* - The exterior side of vertical faces that have an edge that lies over the floor plan's footprint.
- *Inside Surfaces*
  - *Perimeter Walls* - The inner side of faces that qualify as *Outside Surfaces*→*Walls*.
  - *Interior Walls* - Vertical faces that do not qualify as *Outside Surfaces*→*Walls*.
  - *Ceilings* - The underside of faces that qualify as being a part of *Outside Surfaces*→*Roof*.
  - *Surfaces* - The top side of any faces that have not qualified for any of the other categories.
  - *Coverage Map* - This will add a virtual polygon that matches the footprint of each selected sub-structure with the indicated *Coverage Elevation* above its floor.

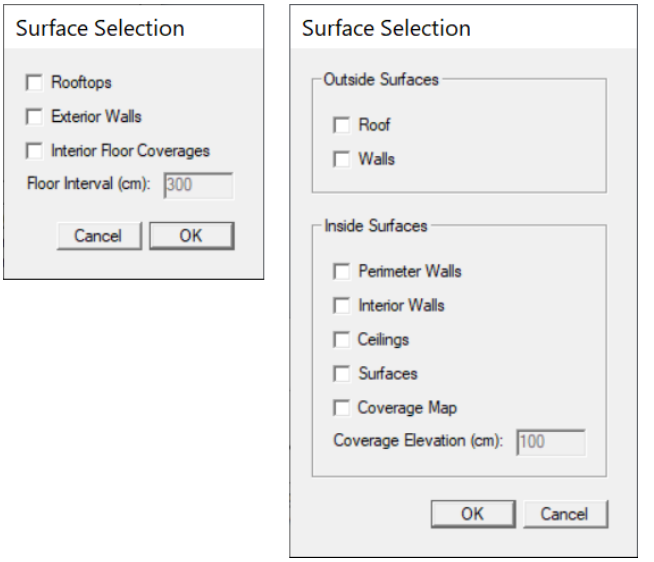


Figure 13.12: Options for covering faces in selected groups.

The LAYOUT tab of the SURFACE ILLUMINATION SET PROPERTIES window contains the *Offset* parameter which is used to adjust the placement of the points with respect to the faces surface. A positive *Offset* will move *Roof(top)* surfaces upward, *Ceilings* downward, *(Exterior) Walls* away from the interior of the *Floor plan*, and *Interior Walls* away from the face on both sides. The *Surfaces* and *Coverage* categories will be moved upward. A negative *Offset* will move the surface away from the face in the opposite direction.

Figures 13.13 and 13.14 show a *Cover with points* action being applied to faces selected as a part of a sub-structure (upper-left), and structure (upper-right), along with two stand-alone faces (center and lower-left) using the default 1 cm offset.

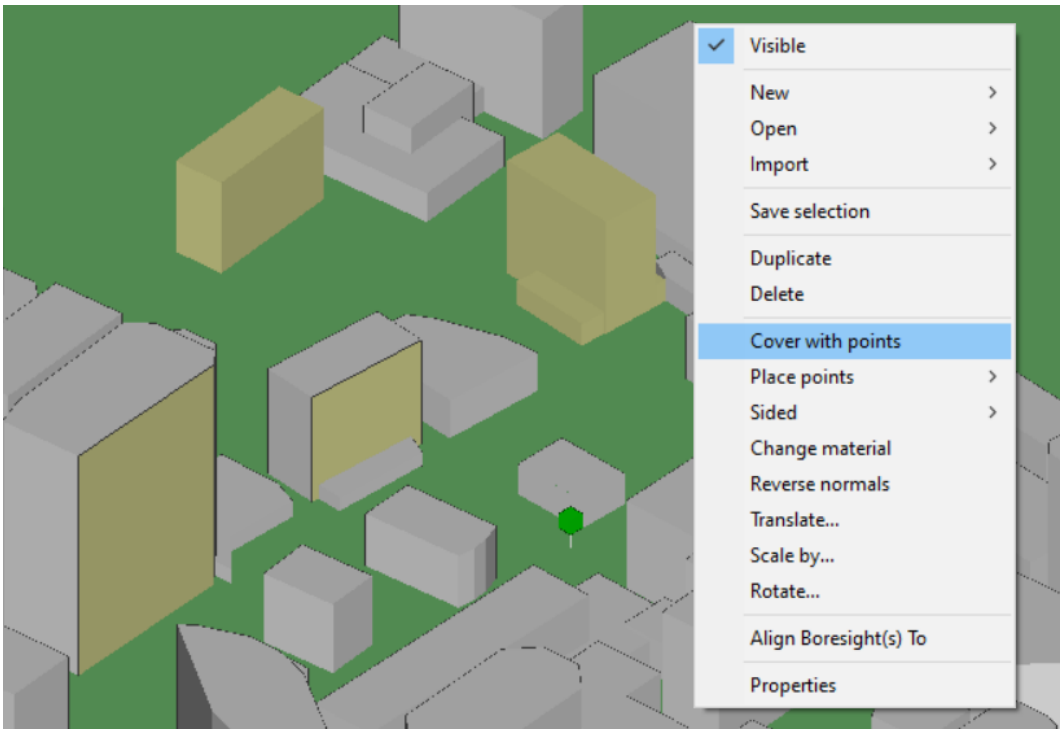


Figure 13.13: Selecting faces for covering.

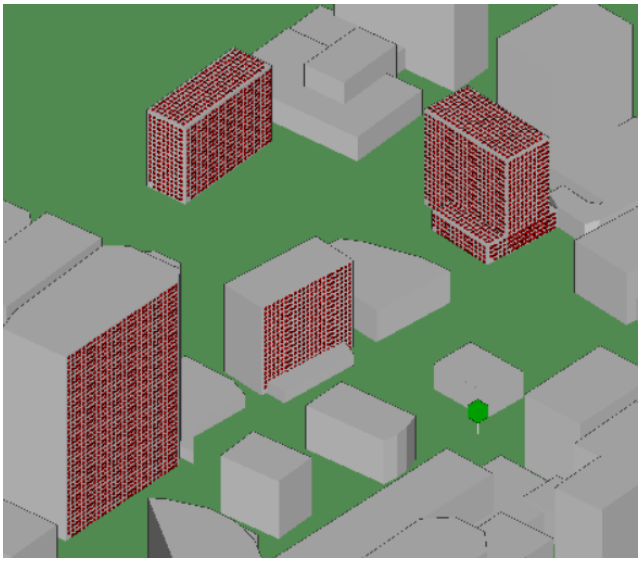


Figure 13.14: Surface Illumination applied to the selection above.

13.1.11 Vertical Surface

A *Vertical Surface* is a *Route* that has been extruded in the Z-direction to form a curtain of points in space. To create a *Vertical Surface* set, click *File*→*New*→*Receiver Set*→*Vertical Surface*. Draw the 2D contour of the *Vertical Surface* as instructed for a *Route*. When completed, the user is prompted to enter base and top heights to completely define the *Vertical Surface*.

The LAYOUT tab of the VERTICAL SURFACE SET PROPERTIES window contains two parameters that control the *Vertical Surface*. The space between points of adjacent rows is determined by *Vertical spacing* and the height that the *Vertical Surface* extends above its base is determined by *Height*.

13.1.12 Points On Face

TRANSMITTER and RECEIVER points may also be placed individually on a face. First select a face, and right-click. In the context menu that appears select *Place points*→*Receivers*. In the editor that appears, click within the outlined face to define receiver locations. Right-click when all desired points have been placed. By right-clicking in the editor, the user may add points or change the height above or below the face of all the points. If the mouse cursor is on top of an existing point when right-clicked, additional options will appear in the context menu which allow the position of that point to be changed by selecting *Edit position*. The point can also be deleted.

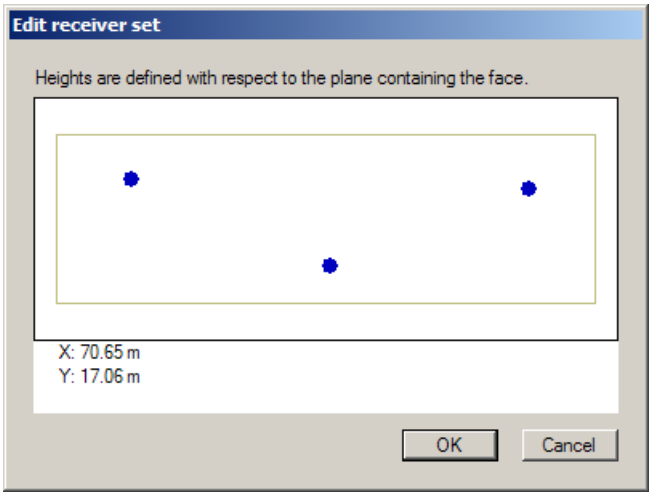


Figure 13.15: Placing Receiver points on a face

When a “Points On Face” set is created, a relation is made between the Tx/Rx set and the face it is mounted on. The sets can not exist without their referenced face. This relation allows the Tx/Rx points to move with the face. In addition, because there is a relationship between the face and the set, if the face is deleted then the points on face sets will also be deleted. Also, when a GEOMETRY that possesses “Points On Face” sets is made inactive, the sets will also be made inactive. When the geometry is set back to active, a prompt will query whether to make the sets active once again.

13.1.13 Plane Waves

For TRANSMITTERS only, one or more incident plane waves can be modeled in Wireless InSite by defining *Plane Wave* transmitter sets. To add a Plane Wave to the current project, select *File→New→ Transmitter Set→Plane wave*.

The PLANE WAVE PROPERTIES Window, seen in Figure 13.16, permits each plane wave to be specified by the following parameters:

- *Waveform* - The spectral characteristics of the Plane Wave
- *Polarization* - Theta, phi, left-hand circular, or right-hand circular
- *Power density/ Received power / RMS E-field* - The amplitude of the Plane Wave may be defined by specifying one of the above quantities. If received power is chosen, a power density will be determined by assuming the presence of an *Isotropic* Antenna whose effective aperture is determined by $\frac{\lambda^2}{4\pi}$, where λ is the wavelength at the carrier frequency of the selected waveform.
- *Incident direction* - Defines the direction from which the Plane Wave originated.

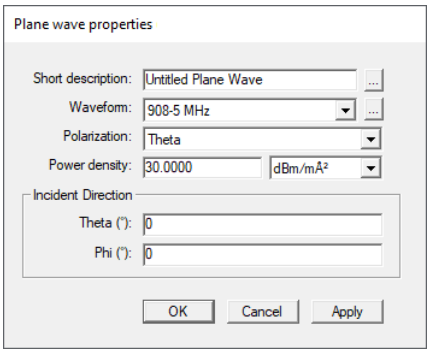


Figure 13.16: Properties of a Plane Wave

The size and position of the *Plane Wave* in the GEOMETRY VIEW is determined automatically based on its specified incident direction and the size and position of any active GEOMETRY and RECEIVER sets, so that it illuminates the entire project. The Plane Wave is rendered transparent so that it does not obscure elements of the project, and the green normal arrow indicates the Plane Wave's direction of propagation.

Most other types of results can be generated for *Plane Wave* transmitters. Only path loss and path gain will not be generated by Plane Waves since the plane wave source is assumed to be infinitely far away. This assumption also complicates the calculation of time of arrival; thus, time of arrival is defined to be the propagation time from the position of the rendered Plane Wave to the RECEIVER. The distance of the Plane Wave away from the project is chosen arbitrarily, and therefore the time of arrival is also somewhat arbitrary. Interaction with results generated for a Plane Wave is otherwise identical to results generated for sets of TRANSMITTER points, and consists of displaying in the GEOMETRY VIEW, creating plots, and using the MOVIE PLAYER.

The use of *Plane Waves* is limited to STUDY AREAS, which use Shooting and Bouncing Ray (SBR) as the ray tracing method. The use of the Eigenray ray tracing method is only permitted if no active Plane

Waves are present in the project.

- See Section 16.1 for more on the SBR and eigenray methods.

13.1.14 User-Defined Files

TRANSMITTER , RECEIVER and TRANSCEIVER sets can be specified in a text file and added to a project by selecting *File→Import→Transceivers→PTS*. Select the text file in the window that appears which should have the extension `*.pts`. The ADVANCED TX/RX PROPERTIES window appears to complete the creation of the set. Here the user may specify the local origin of the set.

An example of a `*.pts` file is shown below:

```
format tx
set_type route
system cartesian
description MyCustomRoute
elevations sealevel
1 100 100 0
2 110 110 0
3 120 120 0
```

The first three lines are required in order to provide information about the type of set. The description and the elevations lines are optional. By default, elevations are relative to the ground.

- `format` - Must be "tx" for a transmitter set, "rx" for a receiver set, or "txrx" for a transceiver set.
- `set_type` - Indicates what type of set to create. The currently supported options are `route`, `vertical surface`, `polygon` and `points`.
- `system` - This indicates the coordinate system that the points are listed in. "Cartesian" indicates they are metric offsets from the sets origin, and "geographic" indicates they are absolute earth coordinates that must be converted into the final Cartesian offsets.
- `description` - Provides a short description for the set.

Each line thereafter specifies, in order:

- `Point_number X_coordinate Y_coordinate Z_coordinate`

The points may be placed in any arrangement, and are not required to be equally spaced. Point number should be an integer.

13.2 Properties

The SET PROPERTIES window as seen in Figure 13.17 allows the user to define and modify the characteristics of a TRANSMITTER , RECEIVER or TRANSCEIVER set. The window appears after the creation of a new Tx/Rx set, or it can be accessed by right-clicking on the set in the MAIN WINDOW and selecting *Properties*.

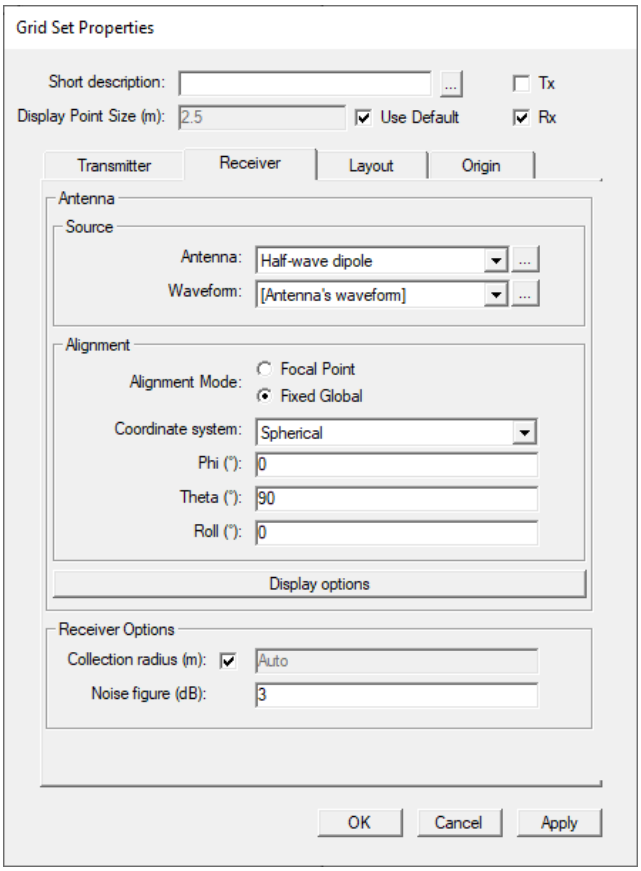


Figure 13.17: The properties window for a receiver set

The contents of the tabs are described in more detail in the following sections, while the following properties are displayed above the tabs and are common to all set types.

- *Short description* - Defines the name which will refer to this Tx/Rx set throughout the GUI.
- *Display Point Size* - Controls the size of the boxes that are rendered at each point in the set.
- *Use Default* - When activated the *Display Point Size* will be set to a size that is appropriate for the type of set.
- *Tx* - Activates this set for use as a TRANSMITTER.
- *Rx* - Activates this set for use as a RECEIVER.

13.2.1 Transmitter/Receiver Properties

Clicking on the TRANSMITTER or RECEIVER tab in the SET PROPERTIES window will access properties that are common to both with a section at the bottom that is specfic to either. The RECEIVER tab is shown in Figure 13.17.

The following parameters are common to both the TRANSMITTER and RECEIVER

- *Antenna*
  - *Source*
    - *Antenna* - Specifies the ANTENNA positioned at each point in the set.
    - *Waveform* - Specifies the WAVEFORM fed into the *Antenna*. The default choice is *[Antenna's waveform]*, which is the waveform that was assigned to the antenna in its ANTENNA PROPERTIES Window. If the user wishes to override the waveform that was assigned to the antenna, any waveform in the project may be selected.
  - *Alignment* - See Section 13.3 for more information about this feature.
  - *Display options* - Access to the antenna viewing options described in 13.4.
- *Transmitter Options* - These parameters are available in the TRANSMITTER tab.
  - *Input power* - Defines the total input power at each TRANSMITTER point. The default value is 0.0 dBm.
    - See Section 21.2 for more on input power and how it relates to radiated power.
  - *Input Power Monte Carlo* - Access to Monte Carlo settings for a transmitter's input power.
    - See Section 19.3 for more on how Monte Carlo is used with respect to input power.
  - *Total Array Power* - This checkbox can be activated when the TRANSMITTER is using a MIMO ANTENNA. If the user checks this, the *Input Power* will be split evenly over the elements of the MIMO array, giving each element power/N of the power, where N is the number of elements in the MIMO antenna. If left left unchecked, each of the N elements will receive the total input power entered.
- *Receiver Options* - These parameters are available in the RECEIVER tab.
  - *Use bounding box*, *Bounding box length* - Defines properties for the simulation related to intersections between propagation paths and Rx sets.
    - These options are more fully described in Section 13.8.
  - *Collection surface radius* - Sets the size of the collection surface constructed around receiver locations. Manually setting this value needs to be done in conjunction with the ray spacing to ensure accurate results. See section 16.1 for more details.
  - *Noise figure* - Represents the contribution of the receiver itself to thermal noise at its output. The value indicates the ratio of the thermal noise power at the output to that at the input, expressed in dB.
    - See Chapter 14 for more on how this quantity is used in communication system analysis.

13.2.2 Layout Properties

The LAYOUT tab displays properties describing the physical characteristics of the set, as seen in Figure 13.18.

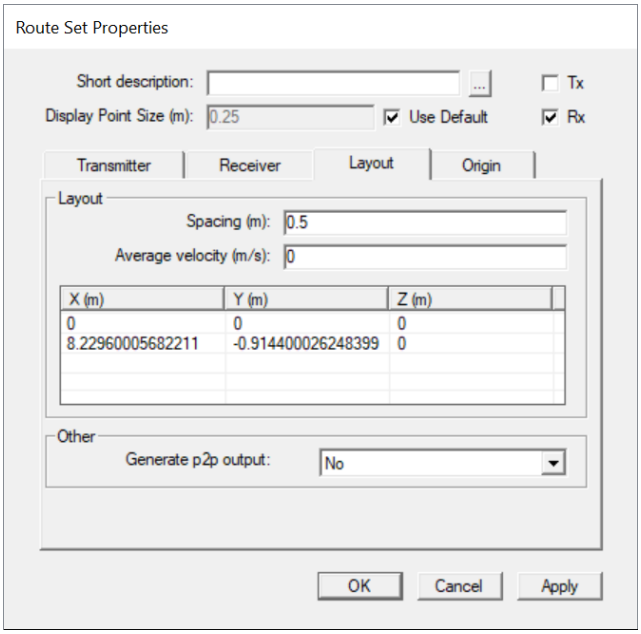


Figure 13.18: The Tx/Rx Layout properties tab for a route set.

The following parameters appear based on the type of set and the information it needs to arrange its points:

- *Layout*
  - *Area lengths (XY Grid only)* - Defines dimensions of the grid in the X- and Y-direction with respect to the global coordinate system.
  - *Average velocity (Route, Arc and Trajectory sets only)* - Describes the speed at which an object would be moving along the path of the set.
  - *Direction (Arc, Vertical Arc and Cylinder sets only)*: including *Arc radius*, *Arc start angle*, and *Arc stop angle* - Control which direction points move around the perimeter of the set, and the segment of a circle with the given radius that the points lie within. For an *Arc*, the angle starts at the positive X-axis moves counter-clockwise in the XY plane from that point. For a *Vertical Arc*, the angle starts at the positive Z-axis and moves counter-clockwise in the YZ plane from that point.
  - *Draw mast (Point Sets Only)* - If set to “Yes”, then a vertical line is drawn from each point in the set to the XY plane at Z=0.
  - *Height (Vertical Surface and Cylinder sets only)* - Indicates the height from the bottom row of points in the set to the top row.
  - *Include roll angles (Trajectory sets only)* - Applies a roll angle to the points in the *Trajectory*.
    - ▶ See Section 13.1.3 for more details on the roll angle.

  - *Offset (Polygon set only)* - Indicates how far the actual Tx/Rx points are placed above or below the *Polygon* defined by the control points.
  - *Phi start/stop angle* and *Theta start/stop angle (Sphere sets only)* - These options control the shape of the *Sphere* or partial sphere of points. The phi angles represent orientation in the XY plane. The theta angles represent orientation in the YZ plane. For a full *Sphere*, phi should range from 0° to 360°, and theta should range from 0° to 180°.
  - *Rotation (Vertical Arc and XY Grid sets only)* - Rotates the Tx/Rx set location in the project about the Z-axis. It does not affect the orientation of the antenna patterns in that set.
  - *Spacing* - Defines the spacing in meters between adjacent points along a *Route* and other Tx/Rx sets which require it. This includes in the plane of the *XY Grid* or *Polygon*, and around the *Arc* or *Cylinder*. The spacing for circular sets can also be specified as a constant angle, which results in the distance between points varying with the sets' radius.
  - *Start time (Trajectory sets only)* - Defines the time offset used when plotting results versus time.
  - *Vertical spacing (Vertical Surface only)* - Since a *Vertical Surface* is considered to be a stack of *Route* sets, this option controls how far apart each route is from the other in the vertical direction.

- *Other*
  - *Enable APG for X3D* - Activates the use of APG for X3D.
  - *Adjacency Distance* - The spacing of the sparse grid when using APG.
  - *Generate p2p results* - Begins generating p2p results between the points of a Tx and Rx set that have this flag activated.

13.2.3 Origin Properties

This ORIGIN tab is used to modify the position of the set in the project's scene.

- *Coordinate System* - Defines the display format of the Tx/Rx locations.
- *Elevations relative to* - Defines the Z-values of the control points for the Tx/Rx set. When set to "Terrain", the Z-values are measured with respect to ground; when set to "Sea level", the Z-values are absolute.
- *Show origin when active* - Displays the Tx/Rx set's local origin in the GEOMETRY VIEW.

13.3 Antenna Alignment

When a Tx/Rx set is first created its alignment corresponds to the project's global triad. The red vector points along the X-axis, green along the Y-axis and blue along the Z-axis. In most cases, the X-axis vector corresponds to the direction that the electrical boresight of the antenna is aligned to and the Z-axis acts as a polarization indicator. If an antenna is vertically polarized the polarization is parallel to the Z-axis and horizontal is perpendicular to it.

✓ An exception to this general rule is when the antenna is not vertically polarized by design. For example, if the antenna is an imported pattern, then the boresight and polarization lie in the default directions of a new antenna, as indicated above.

▶ To see if an antenna is not vertically polarized by default, see Chapter 12 to make this determination for the type of antenna you will be using.

There are two ways to specify the alignment of an antenna:

- *Focal Point* - Each point has a unique alignment so that it will point to a common location in the global Cartesian space.
- *Fixed Global* - All points in the set have the same alignment in the global reference frame. This results in the alignment at each point being parallel.

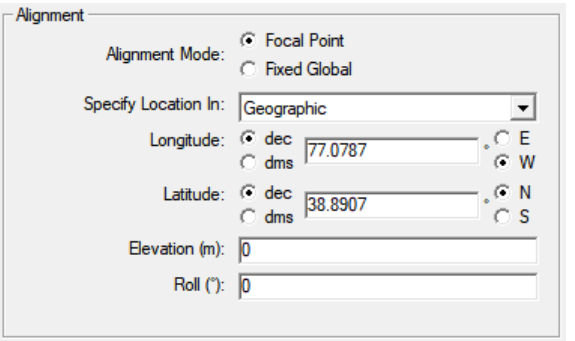


Figure 13.19: Setting the antenna alignment to a focal point.

A *Focal Point* alignment can be specified as either a global Cartesian location or using geographic coordinates. Switching the *Specify Location In* field between these two will convert the current value to the new selection.

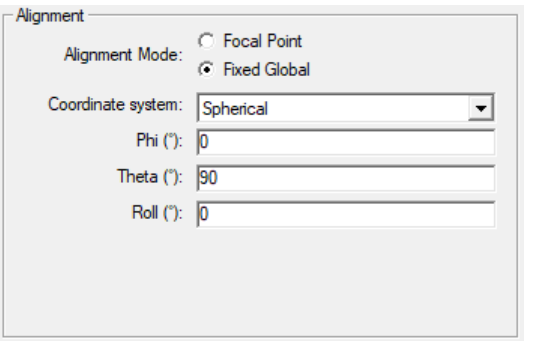


Figure 13.20: Setting the antenna alignment to a fixed alignment using spherical values.

A *Fixed Global* alignment can be specified using a nautical, spherical, or sequential coordinate system. These systems work as follows:

- *Nautical* - The (0,0,0) bearing is North. *Bearing* goes from North to East and positive *Pitch* lifts the directional X-axis into the sky.

- *Spherical* - This is the mathematical spherical coordinate system. As a courtesy *Theta* is initialized to 90 degrees so that it aligns with the project's global triad which is also the starting point for the sequential coordinate system. This can be seen in 12.1.
- *Sequential* - This alignment applies rotations in an X-Y-Z ordered sequence. The starting point for the alignment is the project triad. Roll is first applied about the X-axis, then pitch about the Y-axis, and finally a rotation is applied about the Z-axis.

In all cases the roll is applied around the red X-axis using the right hand rule the amount of which is indicated by the direction of the blue Z-axis. The initial roll of 0 is the vector that is the cross product of the X-axis with the vector that is orthogonal to it in the XY-plane such that the final vector has a positive Z-value. Due to how the *Sequential* alignment works the roll is performed while the Z-axis is still aligned with the project triad.

13.4 Viewing Transmitter/Receiver Control Vectors and Antenna Patterns

Every TRANSMITTER and RECEIVER set has a set of associated vectors, called "Control Vectors", which can be rendered in the GEOMETRY VIEW. These vectors indicate how the antenna used by the set is aligned with respect to the global triad within the project.

The ANTENNA VIEWING PROPERTIES Window, as seen in Figure 13.21, is accessible by clicking on the DISPLAY OPTIONS button in the properties window for either the Transmitter or Receiver portion of a Transceiver set. This window allows the user to customize the appearance of the control vectors for each set individually.

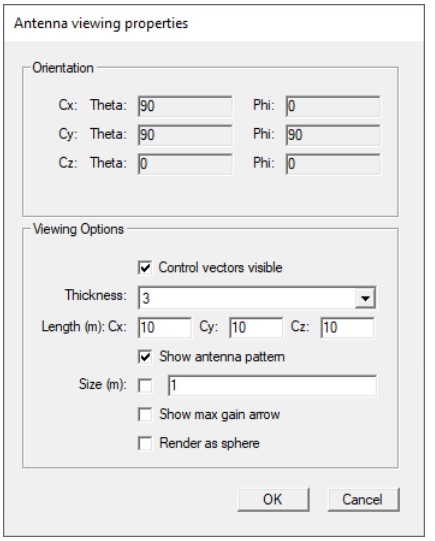


Figure 13.21: Properties of a Tx/Rx sets' antenna boresight

The following are descriptions of the parameters found in the viewing properties window:

- *Orientation* - Displays the current direction of all three indicators based on the rotation values that were given in the TRANSMITTER/RECEIVER PROPERTIES Window. These values are read-only and can only be changed by changing the transmitter or receiver's alignment.
- *Viewing Options* - Modifies how the boresight indicators are drawn on the screen.
  - *Control vectors visible* - Toggles the display of the boresight in the GEOMETRY VIEW.
  - *Thickness* - Alters the thickness of the lines drawn in the GEOMETRY VIEW.
  - *Length* - Lengthens the boresight indicators to help the user determine if they are pointed at specific points in the project. Each indicator can be adjusted independently.
  - *Pattern visible* - Renders the antenna pattern of the Tx/Rx sets' antenna at the currently selected point in the set. If no point is selected, then the first/only point will be used for this purpose. The Tx/Rx set will generate a pattern for the antenna at its waveform frequency.

    If the waveform that the Tx/Rx sets' antenna is using is changed or replaced, then the pattern will automatically be regenerated.
  - *Size* - Defines the size of the antenna pattern when it is rendered. When the pattern is first created for display, it is set to have a maximum size of 1 meter. This value is the largest distance from the Tx/Rx point to the point on the pattern with the maximum gain.

    When the antenna pattern is first rendered, there will be a slight delay as the pattern is calculated and created for display.

Figure 13.22 shows the use of the boresight with the antenna pattern activated. A TRANSMITTER point (left side) has been aligned with a RECEIVER (right side) on another rooftop. The transmitter uses a *Pyramidal Horn* and the receiver is using a *Parabolic Reflector*. The pattern is colored so that red represents the higher values and blue the lower. The electrical boresight of the transmitter point is visible and the polarization indicator shows the direction of vertical polarization for the antenna based on its alignment.

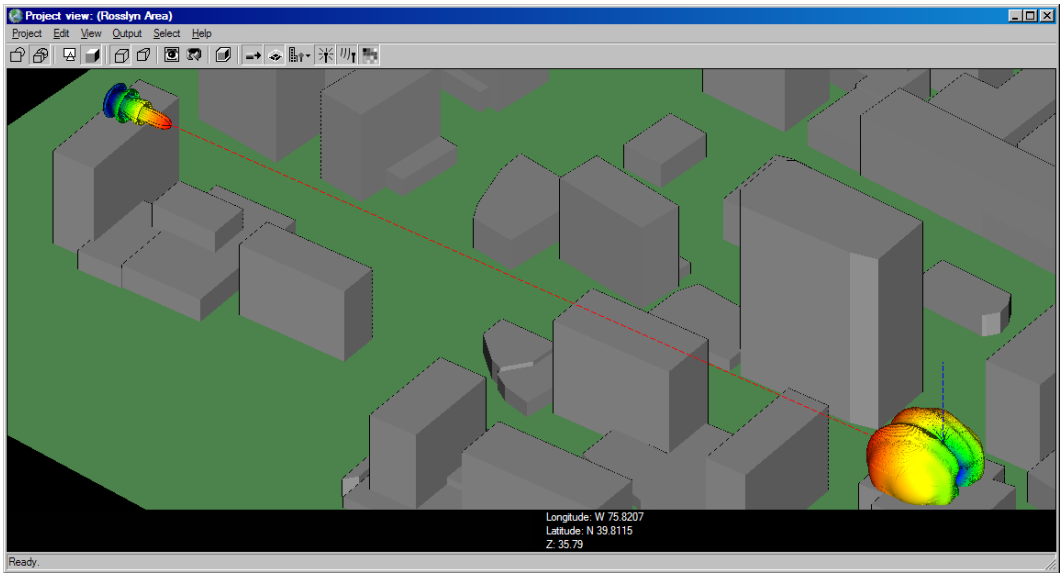


Figure 13.22: The main beam of the transmitting antenna has been aimed directly at the receiving antenna on another rooftop

When a set is first created, the antenna pattern appears at the first point in the set. For Tx/Rx sets that contain multiple points, the antenna pattern for the set will be displayed at the last selected point in the set. In Figure 13.23, this is done in the Rosslyn streets project for a point in the receiver route along Nash Street.

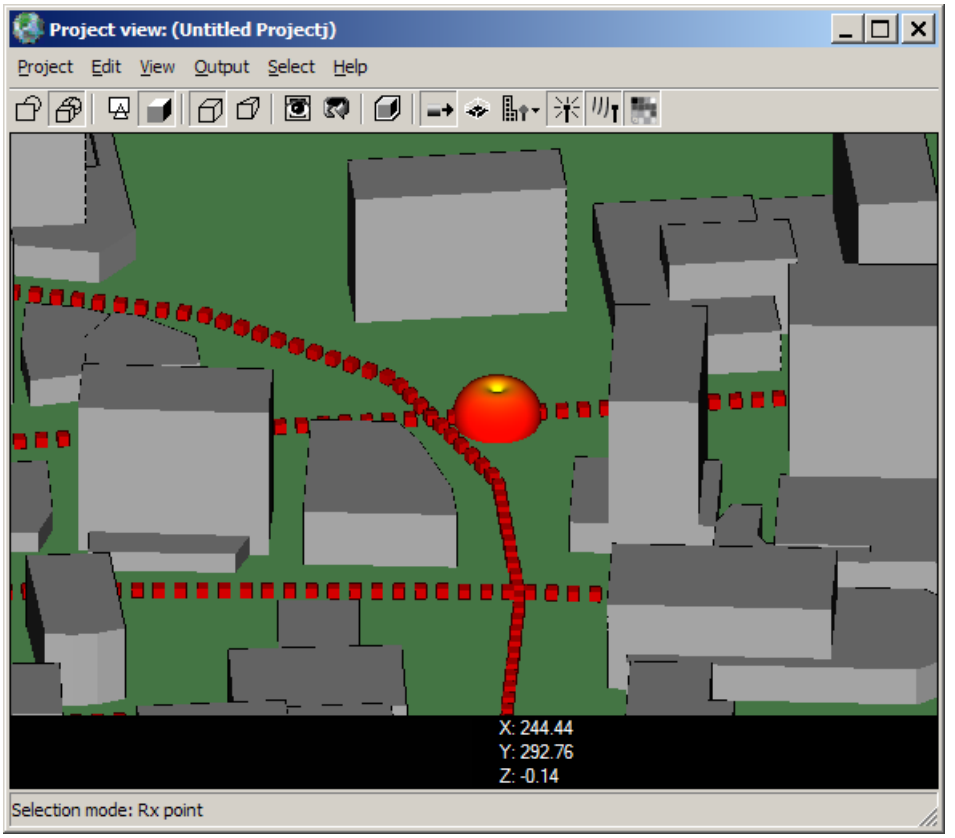


Figure 13.23: Receiver point selected with antenna pattern displayed

13.5 Aligning Multiple Transmitting and Receiving Antennas

Multiple Transmitter and Receiver points and sets can have a common alignment assigned by selecting them in the GEOMETRY VIEW and then selecting the ALIGN BORESIGHT(S) context menu option. Geometry can also be selected in multiple pieces at any level so that the center of the selected region can be used as a pre-calculated point for specifying a *Focal Point* alignment.

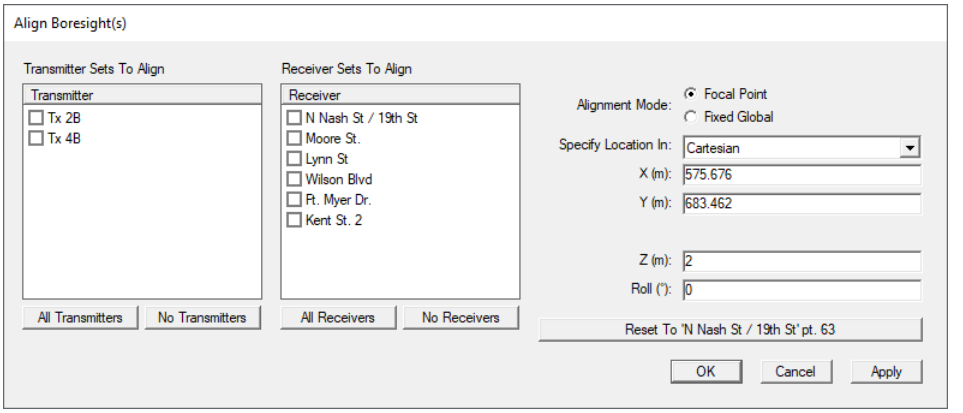


Figure 13.24: Aiming the boresight of the transmitting or receiving antenna to a selected transmitter point

The following are descriptions of the parameters found in the ALIGN BORESIGHT(S) Window:

- *Transmitter Sets To Align* - Select the transmitter side of transceiver sets in the project to apply the alignment to. Below this table are buttons to select *All Transmitters* or *No Transmitters*.
- *Receiver Sets To Align* - Select the receiver side of transceiver sets in the project to apply the alignment to. Below this table are buttons to select *All Receivers* or *No Receivers*.

  Trajectory and surface mounted (points on face) transceiver sets are not allowed to be aligned in this way so they will not appear in either table. Transmitters or receivers that are using a surface mounted antenna are also excluded from being configurable through this window.
- Alignment Fields - See 13.3 for details on how these fields specify the alignment.
- *Reset To ...* - If this window is brought up through the context menu of a selected *Tx/Rx Point* or a composite selection of geometry geometries at various levels, then that location will be loaded by default. This button will reset the alignment to that position. This facilitates performing multiple focal alignment applications to various offsets from this location. Performing this reset will change the *Alignment Mode* to *Focal Point* and change the values to the default location.
- *OK* - Apply changes and exit the window.
- *Cancel* - Exit the window without applying changes.
- *Apply* - Applies the current alignment settings to any selected transmitter or receiver sets.

The alignment between a transmitter and receiver pair can be quickly modified using the ALIGN BORESIGHT(S) option. To do so, choose *Select→Transmitter ->Receiver pair* in the GEOMETRY VIEW, click on the transmitter and receiver points. A this point the right-click context menu will have options to *Align boresights* as either a *Fixed* or *Focal* alignment. In both cases the opposite point is used to determine the direction when calculating the settings and the roll about the X-axis will be preserved.

13.6 Editing Transmitter and Receiver Sets

Once a TRANSMITTER or RECEIVER set has been created, several new options are available in the context menu. To access this menu, select it in either the PROJECT TREE or MAIN WINDOW and right-click. The options are as follows:

- *Edit* - Allows for graphical editing of the sets control points as described below.
- *Translate/Rotate/Scale* - All of the control points of a set can be translated, rotated and scaled in the same way that GEOMETRY are. All of these operations work with respect to the sets' local origin. Operations that are available for each set are as show in Table 13.1.

Table 13.1: Control Point Set Operations

Set Type	Rotate	Scale	Translate
Arc			✓
Cylinder			✓
XY Grid	✓ (soft)		✓
Points	✓	✓	✓
Polygon	✓	✓	✓
Route	✓	✓	✓
Sphere			✓
Trajectory			✓
Vertical Arc	✓ (soft)		✓
Vertical Surface	✓		✓

  Sets marked with the word "soft" in this table refer to the *Rotation* field in the properties window which rotates the set in the XY-plane. This is used to create a grid that is not axis-aligned to the project's X- and Y-axes, and to specify the angle of a *Vertical Arc* as it appears from above in the XY-plane. User-defined sets can only be modified by editing the `*.pts` file.

- *Duplicate* - Creates an identical set and appends a number to the Short description to differentiate the two sets.
- *Duplicate as* - Creates an identical set of the opposite type; i.e. a transmitter set is duplicated as a receiver set and vice versa.
- *Delete* - Removes the selected set from the project.
- *Save* - Commits all changes to every transmitter or receiver set, depending on which type was selected.
- *Save As* - Allows the selected set to be written to a `*.tx` or `*.rx` file that is not related to the project.
- *Align Boresight(s)* - Brings up the ALIGN BORESIGHT(S) window. See Section 13.5 for more information about this feature.
- *Export Rx Focal Alignments* - Writes out the rotational values for each receiver point in the set if it is using a focal point alignment for its receiver. The file is written to the project folder and is in the form

of *set description*_rxAlignments.csv.

- *Export Tx Focal Alignments* - Writes out the rotational values for each transmitter point in the set if it is using a focal point alignment for its transmitter. The file is written to the project folder and is in the form of *set description*_txAlignments.csv.
- *Properties* - Opens the properties window for the selected set. This window may also be accessed by double-clicking on the set in the MAIN WINDOW.

If a point is selected for the set the following options will also appear in the context menu.

- *Align Boresight(s) To* - Brings up the ALIGN BORESIGHT(S) window with the selected point loaded as the focal alignment point. See Section 13.5 for more information about this feature.
- *Create radial from point* - Create a *Vertical Surface* Receiver set radiating out from the selected point. This is very useful when setting up VERTICAL PLANE simulations, as the entire set is guaranteed to fall on the same radial.

In edit mode, the control points of a set are displayed in the editor as blue dots, which may be moved or altered by right-clicking on them. The menu that is displayed varies for different types of sets. Common functions include:

- Continuing a *Route* or *Vertical surface*
- Adding points to a set
- Changing the height or location of one or all of the control points
- Deleting the point that was clicked

The shape of a *Route* or *Vertical surface* may be changed by adding control points; simply click on the line defining the *Route* to add a point.

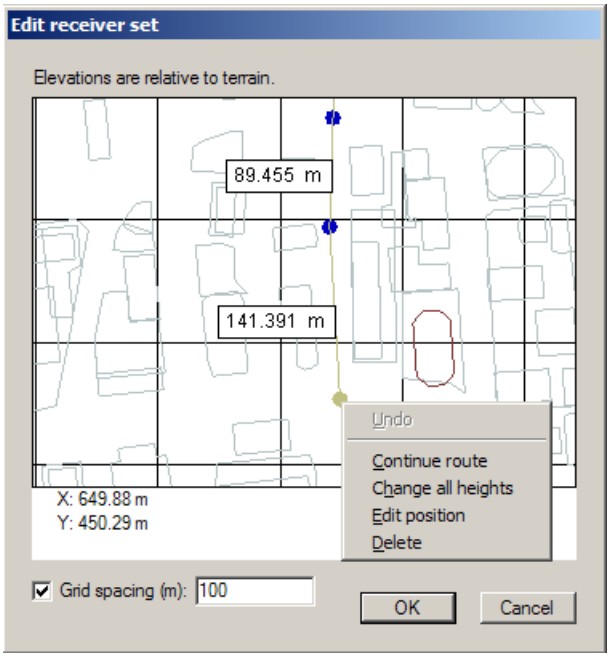


Figure 13.25: The options for editing an existing Route are shown.

Arcs, *Vertical Arcs* and *Cylinders* can be edited by clicking on and moving the point which defines either the center or outer radius of the set. To move the circle, click and move the center; to increase or decrease the radius, click and move the radial point. The displayed radial distance will update when changed, as seen in Figure 13.26.

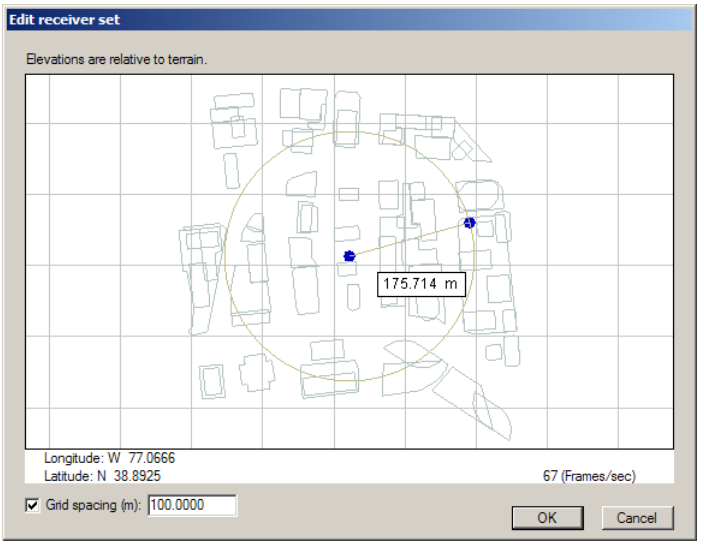


Figure 13.26: Editing the radial point of a Sphere set

The EDIT CONTROL POINTS button in the properties window allows the user to view and edit the control points, which are used to define the TRANSMITTER or RECEIVER set, without entering the graphical editor. The number of control points and their significance is dependent upon the set type. Brief descriptions of how control points are used for each type of set is described below:

- *Route* - Defined by at least two control points, each control point along a Route represents the beginning and/or end of a line segment. The first point is always (0,0), which corresponds to the local origin of the set.
- *XY Grid* - A single control point (0,0,Z) marks the lower left-hand corner of the grid, with Z determining the constant Z-coordinate for all points in the grid. The lower left-hand corner before applying the rotation is also the local origin of the set.
- *Arc* - A single control point (0,0 Z) marks the center of the Arc, with Z determining the constant Z-coordinate for all points in the Arc. The center is also the local origin of the set.
- *Vertical Arc* - A single control point (0,0,Z) marks the center of the Vertical Arc, which is also the local origin of the set. The angle of the set as viewed from above is indicated by the set's rotation value.
- *Cylinder* - Two control points define the endpoints of the central axis of the Cylinder. The X- and Y-coordinates must be equal and the Z values determine the top and bottom of the Cylinder.
- *Polygon* - The control points are identical to the coordinates entered when the Polygon was first defined. If the Polygon was created by selecting *Cover with→Receiver points* from a FACES context menu, then the control points are the same as the vertices of the face being covered. Graphical editing is only available for Polygon sets that lie in the XY-plane. The set can be rotated out of the XY-plane after the editing is complete.
- *Vertical Surface* - Identical to *Route*.

- *Sphere* - A single control point (0,0,Z) marks the center of the Sphere, with Z determining the height of the center of the sphere. The locations of the points are determined by the spheres radius and the subset that is specified by the starting and ending phi/theta angles.
- *Trajectory* - Defined by at least three control points, each control point between the beginning and ending point of a Trajectory represents a transition point in the path that is defined by the set.
- *Points* - Each control point represents the offset of a point from the sets origin.
- "Points-on-face" - The control points define, in order, the points placed on the face when the set was created. The coordinates are defined as offsets in the X- and Y-directions from the first vertex of the face. The Z component defines the distance of the point above or below the face, along the face's normal vector.

13.7 Display Options

The GEOMETRY VIEW displays TRANSMITTERS as green cubes, RECEIVERS as red cubes, and TRANSCEIVERS as blue cubes. The size of the cubes can be changed for a set, as seen in Figure 13.27, by selecting the set and choosing *Properties* from the context menu. In the properties window, select LAYOUT PROPERTIES and change the value labeled *Rendered size*.

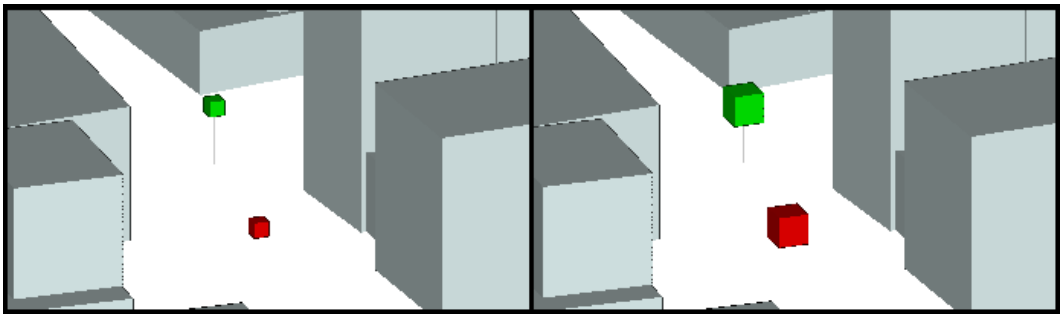


Figure 13.27: Adjusting the rendered size of a transmitter and receiver

Each transmitter set can also be made visible or invisible, as desired, to ease viewing other elements of the project. To change the visibility of a set, select it an click on *Visible* in the context menu to toggle its visibility in the GEOMETRY VIEW. To change the visibility of other aspects of the set, you can also select *Show origin*, *Show control vectors*, *Show antenna pattern* and *Show Description*. Each set can also be deactivated so they are not included in future simulations. To toggle the set active or inactive, select *Active* in the context menu.

13.8 Receiver Bounding Boxes

The RECEIVER bounding box is used to reduce the computation time for the ray tracing. It does so by grouping receiver points within larger bounding boxes, considering the intersections of rays with the bounding boxes, and then checking for the intersection of each ray with the small collection surfaces surrounding each receiver point.

The bounding box should be used in most situations for all receiver set types except *Points* and "Points-on-Face". If there are a large number of points in the set (> 25), then it may be helpful to use bounding boxes.

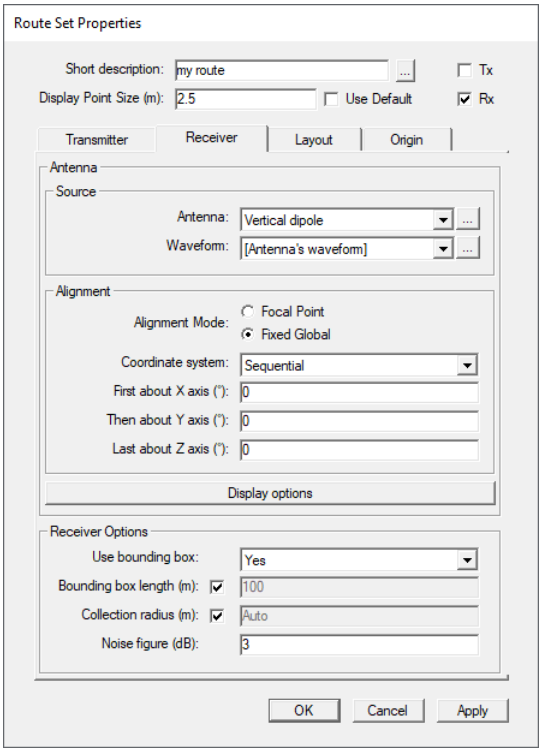


Figure 13.28: Properties for a receiver route

Receiver bounding boxes properties can be set in the RECEIVER PROPERTIES Window, as seen in Figure 13.28. To define a bounding box around a RECEIVER set, first set *Use bounding box* to "Yes". The size of the bounding box can be automatically set by the simulation, or it can be entered manually by un-checking the *Auto* box and entering a value in the *Bounding box length* field. The size of auto-determined bounding boxes are written to the "Receiver Sets" section of the `*.diag` file.

When manually setting the bounding box size, some care must be taken in selecting the length. Incorrectly sizing a bounding box can lead to increased run times.

Below are general rules of thumb for manually setting the bounding box length based on the type of Tx/Rx set:

- For *XY Grids*, *Vertical Surface*, *Polygon*, and *Cylinder* sets, define the bounding box size as 10 times the spacing between points.
- For *Routes*, *Arcs*, *Vertical Arcs*, *Trajectory*, and *Spheres*, set the box size to 20 times the spacing between points.
- For *Points*, set the box size so that encloses about 25 points.

Chapter 14

Communication Systems

In this chapter, you will learn...

→ how to define a communication system

→ how to set the properties of a communication system

Wireless InSites Communication System Analyzer is a collection of post-processing calculations that analyze Wireless InSite propagation results:

- Bit error rate
- Throughput
- Shannon capacity
- Various standard signal quantities related to power, noise, and interference

Each communication system operates independently and can be used to observe the effect of changing system parameters on the quality of the communications environment.

When a COMMUNICATION SYSTEM is defined in a project, Wireless InSite automatically runs a communication system analysis after it performs any propagation simulations. In addition, once a propagation simulation is complete, users can perform a communication system analysis at any time by selecting →*Run Communication system analysis* from the RUN menu. If the results required for the calculation do not exist, Wireless InSite will halt the analysis and write data to the SIMULATION LOG, indicating what result it was unable to locate. The result is accessible from the RESULTS tree of each study area under the *Communication systems* heading, as seen in Figure 14.1.

✓ Results generated by a communication system are stored in `..\Project directory\Study area name\<Communication system name>`.

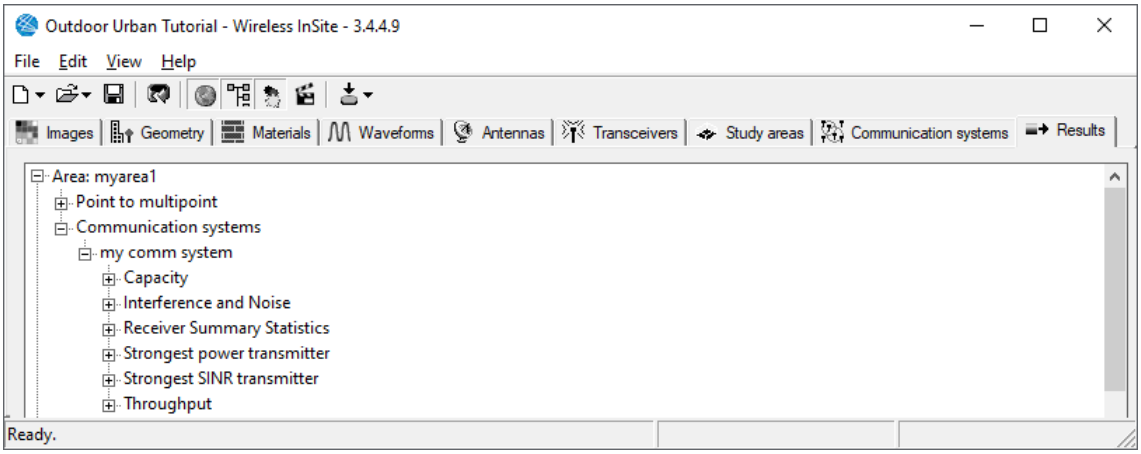


Figure 14.1: The communication system results as they appear in the Results tree

14.1 Creating a Communication System

To add a new COMMUNICATION SYSTEM to the project, select *File→New→Communication systems*.

14.2 Communication System Properties

After creating a COMMUNICATION SYSTEM, the properties window will appear, allowing users to select the TRANSMITTER and RECEIVER sets in the system, request the type of analysis, and enter system input parameters.

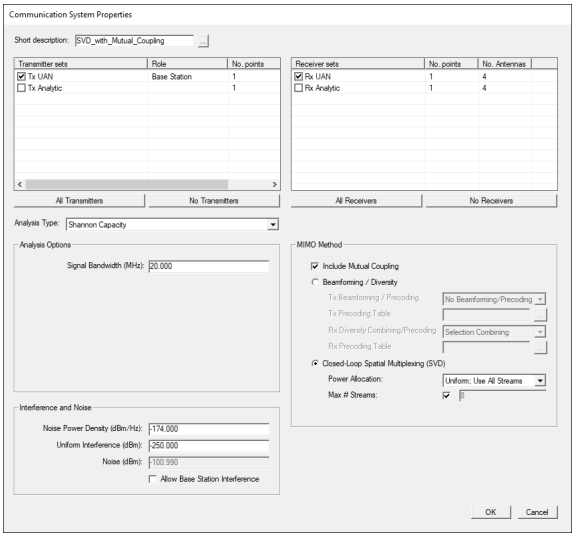


Figure 14.2: The communication system properties window

14.2.1 Transmitter and Receiver Selection

The upper portion of the COMMUNICATION SYSTEM PROPERTIES window controls which TRANSMITTER and RECEIVER sets to analyze and designates what sort of role each will play in the analysis. A TRANSMITTER set can be designated as either a *Base Station* or an *Interferer* via the context menu of each set, represented in the table with a role of either “Base Station” or “Interferer”. To activate or deactivate receiver sets, the user can check or uncheck the box next to each receiver set’s name in the table. If a transmitter or receiver set is inactive, it will be excluded from all post-processing calculations. Additionally, all transmitter or receiver sets can be activated or deactivated at once using the buttons located under the transmitter and receiver set tables.

- At least one transmitter set and one receiver set must be selected to define a valid communication system.
- Using the ALL TRANSMITTERS button defaults all TRANSMITTER sets to be *Base Stations*. Be sure to make desired adjustments before saving the COMMUNICATION SYSTEM.

When a communication system contains multiple Base Station antennas, receivers will choose to establish a link with the transmitter that delivers the highest received power.

The time required to perform the communication system analysis is linearly dependent on the number of transmitter and receiver points included in the analysis. Because of this run-time dependence, the number of points in each set is shown in the transmitter and receiver tables as a reference.

14.2.2 Analysis Type

The types of analyses that a COMMUNICATION SYSTEM can calculate are organized into three groups: *BER* (Bit Error Rate), *Throughput*, and *Shannon Capacity*. After selecting one of the options using the drop down menu under the *Analysis Type* section of the COMMUNICATION SYSTEM properties window, users can enter input parameters and request results for the desired analysis.

14.2.3 Interference and Noise Settings

COMMUNICATION SYSTEMS provide the ability to control the level and sources of interference and noise for a given project. The interference and noise levels are used with the received power from base station transmitters to calculate signal-to-interference-plus-noise ratios (SINR) used in the prediction of bit error rate and throughput.

The interference and noise are defined using the following parameters:

- *Interferring Transmitters* - Received power from transmitters designated as interferers in the transmitter selection window is treated as a contribution to the overall interference level at a given receiver point.
- *Noise Power Density* - Measure of the noise power per Hz of bandwidth in the propagation environment.
- *Uniform Interference* - Specifies a constant level of interference for all receiver locations.
- *Noise* - The level of noise in the propagation environment calculated from the *Noise Power Density* and *Signal Bandwidth*.

When *Allow Base Station Interference* is checked, base stations in the communication system will be allowed to interfere with each other, such that for each base station, all other operational base stations will be considered as additional interferers. When the setting is unchecked, no interference between base stations is included in the analysis.

Total interference and noise levels are calculated by summing the power values of the contributors listed above. The total interference and noise level is then compared against the Receiver Threshold setting in the antenna properties window, and the maximum value is used as the interference plus noise level for the receiver.

Additionally, the RECEIVER tab in the SET PROPERTIES window also contains a *Noise figure* (dB) which represents the contribution of the receiver's circuitry to noise at its output. It is the ratio of the interference plus noise power at the output to that at the input, expressed in dB [42].

- See Section 13.2.1 for a description of the RECEIVER tab containing the *Noise figure* field.

14.2.4 MIMO Method

The settings in the *MIMO Method* section allow users to specify the specific MIMO techniques that will be applied to communications between transmitters and receivers in the COMMUNICATION SYSTEM that have MIMO antennas. The *Beamforming / Diversity* option allows a beamforming or other precoding method to be applied to the transmitter and a diversity option to be applied at the receiver. Alternatively, the Closed-Loop Spatial Multiplexing option assumes coordination between the transmitter receiver antennas.

Detailed descriptions of these methods are provided in Chapter 17.

In cases where an S-parameter file has been associated with one or both of the MIMO antennas involved in the analysis, the user has the option to include the effect of mutual coupling between the MIMO antenna elements through non-zero mutual impedances defined via the S-parameter file. If the *Include Mutual Coupling* checkbox is enabled, the effect of these mutual coupling impedances are included as prescribed in the literature (see [43], [44], and [45]). If the *Include Mutual Coupling* checkbox is disabled, the mutual impedances between the antenna elements are assumed to be exactly zero. However, even in this case, the antenna element self-impedance and transmission line impedance defined in the S-parameter files associated with a MIMO antenna will still be used to calculate effects of impedance mismatch (e.g. VSWR).

14.2.4.1 Beamforming / Diversity Settings

When *Beamforming / Diversity* is selected and MIMO antennas are assigned to the transmitter and/or the receiver, the selected MIMO methods will be used to generate one stream of data between them.

Transmitter *Beamforming* settings include:

- *No Beamforming / Precoding* - All MIMO elements in the transmitter's antenna operate at equal weighting and phase settings.
- *Max Ratio Trans (MRT)* - Uses the maximum ratio transmission technique to calculate MIMO antenna element weightings based on the channel vector between the transmitter and receiver pair.

- *Precoding / Beamforming Table* - Applies MIMO antenna element weightings from a user-defined file to the transmitter. The file can contain multiple entries representing different possible beams of the transmitting antenna, or alternatively, one or more sets of precoding weights for a transmitter diversity method. When this option is selected, the analysis will use the weightings that maximize the SINR for the given the transmitter and receiver pair. The format of the user-defined throughput file is described in section K.1 of Appendix K.

Receiver *Diversity*/*Precoding* settings include:

- *Selection Combining* - The receiver selects the MIMO antenna element with the strongest SINR.
- *Equal Gain Combining* - Receiver antenna elements are weighted to equalize voltage phases of each element in order to increase total received voltage at the receiver.
- *Maximal Ratio Combining* - Uses the maximal ratio combining technique to calculate MIMO element weightings for the receivers antennas based on the channel vector between the transmitter and receiver. This technique adjusts both the magnitudes and phases of the elements in order to maximize the total received power from the combination of all receiver elements.
- *Precoding/Beamforming Table* - Sets of receive combining weights are defined in a user defined precoding table. The receiver selects the receive combining weight set from the precoding table (i.e. codebook) with the strongest SINR.

14.2.4.2 Closed-Loop Spatial Multiplexing

Closed-Loop Spatial Multiplexing uses Singular Value Decomposition (SVD) to calculate precoding weights applied at the transmitter and combining weights at the receiver to generate multiple, non-interfering, data streams. For throughput and capacity calculations, the total throughput or capacity is calculated by summing the individual throughputs and capacities of each stream. For other communication system results the best value from all streams is returned. For example, for bit error rate analysis, the lowest bit error rate of all the streams is returned. Similarly, in the interference and noise summary results for each analysis type, the highest SNR and SINR of all generated data streams is returned.

Closed-Loop Spatial Multiplexing (SVD) settings include:

- *Power Allocation* - Select from one of three methods for allocating power amongst the data streams:
  - *Uniform; Use All Streams* enforces the use of all allowed streams with the total transmit power divided equally between them.
  - *Uniform; Optimize Throughput/Capacity* allows for the optimization of throughput/capacity by re-allocating power to a reduced number of streams with power equally divided amongst the used streams and zero power allocated to unused streams.
  - *Waterfilling* allows for the optimization of throughput/capacity by allowing unequal power amongst the allowed streams.

  Notes: Only the *Uniform; Use All Streams* method is available for BER analysis; *Equal MCS Across Streams* must be unchecked for the *Waterfilling* method to be available for the Throughput analysis.
- *Max # Streams* - Sets a maximum limit on the number of data streams that can be used.
- *Equal MCS Across Streams* - (Throughput analysis only) Checkbox allows or forbids the use of different modulation and coding schemes (MCS) for each individual data stream.

14.2.5 Bit Error Rate (BER)

Wireless InSite offers three methods for Bit Error Rate analysis: *AWGN*, which assumes an additive white Gaussian noise (AWGN) channel, *Theoretical Fading*, which assumes a Rayleigh or Rician fading channel, or *Semi-Analytic*. These calculations are based on the Matlab Communication System Toolbox Bit Error Rate functions: `berawgn`, `berfading` and `semianalytic`. Bit Error Rate is activated by selecting *Bit Error Rate* from the *Analysis Type* drop down menu in the *Analysis Options* section below the transmitter and receiver tables.

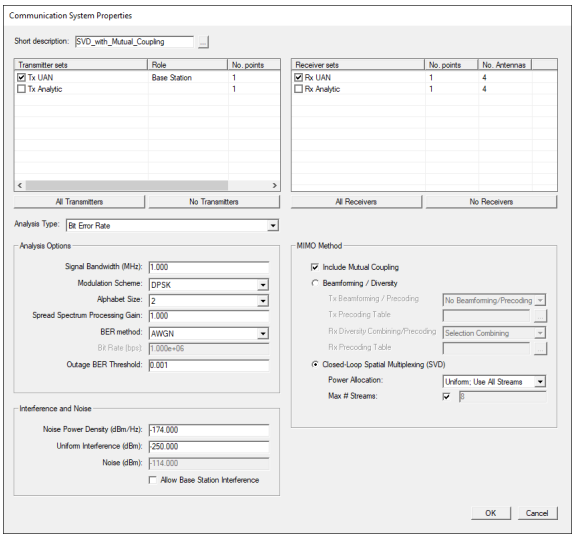


Figure 14.3: The communication system properties window when BER is selected.

14.2.5.1 Communication Parameters

The performance of the communication system is dependent on the modulation scheme employed and the size of the alphabet. The available *Modulation schemes* are:

- Pulse Amplitude Modulation (PAM)
- Quadrature Amplitude Modulation (QAM)
- Phase Shift Keying (PSK)
- Differential Phase Shift Keying (DPSK)
- Frequency Shift Keying (FSK)
- Minimum-Shift Keying (MSK)

Some modulation schemes are not available when performing certain BER methods. *Alphabet size* ranges from 2 to 1024, but not all values are available for every modulation scheme and BER analysis method.

▶ BER analysis methods are discussed in Section 14.2.5.2.

The *Bit rate*, in bits per second, specifies the communication rate that is being used over the channel. This field is used only when performing a semi-analytic simulation which considers the delay spread of the communication channel. Since the communication analysis is performed at baseband, none of the wideband waveform characteristics are included. If a spread spectrum system is being modeled, however, the *Spread Spectrum Processing Gain*, L, can be specified to increase the SINR at each RECEIVER point by $10log_{10}(L)$.

14.2.5.2 BER Methods

Several options determine how Wireless InSite carries out the analysis. The available *BER Methods* are:

- AWGN
- Theoretical Fading
- Semi-Analytic

Currently, only the AWGN option is available for COMMUNICATION SYSTEM that include a MIMO antenna. *Theoretical Fading* and *Semi-Analytic* analysis requires all active transmitter and receivers be assigned SISO antennas.

AWGN applies the known analytical expressions for BER in the presence of AWGN without considering any dispersive effects of the channel. E_b/N_0 is found by

$$\frac{E_b}{N_0}(dB) = 10log_{10}\left(\frac{P_R}{P_I + N_0}\right) + 10log_{10}(L) \tag{14.1}$$

where

- P_R is the received power
- P_I is the power of the interference sources
- N_0 is the power of the noise sources

Supported modulation schemes and alphabet sizes for AWGN analysis are:

- PAM (2 to 1024)
- QAM (4 to 1024)
- PSK (2 or 4)
- DPSK (2 to 1024)
- FSK (2 to 64)
- MSK (2)

▶ For more information, refer to the Matlab Communication Toolbox command `berawgn`.

Theoretical Fading analysis assumes the envelope of the fading channel is either a Rayleigh or Ricean distribution. The Ricean K factor, which is the ratio of the specular and line-of-sight energy to diffuse scattered energy, is used to determine which distribution to use. Supported modulation schemes and alphabet sizes for theoretical fading analysis are:

- PAM (2 to 1024)
- QAM (4 to 1024)
- PSK (2 or 4)
- DPSK (2 to 1024)
- FSK (2 only)

The use of FSK will always assume a Rayleigh channel due to an implementation limitation.

▶ For more information, refer to the Matlab Communication Toolbox command `berfading`.

Semi-Analytic employs a semi-analytic approach to the analysis. A random complex baseband signal is constructed of length $log_2(M) \cdot M^L$, where M is the alphabet size and L is the integer length of the multi-path channel's impulse response in symbol durations. This signal is then modulated using the selected scheme and passed through a linear Finite-Impulse Response (FIR) filter, whose inputs are the relative magnitudes and phases of the individual multi-path components. A semi-analytic analysis is then performed on the signal. This analysis predicts the likelihood of bit error given both the original and faded signals, and the type and alphabet size of the modulation applied.

▶ The semi-analytic approach is described in [46].

▶ For more information, refer to the Matlab Communication Toolbox command `semianalytic`.

Supported modulation schemes and alphabet sizes for semi-analytic analysis are:

- QAM (4 to 1024)
- PSK (2 or 4)
- DPSK (2 or 4)
- MSK (2 only)

As mentioned in Section 14.2.1, a file for each RECEIVER set is created which contains the combined BER from all the TRANSMITTERS based on the choice for the *Select Transmitter Based On* field: Highest Power or SINR.

A final outage probability for each receiver set is also computed. By specifying an *Outage BER threshold* between 0 and 1, the number of receivers in the set which exceed this threshold is found and displayed as a percentage of the number of receivers in the set. The final outage probability is written to the SIMULATION LOG window.

14.2.6 Throughput

The throughput analysis for Wireless InSite provides throughput estimates for LTE, 5G NR FR1, 5G NR FR2, WiMax, 802.11n, 802.11ac, 802.11ax as well as any number of other wireless access methods through the use of user-defined throughput tables. After running a full propagation simulation, users can define a COMMUNICATION SYSTEM and post-process the received power results to generate throughput results. Wireless InSite projects may contain multiple communication systems, enabling users to quickly compare many configurations.

The throughput analysis options, as seen in Figure 14.4, are accessible by selecting *Throughput* from the *Analysis Type* drop down menu in the *Analysis Options* section under the transmitter and receiver tables.

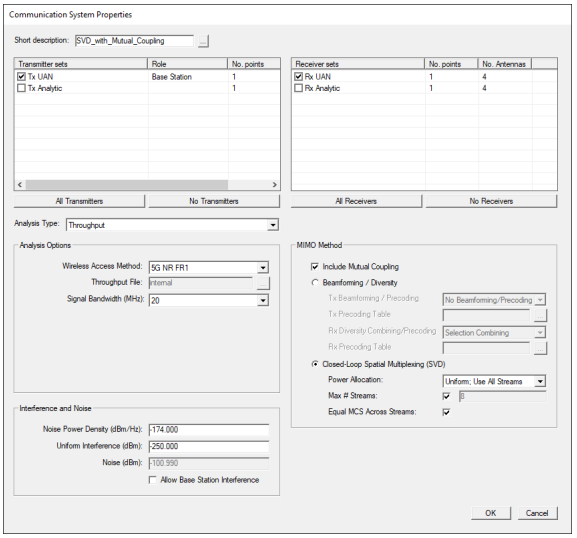


Figure 14.4: Communication System properties for throughput

- *Wireless Access Method* - Defines the protocol used in the throughput analysis. Users have a choice between 5G NR FR1, 5G NR FR2, LTE, 802.11ax, 802.11ac, 802.11n, WiMax and user-defined systems.
- *Throughput File* - Contains data to define the Wireless Access Method table. This parameter can only be modified when using a User-Defined Wireless Access Method. The format of the user-defined throughput file is described in K.2 of Appendix K.
- *Signal Bandwidth* - Defines the bandwidth of the signal used in the throughput analysis. The available options are dependent on the selected Wireless Access Method. Generally, larger bandwidths allow for higher throughput rates.

  The bandwidth set in this field overrides the bandwidth in the WAVEFORM PROPERTIES Window. The WAVEFORM still defines the carrier frequency of the signal.

14.2.6.1 Throughput Calculations

Prior to calculating throughput, a project must have valid received power results for the TRANSMITTERS and RECEIVERS selected in the COMMUNICATION SYSTEM PROPERTIES window. For each receiver location, the transmitter that provides the highest SINR is identified. Using the SINR, each system will then select a modulation and coding scheme (MCS) based on the *Wireless Access Method*. The throughput rate at the receiver is directly related to the selected MCS.

When throughput analysis is used with the closed-loop spatial multiplexing MIMO option, SINR values and corresponding throughput rates will be calculated for each independent data stream. The total throughput between the transmitter and receiver is the sum of the throughput of each data stream.

Supported modulation and coding rates for **LTE** systems are:

- QPSK (9 coding rates): 1/10, 1/6, 1/5, 1/4, 1/3, 2/5, 1/2, 3/5, 2/3, 3/4
- 16QAM (7 coding rates): 2/5, 9/20, 1/2, 11/20, 3/5, 2/3, 3/4, 4/5, 5/6
- 64QAM (12 coding rates): 1/2, 3/5, 5/8, 2/3, 17/24, 3/4, 4/5, 5/6, 7/8, 9/10

Supported modulation and coding rates for **5G NR FR1** and **5G NR FR2** systems are:

- QPSK (5 coding rates): 0.117, 0.188, 0.301, 0.438, 0.588
- 16QAM (6 coding rates): 0.369, 0.424, 0.479, 0.540, 0.602, 0.643
- 64QAM (9 coding rates): 0.455, 0.505, 0.554, 0.602, 0.650, 0.702, 0.754, 0.803, 0.853
- 256QAM (8 coding rates): 0.667, 0.694, 0.736, 0.778, 0.821, 0.864, 0.895, 0.926

Supported modulation and coding rates for **WiMax** systems are:

- QPSK (4 coding rates): 1/12, 1/8, 1/4, 1/2
- 16QAM (1 coding rates): 1/2
- 64QAM (3 coding rates): 1/2, 2/3, 3/4

Supported modulation and coding rates for **802.11n** systems are:

- BPSK (1 coding rate): 1/2
- QPSK (2 coding rates): 1/2, 3/4
- 16QAM (2 coding rates): 1/2, 3/4
- 64QAM (3 coding rates): 1/2, 2/3, 3/4

Supported modulation and coding rates for **802.11ac** systems are:

- BPSK (1 coding rate): 1/2
- QPSK (2 coding rates): 1/2, 3/4
- 16QAM (2 coding rates): 1/2, 3/4
- 64QAM (3 coding rates): 1/2, 2/3, 3/4
- 256QAM (2 coding rates): 3/4, 5/6

Supported modulation and coding rates for **802.11ax** systems are:

- BPSK (1 coding rate): 1/2
- QPSK (2 coding rates): 1/2, 3/4
- 16QAM (2 coding rates): 1/2, 3/4
- 64QAM (3 coding rates): 2/3, 3/4, 5/6
- 256QAM (2 coding rates): 3/4, 5/6
- 1024QAM (2 coding rates): 3/4, 5/6

For each wireless access method, the relationship between SINR and the modulation and coding scheme that can be supported, in terms of modulation order and coding rate, is based on results from link level simulations in the technical literature [47], [48]. Alternatively, the user-defined table can be used to define a custom relationship between SINR and MCS, along with the resultant throughput.

To summarize the throughput assumptions:

- Noise distribution is constant over the signal bandwidth.
- Throughput estimates are for the peak downlink data transfer rates.
- Estimates are based on the MCS that can be supported at the calculated SINR for each channel or data stream.

14.2.6.2 Generating Throughput Results

If an active COMMUNICATION SYSTEM that is configured to perform a *Throughput* analysis is in the project when a new simulation is initiated throughput values will be calculated when it completes. Users can also post-process simulation results into throughput. In any project with valid received power results, new communication systems can be analyzed by selecting *Run Communication system analysis* from the RUN drop-down menu.

Throughput analysis generates the following results:

- Throughput for LTE, 5G NR FR1/FR2, WiMAX, 802.11n, 802.11ac, 802.11ax and user-defined systems
- Channel capacity using the Shannon-Hartley theorem
- Reference signal received power (RSRP) for LTE systems
- Reference signal received quality (RSRQ) for LTE systems

14.2.6.3 Viewing and Plotting Throughput Results

Like other results throughput will be calculated for each RECEIVER point in the project. After the communication system analysis is complete, throughput will be available for viewing in the GEOMETRY VIEW or plotting from the RESULTS tab in the MAIN WINDOW.

Throughput results can be viewed in the GEOMETRY VIEW using the scale bar, as seen in Figure 14.5, or plotted in a graph, as seen in Figure 14.6.

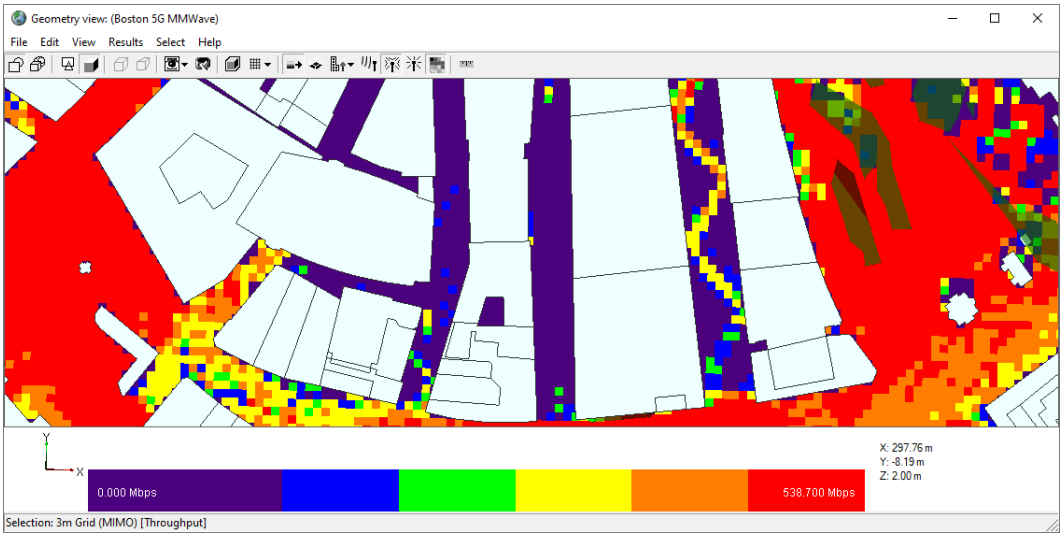


Figure 14.5: Throughput results displayed in the Geometry View

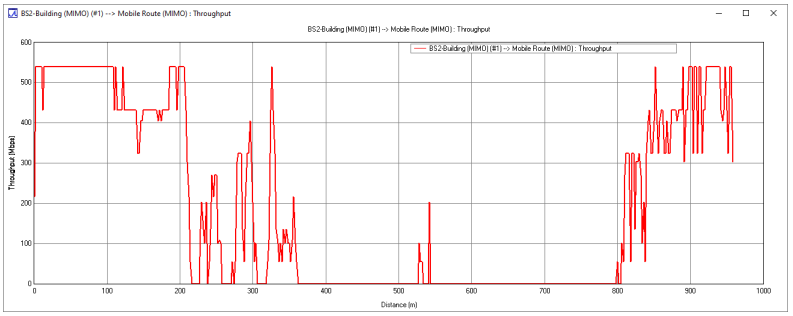


Figure 14.6: A graph of throughput results

14.2.7 Shannon Capacity

The Shannon capacity analysis for Wireless InSite calculates the maximum theoretically allowed capacity based on the SINR value for a given transmitter/receiver pair. After running a full propagation simulation, users can define a COMMUNICATION SYSTEM and post-process the received power to generate capacity. Wireless InSite projects may contain multiple communication systems, enabling users to quickly compare many configurations.

The only additional user input used for Shannon capacity analysis is the *Signal Bandwidth*. A field that allows for specifying *Signal Bandwidth* becomes accessible when *Shannon Capacity* from the *Analysis Type* drop down menu in the *Analysis Options* section is selected.

- *Signal Bandwidth* - Defines the bandwidth of the signal used in the capacity calculations. Larger bandwidths allow for higher capacity limits.

The bandwidth set in this field overrides the bandwidth in the WAVEFORM PROPERTIES window. The WAVEFORM still defines the carrier frequency of the signal.

14.2.7.1 Shannon Capacity Calculations

Prior to calculating capacity, a project must have valid received power for the TRANSMITTERS and RECEIVERS selected in the COMMUNICATION SYSTEM PROPERTIES window. For each receiver location, the transmitter that provides the highest SINR is identified. The capacity for that transmitter/receiver pair is then simply calculated based on the Shannon-Hartley theorem:

$$C = B \log_2 (1 + \mathrm{SINR}) . \tag{14.2}$$

where B is the signal bandwidth.

When capacity analysis is used with the closed-loop spatial multiplexing MIMO option, SINR values and corresponding capities will be calculated for each independent data stream. The total capacity between the transmitter and receiver is the sum of the capacity of each data stream. See 17.6 for more details.

14.2.7.2 Generating Capacity Results

If an active COMMUNICATION SYSTEM that is configured to perform a *Shannon Capacity* analaysis is in the project when a new simulation is initiated capacity values will be calculated when it completes. Users can also post-process simulation results into capacity. In any project with valid received power results, new communication systems can be analyzed by selecting *Run Communication system analysis* from the RUN drop-down menu.

Shannon capacity analysis generates the following results:

- Channel capacity using the Shannon-Hartley theorem

14.2.7.3 Viewing and Plotting Capacity Results

Like other results capacity will be calculated for each RECEIVER point in the project. After the communication system analysis is complete, capacity will be available for viewing in the GEOMETRY VIEW or plotting from the RESULTS tab in the MAIN WINDOW.

The capacity results can be viewed in the GEOMETRY VIEW using the scale bar or plotted in a graph.

14.2.8 Interference and Receiver Summary Files

A Wireless InSite COMMUNICATION SYSTEM analysis always generates several files containing data which summarize basic properties about the received power, interference and noise for each active receiver. These include consolidated results that combine signals from several transmitters. The COMMUNICATION SYSTEM will use only the selected transmitter points and receiver sets, and generate the following result types:

- Interference

- Noise
- Signal-to-noise ratio (SNR)
- Signal-to-interferer ratio (SIR)
- Signal-to-interference-plus-noise ratio (SINR)
- Strongest Power
- Total Power summed without phase
- Total Power summed with phase
- Best SINR
- Receiver signal strength indicator (RSSI)
- Receiver's strongest transmitter
- Strongest SINR transmitter

▶ Refer to 21.1.2 for information about how these results are calculated.

Chapter 15

Study Areas

In this chapter, you will learn...

→ how to create and define a study area

→ how to set up the study area to meet the requirements of your Wireless InSite project

A STUDY AREA defines three important aspects that determine what information will be used during a simulation: the region of the project in which to perform a simulation, the propagation model to be used in calculating the power transferred from active TRANSMITTERS to active RECEIVERS within the study area, and the set of results to generate during the simulation. Multiple study areas can be defined in a project, each with a unique short description, independent propagation models, and parameters. Through this mechanism, results from various simulation techniques can be compared. TRANSMITTERS RECEIVERS and *Objects* falling outside the study area boundary are not considered.

15.1 Creating a Study Area

To create a STUDY AREA, right click in the GEOMETRY VIEW and choose *New→Study area*. The window shown in Figure 15.1 will appear.

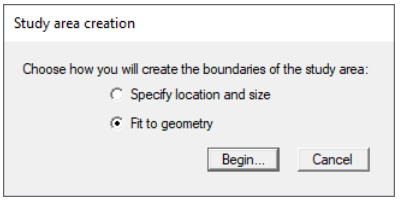


Figure 15.1: Creating a new study area

Specify the method the application should use to create the STUDY AREA boundaries. The first option, *Specify location and size*, allows the user to draw the boundary perimeter manually and specify the

bottom and top heights. To do so, use the mouse to select vertices along the desired boundary in the GEOMETRY VIEW. These points are connected by straight lines, and any number of points can be selected, allowing the user to define irregularly-shaped study areas. After clicking on the last point of the boundary, right-click to connect the last point to the first. The base and top height of the study area are specified in the window seen in Figure 15.2.

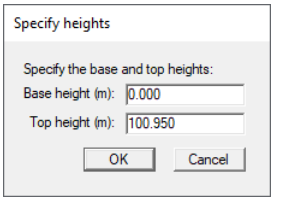


Figure 15.2: Finishing the creation of a new study area

- ✓ By default, these height values are relative to the terrain. However, they can be specified as height relative to sea level in the STUDY AREA PROPERTIES Window (Figure 15.3).
- ✓ It is recommended that the boundary of the STUDY AREA be convex. Concave STUDY AREA boundaries can lead to unintended results, by eliminating ray paths that exit and re-enter the studyarea region.

The second option, *Fit to Geometry*, automatically fits the study area boundary around all GEOMETRY, TRANSMITTERS, and RECEIVERS in the project. The size expands and contracts as items are added, deleted, or repositioned.

15.2 Study Area Properties

After clicking OK in the SPECIFY HEIGHTS Window, the STUDY AREA PROPERTIES Window will appear. Figure 15.3 shows the default study area properties window. Once created, this window can also be accessed by right-clicking on the study area entry in either the table under the STUDY AREA tab on the MAIN WINDOW or in the PROJECT TREE.

Note that the contents and layout of the STUDY AREA PROPERTIES window will change to reflect relevant parameters associated with the selected *propagation model*. After selecting the desired *Propagation model*, enter a *Short description*; this will be used as the folder name where the result files will be written during the simulation. Only valid filename and directory characters may be used for a short description. If one is not entered, folders will be named after the selected model. For example x3d, full3d, full3d(2) etc.

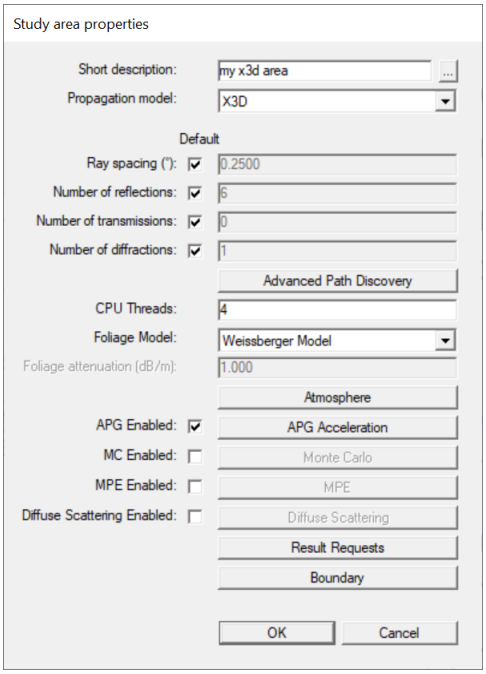


Figure 15.3: Study area properties window

The X3D model provides high fidelity, GPU accelerated, 3D ray tracing, suitable for indoor or outdoor scenes. It accounts for atmospheric attenuation, and supports several special capabilities including Monte Carlo parameter variation, adjacent path generation for highly efficient prediction for dense receiver sets, and MIMO system analysis. For urban scenes with transmitter and receiver heights considerably lower than the building heights, the URBAN CANYON model should be used. Other models include the 2D ray-based VERTICAL PLANE model for irregular terrain, and several fast ray based and empirical Wireless InSite Real Time models.

The ray-tracing models, X3D or FULL 3D, URBAN CANYON, and VERTICAL PLANE enable the user to set the ray spacing and the maximum number of reflections, diffractions and (when applicable) transmissions, as well as set other model-specific inputs, edit the boundary (for a manually defined STUDY AREA), and request specific results. The Real Time models are typically much simpler and generally produce a limited number of results and do not support *Result Filters* or editing the STUDY AREA boundary.

Table 15.1 lists the inputs available for each propagation model.

Table 15.1: Summary of Study Area inputs by Propagation Model

Model	Model Input Parameters	Result Requests	Result Filters	Boundary
X3D	Ray spacing Ray interactions Terrain Diffractions Ray Casting Limits EES Diffuser Limits CPU Threads Use GPU Foliage Model Atmosphere APG* Monte Carlo MPE Diffuse Scattering	✓	No	✓
FULL3D	Ray spacing Ray interactions Raytracing Method Allowed Interactions Advanced	✓	✓	✓
URBAN CANYON	Ray spacing Ray interactions Allowed Interactions Advanced	✓	✓	✓
VERTICAL PLANE	Ray spacing Ray interactions Linearize Terrain Effective Earth Radius Allowed Interactions Advanced	✓	✓	✓
VPUP TPG	Ground bounce	✓	No	No
HATA COST HATA	Environment	No	No	No
WALFISCH-IKEGAMI OPAR FREESPACE	No	No	No	No

The *Automatic* checkboxes next to the parameter entries are used to request that the simulation use default values for a parameter. Each parameter can be set independently by turning off the automatic option and entering a number or selecting the desired option.

✓ The novice user is encouraged to make use of these options when initially performing simulations. After acquiring more experience with Wireless InSite, the user may find that setting some of the properties manually may improve results.

The remaining parameters in the STUDY AREA PROPERTIES Window are described below.

Model Input Parameters

The different propagation models included in Wireless InSite support different input parameters. The simplest models do not require any model inputs.

- For ray-tracing models these include:
  - *Ray spacing*: used during the ray tracing portion of a simulation. The proper setting for this parameter depends on several factors, including the size of the project, the size of the facets in the project and the distance of facets from the transmitter.
    - ✓ As a rule of thumb, for a 500 m x 500 m area, set the ray spacing to 0.2° or smaller.
  - *Number of reflections*: this indicates the maximum number of reflections which a path can undergo. When diffractions are present, additional logic determines when reflections can occur before, after and between diffractions. FULL 3D has additional settings in the advanced parameters tab to control reflections for diffracted paths.
    - ✓ In our experience, six reflections are usually sufficient, although areas with highly reflective buildings or narrow streets may require more.
  - *Number of transmissions*: indicates the maximum number of transmissions which a non-diffracted path can undergo (other settings in the advanced parameters tab control transmissions for diffracted paths). Transmissions through building walls are only used by the X3D and FULL 3D models. The URBAN CANYON model cannot be used to find propagation paths to points inside a building.
    - The user should be aware that the run time will increase as the number of reflections and transmissions increase, and, in some cases, it may increase dramatically.
    - For the X3D model, transmissions are limited to 30 (independently of the number of reflections). For the FULL 3D and URBAN CANYON models, the combined number of reflections and transmissions is limited to 30. If they exceed this value, they are reduced proportionately so that they sum to 30.
  - *Number of diffractions*: indicates the maximum number of diffractions which a path can undergo.
    - ✓ In our experience, one diffraction is usually sufficient.
    - ✓ In general, the maximum number of diffractions is 3. For the VERTICAL PLANE model, it is 4.
    - Additional diffractions can dramatically increase run time, even more so than additional reflections and transmissions.
- For Real Time models:
  - For VERTICAL PLANE URBAN PROPAGATION (VPUP), and TRIPLE PATH GEODESIC (TPG), model inputs are limited to toggling *ground bounce* on and off.
  - For HATA and COST HATA, the model inputs consist of selecting a propagation environment from a choice of *Large-sized city*, *Medium-sized city*, *Suburban*, or *Rural*.

▶ For model-specific input parameters, please refer below to Section 15.3 or to the appropriate section of Chapter 16.

Result Requests

All ray-based models have an *Result Requests* button, which allows the user to select which results will be generated by the simulation. The propagation models that generate a limited number of results do not have this button and generate all results.

Result Filters

A limited number of propagation models, FULL 3D, URBAN CANYON, and VERTICAL PLANE support RESULT FILTERS. RESULT FILTERS allow paths with certain interactions to be filtered out when generating results. For these models, it is possible to change a result filter and quickly recalculate the results using the *Added requested results* simulation mode.

Boundary

The BOUNDARY button provides two ways for the user to redefine the region contained within a manual STUDY AREA either by editing the control points or by graphically moving the vertices of the STUDY AREA within the GEOMETRY VIEW. It also provides an interface for defining the *Coordinate system* and origin for the STUDY AREA. The available coordinate systems are *Cartesian*, *Longitide/Latitude*, and *UTM*.

Edit Control Points

The vertices of the STUDY AREA can be edited using the EDIT CONTROL POINTS button. Clicking on the button causes the VIEW/EDIT VERTICES Window to appear, as seen in Figure 15.4. Point coordinates are given in meters relative to the global origin. Vertices' locations can be modified by double-clicking on the entry for a vertex and entering new X-, Y-, and Z-coordinates. The Z-values are heights described relative to the terrain or sea level, as determined by the *Elevations relative to* drop-down menu in the STUDY AREA PROPERTIES Window. Selecting a vertex and then right-clicking allows users to add a *New vertex*, *Delete vertex*, or move a vertex up or down in the list.

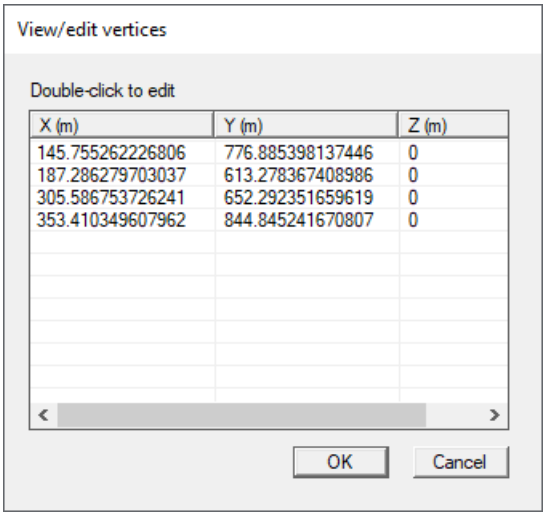


Figure 15.4: The vertices describing the outline of a study area

Edit Boundary

The EDIT BOUNDARY button allows the user to graphically reposition the vertices of the STUDY AREA. Clicking on the button causes the EDIT STUDY AREA BOUNDARY Window to appear, as seen in Figure 15.5. Each of the blue points represents a vertex. To change the position of the vertex, click and drag the point to its new location. Also from this window, the *Auto-boundary* checkbox can be clicked to switch to the automatic boundary mode described above. If this checkbox is selected and a vertex is moved, the study area reverts back to the manual mode.

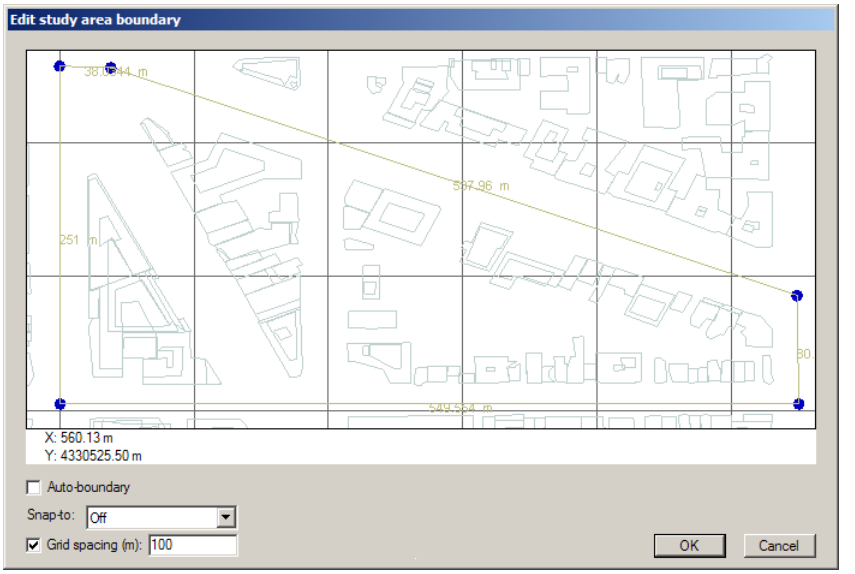


Figure 15.5: Editing the boundary of a study area

When the user clicks OK and the *Auto-boundary* checkbox is unchecked, the SPECIFY HEIGHTS Window will appear, allowing the specification of the bottom and top heights of the study area.

15.3 Model-Specific Study Area Inputs

- For more information on propagation models and their parameters, see Chapter 16.

15.3.1 Full 3D, Urban Canyon, and Vertical Plane Study Area Inputs

✓ The ADVANCED STUDY AREA PROPERTIES Window allows the user to further adjust these values for diffracted rays when necessary.

Raytracing Method

Choose between Shooting and Bouncing Ray (SBR) and Eigenray (ER).

- For the ER method, the sum of the number of interactions (reflections, transmissions, and diffractions) must be no greater than 3.
- Ray tracing methods are explained in Section 16.1.

Allowed Interactions

This option applies to the FULL 3D, VERTICAL PLANE, and URBAN CANYON models. It allows each kind of interaction to be turned off for each GEOMETRY type. For example, it is possible to turn off reflections from the *Terrain* or transmissions through an *Object* using this feature. By default, all interactions are enabled when a new STUDY AREA is created. In Figure 15.6, the allowed interaction types are checked, and the disallowed types are unchecked.

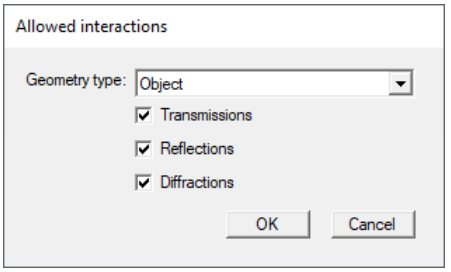


Figure 15.6: Setting types of interactions that are allowed in the study area

This capability is similar in some respects to the RESULT FILTERS. However, when an interaction in a STUDY AREA is disallowed, the specified ray paths are never found; but the unwanted interactions in result filters are filtered out when the results are generated. It is possible to change a result filter and quickly recalculate the results using the *Added requested results* simulation mode, whereas if the allowed interaction settings in a study area are changed it is necessary to run a new simulation.

One application of this feature would be to disable one or more interactions in order to reduce the computation time by eliminating the search for propagation paths that are of no interest. For example, if the *Terrain* is nearly flat but still consists of many facets, it may be helpful to eliminate the search for rays that diffract from the terrain. Simply setting the terrain to *Inactive* is another alternative. This option would not usually be acceptable if building or antenna heights have been specified relative to the terrain, because the heights would no longer be set correctly.

Advanced Model Parameters

The ADVANCED button provides additional options for customizing the simulation model. Figure 15.7 shows the ADVANCED MODEL PARAMETERS window for Full 3D. For FULL 3D, URBAN CANYON, and VERTICAL PLANE the user can specify several ray tracing parameters to set the maximum number of reflections and transmissions for ray paths which undergo diffractions. The maximum number of reflections and transmissions can be specified independently; before the first diffraction, between diffractions, and after the last diffraction.

✓ We suggest that the maximum number of interactions (reflections and transmissions) that can occur before the first diffraction be the same as the number after the last diffraction, and that the number between diffractions be 0, 1 or 2.

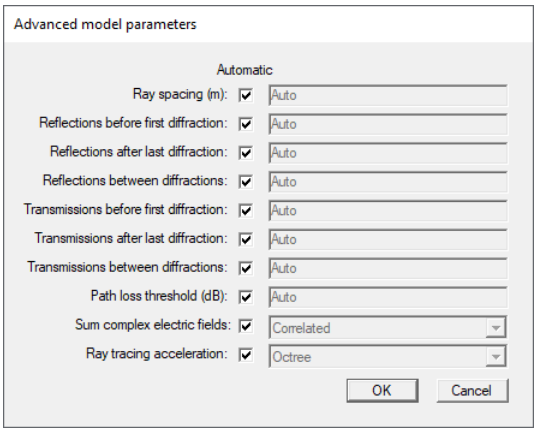


Figure 15.7: Advanced model parameters for the Full3D model

- *Plane Wave Ray Spacing*

  The spatial separation of rays shot from *Plane Wave* transmitters can be set manually. Using the *automatic* option allows the simulation to calculate an optimal value at run time.

- *Number of Reflections*

  The first three fields on the window above allow the maximum number of reflections before, after and between diffractions to be set manually. If the *Automatic* box is checked, the maximum number before and after diffractions will be half the maximum number for non-diffracted paths set in the STUDY AREA PROPERTIES Window, with fractions rounded up to the next whole number. For example, if the maximum number of reflections is set to 5, the maximum number before and after a diffraction will be 3. The maximum number between diffractions will be set to 0, 1 or 2 if the *Automatic* box is checked, depending on the maximum number of reflections.

- *Number of Transmissions*

  Propagation paths with transmissions through walls are usually only found in the FULL 3D model, the only exception being when the material type is set to *Free Space*. The maximum numbers of transmissions are set in the basically the same way as the numbers of reflections. If the *Automatic* box is checked, the maximum number before and after diffractions will be one half the maximum number for non-diffracted paths set in the STUDY AREA PROPERTIES Window, and with fractions rounded up to the next whole number. For example, if the maximum number of transmissions is set to 3, the maximum number before and after a diffraction will be 2. The maximum number between diffractions will be set to 0, 1 or 2 if the *Automatic* box is checked, depending on the maximum number of transmissions.

- *Path Loss Threshold*

  This parameter provides a means to omit ray paths with very small power levels from the latter stages of the computations, which can help to reduce the computation time. For models which use a path loss threshold, the default setting is -250 dB, with the exception of Urban Canyon, which is -100 dB. The simulation will ignore all paths for which the path loss exceeds the free space loss by the amount specified. This threshold is applied to the electric field assuming an *Isotropic* source,

so any antenna pattern effects at either the TRANSMITTER or RECEIVER are not taken into account.

In general, the *Automatic* box should be left checked since it will be set to a very large value. However, if the user is concerned that extremely weak paths may be missed, this value can be adjusted. In the VHF and UHF bands an excess path loss above 60 dB would be rare, so setting this value too much higher will usually have no effect. The drawback to setting it too high is that this can lead to significantly longer run times as computations are performed on a large number of very weak paths which make no contribution to the final results.

✓ For each simulation, the geometrical path data is retained, and this threshold can be reduced and new predictions generated without repeating the ray tracing procedures.

In some situations, the user may wish to set the threshold even lower as a way of screening out all weak paths. An alternative means of doing this is to use the *Minimum received power* setting in the ANTENNA PROPERTIES Window. In the latter case all antenna pattern effects as well as the input power to the transmitter will be considered.

- *Sum Complex Electric Fields*

  This setting controls whether the phase of the electric field of each ray path is retained when combining the fields to predict the received power and path loss. The differences among the options have to do with whether ray paths are combined by adding complex-valued electric fields or by summing powers.

  – *None* - The received power is found by adding the power of each path. The phase difference between rays is ignored.

  – *All* - All fields are first summed with phase and then Wireless InSite finds the total power from the magnitude of the total field.

  – *Correlated* - The ray paths which follow nearly the same path through the environment (the correlated paths) are combined with phase, and then the powers of all the correlated groups are added.

  ▶ For a more thorough explanation on computing received power, see Section 21.1.

- *Ray Tracing Acceleration*

  – *Partitions* - A visibility map is constructed from the geometry and used when searching for ray path interactions with geometry. This acceleration tends to work best when the overall face count is below 2500.

  – *Octree* - Face geometry is organized into an octree data structure that is then used to locate ray path interactions in the geometry. Octree acceleration only out performs *Partitions* when the project face count exceeds 2500.

  – *None* - No ray tracing acceleration is used.

15.3.2 X3D Specific Study Area Inputs

Advanced Path Discovery

When using X3D, the ADVANCED PATH DISCOVERY button allows the user to limit where the ray tracer shoots rays, control whether to allow terrain diffractions, and to limit interactions with engineered surface diffusers. Ray-casting can be limited by setting the start and end phi and theta angles. The EES diffuser limit defaults to 1, and controls the number of times each ray path can interact with an engineered surface diffuser. The *Include Terrain Diffractions* checkbox controls whether paths will diffract off of terrain edges.

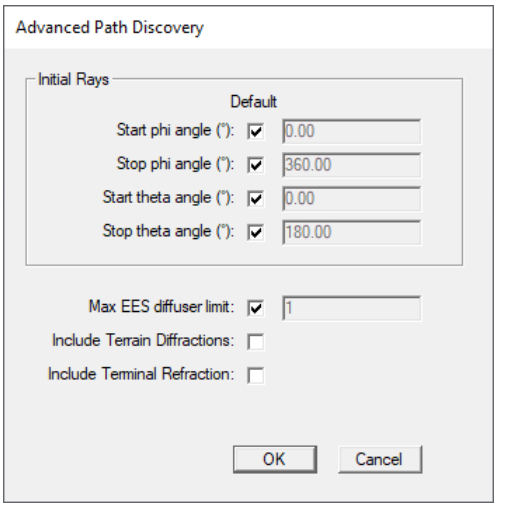


Figure 15.8: Advanced Path Discovery Parameters

Foliage Model

X3D provides two options for modeling foliage: the Weissberger model and a simple attenuation model. There is a drop-down to select which model to use, and a field for inputting the dB/m attenuation due to foliage.

- ✓ Note that *foliage* geometry must be defined in the project for this to impact propagation.
- When running X3D, all foliage geometry will be modeled using the selected method, regardless of what material properties were assigned to each geometry.

Atmosphere

The X3D model supports atmospheric absorption for oxygen and water. Input values for *Temperature*, relative *Humidity*, and *Pressure*, that are used by the X3D model to calculate atmospheric absorption. Details of the algorithm are provided in Section 16.2.

APG Acceleration

Checking the *APG Enabled* button activates the X3D model *Adjacent Path Generation* acceleration capability. APG uses X3D's exact path algorithm and the paths found to a coarse set of receivers, to quickly find exact paths to a denser set of receivers. In the APG ACCELERATION PROPERTIES window, specify *Adjacency Distance* and *Paths to Consider*.

Monte Carlo

The X3D model includes a Monte Carlo capability, useful for simulations where there is some uncertainty about environmental parameters. In the STUDY AREA PROPERTIES window check the *MC Enabled* box. This will automatically request statistical results related to the capability and additional Monte Carlo parameters can be set in the MONTE CARLO PROPERTIES window.

✓ Note that Monte Carlo *materials* or transmit *Input power* must be defined in the project for this to impact propagation.

▶ For more information on parameter uncertainty and Monte Carlo capabilities, see Chapter 19.

MPE

Checking the *Maximum Permissible Exposure* box activates the MPE calculation in X3D. User can choose between IEEE Controlled, IEEE Uncontrolled or user defined environments.

▶ For more information on Maximum Permissible Exposure, see Chapter 18.

Diffuse Scattering

Checking the *Diffuse Scattering Enabled* box enables Diffuse Scattering interactions.

▶ For more information on Diffuse Scattering, see Chapter 20.

Partitioning and Queuing

The PARTITIONING AND QUEUING PROPERTIES window, shown in one configuration of parameters in Figure 15.9, exposes two additional options for running a simulation.

- *Partitioning type*

  The *Partitioning type* option allows users to split the simulation into multiple parallel processes based on *Partitioning type*. There are two partitioning types: *Partition by transmitter point* and *Partition by receiver set*; however, the default option is not to partition. When partitioning by transmitter point, the user must choose a number of partitions to create. The total number of active transmitter points in the scenario will be split across those partitions. When using local queuing in conjunction with partitioning, the number of CPU threads specified in the study area properties window will be used in each parallel process that launches when the simulation begins. Wireless InSite will not use more CPU threads than what has been specified in the application's SIMULATION preferences.

- *Queue type*

  The *Queue type* option allows users to specify whether to run on the user's workstation (*Local* queuing) or on a cluster (*External* queuing). The default option is *Local* queuing.

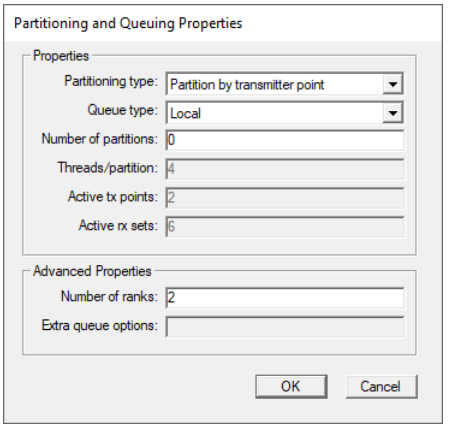


Figure 15.9: Partitioning and Queuing Properties for a Sample Configuration

15.4 Study Area Operations

A set of operations are available to modify the STUDY AREA through its context menu. Multiple study areas can be selected and modified by holding the control key while clicking on their entries in the PROJECT TREE.

- *Duplicate* - Creates a duplicate of the study area, including the model type and all related parameters.
- *Delete* - Removes the study area from the project.
- *Edit Boundary* - Opens the boundary editor so that the vertices of its footprint can be edited manually.
- *Crop geometry to* - Removes the following geometry:
  - Faces of the *Terrain* that do not lie within the study area boundary.
  - Any building in any *City* that lies outside of the study area boundary.
  - Any object in any *Object* that lies outside of the study area boundary.
  - Any floor plan in any *Floor plan* that lies outside of the study area boundary.
  - Faces of a *Foliage* geometry that do not lie within the study area boundary.

Chapter 16

Propagation Models

In this chapter, you will learn...

→ the characteristics, limits and electric field evaluation of each Wireless InSite propagation model

Wireless InSite provides several ray-based propagation models: X3D, FULL 3D, URBAN CANYON, and VERTICAL PLANE. These models all combine ray-tracing algorithms with Geometric Optics (GO) and the Uniform Theory of Diffraction (UTD) [7] [49] [50]. Ray-tracing methods are used to find the propagation paths between each transmitter and receiver point. The physics methods (GO, UTD) are then used to evaluate the complex electric fields associated with each ray path.

- Chapter 21 describes how the propagation predictions for quantities, such as received power, and path loss, are computed from the electric fields and propagation paths.

In addition to the ray-based methods, Wireless InSite's Real Time models provide several fast-running models. The simplest of these are FREE SPACE, HATA [51], and COST-HATA [52]. Four additional fast options provide higher fidelity results than empirical models while drastically reducing simulation times from that of the full physics ray-tracing models. The Opnet Attenuation Routine (OPAR), and the Walfisch-Ikegami models are semi-deterministic models that account for building attenuation and roof top diffraction respectively. The Vertical Plane Urban Propagation (VPUP) model, and the Triple-Path Geodesic (TPG) models are based on convex hull diffraction and use accurate UTD calculations to determine received power and all other results.

Each section of this chapter starts with an overview describing how the model works, followed by a short summary of the model's capabilities and limitations. Then the under-lying algorithms used for that model are described, and the section concludes with a discussion of how the electric fields are calculated. The angles used in discussions of the electric field components or antenna gain are specified in terms of the spherical components at the TRANSMITTER or RECEIVER point using a spherical coordinate system, shown in Figure 16.1. The chapter concludes with a summary of model capabilities and inputs.

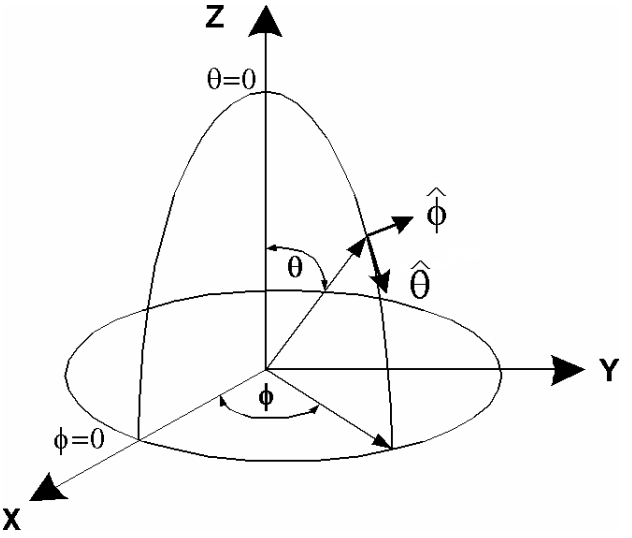


Figure 16.1: The Wireless InSite spherical coordinate system

16.1 Full 3D Model

Overview

The FULL 3D model propagates rays through the project's urban geometry and includes the effects of reflections, transmissions, and diffractions on the electric field. FULL 3D places no restriction on object shape, allowing buildings to have flat or sloped roofs. It also includes transmission through surfaces allowing it to model propagation in indoor environments.

The combined number of reflections and transmissions cannot exceed 30. If the combined number of reflections and transmissions is greater than 30, each value is reduced proportionately so that they sum to 30. When transmissions are included in a project, all facets, except those comprising the *Terrain* and *Foliage*, should typically be doubled-sided.

- Refer to Chapter 3 for more information on the differences between doubled-sided and single-sided facets and the options for changing this attribute.

Summary

The following list summarizes the capabilities of the FULL 3D model:

- Maximum reflections: 30 (SBR,assuming no transmissions), 3 (Eigenray, assuming no diffractions)
- Maximum transmissions: 30 (SBR,assuming no reflections), 30 (Eigenray)
- Maximum diffractions: 4 (SBR), 3 (Eigenray, assuming no reflections)
- Environments: all
- Terrain: all

- Urban: all
- Foliage: direct waves, no lateral wave
- Indoor: all, facets should usually be double-sided
- Objects: all
- Range: depends on application
- Antenna heights: all
- Antenna types: all
- Ray tracing: SBR or Eigenray
- Minimum frequency: 100 MHz
- Maximum frequency: depends on application

Ray Tracing

Two ray tracing methods are available with the FULL 3D model: the Shooting and Bouncing Ray (SBR) method and the Eigenray method.

Shooting and Bouncing Ray (SBR) Method

The SBR method is first employed to trace ray paths through the three-dimensional building geometry without regard for the location of specific field points, using the procedure described in [53] and [54]. Rays are first traced from the source points with the rays reflecting specularly from the building walls. The rays that hit building walls are reflected specularly and continue to be traced up to the maximum number of reflections, or when the rays hit the STUDY AREA boundary.

Figure 16.2 shows the rays shot from the source point that end at the first reflection point on a building wall or that intersect the STUDY AREA boundary. The ray density shown in the figure is much less than the density used in a typical simulation.

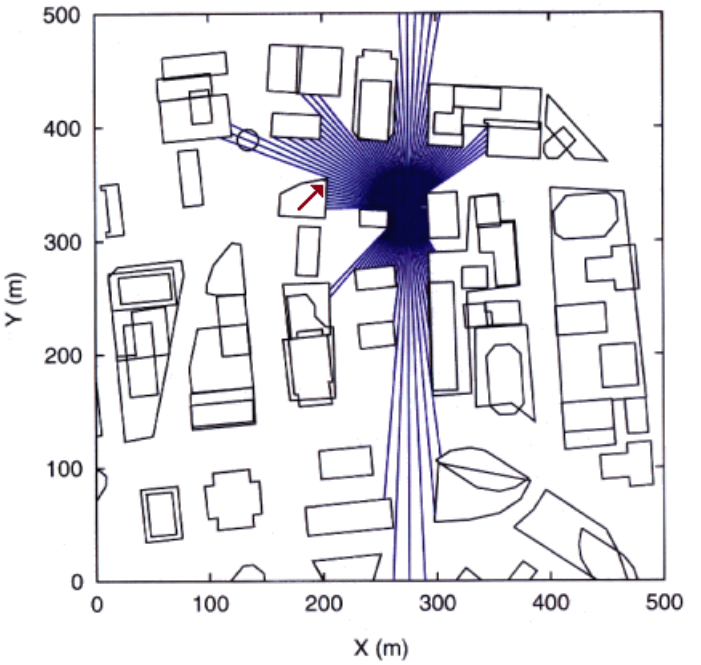


Figure 16.2: Rays, shot from a source, end on the reflection point on a building surface or terminate on the outer boundary.

Before describing how the SBR paths are used to find the geometrical propagation paths from the transmitter to the receiver points, the procedure for identifying diffracting edges will be described. In the Geometrical Theory of Diffraction (GTD), diffractions occur at the points where the field becomes discontinuous. The first order diffracting edges are found by searching for adjacent rays which follow different paths through the geometry, since such occurrences identify discontinuities in the Geometrical Optics (GO) fields. A diffracting edge can then be located between these rays. The arrow near the center of Figure 16.2 points to a diffracting edge identified in this way.

Figures 16.3 and 16.4 show examples of how the rays shot from the transmitter are used to identify diffracting edges. For example, the two adjacent SBR rays in Figure 16.3 both reflect from facet 3, but only one reflects from a facet 55. This means that a diffracting edge lies between these two rays, and it is then quite simple to locate a diffraction point on edge 55-56 and to construct the path followed by the incident field.

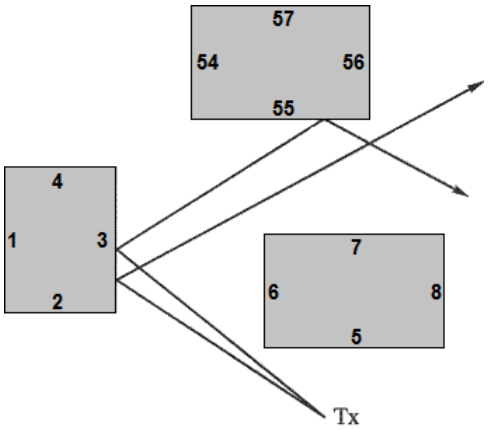


Figure 16.3: Rays which identify edge 55-56 as a diffracting edge

A different diffraction situation is shown in Figure 16.4, where the edge 10-11 would be identified as a diffracting edge for the incident field, which first reflects from facet 13.

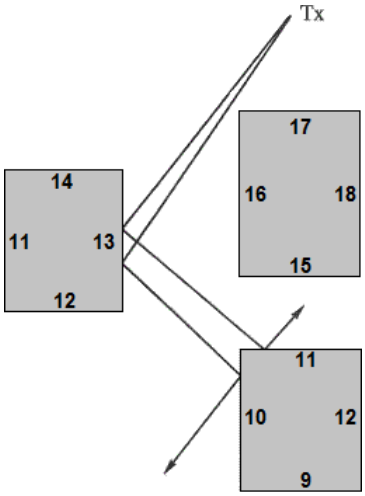


Figure 16.4: Rays which identify edge 10-11 as a diffracting edge

The two situations differ in that the latter example has two reflection shadow boundaries, whereas the former example has an incident shadow boundary and a reflection shadow boundary. In an urban environment, either situation can give rise to a strong propagation path into a non-line-of-sight street.

Once the rays have been shot and bounced from all the active TRANSMITTERS, rays are then shot and traced from all the diffracting edges. This procedure is then repeated if higher order diffractions are to be included. Figure 16.5 shows an example of rays shot from the diffracting edge indicated in Figure 16.2.

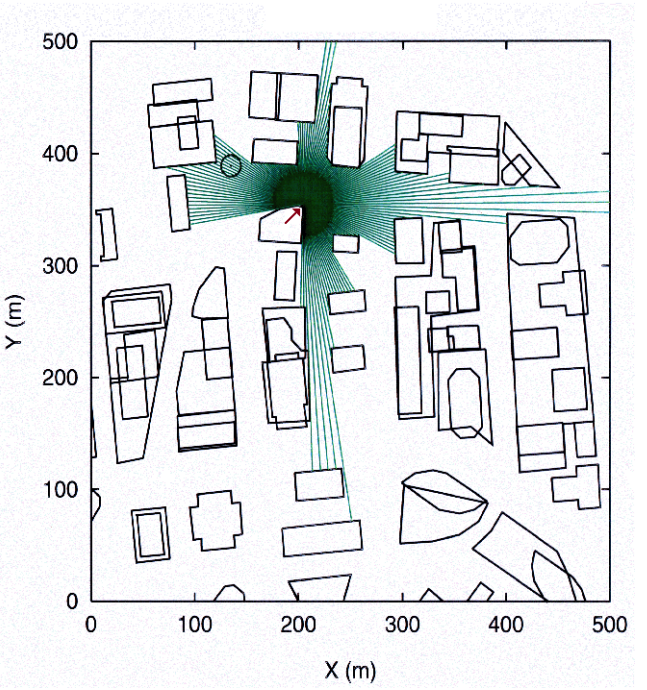


Figure 16.5: Rays to the first reflection points shot from the diffracting edge

Once the SBR paths have been traced from a transmitter or edge, the next step is to construct the specific geometrical paths to each field point from the transmitter or diffracting edge. Because SBR launches rays at discrete angles, it is unlikely that any ray will pass exactly through a field point. To compensate for the spacing between rays, a collection sphere is constructed around the field point. Rays that pass through this surface are used to construct the specific GO and GTD ray paths to the field point. The circle near the point (125, 400) in Figures 16.2 and 16.5 represents this collection sphere.

The circle shown here is 2 to 5 times larger than the collection surface used in a typical urban simulation.

For valid ray spacing and collection surface sizes, the SBR method will commonly find a number of rays that have followed essentially the same path through the project geometry. These types of rays represent the same wave front and therefore contain duplicate energy. To avoid over predicting energy at the field point, the rays are sorted according to the geometry faces they interacted with on the way to the field point. Rays that have similar interactions with the same geometry faces are identified as duplicates.

For example, several line-of-sight rays intersect the circular collection surface shown in Figure 16.2. The collection process finds all of these. Sorting the rays identifies duplicate paths, and from the duplicates, one unique path is selected. Usually the ray passing closest to the field point is selected.

Many ray paths will have a companion ray that follows nearly the same path but reflects from the ground. The ground reflection points on these rays are found analytically using the method of images. Once the full 3D paths are constructed, the endpoints of each segment of the path are stored to a file in the study area folder, along with the sequence of interactions. These ray paths are then used to evaluate the electric field strength.

- Section 16.1 describes the evaluation of electric field strength.

Before describing how the SBR paths are used in the evaluation of the electric field, the question of how to set the angular spacing between rays to shoot should be addressed. There are two considerations when determining the number of rays to shoot. The primary is to set the spacing of the SBR rays dense enough to ensure that at least two rays for each unique ray path intercept the collection surface surrounding each field point. The estimate of the angular spacing in radians is:

$$\triangle \phi \approx \frac{R_{collect}}{D_{max}} \tag{16.1}$$

where D_{max} is the maximum distance across the area being considered, and $R_{collect}$ is the radius of the collection sphere.

The secondary consideration is to consider the ray spacing relative to the geometry. The spacing should be set small enough that at least two rays intersect the majority of faces in line of sight of the receiver. This condition is commonly satisfied when considering the ray spacing relative to the collection surface, but in some instances, it can be the determining factor in choosing a ray spacing.

✓ Experience has shown that a ray spacing of 0.2° and a collection surface 2.5 m in radius works well in most situations.

The SBR method can construct ray paths with up to 30 total reflections and transmissions. The ray paths can also undergo diffractions.

Although the method allows for up to 30 reflections and transmissions on a ray path, the computation time can become large when both reflections and transmission are requested. The computation time is roughly proportional to:

$$\frac{(N_R + N_T + 1)!}{N_R! N_T!} \tag{16.2}$$

where N_R is the number of reflections and N_T the number of transmissions. If no diffractions are requested, the computation time will roughly be proportional to the number of facets in the geometry, with double-sided facets counting as two facets. When a single diffraction is requested, the computation time is roughly proportional to the number of facets squared. Requesting additional surface diffractions does not greatly affect the run time due to the limitation to coplanar edges.

Eigenray Method

The other ray tracing method is the called the Eigenray method. This approach involves an explicit construction of the ray paths between each TRANSMITTER and RECEIVER which satisfy Fermat's Principle of least time, except that there is no refraction at transmissions. The Eigenray method is limited to ray paths with up to three reflections and diffractions. Like the SBR method, it finds up to 30 transmissions per ray path. However, this method has the following path-finding restrictions:

- No more than a combined total of 3 reflections and diffractions per ray path are found (Tx-D-D-R-Rx or Tx-R-R-R-Rx).

- Paths which diffract, reflect, and then diffract (Tx-D-R-D-Rx) are not constructed.
- Paths with 3 diffractions (Tx-D-D-D-Rx) must diffract off of parallel edges (typically all vertical or all horizontal edges).

The computation time with the Eigenray method is roughly proportional to $(N_F)^{\chi}$, where N_F is the number of facets, and χ is the maximum number of reflection plus diffraction interactions on any path, with the maximum being three.

✓ Because the computation time does not increase significantly with the number of transmissions, the Eigenray method will often be a good choice for applications requiring a large number of transmissions, as long as the restrictions on the number of reflections and diffractions is acceptable.

Electric Field Evaluation

The first step in evaluating the electric field is to find the electric field in the far zone of the transmitting antenna. At present, Wireless InSite does not consider any near zone fields. In free space the electric field in the direction (θ, ϕ) in the far field of the transmitting antenna at a distance r can be written as:

$$E(r,\theta,\phi) = (A_\theta(\theta,\phi)\hat{e}_\theta \ + \ A_\phi(\theta,\phi)\hat{e}_\phi) \ \frac{e^{-j\beta r}}{r} \tag{16.3}$$

where

$$A_\theta(\theta,\phi) = \sqrt{\frac{P_T\eta_0}{2\pi}}g_\theta(\theta,\phi) \tag{16.4}$$

$$A_\phi(\theta,\phi) = \sqrt{\frac{P_T\eta_0}{2\pi}}g_\phi(\theta,\phi) \tag{16.5}$$

$$g_\theta(\theta,\phi) = \sqrt{\mid G_\theta(\theta,\phi) \mid}e^{j\psi_\theta} \tag{16.6}$$

$$g_\phi(\theta,\phi) = \sqrt{\mid G_\phi(\theta,\phi) \mid}e^{j\psi_\phi} \tag{16.7}$$

$$\beta = \frac{\omega}{c} \tag{16.8}$$

where

$G_\theta(\theta,\phi)$ is the θ component of the gain of the transmitting antenna

ψ_θ is the relative phase of the θ component of the far zone electric field

P_T is the time-averaged power radiated by the transmitter

r is the distance from the transmitter to the field point

In a fully three-dimensional simulation, the equations for the electric field amplitudes are somewhat complicated. This is primarily because the polarization of the incident electric field at each reflection is neither entirely parallel or perpendicular to the plane of incidence but is some combination of the two, and therefore must be broken down into these components by using a ray-fixed coordinate system at each reflection. The reflected field is then calculated from the dyadic reflection coefficient, where the components of the reflected electric field parallel and perpendicular to the reflection plane are given by:

$$\begin{pmatrix} E_{\parallel}^{r} \\ E_{\perp}^{r} \end{pmatrix} = \begin{pmatrix} R_{\parallel} & 0 \\ 0 & R_{\perp} \end{pmatrix} \begin{pmatrix} E_{\parallel}^{i} \\ E_{\perp}^{i} \end{pmatrix} \tag{16.9}$$

where

$$E_{\perp}^{i} = \hat{e}_{\perp} \cdot E^{i}$$

$$E_{\parallel}^{i} = \hat{e}_{\parallel} \cdot E^{i}$$

$$\hat{e}_{\perp} = \frac{k \times \hat{n}}{|\, k \times \hat{n}\,|}$$

$$\hat{e}_{\parallel} = \frac{k \times \hat{e}_{\perp}}{|k \times \hat{e}_{\perp}|}$$

$$\hat{e}'_{\parallel} = \frac{k' \times \hat{e}_{\perp}}{|k' \times \hat{e}_{\perp}|}$$

The reflected field in the original reference frame is $E^{r} = \hat{e}'_{\parallel} \cdot E_{\parallel}^{r} + \hat{e}_{\perp} \cdot E_{\perp}^{r}$.

The vector **k** is in direction of propagation of the incident field, **k'** is in the direction of propagation of the reflected field, and **n** is the unit vector normal to the reflecting face.

Equation 16.9 gives the amplitude for a ray which undergoes a single reflection. For rays undergoing multiple reflections, the amplitude no longer simply depends on a product of the reflection coefficients (as it does in the URBAN CANYON model), but must be evaluated by transforming to the new ray-fixed frame at each reflection.

The diffraction coefficients used for the FULL 3D model are similar to those given in [55]. These coefficients include the angle-dependent reflection coefficients for the particular building face and thus keep the fields continuous at the reflection boundaries. These diffraction coefficients are for fields polarized either parallel or perpendicular to the diffracting edge. In two dimensions this always holds since the fields are always incident normal to the edge, whereas in three dimensions the fields may be obliquely incident and the field can be polarized in any direction relative to the edge. In order for the fields to remain continuous at the reflection boundaries, it is necessary to generalize the coefficients in Luebbers [55] using the results in Burnside [56] to allow for arbitrary incident direction and polarization.

First, the incident and diffracted fields are expressed in terms of an edge-fixed coordinate system [56]. The unit vector **l** is parallel to the diffracting edge, **k** is the propagation vector for the diffracted field, and **k'** is the propagation vector for the incident field. The following unit vectors are then defined by:

$$\hat{\phi}' = \frac{-\hat{l} \times \hat{k}'}{\left|\hat{l} \times \hat{k}'\right|} \tag{16.10}$$

$$\hat{\beta}' = \hat{\phi}' \times \hat{k}' \tag{16.11}$$

The components of the incident field in this edge-fixed coordinate system are $E^i_{\phi'} = E^i \cdot \hat{\phi}'$ and $E^i_{\beta'} = E^i \cdot \hat{\beta}'$. The components of the diffracted field at the receiver location (omitting any reflections following the diffraction) in this coordinate system are given by:

$$\begin{pmatrix} E^d_\beta \\ E^d_\phi \end{pmatrix} = \begin{pmatrix} D_{ss} & D_{sh} \\ D_{hs} & D_{hh} \end{pmatrix} \begin{pmatrix} E^d_{\beta'} \\ E^d_{\phi'} \end{pmatrix} \sqrt{\frac{r_{TD} \cdot r_{DR}}{r_{TD} + r_{DR}}} \frac{e^{-j\beta\, r_{DR}}}{r_{DR}} \tag{16.12}$$

The diffracted field is then given by:

$$E^d = E^d_\beta \, \hat{\beta} + E^d_\phi \hat{\phi} \tag{16.13}$$

The dyadic equation allows for the fields to be arbitrarily polarized and for any angle-dependent reflection coefficient to be used. The elements of the diffraction dyadic are given by:

$$\begin{aligned} D_{ss} = \; & D^+(\phi - \phi') + D^-(\phi - \phi') - \\ & \left[R^0_\parallel cos^2\alpha_1 - R^0_\perp sin^2\alpha_1\right] D^-(\phi + \phi') - \left[R^n_\parallel cos^2\alpha_2 - R^n_\perp sin^2\alpha_2\right] D^+(\phi + \phi') \end{aligned} \tag{16.14}$$

$$\begin{aligned} D_{hh} = \; & D^+(\phi - \phi') + D^-(\phi - \phi') + \\ & \left[R^0_\parallel sin^2\alpha_1 - R^0_\perp cos^2\alpha_1\right] D^-(\phi + \phi') + \left[R^n_\parallel sin^2\alpha_2 - R^n_\perp cos^2\alpha_2\right] D^+(\phi + \phi') \end{aligned} \tag{16.15}$$

$$D_{sh} = -\left[R^0_\parallel + R^0_\perp\right] cos\alpha_1 sin\alpha_1 D^-(\phi + \phi') - \left[R^n_\parallel + R^n_\perp\right] cos\alpha_2 sin\alpha_2 D^+(\phi + \phi') \tag{16.16}$$

$$D_{hs} = \left[R^0_\parallel + R^0_\perp\right] cos\alpha_1 sin\alpha_1 D^-(\phi + \phi') + \left[R^n_\parallel + R^n_\perp\right] cos\alpha_2 sin\alpha_2 D^+(\phi + \phi') \tag{16.17}$$

The 0 and n superscripts are used to designate the two sides of the wedge as shown in Luebbers [55] and in Holm [57] as well as other papers on the UTD; ϕ and ϕ' are the observation and incidence angles as measured in Balanis [7] and Luebbers [55].

The angles α_1 and α_2 are determined from $sin\,\alpha_1 = \hat{e}^0_\parallel \cdot \hat{\phi}'$ and $sin\,\alpha_2 = \hat{e}^n_\parallel \cdot \hat{\phi}'$, where the unit vectors $\hat{e}^0_\parallel$ and $\hat{e}^n_\parallel$ are the ray-fixed vectors parallel to the plane of incidence.

The functions $D^\pm(\phi \pm \phi')$ are given by:

$$D^\pm(\phi \pm \phi') = \frac{-e^{-j\pi/4}}{2n\sqrt{2\pi k}} cot\left(\frac{\pi \pm (\phi \pm \phi')}{2n}\right) \cdot F(kLa^\pm(\phi \pm \phi')) \tag{16.18}$$

where $F(kLa^{\pm}(\phi \pm \phi'))$ is the Fresnel integral, defined in several references ([7],[50],[55]).

When the field is incident normal to the edge $(\alpha_1 = \alpha_2 = 90°)$, the off-diagonal elements vanish $(D_{sh} = D_{hs} = 0)$ and the diagonal elements D_{hh} and D_{ss} reduce to the diffraction coefficients in Luebbers [55]. For perfect conductor $\left(R_{\parallel} = 1,\ R_{\perp} = -1\right)$, again $D_{sh} = D_{hs} = 0$, and the diagonal elements reduce to the coefficients in [7] and [50].

After each reflection or diffraction is applied, the electric field is transformed back into Cartesian components. The final step is to determine the spherical components of the electric field at the receiver point from the Cartesian components using:

$$\begin{pmatrix} E_r \\ E_\theta \\ E_\phi \end{pmatrix} = \begin{pmatrix} sin(\theta_A)cos(\phi_A) & sin(\theta_A)sin(\phi_A) & cos(\theta_A) \\ cos(\theta_A)cos(\phi_A) & cos(\theta_A)sin(\phi_A) & -sin(\theta_A) \\ -sin(\phi_A) & cos(\phi_A) & 0 \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix} \tag{16.19}$$

where (θ_A, ϕ_A) is the direction from which the ray arrives at the receiver. For TEM fields, the radial component E_r calculated from the above equation should be always be zero.

16.2 X3D Ray Model

Overview

The X3D ray model was developed to provide a highly accurate, full 3D propagation model capable of running on a graphics processing unit (GPU) and using multi-threading to take advantage of multi-core processors. This full 3D ray-tracing model also uses Remcom's depth-first and exact path algorithms to overcome some of the shortcomings of the traditional shooting and bouncing ray method. Key benefits of the X3D model are the speed-ups achieved by the GPU acceleration and multithreading, and the accuracy achieved through the exact path calculations.

Wireless InSite's X3D model also provides the following capabilities that other models do not support:

- MIMO simulation capability
- APG (Adjacent Path Generation) Acceleration for dense receiver sets
- Atmospheric absorption
- Monte Carlo simulations with varying material properties, transmit power, and frequency
- Diffuse Scattering calculations
- Engineered Electromagnetic Surfaces
- Maximum Permissible Exposure calculations
- Modified COST Building Penetration Model for outdoor to indoor propagation

This model is significantly different from Full 3D and will produce different results in some cases.

Summary

The following list summarizes the capabilities of the X3D model:

- GPU Acceleration and multithreading
- Maximum reflections: 30
- Maximum transmissions: 8
- Maximum diffractions: 3
- Environments: urban, indoor
- Terrain: any
- Urban: all
- Foliage: Weissberger or dB/m
- Indoor: all
- Objects: all
- Range: depends on application
- Antenna heights: all
- Antenna types: all
- Ray tracing: SBR with Exact Path correction
- Minimum frequency: 100 MHz
- Maximum frequency: depends on application, but extends into the millimeter wave frequencies

Ray Tracing

The X3D SBR algorithm is similar to the algorithm as described for the Full 3D model, with the following differences:

- Diffractions are not identified by looking for direction differences between adjacent rays from similar reflections. Instead, when a reflection occurs close enough to an edge, a discrete set of points along that edge are considered. The reflected path off the face, and the diffracted paths from the edge points are traced further.
- After finding the SBR paths, X3D implements an Exact Path Calculator (EPC) algorithm. As discussed in Section 16.1, ray paths intersect receiver targets within a tolerance defined by a sphere. EPC then adjusts the actual interaction points (reflections, diffractions and transmissions) so that the path hits the exact receiver center. EPC also validates that all of the diffraction and reflection angles are valid and that the corrected path does not get blocked by any intervening faces. Paths that fail this validation are discarded.
- X3D lets rays hit either side of a face, ignoring the single-sided versus double-sided property of a face.

▶ FULL 3D 's SBR ray tracing method is described in Section 16.1.

Exact Path calculations

The shooting and bouncing method requires a collection radius to be constructed around receiver locations to compensate for the discrete ray shooting. Rays that intersect this sphere are considered to reach the receiver. Exact path corrects SBR ray paths so that they end at the exact receiver location. This correction reduces errors in calculated power and phase associated with SBR, but without the longer run times required by methods based on image theory.

Electric Field Evaluation

Electric fields in X3D are evaluated in the same manner as FULL 3D.

- See description in Section 16.1.

Atmospheric Absorption

Simulations made with the X3D Ray Model include frequency dependent absorption due to oxygen and water content of the atmosphere. Users can specify the temperature, relative humidity, and pressure within the X3D ATMOSPHERICPROPERTIES window, accessed through the ATMOSPHERE button in the X3D study area properties window. Path loss and received power calculations will be reduced based on the distance a ray travels, frequency, and specified atmospheric properties. Setting each field to zero will turn off the absorption loss.

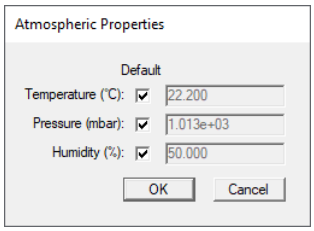


Figure 16.6: The X3D Atmospheric Properties Window

The atmospheric absorption model used in the software was adapted from a public domain model presented in [58]. The behavior of the specific attenuation of the model as a function of frequency conforms to ITU recommendation ITU-R P.676-9 [59]. Results from the atmospheric absorption model integrated into Wireless InSite are shown in Figure 16.7, below, while the loss described in the ITU recommendation is shown in Figure 16.8.

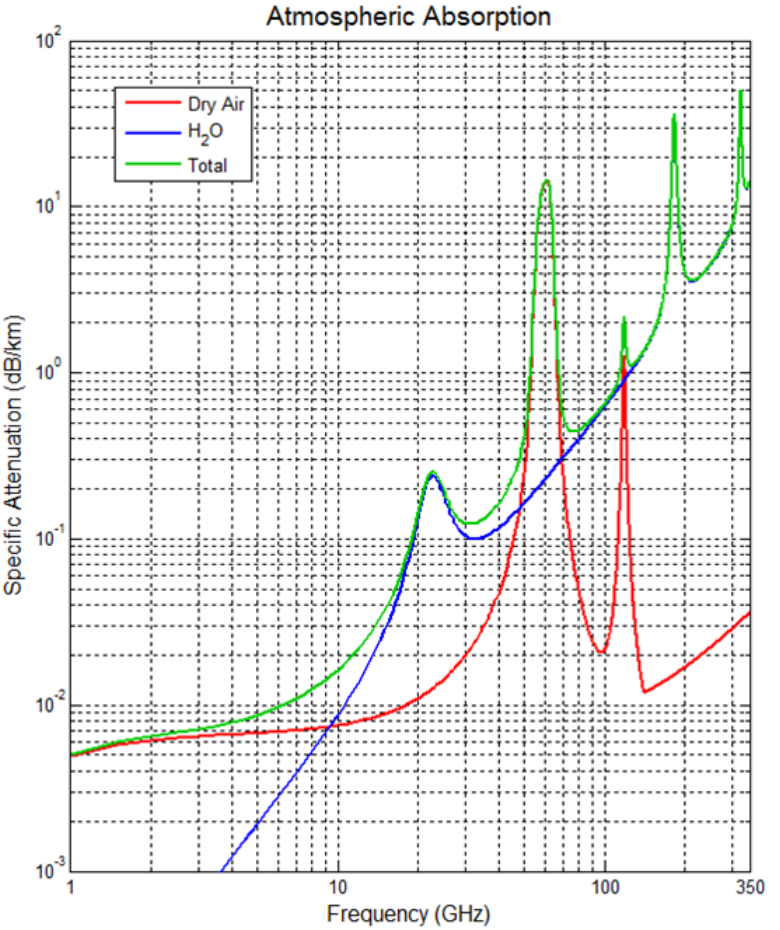


Figure 16.7: Specific Attenuation as predicted by Wireless InSite's X3D

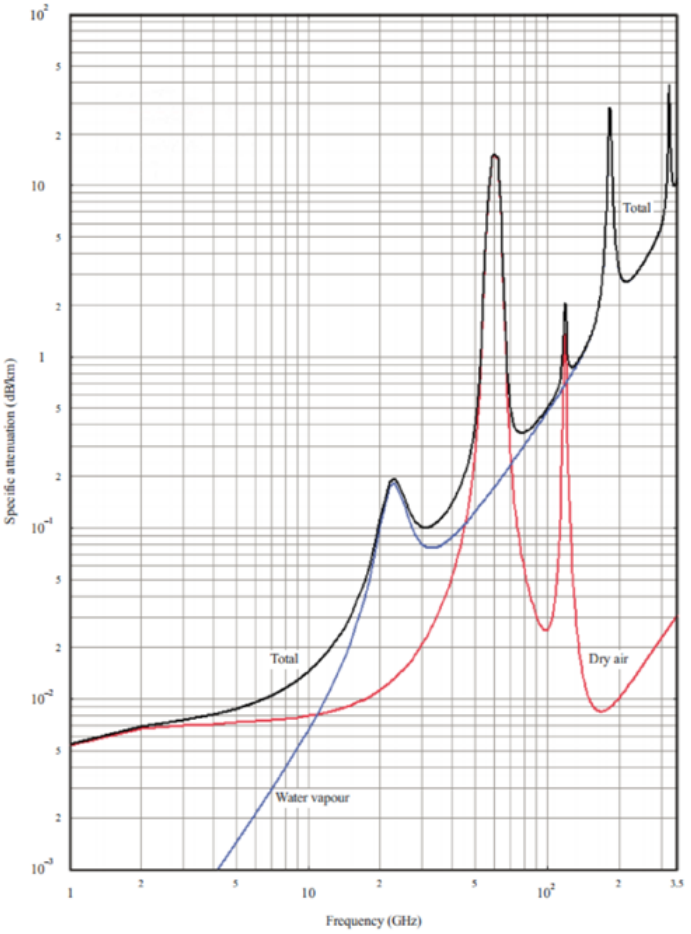


Figure 16.8: ITU-R P.676-9 Specific Attenuation due to Oxygen and Water Vapor

MIMO Simulations

The X3D model provides a unique ray tracing capability for simulating MIMO antennas for 5G, WiFi and other applications. Wireless InSiteMIMO simulates the detailed multipath of large numbers of MIMO channels while overcoming the increased level of computations required for traditional ray tracing methods.

Wireless InSite's MIMO ARRAY BUILDER gives users the ability to create 1D, 2D, and 3D antenna arrays. Users have control over every aspect of their model and can define unique antenna patterns and orientations for each element.

- The MIMO ARRAY BUILDER is described in Section 12.6.

All data is accessible, allowing results to be visualized within the context of the scene, as a 2D plot, or exported to a file. Post-processing options include received power, complex channel matrix or H-matrix, complex impulse response, times of arrival, directions of arrival, and directions of departure for multipath.

- The MULTI-CHANNEL RESULTS BROWSER is described in Section 22.4.

Adjacent Path Generation (APG)

X3D's Adjacent Path Generation capability reduces run time and memory footprint for scenarios involving large numbers of receivers, or dense receiver sets. This is particularly useful for MIMO simulations. APG limits ray tracing to a coarsely spaced set of points within a *route*, *grid*, or *arc* receiver set, then uses the interactions among coarse paths to determine exact paths to the dense set of receivers defined by the user. Figure 16.9 shows paths to a coarse set of points along a *route* representing a mobile device, and shows how the interactions of the paths to the coarse set are used to find exact paths to the dense set of receivers along the *route*.

✓ Run time reduction may be order of magnitude or more depending on the receiver spacing within the route, arc or grid.

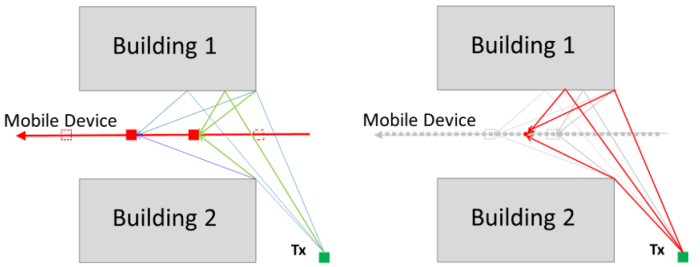


Figure 16.9: APG limits ray trace to coarse set of points, then rapidly and accurately finds exact paths to dense set of receivers

To use ADJACENT PATH GENERATION, the APG ENABLED button in the X3D STUDY AREA window must be checked, and at least one TRANSMITTER or RECEIVER set must have APG active. Clicking on the APG ENABLED button opens the APG PROPERTIES window, where the *Adjacency distance* and *Paths to consider* are specified. *Adjacency distance* refers to the spacing of the coarse points to which the ray-tracing finds potential paths, while *Paths to consider* refers to the number of paths per point that the APG algorithm considers.

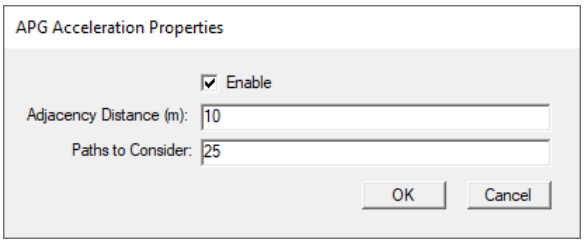


Figure 16.10: APG Properties window with fields for adjacency distance and number of paths to consider

APG settings for TRANSMITTER or RECEIVER sets are accessed through the TX/RX LAYOUT PROPERTIES window. By default, APG is active for *route* *grid* and *arc* sets, and uses the APG settings defined in the STUDY AREA. If the user wishes to have different *Adjacency Distance*s defined for different layout sets, this can be defined in the TX/RX LAYOUT PROPERTIES window.

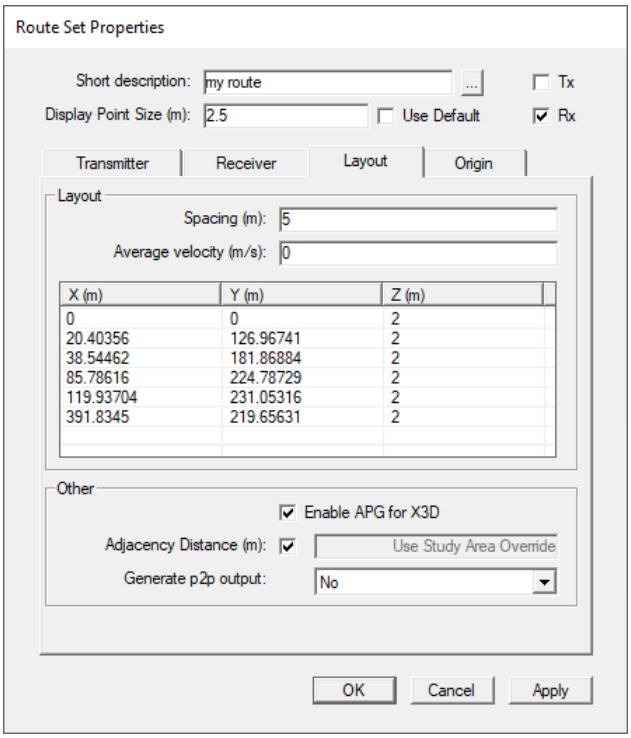


Figure 16.11: Tx/Rx layout properties window showing APG enabled.

Modified COST Building Penetration Model

- The modified COST Building Penetration models outdoor to indoor propagation when the interior details of a building are not explicitly defined. A full description of this capability is available in the Appendix B, Section B.6.

Parameter Uncertainty Simulations

- X3D allows certain model inputs like frequency, permittivity, and conductivity, to be defined with a distribution. Using a Monte Carlo approach, the results are generated by sampling the input distribution to estimate the mean, median, minimum, maximum and standard deviation of the results. A full description of this capability is available in Chapter 19.

Maximum Permissible Exposure Calculations

- Maximum Permissible Exposure Calculations are described in Chapter 18.

Diffuse Scattering Calculations

To enable DIFFUSE SCATTERING in X3D, check the *Diffuse Scattering Enabled* checkbox in the X3D STUDY AREA PROPERTIES window. Note that in order to run DIFFUSE SCATTERING calculations, it must be enabled in both the relevant MATERIAL and STUDY AREA properties. By default, it is disabled in both. For information about DIFFUSE SCATTERING properties, go to Chapter 20.

Engineered Electromagnetic Surfaces

Wireless InSite's X3D Model has integrated the Communications Research Centre Canada's (CRC) Ray-optical Engineered Electromagnetic Surface (EES) scattering model (EES Scattering Model) in order to calculate EM propagation in environments that include metasurfaces with electromagnetic material properties that do not adhere to Snell's law. An EES is constructed by printing conductive, dielectric, and/or ferromagnetic ink on a substrate such as plastic, glass, or drywall. The pattern printed on the substrate causes the incident electromagnetic field to reflect, transmit, and/or diffract in directions that do not generally obey conventional geometrical constraints (i.e., Snells law and the Keller cone). [60]

The EES Scattering model provides a uniform ray description of electromagnetic wave scattering by an EES, or more generally by locally periodic metasurfaces of polygonal shape. It is valid for metasurfaces whose bulk scattering coefficients are locally periodic functions of a phase parameter. The gradient of the phase parameter on the surface is represented by two continuous and smooth functions of surface coordinates, one for each direction on the surface. The scattered field is expressed in terms of reflected, transmitted, and diffracted rays that propagate in directions determined by the incident direction and the phase gradient functions. [60]

The integration of the EES Scattering model into Wireless InSite's X3D model enables ray-based EM propagation prediction in complex, electrically large environments, such as office buildings and urban areas equipped with metasurfaces. [61] The EES materials are described in terms of the material's electric and magnetic sheet impedances and phase gradients. Each EES material's sheet impedances and phase gradients are specified by a file in a format given in Appendix G

Although it is possible for a user to construct an EES definition file with an external process, most users will opt for an automated tool in Wireless InSitethat allows description of the EES properties at a higher level. Inputs to this tool include the horizontal deflection angles of a normally incident ray at the right and left edges, the vertical deflection angles at the top and bottom edges, and the desired reflection and transmission coefficients.

Geometry Processing and Caching

Before the X3D model runs the simulation it must first load and process all active geometry GEOMETRY. Wireless InSite maintains a set of cache files of the processed geometry in order to bypass this potentially time consuming step in subsequent simulations.

There are two types of cache files that are maintained for use by the X3D model: one with and one without terrain edges. Wireless InSite will pass the correct one to the simulation based on whether or not the STUDY AREA includes terrain diffractions. The *C* column in the STUDY AREAS properties tab of the MAIN WINDOW indicates if the cache file associated with that STUDY AREA is still valid. If the associated cache file is invalid, then Wireless InSite will run a preprocessing job to generate it before launching the X3D simulation.

Because the cache represents the state of all active geometry GEOMETRY for an X3D simulation, making modifications to geometry in an active GEOMETRY causes it to become invalid. The following list outlines the actions that a user can invoke that cause the cache to become invalid.

- Operations that are performed at the GEOMETRY level.
  - Adding a new GEOMETRY to the project.
  - Deleting an existing GEOMETRY from the project.
  - Toggling the active flag.
  - Modifying the origin and its orientation.
  - Modifying what the elevations of its faces are relative to terrain or sea-level.
  - Adding or deleting MATERIALS even if they are not used.
- Operations that can be performed on faces at any level of the GEOMETRY hierarchy. This includes the GEOMETRY, structure groups, structures, substructures, or individual facets.
  - Translating.
  - Rotating.
  - Scaling.
  - Joining co-planar faces.
  - Lowering or extruding to the terrain.
  - Toggling the sidedness of faces.
  - Replacing MATERIALS.
- Operations that can only be performed at the face level.
  - Specifying exact locations of vertices.

Invalidation of the geometry cache is not permanent until the changes that caused it are saved.

16.3 Urban Canyon Model

Overview

The URBAN CANYON model is intended for high-rise urban environments where the transmitting and receiving antennas are located close to the ground relative to the building heights. In this model, buildings are approximated as being infinitely tall, and the interactions with the buildings are entirely determined by

their 2D ground level perimeters, or footprints. Any energy that diffracts over the rooftops of buildings is assumed to be negligible compared with that carried by the rays that propagate between buildings.

The Urban Canyon model includes:

- Rays that diffract from the corners of the buildings
- Paths with up to three diffractions
- Users can specify the maximum number of diffractions
  - ✓ Two diffractions are usually enough to obtain accurate results, but one is often sufficient.
  - The run time will increase substantially for three diffractions, so this number should not usually be used for a large number of receiver points.

The Shooting and Bouncing Ray method (SBR) is used to find the propagation paths from the TRANSMITTERS to the RECEIVER points. Once the propagation paths are determined, the UTD is used to evaluate the electric field for each path.

▶ More information on this model can be found in [53] and [54].

Summary

The following list summarizes the capabilities of the URBAN CANYON model:

- Maximum reflections: 30
- Maximum transmissions: N/A
- Maximum diffractions: 3
- Environments: urban
- Terrain: flat or slightly hilly, maximum of 50 faces
- Foliage: direct waves, no lateral wave
- Indoor: N/A
- Objects: N/A
- Range: usually 1 km, but can depend on application
- Antenna heights: lower than most buildings
- Antenna types: all
- Ray tracing: SBR for horizontal plane, image method for ground reflection
- Minimum frequency: 100 MHz
- Maximum frequency: depends on application

Ray Tracing

The SBR method is employed to trace ray paths, as described in Section 16.1.

Electric Field Evaluation

Once the diffracted paths have been found, the electric field amplitudes are calculated using the UTD diffraction coefficients modified for finite conductivity ([53], [54], [55]). Several assumptions are made to reduce run time.

1. The difference in the RECEIVER and TRANSMITTER heights is small compared to the path length. This allows us to avoid using dyadic reflection and diffraction coefficients.
2. The vertically polarized components will remain nearly perpendicular to the plane of propagation and the horizontal components will remain parallel to the plane of propagation. This assumption means that any possible depolarization of the field is neglected.
3. The vertical components due to the height difference and/or the ground reflection are included.

The first step in evaluating the electric field is to find the electric field in the far zone of the transmitting antenna. At present, Wireless InSite does not consider any near zone fields.

In free space the electric field in the direction (θ, ϕ) in the far field of the transmitting antenna at a distance r can be written as:

$$E(r,\theta,\phi) = (A_\theta(\theta,\phi)\hat{e}_\theta \ + \ A_\phi(\theta,\phi)\hat{e}_\phi) \ \frac{e^{-j\beta r}}{r} \tag{16.20}$$

where

$$A_\theta(\theta,\phi) = \sqrt{\frac{P_T\eta_0}{2\pi}}g_\theta(\theta,\phi) \tag{16.21}$$

$$A_\phi(\theta,\phi) = \sqrt{\frac{P_T\eta_0}{2\pi}}g_\phi(\theta,\phi) \tag{16.22}$$

$$g_\theta(\theta,\phi) = \sqrt{|\ G_\theta(\theta,\phi)\ |}e^{j\psi_\theta} \tag{16.23}$$

$$g_\phi(\theta,\phi) = \sqrt{|\ G_\phi(\theta,\phi)\ |}e^{j\psi_\phi} \tag{16.24}$$

$$\beta = \frac{\omega}{c} \tag{16.25}$$

where

$G_\theta(\theta,\phi)$ is the θ component of the gain of the transmitting antenna

ψ_θ is the relative phase of the θ component of the far zone electric field

P_T is the time-averaged power radiated by the transmitter

r is the distance from the transmitter to the field point

This equation gives the electric field associated with any line-of-sight (LOS) rays to the receiver.

Next, consider a ray that leaves the transmitter in the direction (θ, ϕ) and reflects N times from the vertical faces of the buildings as it travels to the receiver. The θ component of the electric field for this ray is given by:

$$E_\theta = A_\theta(\theta, \phi)\frac{e^{j\beta r_{TR}}}{r_{TR}}\prod_{n=1}^{N} R_n^{\perp} \tag{16.26}$$

where $R_n^{\perp}$ is the reflection coefficient at the nth reflection for the field component in the plane perpendicular to the plane of incidence. The ϕ component of the electric field for the same ray path is given by:

$$E_\phi = A_\phi(\theta, \phi)\frac{e^{j\beta r_{TR}}}{r_{TR}}\prod_{n=1}^{N} R_n^{\parallel} \tag{16.27}$$

where $R_n^{\parallel}$ is the reflection coefficient for the component parallel to the plane of incidence.

- More information about the calculation of reflection coefficients can be found in Chapter 10.

The total field associated with this ray path will be $E = E_\theta \hat{e}_\theta + E_\phi \hat{e}_\phi$. One component will often be much larger than the other. For example, when a vertically polarized antenna is used, E_θ will be much greater than E_ϕ, and for a horizontally polarized antenna E_ϕ will typically, but not necessarily, be much larger than E_θ.

Next, consider a ray that undergoes a ground reflection in addition to the N reflections from the building walls. The complex electric field for this ray path is given by:

$$E_\theta = A_\theta(\theta, \phi)\frac{e^{j\beta r_{TR}}}{r_{TR}} R_G^{\parallel}\prod_{n=1}^{N} R_n^{\perp} \tag{16.28}$$

where $R_G^{\parallel}$ is the reflection coefficient from the ground for the field component parallel to the plane of incidence.

As for rays that have diffracted from a single building corner, consider that a particular building edge has been identified as the source of a diffracted ray. Further, let this building edge be illuminated by a ray reflected N times. Then assume that this ray is further reflected M times as it travels from the diffracting building edge to the RECEIVER point. Using the same notation as for the non-diffracted rays above, the θ component of the complex electric field for such a path is given by:

$$E_\theta = A_\theta(\theta, \phi)\frac{e^{j\beta r_{TR}}}{r_{TD}}\prod_{n=1}^{N} R_n^{\perp} \cdot D_s \cdot \sqrt{\frac{r_{TD}}{r_{DR}(r_{TD} + r_{DR})}} \cdot \prod_{m=1}^{M} R_m^{\perp} \tag{16.29}$$

where

r_{TD} is the distance from the transmitter to the diffraction point

r_{DR} is the distance from the diffraction point to the receiver

D_s is the UTD wedge diffraction coefficient for what is commonly referred to as the “soft” component, which in this case is the component parallel to the diffracting edge.

The ϕ component of the complex electric field is given by:

$$E_\phi = A_\phi(\theta,\phi)\frac{e^{j\beta r_{TR}}}{r_{TD}}\prod_{n=1}^{N} R_n^{\parallel}\cdot D_h\cdot\sqrt{\frac{r_{TD}}{r_{DR}(r_{TD}+r_{DR})}}\cdot\prod_{m=1}^{M} R_m^{\parallel} \tag{16.30}$$

where D_h is the diffraction coefficient for what is commonly referred to as the “hard” component, which in this case is the component perpendicular to the diffracting edge.

The complex electric field for the component of the ground reflected ray is given by:

$$E_\theta = A_\theta(\theta,\phi)\frac{e^{j\beta r_{TR}}}{r_{TD}}\cdot R_G^{\parallel}\cdot\prod_{n=1}^{N} R_n^{\perp}\cdot D_s\cdot\sqrt{\frac{r_{TD}}{r_{DR}(r_{TD}+r_{DR})}}\cdot\prod_{m=1}^{M} R_m^{\perp} \tag{16.31}$$

with the analogous equation for the ϕ component of the field.

Although the UTD is not strictly accurate for multiple diffractions, these equations can be extended to include double diffraction and higher order diffractions quite simply by using the approach described in [62]. The amplitude of the θ component of the double diffracted ray which reflects N times before the first reflection, Q times between diffractions, and M times after the second diffraction, is given by:

$$E_\theta = A_\theta(\theta,\phi)\frac{e^{j\beta r_{TR}}}{r_{T1}}\prod_{n=1}^{N} R_n^{\perp}\cdot D_{s1}\cdot\sqrt{\frac{r_{T1}}{r_{12}(r_{T1}+r_{12})}}\prod_{q=1}^{Q} R_q^{\perp}\cdot D_{s2}\sqrt{\frac{r_{T1}+r_{12}}{r_{2R}(r_{T1}+r_{12}+r_{2R})}}\cdot\prod_{m=1}^{M} R_m^{\perp} \tag{16.32}$$

where

r_{T1} is the distance from the transmitter to the first diffraction point

r_{12} is the distance from the diffraction point to the second

r_{2R} is the distance from the second diffraction to the receiver

D_{s1} and D_{s2} are the diffraction coefficients for the first and second diffractions

Analogous equations are used for the ϕ component of the field and the field of the ground reflected rays. A finite conductivity slope diffraction contribution found using the formulation described in [63] is also included for paths undergoing two or more diffractions.

16.4 Vertical Plane Model

Overview

The VERTICAL PLANE model is a ray-based propagation model that traces all paths from the transmitter within a two-dimensional vertical plane. It is primarily intended for predicting propagation over irregular

terrain at VHF and UHF. Vertical Plane keeps information about all propagation paths, including time and direction of arrival, for various contributions.

For the free space cases, the VERTICAL PLANE model results are ideal, with only the antenna radiation pattern and distance impacting the results. For cases with ground or water, Vertical Plane applies material properties through the reflection coefficients when the ray paths intersect and interact with a surface. It handles reflections and multiple diffractions through the application of Geometric Optics (GO) and the Uniform Theory of Diffraction (UTD).

✓ The VERTICAL PLANE model is very similar to the URBAN CANYON model, except that the 2D geometry is created by finding the intersection of the 3D geometry with a vertical plane instead of a horizontal plane.

Summary

The following list summarizes the capabilities of the VERTICAL PLANE model:

- Maximum reflections: 30
- Maximum transmissions: Only allows transmissions through foliage geometry
- Maximum diffractions: 6
- Environments: irregular terrain with optional urban and foliage geometry
- Terrain: all
- Foliage: direct waves, no lateral wave
- Indoor: N/A
- Objects: N/A
- Range: Validated to 20 km, possible 100 km with properly set ray spacing
- Antenna heights: all
- Antenna types: all
- Ray tracing: SBR for ray paths with more than 2 interactions, image method for ray paths with less than 3 interactions
- Minimum frequency: 50 MHz
- Maximum frequency: depends on application
- Earth model: flat, 4/3 Earth, or true Earth curvature

Ray Tracing

For ray paths with two or fewer interactions, image method is used. The SBR ray tracing method described for the URBAN CANYON model is employed for generating ray paths with more than two interactions.

▶ The SBR ray tracing method is described in Section 16.1.

Electric Field Evaluation

The VERTICAL PLANE electric field is evaluated in much the same way as the URBAN CANYON electric field, except that the components of the reflection and diffraction coefficients are opposite to those used in the Canyon model. The θ component of the electric field for this ray is given by:

$$E_\theta = A_\theta(\theta,\phi)\frac{e^{-j\beta r_{TR}}}{r_{TR}}\prod_{n=1}^{N} R_n^{\parallel} \tag{16.33}$$

where $R_n^{\parallel}$ is the reflection coefficient for the nth reflection for the field component in the plane parallel to the plane of incidence. The ϕ component of the electric field for the same ray path is given by:

$$E_\theta = A_\theta(\theta,\phi)\frac{e^{-j\beta r_{TR}}}{r_{TR}}\prod_{n=1}^{N} R_n^{\perp} \tag{16.34}$$

where $R_n^{\perp}$ is the reflection coefficient for the component perpendicular to the plane of incidence.

Next, consider rays which have diffracted from a single edge. Consider that a particular edge has been identified as the source of a diffracted ray. Further, let this edge be illuminated by a ray reflected N times. Then assume that this ray is further reflected M times as it travels from the diffracting edge to the receiver point. Using the same notation as for the non-diffracted rays above, the θ component of the complex electric field for such a path is given by:

$$E_\theta = A_\theta(\theta,\phi)\frac{e^{-j\beta r_{TR}}}{r_{TD}}\prod_{n=1}^{N} R_n^{\parallel}\cdot D_h\sqrt{\frac{r_{TD}}{r_{DR}(r_{TD}+r_{DR})}}\cdot\prod_{m=1}^{M} R_m^{\parallel} \tag{16.35}$$

where

r_{TD} is the distance from the transmitter to the diffraction point

r_{DR} is the distance from the diffraction point to the receiver

D_h is the UTD wedge diffraction coefficient for what is commonly referred to as the “hard” component, which in this case is the component perpendicular to the diffracting edge.

$$E_\theta = A_\theta(\theta,\phi)\frac{e^{-j\beta r_{TR}}}{r_{TD}}\prod_{n=1}^{N} R_n^{\perp}\cdot D_s\sqrt{\frac{r_{TD}}{r_{DR}(r_{TD}+r_{DR})}}\cdot\prod_{m=1}^{M} R_m^{\perp} \tag{16.36}$$

where D_s is the diffraction coefficient for what is commonly referred to as the “soft” component, which in this case is the component parallel to the diffracting edge. Higher order diffractions are evaluated in much the same way as in the URBAN CANYON model.

The VERTICAL PLANE irregular terrain propagation model also includes the effects of Earth’s curvature in its calculations by employing a $4/3$ Earth radius correction. This is the default Earth curvature setting, and may be deactivated by changing the setting to *Flat*.

16.5 Free Space Model

Overview

The FREE SPACE model assumes the electric field decreases as $1/r$ with distance in all directions. The predicted electric field includes the effect of the pattern of the transmitting ANTENNA. For receivers blocked by terrain, building or object GEOMETRY, this model returns a path loss of 250 dB. No blockages due to floor plans are taken in to account.

Summary

The following list summarizes the capabilities of the FREE SPACE model:

- Maximum reflections: N/A
- Maximum transmissions: N/A
- Maximum diffractions: N/A
- Environments: all
- Terrain: N/A
- Foliage: N/A
- Urban: N/A
- Indoor: N/A
- Objects: N/A
- Range: unlimited
- Antenna heights: all
- Antenna types: all
- Ray tracing: single ray from Tx to Rx
- Minimum frequency: depends on application
- Maximum frequency: depends on application

Ray Tracing

A single ray from the TRANSMITTER to the RECEIVER point is constructed. The ray path is used to determine the distance, time-of-arrival, direction-of-arrival and direction-of-departure. The directions are used to find the gains of the transmitting and receiving ANTENNAS.

Electric Field Evaluation

The first step in evaluating the electric field is to find the electric field in the far zone of the transmitting antenna. At present, Wireless InSite does not consider any near zone fields. In free space the electric field in the direction (θ, ϕ) in the far field of the transmitting antenna at a distance r can be written as:

$$E(r,\theta,\phi) = (A_\theta(\theta,\phi)\hat{e}_\theta \ + \ A_\phi(\theta,\phi)\hat{e}_\phi \ ) \ \frac{e^{-j\beta \ r}}{r} \tag{16.37}$$

where

$$A_\theta(\theta,\phi) = \sqrt{\frac{P_T\eta_0}{2\pi}}g_\theta(\theta,\phi) \tag{16.38}$$

$$A_\phi(\theta,\phi) = \sqrt{\frac{P_T\eta_0}{2\pi}}g_\phi(\theta,\phi) \tag{16.39}$$

$$g_\theta(\theta,\phi) = \sqrt{|\ G_\theta(\theta,\phi)\ |}e^{j\psi_\theta} \tag{16.40}$$

$$g_\phi(\theta,\phi) = \sqrt{|\ G_\phi(\theta,\phi)\ |}e^{j\psi_\phi} \tag{16.41}$$

$$\beta = \frac{\omega}{c} \tag{16.42}$$

where

$G_\theta(\theta,\phi)$ is the θ component of the gain of the transmitting antenna

ψ_θ is the relative phase of the θ component of the far zone electric field

P_T is the time-averaged power radiated by the transmitter

r is the distance from the transmitter to the field point

16.6 Hata Model

Overview

The HATA model's path loss formulas [51] are derived from the path loss measurements reported by Okumura [64]. The model can be applied over the frequency range of 150 MHz to 1500 MHz. The correction factors for different types of environments reported in Rappaport [52] are included. The current implementation of this model in Wireless InSite assumes the TRANSMITTER is the base station and the RECEIVER is the mobile.

This model has been adapted in a straightforward way to provide free space loss in line of sight regions, and return the HATA calculation in areas where the receiver is blocked by terrain, urban geometry or object GEOMETRY. Inside buildings, a path loss of 250 dB is returned.

A single direct ray from the transmitter to the receiver is constructed and treated as the ray path. The patterns of the transmitting and receiving ANTENNAS are evaluated using the direction of this ray; therefore, this model will not capture multi-path effects. Because this single direct ray is assumed, this model is not recommended for directional receiving antennas in urban areas.

Summary

The following list summarizes the capabilities of the HATA model:

- Maximum reflections: N/A
- Maximum transmissions: N/A
- Maximum diffractions: N/A
- Environments: urban, suburban and rural
- Terrain: flat or slightly hilly
- Foliage: N/A
- Urban: correction factors for large city, medium city, suburban, rural environments
- Indoor: N/A
- Objects: N/A
- Range: 1-20 km recommended
- Antenna heights: Tx (Base Station):30-200 m; Rx (Mobile):1-10 m
- Antenna types: any; isotropic or omni-directional recommended for receiving antennas
- Ray tracing: single ray from Tx to Rx
- Minimum frequency: 150 MHz
- Maximum frequency: 1500 MHz

Ray Tracing

A single ray from the TRANSMITTER to the RECEIVER point is constructed. The ray path is used to determine the distance, time-of-arrival, direction-of-arrival and direction-of-departure. The directions are used to find the gains of the transmitting and receiving ANTENNAS.

Electric Field Evaluation

The path loss of the HATA model in decibels is given by:

$$L_{Path} = \begin{array}{l} 69.55 + 26.16log_{10}(f) - 13.82log_{10}(h_T) - a(h_R) + \\ (44.9 - 6.55log_{10}(h_T)) \cdot (log_{10}(r_{TR}) - 3) + C_{Environment} \end{array} \quad (16.43)$$

where $a(h_R)$ is given by:

$$a(h_R) = \begin{cases} (1.1log_{10}(f) - 0.7)h_R - (1.56log_{10}(f) - 0.8)\ dB & \text{for small to medium-sized cities} \\ 8.29(log_{10}(1.54h_R))^2 - 1.1\ dB & \text{for large cities where } f < 300 \text{ MHz} \\ 3.2(log_{10}(11.75h_R))^2 - 4.97\ dB & \text{for large cities where } f > 300 \text{ MHz} \end{cases} \quad (16.44)$$

and $C_{Environment}$ is given by:

$$C_{Environment} = \begin{cases} -2\left[log\left(\frac{f}{28}\right)\right]^2 - 5.4 & \text{for medium and large-sized cities, and for suburban areas} \\ -4.78(log_{10}(f))^2 + 18.33log_{10}(f) - 40.94 & \text{for open rural areas} \end{cases} \quad (16.45)$$

where

f is the frequency in MHz

r_{TR} is the distance in meters from the transmitter to the receiver

h_T is the height of the transmitter above the local ground in meters

h_R is the height of the receiver above the local ground in meters

The electric field is found by replacing the $1/r$ spatial attenuation in free space by a factor derived from the path loss:

$$E_\theta = \sqrt{\frac{P_T\eta_0}{2\pi}} \cdot g_\theta(\theta,\phi)\sqrt{\frac{4\pi}{\lambda^2}} \left(10^{\left(\frac{-L_{Path}}{20}\right)} \right) e^{-j\beta r_{TR}} \quad (16.46)$$

where

$$g_\theta(\theta,\phi) = \sqrt{|G_\theta(\theta,\phi)|}e^{j\psi_\theta} \quad (16.47)$$

and

$G_\theta(\theta,\phi)$ is the θ component of the gain of the transmitting antenna

ψ_θ is the relative phase of the θ component of the far zone electric field

λ is the wavelength

$$\beta = \frac{\omega}{c}$$

P_T is the time averaged power radiated by the transmitter

The analogous equation is used to find the ϕ component of the field. The electric field is set to zero at all points located inside of buildings.

16.7 COST-Hata Model

Overview

The COST-HATA model is an empirical model that is an extension of the HATA model to higher frequencies (1500-2000 MHz). The implementation in Wireless InSite was adapted from the formulas given in [52]. The model is intended for urban and suburban areas, and other than the higher frequency range, this model is very similar to the HATA model and has the same limits on range and antenna heights. The current implementation of this model assumes the TRANSMITTER is the base station and the RECEIVER is the mobile.

This model has been adapted in a straightforward way to provide free space loss in line of sight regions, and return the COST-HATA calculation in areas where the receiver is blocked by terrain, urban geometry or object GEOMETRY. Inside buildings, a path loss of 250 dB is returned.

A single direct ray from the transmitter to the receiver is constructed and treated as the ray path. The patterns of the transmitting and receiving ANTENNAS are evaluated using the direction of this ray; therefore, this model will not capture multi-path effects. Because this single direct ray is assumed, this model is not recommended for directional receiving antennas in urban areas.

Summary

The following list summarizes the capabilities of the COST-HATA model:

- Maximum reflections: N/A
- Maximum transmissions: N/A
- Maximum diffractions: N/A
- Environments: urban
- Terrain: flat or slightly hilly
- Foliage: moderate to none
- Urban: correction factors for large city and medium city/suburban
- Indoor: N/A
- Objects: N/A
- Range: 1-20 km recommended
- Antenna heights: Tx (Base Station):30-200 m; Rx (Mobile):1-10 m

- Antenna types: any; isotropic or omni-directional recommended for receiving antennas
- Ray tracing: single ray from Tx to Rx
- Minimum frequency: 1500 MHz
- Maximum frequency: 2000 MHz

Ray Tracing

A single ray from the TRANSMITTER to the RECEIVER point is constructed. The ray path is used to determine the distance, time-of-arrival, direction-of-arrival and direction-of-departure. The directions are used to find the gains of the transmitting and receiving ANTENNAS.

Electric Field Evaluation

The path loss of the COST-HATA model in decibels is given by:

$$\begin{aligned} L_{Path} = \quad & 46.3 + 33.9log_{10}(f) - 13.82log_{10}(h_T) - a(h_R) + \\ & (44.9 - 6.55log_{10}(h_T)) \cdot (log_{10}(r_{TR}) - 3) + C_{Environment} \end{aligned} \tag{16.48}$$

where $a(h_R)$ is given by:

$$a(h_R) = (1.1log_{10}(f) - 0.7)h_R - (1.56log_{10}(f) - 0.8)\ dB \tag{16.49}$$

and $C_{Environment}$ is given by:

$$C_{Environment} = \begin{cases} 0 & \text{medium-sized cities, suburban centers, and rural areas} \\ 3 & \text{for large cities} \end{cases} \tag{16.50}$$

where

f is the frequency in MHz

r_{TR} is the distance in meters from the transmitter to the receiver

h_T is the height of the transmitter above the local ground in meters

h_R is the height of the receiver above the local ground in meters

In Wireless InSite the Path Loss is modified in the following way:

$$L_{Path} = \begin{cases} 250 & \text{if Rx is indoors} \\ L_{FS} & \text{if Rx is within LOS} \\ L_{CH} & \text{if Rx is blocked} \end{cases} \tag{16.51}$$

where L_{CH} is given by the expression in 16.48 above.

The electric field is found by replacing the $1/r$ spatial attenuation in free space by a factor derived from the path loss:

$$E_\theta = \sqrt{\frac{P_T \eta_0}{2\pi}} \cdot g_\theta(\theta,\phi) \sqrt{\frac{4\pi}{\lambda^2}} \left(10^{\left(\frac{-L_{Path}}{20} \right)} \right) e^{-j\beta r_{TR}} \tag{16.52}$$

where

$$g_\theta(\theta,\phi) = \sqrt{|G_\theta(\theta,\phi)|} e^{j\psi_\theta} \tag{16.53}$$

and

$G_\theta(\theta,\phi)$ is the θ component of the gain of the transmitting antenna

ψ_θ is the relative phase of the θ component of the far zone electric field

λ is the wavelength

$\beta = \dfrac{\omega}{c}$

P_T is the time averaged power radiated by the transmitter

The analogous equation is used to find the ϕ component of the field. The electric field is set to zero at all points located inside of buildings.

16.8 Wireless InSite Real Time Models

The Wireless InSite REAL TIME module is a collection of deterministic and semi-deterministic wireless propagation prediction solvers made for ultra-fast response times in environments with urban structures or rough terrain. Its goal is to provide higher fidelity results than empirical models while drastically reducing simulation times from that of full physics-based models.

16.8.1 Real Time - Vertical Plane Urban Propagation (VPUP)

Overview

REAL TIME - VPUP is a deterministic, ray-based model designed to provide more accurate results than empirical models at a fraction of the time of conventional ray-based models. It achieves fast simulation times by constructing ray paths as follows:

- For receivers with line of sight to the transmitting antenna, a direct ray between the TRANSMITTER and RECEIVER is used to calculate results. The model can also add an empirical ground bounce contribution for line of sight receivers

- For receivers blocked by terrain or building geometry, the model constructs a 2D profile of the obstructing geometry that lies in the vertical plane containing the TRANSMITTER and RECEIVER . An over the rooftop ray path is constructed from the convex hull of the 2D profile including the TRANSMITTER and RECEIVER locations. Additional diffracted paths from the first and last obstruction edges are also added if they differ from the path that follows the convex hull.

The electric field for each diffracted path is evaluated using UTD assuming a vertical polarization. The contribution of each path at the receiver location are combined to calculate the same set of results as the X3D model.

Summary

The following list summarizes the capabilities of the VPUP model:

- Maximum reflections: N/A
- Maximum transmissions: N/A
- Maximum diffractions: 64
- Environments: Urban and Rural
- Terrain: all
- Urban: all
- Foliage: N/A
- Indoor: N/A
- Objects: N/A
- Range: depends on application
- Antenna heights: all
- Antenna types: all
- Polarization: all
- Ray tracing: diffracted paths in vertical plane (image theory)
- Electric field summation: Power sum
- Minimum frequency: 100 MHz
- Maximum frequency: depends on application

REAL TIME - VPUP returns the *invalid* data value if the TRANSMITTER and RECEIVER are inside of buildings or underground.

Two-Dimensional Ray-Tracing

VPUP calculates ray paths in a two-dimensional vertical plane using the image method. For line-of-sight points, it calculates the field propagation using free-space path loss. For points that are shadowed by

terrain or structures, it uses the Uniform Theory of Diffraction (UTD) to calculate the loss from diffractions around the obstructions within the plane.

Figure 16.12 shows a sample vertical plane traced between a transmitter and receiver with four buildings between them. Figure 16.13 shows paths that would be found for various different receiver locations within this plane. As shown, a receiver within LOS would simply get the direct path, while receivers beyond obstructions would get one or more diffracted paths.

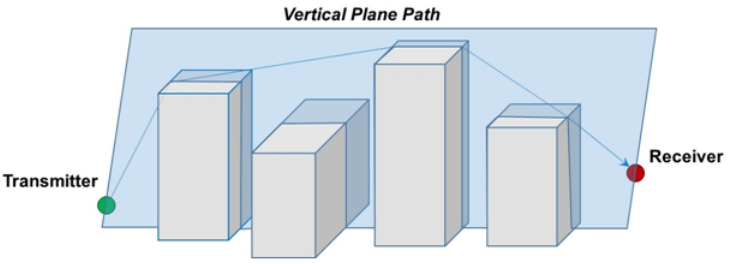


Figure 16.12: Extracting the Vertical Plane between a Transmitter and Receiver

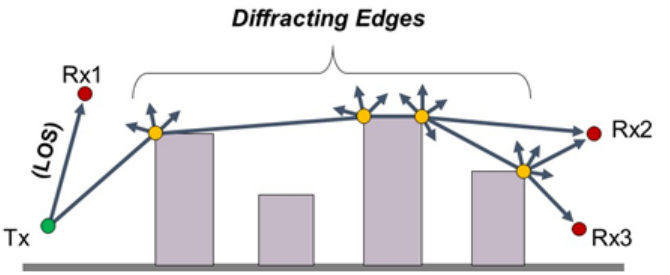


Figure 16.13: Sample paths to receivers in LOS and shadowed locations

Once the paths have been found, the electric fields are calculated as described in Section 16.9, using the Uniform Theory of Diffraction (UTD) for all diffracted paths, and assuming that all edges are perpendicular to the vertical plane. See Equations 16.3 - 16.8 for the equations used in the electric field calculations. Received power is then calculated through a coherent sum of the quantities from each path to the receiver using Equation 21.6.

Optional Ground Reflection Interference

REAL TIME - VPUP has an optional flag that allows a ground-bounce interference calculation to be performed for all line of sight points. When enabled, the model determines an approximate ground plane between the transmitter and receiver by averaging the segments of terrain that are within view of the transmitter. It then generates a ground bounce path that reflects from this plane and coherently sums this with the direct path using the formula provided in the previous section. When the ground is flat, this model resolves to a two-ray ground-bounce model, generating the banding effect often observed over generally

flat terrain. As terrain becomes rough, the interference pattern varies, as the ground profiles between transmitters and receivers change with each receiver position. Because of the approximations in this model, the ground bounce interference is most accurate when the ground is flat or nearly flat, but provides approximate effects for more rough conditions as well.

16.8.2 Real Time - Triple Path Geodesic

Overview

REAL TIME - TRIPLE PATH is a deterministic, ray-based model designed to enhance the vertical plane urban calculations of the REAL TIME - VPUP model by including the additional energy that diffracts around the sides of buildings that obscure line of sight between the TRANSMITTER and RECEIVER . The side paths are determined by constructing a convex hull in the plane that contains the TRANSMITTER and RECEIVER, and that is perpendicular to the vertical plane containing them. Apart from these two additional side paths, the capabilities of the TPG model are identical to those described for the REAL TIME - VPUP model.

Summary

The following summarizes the capabilities of the TPG Model:

- Maximum reflections: N/A
- Maximum transmissions: N/A
- Maximum diffractions: 64
- Environments: Urban and Rural
- Terrain: all
- Urban: all
- Foliage: N/A
- Indoor: N/A
- Objects: N/A
- Range: depends on application
- Antenna heights: all
- Antenna types: all
- Polarization: all
- Ray tracing: diffracted paths in vertical plane and around convex hull
- Electric field summation: Power sum
- Minimum frequency: 100 MHz
- Maximum frequency: depends on application

REAL TIME - TRIPLE PATH returns the *invalid* data value if the TRANSMITTER and RECEIVER are inside of buildings or underground.

Ray-tracing

The Triple-Path Geodesic model calculates ray paths, first in the vertical plane, directly calling the VPUP model and using the calculations described in Section 16.10.1. When the points is within LOS of the transmitter, this will include the ground bounce interference calculation if it is enabled. When the receiver is obstructed by buildings, the model uses techniques similar to the vertical plane trace to find the convex hull of obstructing buildings that are between the transmitter and receiver. This is used to generate a second, two-dimensional slant plane that is perpendicular to the vertical plane and contains the transmitter, receiver, and planar cuts of the convex hull of obstructing buildings. Figure 16.14 shows an example of the vertical and slant planes, along with 3 paths between the transmitter and receiver. As described earlier, the vertical plane may contain more than one diffracted path; the horizontal plane, on the other hand, will just contain the two principle diffracted paths around each side of the convex hull. The electric fields for each of the side paths are calculated as described in Section 16.8.1 for VPUP. The total received power is then the coherent sum of all of the paths from both the vertical and slant planes as given by Equation 21.6.

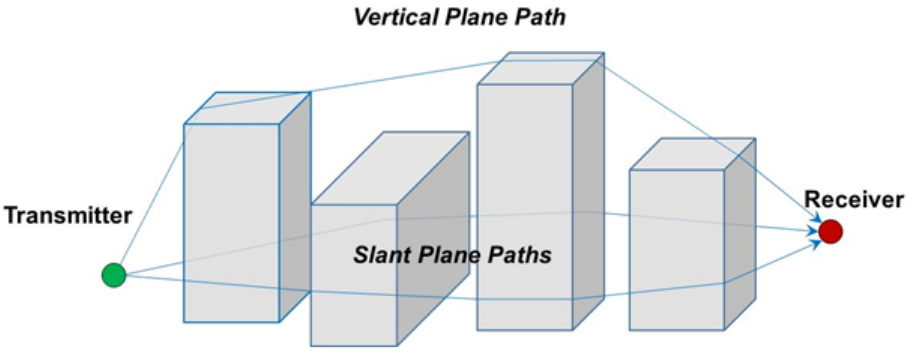


Figure 16.14: Sample Vertical Plane and Convex Hull in Slant Plane between Transmitter and Receiver

16.8.3 Real Time - OPNET Path Attenuation Routine (OPAR)

Overview

REAL TIME - OPAR [65] is a semi-deterministic urban path loss algorithm that uses the building depth between the transmitting and receiving ANTENNAS to enhance attenuation predictions. Loss is calculated according to whether the path is line-of-sight or obstructed, and is attenuated according to the total depth penetrated through obstructions.

Limitations

- Through extensive testing of this algorithm, REAL TIME - OPAR has been determined to be valid only when considering relatively low percentages of building depth over the entire path length of the calculation.

Model Calculations

For line-of-sight paths, the path loss is calculated as the greater of the free space path loss and the plane-earth loss, which is a basic calculation of loss due to ground interference when the transmitter and receiver are close to the earth relative to the distance between the two. The plane-earth loss is given in dB as:

$$L_p = 40\ Log(D) - 20\ Log(H_t) - 20\ Log(H_r) \tag{16.54}$$

where D is the total source-to-destination path length in meters, and H_t and H_r are the transmit and receive antenna heights above ground, respectively, in meters. The free space loss is given as:

$$L_{fs} = 32.45 + 20\ Log(D) + 20\ Log(f) \tag{16.55}$$

where D is again the total source-to-destination path length in meters, and f is frequency (GHz).

When the path is obstructed by buildings, the loss is calculated according to the total amount of penetration through buildings along the direct path. This loss is calculated as:

$$L_B = K_1(0.6)^f + K_2 D_B \tag{16.56}$$

where f is the frequency (GHz), K_1 and K_2 are empirically-derived constants to account for penetration loss, and D_B is the total distance traveled through buildings. See reference [65] for further details on the methodology and the empirically-derived terms and constants referred to above.

16.8.4 Real Time - Walfisch-Ikegami

Overview

REAL TIME - WALFISCH-IKEGAMI [66] is a semi-deterministic model with empirically-derived coefficients, that is useful in predictions where the dominant energy is contributed by over-the-rooftop diffractions. Buildings in the vertical plane between the transmitting and receiving ANTENNAS are used to seed the equations.

Limitations

- REAL TIME - WALFISCH-IKEGAMI returns the *invalid* data value if the following restraints are *not* met:

- The frequency is between 800 MHz and 2000 MHz
- The distance from TRANSMITTER to RECEIVER is between 20 m and 5 km
- The transmitting ANTENNA height is between 4 m and 50 m
- The receiving ANTENNA height is between 1 m and 3 m

Empirical Calculations

The Walfisch-Ikegami model was developed during the COST 231 [67] project by combining the Walfisch-Bertoni and Ikegami models in an attempt to improve predictions through inclusion of additional input data about the environment. There are four major inputs to the model:

- Average heights of buildings
- Average width of roads
- Average building separation
- Road orientation with respect to the direct path

For each of these inputs, Wireless InSite Real-Time evaluates the actual urban geometry between the transmitter and receiver, determines the average characteristics, and uses those to seed the model. Figure 16.15 shows the deterministic parameters (the range and heights of transmitter and receiver) and the statistical parameters (average building distances, road widths, and building heights) for the Walfisch-Ikegami model.

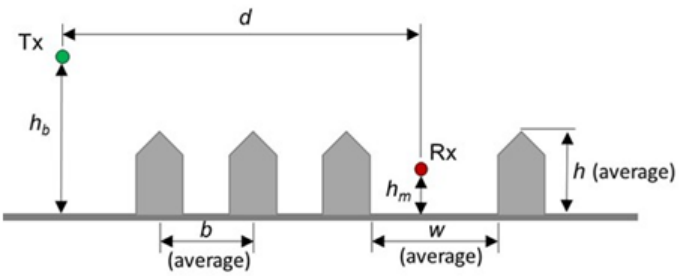


Figure 16.15: Deterministic and Statistical (Average) parameters for Walfisch-Ikegami Model

For line-of-sight paths, the model calculates the path loss as:

$$L_b(dB) = 42.6 + 26\,log(d) + 20\,log(f_c) \quad (d \gg 20m) \tag{16.57}$$

where d is distance in kilometers, and f_c is frequency in Megahertz.

For non-LOS paths, the loss is computed as the sum of the free space path loss as defined below, L_B, multiple-screen diffraction loss from rooftops, L_{msd}, and the rooftop-to-street diffraction loss, L_{rts}:

$$L_b = \begin{cases} L_B + L_{msd} + L_{rts} & L_{msd} + L_{rts} > 0 \\ L_B & L_{msd} + L_{rts} < 0 \end{cases} \tag{16.58}$$

Free-space path loss is given as:

$$L_B(dB) = 32.44 + 20\,log_{10}(f_{MHz}) + 20\,log_{10}(d_{km}) \tag{16.59}$$

Where f is frequency in Megahertz and d is distance in kilometers. The multiple-screen diffraction for propagation over rooftops is given by:

$$L_{msd} = L_{bsh} + K_a + K_d\,log(d) + K_f\,log(f_c) - 9\,log(b) \tag{16.60}$$

where L_{bsh}, K_a, and K_d are factors related to the height of the base station relative to the roof heights; K_f is set based on the type of urban or suburban environment; and b is the mean distance between buildings. Finally, the rooftop-to-street diffraction loss is given as:

$$L_{rts} = -16.9 - 10\,log(w) + 10\,log(f_c) + 20\,log(h - h_m) + L_{ori} \tag{16.61}$$

where w is the average street width in meters, h is the average rooftop height in meters, h_m is the height of the mobile handset (receiver), and L_{ori} is an empirically derived factor that corrects for variations in the orientation of the path relative the normal to the building walls. See reference [66] for further details on the methodology and the empirically-derived terms and constants referred to above.

16.9 Summary of Model Capabilities and Inputs

In this section, you will learn...

- what inputs are relevant for each model
- valid input ranges for each model
- what propagation factors and scene elements are taken into account by each model

It is useful to remember that Wireless InSite includes two basic families of model: ray-tracing (based on SBR or image theory), and semi-deterministic/empirical. Some generalizations can be made about each family of models. Table 16.1 shows the relevant inputs and thier valid ranges for each of Wireless InSite's models. Also listed is information about the underlying algorithm on which each model is based.

* For the FULL 3D model, the sum of the number or reflections and transmissions is limited to 30.

Table 16.1: Summary of Model Capabilities and Inputs

Propagation Model	Max Reflections	Max Transmissions	Max Diffractions	Environments	Objects	Ground Range	Antenna Heights	Antenna Types	Algorithm	Frequency	Propagation Domain
Full 3D	$\leq$ 30*	$\leq$ 30*	3	Urban Indoor Terrain Foliage	All	$\leq$ 10 km	All	All	SBR, Eigen Ray	100 MHz - 20 GHz	3D
X3D	30	8	3	Urban Indoor Terrain Foliage	All	$\leq$ 10 km	All	All	SBR + Exact Path correction	100 MHz - 100 GHz	3D
Vertical Plane	30	Foliage only	6	Terrain Foliage	N/A	$\leq$ 100 km	All	All	SBR	100 MHz - 20 GHz	2D Vertical
Urban Canyon	30	N/A	3	Urban Flat Terrain Foliage	N/A	1 km	Below most roof tops	All	SBR + Image theory ground bounce	100 MHz - 20 GHz	2D Horizontal
Free Space	N/A	N/A	N/A	N/A	N/A	No limit	All	All	Single Ray	No limitations	point-to-point
Hata	N/A	N/A	N/A	Urban, Suburban, Rural	N/A	1-20 km	Tx 30-200m; Rx 1-10m	All	Empirical	150 MHz - 1.5 GHz	point-to-point
COST Hata	N/A	N/A	N/A	Urban, Suburban, Rural	N/A	1-20 km	Tx 30-200m; Rx 1-10m	All	Empirical	1.5 GHz - 2 GHz	point-to-point
VPUP	N/A	N/A	64	Urban Terrain	All	Any	Any	All	Convex hull diffraction	100 MHz - 100 GHz	2D Vertical
TPG	N/A	N/A	64	Urban Terrain	All	Any	Any	All	Convex hull diffraction	100 MHz - 100 GHz	3D
Walfisch-Ikegami	N/A	N/A	N/A	Urban	N/A	20m - 5 km	Tx 4-50m; Rx 1-3m	Isotropic	Semi-Deterministic, Statistical	800 MHz - 2 GHz	point-to-point
OPAR	N/A	N/A	N/A	Urban	N/A	No Restrictions	No Restrictions	Isotropic	Semi-Deterministic, Empirical	No restrictions	point-to-point

Chapter 17

MIMO Beamforming, Diversity, and Spatial Multiplexing

In this chapter, you will learn...

→ The techniques that Wireless InSite MIMO provides to support beamforming, diversity, and spatial multiplexing

MIMO techniques allow a system that uses multiple antennas at the transmitter and/or receiver to take advantage of spatial diversity in order to increase the reliability, the signal-to-interference-plus-noise ratio (SINR), and the throughput of a channel. Wireless InSites MIMO capability includes several techniques, falling into the categories of beamforming, diversity, and spatial multiplexing. Beamforming and diversity techniques can increase the power and SINR of the received signal or reduce effects of multipath fading, which benefits other communications metrics. Spatial multiplexing can allow multiple independent, parallel data streams to be transmitted, increasing the overall throughput of a system. This chapter describes the basic theory behind each of the supported methods in terms of its impact to the received signal power, interference, and noise. The actual panels and options in the graphical user interface for enabling each of these methods are described in more detail in Chapter 14.

17.1 The MIMO Channel Matrix

The channel matrix for a MIMO system, typically referred to as the *H-matrix*, is a matrix that defines the complex channel gains between the antenna elements of the transmitter and the antenna elements in the receiver. It has dimensions $N_r \times N_t$, where N_t is the number of transmit antennas, and N_r is the number of receiver antennas. Each value in the matrix is a complex number that represents the magnitude and phase of the channel gain between one pair of transmitter-receiver antenna elements.

There are different definitions of the H-matrix, which vary depending on how the H-matrix is normalized, and whether each element represents the channel gain, or the complex conjugate of the channel gain. For

the discussion in this section, the matrix H is assumed to follow the definition in [68] [69]. The signal at the receiver is computed as

$$y = Hx + n \tag{17.1}$$

Where x is the $N_t \times 1$ vector containing the transmitted signal, y is the $N_r \times 1$ vector containing the received signal, n is a vector of noise, and H is the $N_r \times N_t$ matrix of complex channel gains, applied as voltage quantities. In the sections that follow, sometimes the matrix is reduced to a one-dimensional vector, designated with a lower-case h. This is used for cases where either a single antenna is assumed on one side of the system, or when the contributions from multiple antennas have been combined, such as in the case of receiver diversity techniques, where only the totals at each receiver element are considered.

In cases where an S-parameter file has been associated with a transmitting/receiving MIMO antenna (or both), the S-parameter file(s) may specify that the elements of the antenna array are mutually coupled by stipulating non-zero mutual impedances between elements. In such cases, an H-matrix which includes the effects of mutual coupling can be computed. In this case, x and y in 17.1 are still understood to represent the vector of transmitted or received signals measured at the antenna ports, but with the understanding that a signal injected into/received from a given port may involve radiation from/reception by other elements in the array due to mutual coupling between the elements. Calculation of the H-matrix with mutual coupling follows treatments presented in the literature (see [43], [44], and [45]).

17.2 Transmit Beamforming using Maximum Ratio Transmission

Wireless InSite provides two methods for transmit beamforming: the first is an adaptive technique called maximum ratio transmission (MRT). The second, described in the next section, is to use a precoding table to specify a collection of predefined beams. The MRT technique uses the channel matrix between the transmitting and receiving antennas to optimize the power at the receiver. The beamforming weighting matrix between a transmitter and a particular receiver antenna element, k, is defined as [70]

$$w_k = \frac{h_k^T}{|h_k|} \tag{17.2}$$

Where h_k is the $1 \times N_t$ channel vector between the transmitting array and the receiver antenna, and the superscript "T" denotes the conjugate transpose. The quantity in the denominator is the absolute magnitude of the channel vector, which is the square root of the sum of the square magnitudes of each of the complex elements of the channel vector. This weighting matrix maximizes the beam to the specific receiver antenna based entirely on the channel vector, scaling both magnitude and phase to form an optimal beam. In an environment with a great deal of multipath, this beam will likely not represent a specific direction, and it may include multiple lobes taking advantage of the multipath for the specific transmitter-receiver element combination. When the receiver has more than one element, MRT will be applied to form an optimal beam to the first element, and receiver diversity techniques can then be applied to further enhance the gain using the additional receiver elements.

17.3 Transmit Precoding for Beamforming or Diversity

The model also provides a mechanism for defining a table of precoding weights that can be applied to the *H-matrix* to perform transmit beamforming or diversity. As an example, Alamouti and pseudo-Alamouti codes provide precoding weights, usually for smaller numbers of antennas (e.g., 2 or 4) that increase the probability that a receiver employing MIMO techniques will receive a signal with spatial diversity that it can take advantage of. Precoding tables can also be used to define codebooks or collections of beamforming weights that can be used to support a beamforming method.

When the precoding table is used to define multiple sets of precoding weights, Wireless InSite will attempt to use each set of weights and will select the set that provides the best SINR. As an example, a precoding table can be defined that provides weights for each of a number of beams for a massive MIMO antenna array. During a simulation, for each receiver, the model will attempt to use each beam and select the beam that provides the best SINR. This simulates a base station that has a fixed set of beams to choose from, which uses reference signals and feedback from each UE to select the best beam for transmission to that UE.

17.4 Receiver Diversity Techniques

Wireless InSite supports three diversity techniques that can be applied to receivers that have MIMO antennas in order to enhance their received power:

- *Selection Combining*: choose signal from receive antenna with strongest SINR
- *Equal Gain Combining*: adjust phase of voltages from each Rx antenna and combine to increase total received voltage
- *Maximal Ratio Combining*: apply weights to each channel to align the phase for voltage from each antenna, while also adjusting magnitude, normalizing by SINR to increase weaker channels

These techniques are described further below.

Selection Combining

In this technique, the receiver selects the antenna element with the highest SINR. To calculate this, each receiver antenna should be treated as if it were an individual receiver. From channel gain, the received power for receiver antenna element, k, is calculated as follows:

$$Pr_k = \frac{P_t}{N_t} |h_k|^2 \, mW \tag{17.3}$$

Where P_t is the total transmitter power in milliwatts, and N_t is the number of transmitter antennas over which this power has been divided. The same equation can be used to calculate the interference power to this antenna using the corresponding h-vectors from transmitters identified as interferers. SINR is then calculated for each receiving antenna, and the highest is selected. The capacity, throughput, and BER can then be calculated from this SINR value as if this were a SISO antenna.

Equal Gain Combining

With this technique, the receiver combines the magnitudes of the received voltages from all antenna elements, applying a weighting vector to align phases before combining the channel gains from each receiver antenna. To calculate the total received power, a weighting vector is calculated that has unit gain and a phase equal to that of the h vector for each element [71][72][73]:

$$w_k = e^{j\angle h_k} \tag{17.4}$$

This weighting vector is then applied to the h-vector for the receiver elements to compute the total received power:

$$Pr = \frac{P_t}{N_t}\left|w^T h\right|^2 = \frac{P_t}{N_t}\left[\sum_{k=0}^{n_R-1}|h_k|\right]^2 \tag{17.5}$$

Where the superscript "T" denotes the Hermitian transpose of the weighting vector. Noise is also multiplied by the absolute value of the determinant of the weighting vector as follows:

$$P_{noise} = w^T w\sigma^2 = N_r\sigma^2 \tag{17.6}$$

For interference, the interfering base stations are assumed to generate a signal similar to noise in that it will not coherently interfere with the signal from this base station due to differences in the specific modulations in use at any given instant in time. Therefore, the H-matrix transmit-receive channels from all interfering base stations are power summed to provide an overall interference power and then averaged over the receiver channels to provide a mean interference level.

$$P_{I,avg} = \sum_{m=0}^{M-1}\frac{P_{t,m}}{N_{t,m}}\left[\sum_{k=0}^{N_R-1}\sum_{i=0}^{N_t-1}\left|H_{m,k,i}\right|^2\right]/N_r \tag{17.7}$$

This single scalar interference power is then treated like noise and multiplied by the weighting vector in a similar manner to provide the total, weighted interference power.

$$P_{I,\text{total}} = w^T w P_{I,avg} = N_r P_{I,avg} \tag{17.8}$$

From these quantities, a single SINR is computed, and the capacity, throughput, and BER can then be calculated as if this were a SISO channel.

Maximal Ratio Combining (MRC)

With this technique, the receiver optimally combines the received voltages from all antenna elements using a weighting vector that adjusts both the phase (like equal gain combining) and the magnitude to maximize the total SNR [71][72][73]. To achieve this, the optimal weighting vector is linearly proportional to h, e.g.,:

$$w = h/norm \tag{17.9}$$

Where $norm$ is a normalization factor that scales the weighting vector such that the sum of the squares of the magnitudes is equal to N_r (just as it would be for the equal gain combining, described in the previous section).

This weighting vector is then applied to the h-vector to compute the total received power:

$$Pr = \frac{P_t}{n_T}\left|w^T h\right|^2 = \frac{P_t}{N_t}\left[\sum_{k=0}^{n_r-1}\left|h_k\right|^2/norm\right]^2 = \frac{P_t}{N_t}\sum_{k=0}^{N_r-1}\left|h_k\right|^2 * N_r \tag{17.10}$$

Noise and interference are then calculated as described in the previous section as

$$P_{noise} = w^T w\sigma^2 = N_R\sigma^2 \tag{17.11}$$

$$P_{I,\text{total}} = w^T w P_{I,avg} = N_r P_{I,avg} \tag{17.12}$$

Note that because the weighting matrix is normalized as described above, the product of the weighting matrix and its conjugate is equal to N_r, as shown in the final term for each equation.

17.5 Receive Combining with Precoding Table

Wireless InSite also provides a mechanism for defining a table of precoding weights that can be applied to the h-vector to perform receive combining based on a codebook. In an analoguous manner to transmit beamforming, precoding tables can be used to define codebooks or collections of receive combining weights that can be used to support a beamforming method.

When the precoding table is used to define multiple sets of receive combining weights, Wireless InSite will attempt to use each set of weights and will select the set that provides the best SINR. As an example, a precoding table can be defined for a MIMO antenna array that provides sets of weights where a given set supports higher gain reception from a given direction.

17.6 Spatial Multiplexing

The last MIMO method described in this section is the use of closed-loop spatial multiplexing, in which the transmitter and receiver coordinate to generate an optimal set of independent data streams. This method uses a technique called Singular Value Decomposition (SVD), which applies precoding weights at the transmitter and combining weights at the receiver to generate multiple, non-interfering, data streams. The calculation follows the form [74]:

$$H = U\Sigma V^T \tag{17.13}$$

V is the precoding matrix applied at the transmitter, and is a square matrix of dimension $N_t \times N_t$. U is the matrix applied to the received signals by the receiver, and is a square matrix with dimension $N_r \times N_r$. Σ is the singular value matrix, which is a diagonal matrix in which each non-zero element represents an independent channel, while the 0's along the non-diagonal values ensure no interference between channels. For a single user, this provides multiple, independent streams (diagonals of the Σ matrix) that can each contribute to the total throughput or capacity.

Once the precoding and post-process matrices, U and V, have been applied the equivalent channel matrix can be described by the singular value matrix Σ instead of H.

$$\Sigma = U^T(U\Sigma\, V^T)V = U^T H V \tag{17.14}$$

This matrix has the same dimensions as H, $N_r \times N_t$, but contains values only along the diagonal for a number of singular values equal to the minimum of the two dimensions of the matrix, N_r and N_t.

Some examples are:

$$\Sigma = \begin{bmatrix} h_0 & 0 & 0 \\ 0 & h_1 & 0 \\ 0 & 0 & h_2 \end{bmatrix} \quad \Sigma = \begin{bmatrix} h_0 & 0 \\ 0 & h_1 \\ 0 & 0 \end{bmatrix} \quad \Sigma = \begin{bmatrix} h_0 & 0 & 0 \\ 0 & h_1 & 0 \end{bmatrix} \tag{17.15}$$

Each singular value represents the channel gain for one stream of data, and can be processed independently, so the singular value matrix can be collapsed into a single h-vector representing $N_S = \min(N_r, N_t)$ independent streams, and each stream can be processed as if it were an independent SISO channel. Equations for received power ($\tilde{P}_r$), noise ($\tilde{P}_n$), and interference ($\tilde{P}_I$) for each data stream s can be expressed as

$$\tilde{P}_r(s) = \frac{P_t}{N_S} |h_s|^2 \tag{17.16}$$

$$\tilde{P}_n(s) = \sum_j \left|U^T_{s,j}\right|^2 \sigma^2(j) \tag{17.17}$$

$$\tilde{P}_I(s) = \sum_j \left|U^T_{s,j}\right|^2 P_I(j) \tag{17.18}$$

Here, the " ˜ " signifies that these power levels are for a stream which generally results from the linear superposition of signals from multiple antenna elements. For each stream, the capacity and throughput are calculated from the streams SINR as if it were a SISO stream. The total capacity and throughput are then calculated as the sum totals from all streams.

17.6.1 Optimizing Capacity/Throughput with Power Allocation

The discussion above assumes that the total transmit power is divided equally between each of the allowed data streams. However, certain data streams may have poor channel conditions indicated by a singular value with a small magnitude relative to the others. By re-allocating power such that poor

performing channels have a reduced share of the power while high-performance channels are allocated a higher fraction of the power, the overall capacity or throughput can typically be increased.

As a specific example, it is straightforward to derive the optimal power allocation that maximizes the total capacity (see e.g., [75]). We briefly summarize the derivation here. For the purposes of this discussion it will be useful to define a quantity $\gamma(s)$ for each stream s such that

$$\gamma(s) = \frac{|h_s|^2 P_{t,\text{total}}}{\tilde{P}_I(s) + \tilde{P}_n(s)}. \tag{17.19}$$

Physically, $\gamma(s)$ represents the SINR for stream s if the total power of the MIMO antenna $P_{t,\text{total}}$ is devoted to this single stream and no others. Given this definition, the combined capacity across all streams is given according to the Shannon-Hartley theorem by:

$$C = \sum_s B \log_2 \left(1 + \frac{\gamma(s)\,\tilde{P}_t(s)}{P_{t,\text{total}}}\right). \tag{17.20}$$

where $\tilde{P}_t(s)$ is the power allocated to each stream and B is the bandwidth.

The power allocation distribution which maximizes the total capacity is readily obtained by applying the method of Lagrange multipliers to maximize the capacity C subject to the constraint that the total power is given by the sum over powers allocated to each stream (i.e., $P_{t,\text{total}} = \sum_s \tilde{P}_t(s)$). Applying the method of Lagrange multipliers with the additional implicit constraint that the power for each stream is positive (i.e. $P_s \geq 0$ for all s), yields the so-called waterfilling power allocation solution

$$\frac{\tilde{P}_t(s)}{P_{t,\text{total}}} = \left[\frac{1}{\gamma_0} - \frac{1}{\gamma(s)}\right]^+ \tag{17.21}$$

where $[z]^+ = \max(0, z)$ and the parameter γ_0 is chosen to satify the constraint on the total power which equivalent to satisfying the expression

$$\sum_s \left[\frac{1}{\gamma_0} - \frac{1}{\gamma(s)}\right]^+ = 1. \tag{17.22}$$

Equations 17.21 and 17.22 are used by the COMMUNICATION SYSTEM to assign transmit powers $\tilde{P}_t(s)$ to each of the streams such that the total Shannon capacity is maximized given a calculated $\gamma(s)$ for each stream. Some streams may get zero power in which case they are unused.

Under the assumption that modulation and coding schemes (MCSs) employed by modern wireless access methods achieve a constant fraction of the maximum theoretical capacity given by the Shannon-Hartley theorem, the same waterfilling power allocation will also maximize throughput for a given set of $\gamma(s)$ values. Thus, equations 17.21 and 17.22 are also used by the COMMUNICATION SYSTEM to assign transmit powers $\tilde{P}_t(s)$ to each stream in order to optimize total throughput. The only added nuance in applying waterfilling power allocation for throughput is that if the lowest SINR stream s_{lowest} is found to be in outage following application of the waterfilling approach, that stream is forced to be unused by setting $\gamma(s_{\text{lowest}}) = 0$ for that stream before re-allocating power with a subsequent application of the waterfilling approach.

The *Closed-Loop Spatial Multiplexing (SVD)* settings found in the COMMUNICATION SYSTEM allow the user to select from several *Power Allocation* schemes for both *Shannon Capacity* and *Throughput* analyses:

- *Uniform; Use All Streams* – no power allocation – enforces the use of all allowed streams with the total transmit power divided equally between them.
- *Uniform; Optimize Throughput/Capacity* – rudimentary power allocation – iteratively finds the optimum throughput/capacity that can be obtained by simply dropping the lowest SINR streams and equally dividing power amongst the remaining streams.
- *Waterfilling* – fully optimized power allocation – allows for the optimization of throughput/capacity by allowing unequal power amongst all available streams and using the waterfilling approach.

Note that the *Waterfilling* power allocation method is only available for a *Throughput* analysis if *Equal MCS Across Streams* is unchecked since the waterfilling approach tacitly assumes that the MCS can be appropriately adjusted on a stream-by-stream basis to attain the maximum throughput achievable for a given SINR.

Chapter 18

Maximum Permissible Exposure

In this chapter, you will learn...

- → what standards can be used for assessing MPE in Wireless InSite
- → how to set up a job to assess MPE
- → how to view field values, power densities, and their respective percent of threshold values
- → how to input user-defined standards for assessing MPE

Wireless InSite's Maximum Permissible Exposure (MPE) module provides the ability to calculate MPE values and compare them to the safety thresholds defined by the IEEE Standard C95.1-2005 [76]. The user may conduct a hazard assessment and visually display the MPE relevant quantities, as well as their relation to the MPE thresholds. This capability allows for both stationary and moving, multi-pulse transmitters, and includes the aggregation of exposure over time.

MPE calculations must use the X3D model and require the transmitter to be defined using a Trajectory set or Point sets containing a single transmitter.

- MPE results are only generated for study areas using the X3D model.
- Transmitters must be defined using a Trajectory set or a Point set containing a single transmitter.

18.1 Definitions

The IEEE standard uses the following definitions and abbreviations:

Specific Absorption Rate (SAR)

The time derivative of the incremental energy (dW) absorbed by (dissipated in) an incremental mass (dm) contained in a volume element (dV) of a given density (ρ):

$$SAR = \frac{d}{dt}\left(\frac{dW}{dm}\right) = \frac{d}{dt}\left(\frac{dW}{\rho dV}\right) \tag{18.1}$$

The SI unit of SAR is the Watt per kilogram (W/kg).

SAR can be related to the electric field at a point by

$$SAR = \frac{\sigma E^2}{\rho} \tag{18.2}$$

where

- σ is conductivity of the tissue $\frac{S}{m}$
- ρ is mass density of the tissue (kg/m^3)
- E is the rms field strength in the tissue (V/m)

SAR can be related to the increase in temperature at a point by

$$SAR = \left.\frac{c\Delta T}{\Delta t}\right|_{t=0} \tag{18.3}$$

where

- ΔT is the change in temperature
- Δt is the duration of the exposure
- c is the specific heat capacity $(\frac{J}{kgC^\circ})$

This assumes the measurements are made under ”ideal” non-thermodynamic circumstances, i.e. no heat loss by thermal diffusion, radiation, or thermoregulation (blood flow, sweating, etc.).

Basic Restrictions (BRs)

Exposure restrictions that are based on established adverse health effects that incorporate appropriate safety factors and are expressed in terms of the in situ electric field (3 kHz to 5 MHz), specific absorption rate (100 kHz to 3 GHz), or incident power density (3 GHz to 300 GHz). Only these last two standards are relevant to Wireless InSite’s X3D model.

Maximum Permissible Exposure (MPE)

MPE values are defined as the highest rms or peak electric or magnetic field strengths, their squares, or the plane-wave equivalent power densities associated with those magnetic fields, or the induced contact currents to which a person may be exposed without incurring an established adverse health effect with an acceptable margin of safety [76]. The MPEs are derived or estimated from the basic restrictions (induced electric field, SAR, or power density). The MPE tables represent more stringent restrictions than the basic restrictions do. Compliance with the MPE thresholds, insures that the BRs are not exceeded. Conversely, even when an exposure is proven to be below the basic restrictions, it may exceed the MPE thresholds. MPEs are sometimes called *reference levels, derived limits,* or *investigation levels*.

Equivalent Plave-wave Power Density (S)

A commonly used term associated with any electromagnetic wave equal in magnitude to the power density of a plane wave having the same electric (E) or magnetic (H) field strength. Specifically, the normalized value of the square of the electric or the magnetic field strength at a point in the near field of a radiating source. The SI unit of equivalent plane-wave power density is the watt per square meter (W/m^2) and is computed as follows:

$$S = \frac{E^2}{\eta} = \eta H^2 \tag{18.4}$$

where

E and H are the root mean square (rms) values of the electric and magnetic field strengths, respectively, and

η is the wave impedance ($\approx$ 377 ohms in free space)

Average (Temporal) Power (P)

The time-averaged rate of energy transfer

$$\bar{P} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} P(t) dt \tag{18.5}$$

where P(t) is the instantaneous power.

NOTE - The time duration (t_2-t_1) can be source related (e.g. the source repetition period, duty cycle) or use related (e.g. the averaging time specified in exposure guidelines).

Averaging Time (t_{avg})

The appropriate time period (t_{avg})over which exposure is averaged for the purposes of determining compliance with a maximum permissible exposure (MPE) or reference level.

Averaging Distance

The distance over which the in situ electric field is averaged when determining compliance with basic restrictions.

Averaging Volume

The distance over which the peak spatial-average specific absorption rate is averaged when determining compliance with basic restrictions. For frequencies up to 3 GHz, the values are based on spatial averaging over the projected cross-section area of a human body. From 3-30 GHz, the spatial averaging is

over a projected area of 100 λ^2 cm^2 where λ is the wavelength in cm. For frequencies over 30 GHz, the averaging is over a projected area of 100 cm^2 but not to exceed a maximum of 1000 W/m^2 in any one square centimeter.

18.2 Maximum Permissible Exposures - MPEs

Wireless InSite can perform MPE assessments for frequencies between 100 MHz and 100 GHz. These MPEs are simpler to evaluate than the BRs on which they are based. In particular, they do not require modeling of the interaction of EM fields with the human body. The IEEE C 95.1-2005 standard defines two hazard tables: one for people in a controlled environment with an established RF hazard safety program, and a second one for the general public or when an RF hazard safety program is unavailable. The second table specifies lower levels to allow for greater safety factors. In both of the tables, f$_M$ is the frequency in MHz and f$_G$ is the frequency in GHz.

Table 18.1: MPEs for People in Controlled Environments

Frequency Range	**RMS Electric Field Strength**	**RMS Magnetic Field Strength**	**Average Power Density**
100 - 300 MHz	61.4 V/m	0.163 A/m	10 W/m^2 in 6 min
0.3 - 3 GHz	N/A	N/A	F$_M$/30 W/m^2 in 6 min
3 - 30 GHz	N/A	N/A	100 W/m^2 in 19.63/f$_G^{1.079}$ min
30 - 100 GHz	N/A	N/A	100 W/m^2 in 2.524/f$_G^{0.476}$ min

Table 18.2: MPEs for the General Public (Uncontrolled Environments)

Frequency Range	**RMS Electric Field Strength**	**RMS Magnetic Field Strength**	**Average Power Density**
100 - 400 MHz	27.5 V/m	0.0729 A/m	2 W/m^2 in 30 min
0.4 - 2 GHz	N/A	N/A	(F$_M$/200) W/m^2 in 30 min
2 - 5 GHz	N/A	N/A	10 W/m^2 in 30 min
5 - 30 GHz	N/A	N/A	10 W/m^2 in 150/f$_G$ min
30 - 100 GHz	N/A	N/A	10 W/m^2 in 25.24/f$_G^{0.476}$ min

In addition to the above thresholds, the IEEE standard also requires:

- The peak value of electric field to be less than 100 kV/m.

- The total incident energy density during any one-tenth second period within the averaging time may not exceed one-fifth of the total energy density permitted during the entire averaging time.

18.2.1 Assumptions

- Tx trajectory sets have a constant speed (m/s), a fixed distance, d, between points, a fixed time interval, t = d/v, between points, and a fixed pulse repetition frequency (prf). The trajectory set will be made up of N transmitter points, each representing a time step, $t_j = t_1, t_2, \ldots t_{Ntx}$ along the trajectory.
- Pulse durations will be short compared to t, the time between two Tx positions.

18.2.2 Setting up an MPE Calculation

To set up an MPE analysis, the user must specify the following:

In the X3D STUDY AREA PROPERTIES window click on the MPE PROPERTIES button and select *MPE Active*.

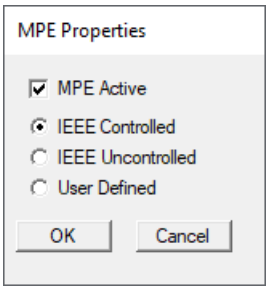


Figure 18.1: The MPE Porperties Window

In the MPE PROPERTIES window select one of the following threshold types:

- *IEEE Controlled*
- *IEEE Uncontrolled*
- *User Defined*

✓ The *User Defined* option allows the user to set averaging times and threshold values for each of Wireless InSite's MPE-related results.

▶ See Chapter 15 for more information on study areas.

Define the transmitter set to be one of the following:

- *Trajectory Set*
- *Point Set*

❗ Any *Point Set* used for MPE calculations must contain *only one* point.

- Multiple Transmitter sets will act in isolation.

▶ See Section 13.1 for more information about transmitter sets.

Define the waveform to be *Sinusoid*, *Blackman* or *Gaussian* and specify the Carrier frequency. In the case of *Blackman* or *Gaussian*, the user has the option of also specifying:

- Pulse Repetition Frequency (prf)
- Pulse width

Other wide band waveforms (Tukey, Gaussian Derivative, Hamming, Hanning, Chirp, Raised Cosine or Root Raised Cosine) will be treated as rectangular waves for the purposes of the MPE calculation.

▶ See Chapter 11 for more information on waveforms.

18.2.3 MPE Results

Result Files

The MPE result files follow the same convention as Wireless InSite's Point-to-Multipoint files, with one exception. Ordinarily, the p2m file names include the transmitter set and the transmitter number as in project.type.txxx_yy.rz.p2m, where xxx is the transmitter set and yy is the transmitter number. MPE calculations that include a Tx Trajectory set, return the aggregate exposure at each Rx point represented by the entire trajectory of transmitters; the contribution from each Tx point along the trajectory set is not considered individually. MPE result files always represent all transmitters in a transmitter set, therefore the transmitter number is omitted from the filename.

Table 18.3: MPE Results and Filenames

IEEE Quantity	**Wireless InSite name**	**Result Filename**	**units**
RMS Electric Field Strength (V/m)	IEEEC95.1-2005:rms E-field	NNN.rhazrmsefield.tXXX.rYYY.p2m	(V/m)
RMS Magnetic Field Strength (A/m)	IEEEC95.1-2005-rms H-field	NNN.rhazrmshfield.tXXX.rYYY.p2m	(A/m)
Average Power Density (W/m_2)	IEEEC95.1-2005-Average Power Density	NNN.rhazpowerd.tXXX.rYYY.p2m	(W/m_2)
Instantaneous Peak E-field	IEEEC95.1-2005-Peak E-Field	NNN.rhazpkefield.tXXX.rYYY.p2m	(dBV/m)
Average Power Density during 1/10 second	IEEEC95.1-2005-Average Power Density Short Interval	NNN.rhazpowerdshrt.tXXX.rYYY.p2m	(W/m_2)

Note: The MPE module will produce one set of 5 result files for each transmitter set. For each of these the user may view or plot the raw data, each parameter's percent of MPE threshold, or the aggregate percent MPE.

Viewing MPE Results

Once the calculation has finished, the IEEEC95.1-2005 MPE-related results will be available for viewing, and be accessible from either the RESULTS tab in the MAIN WINDOW or the PROJECT TREE window. As with most of Wireless InSite's results, the five new IEEEC95.1-2005 MPE-related results can be viewed in either the GEOMETRY VIEW window as colored field values at the location of each receiver, or can be plotted on a graph.

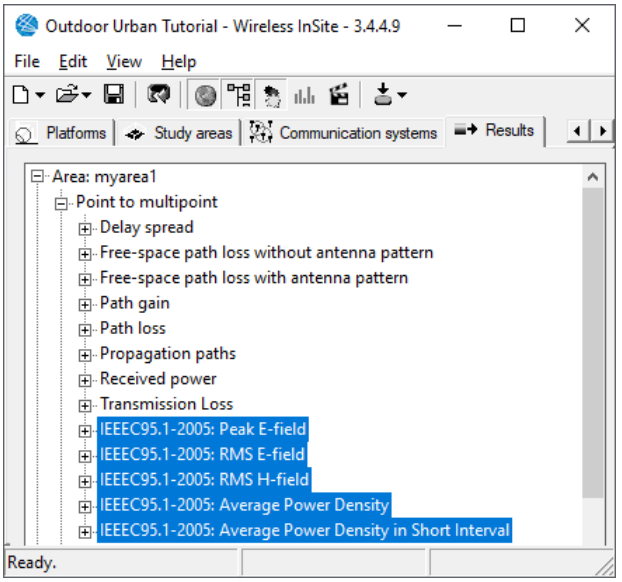


Figure 18.2: IEEEC95.1-2005 MPE-related results in the Results tab

Unlike for other Wireless InSite results the user has the additional option of rendering the percent threshold represented by the IEEEC95.1-2005 MPE-related data. Right clicking on any of the C95.1-2005 results give the user four new rendering options (see Figure 18.3).

- **Plot MPE Threshold** Allows the user to plot the percent threshold represented by the selected results on a line plot.
- **View MPE Threshold** Allows the user to view a colorscale rendering of the percent threshold represented by the selected results in the GEOMETRY VIEW window.
- **Plot Aggregated MPE Threshold** Allows the user to plot the maximum percent threshold represented by any of the IEEEC95.1-2005 results associated with that particular simulation.
- **View Aggregated MPE Threshold** Allows the user to view a colorscale rendering of the maximum percent threshold represented by any of the IEEEC95.1-2005 results associated with that particular simulation.

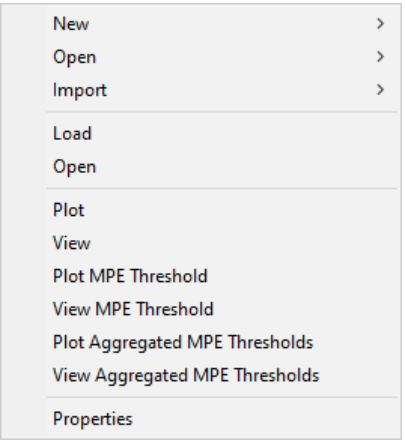


Figure 18.3: Options available for rendering IEEEC95.1-2005 MPE-related results

Wireless InSite uses a four-color discrete color scale to render the percent of threshold. Purple indicates less than 0.001% of the threshold has been met. Green corresponds to values between 0.001 and 50%. Yellow represents values between 50 and 100%, and Red indicates values above the threshold. Figure 18.4 shows the raw IEEEC9.5.1-2005 Average Power Density in Short Interval for a grid of receivers. Right clicking on the same result, and selecting *View MPE Threshold* renders the percent of threshold for the IEEEC9.5.1-2005 Average Power Density in Short Interval as shown on the right.

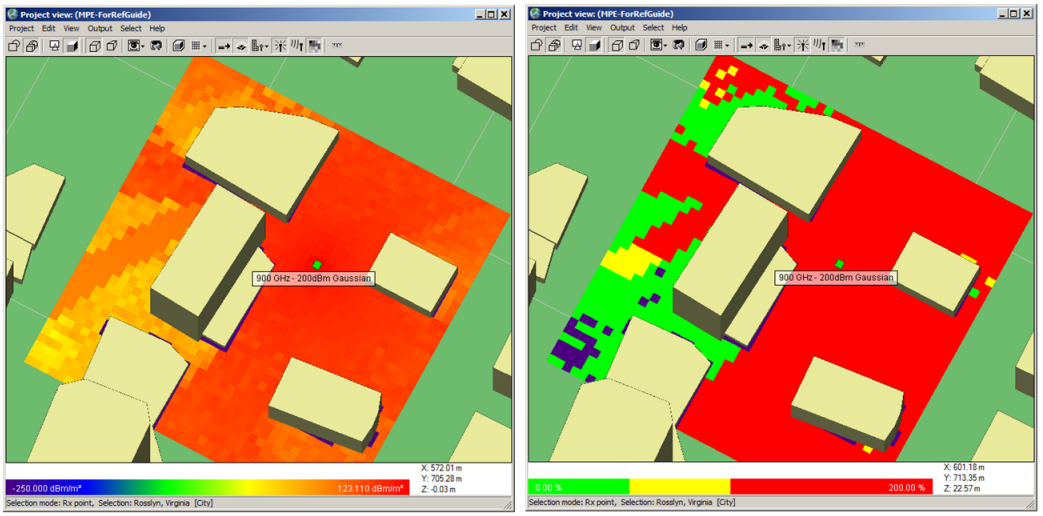


Figure 18.4: Average Power Density in Short Interval (dBm/m^2) and the corresponding percent of IEEEC95.1-2005 MPE Threshold.

Figure 18.5 shows the ”Aggregated threshold” for the same scenario as above. This represents the maximum percent of IEEEC95.1-2005 Threshold for all quantities (RMS Electric Field Strength, RMS Magnetic Field Strength, Average Power Density, Instantaneous Peak E-field, and Average Power Density during 1/10 second). In this particular example, the percent threshold reached by the peak E-field was

higher than that of the Average Power Density, which is why the aggregated threshold is different than the Average Power Density percent threshold. The aggregated threshold can be rendered by right-clicking on any one of the IEEEC95.1-2005 results and selecting *View Aggregated MPE Threshold*.

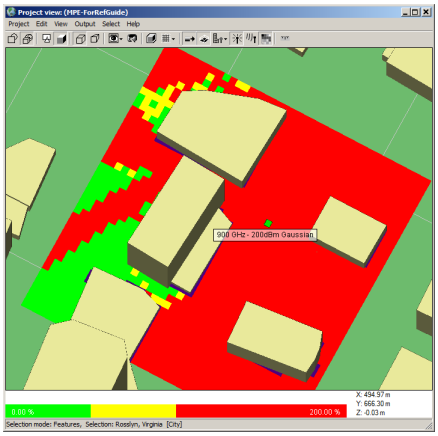


Figure 18.5: Aggregated percent of IEEEC95.1-2005 MPE threshold for all quantities

✓ It is often convenient to first render either *View Aggregated MPE Threshold* or *Plot Aggregated MPE Threshold*. If the aggregated quantities lie below the MPE threshold, this indicates that none of the IEEEC95.1-2005 quantities have surpassed their given threshold.

When receiver routes are used, data is often presented more clearly by using a line plot. This is done in the same way as described in Chapter 22, with the addition of being able to plot the MPE individual and aggregate thresholds. Figure 18.6 shows a line plot of the IEEEC95.1-2005 percent of MPE threshold for Average Power Density in the Short Interval, Peak E-Field, and the Aggregated percent of IEEEC95.1-2005 MPE threshold.

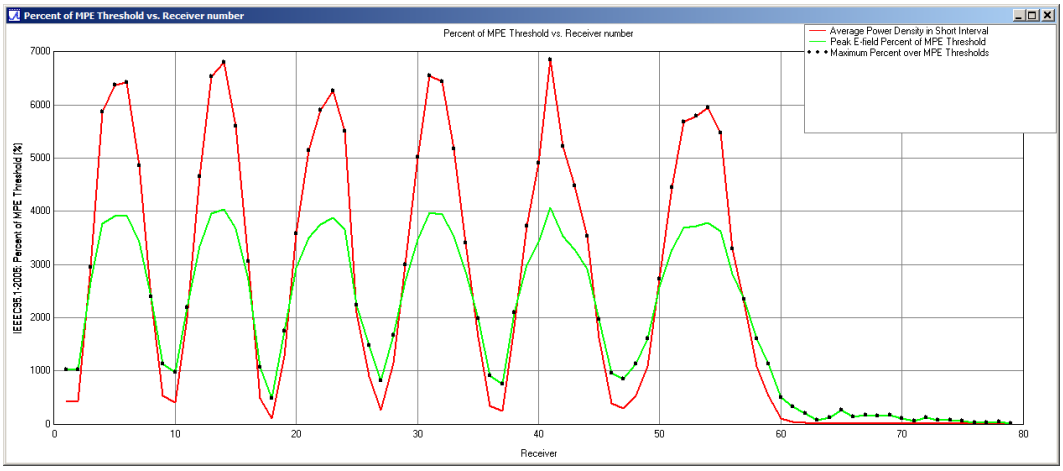


Figure 18.6: Line Plot showing IEEEC95.1-2005 percent of MPE threshold vs. receiver number

▶ For more information about rendering results see Chapter 22 on Data Visualization.

Chapter 19

Parameter Uncertainty using Monte Carlo Simulations

In this chapter, you will learn...

- → how to use Wireless InSite's Monte Carlo capability to capture uncertainty about scenario inputs
- → how to define Monte Carlo distributions for materials and input power
- → how to set up a Monte Carlo project, run simulations, and view results

Wireless InSite's Parameter uncertainty capability enables the user to define distributions for various aspects of the propagation environment (material properties and power) in order to efficiently assess the impact of these uncertainties. Wireless InSite performs the ray trace only once, and need only perform its UTD calculations for each of the Monte Carlo instances. Wireless InSite `*.p2m` result files are created for aggregated maximum, minimum, mean and median path loss, path gain and received power, which are available via the PROJECT TREE. Access to each Monte Carlo instance is available via the MULTI-CHANNEL RESULTS BROWSER.

- This capability is only compatible with the X3D model.
- ✓ The Monte Carlo enabled flag must be checked in the X3D STUDY AREA PROPERTIES window to be active.

19.1 Monte Carlo Multi-Material

The Monte Carlo Multi-Material capability allows the user to define a uniform distribution of a discrete set of existing materials. This is useful to assess the impact made by a variety of possible material types. Note that when Monte Carlo has not been activated in the study area, the first material listed is used as the default material. The user should create the Monte Carlo material before assigning it to any geometry in the scene.

19.1.1 Creating a Monte Carlo Multi-Material

To create a *Monte Carlo Multi-Material*,

- Right click *New* →*Material*. User will be prompted to select a geometry in which to assign the material.
- This will bring up the CHOOSE GEOMETRY window
- Double click on the desired geometry.
- This will bring up the GOMETRY PROPERTIES window.
- Click OK.
- This will return to the CHOOSE GEOMETRY window.
- Click OK.
- This will bring up the CREATE NEW MATERIAL window.
- Select *Monte Carlo multi-material* from drop down menu.
- Click OK.
- This will bring up the MONTE CARLO MULTI-MATERIAL PROPERTIES Window.
- Enter a short description, such as ”Monte Carlo Multi-Material Terrain”.
- Click the ADD MATERIAL Button
- This will bring up the CHOOSE MATERIAL window.
- With each material added, the MONTE CARLO MULTI-MATERIAL PROPERTIES window will come back up, indicating the materials that have been added under the *Material Included* list.
- Continue to add materials to make up the desired distribution of materials. (E.g. Wet Earth, Dry Sand, Sea water, etc.) Currently, the system will not allow you to choose another Monte Carlo Multi Material
- When all materials have been chosen, click OK to exit the MONTE CARLO MULTI-MATERIAL PROPERTIES window.
- Within the *Materials* Tab, all materials referenced by a Monte Carlo Multi Material will register as ”In Use,” even if the Monte Carlo Multi Material itself is not assigned to any geometry.

19.1.2 Limitations

Wireless InSite’s Monte Carlo Multi-Materials may not include the following Material types:

- Free space materials
- Foliage materials
- Other Monte Carlo materials
- User-defined materials

▶ See Chapter 10 for more information on Wireless InSite's material types.

19.2 Monte Carlo Variable Parameter Material

This capability allows the user to define either a uniform or a normal distribution for the dielectric properties (permittivity and conductivity), thickness and roughness of a material. These parameters can be varied independently, singly or in combination with one another.

To create a Monte Carlo Variable Parameter Material:

- Right click *New →Material*
- Select Monte Carlo variable parameter single layer from drop down menu
- Click OK, and provide a name for the new material
- Click the MONTE CARLO button next to whichever parameter you would like to vary.

Before you can define the distribution for a parameter, Monte Carlo must be activated by checking the "Activate Monte Carlo" box. Once this is checked the Distribution field will become active and "Uniform" or "Normal" can be selected. In either case, the minimum and maximum desired values for the perameter must be confirmed. If using a Normal Distribution provide a Mean and a Standard Deviation must also be specified. Default values for each parameter are listed below:

- *Permittivity*
  - mean = 15
  - minimum = 3
  - maximum = 15
- *Conductivity*
  - mean = 0.015
  - minimum = 0.005
  - maximum = 0.025
- *Roughness*
  - mean = 0.01
  - minimum = 0
  - maximum = 0.02
- *Thickness*
  - mean = 0.3
  - minimum = 0.1
  - maximum = 0.5

19.3 Monte Carlo Variable Input Power

Wireless InSiteś Monte Carlo Variable Input Power enables the user to define a uniform or normal distribution for the transmitter's input power.

To set up a Monte Carlo Input Power Sweep:

- Select an active TX set or create one.
- Select the TRANSMITTER tab.
- Open the TRANSMITTER PROPERTIES window and click on the INPUT POWER MONTE CARLO button.

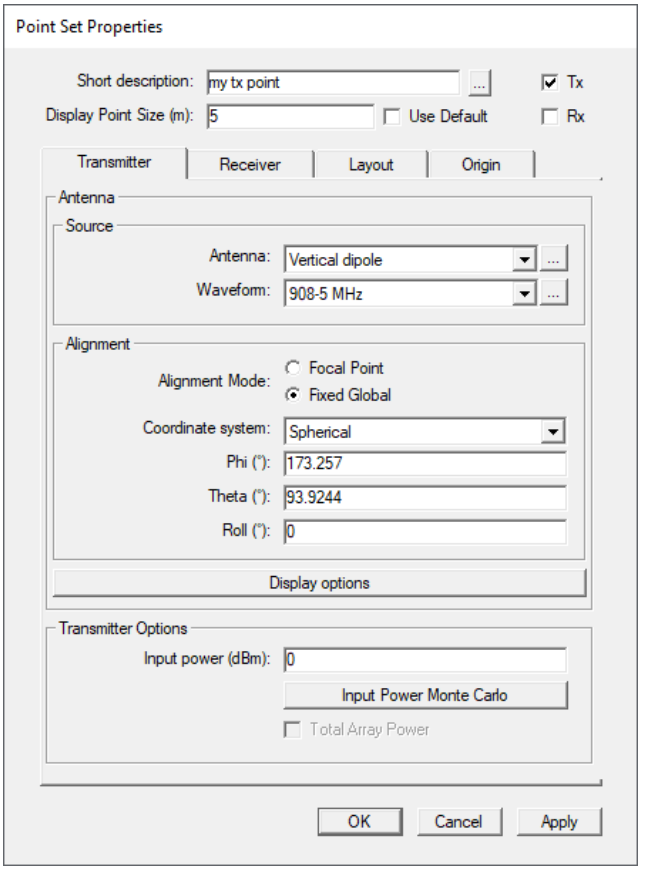


Figure 19.1: Transmitter properties window with Monte Carlo button

- This will bring up the MONTE CARLO PARAMETER window.
- Click on the box next to *Activate Monte Carlo for Input Power*.
- Select a Normal Distribution, and specify the desired mean input power and standard deviation.

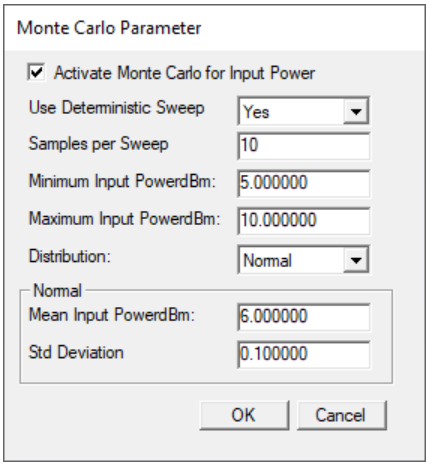


Figure 19.2: Activate Monte Carlo for Input Power

- Click OK.
- Click OK to close the Transmitter Properties Window.

▶ See Section 13.2 for more information on transmitter properties.

19.4 Activating Monte Carlo

After the desired distributions have been defined, the user must activate Monte Carlo in the Study Area window.

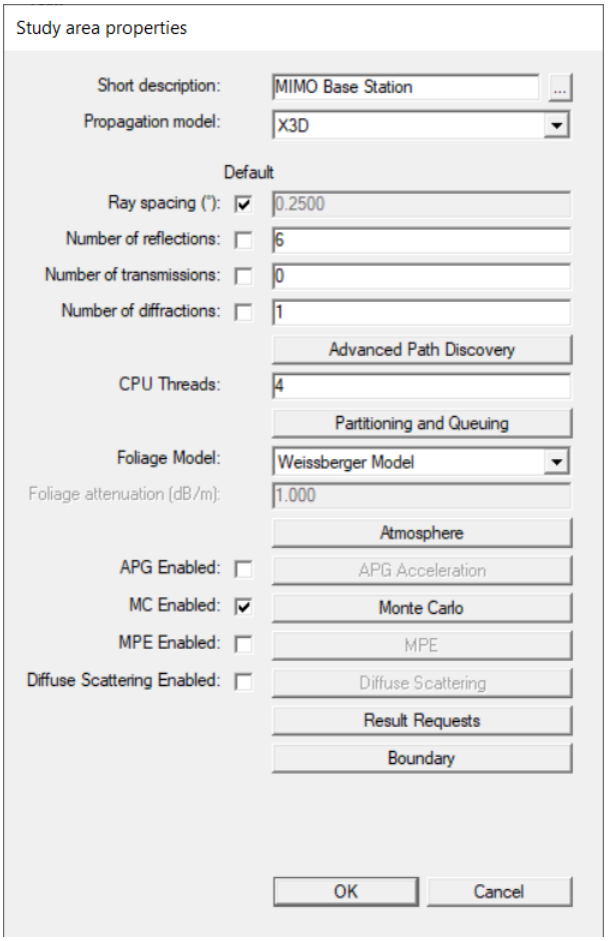


Figure 19.3: Study Area properties window with Monte Carlo Activation button

- Create a new X3D study area.
- Check the *MC Enabled* box to set Monte Carlo as active, and double click on the MONTE CARLO button to access the MONTE CARLO PROPERTIES window.
- In the MONTE CARLO PROPERTIES window:
  - Specify the number of Monte Carlo instances desired.
  - Click on the vertical bar on the right hand side of the window to expand it to reveal the Randomizer seed.

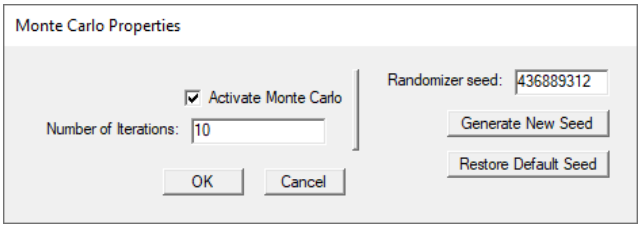


Figure 19.4: Monte Carlo properties window showing Randomizer Seed inputs

The seed is used to create the sequences of random numbers that are used to compute the varied values used in each instance of the Monte Carlo simulation. By using the same seed over and over for the same job, you can run it repeatedly and get the same results. Therefore, you can change other parameters about the job, rerun it, and see if those changes affected your results, and be assured that the Monte Carlo behavior is the same. Conversely, you can change the seed and rerun a Monte Carlo case to see if you get the same distribution behavior over multiple trials with different randomly generated values.

- Click OK.
- Click OK to exit STUDY AREA PROPERTIES window.

▶ See Section 15.2 for more information on setting Study Area properties.

19.5 Monte Carlo Results

Once the simulation finishes, the following statistical results aggregated from the Monte Carlo iterations are automatically generated and will appear in the results tree under the STUDY AREA in the RESULTS tab:

- Minimum
- Maximum
- Mean
- Median
- Standard Deviation

These results can be viewed following the same conventions as all other Wireless InSite results. RESULTS can be plotted on line graphs or displayed in the GEOMETRY VIEW.

▶ See Section 22.1 for more on visualizing results.

Chapter 20

Diffuse Scattering Calculations

In this chapter, you will learn...

- → how the Diffuse Scattering X3D model works
- → how to enable and modify Diffuse Scattering material properties
- → how to enable Diffuse Scattering calculations in X3D
- → how to read and visualize Diffuse Scattering results

Wireless InSite's Diffuse Scattering model allows for the simulation of scattered paths caused by inherent irregularities in real-world structures and materials. Approximate results that account for this effect can be calculated without the need for intricate details in the project geometry.

- Diffuse Scattering results are only generated for study areas using the X3D model.
- Diffuse Scattering must be enabled in the MATERIAL properties and STUDY AREA properties in order to run Diffuse Scattering calculations.
- Diffuse Scattering can significantly impact run times, especially for large scenes or scenes where allowed interaction limits are high.

20.1 Overview of X3D Diffuse Scattering Model

Wireless InSite's diffuse scattering model is based on the model developed by Degli-Esposti (2007) [77], with supplemental modifications for cross-polarization from his later work [78] in 2011. The general concept is that electromagnetic waves scatter from a building surface in all directions, not just specular reflection, according to a stochastic process termed *diffuse scattering*. The usual Fresnel coefficient for specular reflection is reduced by a factor that accounts for diffuse scattered power. Figure 20.1 shows the geometric configuration for diffuse scattering.

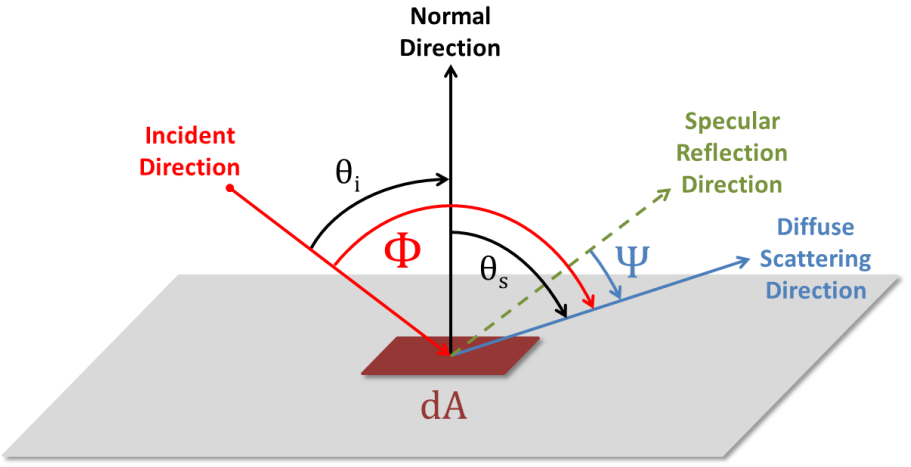


Figure 20.1: Reflection and Diffuse Scattering from a Surface Element

The fraction of reflected power that is scattered diffusely from a surface rather than reflecting in the standard specular direction is defined through the diffuse scattering coefficient, S, given by [77]:

$$S = \frac{\left|\vec{E}_S\right|}{\left|\vec{E}_i\right|} \tag{20.1}$$

or an alternative definition

$$S = \frac{\left|\vec{E}_S\right|}{\left|\vec{E}_R\right|} \tag{20.2}$$

where E_S is the diffuse scattered field, E_i is the incident electric field, and E_R is the reflected field that would exist for a smooth surface in the absence of diffuse scattering. The difference between the two formulations is that in the first, the diffuse scattered power density is diverted from the incident field, while in the second it is diverted from the reflected field. In the second option, the diffuse scattered power depends on both S and the specular reflection coefficient, Γ. Wireless InSite allows the user to choose the expression for S through an Advanced Parameter when defining diffuse scattering properties for a material. The default is to use the first expression involving E_i. The reasoning behind this default is that the scattering is due to rough geometry which may involve structures or attached objects, and in many cases may not be directly linked to normal material characteristics such as the *permittivity* and *conductivity*.

To calculate the total diffuse scattered field reaching a given receiver from a given facet, the model integrates over the surface numerically, subdividing the surface into a number of integration points, using an iterative process to achieve convergence in the final solution. The expression for the magnitude of the

scattered field from each integration point depends on the specific scattering model selected. Available scattering models are defined in the next section.

Diffuse Scattering Models

Wireless InSite supports three models for calculating diffuse scattered fields, as defined in reference [77].

Model 1 - Lambertian Model

The Lambertian model is a simple model in which the diffuse scattering is concentrated in the direction normal to the scattering surface, proportional to $cos(\theta_s)$. The expression for the magnitude of the scattered field at the receiving antenna (in line of sight) due to one point of integration is

$$\left|\vec{E}_S\right|^2 = \left(\frac{KS\Gamma}{r_i\, r_s}\right)^2 \frac{\cos\theta_i \cos\theta_s}{\pi} dA \tag{20.3}$$

where dA is the area of the surface element during integration, and Γ is the specular reflection coefficient (magnitude) for the second definition of S, or set to 1 for the first. K is a factor defined as follows [79], when the surface is in line-of-sight of the transmitter:

$$K = \sqrt{\frac{ZG_{Tx}P_{Tx}}{2\pi}} \tag{20.4}$$

where G_{Tx} is the gain of the transmitter on the path hitting the surface and P_{Tx} is the radiated power of the transmitter. For the more general case of a path incident on a surface that may not be within line-of-sight, this can be simplified to

$$K = E_i\, r_i \tag{20.5}$$

where E_i is the magnitude of the field incident on the surface at a distance r_i traveled from the transmitter.

Model 2 - Directive Model

This model creates a forward lobe of diffuse scattered power centered on the direction of specular reflection. There is an adjustable integer parameter, α, which determines the narrowness of the lobe - the larger the value of α, the narrower. The expression for $\left|E_s\right|^2$ requires normalization dependent on the values of α and θ_i, which is done using the following factors:

Let Ψ be the angle between the specular direction and the diffuse scattered direction. (See Figure 20.1, above.) A term from integration, I_j, is given by

$$I_j = \frac{2\pi}{j+1}\left[\cos\theta_i \sum_{w=0}^{\frac{j-1}{2}} \binom{2w}{w} \frac{\sin^{2w}\theta_i}{2^{2w}}\right]^{\left(\frac{1-(-1)^j}{2}\right)} \tag{20.6}$$

I_j is evaluated for values of j from 0 to α. The normalization factor is F_α:

$$F_\alpha = \frac{1}{2^\alpha} \sum_{j=0}^{\alpha} \binom{\alpha}{j} I_j \tag{20.7}$$

The expression with over/under integers inside parentheses is the combinatorial function:

$$\binom{n}{m} = \frac{n!}{m!(n-m)!} \tag{20.8}$$

The value of the contribution at each integration point is

$$\left|\vec{E}_S\right|^2 = \left(\frac{KS\Gamma}{r_i\, r_s}\right)^2 \frac{\cos\theta_i}{F_\alpha} \left(\frac{1+\cos\Psi}{2}\right)^\alpha dA \tag{20.9}$$

Model 3 - Backscatter Model

This model is similar to the *directive* model, except that it allows diffuse backscatter with the same mathematical form of lobe as the forward diffuse scatter. The fraction of power in the forward lobe is Λ and in the backscatter lobe $(1-\Lambda)$. The backscatter lobe width parameter is β and the angle between the backscatter direction (the reverse of the incident direction) and the diffuse scattering direction is Φ. The composite diffuse scattered field expression is

$$\left|\vec{E}_S\right|^2 = \left(\frac{KS\Gamma}{r_i\, r_s}\right)^2 \frac{\cos\theta_i}{\Lambda F_\alpha + (1-\Lambda)F_\beta} \left[\Lambda \left(\frac{1+\cos\Psi}{2}\right)^\alpha + (1-\Lambda)\left(\frac{1+\cos\Phi}{2}\right)^\beta\right] dA \tag{20.10}$$

If Λ = 1, all of the power scatters in the forward direction, and this model reduces to the forward scattering model, Model 2. Note that β may or may not be equal to α.

Co-Polarized and Cross-Polarized Terms

In [77], the authors make the simplifying assumption that half of the diffuse scattered power is co-polarized with the reflected field polarization and half is cross-polarized. This idea can be generalized to factors for co-polarized and cross-polarized parts of the diffuse scattered power, as was done in [78]. The respective factors are K_{xpol}, and $K_{copol} = 1 - K_{xpol}$. With this cross-polarization factor defined, the scattered electric field can be expressed as [78]

$$\vec{E}_S = \left[\sqrt{1-K_{xpol}}\left|\vec{E}_S\right|\right] \widehat{i}_{\,copol} + \left[\sqrt{K_{xpol}}\left|\vec{E}_S\right|\right] \widehat{i}_{\,xpol} \tag{20.11}$$

where $\widehat{i}_{\,copol}$ is the polarization vector that would result from a specular reflection, and $\widehat{i}_{\,xpol}$ is the polarization vector normal to it (the cross-polarization term).

Generation of Diffuse Scattered Paths

When diffuse scattering is enabled, the diffuse scattering model generates scattered paths for each path that is incident on a surface whose material properties have been defined to include diffuse scattering characteristics according to one of the previously defined diffuse scattering models. Diffuse scattering contributions are not restricted to the specular direction, and can therefore result in paths in any direction that can ultimately (after further interactions) reach a receiver point. This can result in a large increase in the potential propagation paths between transmitters and receivers, particularly if there are many surfaces with diffuse scattering properties. To keep this number manageable, Wireless InSite has provided some controls:

1. Each propagation path can have at most one diffuse scattering interaction.
2. The user can specify the total number of reflections and diffractions that diffuse scattering paths may incur, and this number can be different than the totals specified for specular paths. This allows the user some control over the number of paths that may be generated.
3. The user can specify which materials have diffuse scattering properties, and depending on the frequency being simulated, may choose to limit these to particularly rough surfaces.

Simulations that have diffuse scattering enabled, generate a collection of diffuse scatted paths, which are recorded in addition to the normal specular paths. From the complete set of paths, the *specular power*, *diffuse scattered power*, and *total power* are calculated for each receiver. The specific results that are available are described later, in Section 20.4.

20.2 Diffuse Scattering Material Properties

To run Diffuse Scattering calculations, Diffuse Scattering must be enabled in each desired material in the scene. To enable Diffuse Scattering, open the MATERIAL PROPERTIES window by double-clicking on an entry in the MATERIALS tab in the MAIN WINDOW. Then check the *DS Enabled* box to activate Diffuse Scattering. Individual Diffuse Scattering properties can be specified by clicking on the DIFFUSE SCATTERING button (Figure 20.2).

The DIFFUSE SCATTERING material properties are:

- *Scattering model* - There are three scattering models to choose from in Wireless InSite:
  - *Lambertian* - Fully statistical, non-directional diffuse scattering model
  - *Directive* - Directional diffuse scattering model with one forward-facing lobe
  - *Directive with Backscatter* - Directional diffuse scattering model with one forward-facing lobe and one rear-facing backscatter lobe
- *Scattering Factor,* S - The fraction of the incident electric field scattered diffusely. Valid values are from 0 (no diffuse scattering) to 1 (all diffuse scattering). S^2 is the fraction of the incident power density scattered diffusely.
- *Cross-pol fraction,* K_{xpol} - The fraction of diffuse power that becomes cross-polarized relative to the incoming ray polarization. Valid values are from 0 to 0.5.

- *Alpha,* α - An integer value between 1 and 10 that controls the shape of the forward-facing lobe. Raising *Alpha* makes this lobe narrower. This parameter only applies when using the *Directive* or *Directive with Backscatter* Diffuse Scattering models.
- *Beta,* β - An integer value between 1 and 10 that controls the shape of the rear-facing backscatter lobe. Raising *Beta* makes this lobe narrower. This parameter only applies when using the *Directive with Backscatter* Diffuse Scattering model.
- *Lambda,* Λ - The mixing fraction between the forward and backscatter lobes. Raising *Lambda* directs more power forward. Valid values are from 0 to 1. This parameter only applies when using the *Directive with Backscatter* Diffuse Scattering model.
- *Advanced Parameters* - It is recommended that these parameters only be modified by users with a thorough understanding of the Diffuse Scattering model, and are generally only used in specialty cases.
  - *Use reflection coefficient* - Disabled by default, which makes Diffuse Scattering materials divert the specified fraction (S^2) of the *incident* power density into diffuse scattering, which reduces both the specular reflected power density and the transmitted power density (if any). When enabled, the specified fraction (S^2) of the *reflected* power density is diverted to diffuse scattering, leaving the transmitted power density unchanged. Note that for most materials, enabling this option will reduce the amount of diffusely scattered power. There are two additional options when enabled:
    - *Incidence Angle* - Uses the incoming ray's reflection coefficient
    - *Normal Angle* - Uses normal incidence to calculate the reflection coefficient

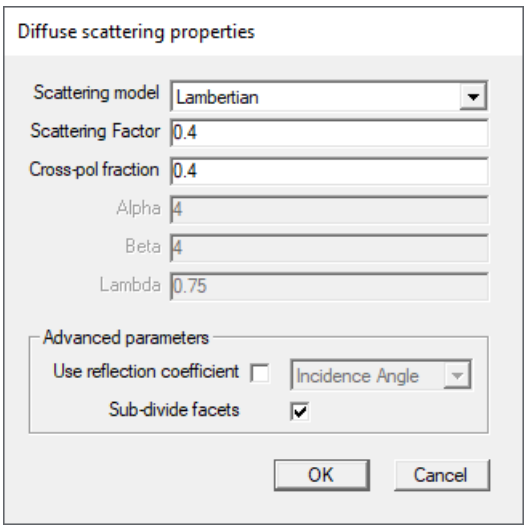


Figure 20.2: The Diffuse Scattering Properties window

20.3 X3D Study Area Diffuse Scattering Properties

To enable DIFFUSE SCATTERING for the X3D model, check the *Diffuse Scattering Enabled* checkbox in the X3D STUDY AREA PROPERTIES window. Note that in order to run DIFFUSE SCATTERING calculations, it must be enabled in both the relevant MATERIAL and STUDY AREA properties. By default, it is disabled in both.

The study area properties for DIFFUSE SCATTERING are accessed by clicking the DIFFUSE SCATTERING button in the X3D STUDY AREA PROPERTIES window.

- *Max Reflections, Transmissions, and Diffractions* - These properties specify the maximum number of each interaction type to be calculated for paths that include a diffuse scattering (DS) interaction.
- *Final Interaction Only* - This restricts the diffuse scattering interaction to the final interaction along a path before reaching a receiver. This is checked by default and can significantly reduce run time for diffuse scattering calculations. Paths that do not encounter diffuse scattering interactions use the main study area interaction limits.
- *Spatial Resolution* - This allows the user to control the resolution of diffuse scattering integration points over scattering surfaces. Modification to this parameter should be done with care as a tighter resolution could significantly increase the computation time for diffuse scattering surface integrations.

Diffuse scattering can significantly increase the number of paths, which has a major impact on the memory and execution time for a simulation. It is therefore recommended that the interaction limits for diffuse scattering paths be set fairly low (e.g., up to 2 reflections and 1 diffraction as shown in Figure 20.3), and at most no greater than the main study area interaction limits.

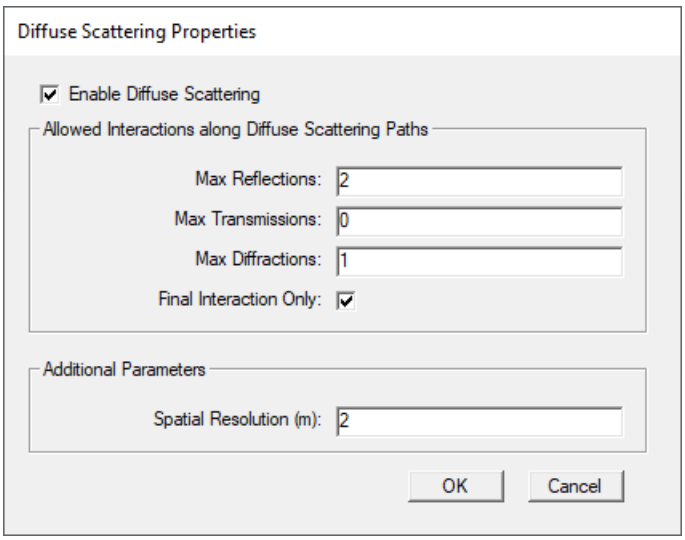


Figure 20.3: Diffuse Scattering interaction limits in the Diffuse Scattering Properties window.

20.4 Diffuse Scattering Results

Diffuse Scattering calculations produce *Received power with diffuse scattering* files (`*.dspower`), which contain the following results:

- *Specular Power* - the total specular power calculated before diffuse scattered paths are considered. It is calculated as the coherent sum of the specular paths, which include line-of-sight, reflected, and diffracted paths. Results include magnitude and phase.
- *Diffuse Power (coherent sum)* - the total diffuse scattered power when scattered paths are assumed to be coherent. It is calculated through the coherent sum of diffuse scattered paths to a given receiver. Results include magnitude and phase.
- *Diffuse Power (power sum)* - the total diffuse scattered power when scattered paths are assumed to have lost phase coherence. It is calculated through a power sum (ignoring phase). This results in a single power magnitude.
- *Total Power (coherent sum)* - the total power when the specular power and phase-coherent diffuse power are summed coherently. Results include magnitude and phase. This result is also reported as the “received power” in the standard received power result.
- *Total Power (power sum)* - the total power when scattered paths are assumed to have lost phase coherence, calculated as the scalar sum of the magnitude of the specular power with the magnitude of the “power-summed” diffuse power. This results in a single power magnitude.

Viewing Diffuse Scattering Results

Each of the result types contained in the Diffuse Scattering result file (`*.dspower`) can be visualized in Wireless InSite like any other standard form of result. For more information on viewing and plotting the results of a simulation, see Section 22.1.

Chapter 21

Results

In this chapter, you will learn...

- → the available result types in Wireless InSite
- → how to define result properties
- → how to request specific results
- → about result files and their units

With a few exceptions, all propagation predictions are generated by Wireless InSite's simulation engine. The engine is a separate executable from the GUI. The engine writes all results to specific files, and when the user chooses to view or plot the results, the GUI loads the data. Whenever a RUN is made the result files for all active TRANSMITTER and RECEIVER sets are updated and all previously loaded RESULTS is refreshed automatically when the run is completed.

▶ Refer to Chapter 22 for more information on how to display the predictions using Wireless InSite's line plotting tools or graphically within the GEOMETRY VIEW.

This chapter describes the following features of the RESULTS generated by the simulation engine:

- Result Types
- Settings That Affect Results
- Result Units
- Result Properties
- Requesting Results
- Result Files

21.1 Result Types

21.1.1 Propagation Model Result Types

Received Power

The way in which the power of each ray path is combined to determine the total received power depends on the setting of the “Sum complex electric fields” option in the ADVANCED STUDY AREA PROPERTIES Window.

- See Chapter 15 for more on the ADVANCED STUDY AREA PROPERTIES Window.
- Refer to Chapter 16 for more on how the electric field is calculated by different propagation models.

If the electric fields are summed without phase information, the time averaged received power in watts is given by:

$$P_R = \sum_{i=1}^{N_P} P_i \tag{21.1}$$

where

N_P is the number of paths

P_i is the time averaged power in watts of the ith path

P_i is given by:

$$P_i = \frac{\lambda^2 \beta}{8\pi\eta_0} \left| E_{\theta,i} g_\theta(\theta_i, \phi_i) + E_{\phi,i} g_\phi(\theta_i, \phi_i) \right|^2 \tag{21.2}$$

where

λ is the wavelength

η_0 is the impedance of free space (377 Ω)

$E_{\theta,i}$ and $E_{\phi,i}$ are the theta and phi components of the electric field of the ith path at the receiver point

θ_i and ϕ_i give the direction of arrival

The direction of arrival is given by:

$$g_\theta(\theta, \phi) = \sqrt{|G_\theta(\theta, \phi)|} e^{j\psi_\theta} \tag{21.3}$$

where

G_θ is the theta component of the receiving antenna gain

ψ_θ is the relative phase of the θ component of the far zone electric field

The direction of arrival in terms of g_ϕ has an analagous definition.

The quantity β is the overlap of the frequency spectrum of the transmitted waveform $S_T(f)$ and the spectrum of the frequency sensitivity of the receiver $S_R(f)$:

$$\beta = \frac{\int_{f_T-(B_T/2)}^{f_T+(B_T/2)} S_T(f)S_R(f)\,df}{\int_{f_T-(B_T/2)}^{f_T+(B_T/2)} S_T(f)\,df} \tag{21.4}$$

where f_T and B_T are, respectively, the center frequency and bandwidth of the transmitted waveform.

At present, a flat distribution is always assumed for narrowband waveforms, such that:

$$S(f) = \begin{cases} 1 & \text{for } f_0 - \dfrac{B}{2} < f < f_0 + \dfrac{B}{2} \\ 0 & \text{otherwise} \end{cases} \tag{21.5}$$

where f_0 is the center frequency and B is the bandwidth.

When all fields are combined coherently (with phase) the total received power is:

$$P_R = \frac{\lambda^2\beta}{8\pi\eta_0}\left|\sum_{i=1}^{N_p}\left[E_{\theta,i}g_\theta(\theta_i,\phi_i) + E_{\phi,i}g_\phi(\theta_i,\phi_i)\right]\right|^2 \tag{21.6}$$

If the correlated option for combining paths is used, the time averaged power due to each group of correlated paths is found first using:

$$Q_j = \frac{\lambda^2\beta}{8\pi\eta_0}\left|\sum_{i=1}^{M_j}\left[E_{\theta,i}g_\theta(\theta_i,\phi_i) + E_{\phi,i}g_\phi(\theta_i,\phi_i)\right]\right|^2 \tag{21.7}$$

where Q_j is the power due to the jth group of paths and M_j is the number of paths in the group.

The total power is then found by summing the power of each group:

$$P_R = \sum_{i=1}^{N_c} Q_i \tag{21.8}$$

where N_c is the number of groups.

Once the received power in watts is found, the power in dBm is determined from:

$$P_R(dBm) = 10log_{10}[P_R(W)] \; + \; 30dB \; - \; L_S(dB) \tag{21.9}$$

where L_S is any additional loss in the system which can be specified through the cable loss field in the ANTENNA PROPERTIES Window. A value of -250 dBm is written to the result file if $P_R \leq 1 \times 10^{-25}\, mW$.

Received Power in Free Space with Antenna Patterns

Results for the received power due only to the free space $\left(\frac{1}{R^2}\right)$ loss are given with and without the effect of the antenna patterns. When the antenna patterns are included,

$$P_R = \frac{\lambda^2 P_T \beta}{(4\pi)^2 R^2} \left| g_{T,\theta}(\theta_D, \phi_D) g_{R,\theta}(\theta_A, \phi_A) + g_{T,\phi}(\theta_D, \phi_D) g_{R,\phi}(\theta_A, \phi_A) \right|^2 \tag{21.10}$$

where

P_T is the time averaged radiated power

g_T and g_R are defined in the generic Equation 21.3

θ_D and ϕ_D give the direction in which the ray leaves the transmitter

θ_A and ϕ_A give the direction from which the ray arrives at the receiver

R is the distance between the transmitter and receiver

Once the received power in watts is found, the power in dBm is determined from

$$P_R(dBm) = 10 log_{10}[P_R(W)] + 30dB - L_S(dB) \tag{21.11}$$

Received Power in Free Space without Antenna Patterns

The received power without considering the antenna patterns is equivalent to assuming polarization matched isotropic patterns, and is given by:

$$P_R = \frac{\lambda^2 \beta P_T}{(4\pi)^2 R^2} \tag{21.12}$$

Received Power with Diffuse Scattering

DIFFUSE SCATTERING calculations produce a variant of received power result that contains specular power magnitude and phase, diffuse scattered power magnitude and phase, total power (coherent sum) magnitude and phase, diffuse scattered (power sum) magnitude, and total power (power sum) magnitude. For more information about each individual type of result contained in the received power with diffuse scattering file, go to section 20.4.

Path Loss

Path loss factors out the transmitter power and the maximum gains of the transmitter and receiver to yield a number that represents signal loss in the transmission budget due to multipath propagation: distance and interactions along each path, individual paths being off max gain and individual path polarization mismatch, constructive and destructive interference, atmospheric absorption, etc. It does not include system losses such as VSWR, and Transmission Line loss which affect received power.

The most commonly used definition of path loss is:

$$L_{Path}(dB) = P_T(dBm) - P_R(dBm) + G_{T,Max}(dBi) + G_{R,Max}(dBi) - L_S(dB) \quad (21.13)$$

where

$G_{T,Max}$ and $G_{R,Max}$ are the maximum gains of the transmitting and receiving antennas, respectively

L_S is the sum of all other losses in the system (in dB), including the bandwidth overlap factor

For *Directional* Antennas, the path loss depends on the orientation of the antenna. Keep this in mind when making use of the path loss as a term in a link budget. It will not depend solely on the frequency and the environment as it does for *Omni-directional* antennas.

Path Gain

Path gain is equivalent to path loss, but is represented with the opposite sign.

$$G_{Path}(dB) = -L_{Path}(dB) \quad (21.14)$$

The path gain is a less commonly encountered measure than path loss, but it sometimes used instead of the path loss when it provides a more convenient way to present results. In fact, the terms are sometimes used almost interchangeably, and one commonly encounters plots in literature labeled as path loss which are technically path gain. This generally does not present any confusion to the knowledgeable reader with a background in RF propagation, but may be confusing to readers without this background.

Free Space Path Loss with Antenna Patterns

When the antenna patterns are included, the free path loss is calculated using above definition of the path loss (Equation (21.13)) and the free space received power (Equation (21.10)). Assuming polarization of the transmitter and receiver match, the path loss in free space reduces to:

$$L_{FS}(dB) = -10log_{10}\left(\frac{\lambda^2 G_T G_R}{(4\pi)^2 R^2}\right) + G_{T,Max}(dBi) + G_{R,Max}(dBi) \quad (21.15)$$

where

G_T is the gain of the transmitting antenna in the direction of the receiver

G_R is the gain of the receiving antenna in the direction of the transmitter

R is the distance between the transmitter and receiver

Free Space Path Loss Without Antenna Patterns

If the antenna patterns are ignored and isotropic patterns are assumed, the free space path loss reduces to:

$$L_{FS}(dB) = -10log_{10}\left(\frac{\lambda^2}{(4\pi)^2R^2}\right) \tag{21.16}$$

The previous definition (Equation (21.15)) includes any loss due to a mismatch of the transmitting and receiving antenna polarizations, whereas the current one (Equation (21.16)) does not.

Excess Path Loss with Antenna Patterns

The excess path loss is a measure of the loss above that due to free space loss,

$$L_X(dB) = L_{Path}(dB) - L_{FS}(dB) \tag{21.17}$$

where $L_{FS}(dB)$ is the free space loss with the antenna patterns.

Excess Path Loss Without Antenna Patterns

The excess path loss is a measure of the loss above that due to free space loss,

$$L_X(dB) = L_{Path}(dB) - L_{FS}(dB) \tag{21.18}$$

where $L_{FS}(dB)$ is the free space loss *without* the antenna patterns.

Propagation Paths

Results of this type consists of the geometrical ray paths through the environment from the transmitter to the receiver, and the type of interactions each ray undergoes. The geometrical data for each path consists of the Cartesian coordinates of the endpoints of one or more connected line segments. With the exception of the last segment, which ends on the receiver, each line segment ends at a point where an interaction with the environment occurred. This interaction will typically be a reflection, transmission or diffraction, although other interactions are possible at foliage surfaces. The interactions undergone by each path are stored in sequence form, such as Tx-R-R-D-T-Rx, which in this case indicates that a ray left the transmitter, reflected twice, diffracted from an edge, then underwent a transmission through a wall before finally arriving at the receiver. A list of the interaction designations is provided in Table 21.1.

Table 21.1: Propagation Path Interactions

Interaction Description	X3D	Full 3D	Urban Canyon	Vertical Plane
Transmitter	Tx	Tx	Tx	Tx
Receiver	Rx	Rx	Rx	Rx
Reflection	R	R	R	R
Ground Bounce	R	G	G	G
Transmission	T	T	T	T
Diffraction	D	D or d*	D	D
Enter/leave foliage	F/X	F/X	–	–
Diffuse Scattering	DS	–	–	–

*The Full 3D propagation model uses **D** to represent a diffraction from a vertical or near vertical edge and **d** to represent a diffraction from a horizontal or near horizontal edge.

- An example of a propagation paths result file can be found in Appendix L, along with a brief description of the format.

Time of Arrival

The time of arrival for each propagation path is given by:

$$t_i = \frac{L_i}{c} \tag{21.19}$$

where

L_i is the total geometrical path length

c is the speed of light in free space (or in the defined air medium for the X3D model)

Mean Time of Arrival

The mean time of arrival is given by:

$$\bar{t} = \frac{\sum_{i=1}^{N_P} P_i t_i}{P_R} \tag{21.20}$$

where P_R and P_i are defined by Equations 21.1 and 21.2, respectively.

Delay Spread

The delay spread is a useful measure of a variety of multipath-related effects. It is a power-weighted rms of the time delays, calculated assuming a narrowband signal at the carrier frequency, using the following formula [80]:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N_P}(t_i - \bar{t})^2 P_i}{P_R}} \tag{21.21}$$

where t_i and $\bar{t}$ are defined by Equations 21.19 and 21.20, respectively.

Electric Field Magnitude

Magnitude of the total electric field at the receiver points is found by summing the complex valued electric field contributed by each ray path:

$$|\, E_{Tot}| = \left| \sum_{i=1}^{N_P} E_i \right| \tag{21.22}$$

where E_i is the complex valued electric field due to the ith ray path. The field strength given here does not take into account the antenna pattern of the receiving antenna.

Electric Field Phase

The phase of each Cartesian component of the total electric field, for example the phase of the X-component of the total field, is given by:

$$\phi_x = tan^{-1}\left(\frac{Im(E_{Tot,x})}{Re(E_{Tot,x})}\right) \tag{21.23}$$

with the phase set to 0 if $E_x = 0$.

Poynting Vector

The time-averaged Poynting vector magnitude and its components at a receiver point are given by the cross product of the electric field, $\mathbf{E}$, and the complex conjugate of the magnetic field, $\mathbf{H}^*$.

$$\mathbf{S}_{Av} = \frac{1}{2}(\mathbf{E} \times \mathbf{H}^*) \tag{21.24}$$

Animated Fields

The magnitude of the Cartesian electric field components versus time are found by combining complex E-fields with time-of-arrival, direction-of-arrival and the shape of the radiated waveform.

- Results may be viewed using the MOVIE PLAYER, as described in Chapter 22.

Complex Electric Field

The complex electric field result provides the spherical and Cartesian components of the electric field for individual ray paths between transmitter and receiver points. Because the Complex E-Field is provided for each ray path, the organization of the data differs from a common *.p2m file and this result type cannot be visualized from the graphical user interface. However, users can access this data from an ASCII file written to the study area directory.

Additional comments about the complex e-field result:

- The complex e-field values are the electric field values at the receiver point location and do not include any effects of the receiver antennas.
- Paths are organized by received power.
- The number of paths written to the file is controlled by the *Maximum rendered paths* field in the PROJECT RESULT PROPERTIES window. By default, 25 paths are written to the file.

The general format of the complex e-field result file consists of a header, the number of receiver points listed in the set, identification of an individual rx point and the complex e-field values of the paths reaching that point.

```
# Receiver Set:  rx short description
# Path TOA Mag(E_φ) Phase(E_φ) Mag(E_θ) Phase(E_θ) Mag(E_x) Phase(E_x) Mag(E_y) Phase(E_y) ...
     ...Mag(E_z) Phase(E_z)
<rx point 1> <number of paths to rx point 1>
<complex e-field data for path 1 to rx point 1>
<complex e-field data for path 2 to rx point 1>
...
<complex e-field data for last path to rx point 1>

<rx point 2> <number of paths to rx point 2>
<complex e-field data for path 1 to rx point 2>
<complex e-field data for path 2 to rx point 2>
...
<complex e-field data for last path to rx point 2>


.
.
.
<last rx point> <number of paths to last rx point>
<complex e-field data for path 1 to last rx point>
<complex e-field data for path 2 to last rx point>
...
<complex e-field data for last path to last rx point>
```

Direction of Arrival

The angles θ_A and ϕ_A, with reference to the spherical coordinate system, give the direction from which the propagation path arrives at receiver point.

▶ See Figure 12.1 for an image of the spherical coordinate system.

The direction in Cartesian coordinates is given by the unit vector:

$$\hat{\mathbf{a}} = sin(\theta_A)cos(\phi_A)\hat{\mathbf{x}} \; + \; sin(\theta_A)sin(\phi_A)\hat{\mathbf{y}} \; + \; cos(\theta_A)\hat{\mathbf{z}} \tag{21.25}$$

Mean Direction of Arrival

The mean direction of arrival from which energy arrives at the receiver is in the direction:

$$\bar{\theta}_A = tan^{-1}\left(\frac{\sqrt{A_x^2 + A_y^2}}{A_z}\right) \tag{21.26}$$

$$\bar{\phi}_A = tan^{-1}\left(\frac{A_y}{A_x}\right) \tag{21.27}$$

where

$$A = \sum_{i=1}^{N_P} P_i \, \hat{\mathbf{a}}_i$$

P_i is the power carried by i^{th} ray path

$\hat{\mathbf{a}}_i$ is the unit vector in the direction from which the ith path arrives at the receiver point.

Direction of Departure

The angles θ_0 and ϕ_0, with reference to the spherical coordinate system, give the direction in which the propagation path leaves the transmitter.

- See Figure 12.1 for an image of the spherical coordinate system.

The direction in Cartesian coordinates is given by the unit vector:

$$\hat{\mathbf{d}} = sin(\theta_D)cos(\phi_D)\hat{\mathbf{x}} + sin(\theta_D)sin(\phi_D)\hat{\mathbf{y}} + cos(\theta_D)\hat{\mathbf{z}} \tag{21.28}$$

Mean Direction of Departure

The mean direction of departure from which energy leaves the transmitter is given by the angles:

$$\bar{\theta}_D = tan^{-1}\left(\frac{\sqrt{D_x^2 + D_y^2}}{D_z}\right) \tag{21.29}$$

$$\bar{\phi}_D = tan^{-1}\left(\frac{D_y}{D_x}\right) \tag{21.30}$$

where

$$D = \sum_{i=1}^{N_P} P_i \, \hat{\mathbf{d}}_i$$

P_i is the power carried by the i^{tdsh} ray path

$\hat{\mathbf{d}}_i$ is the unit vector in the direction in which the ith path leaves the transmitter.

Complex Impulse Response

The complex voltage at the feed point of the receiving antenna due to the ith propagation path is proportional to:

$$V_i = E_{\theta,i} g_\theta(\theta_i, \phi_i) + E_{\phi,i} g_\phi(\theta_i, \phi_i) \tag{21.31}$$

where E and g are defined under "Received Power" in Section 21.1. The complex impulse response is considered to be:

$$s_i = P_i e^{j\psi_i} \tag{21.32}$$

where

P_i is the power carried by the ith ray path

$\psi_i = tan^{-1}\left(\frac{Im(V_i)}{Re(V_i)}\right)$ is the phase

Power Delay Profile

This result type is only available for the *tPoint* Receiver set. The evaluation of this result type depends on whether the waveform is non-dispersive or dispersive.

For **dispersive waveforms**, the voltage $V_0(f)$ on a hypothetical transmission line with a constant real impedance of Z_L is first found from:

$$V_0(f) = \sqrt{Z_L P_0(f)} \tag{21.33}$$

where $P_0(f)$ is the total time averaged received power, calculated for a time averaged radiated power (P_T) of $1\,W$. The phase of $V_0(f)$ is equal to the phase of $\sum V_i$, where V_i is given by Equation 21.31.

The instantaneous voltage on the transmission line as a function of time is found from:

$$V(t) = F^{-1}[V(f)] \tag{21.34}$$

where

$V(f) = V_0(f)S(f)$

$S(f)$ is the radiated waveform scaled to give the correct total energy radiated from the transmitter

F^{-1} designates the inverse Fourier transform

Finally, the instantaneous power is found using:

$$P(t) = \frac{V(t)^2}{Z_L} \tag{21.35}$$

For **non-dispersive waveforms**, the power delay profile is the envelope of the instantaneous received power as a function of time:

$$p(t) = \frac{\lambda^2 \beta}{8\pi\eta_0} \left| \sum_{i=1}^{N_P} e(t_i - t) V_i \right|^2 \tag{21.36}$$

where

V_i is the complex voltage, defined by Equation 21.31

$e(t)$ is the envelope of the transmitted waveform

t_i is the time-of-arrival

The bandwidth is assumed to be much less than the carrier frequency.

Electric Field vs. Frequency

This result type is only available for dispersive waveforms. It gives the Cartesian components of the complex electric field as a function of frequency.

- See Chapter 11 for more information on creating dispersive waveforms.

For all waveforms which are defined as a modulation of a carrier frequency, the electric field is calculated from:

$$\mathbf{E}(f) = \mathbf{E}_0(f) S(f) \tag{21.37}$$

where

$\mathbf{E}_0(f)$ is the complex electric field calculated at a frequency f, assuming a unit amplitude at a distance of a meter from the source

$S(f)$ is the inverse Fourier transform of the actual time domain electric field, $\mathbf{E}(t)$, radiated from the source in free space

$S(f)$ is evaluated at a distance of one meter from the source. The function $\mathbf{E}(t)$ is defined to give the correct radiated power as described in Section 21.2.

Electric Field vs. Time

The evaluation of this result type depends on whether the waveform is non-dispersive or dispersive.

For **dispersive waveforms**, the electric field as a function of time is found from:

$$E(t) = F^{-1}[E(f)] \tag{21.38}$$

where F^{-1} designates the inverse Fourier transform. The evaluation of $E(f)$ is described by Equation 21.37.

For **non-dispersive modulated waveforms**, the total electric field is calculated from:

$$E(t) = Re\left[\sum_{i=1}^{N_P} A_i m(t - t_i) e^{j\omega(t-t_i)}\right] \tag{21.39}$$

where

$m(\tau)$ is the normalized modulation

N_P is the number of paths to the field point

A_i is the complex electric amplitude of the ith path evaluated at the carrier frequency

t_i is the time-of-arrival of the ith path

Doppler Shift

Apparent change in frequency of the ith propagation path due to the motion of the transmitter and/or receiver is given by:

$$\triangle f_i = f_0\left[\frac{\mathbf{d}_i \cdot \mathbf{v}_T}{c} + \frac{\mathbf{a}_i \cdot \mathbf{v}_R}{c}\right] \tag{21.40}$$

where

$\mathbf{v}_R$ and $\mathbf{v}_T$ are the velocities of the receiver and transmitter

$\mathbf{a}_i$ and $\mathbf{d}_i$ are the directions of arrival and departure of the ith ray as defined above

f_0 is the carrier frequency

Routes, *Arcs* and *Vertical Arcs* are the only types of TRANSMITTER or RECEIVER sets that allow users to enter a velocity in the TX/RX LAYOUT PROPERTIES Window. For *Routes*, the direction of the velocity is along the Route segment in the order the points were defined. For *Arcs*, the direction is tangential to the Arc in the direction listed in the TX/RX LAYOUT PROPERTIES Window. Entering a negative velocity reverses the direction along the Route or Arc without redefining the transmitter or receiver set.

✓ Doppler shift data is generated when a *Time of Arrival* result is requested.

where

the subscripts μ and ν can be either the linear polarization components θ, ϕ or circular polarization components $rhcp$, $lhcp$

The phase of the scattering amplitude matrix is the global origin of the project.

Terrain Profiles

Terrain profile result files are only created when the VERTICAL PLANE propagation model is used. All vertical profiles containing active RECEIVER points are written to separate result files. The location and file naming scheme for these files is described under **Diagnostic Data**.

Diagnostic Data

When diagnostic data is selected as a result type, several files containing building data, reflection/transmission coefficients, and antenna pattern data are written to a folder called “diag”, which is located in the same folder as the project file (`*.setup`). The name of the project is at the beginning of all file names. This data cannot be viewed or plotted by Wireless InSite and is only provided for diagnostic and troubleshooting purposes.

- The file naming schemes are described in Section 21.6.

21.1.2 Communication System Result Types

Signal-to-Interferer Ratio

The signal-to-interferer ratio (SIR) is the ratio of the received power from the transmitter to the sum of power from all interference sources. The ratio is given by:

$$SIR(dB) = 10log_{10}(P_R(i)) - 10log_{10}\left(I_{total}\right) \tag{21.41}$$

where

P_R is the received power from the transmitter

I is the total interference

Signal-to-Noise Ratio

The signal-to-noise ratio (SNR) is the ratio of the received power from the transmitter to the sum of power from all noise sources. The ratio is given by:

$$SNR(dB) = 10log_{10}(P_R(i)) - 10log_{10}\left(N_{total}\right) \tag{21.42}$$

where

P_R is the received power from the transmitter

N_{total} is the total noise

Signal-to-Interferer-Plus-Noise Ratio

The signal-to-interferer-plus-noise ratio (SINR) is the ratio of the received power from the transmitter to the sum of power from all interference sources and all noise sources. The ratio is given by:

$$SINR(dB) = 10log_{10}(P_R(i)) - 10log_{10}\left(I_{total}\right) - 10log_{10}\left(N_{total}\right) \tag{21.43}$$

where

P_R is the received power from the transmitter

I_{total} is the total interference

N_{total} is the total noise

Receiver Signal Strength Indicator

The receiver signal strength indicator (RSSI) is the sum of the received power, interference sources and noise sources:

$$RSSI = P_R + I_{total} + N_{total} \tag{21.44}$$

where

P_R is the received power from the transmitter

I_{total} is the total interference

N_{total} is the total noise

Reference Signal Received Power

The reference signal received power (RSRP) for an LTE system is given by:

$$RSRP(dBm) = RSSI(dBm) - 10log_{10}(12 \times N_{blocks}) \tag{21.45}$$

where

$RSSI$ is the reference signal strength indicator

N_{blocks} is the number of resource blocks which varies according to the bandwidth of the LTE system.

Table 21.2: Number of Resource Blocks for LTE Systems

Bandwidth	1.4 MHz	3 MHz	5 MHz	10 MHz	15 MHz	20 MHz
N_{blocks}	6	15	25	50	75	100

Reference Signal Received Quality

The reference signal received quality (RSRQ) for an LTE system is given by:

$$RSRQ = N_{blocks} \frac{RSRP}{RSSI} \tag{21.46}$$

where

N_{blocks} is the number of resource blocks which varies according to the bandwidth of the LTE system.

$RSRP$ is the reference signal received power

$RSSI$ is the reference signal strength indicator

Total Received Power

Total received power from multiple transmitters to a single receiver is calculated both with and without phase. When the electric fields are combined without phase, or as a power sum, the total received power from all active transmitters is:

$$P_{Tot} = \sum_{i=1}^{N_T} P_R(i) \tag{21.47}$$

where

$P_R(i)$ is the received power due to the ith transmitter

N_T is the number of active transmitters

When the electric fields are combined with phase, the total power is:

$$P_{Tot} = \frac{1}{Z_L} \left| \sum_{i=1}^{N_T} V_i \right|^2 \tag{21.48}$$

where

V_i is the total complex voltage at the feed point of the receiving antenna due to the ith transmitter

Z_L is the impedance of the transmission line attached to the receiving antenna

A value of -250 dBm is written to the result file if $P_{Tot} \leq 1 \times 10^{-25}\, mW$.

Receiver's Strongest Transmitter

This result type identifies the transmitter which contributes the largest received power at each receiver point. The transmitter is identified by the set number and the number of the point within the set.

Receiver's Strongest SINR Transmitter

This result type identifies the transmitter which produces the highest SINR value at each receiver point. The transmitter is identified by the set number and the number of the point within the set.

Channel Capacity

The channel capacity represents the maximum possible data transmission rate for a communication channel and is calculated using the Shannon-Hartley theorem [81]:

$$Capacity = B\, log_2(1 + SINR) \tag{21.49}$$

where

B is the bandwidth of the channel

$SINR$ is the linear signal-to-interference-plus-noise ratio

21.1.3 MIMO-Specific Result Types

This section describes additional results that are available from MIMO channel data outputs.

H-Matrix

$G_k[n]$ is the ratio of the power received by receiver antenna element k divided by the power radiated by transmitter antenna element n with all other elements radiating zero power. $\theta_k[n]$ is the phase in radians of the voltage across a matched load at k under the same conditions. Note that $G_k[n]$ and $\theta_k[n]$ include all of the propagation paths in a complex multi-path environment from transmitter antenna element n to receiver antenna element k summed coherently. The propagation factor, $g_k[n]$, is defined as

$$g_k[n] = \sqrt{G_k[n]}e^{i\theta_k[n]} \tag{21.50}$$

When written in **bold** without the $[n]$, $\mathbf{g}_k$ is an N-dimensional complex (real, imaginary) row vector $(1 \times N)$. The complex vector $\mathbf{g}_k$ is often referred to as the *channel vector* which plays an important role in MIMO systems. However, in such contexts, the channel vector is typically denoted as $\mathbf{h}_k$ rather than $\mathbf{g}_k$ but has the same definition (i.e., $\mathbf{h}_k = \mathbf{g}_k$). The H-matrix is then the aggregation of the channel vectors for each of the K receiver antennas, and has dimensions $K \times N$. The H-matrix is used in communications theory as the basis for modeling various optimization strategies such as Maximum Ratio Combining (MRC) and Zero-Forcing Beam Forming (ZFBF). (*Note*: Occasionally, some literature sources, particularly in the context of multi-user MIMO, define the channel vector $\mathbf{h}_k$ such that $\mathbf{h}_k = \mathbf{g}_k^*$, where * denotes the conjugate transpose, in contrast to the definition used here. Correspondingly, in those cases, the H-matrix is the conjugate transpose of the definition used here.)

RMS Delay Spread

The RMS Delay Spread is a useful measure of a variety of multipath-related effects. It is a power-weighted rms of the sum over all the MIMO elements and paths to the elements of the time delays given by:

$$\sigma = \sqrt{\frac{\sum_{e=1}^{N_E}[\sum_{p=1}^{N_{PE}}(P_{pe} * (t_{pe} - t_{avg})^2)]}{P_R}} \tag{21.51}$$

where

N_E is the number of elements

N_{PE} is the number of paths for element e

P_{pe} is the power of path p for element e

t_{pe} is the time of arrival of path p of element e

t_{avg} is the mean time of arrival over all paths and elements

P_R is sum of the power over all paths and elements

P_R is given by:

$$P_R = \sum_{e=1}^{N_E}[\sum_{p=1}^{N_{PE}}(P_{pe})] \tag{21.52}$$

where

N_E is the number of elements

N_{PE} is the number of paths for element e

P_{pe} is the power of path p for element e

t_{avg} is given by:

$$t_{avg} = \frac{\sum_{e=1}^{N_E}[\sum_{p=1}^{N_{PE}}(P_{pe} * t_{pe})]}{P_R} \tag{21.53}$$

where

N_E is the number of elements

N_{PE} is the number of paths for element e

P_{pe} is the power of path p for element e

t_{pe} is the time of arrival of path p of element e

P_R is sum of the power over all paths and elements

RMS Angle Spread of Arrival

The RMS Angle Spread of Arrival is the vertical, horizontal, and total angular spread of the angle of arrival over all MIMO elements and paths to the elements. The RMS Angle Spread of Arrival is given by:

$$\sigma_V = \sqrt{\frac{\sum_{e=1}^{N_E}[\sum_{p=1}^{N_{PE}}(P_{pe} * (\Delta_V)^2)]}{P_R}} \tag{21.54}$$

$$\sigma_H = \sqrt{\frac{\sum_{e=1}^{N_E}[\sum_{p=1}^{N_{PE}}(P_{pe} * (\Delta_H)^2)]}{P_R}} \tag{21.55}$$

$$\sigma_\alpha = \sqrt{(\sigma_V)^2 + (\sigma_H)^2} \tag{21.56}$$

where

σ_V is the vertical spread of arrival

σ_H is the horizontal spread of arrival

σ_α is the total spread of arrival

N_E is the number of elements

N_{PE} is the number of paths for element e

P_{pe} is the power of path p for element e

Δ_V is defined by the equation 21.57

Δ_H is defined by the equation 21.58

P_R is sum of the power over all paths and elements and defined by the equation 21.52

Δ_V is given by:

$$\Delta_V = (\theta_{pe} - \theta_{Aavg}) \tag{21.57}$$

where

θ_{pe} is the theta angle of arrival of path p for element e

θ_{Aavg} is defined by the equation 21.63

Δ_H is given by:

$$(\Delta_H)^2 = \alpha^2 - (\Delta_V)^2 \tag{21.58}$$

where

α is defined by the equation 21.59

Δ_V is defined by the equation 21.57

α is given by:

$$\alpha = cos^{-1}(B * a_{pe}) \tag{21.59}$$

where

B is defined by the equation 21.60, 21.61, 21.62

a_{pe} is defined by the equation 21.25

B is given by:

$$B_x = \frac{A_x}{A_{mag}} \tag{21.60}$$

$$B_y = \frac{A_y}{A_{mag}} \tag{21.61}$$

$$B_z = \frac{A_z}{A_{mag}} \tag{21.62}$$

where

A_x is defined by the equation 21.65

A_y is defined by the equation 21.65

A_z is defined by the equation 21.65

A_{mag} is defined by the equation 21.64

θ_{Aavg} is given by:

$$\theta_{Aavg} = cos^{-1}\left(\frac{A_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}}\right) \tag{21.63}$$

A_{mag} is given by:

$$A_{mag} = \sqrt{A_x^2 + A_y^2 + A_z^2} \tag{21.64}$$

where

A_x is defined by the equation 21.65

A_y is defined by the equation 21.65

A_z is defined by the equation 21.65

A is given by:

$$A = \sum_{e=1}^{N_E} [\sum_{p=1}^{N_{PE}} (P_{pe} * a_{pe})] \tag{21.65}$$

where

N_E is the number of elements

N_{PE} is the number of paths for element e

P_{pe} is the power of path p for element e

a_{pe} is the direction of arrival defined by 21.25

RMS Angle Spread of Departure

The RMS Angle Spread of Departure is the vertical, horizontal, and total angular spread of the angle of departure over all MIMO elements and paths to the elements. The RMS Angle Spread of Departure is given by:

$$\sigma_V = \sqrt{\frac{\sum_{e=1}^{N_E} [\sum_{p=1}^{N_{PE}} (P_{pe} * (\Delta_V)^2)]}{P_R}} \tag{21.66}$$

$$\sigma_H = \sqrt{\frac{\sum_{e=1}^{N_E} [\sum_{p=1}^{N_{PE}} (P_{pe} * (\Delta_H)^2)]}{P_R}} \tag{21.67}$$

$$\sigma_\alpha = \sqrt{(\sigma_V)^2 + (\sigma_H)^2} \tag{21.68}$$

where

σ_V is the vertical spread of departure

σ_H is the horizontal spread of departure

σ_α is the total spread of departure

N_E is the number of elements

N_{PE} is the number of paths for element e

P_{pe} is the power of path p for element e

Δ_V is defined by the equation 21.69

Δ_H is defined by the equation 21.70

P_R is sum of the power over all paths and elements and defined by the equation 21.52

Δ_V is given by:

$$\Delta_V = (\theta_{pe} - \theta_{Davg}) \tag{21.69}$$

where

θ_{pe} is the theta angle of departure of path p for element e

θ_{Davg} is defined by the equation 21.75

Δ_H is given by:

$$(\Delta_H)^2 = \alpha^2 - (\Delta_V)^2 \tag{21.70}$$

where

α is defined by the equation 21.71

Δ_V is defined by the equation 21.69

α is given by:

$$\alpha = cos^{-1}(B * d_{pe}) \tag{21.71}$$

where

B is defined by the equation 21.72, 21.73, 21.74

d_{pe} is defined by the equation 21.28

B is given by:

$$B_x = \frac{D_x}{D_{mag}} \tag{21.72}$$

$$B_y = \frac{D_y}{D_{mag}} \tag{21.73}$$

$$B_z = \frac{D_z}{D_{mag}} \tag{21.74}$$

where

D_x is defined by the equation 21.77

D_y is defined by the equation 21.77

D_z is defined by the equation 21.77

D_{mag} is defined by the equation 21.76

θ_{Davg} is given by:

$$\theta_{Davg} = cos^{-1}\left(\frac{D_z}{\sqrt{D_x^2 + D_y^2 + D_z^2}}\right) \tag{21.75}$$

D_{mag} is given by:

$$D_{mag} = \sqrt{D_x^2 + D_y^2 + D_z^2} \tag{21.76}$$

where

D_x is defined by the equation 21.77

D_y is defined by the equation 21.77

D_z is defined by the equation 21.77

D is given by:

$$D = \sum_{e=1}^{N_E}[\sum_{p=1}^{N_{PE}}(P_{pe} * d_{pe})] \tag{21.77}$$

where

N_E is the number of elements

N_{PE} is the number of paths for element e

P_{pe} is the power of path p for element e

d_{pe} is the direction of departure defined by 21.28

21.2 Settings That Affect Results

Radiated Power

Radiated power is the power emitted by the transmitting ANTENNA. It is related to input power, or the power going into the transmitting antenna, by the following equation:

$$P_{rad} = P_{in} + L_{mismatch} + L_{cable} \tag{21.78}$$

where

P_{rad} is the radiated power, in dBm

P_{in} is the input power, in dBm

$L_{mismatch}$ is the mismatch loss between the input cable and the antenna (characterized by the VSWR value in the antenna properties window) in dB, always ≥ 0

L_{cable} is the general loss of the antenna lines in dB, always ≥ 0

Input power is set as one of the TRANSMITTER properties. The way in which the specified value is used to determine the electric field depends on the type of WAVEFORM assigned to the transmitter. There are currently three kinds of waveforms in Wireless InSite:

- *Sinusoid*
- Modulated sine waves
- Waveforms of a more general shape, including *Chirp*, *Gaussian Derivative* and *User-Defined* waveforms (either frequency or time domain)

For *Sinusoidal* waveforms, the radiated power is simply interpreted as the power averaged over one cycle of the sine wave. For modulated carrier waveforms, the radiated power is interpreted to be the peak time-averaged power, which is averaged over one period of the carrier wave. The electric field at a distance R from an isotropic source is given by:

$$E(t) = \sqrt{\frac{\bar{P}\eta_0}{2\pi}}\,\frac{1}{R}m(t)sin(\omega t) \tag{21.79}$$

where

- $\bar{P}$ is the radiated power
- $m(t)$ is the modulation, defined to have a maximum value of 1
- η_0 is the impedance of free space
- ω is the angular frequency of the carrier in radians/second

For *Chirp*, *Gaussian Derivative* and *User-Defined* waveforms, the radiated power is interpreted to mean the time-averaged power over the entire duration of the pulse. Assuming $E(t)$ is the electric field in V/m at time t, at a distance R from an isotropic source, the time averaged radiated power in Watts is then given by:

$$\bar{P} = \frac{4\pi R^2}{\eta_0 T_D}\int_0^{T_D}[E(t)]^2\,dt \tag{21.80}$$

where T_D is the duration of the pulse, such that $E(t) = 0$, for $t > T_D$.

Conversely, if $s(t)$ is a normalized time domain pulse, the electric field at a distance R from an isotropic source is given by:

$$E(t) = A_0 s(t) \tag{21.81}$$

where

$$A_0 = \frac{\sqrt{\frac{\bar{P}\eta_0}{4\pi R^2}}}{\sqrt{\frac{1}{T_D}\int_0^{T_D}[s(t)]^2\,dt}} \tag{21.82}$$

Dispersive Waveforms

The calculation of a few RESULTS types will depend on whether the transmitter waveform is dispersive or non-dispersive, but many other RESULTS types will not. By selecting the dispersive option, the calculation of electric field as a function of time will include the dispersion on the broadband waveform pulse as it propagates. The electric field is also calculated as a function of frequency for dispersive waveforms. The electric field vs. time and electric field vs. frequency result files are only created for *Points* and "Points-On-Face" RECEIVER sets.

Even when a WAVEFORM is designated as dispersive many RESULTS types will still only be evaluated at a single frequency. The carrier frequency, or center frequency for a *Gaussian Derivative* and *User-Defined* waveform, will be used in these cases. RESULTS types in this category include TOTAL ELECTRIC FIELD and COMPLEX ELECTRIC FIELD. The RECEIVED POWER, PATH LOSS/GAIN, and COMPLEX IMPULSE RESPONSE are also only evaluated at a single frequency, but in these cases the bandwidth is also taken into account through the frequency response of the receiver.

▶ See Section 21.1 for more on the calculation of the received power.

21.3 Result Units

The units listed below are the ones used for the numerical data in all result files. In some cases other units many be available when viewing results in the GEOMETRY VIEW or in line plots, but selecting one of these other options will not change the numerical data in the files.

- Received Power: dBm
- Time: seconds
- Frequency: Hz
- Length: meters
- Phase: degrees (-180 to 180)
- Direction: degrees (0 to 360)
- Electric Field: V/m
- Path Loss: dB

21.4 Result Properties

The following properties can be set in the PROJECT RESULTS PROPERTIES window as seen in Figure 21.1. This is accessed by first selecting *Edit→Project Properties* in the MAIN WINDOW or GEOMETRY VIEW and then clicking on *Project results properties* in the PROJECT PROPERTIES window.

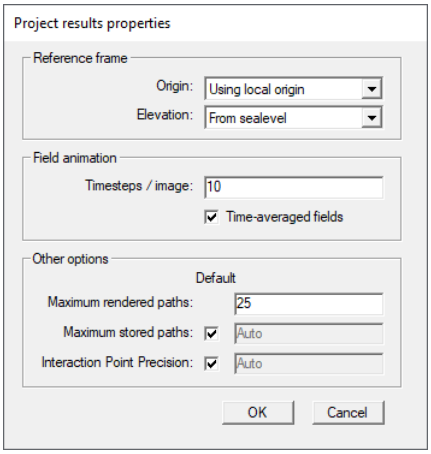


Figure 21.1: The result properties for the project

Reference Frame

Origin - Determines whether the X- and Y-coordinates are given relative to the *Project Origin* or the *Local Origin* (relative to the origin of the TRANSMITTER set).

Elevation - Determines whether the Z-coordinate is the elevation relative to *Sea Level* or the *Terrain*.

Field animation

Timesteps/image - Determines how frequently timesteps are saved as a `jpeg` image for movie display purposes.

Time-averaged fields - Indicates that time-averaging is performed on the fields for movie display purposes.

Other options

Maximum rendered paths - Allows the user to determine how large a subset of the *Maximum stored paths* are available for viewing between a transmitter and a receiver point. The strongest paths, up to this maximum, can be viewed from the user interface.

Maximum stored paths - Allows the user to set the maximum number of paths to be computed and stored between a transmitter and a receiver point. Many more paths may actually exist, however, only the strongest paths are stored up to this maximum.

21.5 Requesting Results

In order to minimize simulation time and disk usage, only the REQUESTED RESULTS are written to the result files. By default, only received power, path loss and propagation paths are selected. The default result requests for a project can be set by checking items on the REQUESTED RESULT CATEGORIES list, shown in Figure 21.2, which can be accessed through the DEFAULT RESULT REQUESTS button in DEFAULTS tab of the APPLICATION PREFERENCES window. Some propagation models allow specific result requests to be made per study area. This can be done via the RESULT REQUESTS button in the STUDY AREA PROPERTIES window. Models that generate a limited number of results do not have this button, and will generate all results for every simulation.

✓ Some models do not produce all results on the project-level default REQUESTED RESULT CATEGORIES list. The REQUESTED RESULT CATEGORIES listed in each study area show what results are supported by that model.

FULL 3D, VERTICAL PLANE, and URBAN CANYON share their REQUESTED RESULT CATEGORIES list. Results requested in one of these study areas will automatically be requested for all other study areas with these models.

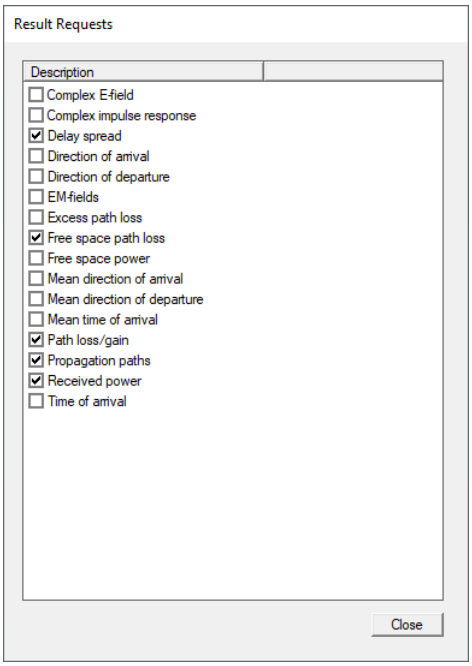


Figure 21.2: The requested result categories

Additional consolidated RESULTS types can be selected by creating a COMMUNICATION SYSTEM that allows filtering of the transmitter and receiver points to use when generating them. See chapter 14 for more information about how to request these types of results.

21.6 Result Files

Location of Result Files

When a simulation for a project is ran for the first time, a folder is created for each active STUDY AREA and the RESULTS are written to this folder. The folder names are derived from the first 50 characters of the short description given to the study areas. If the user does not enter a short description, a default description is created from the propagation model name. Results generated as a post process, such as `.csv` files generated from MIMO data, or COMMUNICATION SYSTEM results, are placed in a sub-folder within the study area folder.

Result File Names

Point-To-Multipoint Files

These files follow the format `*.p2m` (short for point-to-multipoint). Most of the `*.p2m` files contain the predictions at all RECEIVER points on a set due to the energy radiated from a single TRANSMITTER point. The names of these files are of the form: `project.type.tx_y.rz.p2m`, where:

- `project` stands for the name of the project
- `type` is the keyword for the result type
- `x` is the number of the transmitter site within the set
- `y` is the number of the transmitter set
- `z` is the number of the receiver set

For example, the name of the file containing the received power for all points on “Receiver set 2” due to the “transmitter at point 3 of set 4” would be `project.power.t003_04.r002.p2m`.

The RESULTS type keywords along with a brief description of the result are listed below:

- `cef` - Complex electric field for strongest paths
- `cir` - Complex impulse response for strongest paths
- `doa` - Direction of arrival for strongest paths
- `dod` - Direction of departure for strongest paths
- `doppler` - Change in frequency due to motion of transmitter and receiver
- `dspower` - Received power with diffuse scattering
- `e?mag` - Magnitude of E-field components, where ? = X, Y or Z
- `e?phs` - Phase of E-field components, where ? = X, Y or Z
- `erms` - Time averaged E-field magnitude
- `fspl` - Path loss in free space in dB
- `fspl0` - Free space path loss with *Isotropic* Antennas

- `fspower` - Received power in free space in dBm
- `fspower0` - Received power in free space with *Isotropic* Antennas
- `h?mag` - Magnitude of H-field components, where ? = X, Y or Z
- `h?phs` - Phase of H-field components, where ? = X, Y or Z
- `hrms` - Time averaged H-field magnitude
- `mdoa` - Mean direction of arrival at receiver
- `mdod` - Mean direction of departure from the transmitter
- `mtoa` - Mean time of arrival in seconds
- `mxtoa` - Mean delay
- `paths` - Geometrical propagation paths
- `pg` - Path gain in dB
- `pl` - Path loss in dB
- `power` - Received power in dBm
- `s?` - Cartesian components of the Poynting vector, where ? = X, Y or Z
- `savg` - Time averaged (RMS) magnitude of the Poynting vector
- `srms` - Time averaged propagation field magnitude
- `spread` - Delay spread in seconds
- `toa` - Time of arrival of strongest paths in seconds
- `xpl` - Excess path loss in dB
- `xpl0` - Excess path loss using *Isotropic* Antennas for free space loss
- `xtoa` - (Excess time-of-arrival) delay of strongest paths in seconds

▶ See Appendix L for more on `*.p2m` files.

Point-To-Point Files

These files end in the extension `*.p2p` (short for point-to-point) and can be produced for *Point* Receiver sets. All of the `*.p2p` files contain the predictions at a single RECEIVER point due to the energy radiated from a single TRANSMITTER point. The names of these files are of the form: `project.type.tx_y.rw_z.p2p`, where:

- `type` is the keyword for the result type
- `y` is the number of the transmitter set
- `x` is the number of the transmitter point within the set
- `z` is the number of the Receiver set
- `w` is the number of the receiver point within the set

For example, the name of the file containing the time domain electric field for “Receiver point 5 on Receiver set 2” due to the “transmitter at point 3 of set 4” would be `project.tdef.t003_04.r005_02.p2p`.

The currently available `*.p2p` RESULTS types are:

- `fdef` - Cartesian components of electric field vs. frequency
- `pdp` - Power delay profile (received power vs. time)
- `tdef` - Cartesian components of electric field vs. time

Communication System Result Files

Bit error rate, throughput, noise, interference, and results dependent on multiple transmitters are only generated when a project contains an active COMMUNICATION SYSTEMS. These result files follow the general format and naming convention of `p2m` files. When the results in the file are dependent on the combined effects of the selected transmitters, the specific transmitter set and point numbers are not included in the file name. The file name for data of this type is of the form `project.type.rz.p2m`, where `z` is the number of the receiver set.

The currently available RESULTS types from the COMMUNICATION SYSTEMS include:

- `noise` - Total interference, total noise, signal-to-noise ratio (SNR), signal-to-interference ratio (SIR), signal-to-interference-plus-noise ratio (SINR)
- `rsum` - Strongest power, total power summed without phase, total power summed with phase, best SINR, receiver signal strength indicator (RSSI)
- `rsumlte` - Specific to throughput of LTE systems, contains the same values as the `rsum` plus received signal received power (RSRP) and received signal received quality (RSRQ)
- `st2r` - Receiver’s strongest transmitter
- `st2rsinr` - Transmitter providing the best signal-to-noise-plus-interference ratio
- `ber` - Bit error rate
- `tp2` - Throughput, capacity, and modulation and coding scheme
- `cap` - Capacity

MIMO-generated Binary Files

When running a scenario with MIMO antennas present, Wireless InSite will produce several binary and non-binary result files. These files are used by Wireless InSite to generate the final results requested by the user, such as received power, H-Matrix, and Angle of Arrival.

The following binary files are generated:

```
<project name>.antenna.txXX.bin.ant

<project name>.bin.idx

<project name>.bin.inst
```

`<project name>.locations.txrxXX.bin.loc`

`<project name>.paths.txxx_yy.rzzz.bin.p2m`

The following non-binary MIMO results are generated:

`<project name>.inputpower.tXX.p2m`

Using the MULTI-CHANNEL RESULT BROWSER, the user can also request the following results to be exported to a `.csv` file. See Appendix M for details on the format of these files.

Complex Impulse Response

Direction of Arrival

Direction of Departure

H-Matrix

H-Matrix w/ Mutual Coupling

Received Power

Path Loss

Path Gain

RMS Angle Spread of Arrival

RMS Angle Spread of Departure

RMS Delay Spread

Terrain Profiles

This result is only available with the VERTICAL PLANE propagation model. File names are of the form `project.profile.tx_y.sz.dat`, where `x` and `y` are as described in Section 21.6 and `z` is the number of the sector. At present the sectors are at one-degree spacing, so the direction of the profile in degrees from the X-axis will be one less than the sector number. There is also a file of type `*.lprofile` which contains the simplified profile when this option is used.

Diagnostic Data

Diagnostic data files are ASCII files written in a variety of formats. Many of the files have headers describing the format and the file names are generally descriptive of the contents. For example, the file `project.ant05.tht135.dat` contains the antenna pattern vs. phi at a constant theta of 135° for antenna number 5. Other files include:

- Reflection coefficients (`project.refl.xx.dat`)
- Transmission coefficients (`project.trans.xx.dat`)
- GEOMETRY geometry data (e.g., `project.foliage.dat`, `project.terrain.dat`, etc.)

21.7 Summary of Results generated by Wireless InSite

Input files necessary for a Wireless InSite project are stored in the project file, while subdirectories for each study area contain result files associated with that study area. This section explains how the result files are related to the requested results and which results will appear in the RESULTS tree for a given simulation.

Wireless InSite produces different types of result files: diagnostic files, database files, point-to-many (p2m), and point-to-point (p2p) files. The model chosen, the waveform, the receiver set type, and the requested results determine which of these files you can expect to find.

Wireless InSite has two families of propagation models: Shooting and Bouncing Ray (SBR) Models, and Empirical Models. Because of differences in these models' algorithms, they are capable of generating different families of results. In general, the SBR models can generate the greatest number of results since they account for the 3D geometry and use the Uniform Theory of Diffraction (UTD) to determine the physical properties of each ray that reaches each receiver. Empirical models are typically based on simpler algorithms that cannot predict E-field, phase, or other physical properties. The results available from these models are often limited to path loss or gain.

p2m Results

The table below shows the p2m result files that are generated for each requested result for Wireless InSite's Ray-tracing models. These text files are suitable for post processing. Note that under 'File Name Key', the table lists the file keyword for each result type. The full file name for a result with keyword '`KEY`' is of the form `project.KEY.tX_Y.rZ.p2m`, where '`project`' refers to the project name, '`tX_Y`' identifies the `Y`th transmitter point in transmitter set `X`, and `rZ` refers to the receiver set `Z`.

- For more information on result files see Section 21.6.

Table 21.3: Requested p2m Results and Associated Result Names for Ray-tracing Models

Requested Result	Result Tree Label	File Name Key	Full 3D	X3D	Urban Canyon	Vertical Plane
Animated Fields	Animated Fields	tdEF	✓		✓	✓
Complex E-Field	Complex E-field	cef	✓	✓	✓	✓
Complex Impulse Response	Complex Impulse Response	cir	✓	✓	✓	✓
Delay Spread	Delay Spread	spread	✓	✓	✓	✓
Direction of Arrival	Direction of Arrival	doa	✓	✓	✓	✓
Direction of Departure	Direction of Departure	dod	✓	✓	✓	✓
EM-fields and Poynting Vector	E-field RMS	erms	✓	✓	✓	✓
	E-field X, Y, Z Magnitude	e*mag	✓	✓	✓	✓
	E-field X, Y, Z Phase	e*phs	✓	✓	✓	✓
	H-field RMS	hrms	✓		✓	✓
	H-field X, Y, Z Magnitude	h*mag	✓		✓	✓
	H-field X, Y, Z Phase	h*phs	✓		✓	✓
	Poynting vector magnitude	savg	✓		✓	✓
	Poynting vector X, Y, Z	s*	✓		✓	✓
Excess path Loss	Excess Path Loss with antenna	xpl	✓	✓	✓	✓
	Excess Path loss without antenna	xpl0	✓	✓	✓	✓
Free space path loss	Free space Path loss with antenna	fspl	✓	✓	✓	✓
	Free space Path loss without antenna	fspl0	✓	✓	✓	✓
Free space Power	Free-space Power with antenna	fspower	✓	✓	✓	✓
	Free-space Power without antenna	fspower0	✓	✓	✓	✓
Mean Direction of Arrival	Mean Direction of Arrival	mdoa	✓	✓	✓	✓
Mean Direction of Departure	Mean Direction of Departure	mdod	✓	✓	✓	✓
Mean time of Arrival**	Mean time of Arrival	mtoa	✓	✓	✓	✓
Path loss/gain	Path gain	pg	✓	✓	✓	✓
	Path loss	pl	✓	✓	✓	✓
Propagation Paths	Paths	paths	✓	✓	✓	✓
Received Power	Received power	power	✓	✓	✓	✓
	Received power with diffuse scattering	dspower		✓		
Time of Arrival ***	Time of Arrival	toa	✓	✓	✓	✓

* x,y, or z to indicate reported Cartesian component

** FULL 3D, VERTICAL PLANE, and URBAN CANYON models produce Mean Excess Time of arrival (`project.mxtoa.tX_Y.rZ.p2m`) in addition to Mean Time of arrival

*** FULL 3D, VERTICAL PLANE, and URBAN CANYON models produce Excess time of arrival (`.xtoa.tX_Y.rZ.p2m`) in addition to Time of Arrival

The table below shows the p2m result files that are generated for each requested result for Wireless InSite's Real Time models. Note that VPUP and TPG are ray-based models that use UTD, and are therefore able to produce a more complete set of results.

Table 21.4: Requested Results and associated result names for Real Time Models

Requested Result	COST Hata	Free Space	Hata	OPAR	Triple Path	VPUP	Walfisch Ikegami
Complex E-Field					✓	✓	
Complex Impulse Response					✓	✓	
Delay Spread					✓	✓	
Direction of Arrival					✓	✓	
Direction of Departure					✓	✓	
EM-fields and Poynting Vector					✓**	✓**	
Mean Direction of Arrival					✓	✓	
Mean Direction of Departure					✓	✓	
Mean time of Arrival					✓	✓	
Excess path Loss					✓	✓	
Free space path loss					✓	✓	
Free space Power					✓	✓	
Path loss/gain	✓	✓	✓	✓	✓	✓	✓
Propagation Paths					✓	✓	
Received Power					✓	✓	

** E-Fields only - no Poynting Vector

p2p Results

Certain time domain results are limited to point receiver sets. These results typically generate exceedingly large files when receiver sets contain large numbers of receivers (e.g. receiver grids or routes) are used. These results include E-field vs. time, and E-fields vs. frequency. Other results are only produced when both a point receiver set and a dispersive waveform are used, such as the power delay profile.

At this time the X3D does not produce p2p results.

Table 21.5: Requested p2p Results and Associated Result Names for Ray-tracing Models

Requested Result	E-Field vs. Time	File Name Key	Full 3D	Urban Canyon	Vertical Plane
Electric field vs. time	E-Field vs. Frequency	tdef	✓	✓	✓
Electric field vs. frequency	Complex E-field	fdef	✓	✓	✓
Power delay profile	Power delay profile	pdp	✓	✓	✓

Chapter 22

Data Visualization

In this chapter, you will learn...

- → how to choose the correct result to view in Wireless InSite
- → how to display the data in 2D, 3D and/or animated format

Wireless InSite has a wide range of built-in data visualization capabilities. Input data such as ANTENNA patterns and MATERIAL electrical parameters can be plotted to verify model assumptions and check performance. RESULTS data can be plotted for any of the 2D data types that Wireless InSite produces. For data types which are not appropriate for 2D plotting, powerful 3D graphics can be used to display data in various formats. The plots created can be edited to produce high quality graphs for both presentation and archive. It is also possible to create movies of time-domain E-field evolution. What follows is an overview of RESULTS viewing, viewing ray paths, line graphs and associated features, and several examples of how to create various plots.

- ✓ In general throughout the manual, “graph” refers to the figure that includes the title, axes, legends, and data, and “plot” refers to a particular data set being plotted on the graph. Using this terminology, one graph may contain several different plots.

22.1 Viewing Results

Selecting Data

Once a simulation has finished, all of the data selected in the REQUESTED RESULT CATEGORIES list will be available for viewing. The data is accessible by clicking the RESULTS tab in the MAIN WINDOW, which will display a window similar to the one seen in Figure 22.1.

For each STUDY AREA in the project, there will be an entry with the name of the study area. The data is generally organized in a hierarchical format by study area, then dataset type (point-to-point, point-to-multipoint, filtered), followed by result item, then the individual data items. Almost all operations involving data display and visualization will start by drilling down into the dataset through this hierarchy.

Some results are nested under one entry in the RESULTS tree. For example, *received power* and *received power with diffuse scattering* result files contain several types of results. Right click on the RESULTS tree entry to access result options.

✓ The results available under the RESULTS tab are also available in the PROJECT TREE under "Study areas and Results".

✓ The way that the results are organized can be modified in the application preferences. See section 2.7.1 for more information.

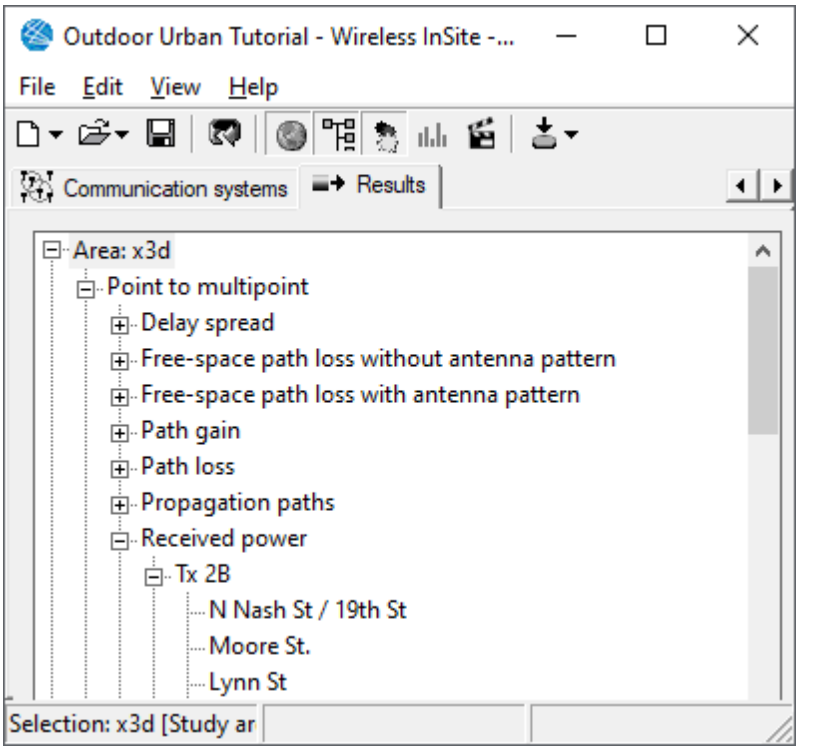


Figure 22.1: Display of results available in a project's study area

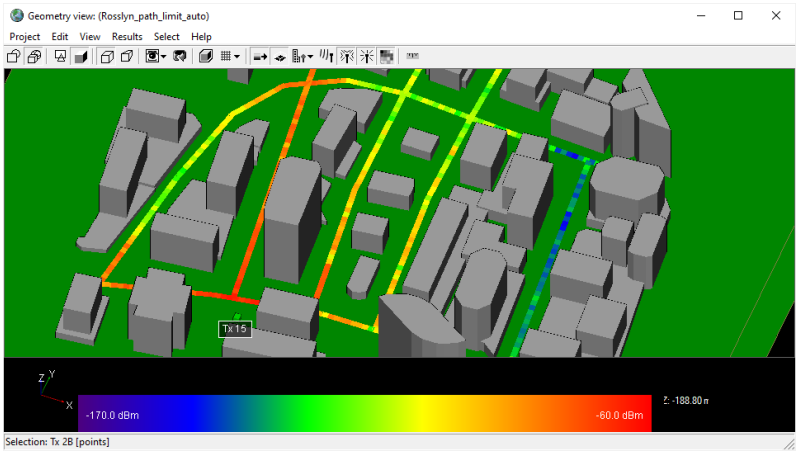


Figure 22.2: Viewing received power in a city section

Viewing Results in the Geometry View

To demonstrate viewing RESULTS in the GEOMETRY VIEW, we will use a simple example from a small section of a city. The example project consists of a TRANSMITTER and *XY Grid*. The project and its received power results are shown in Figure 22.2.

To view results in the GEOMETRY VIEW select *View* from the context menu of a result entry in the RESULTS tab. This will load the data into the GEOMETRY VIEW and display it. A scale bar for that data type will appear in the lower left corner indicating the values of each color.

The scale bar automatically adjusts to fit the range of loaded values. If multiple files are loaded then the range will adjust to encompass the minimum and maximum of each one. This automated adjustment can be overridden with exact values by selecting *Range options→Manual scaling* from the scale bar's context menu. If the mouse pointer is placed at a point on the color bar text will appear showing the numerical value corresponding to that color and the current data type being viewed.

Specific points can be examined by selecting *Select→Results* in the menu bar of the GEOMETRY VIEW. While in this selection mode selecting any data point will create an indicator on the scale bar showing where it falls in the color spectrum and its exact value.

✓ When the selection mode in the GEOMETRY VIEW is *Select→Tx/Rx Point* selecting a receiver point with visible underlying results will not select a result item. This means that its value will not appear on the scale bar as described above. The selection mode must be *Select→Results* to do this.

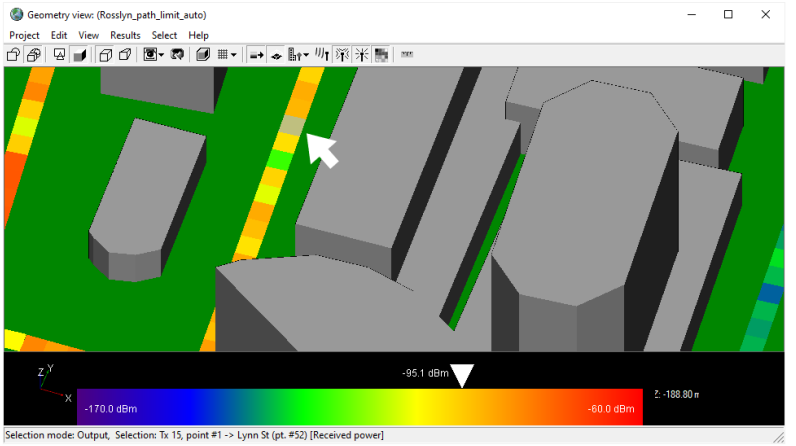


Figure 22.3: Selecting a result item and viewing its value on the scale bar.

The values of individual propagation paths can also be viewed, but they must be selected under the RESULTS tab as shown in Figure 22.4. This will activate the path in the GEOMETRY VIEW and show the value for that path on the scale bar.

More details on viewing propagation paths can be found in section 22.1.

✓ The propagation paths use the same scale bar as the received power RESULTS and uses the range as set for that RESULTS type. This means that paths with small contributions to the overall received power may not appear on the scale bar as it could fall below the minimum for the consolidated values of received power for the currently loaded data sets.

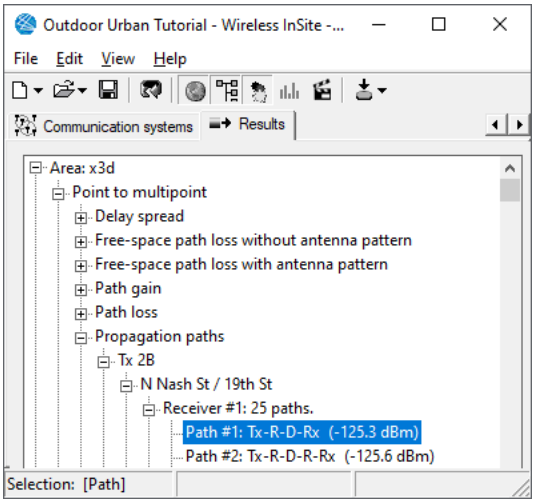


Figure 22.4: Selecting a path in the result tab.

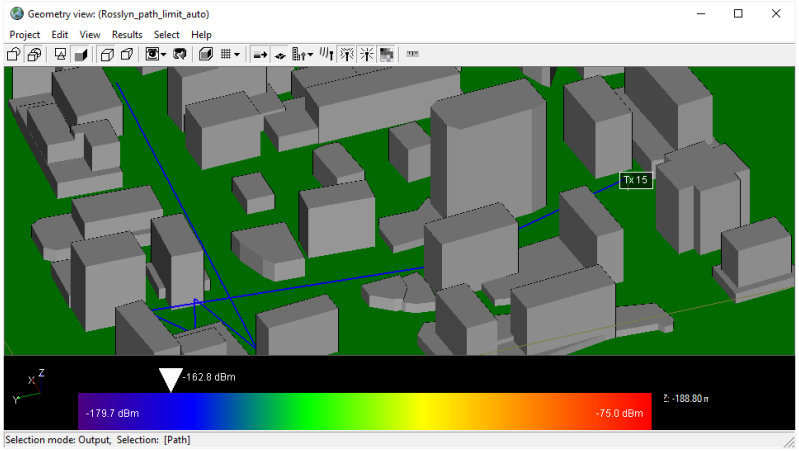


Figure 22.5: A selected path in the Geometry View with its value displayed in on the scale bar.

Scale Bar Properties

Figure 22.6 shows the SCALE PROPERTIES FOR RECEIVED POWER Window. This window is accessed by right clicking on the scale bar and selecting *Properties*. This window allows you to customize all of the various aspects of the scale bar for each result type.

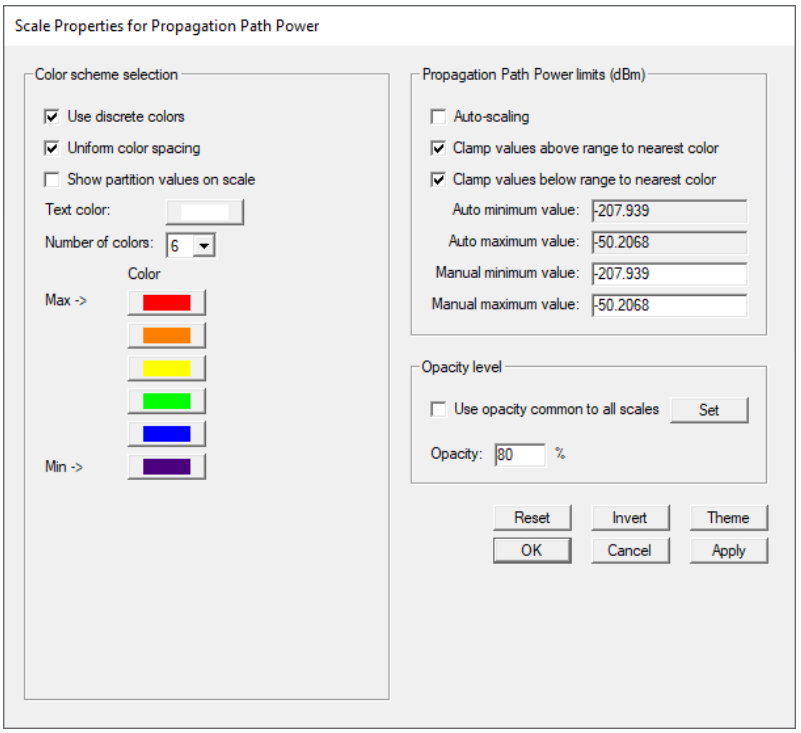


Figure 22.6: The properties of the received power scale bar

The properties set from this window are:

- **Color scheme selection**

  This section allows the user to set up the scale bar to display results in discrete colors as opposed to a continuous color spectrum. The sections within this area are as follows:

  – *Use discrete colors* - Enables or disables the use of discrete colors for displaying results.

  – *Uniform color spacing* - Calculates the range for each color based on the current min/max and cover an equal range of values. If this is deselected, then the points where the result data transitions from one color to the next can be adjusted from the defaults by putting a value in the appropriate partition point box.

  – *Show partition values on scale* - Displays the discrete color values on the scale bar in the *Text color* shown.

  – *Number of colors* - Allows the user to select the number of colors to use for discrete color result display. There can be from two to twelve colors used for discrete results.

  – *Color / partition points* - Determines the number of color buttons and partition points that appear. The color of a color button can be changed by clicking on it.

- **Received Power Limits**

  If the *Auto-scaling* box is checked, then the scale bar will automatically adjust the minimum and

maximum as data is loaded to be inclusive of all of the sample points. The current values for the given scale bar are shown. If *Auto-scaling* is deselected, the minimum and maximum can be set manually. If the minimum is increased or the maximum is decreased, then any points that lie outside the range can either be displayed using the closest color in the range, or will be transparent when the *Clamp values above (or below) to nearest value* box is unchecked.

- **Opacity level**

  This determines the transparency of the rendered results. This is useful in situations where the underlying image still needs to be visible. This is also used to highlight a transmitter against a background of receiver grids when viewing the strongest transmitter result. The user can set the opacity for all of the scale bars by clicking on the SET button, or deselecting the *Use opacity common to all scales* checkbox and entering the desired percentage in the *Opacity* text box.

- RESET

  This option will put the scale bar in its default state. The default options are:

  - Color scheme set to Continuous.
  - Discrete color scheme reset to use uniform spacing. This overrides any applied theme.
  - Mins/maxes set to auto.
  - Clamping reactivated.

- INVERT

  Reverses the order of the colors.

- THEME

  Enables the user to select a color theme. There are a number of theme files in the installation directory in a folder named 'colormaps'.

- APPLY

  Applies all scale bar changes to the project without having to close the properties window. Note that the INVERT, THEME, and RESET actions are applied immediately.

Several of these options are available directly on the context menu for the scale bar. They are:

- *Result type* for selecting which scale bar to use based on what data needs to be viewed.
- *Range options* for selecting either auto or manual scales.
- *Scale options* for putting the scale bar into either continuous or discrete display mode.
- *Set common opacity level* for setting the globally used opacity level.
- *Select color theme* to apply a color theme to the current scale bar.
- *Save current settings* to save the settings for all scale bars in the current project into a scale bar configuration file (`*.sb`)
- *Load current settings* to load the `*.sb` file into the project and overwrite the current scale bar settings for any scale bar listed in the file.

The settings for all of the scale bars that have been modified from the default configuration are saved automatically into the project settings file, but the settings can also be saved individually to a separate scale bar settings file. This is useful for applying the same scale bar settings across multiple projects. The scale bar context menu contains the option to save the current settings.

Viewing Propagation Paths

For many types of simulations it is useful to display the ray paths that it produces. Viewing the individual paths is a powerful tool for gaining knowledge of the physics of the particular situation and also as a diagnostic tool for checking both model assumptions and the accuracy of the results.

By default, Wireless InSite saves the set of paths (up to the maximum number chosen for the project) calculated for every Tx/Rx combination. To access the paths, click the RESULTS tab in the MAIN WINDOW of your project. In this tab one or more STUDY AREAS will be listed. If the simulation completes successfully, it is possible to expand each study area and drill down to view the ray paths.

✓ The maximum number of paths which are viewable in the GEOMETRY VIEW can be set by clicking *Edit→Project properties* in the MAIN WINDOW or GEOMETRY VIEW then *Project results properties* in the window that appears, and entering the desired *Maximum Rendered Paths* value. The default for this setting is 25, which is often sufficient, but may need to be increased under special circumstances.

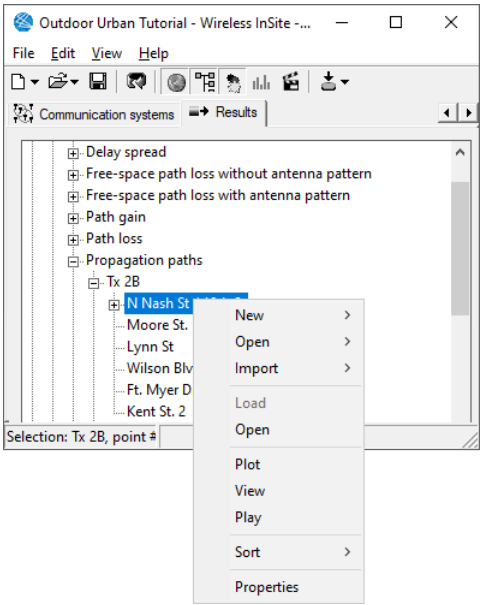


Figure 22.7: The context menu for a propagation path result

Select the *Propagation Paths* branch in the RESULTS tree and right-click the desired result, as seen in Figure 22.7. Clicking on *Load* will load all of the path data for the particular chosen path and make it available for viewing or plotting. Once loaded, the expand sign will appear next to the data set. Clicking it will expand the data set to show the individual receiver points. Expanding a point will show each ray path from the transmitter point to the receiver point, as seen in Figure 22.8.

✓ Clicking *Load* is not required if the user wants to directly view the ray paths. Selecting *View* will automatically load all of the paths for the receiver set selected and display them. This could potentially display thousands of rays for large receiver sets, so it is good practice to select *Load* and then *View* to select the ray paths.

✓ The result available in the RESULTS tab is also available in the PROJECT TREE under *Study Areas and Results*.

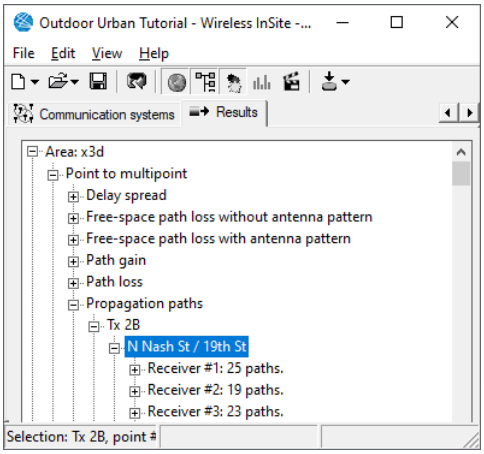


Figure 22.8: Viewing the receiver points and associated paths in the Results tab

At each of these levels you can view the paths by selecting *View* from the context menu at that point.

The context menu of a specific receiver point in the list provides an option to *View All Paths* that will display the paths from the transmitter point in the GEOMETRY VIEW, as seen in Figure 22.9.

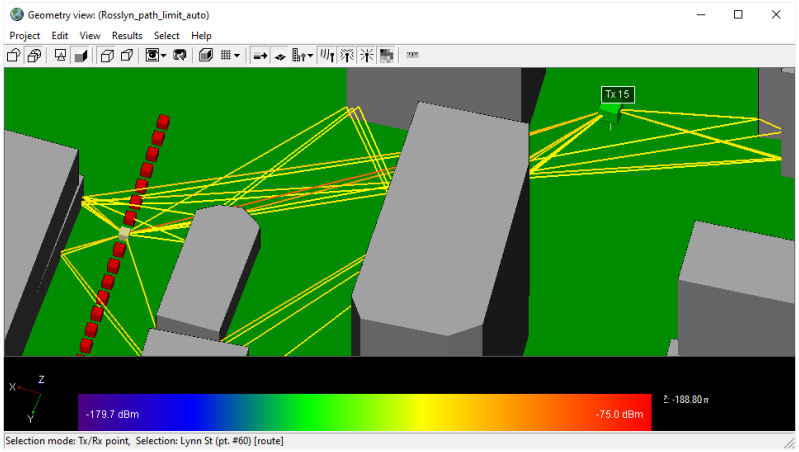


Figure 22.9: Viewing the propagation paths to the receiver point

If the user wants to view a certain subset of the paths for a given RECEIVER, clicking *View Some Paths* brings up the VIEW PATHS Window for choosing the way the paths are selected for display, as seen in

Figure 22.10. Using this method, the user can choose a specific set of rays with maximum or minimum power or time-of-arrival (TOA).

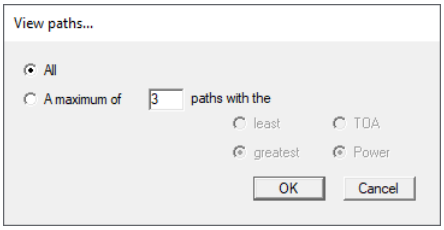


Figure 22.10: Filtering the paths to display

It is also possible to view single paths by selecting one or more of the paths listed under the desired receiver, as seen in Figure 22.11. The single ray chosen is displayed in Figure 22.12.

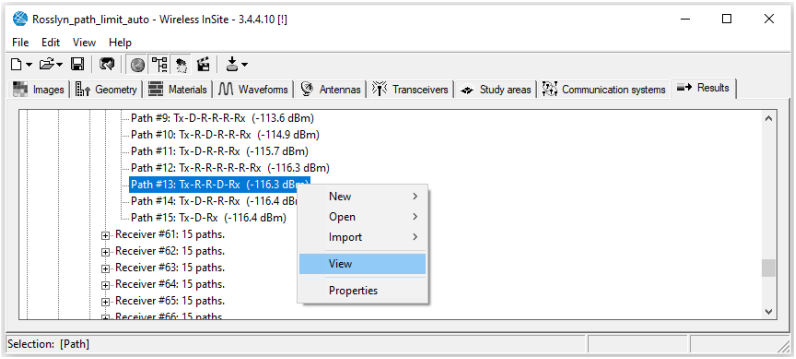


Figure 22.11: Selecting a single ray path to a receiver point

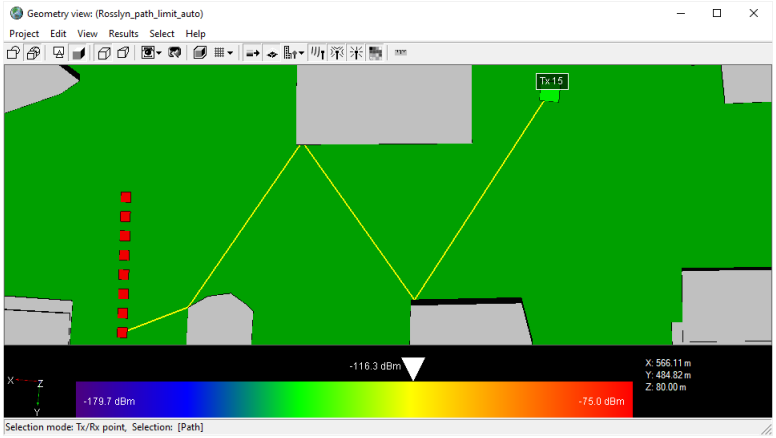


Figure 22.12: Display of single ray path to receiver point in Geometry View

22.2 Line Plotting

Wireless InSite has two types of 2D graphs: Rectangular and Polar. Users can plot input data, such as ANTENNA patterns and MATERIAL electrical parameters, to verify model assumptions and check validity.

In the following section, references to “graphs” mean the entire 2D graph, including the individual traces, the grid, the axes, and any titles and labels. The individual traces representing different data sets are referred to as “plots”.

Most of the types of RESULTS data can be plotted on a 2D graph. For certain types of data it may make more sense to use one type of graph or another. Wireless InSite gives the user great control over how the data is displayed.

✓ Graphs are stored as part of the project and are available in the RESULTS tab or PROJECT TREE *Study Areas and Results* branch. Once a graph is created, all of the modifications and formatting to the graph are stored. If the data in the graph is re-calculated, the graph will update the associated plots (traces) automatically.

Input Data

The primary types of input data for 2D plotting are ANTENNA patterns and MATERIAL electrical parameters.

Input and Output Antenna Radiation Patterns

To verify the antenna pattern which will be used in the ray tracing calculation for each TRANSMITTER, RECEIVER, or Tx/Rx set, the user may plot the antenna pattern for any antenna in the project.

▶ This procedure of plotting antenna patterns is described in detail in Chapter 12.

Material Parameters

All GEOMETRY in a Wireless InSite project have an associated MATERIAL definition. These materials have reflection and transmission coefficients which are used in the ray tracing and dictate how the ray will interact with a given GEOMETRY' surface. The reflection and transmission coefficients can be functions of frequency as well as angle of incidence. Plots of the effective permittivity and the attenuation are available for foliage materials.

- Chapter 10 describes plotting reflection and transmission coefficients of materials.

Result Data

In general, graphing begins the same way for any data type. The RESULTS tab in the MAIN WINDOW contains the data available for plotting. Expanding the study area which contains the data of interest will show the list of available point-to-multipoint, point-to-point, or filtered data sets. By expanding the data set of interest, the available result will be listed. Choosing *Plot* from the context menu will open the CHOOSE PLOT Window, as seen in Figure 22.13. Here the user will be able to choose the ordinate and abscissa of the graph.

✓ If *Plot* is not an available option after right-clicking on a dataset, this indicates that the particular dataset is not applicable to 2D plotting.

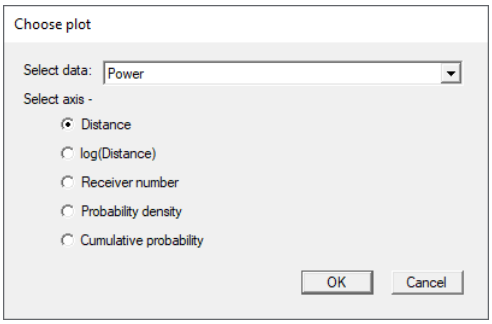


Figure 22.13: Determining the axis of the plot for received power. In this case the distance along the receiver Route is chosen.

From the example project (Section 22.1), the plot of the received power along the receiver *Route* is shown in Figure 22.14. In this plot, the received power was plotted versus distance. In this case, "Distance" is the distance along the receiver route measured from the first receiver.

✓ For *Routes*, *Trajectory*, *Arcs*, and *Vertical Arc* Receiver sets, the distance given in the result file is the distance along the receiver set measured from the first point in the receiver set to the point of the sample. For *Points*, *XY Grids*, *Cylinders*, *Spheres*, *Vertical Surfaces*, and *Polygon* sets, the distance associated with each receiver is the distance from the transmitter.

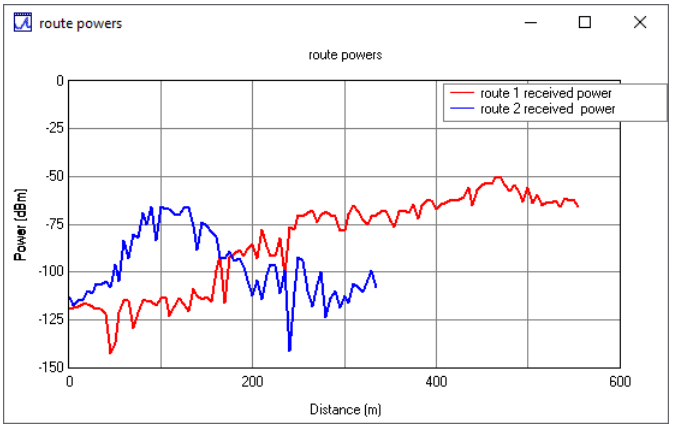


Figure 22.14: A plot of received power as a function of distance along the receiver route

Results can also be plotted as a probability density function (`*.pdf`) or a cumulative distribution function (`*.cdf`) of the values within the file. For the `*.pdf`, as seen in Figure 22.15, a "bin size" is specified that indicates the range that each point in the plot covers. For example, if the data in a file has a range from 1 to 5 and the bin size is 1, then the first point in the plot will be the percentage of points that lie in the range of (0,1). For the `*.cdf` function, each point represents the number of points that are within the current range from $-\inf$ to the current value.

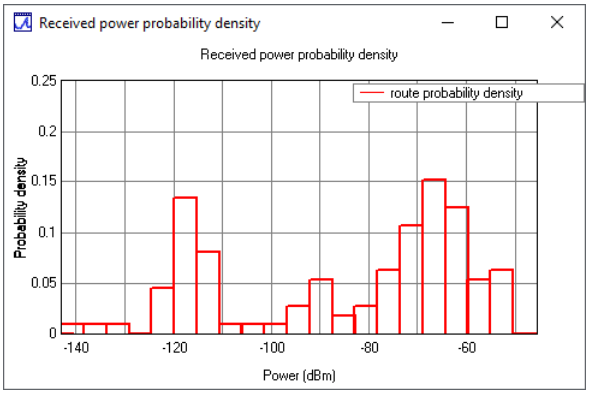


Figure 22.15: The pdf of the received power data in Figure 22.14

Graph Properties

After a graph is created, there are several options for controlling its display. These options are accessible under the RESULTS tab in the context menu of a graph, as shown in Figure 22.16. It is also possible to edit the graph through the PROPERTIES Window, shown in Figure 22.17.

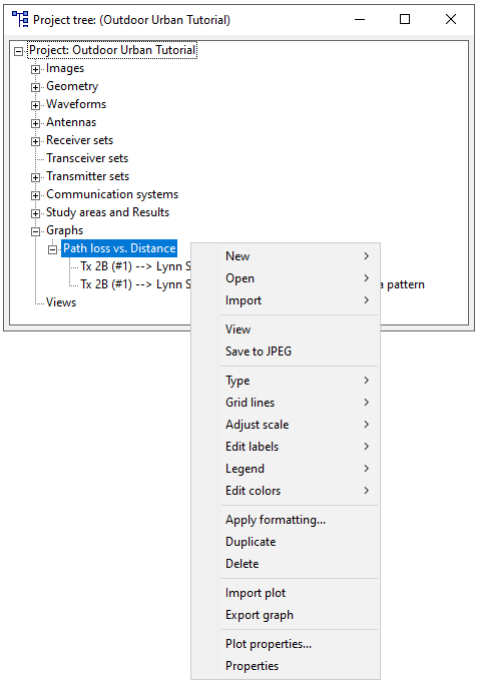


Figure 22.16: Selecting Graph Properties

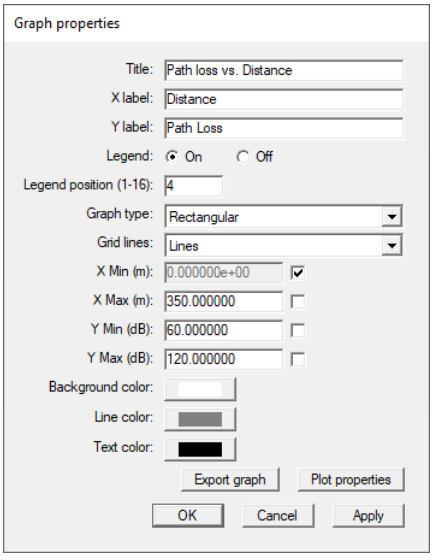


Figure 22.17: The Graph Properties Window

The items available in the context menu are:

- *New, Open and Import* - Project control items as described in Chapter 2.
- *View* - Toggles the view of the graph.
- *Save to JPEG* - Saves a graph to a `*.jpeg` image file. To save `*.jpeg` files for multiple graphs, the same procedure is followed and the `*.jpeg` images are numbered sequentially based on the filename entered, e.g. `Image1.jpg`, `Image2.jpg`, etc.
- *Type* - Determines the type of graph (Rectangular or Polar).
- *Grid Lines* - Determines whether to show lines, ticks, or nothing for the graph gridlines.
- *Adjust Scale* - Toggles auto scaling for each axis and determines manual scaling.
- *Edit Labels* - Defines the axis labels and graph title.
- *Legend* - Toggles the viewing of the legend and sets its position. The position is entered as an integer from 1-16. Position 1 is the upper left corner, position 4 the upper right, and the pattern continues to position 16 in the lower right corner.
- *Marker* (available in GRAPH PROPERTIES Window) - Toggles the display of the marker. The marker can be controlled to follow a trace or choose the nearest trace, or follow mouse clicks. The marker readout is in the lower left of the graph when this option is turned on.
- *Edit Colors* - Defines the background, grid, and label colors.
- *Apply formatting* - Allows the user to copy properties from another graph to apply to the selected graph.
- *Duplicate* - Makes a copy of the graph and adds it to the tree.
- *Delete* - Deletes the open graph.
- *Import Plot* - Launches a file browser to import a `*.plt` file
- *Export Graph* - Launches a SAVE AS window for saving the entire graph (all traces, edits and formats changes) to a user-specified name `*.grf`. Saving a graph requires saving all of the plots on the graph. Each plot will require a unique `*.plt` file.
- *Plot Properties* - Brings up the PLOT PROPERTIES window for choosing the line type and color for each plot.
- *Properties* - Summarizes all of the controls described above in one window (Figure 22.17). It allows the user to APPLY the effects of their changes for immediate review.

22.3 Animated Field and Ray Path Movies

In order to better visualize propagation in the time domain, the Wireless InSite MOVIE PLAYER is capable of displaying and saving movies which depict changing electric fields versus time and the movement of propagation paths within the simulation space. The Movie Player Window contains playback controls, viewing options, and an RESULTS selection interface.

Prior to using the MOVIE PLAYER, predictions must be performed with the REQUESTED RESULT CATEGORIES type “Animated fields” activated.

▶ See Chapter 21 for more on adjusting the settings of the MOVIE PLAYER under PROJECT RESULT PROPERTIES.

The MOVIE PLAYER, as seen in Figure 22.18, is accessible through the MAIN WINDOW by selecting *View-Movie player* or by clicking on the icon in the toolbar. A single E-field or propagation path file may also be viewed quickly in the RESULTS tab in the Main Window or in the PROJECT TREE. To play a particular result, select *Play* from the result’s context menu. The Movie Player will open and the selected result will immediately begin playback.

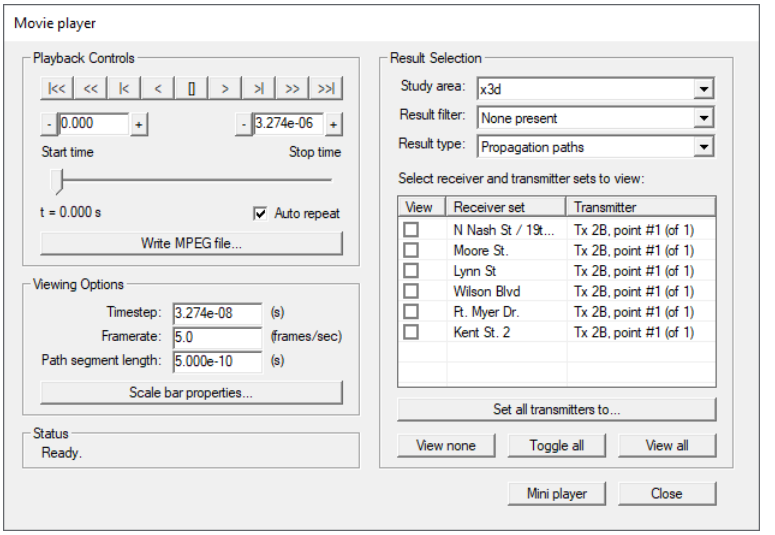


Figure 22.18: The Movie Player control panel

- **Playback Controls**

  The functionality of the Playback Controls resembles a video cassette recorder. The buttons are, from left to right:

  |<< JUMP TO THE BEGINNING

  << REWIND AT 2X SPEED

  |< MOVE BACKWARD ONE FRAME

  < PLAY BACKWARD

  [] STOP PLAYBACK

  > PLAY FORWARD

  >| MOVE FORWARD ONE FRAME

  >> FAST-FORWARD AT 2X SPEED

  >>| JUMP TO THE END

The *Start time* and *Stop time* limit the segment of the movie to play. The default values are determined automatically from the selected result. The *slider* shows the point in the movie which is currently displayed, and moves when the control buttons are pressed, while the movie plays or when dragged by the mouse. Check the *Auto repeat* box to play the movie in a loop. The sequence is rewound to the *Start time* and played again when the *Stop time* is reached.

Click the WRITE MPEG FILE button to write the movie to an `*.MPEG` file. In the SAVE AS window enter the desired directory and filename in which to save the movie. The movie will then play once, during which it is captured to the disk. You can halt the movie export by clicking on STOP MOVIE CAPTURE; the portion of the movie already captured will be saved. While the movie is being saved, user interaction with the GEOMETRY VIEW is disabled.

- **Viewing Options**

  The *Timestep* is determined automatically from the selected result. Reducing this value will not allow E-fields to be viewed with greater resolution in the time domain, as the calculated fields were only saved at the default timestep. Increasing the timestep will cause the movie to skip over some of the saved timesteps, increasing the apparent speed of the playback. When viewing propagation path segments, however, the timestep can be adjusted freely, since path data was calculated in the frequency domain, and may be viewed at any given value of time.

  The *Framerate* controls how rapidly the frames of the movie are displayed. The rate of display is limited by the complexity of the scene being displayed and the speed of your computer hardware. Therefore, you may observe movie playback slower than the desired framerate, or observe the speed of playback changing during the movie as the complexity of the scene changes. This behavior is expected.

  While viewing propagation paths, the physical length of the path segments, in terms of speed of light propagation time, may be controlled by *Path segment length*. This field is disabled during playback of E-field result.

  The scale bar in the GEOMETRY VIEW will appear during playback to provide reference for the colors displayed. The scale bar settings may be changed by clicking the SCALE BAR PROPERTIES button before or during playback. The scale bar shown corresponds to the *Result type* selected in the MOVIE PLAYER.

- **Status**

  The status of the MOVIE PLAYER is displayed here during playback and capture of movies.

- **Result Selection**

  The user must select the desired result here before the movie can play. To select a result, follow these steps:

  1. Select from the *Study areas* and *Result filters* in the project for which to view results. To view unfiltered results, select “None selected”. The *Result filter* selection is disabled when the chosen *Study area* is not based on a UTD propagation model.
  2. Choose what *Result type* will be displayed:
     (a) E-field X-, Y-, or Z-axis directed fields
     (b) E-field magnitude

(c) Propagation path segments

3. Finally, select one or more of the project's receiver sets to view by checking the box next to the receiver set name. Each RECEIVER set can render results for a TRANSMITTER set and point.

To assign a different transmitter set and point to a receiver set, choose *Select transmitter* from the receiver set context menu. The SET ALL TRANSMITTERS TO button may be used to quickly assign a single transmitter set and point to all the receiver sets in the table. The VIEW NONE, TOGGLE ALL, and VIEW ALL buttons below the table may be used to quickly change which receiver sets to view.

- MINI PLAYER

  This button switches the MOVIE PLAYER Window to a smaller format which makes more of the screen visible for viewing results. To return to the original window layout, click the button again when it says NORMAL PLAYER.

- CLOSE

  This button closes the MOVIE PLAYER and stops the movie playback, if in progress. The *Start time*, *Stop time*, *Timestep*, *Framerate*, *Path segment length*, and *Auto repeat* settings are retained the next time the window is opened. Result selection settings are not retained, since study areas, receiver sets, and transmitter sets may have changed since last opening the window.

22.4 Multi-Channel Result Browser

The MULTI-CHANNEL RESULT BROWSER acts as a result manager that simplifies and expedites the navigation of raw data produced by Wireless InSite *MIMO* simulations. Due to the nature of these types of calculations, large amounts of data can be produced, which can be cumbersome to navigate with the traditional `*.p2m` result file format. Additionally, many results can potentially be derived from the basic run time calculation. The MULTI-CHANNEL RESULT BROWSER allows the user to selectively choose which results to generate for specific channels and provides three options for accessing the data (viewing in the GEOMETRY VIEW, plotting, or exporting).

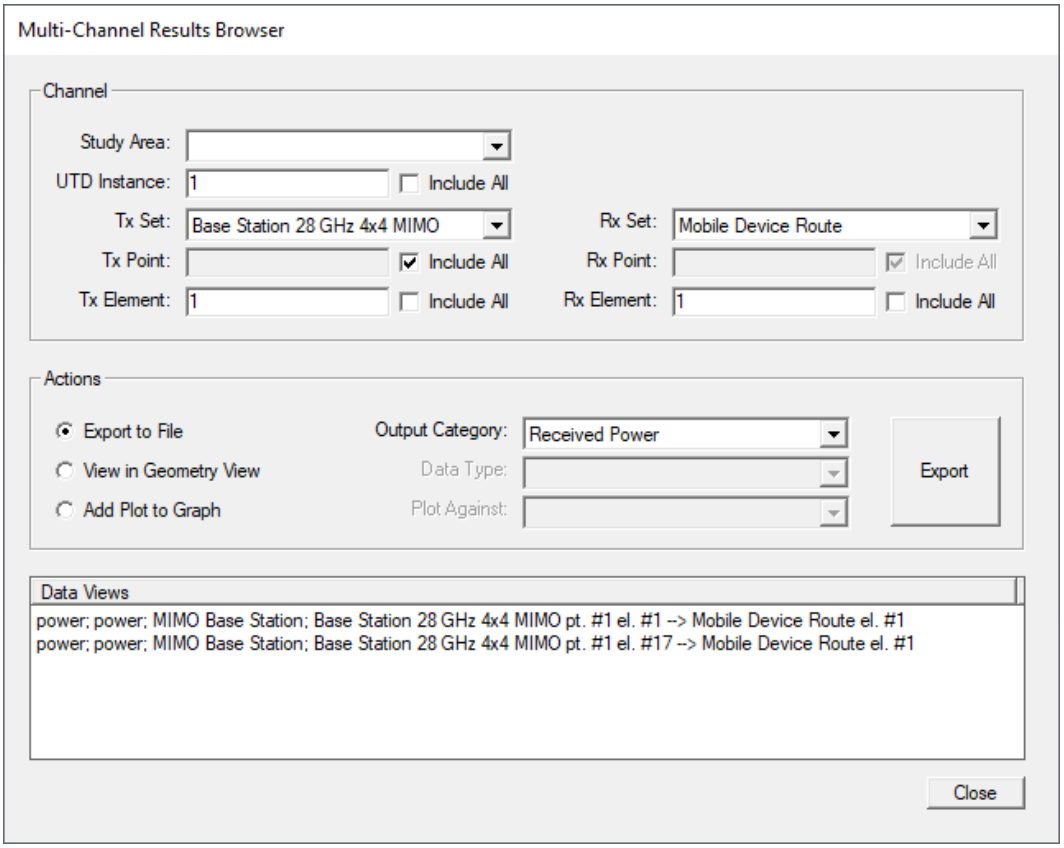


Figure 22.19: The Multi-Channel Result Browser populated with example information.

The following are descriptions of the parameters found in the MULTI-CHANNEL RESULT BROWSER:

Channel

The *Channel* section of the MULTI-CHANNEL RESULT BROWSER contains parameters used to specify the STUDY AREA and TRANSMITTER / RECEIVER pair(s) from which to request various forms of results.

- *Study Area* - Specifies the study area from which to generate analytic results.
- *UTD Instance* - Specifies a UTD instance, or all instances via the “Include All” checkbox, (e.g. frequency samples in a frequency sweep) from which to generate results. This field only applies when multiple UTD instances exist.
- *Tx Set* - Specifies which TRANSMITTER set to request results from.
- *Tx Point* - Specifies which transmitter *Point*, or all transmitter *Points* via the “Include All” checkbox, in the selected *Tx Set* to request results from.
- *Tx Element* - Specifies which *MIMO* transmitter element, or all elements via the “Include All” checkbox, to request results from. This field only applies when the selected *Tx Set* utilizes a

MIMO antenna.

- *Rx Set* - Specifies which RECEIVER set to request results from.
- *Rx Point* - Specifies which receiver *Point*, or all receiver *Points* via the “Include All” checkbox, in the selected *Rx Set* to request results from.
- *Rx Element* - Specifies which *MIMO* receiver element, or all elements via the “Include All” checkbox, to request results from. This field only applies when the selected *Rx Set* utilizes a *MIMO* antenna.

Actions

The *Actions* section of the MULTI-CHANNEL RESULT BROWSER contains the parameters used to specify the type of results to generate and the method in which to view it. The source of the result(s) is specified in the *Channel* section.

- *Export to File* - Creates per-channel `*.csv` files containing raw data for the selected result category. These files are stored in separate subdirectories based on the result category within the folder of the selected STUDY AREA.
- *View in Geometry View* - Displays graphic results (e.g. received power) in the GEOMETRY VIEW.
- *Add Plot to Graph* - Plots the selected results to a graph.

  ✓ Graphs generated this way can be accessed through the RESULTS tree in the Main Window alongside any other graphs from the project.
- *Action Button* - Executes the result request. This button will take the form of “Export”, “View”, or “Plot” based on the selected method of action.

Table 22.1: MIMO Results

	Result Type							Plot Against					
Category	*Received Power*	*Path Received Power*	*Phase*	*Angle Spread*			*Delay Spread*	*Distance*	*Receiver Number*	*Time of Arrival*	*Phase*	*Phi*	*Theta*
				Vertical	*Horizontal*	*Total*							
Received Power	✓		✓					✓	✓				
Path Loss / Path Gain	✓		✓					✓	✓				
H-Matrix													
H-Matrix w/ Mutual Coupling													
Complex Impulse Response	✓	✓								✓	✓		
RMS Delay Spread							✓	✓	✓				
Direction of Arrival / Departure		✓										✓	✓
RMS Angle Spread of Arrival / Departure				✓	✓	✓		✓	✓				

Data Views

The *Data Views* table contains a list of previously-requested views for results created by the “View in Geometry View” function in the *Actions* section. Each view can be toggled on or off as well as deleted through the Context Menu.

Chapter 23

Result Filters

In this chapter, you will learn...

- → how to set up and use result filters
- → how to filter interactions that occur with a geometry

RESULT FILTERS allow the user to isolate ray paths with specific interactions when using the FULL 3D, VERTICAL PLANE, or URBAN CANYON model. A filter defines results based on a subset of the generated paths. This allows a user to identify which objects are the major contributors to the power received by a set of RECEIVERS.

23.1 Creating a Result Filter

When a filter is created it is shared between all STUDY AREAS for simulation models that support them. To create a result filter select *Properties* from a study area and then click the *Result Filters* button in the STUDY AREA PROPERTIES window which will bring up the PROJECT GEOMETRY INTERACTION FILTERS window. Associated with a filter are a number of settings which can be accessed through the context menu, as seen in Figure 23.1.

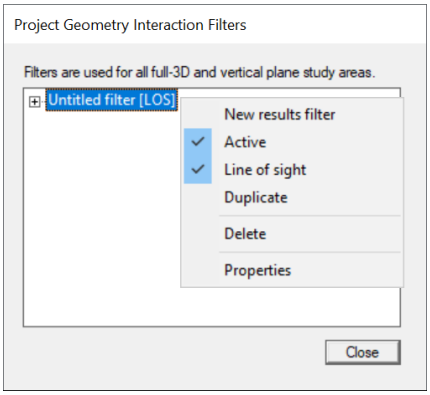


Figure 23.1: The Project Geometry Interaction Filters window with the context menu shown for a filter

- *Active* - When turned off, the filter is not included in simulations. New results are not generated for inactive filters.
- *Line of sight* - When turned off, results associated with the filter will not contain any line-of-sight (LOS) rays, which are those that travel from the TRANSMITTER to the RECEIVER without encountering any obstructions. This setting is not tied to any particular GEOMETRY since LOS rays do not interact with anything.
- *Duplicate* - Duplicates the selected filter and adds it to the list.
- *Delete* - Removes the selected filter from the list.
- *Properties* - Opens the RESULT FILTER PROPERTIES window seen in Figure 23.2. This window allows the user to enter a specific description for the filter entry. The description must be unique and is used to reference the filter in the RESULTS tree.

23.2 Result Filter Properties

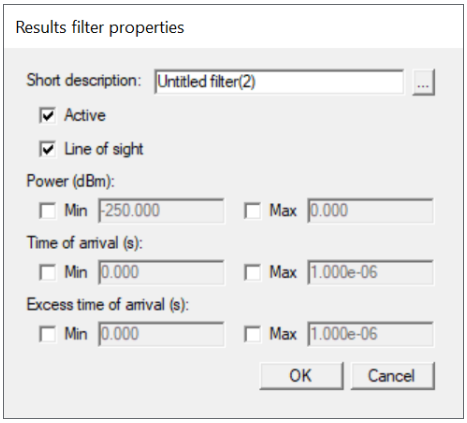


Figure 23.2: Properties of a result filter

From the RESULT FILTER PROPERTIES window, the user can enter filter settings for *Power*, *Time of arrival*, and *Excess time of arrival*. The *Power* and *Time of arrival* settings are applied to the ray paths and do not have any association with geometry in the project. In reference to Figure 23.2, rays whose power is less than -125 dBm and which did not arrive between 0 ns and 100 ns will be excluded from the filtered results. The properties window also contains check boxes to change the *Active* status of the result filter and to exclude *Line of sight* rays.

In each of the study area folders, subfolders are created corresponding to each filter. For this reason, filter names must be valid directory names and cannot contain special characters.

23.3 Filter Settings

In order to filter interactions that occur with a GEOMETRY, it is necessary to accurately set the filter properties for each one. Expanding the filter entry will reveal each geometry in the project as sub-items. Right-click on a geometry and choose the *Properties* menu option to bring up the FILTER ENTRY PROPERTIES Window, as seen in Figure 23.3.

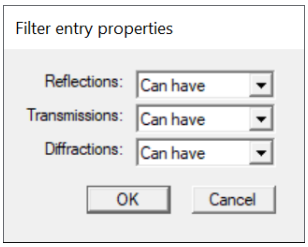


Figure 23.3: Properties of an entry in a result filter

From this window, the user can specify the types of possible interactions with the GEOMETRY: *Reflections*, *Transmissions*, and *Diffractions*. Each geometry contains filtering options: “Must have”, “Can have”, and “Must not have”. The filter behavior is defined through these options.

Filters can also contain settings with the structure group level in the GEOMETRY hierarchy. In order to add settings to a filter for a structure group:

1. Select the structure group.
2. From the context menu, select *Add to results filter...* and select the filter that the structure group should be associated with.

When a filter setting for a structure group conflicts with the setting for the GEOMETRY to which it belongs, the setting for the geometry will be changed to “Can have”. For example, if the reflection setting for a particular structure group within a geometry is set to “Must have”, but the setting for reflections from the geometry is set to “Must not have”, the setting for the geometry will be changed to “Can have” so that the settings do not conflict.

When filters are defined, the content of the RESULTS tree in the PROJECT TREE is slightly different. Beneath the point-to-multipoint tree item appears the filtered and unfiltered result sets. The Unfiltered tree contains all RESULTS as it would be had there been no filters defined.

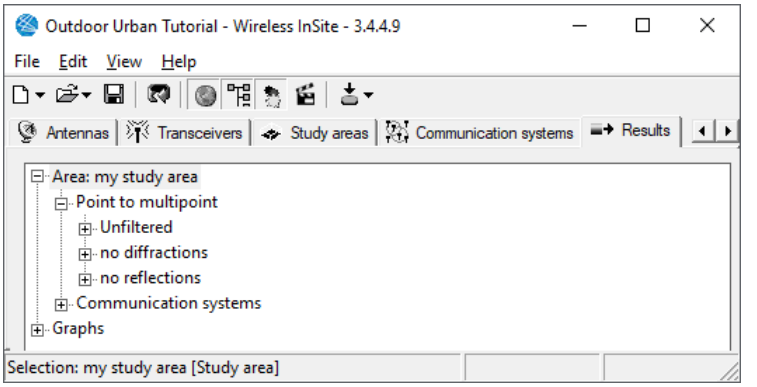


Figure 23.4: Display of unfiltered and filtered results in a study area

Along with the “Unfiltered” tree item, there are corresponding filtered result branches, each identifiable by the short descriptions associated with the filters. Results in each of these branches is a subset of the unfiltered results. The exact rules defined by the filter have been applied, and the resulting results are organized and made available here. Results from the unfiltered and filtered branches can be viewed and plotted together or separately.

Once an initial new RUN is completed, the *Added requested results* run option can be used to generate results for new filters or to make modifications to existing filters.

Chapter 24

Databases

In this chapter, you will learn...

→ how to set up a database

→ how to filter the components of a database

In order to allow the user to define and reuse components in many projects, Wireless InSite contains ANTENNA, MATERIAL, and WAVEFORM databases. These databases are accessible to all projects in Wireless InSite and save the user the trouble of entering the same information repeatedly.

24.1 Using Databases

Databases appear as a table in the ANTENNAS, MATERIALS, and WAVEFORMS tabs. If the database table is hidden in any tab, click on the long rectangular button at the bottom of the window to show the database. Each tab's database can be shown or hidden independently of the others.

The installation of Wireless InSite places default objects in some of the databases. In order to add objects to the database, select an ANTENNA, MATERIAL, or WAVEFORM and right-click. Choose *Copy to personal database* to add a copy of the object to the object's database. If the object references an external file (MATERIAL reflection and transmission coefficients, imported ANTENNA pattern, or *Wideband* waveform), it will be copied to the database automatically.

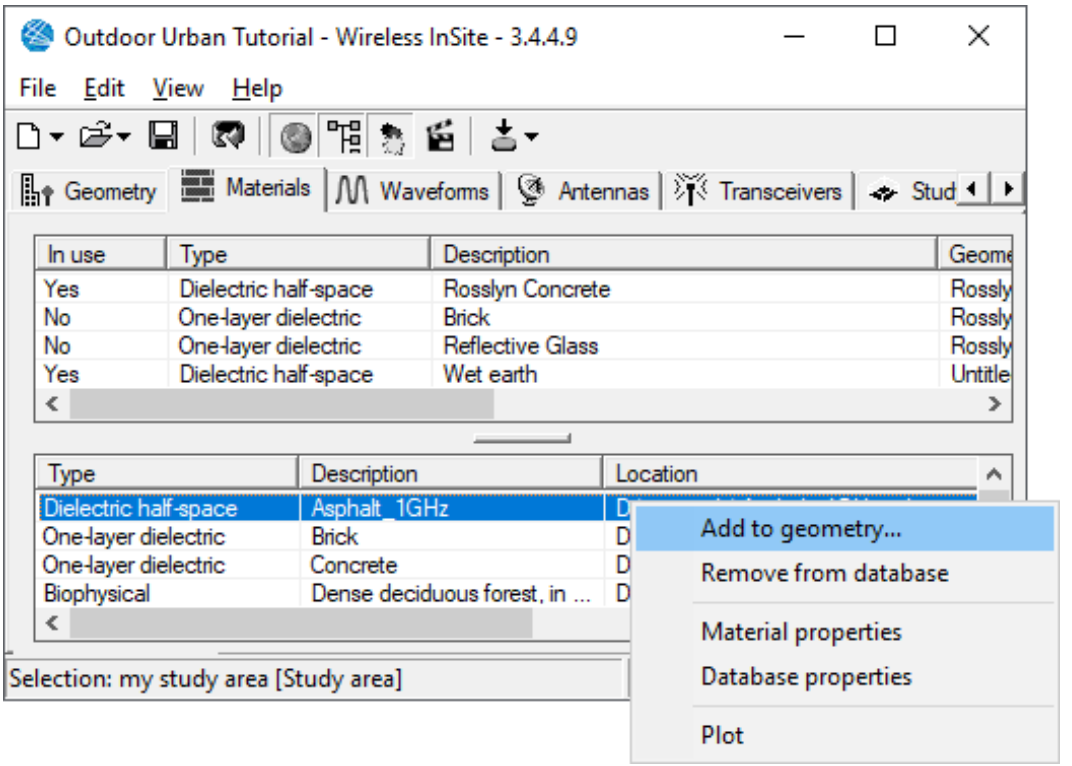


Figure 24.1: The materials database and context menu

The wording of right-click context menu, seen in Figure 24.1, may differ depending on the database, but each has similar function:

- *Add to project* - Places a copy of the selected database object in the current project. The MATERIAL database option *Add to geometry* prompts the user to select a GEOMETRY within the project to which the selected material will be added. External files are copied to the project directory when this command is selected.
- *Remove from database* - Deletes the selected object from the database and deletes the file associated with the object. The user will be prompted before deleting any external files which the object references.
- *Antenna/Material/Waveform properties* - Displays the properties window for the selected object. Any object within a database is considered read-only within Wireless InSite, and can not be modified until it is added to a project.

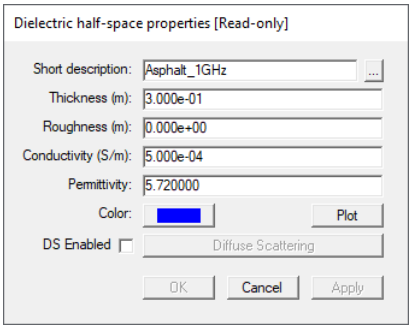


Figure 24.2: A material properties window for a read-only database object cannot be modified.

- *Database properties* - Displays the properties window for the database in which the object is contained. Each database contains a separate list of directories which are scanned when Wireless InSite is loaded. Any object files present in these directories will be accessible through Wireless InSite. However, Wireless InSite does not scan subdirectories contained in these paths. The default location defines the location where new entries are stored.
- *Add* - Allows the selection and addition of a new directory to the database.
- *Edit* - Allows the selected directory to be modified without removing and re-adding a new directory to the database.
- *Remove* - Eliminates the selected directory as a location to search for database objects. The directory is not deleted, and the contents of the directory are not modified.
- *Set as default* - Marks the selected directory as the path in which objects added to the database will be stored. Each database maintains its own default directory.

A database must have at least one path defined, and a default directory set. The DATABASE PROPERTIES Window, seen in Figure 24.3, will not close if all directories have been removed or if a default directory has not been set.

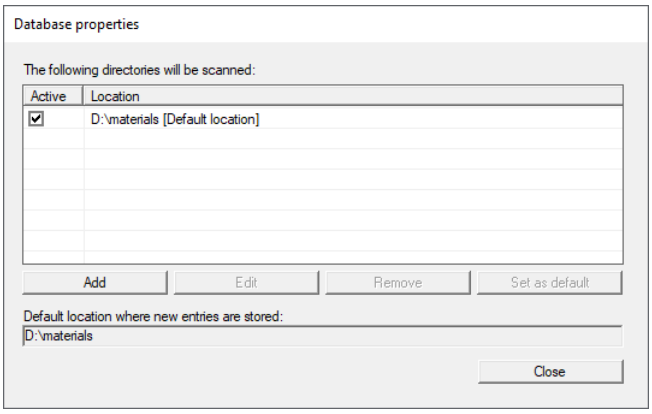


Figure 24.3: The Database Properties Window

24.2 Filtering Databases

In addition to deactivating directories in the databases, it is also possible to reduce the number of items which appear in the database by using the *Filter* field in the ANTENNAS, MATERIALS, and WAVEFORMS tabs in the MAIN WINDOW. For example, under the ANTENNA tab, entering text in the field labeled *Antenna filter* will cause the project's antenna table and the antenna database table to show only those items which match one or more terms in the filter text.

- ✓ The filtering is not case sensitive. Exact expressions can be entered in quotation marks, and exclusion terms preceded by a minus sign (-) to further refine the filtering. For example, the filter text "horn" will show only horn antennas, while the addition of the term "-1920MHz" will eliminate those horns which include 1920MHz in their description. Filtering MATERIALS and WAVEFORMS is performed similarly.

Chapter 25

Command-Line and Batch Execution Modes

In this chapter, you will learn...

- → how to use the command prompt to run the simulation engine
- → how to run simulations faster with batch scripts
- → batch script keywords and constants
- → how to generate cluster scripts to manage batch scripts

It is possible to run Wireless InSite's simulations from the command line. Wireless InSite also has the ability to generate batch scripts which can be submitted on a parallel computer, such as a cluster. Each script instructs the simulation engine to perform the simulation for a single TRANSMITTER point, or alternatively to perform the analysis for a single large RECEIVER set. By submitting these scripts to a load balancing batch manager in a clustered environment running under Linux or Unix, the simulation engine can run on multiple transmitters at the same time and can complete the simulation more quickly.

25.1 Command Line Execution of Simulation Engines

On Linux, command line controls are the primary way of running Wireless InSite. The command line controls are the same for both Windows and Linux, but on Linux, the user must first run a shell script to set up the environment for execution using the following steps:

- Navigate to the installation directory (e.g. `''install_location/Wireless InSite/3.4.4.13 /Linux_x86_64RHEL/bin''`)
- Run `remcom-shell`

On either Windows or Linux, the user then runs a simulation by launching a model engine from the `calc` subdirectory of the `bin` folder. The specific simulation engine depends on the propagation model to be run.

For X3D RAY MODEL and REAL-TIME MODELS (`wibatch`):

To execute models within the `wibatch` engine (X3D RAY MODEL, FREESPACE, HATA, COST HATA, REAL-TIME MODELS: VPUP, TPG, OPAR, WALFISCH-IKEGAMI), the user must call the `wibatch` executable and specify the input `xml` file and study area name as follows:

```
install_location\bin\calc\wibatch.exe -file <inputFilename.xml> -outdir
<studyAreaName>
```

The command line arguments are:

- `-(f)ile` - The input `xml` file will be located in the Wireless InSite project directory. For projects generated by the GUI, the name will be in the format: `ProjectName,StudyAreaName.xml`
- `-(out)dir` - This argument specifies the directory where results will be written. When using the GUI, this directory is created as a subfolder of the project file location.
- `-(p)refix` - This must be used when writing results as `*.p2m` files for the Wireless InSite GUI. It should be the same as the name of the Wireless InSite project so that the GUI can properly detect and load the results. If not properly specified, and the project file is in the format of `project.outdir.xml`, the project portion of the filename will be used.
- `-(outx)ml` - If present, the results will be written to a copy of the input `*.xml` file. The filename of the copy will be the same as the input filename, but with a `.out.xml` extension.
- `-set_licenses` - This argument specifies the license to use during this simulation. If the REMCOMINC_LICENSE_FILE environment variable has not been set, this argument must be specified for each simulation run.
- `-fn` - The function that should be invoked on the node represented in the `-(f)ile`.
- `-type` - The type of the node in the API schema to call the function on. A search is made throughout the file for all nodes of this type.
- `-(s)ave` - Saves the generated results to the file specified with `-(f)ile`.
- `-(v)ersion` - Print a single line with the version of the Wireless InSite API that is being used.
- `-(d)ebug` - Display debug messages from the API.
- `-licenses` - Display the contents of the current license file.
- `-(h)elp` - Display the help documentation.

An example command is as follows:

```
install_location\bin\calc\wibatch.exe -f
exampleProject\exampleProject.myX3DStudyArea.xml -out myX3DStudyArea -p
exampleProject
```

For FULL3D, URBAN CANYON, AND VERTICAL PLANE MODELS (calcprop):

To execute models within the calcprop engine (FULL3D, URBAN CANYON, OR VERTICAL PLANE), the user must call the calcprop executable and specify the project's *.setup file as follows:

```
install_location\bin\calc\calcprop.exe --project=<inputFilename.setup>
[optional arguments]
```

When using calcprop the command line parameters are:

- --calc-mode=<*mode*>

  <mode> must be one of the following: New, AddTransmitters, AddReceivers, ChangeHeights, ChangeFrequency, ChangeAntennas, ChangeMaterials, ChangeWallTypes, AddOutput, or ConsolidateClusterRun.

- --clean-run

  All previously generated data files will be deleted and recreated.

- --delete-temp

  All temporary files will be deleted at the end of the run.

- --set-licenses

  This argument specifies the license to use during this simulation. If the REMCOMINC_LICENSE_FILE environment variable has not been set, this argument must be specified for each simulation run.

- --help

  Print help text showing valid command line options

- --memory=<*memoryspec*>

  A memory specification is a number followed by K, M, or G for Kilobytes, Megabytes, or Gigabytes. The number optionally may have a decimal point. If the --memory command line option is not given under Windows, an algorithm tries to determine a "polite" maximum amount of memory to use. Under Linux, calcprop will attempt to use 800 Megabytes unless the ---memory command line option is specified. Some example valid memory specifications are:

  ```
  --memory=450000K
  --memory=450M
  --memory=1.75G
  ```

- --project=[<*path*>]/<*file*>.setup

  The project command line option is the only required option. A full path to the *.setup file may be given or a relative path may be given. No path is required if the *.setup file is in the current working directory.

- --rank=<*num*>

Transmitters may be split into individual runs for running on a Linux cluster. The rank parameter is used to assign a unique numeric ID, normally an integer starting at zero or one, to each process. Wireless InSite provides tools for generating appropriate cluster scripts.

Use of the `--rank` parameter requires a separate CalcPropNode license.

- `--rx-set=<`*set-num*`>[-set-num];<`*rx-set*`>`

  Receiver sets may be split with one or more receiver sets running on a Linux cluster node. Example valid entries are:

  `--rx-set=1`

  Receiver set 1.

  `--rx-set=3;5;9-11`

  Receiver sets 3, 5, 9, 10, and 11.

  Use of the `--rx-set` parameter requires a separate CalcPropNode license.

- `--tx-set=<`*set-num*`>:<`*point-num*`>[-<`*point-num*`>][,<`*point-num*`>[-<`*point-num*`>]]...;<`*tx-set*`>`

  Transmitters may be split across Linux cluster nodes. An example valid entry is:

  `--tx-set=1;2:5-7;15-17`

  Transmitter set 1 (every point), transmitter set 2 points 5, 6, and 7, and all points in transmitter sets 15, 16, and 17.

  ✓ Transmitter sets can be split by points within sets, while receiver sets must remain whole.

  Use of the `--tx-set` parameter requires a separate CalcPropNode license.

25.2 Batch Calculation Steps

To run a simulation on a cluster, proceed through the following steps.

1. Create a script template, using the example in Section 25.4.
2. Use the CLUSTER SCRIPT GENERATOR Window to generate the scripts that will be used to submit the simulation on the cluster separated by TRANSMITTER point or RECEIVER set.
   - See Section 25.5 for more on the CLUSTER SCRIPT GENERATOR Window.
   - See Section 25.1 for more on sending commands to the simulation engine.
3. Copy the project directory to the location on the cluster running the simulation.

   Remember that if you are specifying directories in the script that they must match the directories that exist on the cluster.
4. If a master submission script was created in the script, execute it to submit all of the batch jobs. Otherwise, each script will have to be manually submitted.
   - Section 25.4 contains an example of a master submission script.

5. Once all of the batch jobs have completed, copy the contents of the study area directory back to the computer running the Wireless InSite GUI.
6. Some result files depend on the results from all TRANSMITTER points to the active RECEIVER sets in the project. For example, the total power cannot be calculated without knowing the power that a given receiver set receives from each transmitter point in the project. Therefore, these types of results will need to be consolidated from the results of all the files involving a given receiver set if the project was split up by transmitter point. To perform this action, select RUN→*Consolidate cluster results*.

25.3 Generating Batch Scripts

In order to submit batch jobs to the cluster's batch management system, there must be a script for each simulation. Generating these scripts by hand would be a tedious task that would need to be repeated each time the variables controlling the simulation or the batch job's environment changed. To simplify the creation and maintenance of the script files, Wireless InSite provides the ability to automatically generate them for the user with a template that works as described in this section. The general format of a template is to:

1. Generate the commands for a driver script
2. Loop through all of the transmitter points or receiver sets in the project
3. Create the script files that will run each simulation

The template includes keywords that the script generator can extract to allow a user to provide values that they may want to change on a regular basis.

25.3.1 Keywords

The scripting engine provides a lot of flexibility through the use of keywords, which are special words recognized by the scripting module and interpreted to perform a custom action. Keywords can appear on any part of a line, nested or in sequence. If there is still text on the line once all keywords within it are processed, it will be written out to the currently opened file (if the script has already opened one). Keywords will be interpreted and added to the text that is written to the currently open file, depending on what action the keyword is intended to perform.

✓ To add blank lines in the final script files, include a line with one space. Blank lines without spaces will be ignored.

To allow the use of keywords for other purposes and prevent situations where the environment uses identical keywords, all keywords recognized by the scripting engine must be enclosed between angle brackets ("<" and ">"). Each keyword that is currently recognized is listed in the following subsections.

Keyword "ask"

This keyword allows the user to specify required input to configure the script for the specific project that it will be used on. This also allows the scripts to be quickly modified without having to locate the information

directly. All questions that are asked will appear in the CLUSTER SCRIPT GENERATOR Window.

- See Section 25.5 for more on the CLUSTER SCRIPT GENERATOR Window.

Format

```
<ask question_number type "question|default answer" variable_name>
```

Parameters

`question_number` - This indicates the index of the question with respect to other questions as they appear in the symbol table.

`type` - In order for the script generator to perform proper error checking, a type must be associated with the question. The types and the error checking they perform are:

`file` - Used for filenames. Filenames should not contain spaces or special characters that the operating system does not allow in filenames. If backslashes are used as directory separators when specifying a filename, they will be converted to forward slashes as they are recognized by both Windows and UNIX.

`int` - Used for integral values.

`real` - Used for floating point values.

`str` - Used for string substitutions.

`question|default answer` - The question will be the text that appears in the *Question* column in the CLUSTER SCRIPT GENERATOR Window. If the writer wants to specify a default answer that should satisfy most cases, it can be specified by following the question with a "|" character and then the default answer. This section must be included inside of double quotations.

`variable_name` - This is the variable name assigned to the "default answer". When enclosed in angle brackets, it makes the answer accessible from the CLUSTER SCRIPT GENERATOR Window.

Keyword "for_each" / "end"

This keyword indicates the beginning of a loop that will go over all of the active TRANSMITTER or RECEIVER sets. This depends on the mode specified through the selection made under the *Create script for each* option in the CLUSTER SCRIPT GENERATOR Window.

Format

```
<for_each>
    ...statements that produce output for each Tx/Rx point...
<end>
```

Keyword “open”

This keyword will open an output stream that all of the following text will be written to. This stream remains the active stream until another `open` command is issued.

Format

```
<open filename>
```

Parameters

`filename` - This is the name of the file to open relative to the project directory. Once variable name substitutions are performed, the path will be re-verified.

The use of `../` and `./` to escape the project directory structure is not allowed.

Keyword “rem”

This keyword allows the user to comment out text that would otherwise be written out to the currently opened file.

Format

```
<rem comment>
```

Parameters

`comment` - The text of the remark.

25.3.2 Constants

Constants are variables that are always defined and are not modifiable by the client. They are:

`<current_pt>` - The current point being processed in a `for_each` loop. This is included separately from the `point_option` so that the `current_pt` can be used for creating filenames, allowing more flexibility.

`<rank_option>` - The switch option to add to the simulation engine command line that includes the current value of the loop counter in a `for_each` loop.

`<point_option>` - The switch option to add to the simulation engine command line that includes the current value of `current_point`.

`<project_name>` - The name of the project `*.setup` file, with all spaces converted to underscores.

25.3.3 Answer Books

Files can be created that have the answers to some or all questions in a given script template. The format of this file will be a set of lines with two columns. The first will indicate the question being answered in the script using the defined variable name, while the second column will contain the answer. Answer books are saved as simple text files by clicking the SAVE ANSWER BOOK button in the CLUSTER SCRIPT

GENERATOR Window. Once an answer book is saved, it can be reloaded by selecting LOAD ANSWER BOOK.

25.4 Example Script Template

The following section demonstrates how a script template can be used to gather information and create all of the required scripts for submitting a multiple-node job using the Portable Batch Scheduler (PBS). While the script template is commented through the use of the `rem` keyword, a line-by-line description is provided to clarify the purpose of the script template following the example.

- A copy of this script template is included with the installation and is located at `install_location\data\Batch Management\script.txt`.

```
1   <ask 1 str "Script sub-directory prefix|script" prefix>

    <rem create the driver file>
2   <open <prefix>/<ask 2 file "Driver filename|sub.pbs" driver> >

3   #!/bin/csh
    <rem create directories for each point to run in labeled:>
    <rem project_dir/script_1_1/>
    <rem project_dir/script_1_2/>
    <rem project_dir/script_2_1/>
    <rem project_dir/script_2_2/>
    .
    .
    <rem project_dir/script_2_n/>
    <rem ...>

4   <for_each>
    mkdir <prefix>_<current_pt>
    cd <prefix>_<current_pt>
    rm -rf *
    qsub ../<project_name>_<current_pt>.pbs
    cd ..
    <end>

5   <rem create the individual scripts for each transmitter point>
    <for_each active_tx>
    <rem this filename matches the qsub command above>

6   <open <prefix>/<project_name>_<current_pt>.pbs>

7   #!/bin/csh

    cd $PBS_O_WORKDIR
```

```
#PBS -j eo
#PBS -m be
#PBS -l walltime=12:00:00
#PBS -l nodes=<ask 3 int "Number of nodes to use|1" nodes>:ppn=<ask 4 int
"Number of processes per node|1" processes>:production

<ask 5 file "Working directory|/cluster/user/project" workdir>

setenv LD_LIBRARY_PATH '<ask 6 file "Location of dlls required for the
simulation engine to run|<workdir>/lib" dll_location>'

<ask 6 str "Memory to use|1.5G" memory>

setenv CALCPROPOPTS
'--project=<workdir>/<project_name>/<project_name>.setup
< point_option > < rank_option > --memory=<memory>'

setenv CALCPROP '<workdir>calcprop'
<ask 7 str "MPI run command|mpiexec" MpiCommand>
<ask 8 str "MPI command line options|-kill" MpiOptions>
setenv MPIEXEC_OPTS '<MpiOptions>'
<MpiCommand> $MPIEXEC_OPTS $CALCPROP $CALCPROPOPTS

<end>
```

Description of activities in the example:

1 A subdirectory for keeping the scripts located in the project directory can be specified here.

2 The name of a driver file that submits all of the individual scripts to the batch manager is gathered from the user and then opened.

3 These lines are written to the driver file as is, minus the remarks.

4 This loop will create lines in the driver file for submitting the scripts for each Tx/Rx point to run the simulation on. The working directory for each script is created. If it already existed, then it is cleaned out before submitting the job to the batch manager.

5 This is the beginning of the loop that will create the scripts for each Tx/Rx point that is submitted by the driver to the batch management system.

6 The script for the current point is opened for writing.

7 Information for configuring the batch manager that is being used is gathered from the user. In this case, the batch manager is a PBS system.

8 The client has decided to define environment variables for creating the strings to use for executing the simulation engine. The LD_LIBRARY_PATH variable MUST be adjusted to

include the path to the .dll files that are required to run the simulation engine. Note the use of the loop constants `point_option` and `rank_option` to correctly identify what part of the project each process will work on.

9 In this example, the client uses MPI for running jobs on their cluster and gathers additional information before running the simulation engine with it.

10 The loop started at step 5 is closed.

25.5 Cluster Script Generator Window

Once a template is created, it can be used to create the scripts for running the simulation through a batch manager. The CLUSTER SCRIPT GENERATOR Window, seen in Figure 25.1, will read the template file and present the options used to control how the scripts will be generated. This is accessible by selecting RUN→*Prepare cluster scripts*.

✓ In Figure 25.1, the template from Section 25.4 has been loaded and is displaying the questions and default answers that were specified in it.

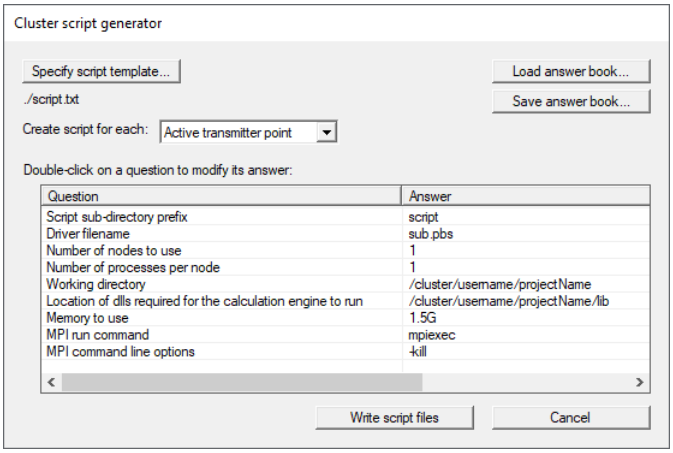


Figure 25.1: The Cluster Script Generator Window

The following describes the options in this window.

- SPECIFY SCRIPT TEMPLATE - Loads the template file that was created using keywords.
  - ▶ See Section 25.3.1 for a list of the keywords available.
- LOAD ANSWER BOOK - Loads an answer book file.
  - ▶ See Section 25.3.3 for more on answer book files.
- SAVE ANSWER BOOK - Saves the current set of answers to the questions in the currently loaded template to a file for future use.

- *Create script for each* - Controls whether the scripts will be created for running the simulation for each active transmitter point or for each active receiver set.
- **Question**/**Answer Table** - Displays all of the questions that were specified in the template through the use of the `ask` keyword. If they need to be changed, an *Edit symbol* option is available through the context menu for a given line.

Appendices

Appendix A

Appendix Notation

In this appendix, you will learn...

→ how to read the notation in the following appendices

All Wireless InSite input files are `ASCII` text files. In addition to the explanation given in the following appendices, users can also set up the problem using the Wireless InSite GUI and open the input file in a text editor to better understand how settings in the user interface translate to text in the input files.

A common format is applied throughout the appendices to describe Wireless InSite file formats.

- `Regular teletype text` represents keywords that are required to be in the file.
- The `<or>` operand separates mutually exclusive keywords, such as "`active <or> inactive`".
- Keywords are commonly followed by a character string, integer, or other value. In these cases, the keyword is listed in teletype, and in the place of the argument is a description of the value that should follow is written in *`italicized teletype text`* and enclosed by brackets, i.e.:

  `StudyAreaNumber[`*`integer ID number of study area`*`]`

- Regular gray text represents a note about the current line.

Appendix B

Wireless InSite Model Extensions

In this appendix, you will learn ...

- → how to activate Wireless InSite model extensions
- → how to allow partitioning and queuing
- → how to allow advanced editing of an Engineered Electromagnetic Surface (EES) materials
- → how to allow deterministic sweeps
- → how to write extra comm system results
- → how to set up and use Wireless InSite's Modified COST Building Penetration Model
- → how to set up and use Wireless InSite's Culvert Modeling capability

Wireless InSite Model extensions allow users access to features that are either less-often used or have not been fully developed. To access the capabilities described in this appendix, from the MAIN WINDOW, click:

- *Edit* →*Application Preferences*
- Click on the *Models* tab
- Check the box to the left of each of the capabilities you wish to enable

Figure B.1 illustrates how to access Wireless InSite's Extended features.

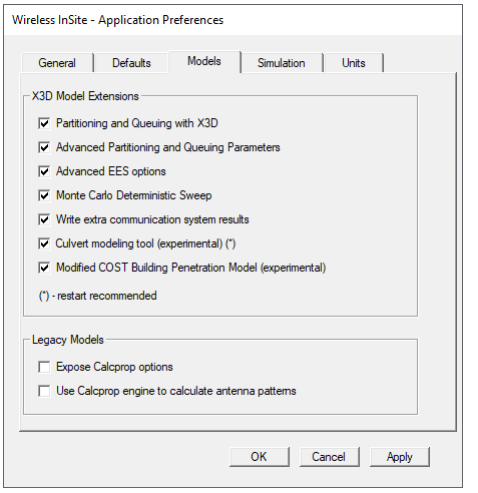


Figure B.1: Activating Model Extensions in the Application Preferences Window

B.1 Partitioning and Queuing

When this is selected, Partitioning and Queuing options are available for the X3D model. The job may be partitioned *by receiver set*, or *by transmitter point*, and may be run either locally or on a cluster, by specifying the desired *Queue type*.

- For more information on partitioning and queuing with the X3D model, see Section 15.3.2.

B.2 Advanced Engineered Electromagnetic Surface (EES) Options

By default Wireless InSite assumes that origin of an EES material file is centered on its data and only allows matching it with the centroid of the face it is attached to. It cannot be edited directly and moves with the face during normal geometric transformations such as translation and rotation. Activating this option will allow the user to edit the materials origin to move it around on the face which may be desirable if the face is not rectilinear.

- For more information about engineered electromagnetic surfaces, see Section 10.7.

B.3 Deterministic Sweep

This option allows the user to specify a number of equally distributed samples to be used instead of stochastic selection for a Monte Carlo distribution.

B.4 Write extra communication system results

When running a communication system simulation that includes MIMO, p2m results for each stream will be written.

B.5 Culvert Modeling

Culvert Modeling is an application, within Wireless InSite, that was created to optimize the accuracy of predicting electric fields inside hollow, tunnel-like structures. This appendix will explain the Culvert Modeling options in an example format, and assumes that the user is already familiar with Remcom's Wireless InSite.

Wireless InSite's Culvert Modeling capability is compatible with FULL3D and X3D. Table B.1, in section B.5.3.1, provides suggested interaction settings to ensure accurate results.

For an explanation of the terms used throughout this example, please refer to the appropriate section of this manual.

B.5.1 Creating a Culvert

For the Culvert Modeling application, a culvert is defined as a tunnel-like structure with an opening at both ends. It may only possess one of the following bends/turns: 90 degree, 45 degree, or 0 degree (no turns). Within the Wireless InSite Graphical User Interface (GUI), a culvert is considered to be an object. There are several options that the user may specify to customize their culvert. These customizations will have an impact on the run-time and results. Before beginning your first Culvert Modeling scenario, Remcom's engineers recommend setting up, simulating, and briefly analyzing, a few different culvert geometries. This will establish a speed/accuracy comfort level for the user to apply toward future projects.

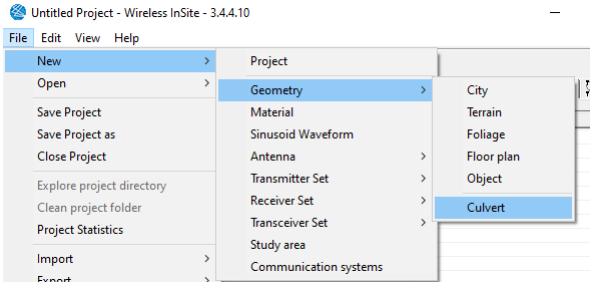


Figure B.2: Selecting a Culvert Geometry

To create a new Culvert geometry, from the Main window:

1. Click on *File→New→Geometry→Culvert* (see Figure B.2, above)
2. The Culvert Geometry window will open up automatically. Specify your culvert geometry in the Culvert Geometry window (shown in Figure B.3, below). The following section explains how to use the Culvert Geometry window.

✓ Do not click on OK until all of your geometry has been specified. You cannot access the Culvert Geometry window to make changes after clicking OK.

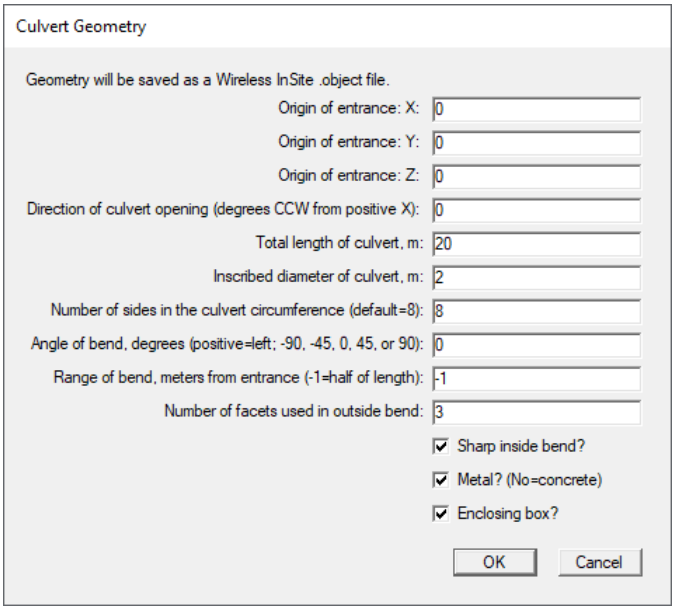


Figure B.3: Culvert Geometry window

3. Click **OK**.

B.5.2 Culvert Geometry Window

The Culvert Geometry window allows the user to control many aspects of their culvert, such as: where it is located, object dimensions, how it is constructed, the type of angles/bends, the material, and much more. This section will explain these options and offer some guidance to set up the culverts. Section **B.5.3**, will offer more suggestions to help obtain optimum results.

The user will begin by determining where the culvert is placed. This is done by selecting the origin and the direction they would like it to face. Once the culvert's geometry is set up it will be saved as a Wireless InSite .object file in the project's directory. The culvert's origin will always be at the center of the culvert's opening entrance. The floating red dot represents the culvert's origin, in Figure B.4, below.

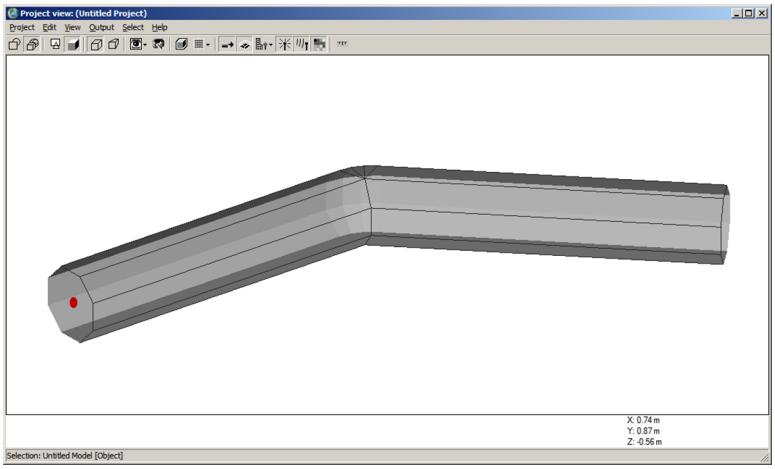


Figure B.4: Shows the object's origin at the center of opening

B.5.2.1 Declaring the Origin

To select the origin, simply enter the proper coordinates in the Culvert Geometry window labeled *Origin of entrance: X:*, *Origin of entrance: Y:*, *and Origin of entrance: Z:*, as seen in Figure B.3.

B.5.2.2 Rotating the Culvert

The user may also rotate the culvert in the Culvert Geometry window by entering an angle (in degrees) in the box labeled: *Direction of culvert opening (degrees CCW from positive X)*. Considering a top view of the culvert (looking toward the -z axis), the rotation angle entered in the box will follow the same orientation as the culvert in Figure B.5, below.

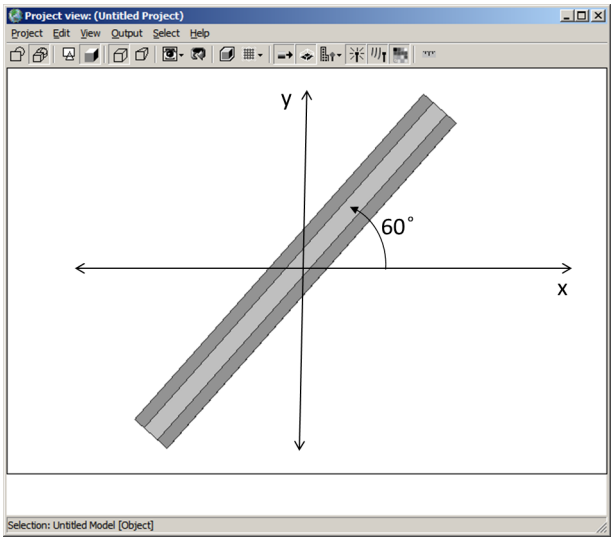


Figure B.5: A culvert rotated 60 degrees

You may also rotate the culvert in the MAIN WINDOW, as you would any Wireless InSite geometry. Please see the reference manual for rotation instructions using the Main window.

B.5.2.3 Setting Dimensions

In the Culvert Geometry Window, *Total length of Culvert, m:*, requires the length of the culvert, measured from the origin and extending to the tail opening of the culvert on the opposite side. This value must be in meters.

Inscribed diameter of culvert, m , represents the diameter, or width, of the culvert's mouth. This value is also in meters.

The option *Number of sides in the culvert circumference (default=8):*, represents the number of segments used to define the culvert's cross section. A culvert requires no fewer than 4 sides, and Remcom recommends using no more than 12.

B.5.2.4 Bending Culvert

Culverts may be created with a single bend. The angle of the bend must be 0, 45, or 90 degrees. The sign convention bends the culvert in the same manner as the rotations were described in section B.5.2.2. When looking at Figure B.5, a negative value will bend from the +y direction toward the +x direction. 0 represents a straight culvert. Bends are created using the box labeled, *Angle of bend, degrees (positive=left: -90, -45, 0, 45, or 90)*. The direction *left* is assuming the geometry is viewed from above.

The user may place the bend at any location of the culvert. This is done by entering a length, in meters, into the box labeled, *Range of bend, meters from entrance (-1=half of length):*. To place the bend at the midpoint of the culvert, enter -1.

The final user defined box is used to create the outer edge of the bend. The value entered in the box labeled, *Number of facets used in outside bend:*, will allow the user to create a more rounded outside bend. More facets will result in a more rounded bend. Zero facets will result in a picture-frame style bend. Below are a few examples of bends that could be created. Notice the difference in facets used on the outside of the bends. Bends with multiple facets may require the user to designate more diffractions in the Study Area, resulting in longer run times.

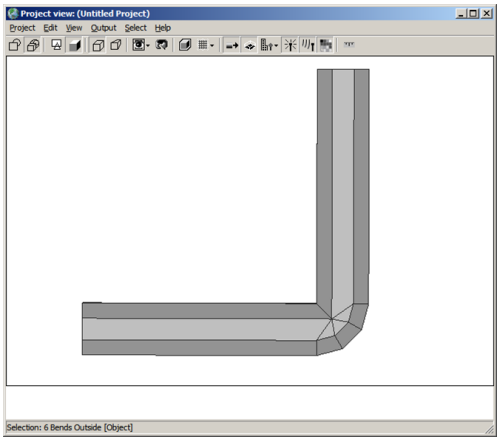


Figure B.6: 90 deg. Bend, with 3 facets on the outside edge of bend.

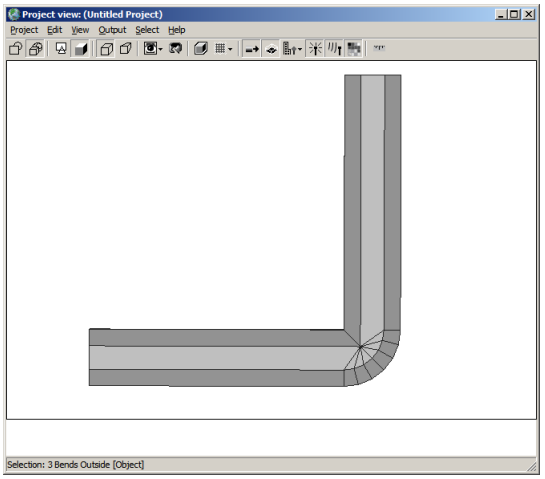


Figure B.7: 90 deg. Bend, with 6 facets on the outside edge of bend.

It is also possible for the user to control the inside bend. To make the bend sharp, click the box beside *Sharp inside bend?*. If this box is not checked, Remcom's Culvert Modeling tool will create the inside bend with as many facets as there are on the outside bend. An example of a sharp inside bend and a gradual inside bend are shown in the two pictures below.

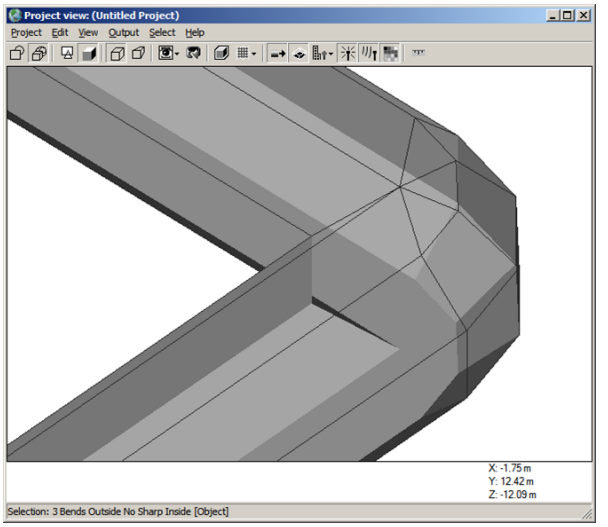


Figure B.8: *Sharp inside bend?* - Checked

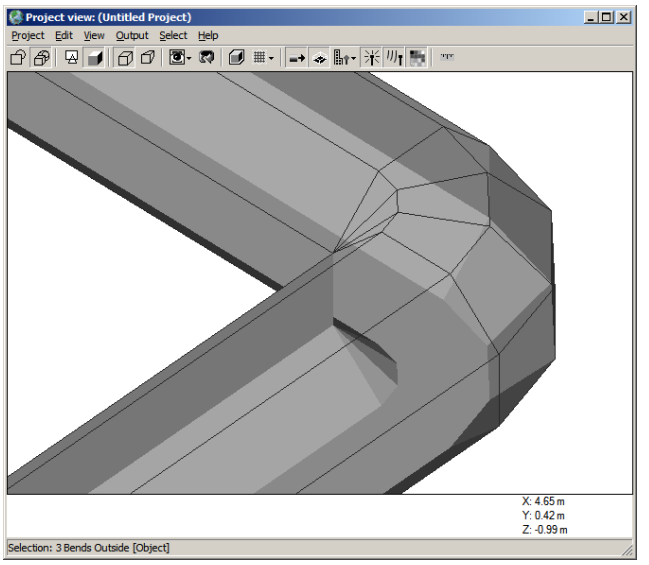


Figure B.9: *Sharp inside bend?* - Unchecked

B.5.2.5 Materials

For now, the user may only select metal or concrete for the culvert's material. To select metal, check the box labeled, *Metal? (No=concrete)*. You may change the culvert's material through the Geometry tab in the Wireless InSite Main window, as you would any other geometry.

B.5.2.6 Enclosing Box

The final step to create the culvert is deciding whether you would like to include an *enclosing box*. The enclosing box is a separate sub-structure that is introduced to prevent unwanted paths during ray-tracing. This feature is useful when modeling propagation near the culvert's entrance or exit. Figure B.10 displays a picture of the culvert without the Enclosing Box. Figure B.11 shows the same culvert with the Enclosing Box.

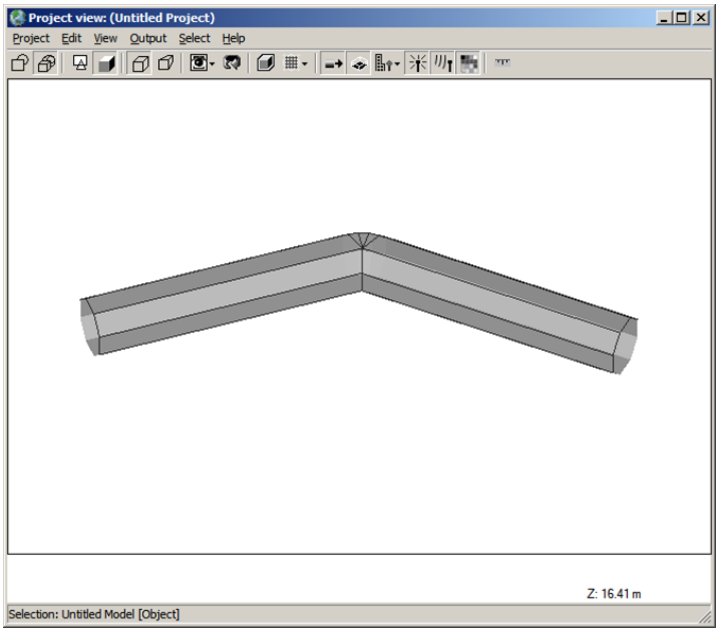


Figure B.10: Culvert without the enclosed box.

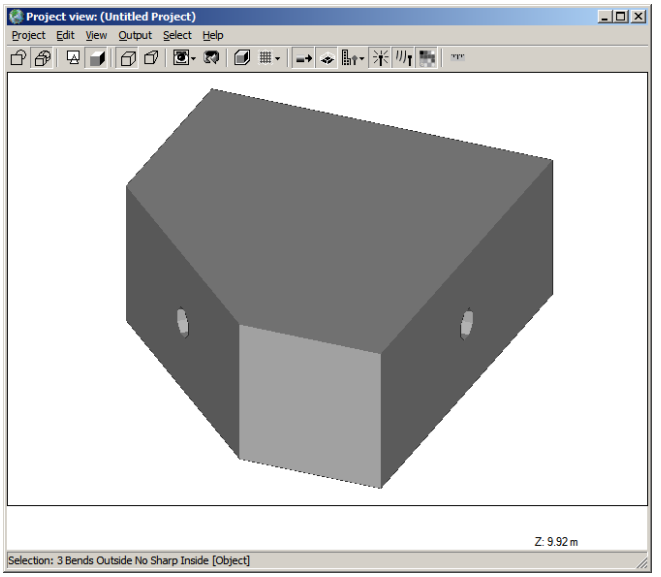


Figure B.11: Culvert with enclosed box.

B.5.3 Study Area Set-Up

B.5.3.1 Study Area Geometry

In situations where it is only a segment of a tunnel that is of interest, the user may wish excluded the entrance or exit of the culvert from the propagation modeling. This can be achieved with careful construction of the STUDY AREA boundary. All ray paths stop once they intersect a STUDY AREA boundary.

The Study Area below intersects the Enclosed Box, so that a path would need to leave the Study Area to make it to the other end of the culvert. Such a path will not exist. As is desired, all paths arriving at the receiver set must go through the culvert. There may also be scenarios where the user wishes for the energy to run across the top of the culvert, or onto the opposite side. Again, with careful STUDY AREA boundary construction, this can easily be done.

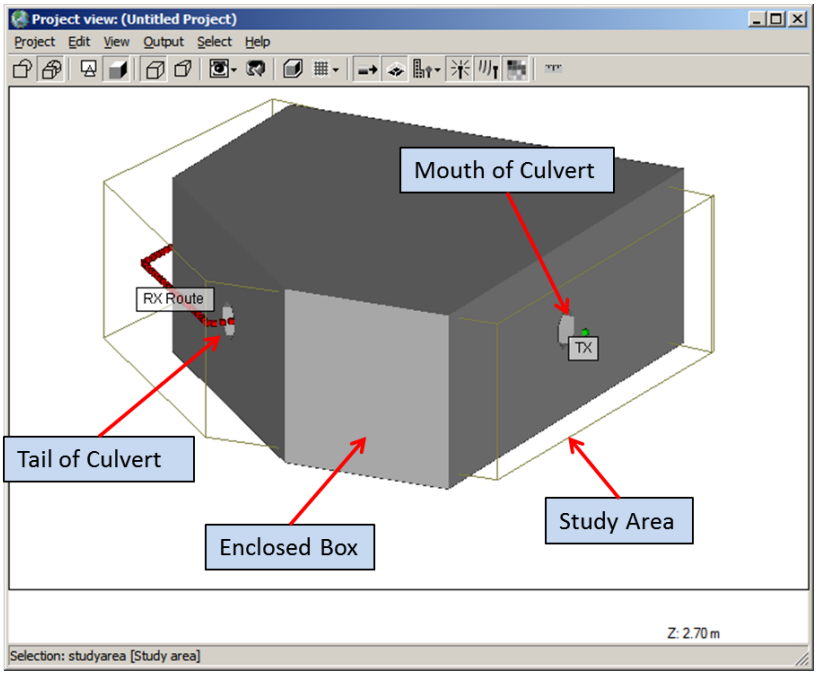


Figure B.12: Shows the Study Area intersecting the Enclosed Box to help eliminate paths from exiting one end of the culvert and re-entering on the opposite end.

B.5.3.2 Propagation Models; Ray Spacing and Suggested Interactions

Selecting adequate interactions and sufficient ray spacing is extremely important to ensure accurate electric field calculations. As in any Wireless InSite project, the user will need to specify the number of interactions and the density of ray spacing. This value will vary for each user because these variables will have a large effect on run-time and accuracy. To obtain results quickly, the user may want to use X3D with 10 reflections, 1 diffraction, and a route of receivers going from the beginning of the culvert, to the end. For a more accurate solution, the user should request 16-20 reflections and as many as 2 diffractions.

✓It is recommended that the user select, at least, 1 diffraction when the culvert has a bend. It is also recommended that the user set 1 diffraction for all scenarios where the user does not choose to utilize the Enclosed Box option.

The run time/accuracy trade-off is ultimately up to the user. Below is a table of the minimal suggested interactions and ray spacing. These values were originally used to validate X3D and Full3D with Remcom's XFdtd. Recent enhancements to X3D have made it possible to use much wider ray spacing (as much as the defaulted 0.25 degrees) to obtain valid results. It is a good idea to periodically compare different ray spacing values for each propagation model, as these values are likely to improve with additional enhancements to each model.

- **Case I. Straight Culvert (no bends)**
  - **A.** All Receivers (Rx) and Transmitters (Tx) are inside culvert.
  - **B.** Tx(s) or Rx(s) outside of culvert.

Table B.1: Minimum Suggested Interactions and Ray Spacing.

Settings	Case I-A	Case I-B	Case II-A	Case II-B
Ray Spacing (deg.)	0.05	0.05	0.05	0.05
Reflections	16	16	16	16
Transmissions	0	0	0	0
Diffractions	0	1∗	1	2∗

- **Case II. 90 degree bend and 45 degree bend**
  - **A.** All Rx and Tx are inside of culvert.
  - **B.** Tx(s) or Rx(s) outside of culvert.

  ∗It is recommended for the user to utilize the Enclosed Box feature and pay close attention when constructing the Study Area when placing antennas outside of the culvert.

✓Our experience suggests the number of reflections and the ray tracing density could be decreased, resulting in faster run-times. Remcom recommends the user to experiment using different settings to obtain a comfortable speed / accuracy trade-off.

B.5.4 Transmitter and Receiver Set-Up

Transmitters and Receivers are set up using the same techniques as any other Wireless InSite project. To obtain results within a considerable amount of time, it is suggested that the user run a receiver route along the culvert and plot the requested result. It is also possible; yet much more time consuming, to run a grid of receivers. By using a grid, the user will be able to see a visual photograph of the coverage area inside a culvert. This could be most helpful for situations where an operator does not have control over the placement of a receiver inside the culvert, but knows the height of the receiver.

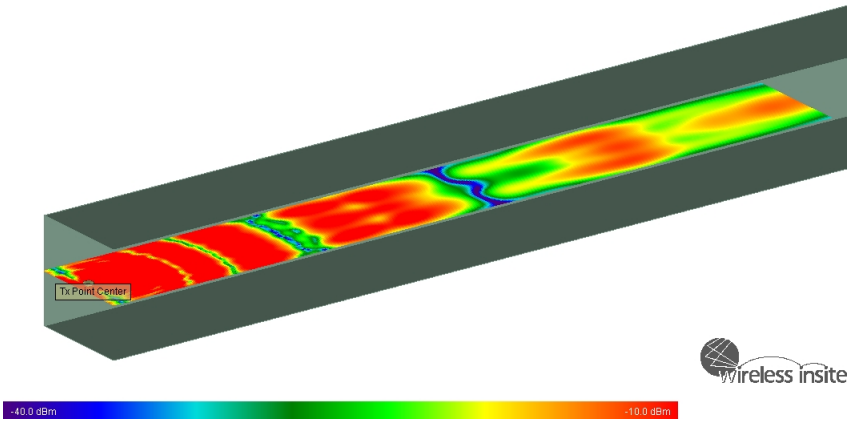


Figure B.13: Shows the Electric field magnitude for a grid of receivers.

B.6 Modified COST Building Penetration Model

Wireless InSite's Modified COST Building Penetration model enables the user to model outdoor-to-indoor propagation when the building's interior floor plan is not known. This module is an enhancement of the empirical outdoor-to-indoor model described in the COST 231 Propagation Prediction Models final report [67]. To use Wireless InSite's Building Penetration model, the user selects one building, or an area within a building, and specifies the empirically derived parameters α and β, which represent an attenuation factor related to the angle of incidence on the building and a loss per meter within the building respectively. The COST report recommends setting α equal to 0.6, whereas β is chosen to represent how "full" the interior of the building is. A horizontal grid of receivers is generated, and the user can specify the spacing of those receivers. Wireless InSite .p2m results are created for the received power.

- This capability is only compatible with the X3D model.
- Each *Building Penetration* receiver set is associated with a single .city structure.

B.6.1 Usage

To create a Building Penetration receiver grid within a city:

1. Load a project that has a *city* within it.
2. Within the GEOMETRY VIEW, choose *Select* → *Structure*
3. Select one of the buildings within the city
4. Right click *Add Building Penetration Rx Set*
5. This will create a new *receiver set* set within the scene. Building Penetration receivers are rendered in purple, whereas normal receivers are rendered in red.
6. Once created, the *Building Penetration Rx Grid* will appear in the RECEIVER tab, and can be edited in the same way as a normal *x-y* grid.
7. Additional fields in the RECEIVER PROPERTIES window, allow the user to change the value of the attenuation parameters alpha and beta. The default values are:
   - $\alpha = 0.6$ dB/m
   - $\beta = 1$ dB/m

The parameter β represents the average loss per meter within the building due to interior objects and structures, and α represents the losses associated with energy penetrating the building at a grazing angle.

- ✓ Switching the view to WIREFRAME will make viewing the grid within the building a little easier.
- *Building Penetration* receiver grids are a special type of *x-y grid*. Receiver points inside the associated building will only return received power, path loss, and path gain as predicted by the COST Building Penetration model, whereas receivers lying outside all buildings will return the full set of results calculated by X3D.
- ▶ See Chapter 21 for more information on viewing received power from a Building Penetration receiver grid.

Appendix C

Project File Format

In this appendix, you will learn...

- how the `*.setup` project file is formatted

C.1 Project File Format

The Wireless InSite project file (`*.setup`) is the primary file used by the GUI and the FULL 3D, URBAN CANYON, and VERTICAL PLANE models. (COMMUNICATION SYSTEMS definitions are included in the project files, but are not fully described here because the analysis is solely a function of Wireless InSite's user interface.) Specifically, the project file contains:

- The location of the global origin
- STUDY AREA description (propagation model, number of reflections, diffractions, transmissions, location of the study area)
- A list of the GEOMETRY files used in the project
- Locations of the TRANSMITTER and RECEIVER files used in the project
- ANTENNA information (type, polarization, associated waveform, etc.)
- WAVEFORM information (type, frequency, etc.)
- RESULT FILTER definitions
- A list of the REQUESTED RESULTS (received power, pathloss, delay spread, propagation paths, etc.)
- A list of graphs referencing RESULTS files
- Color bar display information

- Appendix A describes the notation for these file formats.

C.2 Building a Project File

Project files should be built up from each the following sections in the order the sections appear above. The code has been tabbed for readability.

C.2.1 Project File Header

```
begin_<project> [short description of project, 72 char max]
```

C.2.2 Global Origin Definition

```
begin_<globals>
     offset_mode manual
     longitude 0
     latitude 0
end_<globals>
```

C.2.3 Study Area Definition

```
FirstAvailableStudyAreaNumber [integer ID used for next studyarea]
begin_<studyarea> [short description of study area, 72 char max]
     StudyAreaNumber [integer ID number of study area]
     active <or> inactive
     autoboundary 0 (manual) <or> 1 (auto)
     begin_<model>
          [propagation model type]
          [propagation model parameters]
     end_<model>
     begin_<boundary>
          begin_<reference>
               cartesian <or> longlat
               longitude 0.0
               latitude 0.0
               visible no <or> yes
               sealevel <or> terrain
          end_<reference>
          zmin 0.100000 (bottom of the studyarea)
          zmax 4.900000 (height of the studyarea)
          nVertices 4
          0.000000 0.350000 0.000000
          12.765600 0.350000 0.000000
          12.765600 9.150000 0.000000
          0.000000 9.150000 0.000000
     end_<boundary>
end_<studyarea>
```

The Study Area Definition parameters are described below.

[*propagation model type*] - The keyword indicates the propagation model used. Options are listed in Table C.1.

[*propagation model parameters*] - Keywords and values specific to the type of propagation model. Options are listed in Table C.1.

- `ray_spacing [value in degrees]` - Spacing between adjacent ray paths. Only applies to SBR ray tracing.
- `raytracingmode sbr <or> eigenray` - The ray tracing method.
- `max_reflections [integer]` - Maximum number of reflections per ray path.
- `max_transmissions [integer]` - Maximum number of transmissions per ray path.
- `max_wedge_diffractions [integer]` - Maximum number of wedge diffractions per ray path.

Table C.1: Propagation models and associated properties

	Full 3D	**Urban Canyon**	**Vertical Plane**
Propagation model keyword	full3d	canyon	v_plane
`ray_spacing`	✓	✓	✓
`ray_tracing_mode`	✓		
`max_reflections`	✓	✓	✓
`max_transmissions`	✓		
`max_wedge_diffractions`	✓	✓	✓

C.2.4 Project Geometry

Each GEOMETRY contained in the project is listed in the project file and given a sequential integer ID number. The ID numbers start at 1. Inactive geometries are ignored during the simulation.

▶ Refer to Appendix D for the geometry file format.

```
begin_<feature>  (project files use old "feature" verbiage for geometry elements)
     feature [integer ID number]
     city <or> terrain <or> floorplan <or> foliage <or> object
     active <or> inactive
     filename [full path to the feature file]
end_<feature>
```

C.2.5 Transmitter and Receiver Files

Wireless InSite project files reference a single transmitter and receiver file. All transmitter and receiver definitions must be included in this file.

▶ Refer to Appendix E for the transmitter/receiver file format.

```
begin_<transmitter>
     filename [full path to the transmitter file (<project_name>.tx)]
     FirstAvailableTxNumber [integer ID used for the next Tx set]
end_<transmitter>

begin_<receiver>
     filename [full path to the receiver file (<project_name>.rx)]
     FirstAvailableRxNumber [integer ID used for the next Rx set]
end_<receiver>
```

C.2.6 Waveforms and Antennas

All WAVEFORMS and ANTENNAS used in a project must be defined in the `*.setup` file.

▶ Waveform and antenna files are described in Appendix H and Appendix I, respectively.

C.2.7 Result Filter Definition (optional)

```
FirstAvailableFilterNumber [integer ID used for next filter]
begin_<filter> [short description of result filter, 72 char max]
     FilterNumber [integer ID number of the filter]
     active <or> inactive
     los 1 <or> 0
     PowerMin [minimum ray path power in dBm]
     PowerMax [maximum ray path power in dBm]
     TOAMin [minimum time of arrival in seconds]
     TOAMax [maximum time of arrival in seconds]
     XTOAMin [minimum excess time of arrival in seconds]
     XTOAMax [maximum excess time of arrival in seconds]
     begin_<filter_entry>
          feature [integer ID for the feature] ("feature" refers to geometry element)
          reflections can <or> must <or> must_not
          transmissions can <or> must <or> must_not
          diffractions can <or> must <or> must_not
     end_<filter_entry>
end_<filter>
```

C.2.8 Result Requests

```
begin_<OutputRequests>
     cartesian
     sealevel
     local
     CalculationMode New
```

```
      CEF yes <or> no
      DelaySpread yes <or> no
      DirectionOfArrival yes <or> no
      DirectionOfDeparture yes <or> no
      ElectricFieldVsFrequency yes <or> no
      ElectricFieldVsTime yes <or> no
      FieldAnimation yes <or> no
      MeanDirectionOfArrival yes <or> no
      MeanDirectionOfDeparture yes <or> no
      Paths yes <or> no
      MeanTimeOfArrival yes <or> no
      EField yes <or> no
      EFieldTotal yes <or> no
      FSPathloss yes <or> no
      FSPower yes <or> no
      Power yes <or> no
      Pathloss yes <or> no
      XPathloss yes <or> no
      InputData yes <or> no
      TimeOfArrival yes <or> no
      TotalReceivedPower yes <or> no
      C2I yes <or> no
      ComplexImpulseResponse yes <or> no
      PowerDelayProfile yes <or> no
      StrongestTx yes <or> no
      PoyntingVector yes <or> no
      MaxRenderedPaths 25
      FieldAnimationIncrement 10
      FieldAnimationTimeAveraged yes <or> no
end_<OutputRequests>
```

C.2.9 Scale Bar

Scale bar definitions are only used by the user interface for displaying RESULTS. Their definition does not affect simulation engine results. An example scale bar for displaying ray paths is provided below.

```
begin_<Scales>
      begin_<NPaths>
            AutoScaling 1
            AutoUpdating 1
            Discrete 0
            UseGlobalOpacity 1
            ManualValuesSet 0
            ClampedHigh 1
            ClampedLow 1
            Alpha 1.000e+000
            ManualMin 0.000e+000
```

```
            ManualMax 1.000e+000
            Colors 6
            0.300 0.000 0.500
            0.000 0.000 1.000
            0.000 1.000 0.000
            1.000 1.000 0.000
            1.000 0.500 0.000
            1.000 0.000 0.000
            PartitionValues
            0
        end_<NPaths>
    end_<Scales>

end_<project>
```

Appendix D

Geometry File Format

In this appendix, you will learn...

- how to format a geometry file

D.1 Geometry File Format

Project geometry is stored in geometry files `*.city`, `*.flp`, `*.veg`, `*.obj`, or `*.ter`. Specifically, each geometry file contains:

- a <`begin`> delimiter, indicating the geometry is a *City*, a *Floor plan*, *Foliage*, an *Object*, or *Terrain*, followed by an optional short description of less than 50 characters
- notes (optional), with each line preceded by the keyword `description`
- the location of the geometry origin
- a list of MATERIALS and their parameters that are referenced in the object
- a list of faces which define the geometry, each composed of a list of vertices and an assigned a material.
  - ✓ The outward normal for a face is determined by the order of the vertices and the right hand rule. Related faces can be organized into sub-structures, structures, and structure groups.

- Appendix A describes the notation for the following file formats.

D.2 Building a Geometry File

The basic hierarchical organization of any geometry file is given below with a general description of the data contained in each section. The notes are only shown for the geometry, but descriptions can be given for any section in the file. The code has been tabbed for readability.

The example is for an `*.city` file, but the only difference between the format of `*.city` and that of other geometry files is that the tag `<city>` in the first and last line would be replaced by the desired geometry (`<floorplan>`, `<foliage>`, `<object>`, or `<terrain>`).

Geometry File Header

```
begin_<city> short description
     description [notes] (optional)
     smoothrender yes <or> no
```

where:

- `SmoothRender` - Indicates that the geometry should be rendered in the GEOMETRY VIEW using smooth shading.

Geometry Location

```
begin_<reference>
     cartesian <or> longlat
     longitude 0.0000000000
     latitude 0.000000000
     visible no <or> yes
     terrain <or> sealevel
end_<reference>
```

where:

- `cartesian` - Coordinate system used for the control points. When Cartesian is used, the control points of the set are interpreted as Cartesian offsets from the longitude/latitude origin of the set.
- `longlat` - Control points as listed in longitude and latitude and elevation.
- `longitude` - Longitude of the set's origin.
- `latitude` - Latitude of the set's origin.
- `visible no <or> yes` - Controls the display of the set's origin in the user interface.
- `terrain <or> sealevel` - Reference for the Z-coordinates of the control points. If a terrain does not exist in the project and the terrain keyword is used, the heights are relative to Z=0.

Geometry Material

```
begin_<Material>
     .
     .
end_<Material>
```

One or more materials may be defined in a geometry file.

- See Appendix F for a description of the material definition format.

Geometry Structure Groups

```
begin_<structure_group> [short description]
     begin_<structure> [short description]
          begin_<sub_structure> [short description]
               begin_<face> [short description]
                    double_sided (optional)
                    Material 1
                    nVertices 4
                    [x1],[y1],[z1]
                    [x2],[y2],[z2]
                    [x3],[y3],[z3]
                    [x4],[y4],[z4]
               end_<face>
               .
               .
          end_<sub_structure>
          .
          .
     end_<structure>
     .
     .
end_<structure_group>


end_<city>
```

where:

- `structure_group` - A group of structures (i.e., a city block).
- `structure` - A structure (i.e., a building).
- `sub_structure` - A section of a structure (i.e., the entryway to a building).
- `face` - The faces that define the sub-structure (i.e., the roof of an entryway).
- `Material` - A reference to the material, by number, that makes up the face.
- `nVertices` - The vertices that define each face.
- `double_sided` - Specifies that the face is double-sided. If omitted, the face is single-sided (by default).

Each structure-group can contain multiple structures, each structure multiple sub-structures, and each sub-structure multiple faces.

- See Section 3.1 for a description of geometry hierarchy.

D.3 Geometry File Examples

Object File

```
begin_<object> simple block
   SmoothRender No
   begin_<reference>
      cartesian
      longitude 0.00000000
      latitude 0.00000000
      visible yes
      terrain
   end_<reference>
   begin_<Material> Free Space
      Material 0
      Freespace
      thickness 0.000
      begin_<Color>
         ambient 0.500000 0.500000 0.500000 1.000000
         diffuse 0.500000 0.500000 0.500000 1.000000
         specular 0.500000 0.500000 0.500000 1.000000
         emission 0.000000 0.000000 0.000000 0.000000
         shininess 50.000000
      end_<Color>
   end_<Material>
   begin_<structure_group>
      begin_<structure>
         begin_<sub_structure>
            begin_<face> 1
               Material 0
               nVertices 4
                  -40.0000000000 -30.0000000000 10.0000000000
                  -110.0000000000 -30.0000000000 10.0000000000
                  -110.0000000000 -30.0000000000 0.0000000000
                  -40.0000000000 -30.0000000000 0.0000000000
            end_<face>
            begin_<face> 2
               Material 0
               nVertices 4
                  -40.0000000000 10.0000000000 10.0000000000
                  -40.0000000000 -30.0000000000 10.0000000000
                  -40.0000000000 -30.0000000000 0.0000000000
                  -40.0000000000 10.0000000000 0.0000000000
            end_<face>
            begin_<face> 3
               Material 0
               nVertices 4
```

```
                -110.0000000000 10.0000000000 10.0000000000
                -40.0000000000 10.0000000000 10.0000000000
                -40.0000000000 10.0000000000 0.0000000000
                -110.0000000000 10.0000000000 0.0000000000
            end_<face>
            begin_<face> 4
               Material 0
               nVertices 4
                  -110.0000000000 -30.0000000000 10.0000000000
                  -110.0000000000 10.0000000000 10.0000000000
                  -110.0000000000 10.0000000000 0.0000000000
                  -110.0000000000 -30.0000000000 0.0000000000
            end_<face>
            begin_<face> 5
               Material 0
               nVertices 4
                  -40.0000000000 -30.0000000000 10.0000000000
                  -40.0000000000 10.0000000000 10.0000000000
                  -110.0000000000 10.0000000000 10.0000000000
                  -110.0000000000 -30.0000000000 10.0000000000
            end_<face>
            begin_<face> 6
               Material 0
               nVertices 4
                  -110.0000000000 -30.0000000000 0.0000000000
                  -110.0000000000 10.0000000000 0.0000000000
                  -40.0000000000 10.0000000000 0.0000000000
                  -40.0000000000 -30.0000000000 0.0000000000
            end_<face>
         end_<sub_structure>
      end_<structure>
   end_<structure_group>
   begin_<ControlVectors>
      CVsVisible no
      Stippled no
      CVsThickness 3
      CVxLength 10.0000000000
      CVyLength 10.0000000000
      CVzLength 10.0000000000
      CVsXaxis 1.0000000000 0.0000000000 0.0000000000
      CVsZaxis 0.0000000000 0.0000000000 1.0000000000
   end_<ControlVectors>
end_<object>
```

Terrain File

```
begin_<terrain> simple terrain
```

```
    SmoothRender No
    begin_<reference>
       cartesian
       longitude -0.00000000
       latitude 0.00000000
       visible no
       terrain
    end_<reference>
    begin_<Material> wet earth
       Material 0
       DielectricHalfspace
       begin_<Color>
          ambient 0.350000 0.600000 0.350000 1.000000
          diffuse 0.350000 0.600000 0.350000 1.000000
          specular 0.350000 0.600000 0.350000 1.000000
          emission 0.000000 0.000000 0.000000 0.000000
          shininess 5.000000
       end_<Color>
       begin_<DielectricLayer> wet earth
          conductivity 0.020000
          permittivity 25.000000
          roughness 0.000
          thickness 0.000
       end_<DielectricLayer>
    end_<Material>
    begin_<structure_group>
       begin_<structure>
          begin_<sub_structure>
             begin_<face>
                 Material 0
                 nVertices 3
                    0.0000000000 0.0000000000 0.0000000000
                    1000.0000000000 0.0000000000 0.0000000000
                    1000.0000000000 1000.0000000000 0.0000000000
             end_<face>
             begin_<face>
                 Material 0
                 nVertices 3
                    0.0000000000 1000.0000000000 0.0000000000
                    0.0000000000 0.0000000000 0.0000000000
                    1000.0000000000 1000.0000000000 0.0000000000
             end_<face>
          end_<sub_structure>
       end_<structure>
    end_<structure_group>
 end_<terrain>
```

Appendix E

Transmitter and Receiver File Format

In this appendix, you will learn...

- how to format a transmitter and receiver file

E.1 Transmitter and Receiver File Format

The information for all TRANSMITTERS or RECEIVERS used in the project is stored in a transmitter or receiver file, respectively. Each file shares the same directory and name of the project file followed by `*.tx` or `*.rx` extension. For example, if a project is named `TxRxFormats.setup`, the transmitter set will be named `TxRxFormats.tx` and the receiver set will be named `TxRxFormats.rx`.

TRANSMITTER and RECEIVER sets generally share the same file format. Each file contains:

- Transmitter (Receiver) name
- Set type (*Point*, *Route*, *XY Grid*, etc.)
- Control point locations for defining the set
- Rotations
- Display information
- A reference to the antenna and waveform assigned to the transmitter

- Appendix A describes the notation for the following file formats.

E.2 Building a Transmitter or Receiver File

Within the `*.tx` or `*.rx` file, freestanding transmitter and receiver sets have the following general format given below. The code has been tabbed for readability.

Tx/Rx File Header

```
begin_<[set type]>[short description of the Tx/Rx set, 72 characters max]
     TxSet <or> RxSet [integer ID number for the set]
     active <or> inactive
```

where:

- *set type* - defines the type of Tx/Rx set. Wireless InSite allows the following sets for both Tx and Rx sets:
  - points
  - route
  - grid
  - arc
  - VerticalArc
  - cylinder
  - sphere
  - trajectory
  - polygon
  - VerticalSurface
  - PointsOnFace
  - userdefined
  - planewave (only available for a transmitter set)
  - ▶ The full definition of each set type is available in Chapter 13.
  - ▶ Examples of each type of set are provided in Section E.3.
- TxSet [*integer ID number for the set*] - The ID number is used when naming the result files.
- active <or> inactive - The status of the Tx/Rx sets. Inactive sets are ignored during the simulation.

Tx/Rx Display Properties

These properties are described by a set of keywords which define how the user interface displays the Tx/Rx set.

- vertical_line yes <or> no - Display of a vertical line from the point to height reference.
- cube_size - Size of points in meters
- CVxLength - X control vector length in meters

- `CVyLength` - Y control vector length in meters
- `CVzLength` - Z control vector length in meters
- `AntennaPatternScale` - Display size of the antenna pattern in meters
- `CVsVisible yes <or> no` - Display of the control vectors
- `CVsThickness` - Thickness of the control vectors

Tx/Rx Location

```
begin_<location>
     begin_<reference>
          cartesian <or> longlat
          longitude 0.0000000000
          latitude 0.000000000
          visible no <or> yes
          terrain <or> sealevel
     end_<reference>
     [set properties]
     nVertices [integer number of control points]
     [list of control points]
end_<location>
```

where:

- `cartesian` - Coordinate system used for the control points. When Cartesian is used, the control points of the set are interpreted as Cartesian offsets from the longitude/latitude origin of the set.
- `longlat` - Control points as listed in longitude and latitude and elevation.
- `longitude 0.0` - Longitude of the set's origin.
- `latitude 0.0` - Latitude of the set's origin.
- `visible no <or> yes` - Controls the display of the set's origin in the user interface.
- `terrain <or> sealevel` - Reference for the Z-coordinates of the control points. If a terrain does not exist in the project and the terrain keyword is used, the heights are relative to Z=0.
- *`[set properties]`* - Set properties vary by the type of set used. These are additional properties that define how the points are arranged in the set.
  - ▶ See Section E.3 for set examples with applicable keywords.
- `nVertices [`*`integer number of control points`*`]` - Lists the number of control points that follow.
- `[list of control points]` - Along with the set properties, the control point locations complete the description of the set. The values the control points define vary by set type.
  - For *Points*, each control point is the location of an individual point.

- For *Routes*, the control points define the end points of each segment.
- For *XY Grids*, the control points define opposite corners of a rectangular region.
- For *Arcs* and *Vertical Arcs*, the control point defines the center of the arc.
- For *Polygons*, the control points describe the location of the vertices.
- For *Sphere*, the control point is the center of the sphere.
- For *Trajectory*, the control points define the endpoints of the path and all points in between.
- For *Cylinder*, the control points represent the centers of the bottom and top discs of the cylinder.
- For *Vertical Surface*, the control points represent the bottom line segment of the vertical surface.

Tx/Rx Antenna

```
begin_<antenna>
     antenna [antenna ID number used by the set]
     waveform [waveform ID number used by the set]
     rotation_x [rotation about x-axis in degrees]
     rotation_y [rotation about y-axis in degrees]
     rotation_z [rotation about z-axis in degrees]
     power [transmitter's input power in dBm]
end_<antenna>
```

where:

- `antenna [antenna ID number used by the set]` - Defines the antenna used by the set.
- `waveform [waveform ID number used by the set]` - Defines the waveform used by the set. If this value is -1, the set will use the waveform assigned to the antenna.
- `rotation_x, rotation_y, rotation_z [rotation in degrees]` - Specifies the orientation of the antenna pattern for the particular set in the project. All points in a set will use these rotations. Rotations are applied first about the X-axis, then the Y-axis, and finally the Z-axis. If points in a set need to be oriented in different directions, the set should be divided.
- `power [transmitter's input power in dBm]` - For the transmitter only, specifies the input power of the transmitter set. This keyword is ignored for receiver sets.

Tx/Rx Additional Properties

```
NoiseFigure [additional noise from the receiver in dB]
generate_p2p no <or> yes
[display properties of 3D antenna in Geometry View]

end_<[set type]>
```

where:

- `NoiseFigure [`*`additional noise from the receiver in dB`*`]` - Contribution of the receiver itself to thermal noise at its output. The value represents the ratio of the thermal noise power at the output to that at the input, expressed in dB. This value is only used by the COMMUNICATION SYSTEMS analysis performed by the GUI.
- `generate_p2p no <or> yes` - Allows results restricted to *Point* sets to be generated for this particular receiver set. Results of this type includes *Efield vs. Time*, *Efield vs. Frequency*, and *Power delay profile*.

  These result types require additional processing time. It is unwise to make this request for receivers that contain a large number of points.
- `[`*`display properties of 3D antenna in`* GEOMETRY VIEW`]` - `pattern_show_arrow` toggles the display of an arrow pointing the direction of maximum gain. `pattern_show_as_sphere` determines if the pattern is projected onto a sphere or if the radius is proportional to the gain at a given direction. These properties are not used by the simulation.
- `begin_<PlaneWaveBoundary>` - Begins the definition of the plane wave boundary area. The boundary area represents a plane from which rays are launched toward the project geometry. The ray paths are all parallel to the boundary's face normal. The boundary should be large enough to ensure that all geometry in the project will be illuminated by a ray path. The user interface will automatically size the boundary to meet this condition. The boundary always consists of 4 coplanar control points that are Cartesian offsets from the set's origin and is concluded by the keyword `end_<PlaneWaveBoundary>`.

E.3 Transmitter Examples

Below are examples for TRANSMITTER sets, for each type of set. Receiver sets are nearly identical; the word "TxSet" before the integer ID number is replaced by "RxSet". The code has been tabbed for readability.

Tx Point

```
begin_<points> Tx Point
     TxSet 1
     active
     vertical_line yes
     cube_size 0.25000
     CVxLength 10.00000
     CVyLength 10.00000
     CVzLength 10.00000
     AntennaPatternScale 1.00000
     CVsVisible no
     CVsThickness 3
     begin_<location>
          begin_<reference>
               cartesian
               longitude -0.0001502742
               latitude 0.0000136580
               visible no
               terrain
          end_<reference>
          nVertices 1
          0.00000 0.00000 2.00000
     end_<location>
     begin_<antenna>
          antenna 0
          waveform -1
          rotation_x 0.00000
          rotation_y 0.00000
          rotation_z 0.00000
          power 0.00000
     end_<antenna>
     pattern_show_arrow no
     pattern_show_as_sphere no
     generate_p2p no
end_<points>
```

Tx Route

```
begin_<route> Tx Route
     TxSet 2
     active
     vertical_line no
     CVxLength 10.00000
     CVyLength 10.00000
     CVzLength 10.00000
     AntennaPatternScale 1.00000
     CVsVisible no
     CVsThickness 3
     begin_<location>
          begin_<reference>
               cartesian
               longitude -0.0001417045
               latitude 0.0000139258
               visible no
               terrain
          end_<reference>
          spacing 0.50000
          nVertices 4
          0.00000 0.00000 2.00000
          2.71274 0.02981 2.00000
          4.91870 -1.16260 2.00000
          4.88889 0.98374 2.00000
     end_<location>
     begin_<antenna>
          antenna 0
          waveform -1
          rotation_x 0.00000
          rotation_y 0.00000
          rotation_z 0.00000
          power 0.00000
     end_<antenna>
     begin_<sbr>
          bounding_box
     end_<sbr>
     pattern_show_arrow no
     pattern_show_as_sphere no
     generate_p2p no
end_<route>
```

Tx XY Grid

```
begin_<grid> TxXY Grid
     TxSet 3
     active
     vertical_line no
     CVxLength 10.00000
     CVyLength 10.00000
     CVzLength 10.00000
     AntennaPatternScale 1.00000
     CVsVisible no
     CVsThickness 3
     begin_<location>
          begin_<reference>
               cartesian
               longitude -0.0000886794
               latitude 0.0000040171
               visible no
               terrain
          end_<reference>
          side1 7.12466
          side2 3.36856
          spacing 0.50000
          nVertices 1
          0.00000 0.00000 2.00000
     end_<location>
     begin_<antenna>
          antenna 0
          waveform -1
          rotation_x 0.00000
          rotation_y 0.00000
          rotation_z 0.00000
          power 0.00000
     end_<antenna>
     begin_<sbr>
          bounding_box
     end_<sbr>
     pattern_show_arrow no
     pattern_show_as_sphere no
     generate_p2p no
end_<grid>
```

Tx Arc

```
begin_<arc> Tx Arc
     TxSet 4
     active
     vertical_line no
     CVxLength 10.00000
     CVyLength 10.00000
     CVzLength 10.00000
     AntennaPatternScale 1.00000
     CVsVisible no
     CVsThickness 3
     begin_<location>
          begin_<reference>
               cartesian
               longitude -0.0001320636
               latitude 0.0000390993
               visible no
               terrain
          end_<reference>
          radius 2.66785
          phistart 0.00000
          phistop 360.00000
          spacing 0.50000
          counterclockwise
          nVertices 1
          0.00000 0.00000 2.00000
     end_<location>
     begin_<antenna>
          antenna 0
          waveform -1
          rotation_x 0.00000
          rotation_y 0.00000
          rotation_z 0.00000
          power 0.00000
     end_<antenna>
     begin_<sbr>
          bounding_box
     end_<sbr>
     pattern_show_arrow no
     pattern_show_as_sphere no
     generate_p2p no
end_<arc>
```

Tx Vertical Arc

```
begin_<VerticalArc> Tx Vertical arc
     TxSet 5
     active
     vertical_line no
     CVxLength 10.00000
     CVyLength 10.00000
     CVzLength 10.00000
     AntennaPatternScale 1.00000
     CVsVisible no
     CVsThickness 3
     begin_<location>
          begin_<reference>
               cartesian
               longitude -0.0000980525
               latitude 0.0000570421
               visible no
               terrain
          end_<reference>
          radius 2.25599
          phistart 0.00000
          phistop 360.00000
          spacing 0.50000
          counterclockwise
          nVertices 1
          0.00000 0.00000 2.00000
     end_<location>
     rotation 88.36000
     begin_<antenna>
          antenna 0
          waveform -1
          rotation_x 0.00000
          rotation_y 0.00000
          rotation_z 0.00000
          power 0.00000
     end_<antenna>
     begin_<sbr>
          bounding_box
     end_<sbr>
     pattern_show_arrow no
     pattern_show_as_sphere no
     generate_p2p no
end_<VerticalArc>
```

Tx Cylinder

```
begin_<cylinder> Tx Cylinder
     TxSet 6
     active
     vertical_line no
     CVxLength 10.00000
     CVyLength 10.00000
     CVzLength 10.00000
     AntennaPatternScale 1.00000
     CVsVisible no
     CVsThickness 3
     begin_<location>
          begin_<reference>
               cartesian
               longitude 0.0
               latitude 0.0
               visible no
               terrain
          end_<reference>
          radius 1.58023
          phistart 0.00000
          phistop 360.00000
          spacing 0.50000
          counterclockwise
          nVertices 2
          0.00000 0.00000 1.00000
          0.00000 0.00000 4.00000
     end_<location>
     begin_<antenna>
          antenna 0
          waveform -1
          rotation_x 0.00000
          rotation_y 0.00000
          rotation_z 0.00000
          power 0.00000
     end_<antenna>
     begin_<sbr>
          bounding_box
     end_<sbr>
     pattern_show_arrow no
     pattern_show_as_sphere no
     generate_p2p no
end_<cylinder>
```

Tx Polygon

```
begin_<polygon>Tx Polygon
     TxSet 7
     active
     vertical_line no
     CVxLength 10.00000
     CVyLength 10.00000
     CVzLength 10.00000
     AntennaPatternScale 1.00000
     CVsVisible no
     CVsThickness 3
     begin_<location>
          begin_<reference>
               cartesian
               longitude 0.0000000000
               latitude 0.0000000000
               visible no
               sealevel
          end_<reference>
          spacing 0.50000
          offset 0.00000
          begin_<face>
               nVertices 5
               0.00000 0.00000 2.00000
               2.00000 0.00000 2.00000
               2.00000 3.00000 2.00000
               1.00000 4.00000 2.00000
               0.50000 5.00000 2.00000
          end_<face>
     end_<location>
     begin_<antenna>
          antenna 0
          waveform -1
          rotation_x 0.00000
          rotation_y 0.00000
          rotation_z 0.00000
          power 0.00000
     end_<antenna>
     begin_<sbr>
          bounding_box
     end_<sbr>
     pattern_show_arrow no
     pattern_show_as_sphere no
     generate_p2p no
end_<polygon>
```

Tx Vertical Surface

```
begin_<VerticalSurface> Tx Vertical surface
     TxSet 8
     active
     vertical_line no
     CVxLength 10.00000
     CVyLength 10.00000
     CVzLength 10.00000
     AntennaPatternScale 1.00000
     CVsVisible no
     CVsThickness 3
     begin_<location>
          begin_<reference>
               cartesian
               longitude -0.0001257230
               latitude 0.0000856971
               visible no
               terrain
          end_<reference>
          HorizontalSpacing 0.50000
          VerticalSpacing 0.50000
          Height 4.00000
          nVertices 2
          0.00000 0.00000 0.00000
          4.94851 0.00000 0.00000
     end_<location>
     begin_<antenna>
          antenna 0
          waveform -1
          rotation_x 0.00000
          rotation_y 0.00000
          rotation_z 0.00000
          power 0.00000
     end_<antenna>
     begin_<sbr>
          bounding_box
     end_<sbr>
     pattern_show_arrow no
     pattern_show_as_sphere no
     generate_p2p no
end_<VerticalSurface>
```

Tx Points On Face

```
begin_<PointsOnFace> Tx Points on Face
     TxSet 9
     active
     WasActive Yes
     vertical_line no
     cube_size 0.53385
     CVxLength 10.00000
     CVyLength 10.00000
     CVzLength 10.00000
     AntennaPatternScale 1.00000
     CVsVisible no
     CVsThickness 3
     begin_<location>
          begin_<reference>
               cartesian
               longitude 0.0000000000
               latitude 0.0000000000
               visible no
               terrain
          end_<reference>
          nVertices 1
          9.82264 1.42472 0.00000
          begin_<face>
               double_sided
               Material 0
               nVertices 4
               -18.00000 11.00000 0.00000
               1.00000 11.00000 0.00000
               1.00000 11.00000 3.00000
               -18.00000 11.00000 3.00000
          end_<face>
     end_<location>
     begin_<antenna>
          antenna 0
          waveform -1
          rotation_x 0.00000
          rotation_y 0.00000
          rotation_z 0.00000
          power 0.00000
     end_<antenna>
     pattern_show_arrow no
     pattern_show_as_sphere no
     generate_p2p no
end_<PointsOnFace>
```

Tx Polygon (On Face)

```
begin_<polygon> Tx Polygon (On Face)
     TxSet 10
     active
     vertical_line no
     CVxLength 10.00000
     CVyLength 10.00000
     CVzLength 10.00000
     AntennaPatternScale 1.00000
     CVsVisible no
     CVsThickness 3
     begin_<location>
          begin_<reference>
               cartesian
               longitude 0.0000000000
               latitude 0.0000000000
               visible no
               sealevel
          end_<reference>
          spacing 2.00000
          offset 0.00000
          begin_<face>
               double_sided
               nVertices 4
               -18.00000 0.00000 0.00000
               -18.00000 11.00000 0.00000
               -18.00000 11.00000 3.00000
               -18.00000 0.00000 3.00000
          end_<face>
     end_<location>
     begin_<antenna>
          antenna 0
          waveform -1
          rotation_x 0.00000
          rotation_y 0.00000
          rotation_z 0.00000
          power 0.00000
     end_<antenna>
     begin_<sbr>
          bounding_box
     end_<sbr>
     pattern_show_arrow no
     pattern_show_as_sphere no
     generate_p2p no
end_<polygon>
```

✓ This set is created by selecting *Cover with* in the face's context menu.

Tx Sphere Set

```
begin_<sphere> > Tx Sphere Set
     TxSet 11
     active
     vertical_line no
     CVxLength 10.00000
     CVyLength 10.00000
     CVzLength 10.00000
     AutoPatternScale
     ShowDescription yes
     CVsVisible no
     CVsThickness 3
     begin_<location>
          begin_<reference>
               cartesian
               longitude -0.0002972014
               latitude -0.0003480607
               visible no
               terrain
          end_<reference>
          radius 121.30000
          phistart 0.00000
          phistop 360.00000
          thetastart 0.00000
          thetastop 180.00000
          spacing 0.50000
          counterclockwise
          nVertices 1
          0.00000 0.00000 2.00000
     end_<location>
     begin_<antenna>
          antenna 0
          waveform -1
          rotation_x 0.00000
          rotation_y 0.00000
          rotation_z 0.00000
          power 0.00000
     end_<antenna>
     begin_<sbr>
          bounding_box
     end_<sbr>
     pattern_show_arrow no
     pattern_show_as_sphere no
     generate_p2p no
end_<sphere>
```

Tx Set From File

```
begin_<userdefined> Tx Set from File
     TxSet 12
     active
     vertical_line yes
     cube_size 0.25000
     CVxLength 10.00000
     CVyLength 10.00000
     CVzLength 10.00000
     AntennaPatternScale 1.00000
     CVsVisible no
     CVsThickness 3
     begin_<location>
          begin_<reference>
               cartesian
               longitude 0.0000000000
               latitude 0.0000000000
               visible no
               terrain
          end_<reference>
          filename D:\Wireless InSite_Projects\PointFile.pts
     end_<location>
     begin_<antenna>
          antenna 0
          waveform -1
          rotation_x 0.00000
          rotation_y 0.00000
          rotation_z 0.00000
          power 0.00000
     end_<antenna>
end_<userdefined>
```

Tx Trajectory Set From File

The following is an example of a user-defined *Trajectory* file that can be imported into Rosslyn streets example.

```
begin_<parameters>
     format tx
     minimum_time 0.0
     maximum_time 10.0
     time_increment 0.50
     time_units seconds
     angle_units degrees
     cartesian
     longitude -77.0738143921
     latitude 38.8944740295
     terrain
     from_trajectory
end_<parameters>
[time] [x] [y] [z] [roll] [pitch] [yaw]
0.00 0.0000 0.0000 2.0000 81.07298882 0.0000 0.0000
0.50 0.7759 4.9394 2.0000 81.07300795 0.0000 0.0000
1.00 1.5518 9.8789 2.0000 81.07301847 0.0000 0.0000
1.50 2.3276 14.8183 2.0000 81.07300004 0.0000 0.0000
2.00 3.1035 19.7577 2.0000 81.07285447 0.0000 0.0000
2.50 3.8794 24.6972 2.0000 81.07299357 0.0000 0.0000
3.00 4.6553 29.6366 2.0000 81.07305338 0.0000 0.0000
3.50 5.4312 34.5761 2.0000 81.07298225 0.0000 0.0000
4.00 6.2071 39.5155 2.0000 81.07298225 0.0000 0.0000
4.50 6.9829 44.4549 2.0000 81.07305338 0.0000 0.0000
5.00 7.7588 49.3944 2.0000 81.07281245 0.0000 0.0000
5.50 8.5347 54.3338 2.0000 81.07298225 0.0000 0.0000
6.00 9.3106 59.2732 2.0000 81.07305824 0.0000 0.0000
6.50 10.0865 64.2127 2.0000 81.07298871 0.0000 0.0000
7.00 10.8623 69.1521 2.0000 81.07297093 0.0000 0.0000
7.50 11.6382 74.0915 2.0000 81.07291108 0.0000 0.0000
8.00 12.4141 79.0310 2.0000 81.07298871 0.0000 0.0000
8.50 13.1900 83.9704 2.0000 81.07297093 0.0000 0.0000
9.00 13.9659 88.9098 2.0000 81.07302428 0.0000 0.0000
9.50 14.7418 93.8493 2.0000 81.07297093 0.0000 0.0000
10.00 15.5176 98.7887 2.0000 81.07298871 0.0000 0.0000
```

Tx Plane Wave

Plane Waves can only be modeled for transmitter sets.

```
begin_<PlaneWave> Plane Wave
     TxSet 13
     active
     begin_<location>
          begin_<reference>
               cartesian
               longitude 0.0001325697
               latitude 0.0001325697
               elevation 8.7272539139
               visible no
               sealevel
          end_<reference>
          theta 45.00000
          phi 45.00000
          generate_p2p yes
          begin_<PlaneWaveBoundary>
               nVertices 4
               13.06494 13.89828 10.53077
               16.44883 12.77032 8.93559
               16.44883 15.61550 6.92374
               13.06494 16.74346 8.51892
          end_<PlaneWaveBoundary>
     end_<location>
     begin_<antenna>
          polarization theta
          waveform 0
          power 1.00000
          preferredUnit powerDensity
     end_<antenna>
end_<PlaneWave>
```

Appendix F

Material Definition Format

In this appendix, you will learn...

- how to format a material file

This appendix provides several examples of singular MATERIAL definitions. Material are stored as part of a geometry within a `*.city`, `*.flp`, `*.veg`, `*.obj`, or `*.ter` file. There is no limit on the number of materials that can be defined within a file, and it is permissible for the GEOMETRY to include MATERIALS that are not currently assigned to any face.

- ✓ When used in a geometry file, the material definitions must precede geometrical data.
- Appendix A describes the notation for the following file formats.

F.1 Material Definition Format

The general format for a MATERIAL is:

```
begin_<Material> [short description]
     description [notes] (optional)
     material [ID number]
     [material type]
     thickness [thickness in m]
     [material display information]
     [material parameters]
end_<Material>
```

where:

- *[short description]* - Text entered by the user to describe the material. Description must be less than 72 characters.

- *[ID number]* - An integer number associated with the material. The value associates a material with a geometry.
- *[material type]* - Keyword identifying the type of material.
- *[material display information]* - the keywords used here, including `ambient`, `diffuse`, `specular`, `emission`, and `shininess`, control the color and appearance of the faces in the GUI.
  - ✓ Some of these parameters can be changed once the geometry file is loaded into the interface, but others, such as the "shininess", can only be changed by editing the file.
  - ✓ It is safe to omit these lines from the file if it is created from another software package. In this case, default values will be assigned to the material when the file is loaded into the Wireless InSite GUI and will be included in the properties when the GEOMETRY is saved from the GUI.
- *[material parameters]* - Keywords and values specific to the type of material.

F.2 Material Definition Examples

The code has been tabbed for readability.

Dielectric Half-Space

```
begin_<Material> short description
     description notes (optional)
     Material 1
     DielectricHalfspace
     Thickness 0.30
     begin_<Color>
          ambient 0.500000 0.500000 0.500000 1.000000
          diffuse 0.500000 0.500000 0.500000 1.000000
          specular 0.500000 0.500000 0.500000 1.000000
          emission 0.000000 0.000000 0.000000 0.000000
          shininess 50.000000
     end_<Color>
     begin_<DielectricLayer>
          conductivity 0.0150
          permittivity 15.0000
          roughness 0.000
          thickness 0.300
     end_<DielectricLayer>
end_<Material>
```

Layered Dielectric

```
begin_<Material> short description
     description notes (optional)
     Material 1
     LayeredDielectric
     begin_<Color>
          ambient 0.500000 0.500000 0.500000 1.000000
          diffuse 0.500000 0.500000 0.500000 1.000000
          specular 0.500000 0.500000 0.500000 1.000000
          emission 0.000000 0.000000 0.000000 0.000000
          shininess 50.000000
     end_<Color>
     nLayers 2
     begin_<DielectricLayer> description
          conductivity 0.0010
          permittivity 2.8000
          roughness 0.000
          thickness 0.013
     end_<DielectricLayer>
     begin_<DielectricLayer> description
          conductivity 0.0010
          permittivity 4.4400
          roughness 0.000
          thickness 0.125
     end_<DielectricLayer>
end_<Material>
```

PEC Backed Layer

```
begin_<Material> short description
     description notes (optional)
     Material 1
     PECBackedLayer
     thickness 0.300
     begin_<Color>
          ambient 0.500000 0.500000 0.500000 1.000000
          diffuse 0.500000 0.500000 0.500000 1.000000
          specular 0.500000 0.500000 0.500000 1.000000
          emission 0.000000 0.000000 0.000000 0.000000
          shininess 50.000000
     end_<Color>
     begin_<DielectricLayer>
          conductivity 0.0150
          permittivity 15.0000
          roughness 0.000
```

```
          thickness 0.300
     end_<DielectricLayer>
end_<Material>
```

Constant Coefficient

```
begin_<Material> short description
     description notes (optional)
     Material 1
     ConstantCoefficient
     thickness 0.300
     begin_<Color>
          ambient 0.500000 0.500000 0.500000 1.000000
          diffuse 0.500000 0.500000 0.500000 1.000000
          specular 0.500000 0.500000 0.500000 1.000000
          emission 0.000000 0.000000 0.000000 0.000000
          shininess 50.000000
     end_<Color>
     reflection_pl 1.000000
     reflection_pr 1.000000
     transmission_pl 0.000000
     transmission_pr 0.000000
     roughness 0.000
end_<Material>
```

Perfect Electrical Conductor

```
begin_<Material> short description
     description notes (optional)
     Material 1
     PEC
     thickness 0.000
     begin_<Color>
          ambient 0.500000 0.500000 0.500000 1.000000
          diffuse 0.500000 0.500000 0.500000 1.000000
          specular 0.500000 0.500000 0.500000 1.000000
          emission 0.000000 0.000000 0.000000 0.000000
          shininess 50.000000
     end_<Color>
     roughness 0.000
end_<Material>
```

Free Space

```
begin_<Material> short description
     description notes (optional)
     Material 1
     Freespace
     thickness 0.300
     begin_<Color>
          ambient 0.000000 0.000000 0.000000 1.000000
          diffuse 0.000000 0.000000 0.000000 1.000000
          specular 0.000000 0.000000 0.000000 1.000000
          emission 0.000000 0.000000 0.000000 0.000000
          shininess 50.000000
     end_<Color>
end_<Material>
```

Engineered Surface

```
begin_<Material> short description
     description notes (optional)
     Material 1
     engineered
     thickness 0.000
     begin_<Color>
          ambient 0.500000 0.500000 0.500000 1.000000
          diffuse 0.500000 0.500000 0.500000 1.000000
          specular 0.500000 0.500000 0.500000 1.000000
          emission 0.000000 0.000000 0.000000 0.000000
          shininess 50.000000
     end_<Color>
     local_origin Cartesian position in global reference
     coordinate_rotation in degrees (optional)
     definition_filename location of a valid EES file
end_<Material>
```

User-Defined

```
begin_<Material> short description
     description notes (optional)
     Material 1
     UserDefined
     thickness 0.000
     begin_<Color>
          ambient 0.500000 0.500000 0.500000 1.000000
```

```
            diffuse 0.500000 0.500000 0.500000 1.000000
            specular 0.500000 0.500000 0.500000 1.000000
            emission 0.000000 0.000000 0.000000 0.000000
            shininess 50.000000
      end_<Color>
      reflection_filename path to file with reflection coefficients
      transmission_filename path to file with transmission coefficients
end_<Material>
```

Reflection/Transmission Coefficient File

The reflection and transmission coefficients of a user-defined material must be stored in separate files. Each file contains parameters describing how the data is represented and formatted.

The reflection/transmission coefficient file is formatted in the following manner:

```
begin_<parameters>
      reflection <or> transmission
      frequency_min 1e+09
      frequency_max 2e+09
      frequency_inc 1e+09
      alpha_min 0
      alpha_max 90
      alpha_inc 45
      beta_min 0
      beta_max 90
      beta_inc 10
      magandphase <or> realandimag
      magnitude dB <or> linear
      phase degrees <or> radians
      side 1 <or> rotation 90
end_<parameters>
begin_<frequency> 1 GHz
      begin_<plane> 0 Degrees
            0 0.375 0 0.25 0
            10 0.375 0 0.25 0
            ...
            90 0.265 0 0.17 0
      end_<plane>
      begin_<plane> 45 Degrees
            0 0.358 30 0.315 30
            ...
            90 0.26 30 0.23 30
      end_<plane>
      begin_<plane> 90 Degrees
            ...
      end_<plane>
```

```
end_<frequency>
begin_<frequency> 2 GHz
     begin_<plane> 0 Degrees
     end_<plane>
           ...
     begin_<plane> 90 Degrees
     end_<plane>
end_<frequency>
```

A brief description of the possible keywords follows:

`reflection` - data represents reflection coefficients

`transmission` - data represents transmission coefficients

`frequency_min` - minimum frequency in Hz

`frequency_max` - maximum frequency in Hz

`frequency_inc` - frequency increment in Hz

`alpha_min` - minimum incident plane direction

`alpha_max` - maximum incident plane direction

`alpha_inc` - increment between incident planes

`beta_min` - minimum incident angle

`beta_max` - maximum incident angle

`beta_inc` - increment between incident angles

`magandphase` - magnitude and phase of coefficient

`realandimag` - real and imaginary parts of coefficient

`magnitude` - units for magnitude (linear or dB)

`phase` - units for phase (degrees or radians)

`side` - side on the face relative to which the incident plane angle is measured (used with the FULL3D model)

`rotation` - rotation of reference direction, counter- clockwise from reference direction (used with the X3D model)

`begin_<frequency>` and `end_<frequency>` - data for each frequency is placed between these delimiters. They must be present even if the file only contains data for a single frequency.

`begin_<plane>` and `end_<plane>` - data for each incident plane is placed between these delimiter. They must be present even if only a single incident plane is present. The format of the data for each incident plane is:

`incidence angle`

`magnitude of coefficient` (perpendicular to incident plane)

`phase` (perpendicular)

`magnitude` (parallel)

`phase` (parallel)

The X3D Model

The X3D and FULL3D models use different reference vectors for orienting the reflection and coefficient files. Figure F.1 and Figure F.2 show the orientation of the reference vector for a nearly horizontal facet and for a non-horizontal facet. Figure F.3 shows how the orientation changes when the keyword ”rotation” is used in a coefficient file.

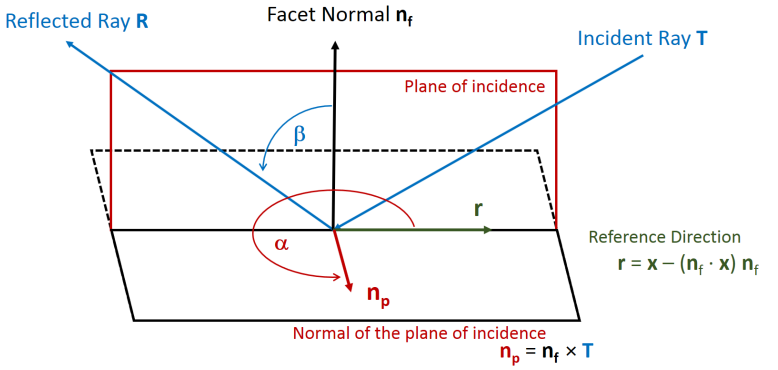


Figure F.1: Orientation of user-defined materials for near-horizontal surfaces, using the X3D model

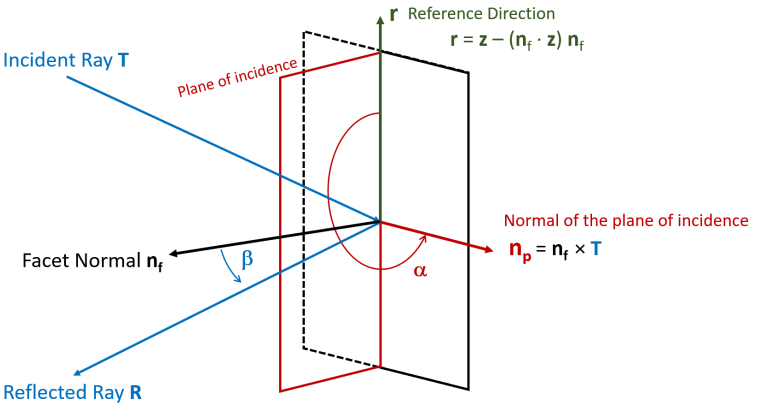


Figure F.2: Orientation of user-defined materials for non-horizontal surfaces, using the X3D model

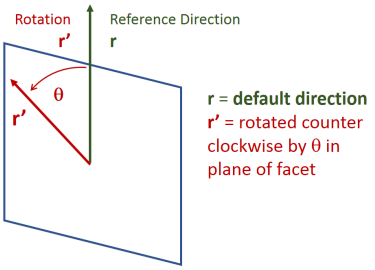


Figure F.3: Rotation of user-defined materials, using the X3D model

- The Plane of Incidence is determined by the face normal $\mathbf{n}_f$ and the incident ray $\mathbf{T}$.
- The Incident Angle, β, is the angle between the incoming ray and the face normal.
- The valid range of angles of incidence (β) is [0,90]. Only degrees are supported. If angle 0° is not included, the coefficients for any incidence angles less than the smallest angle are the same as the smallest angle data. If angle 90° is not included, the coefficients for any incidence angles greater than the largest angle are the same as the largest angle data.
- The angle of the incidence plane, α, is determined by the normal vector of the plane. The angle is measured counter-clockwise around the facet normal from the reference direction to the normal of the Plane of Incidence.
- The valid range of incident plane angles is [0,360]. There is no provision for units other than degrees. It is optional to input a plane for 360°, but it should duplicate the data in the 0° plane if present.
- Incidence plane angles in the range [0,360] that are not represented in data planes will be interpolated, with interpolation going through the branch cut at 360/0 as needed. We make no assumption of reciprocity, and so if the user intends (for example) that the coefficients on the 270° plane should match the 90° plane, the data to support that should be in the file.
- For the near-horizontal case, where $\mathbf{n_f}$ is within 10° of the vertical $\mathbf{z}$ vector, the default reference direction, $\mathbf{r}$, is the projection of the horizontal $\mathbf{x}$ axis onto the facet.

  $$\mathbf{r} = \mathbf{x} - (\mathbf{n}_f \cdot \mathbf{x})\mathbf{n}_f \tag{F.1}$$

  For the non-horizontal case, the default reference direction, $\mathbf{r}$, is the projection of the vertical $\mathbf{z}$ axis onto the facet.

  $$\mathbf{r} = \mathbf{z} - (\mathbf{n}_f \cdot \mathbf{z})\mathbf{n}_f \tag{F.2}$$

- If (real,imag) format is used, the data refer to the coefficient of reflected E field. If linear magnitude is used, it also refers to E field. If dB magnitude is used, it refers to power by convention, or equivalently E field squared. For example, if -20 dB magnitude is specified, the magnitude of reflected E field is 1/10 of the incident E field.
- Phase can be specified in radians for reflection and transmission coefficient data, but that setting only applies to the coefficient data, not to the incidence plane angles (α) nor to the angles of incidence (β).

- A rotation angle for the data planes can be defined in degrees counter-clockwise from the standard reference direction $\mathbf{r}$, using the header entry ”rotation”. If not included the rotation angle is zero, and the default reference is used. If both reflection and transmission data files are used, the rotation angle(s) are separately set in each file and need not be the same.
- The closest frequency match in the data to the transmitter frequency will be used without interpolation between frequencies.

The Full 3D Model

Figure F.4 shows a visual description of the reflection and transmission coefficients.

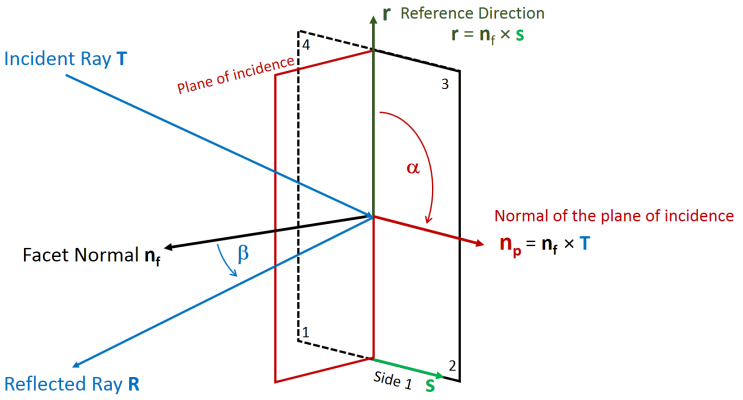


Figure F.4: Orientation of user-defined materials using the Full3D model

Concerning the above figure, note the following:

- The Plane of Incidence is determined by the face normal $\mathbf{n}_f$ and the incident ray $\mathbf{T}$.
- The Incident Angle, β, is the angle between the incoming ray and the face normal.
- α is the angle between the reference direction and the normal of the Plane of Incidence:

$$\alpha = arccos(\mathbf{r} \cdot \mathbf{n}_p) \tag{F.3}$$

- The reference direction, $\mathbf{r}$, is the cross product of the face normal and the unit vector from Vertex 1 to Vertex 2.
- When a project is run at a frequency not listed in the user-defined material file, FULL3D will interpolate reflection and transmission coefficients using the magnitude in dB and phase of coefficients at the adjacent frequency records. If the project frequency is above or below the frequency records contained in the user-defined material file, then the nearest frequency record is used.

✓ The limits on the incidence plane angle (alpha) and the incidence angle (beta) are 0° to 90°, with 0° referenced to the face normal.

In the above example, the “`side 1`” entry specifies that the incident plane is measured with respect to the first side on the face. Unfortunately, at present there is no simple way of using the GUI to determine the ordering of the sides on a particular face. However, all vertical faces created using Wireless InSite’s city

and floor plan editors have a horizontal edge as the first side on the face, and second side is always a vertical edge.

- ✓ The VIEW VERTICES button on the FACE PROPERTIES Window allows the user to determine the direction of the edges on any face.

Attenuation

```
begin_<Material> short description
     description notes (optional)
     Material 1
     FoliageAttenuation
     thickness 0.300
     begin_<Color>
          ambient 0.100000 0.600000 0.200000 0.600000
          diffuse 0.100000 0.600000 0.200000 0.600000
          specular 0.100000 0.600000 0.200000 0.600000
          emission 0.000000 0.000000 0.000000 0.000000
          shininess 5.000000
     end_<Color>
     density 0.600000
     VerticalAttenuation 1.000
     HorizontalAttenuation 1.000
     permittivity_vr 1.000
     permittivity_hr 1.000
end_<Material>
```

Biophysical

```
begin_<Material> short description
     description notes (optional)
     Material 1
     Biophysical
     thickness 0.000
     begin_<Color>
          ambient 0.000000 0.247059 0.000000 0.600000
          diffuse 0.000000 0.247059 0.000000 0.600000
          specular 0.000000 0.247059 0.000000 0.600000
          emission 0.000000 0.000000 0.000000 0.000000
          shininess 50.000000
     end_<Color>
     density 0.600000
     IncidenceAngle 90.0000
     DataType TreeParameters
     LeafRadius 0.0500
     LeafThickness 0.0005
```

```
      LeafDensity 100.0000
      LeafAngle1 0.0000
      LeafAngle2 180.0000
      BranchRadius 0.0160
      BranchLength 0.5000
      BranchDensity 1.0000
      BranchAngle1 0.0000
      BranchAngle2 60.0000
      PermittivityModel Constant
      AttenuationMeas 1.1000
      FrequencyMeas 869000000.0000
      IncidenceMeas 90.0000
      PolarizationMeas Vertical
      RealLeafPerm 26.0000
      ImagLeafPerm -7.0000
      RealBranchPerm 20.0000
      ImagBranchPerm -7.0000
      LeafCond 0.0850
      BranchCond 0.0850
      LeafPerm 40.0000
      BranchPerm 40.0000
end_<Material>
```

Complex permittivity

```
begin_<Material> short description
      description notes (optional)
      Material 1
      ComplexPermittivity
      thickness 0.300
      begin_<Color>
           ambient 0.000000 0.250000 0.000000 0.600000
           diffuse 0.000000 0.250000 0.000000 0.600000
           specular 0.000000 0.250000 0.000000 0.600000
           emission 0.000000 0.000000 0.000000 0.000000
           shininess 50.000000
      end_<Color>
      density 0.600000
      permittivity_vr 1.0300
      permittivity_vi 0.0300
      permittivity_hr 1.0300
      permittivity_hi 0.0300
end_<Material>
```

Lossy dielectric

```
begin_<Material> short description
     description notes (optional)
     Material 1
     LossyDielectric
     thickness 0.300
     begin_<Color>
          ambient 0.000000 0.250000 0.000000 0.600000
          diffuse 0.000000 0.250000 0.000000 0.600000
          specular 0.000000 0.250000 0.000000 0.600000
          emission 0.000000 0.000000 0.000000 0.000000
          shininess 50.000000
     end_<Color>
     density 0.600000
     permittivity_vr 1.0000
     permittivity_hr 1.0000
     conductivity_v 0.000000
     conductivity_h 0.000000
end_<Material>
```

Appendix G

Engineered Electromagnetic Surface Material File Format

In this appendix, you will learn ...

→ the format of the EES file type.

Engineered electromagnetic surface (EES) materials are defined by the user via the EES material specification file. This file will consist of a header followed by two data sections, the first for mode impedances, and the second for phase gradient data. The header contains the following information:

```
begin_<parameters>
     description [generation options]
     ees_type ModeImpedances
     ees_version 1
     frequency_min (optional, default = 0; units Hz)
     frequency_max (optional, default = 10^99; units Hz)
     transmissions (optional, defaults to false)
     grid_increment [grid spacing in meters]
     first_axis_start [u_1 start coordinate in meters from origin]
     first_axis_count [number of grid points in the u_1 direction]
     second_axis_start [u_2 start coordinate in meters from origin]
     second_axis_count [number of grid points in the u_2 direction]
     mode_truncation [number of surface impedance modes]
     desired_mode (optional, defaults to 1; recommended for expert use only)
end_<parameters>
```

Where the fields in the file have the following purpose:

`description` - lists the values of the following options used to generate the material.

Geometry

h = height

w = width

Material

R = reflection coefficient

T = transmission coefficient

Deflection Angles

l = left deflection angle

r = right deflection angle

t = top deflection angle

b = bottom deflection angle

Incident Angles

h = horizontal incident angle

v = vertical incident angle

Advanced

f = center frequency

bw = bandwidth

trunc = mode truncation number

mode = desired mode

`ees_type` - type of the EES format for data entries below

`ees_version` - version of the specified EES format type in use

`grid_increment` - non-zero distance in meters between each data sampling point

`first_axis_start` - distance in meters of first grid point from the origin in the u_1 direction

`first_axis_count` - number of grid points in the u_1 direction - referred to as k

`second_axis_start` - distance in meters of first grid point from the origin in the u_2 direction

`second_axis_count` - number of grid points in the u_2 direction - referred to as l

`mode_truncation` - number of modes represented in impedance data - referred to as M

When `frequency_min` and `frequency_max` are present, the EES will perform only for frequencies within these bounds. Wireless InSite will return an error when an operating frequency is outside of these bounds. When `transmissions` is set to true, the EES material will allow transmissions.

The Mode data will consist of $2M + 1$ lines of mode impedance data, in order from $-M$ to $+M$. The mode impedance values are dimensionless relative to the impedance of free space. Because there are two impedances (Ze and Zm) that are complex numbers, there are four values on every line.

```
begin_<mode_data>
      Re{Ze}, Im{Ze}, Re{Zm}, Im{Zm} (Normalized complex impedances for mode −M)
      ...
      Re{Ze}, Im{Ze}, Re{Zm}, Im{Zm} (Normalized complex impedances for mode +M)
end_<mode_data>
```

The phase gradient section is defined for a rectangular array of interpolation points at which the phase gradient components are specified. There are two directions on the rectangular grid called u_1 and u_2. Unless the EES definition has a non-zero rotation, an EES mounted on a vertical wall will have u_1 in the horizontal direction and u_2 in the vertical direction. Values of u_1 and u_2 are in meters.

u_1 and u_2 values run respectively from u1min to u1max and u2min to u2max. The spacing between points is `grid_increment`, which applies to both the $u1$ and $u2$ directions. Both the minimum and maximum values are data points. For example if `grid_increment` = 0.5, `first_axis_start` = -1, and `first_axis_count` = 5; the $u1$ values are -1, -0.5, 0, +0.5, and +1.

There are two values at each grid point. These values give the EES phase gradient components in the u_1 and u_2 directions respectively. The phase gradient components are dimensionless relative to the wavenumber in free space ($2\pi/\lambda$).

The phase gradient data are given one interpolation point at a time, with two values on a line specifying the two components of the phase gradient. Data are organized in rows of the rectangular grid array, with each row going from u_{1min} to u_{1max} with a constant value of u_2. Row segments are given from the row corresponding to u_{2min} to the row corresponding to u_{2max}.

```
begin_<data>
      begin_<row>
            ... k data lines ...
      end_<row>
      ... l row segments ...
end_<data>
```

Since the material definition is spatially dependent, it is worth noting some assumptions:

- u_1 and u_2 extents are independent
- Samples are spaced along u_1 and u_2 using the same fixed distance (`grid_increment`)

Any interaction with an EES material beyond its u_1 and u_2 extents will be treated as a perfect absorber.

▶ See [60] for details on describing an EES in terms of electric and magnetic sheet impedances.

Appendix H

Waveform Definition Format

In this appendix, you will learn...

- how to format a waveform file

WAVEFORM definitions are located in the project file (`*.setup`). All waveforms except *Sinusoids* require a time domain waveform file. Wireless InSite's user interface automatically generates this file when the simulation engine is called from within the interface. For users who are creating their own project files without aid from the interface, it may be easier to use the *User-Defined* Waveform type with an associated time domain or frequency domain waveform file.

The code has been tabbed for readability.

- Appendix A describes the notation for the following file formats.

H.1 Built-in Waveform File Format

The general format for a built-in WAVEFORM is:

```
begin_<Waveform> [short description]
     [waveform type]
     waveform [ID number]
     [waveform properties]
     TDFilename [path to time domain waveform file (*.wf)]
end_<Waveform>
```

where:

- *[short description]* - Text entered by the user to describe the waveform. Description must be less than 72 characters.
- *[waveform type]* - Keyword identifying the type of waveform. A list of the keywords for built-in waveforms is provided in Table H.1.

- *[ID number]* - An integer number associated with the waveform. The value associates a waveform with antennas, transmitters and receivers.
- *[waveform properties]* - Keywords and values specific to the type of waveform that define the waveforms shape and properties. A description of the keywords is provided below:
  - `CarrierFrequency [`*frequency in Hz*`]` - Center frequency of the waveform
  - `Bandwidth [`*frequency in Hz*`]`
  - `PulseWidth [`*time in seconds*`]` - Time duration of the waveform
  - `Phase [`*degrees*`]` - Initial phase shift for *Sinusoid* waveforms
  - `Rolloff [`*real value between 0.0 and 1.0*`]` - Excess bandwidth roll-off factor
  - `TaperRatio [`*real value between 0.0 and 1.0*`]` - Constant-to-taper ratio used by the *Tukey* Waveform
  - `Dispersive` - When present, the calculation of electric field as a function of time for certain result types will include dispersion on the broadband pulse.

Table H.1: Built-in Waveform Keywords and Associated Properties

Waveform Type	CarrierFrequency	Bandwidth	PulseWidth	Phase	Rolloff	TaperRatio	Dispersive
Blackman	✓		✓	✓			Optional
Chirp	✓*		✓		✓	✓	Always
Gaussian	✓		✓	✓			Optional
GaussianDerivative			✓				Always
Hamming	✓		✓	✓			Optional
Hanning	✓		✓	✓			Optional
RaisedCosine	✓		✓	✓	✓		Optional
RootRaisedCosine	✓		✓	✓	✓		Optional
Sinusoid	✓	✓		✓			N/A
Tukey	✓		✓	✓		✓	Optional

* Chirp Waveforms require both a *Start frequency* and *Stop frequency*, in Hz. They also require a *Frequency Variation* (Linear or Exponential), which describes how the frequency of the Chirp waveform varies between the start and stop frequencies.

H.2 Waveform Examples

Examples of three representative waveforms are given below.

Sinusoid

```
begin_<Waveform> Sinusoid
     Sinusoid
     waveform 0
```

```
     CarrierFrequency 1000000000.000
     phase 0.000
     bandwidth 1000000.000
end_<Waveform>
```

Blackman

```
begin_<Waveform> Blackman
     Blackman
     waveform 1
     CarrierFrequency 1000000000.000
     PulseWidth 1.000000e-007
     Phase 0.00000
end_<Waveform>
```

Dispersive Blackman

```
begin_<Waveform> Dispersive Blackman
     Blackman
     waveform 2
     CarrierFrequency 1000000000.000
     PulseWidth 1.000000e-007
     Phase 0.000
     Dispersive
end_<Waveform>
```

Chirp

```
begin_<Waveform> Chirp
     Chirp
     waveform 3
     CarrierFrequency 1625977050.781
     PulseWidth 1.00e-008
     Dispersive
     TDFilename C:\LocalFiles\Wireless InSite\Chirp\Chirp.waveform0.TD.wf
     Phase 0.000
     Rolloff 0.250
     FrequencyVariation Linear
     StartFrequency 1000.000
     StopFrequency 2000.000
end_<Waveform>
```

H.3 User-Defined Waveforms

In addition to the built-in waveforms, Wireless InSite allows *User-Defined* waveforms. They can be defined in either the time domain (TDFilename) or frequency domain (FDFilename).

User-Defined Waveforms are specified in the <project_name>.setup file using the following format:

```
begin_<Waveform> [short description]
     UserDefinedWaveform
     waveform [ID number]
     FDFilename <or> TDFilename
     [full path to the waveform file (*.wf)]
end_<Waveform>
```

The user may specify time- or frequency-domain samples of a waveform by referencing an external file beginning with the following parameters:

```
begin_<parameters>
     TimeDomain <or> FrequencyDomain
     MinValue 0
     MaxValue 6e-007
     Increment 2.08333e-011
     RealImag <or> MagPhase
     Phase degrees <or> radians
end_<parameters>
```

The first keyword in the parameters section must be either TimeDomain or FrequencyDomain, which indicates that waveform data to follow are samples in time or frequency. The MinValue, MaxValue, and Increment keywords specify the time or frequency range over which the samples were taken and the constant increment between each sample.

The last two possible keywords are only relevant to files which contain frequency-domain samples. RealImag specifies that the last two columns of data represent the real and imaginary components, respectively, of the waveform in the frequency domain. Alternatively, MagPhase specifies that the last two columns of data represent the magnitude and the phase, respectively, of the waveform in the frequency domain. If magnitude and phase are present, the last keyword Phase degrees or Phase radians must be used to specify whether the unit of the phase values is degrees or radians.

After the parameters section, time- or frequency-domain samples must be listed which cover the entire range, MinValue to MaxValue, specified at the top of the file.

If the samples are in the **time-domain**, two columns are required which contain the time and corresponding amplitude of the waveform.

- ✓ Time domain waveforms should start at 0 amplitude at (t=0) and should end at zero amplitude.
- ✓ It may be necessary to ramp up or down the amplitude at the start and end times to avoid high frequency components in the FFT.
- ✓ The minimum sampling rate for time domain waveforms should be the period/10.

```
0.000000e+000 0.000000e+000
```

```
2.500000e-008 9.510565e-001
5.000000e-008 8.090170e-001
.
.
4.500000e-007 8.090171e-001
4.750000e-007 9.510565e-001
5.000000e-007 0.000000e+000
```

If the samples are in the **frequency-domain**, three columns are required which contain the frequency and corresponding real and imaginary, or magnitude and phase of the waveform, e.g.:

```
0.0000000E+00 -0.4422828E-02 0.0000000E+00
0.1725166E+07 0.2102985E+01 0.1144734E+03
0.3450332E+07 0.8423647E+01 0.2288267E+03
.
.
0.3795365E+08 0.9898643E+03 0.2309188E+04
0.3967881E+08 0.1078863E+04 0.2394124E+04
0.4140398E+08 0.1171272E+04 0.2476455E+04
```

Appendix I

Antenna Definition Format

In this appendix, you will learn...

- how to format the antenna section of an Wireless InSite project file

ANTENNA definitions are located in the project file `*.setup`.

The code has been tabbed for readability.

I.1 Antenna File Format

The general format for an antenna definition is:

```
begin_<antenna> [short description of the Tx/Rx set, 72 characters max]
     antenna [integer ID number for the antenna]
     type [antenna type]
     waveform [integer ID number of the assigned waveform]
     power_threshold [receiver threshold in dBm]
     cable_loss [transmission line loss in dB]
     VSWR [voltage standing wave ratio]
     [antenna properties]
     [3D antenna pattern display properties] (optional)
     [array specifications] (optional)
end_<antenna>
```

where:

- `antenna [integer ID number for the set]` - The ID number used to refer to this antenna. Antenna ID numbers are used to assign antennas to transmitter and receiver sets in the `*.tx` and `*.rx` files.
- `type [antenna type]` - The keyword that follows defines the type of antenna. Wireless InSite uses the following keywords for antenna type:
  - `CircularAperture`

  - CircularLoop
  - CircularPatch
  - directional
  - HalfWaveDipole
  - Helical
  - Horn
  - userdefined
  - isotropic
  - linear_dipole
  - linear_monopole
  - omnidirectional
  - ParabolicReflector
  - QuarterWaveMonopole
  - RectangularAperture
  - RectangularPatch
  - short_dipole
  - short_monopole
  - SquareLoop

- `waveform` [*integer ID number of the assigned waveform*] - A reference to the waveform assigned to the antenna.
- `cable_loss` [*transmission line loss in dB*] - Value is used to simulate line losses, feed losses, or other system losses. The value can either be positive or negative.
- `power_threshold` [*receiver threshold in dBm*] - Individual ray paths whose power falls below this value are ignored when evaluating the power at a receiver point.
- `VSWR` [*voltage standing wave ratio*] - VSWR characterizes the impedance match between the antenna and the line. Value must be greater than or equal to 1.
- *[antenna properties]* - Antenna properties vary by the type of antenna used. The properties are generally used in formulas to generate a full 3D antenna pattern. Examples of keywords that appear in this section include `polarization`, `height`, `radius`, etc. Some antennas, such as the *Quarter-Wave Monopole* and the *Square Loop*, do not require any additional properties.

  ▶ See Section I.2 for applicable keywords for each type of set.

- [*3D antenna pattern display properties*] - The following keywords control how the antenna's 3D pattern displays when viewed in the GEOMETRY VIEW or in the ANTENNA PROPERTIES Window. They are not used by the simulation and can be omitted.

```
component TotalGain <or> ThetaComponent <or> PhiComponent
gain_range [gain range for 3D rendered pattern in dB]
show_arrow yes <or> no [arrow points in the direction of max gain]
is_sphere yes <or> no (3D antenna patterns can be projected onto a sphere of
 constant radius, or the radius component can be made a function of gain in that direction.)
```

- [*array specifications*] - An array of antennas can be created by including these keywords in the antenna description. All antennas in the array have the same individual pattern, but relative weights and phases can be assigned to each individual element.

```
begin_<array>
     Nelements [integer number of antennas in the array]
     [amplitude] [phase] [x-offset] [y-offset] [z-offset] (values for element 1)
     [amplitude] [phase] [x-offset] [y-offset] [z-offset] (values for element N)
end_<array>
```

I.2 Antenna File Examples

Below are examples of ANTENNA definitions for the built-in antennas supported by Wireless InSite.

▶ Appendix A describes the notation for the following file formats.

Circular Aperture

```
begin_<antenna> Circular Aperture
     antenna 0
     type CircularAperture
     waveform 0
     cable_loss 0
     power_threshold -250
     radius 1
     EFieldDistribution uniform <or> TE11
end_<antenna>
```

Circular Loop

```
begin_<antenna> Circular Loop
     antenna 1
     type CircularLoop
     waveform 0
     cable_loss 0
     power_threshold -250
     radius 1
end_<antenna>
```

Circular Patch

```
begin_<antenna> Circular Patch
     antenna 2
     type CircularPatch
     waveform 0
     cable_loss 0
     power_threshold -250
     epsilon 1
     radius 1
     height 1
end_<antenna>
```

Directional

```
begin_<antenna> Directional
     antenna 3
     type directional
     waveform 0
     polarization vertical <or> horizontal
     cable_loss 0
     power_threshold -250
     eplane_hpbw 90
     hplane_hpbw 90
     eplane_fnbw 180
     hplane_fnbw 180
end_<antenna>
```

Half-wave Dipole

```
begin_<antenna> Half-wave Dipole
     antenna 4
     type HalfWaveDipole
     waveform 0
     polarization vertical <or> horizontal
     cable_loss 0
     power_threshold -250
end_<antenna>
```

Helical

```
begin_<antenna> Helical
     antenna 5
```

```
      type Helical
      waveform 0
      polarization RightCircular <or> LeftCircular
      cable_loss 0
      power_threshold -250
      radius 1
      length 1
      pitch 1
end_<antenna>
```

Horn

```
begin_<antenna> Horn
      antenna 6
      type Horn
      waveform 0
      polarization vertical <or> horizontal
      cable_loss 0
      power_threshold -250
      HornApertureWidth 1.6489
      HornApertureHeight 0.8244
      FeedApertureWidth 0.1499
      FeedApertureHeight 0.0749
      Length 1.499
end_<antenna>
```

Isotropic

```
begin_<antenna> Isotropic
      antenna 7
      type isotropic
      waveform 0
      polarization vertical <or> horizontal
      cable_loss 0
      power_threshold -250
end_<antenna>
```

Linear Dipole

```
begin_<antenna> Linear Dipole
      antenna 8
      type linear_dipole
      waveform 0
      polarization vertical <or> horizontal
      cable_loss 0
```

```
     power_threshold -250
     length 1
end_<antenna>
```

Linear Monopole

```
begin_<antenna> Linear Monopole
     antenna 9
     type linear_monopole
     waveform 0
     cable_loss 0
     power_threshold -250
     length 1
end_<antenna>
```

Omnidirectional Antenna

```
begin_<antenna> Omnidirectional
     antenna 10
     type omnidirectional
     waveform 0
     polarization vertical <or> horizontal <or> RightCircular <or> LeftCircular
     cable_loss 0
     power_threshold -250
     eplane_hpbw 90
     eplane_fnbw 180
end_<antenna>
```

Parabolic Reflector Antenna

```
begin_<antenna> Parabolic Reflector
     antenna 11
     type ParabolicReflector
     waveform 0
     polarization vertical <or> horizontal
     cable_loss 0
     power_threshold -250
     radius 1
     blockageradius 0.1
     EFieldDistribution Uniform <or> Parabolic <or> ParabolicSquared
     EdgeTaper -20
end_<antenna>
```

Quarter-Wave Monopole Antenna

```
begin_<antenna> Quarter-wave Monopole
     antenna 12
     type QuarterWaveMonopole
     waveform 0
     cable_loss 0
     power_threshold -250
end_<antenna>
```

Rectangular Aperture Antenna

```
begin_<antenna> Rectangular Aperture
     antenna 13
     type RectangularAperture
     waveform 0
     cable_loss 0
     power_threshold -250
     sideA 1
     sideB 1
     EFieldDistribution uniform <or> TE10
end_<antenna>
```

Rectangular Patch Antenna

```
begin_<antenna> Rectangular Patch
     antenna 14
     type RectangularPatch
     waveform 0
     cable_loss 0
     power_threshold -250
     epsilon 1
     length 1
     width 1
     height 1
end_<antenna>
```

Short Dipole Antenna

```
begin_<antenna> Short Dipole
     antenna 15
     type short_dipole
     waveform 0
     polarization vertical <or> horizontal
     cable_loss 0
```

```
     power_threshold -250
end_<antenna>
```

Short Monopole Antenna

```
begin_<antenna> Short Monopole
     antenna 16
     type short_monopole
     waveform 0
     cable_loss 0
     power_threshold -250
end_<antenna>
```

Square Loop Antenna

```
begin_<antenna> Square Loop
     antenna 17
     type SquareLoop
     waveform 0
     cable_loss 0
     power_threshold -250
end_<antenna>
```

User-Defined Antenna

```
begin_<antenna> User Defined Antenna
     antenna 18
     type userdefined
     waveform 0
     cable_loss 0
     power_threshold -250
     gain 7.782
     filename D:\Wireless InSite\UserDefinedAntenna.uan
end_<antenna>
```

- See Section 12.5 for a description of the `*.uan` file format.

Appendix J

Basic Touchstone File Format

In this appendix, you will learn...

- how to format a basic Touchstone file

The code has been tabbed for readability.

J.1 Basic Touchstone File Format

Touchstone files are used to specify the network parameters for N-port linear electrical devices. In the context of Wireless InSite, they can be used to specify the network parameters of a MIMO antenna array, i.e. the self- and mutual-impedances of the antenna elements of an array along with the real-valued characteristic impedance of the transmission line connected to each element. Touchstone format files are ASCII files typically ending with a .snp file extension where the integer value "n" specifies the number of ports (i.e. antenna elements) for the network (e.g. a 4-port network is specified with a .s4p file extension).

While Touchstone file formats can be very flexible, only the most basic example of a version 1.1 Touchstone file format is described here. For a more comprehensive description of the file format see [82] and [83] for descriptions of version 1.1 and version 2.0 Touchstone files respectively, both of which can be read by Wireless InSite. Note, however, that Wireless InSite does *not* support Touchstone files with H- or G-parameter data (i.e. hybrid-H or hybrid-G network parameters), nor does it support mixed-mode network parameter data.

Introduction

The basic structure of a version 1.1 Touchstone file is described in [82]:

> Touchstone data files consist of an *option line* followed by one or more sets of network parameter data, where each set of data is taken at a specific frequency. The option line specifies (among other things) the kind of network data the file contains (S-parameter, Z-parameter, etc.), the format of the data values (magnitude-phase, real-imaginary, etc.) and the normalizing impedance. Data sets are arranged into one or more *data lines*, where the first line of network data is preceded by the

frequency at which the data was taken or derived. Data for a 1-port or 2-port network are contained on a single data line while data for 3-port and above networks are arrayed in a matrix format. The Touchstone format supports matrices of unlimited size.

Only one option-line/data-set pair is allowed per file.

Comment lines can be interspersed in the file as necessary.

Touchstone files are case insensitive.

Descriptions of the option line, data line, and comment line in the sub-sections below are select excerpts from [82] relevant to Touchstone files used with Wireless InSite.

Option Line

Each Touchstone data file must contain an option line (additional option lines after the first one will be ignored). The option line must be the first non-comment line of the file, and is formatted as follows:

<*frequency unit*> <*parameter*> <*format*> R <*n*>

where

#	marks the beginning of the option line.
frequency unit	specifies the unit of frequency. Legal values are GHz, MHz, KHz, Hz. The default value is GHz.
parameter	specifies what kind of network parameter data is contained in the file. Legal values are: S for Scattering parameters, Y for Admittance parameters, Z for Impedance parameters, The default value is S
format	specifies the format of the network parameter data pairs. Legal values are: DB for dB-angle, MA for magnitude-angle, RI for real-imaginary. Angles are given in degrees. The default value is MA.
R n	specifies the reference resistance in ohms, where n is a positive number of ohms (the real impedance to which the parameters are normalized). The default reference resistance is 50 ohms.

If a parameter is missing from the option line, it assumes the default value. With the exception of the opening # (hash mark) symbol and the value following “R”, option line parameters can appear in any order.

Data Line

Following the option line is the data set. Data sets contain the data for the network parameters (S-parameter, Z-parameter, etc.) specified by the option line. Network data for 1-port and 2-port networks is contained on one data line, while data for 3-port and above are arranged on multiple data lines in a matrix format. Each set of network data is preceded by a frequency value (i.e. the first entry in the first [or only] data line of a data set is a frequency value), and the network data itself is formatted as pairs of values (magnitude-angle, dB-angle, or real-imaginary).

There are four general rules for formatting data lines and data sets:

1. No more than four pairs of network data are allowed per data line.
2. Individual entries in a data lines are separated by whitespace.
3. A data line is terminated by a newline character (CR or CR/LF combination).
4. All data sets must be arranged in increasing order of frequency.

Detailed descriptions for arranging the data for various n-port networks follows.

1- and 2-Port Networks

Network parameter data for 1-port and 2-port networks at a single frequency can be contained on a single data line. As shown below, the data line consists of a frequency value followed by either one or four pairs of data values.

1-port data set (line)
<frequency value> <N11>

2-port data set (line)
<frequency value> <N11> <N21> <N12> <N22>

Network parameter data points will be in magnitude-angle, dB-angle or real-imaginary format (i.e. pairs of values) as specified by the option line.

Note the order in which 2-port network data is entered 21 data precedes 12 data.

3- and 4-Port Networks

The network parameter data for a 3-port or 4-port network is arranged in a matrix format, with each line of data representing one row of the matrix. In other words (as shown below), the data for a 3-port network is entered as three lines of data, with each line containing three data pairs (i.e. a 3x3 matrix of network parameter values). Likewise, the data for a 4-port network is entered as four lines with four data pairs per line (a 4x4 matrix). As required by the general rules, the first data line of each network parameter data set is preceded by the frequency value at which the data was taken.

5-Port and Above Networks

The network data for 5-port and above networks is also arranged in a matrix format. However, because the Touchstone format is limited to a maximum of 4 network parameter data points per line, additional entries beyond the first four in the matrix row must be continued on the following line(s). Each row of the matrix must start on a new line. As usual, the first entry in the first data line of a data set is the frequency value.

Comment Line

A Touchstone data file can be documented by preceding a comment with the exclamation mark (!). A comment can be the only entry on a line or can follow the data on any line.

Example 1

```
!2-port S-parameter file, three frequency points,
!frequencies are in GHz, real-imaginary format, 50 ohm reference
# GHZ S RI R 50.0
!freq   ReS11   ImS11   ReS21   ImS21   ReS12   ImS12   ReS22   ImS22
1.0000  0.3926  -0.1211 -0.0003 -0.0021 -0.0003 -0.0021 0.3926  -0.1211
2.0000  0.3517  -0.3054 -0.0096 -0.0298 -0.0096 -0.0298 0.3517  -0.3054
10.000  0.3419  0.3336  -0.0134 0.0379  -0.0134 0.0379  0.3419  0.3336
```

Example 2

```
!4-port S-parameter data, taken at three frequency points
# GHZ S MA R 50
5.00000 0.60 161.24 0.40 -42.20 0.42 -66.58 0.53 -79.34 !row 1
        0.40 -42.20 0.60 161.20 0.53 -79.34 0.42 -66.58 !row 2
        0.42 -66.58 0.53 -79.34 0.60 161.24 0.40 -42.20 !row 3
        0.53 -79.34 0.42 -66.58 0.40 -42.20 0.60 161.24 !row 4
6.00000 0.57 150.37 0.40 -44.34 0.41 -81.24 0.57 -95.77 !row 1
        0.40 -44.34 0.57 150.37 0.57 -95.77 0.41 -81.24 !row 2
        0.41 -81.24 0.57 -95.77 0.57 150.37 0.40 -44.34 !row 3
        0.57 -95.77 0.41 -81.24 0.40 -44.34 0.57 150.37 !row 4
7.00000 0.50 136.69 0.45 -46.41 0.37 -99.09 0.62 -114.19 !row 1
        0.45 -46.41 0.50 136.69 0.62 -114.19 0.37 -99.09 !row 2
        0.37 -99.09 0.62 -114.19 0.50 136.69 0.45 -46.41 !row 3
        0.62 -114.19 0.37 -99.09 0.45 -46.41 0.50 136.69 !row 4
```

Example 3

```
!5-port S-parameter data, at two frequencies
# MHZ S RI R 70
800.000 -0.3842 0.6089 0.0999 0.4079 0.2983 -0.1412 0.3449 0.3193 !row 1
0.2976 -0.2198
0.0999 0.4079 -0.3842 0.6089 0.1018 -0.1950 0.2976 -0.2198 !row 2
0.3449 0.3193
0.2983 -0.1412 0.1018 -0.1950 -0.3842 0.6089 0.1018 -0.1950 !row 3
0.2983 -0.1412
0.3449 0.3193 0.2976 -0.2198 0.1018 -0.1950 -0.3842 0.6089 !row 4
0.0999 0.4079
0.2976 -0.2198 0.3449 0.3193 0.2983 -0.1412 0.0999 0.4079 !  row 5
-0.3842 0.6089
900.0000 -0.3074 0.4871 0.0799 0.3264 0.2386 -0.1129 0.2759 0.2555 !row 1
0.2381 -0.1759
0.0799 0.3264 -0.3074 0.4871 0.0815 -0.1560 0.2381 -0.1759 !row 2
0.2759 0.2555
0.2386 -0.1129 0.0815 -0.1560 -0.3074 0.4871 0.0815 -0.1560 !row 3
0.2386 -0.1129
0.2759 0.2555 0.2381 -0.1759 0.0815 -0.1560 -0.3074 0.4871 !row 4
0.0799 0.3264
0.2381 -0.1759 0.2759 0.2555 0.2386 -0.1129 0.0799 0.3264 !row 5
-0.3074 0.4871
```

Appendix K

Communication System File Formats

In this appendix, you will learn...

- how to format user-defined precoding table files (*.tbf)
- how to format user-defined throughput files (*.wam)

K.1 User-Defined Precoding Table Files

The precoding table file allows users to provide a set of weightings that are applied to a MIMO transmitter/receiver's antenna elements. The file contains multiple entries, specifying the element weightings for each beam direction. The file includes a header section that specifies the format of data in the file, as described below:

```
begin_<parameters>
name (short description)
format mag_phase <or> real_imag
mag_scale decibel <or> linear (only required when mag_phase is specified)
phase_units deg <or> rad (only required when mag_phase is specified)
num_sets 3
num_elements 4
end_<parameters>
```

`num_sets` refers to the number of beams in the file. `num_elements` refers to the number of elements in the MIMO antenna. Default values are: `format mag_phase`, `mag_scale dB`, and `phase_units degrees`.

If the keyword `mag_phase` is included, the format of the data is:

```
element_ID element_magnitude element_phase
```

where mag_scale describes the value of element_magnitude in linear or dB scale, and phase_units describes the units of element_phase.

If format real_imag is listed in the header, the format of the data is:

```
element_ID REAL(element_weighting) Imag(element_weighting)
```

element_ID is not used by Wireless InSite, but is included to make different element record sets easily distinguishable.

An example of a precoding file (*.tbf) is provided below:

```
begin_<parameters>
name 4ElementScanningDipoleArray
format mag_phase
mag_units linear
phase_units deg
num_sets 3
num_elements 4
end_<parameters>
e1 1.0 0.00
e2 1.0 180.0
e3 1.0 360.0
e4 1.0 540.0
e1 1.0 0.00
e2 1.0 155.8846
e3 1.0 311.7691
e4 1.0 467.6537
e1 1.0 0.00
e2 1.0 90.0
e3 1.0 180.0
e4 1.0 270.0
```

Note that the mag value = 1, in the example works well for "mag_scale linear", but should be 0 when "mag_scale dB" is used.

K.2 User-Defined Throughput Files

K.2.1 User-Defined Throughput Files

Users also have the option of defining an SINR-throughput table with the *User-defined table* option. This allows users to provide a text file of SINR and throughput values specific to their wireless access method. An example of the format is shown below. An optional first line of the file can be used to specify the maximum number of data streams that can be transmitted using spatial multiplexing. This optional entry is used to populate the "Max # Streams" field that appears below "Closed-Loop Spatial Multiplexing (SVD)" which defaults to 8 if no specification is given. The file then proceeds with signal bandwidth definitions. User-defined files can contain multiple entries for the bandwidth. The entries for signal bandwidth are

used to populate the options in the “Signal Bandwidth” drop-down menu when the user-defined throughput table is chosen. The rows following the signal bandwidth definition have the following entries: SINR values [dB], throughput value [Mbit/sec], and text string that can be used to identify the modulation and coding scheme.

```
maxStreams 4
bandwidthMHz 20
-2.40 4.584 LTE_MCS_QPSK_1_5
-0.72 7.224 LTE_MCS_QPSK_1_3
1.54 10.296 LTE_MCS_QPSK_1_2
3.53 14.112 LTE_MCS_QPSK_2_3
6.36 17.568 LTE_MCS_16QAM_9_20
6.94 19.848 LTE_MCS_16QAM_1_2
7.92 22.92 LTE_MCS_16QAM_11_20
8.48 25.456 LTE_MCS_16QAM_3_5
9.44 28.336 LTE_MCS_16QAM_2_3
10.84 30.576 LTE_MCS_16QAM_3_4
12.41 32.856 LTE_MCS_16QAM_5_6
13.10 36.696 LTE_MCS_64QAM_1_2
13.78 39.232 LTE_MCS_64QAM_3_5
14.13 43.816 LTE_MCS_64QAM_5_8
14.62 46.888 LTE_MCS_64QAM_2_3
15.64 51.024 LTE_MCS_64QAM_17_24
16.46 55.056 LTE_MCS_64QAM_3_4
17.04 57.336 LTE_MCS_64QAM_4_5
18.15 61.664 LTE_MCS_64QAM_5_6
19.17 63.776 LTE_MCS_256QAM_tbs25
22.01 66.592 LTE_MCS_256QAM_tbs27
26.00 71.112 LTE_MCS_256QAM_tbs28
27.70 73.712 LTE_MCS_256QAM_tbs29
29.51 78.704 LTE_MCS_256QAM_tbs30
30.40 81.176 LTE_MCS_256QAM_tbs31
31.02 84.76 LTE_MCS_256QAM_tbs32
33.30 97.896 LTE_MCS_256QAM_tbs33
```

Appendix L

Propagation Paths Result File Format

In this appendix, you will learn...

→ the format of a propagation paths result file

The endpoint locations that make up the ray path segments between a TRANSMITTER and RECEIVER are accessible in a `*.paths...p2m` result file. These files can be viewed with a text editor. A separate file is created for each active transmitter point and receiver set in the project. The filename includes the transmitter and receiver set ID numbers for the rays that are contained within the file, such as:

```
ThinPlate.paths.t001_01.r001.p2m
(<Project name>.paths.t<tx number>_<tx set number>.r<rx set
number>.p2m)
```

The number of rays saved for each Tx/Rx pair can be set in the PROJECT RESULT PROPERTIES and entering the desired value next to *Maximum number of paths*. The default value is 25, and the maximum number that can be saved depends on the value of *Maximum stored paths*.

- *Maximum stored paths* sets the number of ray paths used in Wireless InSite's calculation of results between a Tx/Rx pair. The default setting is 250 paths.
- *Maximum rendered paths* sets the number of rays that are saved to a file, not the number used in the calculation of results.

▶ Appendix A describes the notation for the following file formats.

L.1 Organization of a Propagation Paths Result File

Propagation Paths File Header

The propagation paths result file begins with a 21-line header describing the format of the file. The file is structured to identify the Tx point and Rx set, list the number of receiver points in the set, then, for each receiver point, the number of paths to that point, and the aggregated data (received power, mean time of

arrival, and delay spread), followed by detailed data for each path to that receiver. The header is as follows:

```
# <Transmitter Set:  Tx:  3 SISO Tx - Point 1>
# <Receiver Set:  Rx:  2 Rx >
# <number of receiver points>
# <receiver point number> <number of paths for this point>
#    rx pt summary:
#    <received power (dBm)>
#    <mean time of arrival (sec)>
#    <delay spread (sec)>
#       path summary:
#       <path number>
#       <total interactions for path> (not including Tx and Rx)
#       <received power(dBm)>
#       <phase(deg)>
#       <time of arrival(sec)>
#       <arrival theta(deg)>
#       <arrival phi(deg)>
#       <departure theta(deg)>
#       <departure phi(deg)>
#          interactions summary:
#          <interaction description> (Tx:transmitter, Rx:receiver,
T:transmission, R:reflection, D:diffraction)
#          <cartesian locations of interactions> (including Tx and Rx)
```

The lines that follow begin with the number of receiver points in the set. The next line gives the first receiver ID and the number of paths to it. The third line is a summary of transmission between the transmitter and the first receiver point in the receiver set.

Following this is the list of the individual ray paths between the transmitter and the first receiver point. The paths are listed in order of strength starting with the strongest ray. The number of rays listed depends of the value of *Maximum number of paths*. Wireless InSite will save this quantity of rays, provided they exist, for the given the number of reflections, transmissions, and diffractions defined in the project's STUDY AREA. The total received power between transmitter and receiver points includes phase information, and therefore cannot be generated by simply summing the received power of the individual rays listed.

This structure is repeated for each subsequent receiver in the set, until the ray paths for all the receiver points in the given receiver set are listed.

Example paths.p2m File

An example of the `*.paths...p2m` for a simple example containing a single transmitter point and an receiver set containing two receiver points is given below. The *Maximum number of paths* was set to 4. (The header is not included.)

```
2
1 4
-46.89 0.12070E-06 0.31116E-07
1 1 -63.7590 0.95380E-07 90.0000 59.3935 90.0000
120.6065
```

```
Tx-R-Rx
-0.3939858E-06 -0.9692657E-06 1.500
-0.1817652E+01 0.2916138E+01 1.500
-0.1456212E+02 -0.1847147E+02 1.500
2 2 -63.8990 0.95904E-07 84.0122 59.3935 84.0122
120.6065
Tx-R-R-Rx
-0.3939858E-06 -0.9692657E-06 1.500
-0.1817654E+01 0.2916138E+01 1.874
-0.7281058E+01 -0.6163047E+01 2.843
-0.1456212E+02 -0.1847147E+02 1.500
3 2 -63.8990 0.95904E-07 95.9878 59.3935 95.9878
120.6065
Tx-R-R-Rx
-0.3939858E-06 -0.9692657E-06 1.500
-0.1817654E+01 0.2916138E+01 1.126
-0.7281058E+01 -0.6163047E+01 0.157
-0.1456212E+02 -0.1847147E+02 1.500
4 3 -64.3611 0.97456E-07 101.8477 59.3935 78.1523
120.6065
Tx-R-R-R-Rx
-0.3939858E-06 -0.9692657E-06 1.500
-0.1817652E+01 0.2916138E+01 2.249
-0.3640530E+01 -0.8833901E-02 2.843
-0.1092159E+02 -0.1231726E+02 0.157
-0.1456212E+02 -0.1847147E+02 1.500
2 4
-49.33 0.12175E-06 0.25628E-07
1 1 -64.8169 0.10706E-06 90.0000 8.3745 90.0000
351.6255
Tx-R-Rx
-0.3939858E-06 -0.9692657E-06 1.500
0.4956527E+01 -0.7527094E+00 1.500
-0.2153662E+02 -0.4675870E+01 1.500
2 2 -64.9173 0.10753E-06 95.3384 8.3745 95.3384
351.6255
Tx-R-R-Rx
-0.3939858E-06 -0.9692657E-06 1.500
0.4956527E+01 -0.7527094E+00 1.017
-0.5655232E+01 -0.2337935E+01 0.157
-0.2153662E+02 -0.4675870E+01 1.500
3 2 -64.9173 0.10753E-06 84.6616 8.3745 84.6616
351.6255
Tx-R-R-Rx
-0.3939858E-06 -0.9692657E-06 1.500
0.4956527E+01 -0.7527094E+00 1.983
-0.5655232E+01 -0.2337935E+01 2.843
-0.2153662E+02 -0.4675870E+01 1.500
4 3 -65.2561 0.10892E-06 100.5857 8.3745 79.4143
351.6255
Tx-R-R-R-Rx
-0.3939858E-06 -0.9692657E-06 1.500
```

```
0.4956527E+01 -0.7527094E+00 2.466
0.2285461E+01 -0.1168969E+01 2.843
-0.1359593E+02 -0.3506904E+01 0.157
-0.2153662E+02 -0.4675870E
```

Appendix M

MIMO-generated .csv Result File Formats

In this appendix, you will learn...

→ the format of .csv files exported through the MULTI-CHANNEL RESULTS BROWSER

The MULTI-CHANNEL RESULTS BROWSER allows users to export several result types to `.csv` format. These files can be viewed with a text editor, opened as a spreadsheet, or manipulated via scripting for customized post-processing.

M.1 Complex Impulse Response

Complex Impulse Response files are saved to the `cir` subfolder within the studyarea folder. The file naming convention is:

`cir.txSetWWW.txPtXXX.rxSetYYY.rxPtZZZ.txElAAA.rxElBBB.instCCC.csv`

A `cir` file is written for each Tx-Rx point/element pair, and contains 5 columns of data: Path ID, Source ID, Received Power (W), Phase (radians), Time of Arrival (s). The file includes a one-line header labeling each column. Each row represents a path between the Tx element and the Rx element. The path ID and source ID are not required for the complex impulse response calculation, but are used by Wireless InSite to identify individual paths.

M.2 Direction of Arrival and Direction of Departure

Direction of arrival and departure files are saved to the `doa/dod` subfolder within the studyarea folder. The file naming convention is:

`angles.txSetWWW.txPtXXX.rxSetYYY.rxPtZZZ.txElAAA.rxElBBB.csv`

An angles file is written for each Tx-Rx point/element pair. The file includes a one-line header labeling each column, followed by rows with 7 columns of data each: Path ID, Source ID, Arrival Phi (radians), Arrival Theta (radians), Departure Phi (radians), Departure Theta(radians), and Received Power (W). Each row represents a path between the Tx element and the Rx element. The path ID and source ID are used internally by Wireless InSite to identify individual paths.

M.3 H-Matrix (and H-Matrix with Mutual Coupling)

H-matrix files are saved in the hmatrix subfolder within the studyarea folder. Two types of H-matrix files may be generated: one type contains data for a receiver at a single point (i.e. a point-to-point result type) while the second type contains data for receivers at multiple points within a set (i.e. a point-to-multipoint result type).

For the point-to-point file type, each H-matrix result file represents the channel matrix between all Tx elements at a single point and all Rx elements at a single point. The file naming convention is:

```
hmatrix.txSetWWW.txPtXXX.rxSetYYY.rxPtZZZ.inst001.csv
```

This hmatrix file contains K rows and $(N \times 2 + 2)$ columns, where N is the number of transmitter elements and K is the number of receiver elements. The first two columns respectively identify the Rx point index and element index for the given row. There are $N \times 2$ additional columns because each n-k pair has a complex gain (real and imaginary parts), requiring two columns. Note that each H-matrix file contains data for all Tx-elements and all Rx-elements for a given Tx/Rx point pair.

For the second file type, a point-to-multipoint file, each H-matrix result file represents the channel matrix between all Tx elements at a single location and all Rx elements for each point in a point set (i.e., at multiple locations). The file naming convention is:

```
hmatrix.txSetWWW.txPtXXX.rxSetYYY.inst001.csv
```

The format is identical to that of the point-to-point file type described above except that the file contains $NP \times K$ rows where NP is the number of points in the point set. As is the case with the point-to-point file type, the first two columns identify the Rx point index and element index for that row.

In cases where H-matrix w/ mutual coupling files are saved, the format of the output files is identical to that described above for H-matrix outputs. The only difference is that the files are saved in the mchmatrix subfolder within the study area folder and the filenames contain the string mchmatrix instead of hmatrix.

M.4 Received Power

Received Power files are saved to the power subfolder within the studyarea folder. The file naming convention is:

```
power.txSetWWW.txPtXXX.rxSetYYY.txElAAA.rxElBBB.inst001.csv
```

A power file is written for each Tx point/element and each Rx set/element pair. The file includes a one-line header labeling each column, followed by rows containing two columns each: Received Power in Watts, and Phase in radians. Each row represents a receiver point in the set. All the rows within a file

represent the same element within the receiver array. (Note: for multiple-point receiver sets results for the nth-array element at all points in the receiver set will be within the same file.)

M.5 RMS Angle Spread of Arrival

RMS Angle Spread of Arrival files are saved to the `rmsaglspreadarr` subfolder within the studyarea folder. The file naming convention is:

```
rmsanglespread.txSetWWW.txPtXXX.rxSetYYY.instZZZ.csv
```

A `rmsaglspreadarr` file is written for each Tx point and Rx set pair. The file includes a one-line header labeling each column, followed by rows containing six columns each: RMS Horizontal Spread of Arrival (radians), RMS Vertical Spread of Arrival (radians), RMS Total Spread of Arrival (radians), RMS Horizontal Spread of Departure (radians), RMS Vertical Spread of Departure (radians), RMS Total Spread of Departure (radians). Each row represents a receiver point in the set.

M.6 RMS Angle Spread of Departure

RMS Angle Spread of Departure files are saved to the `rmsaglspreaddep` subfolder within the studyarea folder. The file naming convention is:

```
rmsanglespread.txSetWWW.txPtXXX.rxSetYYY.instZZZ.csv
```

A `rmsaglspreaddep` file is written for each Tx point and Rx set pair. The file includes a one-line header labeling each column, followed by rows containing six columns each: RMS Horizontal Spread of Arrival (radians), RMS Vertical Spread of Arrival (radians), RMS Total Spread of Arrival (radians), RMS Horizontal Spread of Departure (radians), RMS Vertical Spread of Departure (radians), RMS Total Spread of Departure (radians). Each row represents a receiver point in the set.

M.7 RMS Delay Spread

RMS Delay Spread files are saved to the `rmsdelayspread` subfolder within the studyarea folder. The file naming convention is:

```
rmsdelayspread.txSetWWW.txPtXXX.rxSetYYY.instZZZ.csv
```

A `rmsdelayspread` file is written for each Tx point and Rx set pair. The file includes a one-line header labeling each column, followed by rows containing 1 column each: RMS Delay Spread (s). Each row represents a receiver point in the set.

M.8 Path Loss

Path Loss files are saved to the `power` subfolder within the studyarea folder. The file naming convention is:

power.txSetWWW.txPtXXX.rxSetYYY.txElAAA.rxElBBB.instZZZ.csv

A power file is written for each Tx point/element and each Rx set/element pair. The file includes a one-line header labeling each column, followed by rows containing four columns each: Received Power in Watts, Phase in radians, Path Loss in Decibels, and Path Gain in Decibels. Each row represents a receiver point in the set. All the rows within a file represent the same element within the receiver array. (Note: for multiple-point receiver sets results for the nth-array element at all points in the receiver set will be within the same file.)

M.9 Path Gain

Path Gain files are saved to the power subfolder within the studyarea folder. The file naming convention is:

power.txSetWWW.txPtXXX.rxSetYYY.txElAAA.rxElBBB.instZZZ.csv

A power file is written for each Tx point/element and each Rx set/element pair. The file includes a one-line header labeling each column, followed by rows containing four columns each: Received Power in Watts, Phase in radians, Path Loss in Decibels, and Path Gain in Decibels. Each row represents a receiver point in the set. All the rows within a file represent the same element within the receiver array. (Note: for multiple-point receiver sets results for the nth-array element at all points in the receiver set will be within the same file.)

Appendix N

Troubleshooting

In this appendix, you will learn...

→ how to correct known issues in Wireless InSite

N.1 Rendering Graphics

Some video cards may exhibit problems when rendering graphics. Updating video card drivers or allowing Windows to handle the OpenGL rendering may fix these issues. Some Matrox, Intel and SiS video cards have been known to exhibit problems.

First try to obtain an updated driver from the video card manufacturer. Most manufacturers provide direct downloads from their website. If this does not resolve the issue, it is possible to force Windows to handle OpenGL calls. Doing this will significantly affect program performance; however, it will allow Wireless InSite to function correctly.

To force Windows to handle OpenGL calls, adjust your local Display Properties by navigating to Advanced Settings and setting the *Hardware Acceleration* to “None”. If after restarting Wireless InSite OpenGL errors continue to appear, contact Remcom Professional Support. If this fixes the problem it may be possible to move the slider to a higher level of hardware acceleration to improve rendering speed and avoid rendering issues.

N.2 Running Communication Systems

On some versions of Windows the MATLAB Component Runtime will not be able to initialize in order to run the bit error rate calculations. This is due to the user account restrictions that are enforced in newer versions of Windows. To allow the MATLAB components that are used to create this result type to load properly in Wireless InSite someone with administrative privileges will have to perform the following steps.

1. Right-click on the shortcut in the *Start* menu for *InSite* and select *Properties*.

2. In the Properties window select the *Compatibility* tab.
3. Check the *Run this program as administrator* check box.
4. Run the application once using the shortcut.

   If the user is not the one with administrative privileges you will need to undo the modification before they use the shortcut again.

Doing this will allow the program to run with administrator privileges for all future sessions and all COMMUNICATION SYSTEMS will run correctly.

N.3 GPU Card not detected for X3D

If the X3D model produces slower than expected run-times, check the SIMULATION LOG to make sure that the GPU card is being detected. The SIMULATION LOG provides several lines which acknowledge the detection, and proper use, of a GPU card. They occur before the progress of the ray tracing and E-field calculation begins.

The initial line for GPU information will read ”Detecting X3D GPU Information”. Following this header will be several lines detailing what the X3D simulation engine can detect on the local machine. Two such lines will be the CUDA Runtime version and CUDA Driver version, both useful for troubleshooting X3D difficulties.

The CUDA Runtime version is the version of the CUDA runtime dll that is supplied with Wireless InSite.

The CUDA Driver version is the version of the CUDA supported by the video driver on the local machine. This version should be higher or equal to the version of CUDA supported by the CUDA Runtime for proper execution. If the CUDA Driver version is too low a warning will be issued to update the local machine’s video driver. This is done by downloading the latest driver package from www.nvidia.com and installing it with an administrator account on the local machine.

Below the CUDA version lines will be an enumerated list of CUDA devices detected by the X3D simulation engine. These will be listed similarly to the following:

```
CUDA device #0:  [Quadro K620], 5.0 capability, 3 total cores, 2048 threads
per core
```

For each device listed, there will be an enumeration number. This can be used to tell X3D which devices to enable and disable (see below). This enumeration can change arbitrarily based on the initialization sequence of the computer. After the enumeration for the device the string identifying the device type will be listed.

The next bit of information is the CUDA compute capability version. Wireless InSite only recognizes NVIDIA, CUDA enabled cards with CUDA Compute Capability version 3.0 to 6.1.[1] In order to utilize the GPU version of X3D, a GPU card with the appropriate CUDA compute capability must be installed.

After the compute capability, the number of GPU cores and CPU threads will be listed.

[1] GPU must be a CUDA capable card and compatible with NVIDIA OptiX 4.1.0. Based on the release notes for OptiX 4.1.0 and the timing of its release, CUDA capable cards with compute capability between 3.0 and 6.1 should be supported. Contact NVIDIA support for specific recommendations.

If your card fits the above listed criteria, but is still not recognized by X3D, please review the following recommendations:

1. Check to make sure your GPU drivers are not out of date.
   Solution: Update the GPU drivers.
2. GPU stopped being recognized after a windows update.
   Solution: Reinstall the GPU drivers.

   ✓ Windows Update will sometimes helpfully install the latest WHQL-certified video drivers, which may be an older revision than the drivers available through NVIDIA directly.
3. Your system has 2 GPU cards, but Wireless InSite will only support one of them.
   Solution: Change the System Variable named *CUDA_VISIBLE_DEVICES*. For Windows 7, this can be done through the *Control Panel*.

   (a) Once in the Control Panel select *System and Security*, then *System*.

   (b) From the menu on the left, select *Advanced system settings*.

   (c) Click the *Environment Variables. . .* tab.

   (d) Look under the *System Variables* window, not the window titled *User variables for "your user name"*. If there is a variable named *CUDA_VISIBLE_DEVICES*, change it to the correct device. This is done by specifying the enumeration (listed in the table of X3D detected devices) for each device that you want X3D to use. If this variable is empty, all possible devices will be used for simulation purposes. Multiple devices can be specified using commas to separate enumeration numbers.

   (e) If there is no variable named *CUDA_VISIBLE_DEVICES*, you will need to declare this variable by clicking the *New. . .* tab.

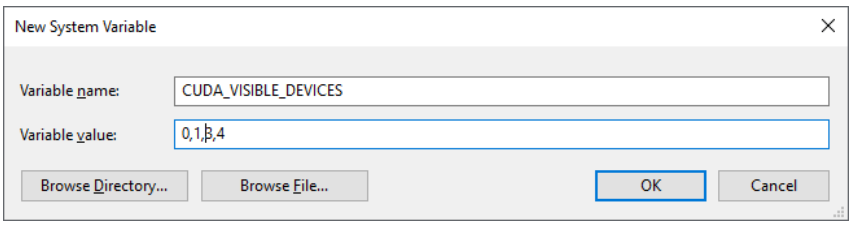


Figure N.1: Changing the CUDA_VISIBLE_DEVICES environment variable

   (f) Once you are finished try running the same job in X3D. You may need to re-log into Windows for User Variables to update.

N.4 Linux Command Line

- "Bad ELF interpreter" when attempting to start `lmgrd` - This error indicates that the Linux Standard Base (LSB) subsystem is not installed.

  - On RedHat-based systems (including CentOS), the redhat-lsb package must be installed using the command: `sudo yum install redhat-lsb`
  - On Suse SLES systems, install the lsb package using the command: `yast -i lsb`
  - On Ubuntu systems, install the lsb-core package using the command: `sudo apt-get install lsb-core`
- "Segmentation Fault" - Make sure the project files have been fully copied to the desired location.
- "Database not open" - Make sure the project files have been fully copied to the desired location, and ensure that the database will be written to a writeable location.

Glossary

AWGN

Additive White Gaussian Noise 214

Antenna

An object associated with a transmitter or receiver which propagates a specific wave pattern 139

BER

Bit Error Rate 214

COST-Hata Propagation Model

An extension of the Hata Propagation Model for higher frequencies 268

Communication Systems

A defined group of transmitters and receivers used to analyze the performance of the communications environment 211

Context Menu

The right-click menu of an item 14

Control Point

The origin of the Control Vectors 200

Control Vectors

A set of basis vectors centered on the control point of a transmitter or receiver, which describes its current orientation within the projects global coordinate system 200

Database

A table of antenna, material or waveform components that are accessible to all projects 375

Diffraction

A ray deflection that occurs at the point where the field becomes discontinuous 244

Diffuse Scattering

281, 307

Double-Sided Face

A FULL 3D, VERTICAL PLANE, and URBAN CANYON specific setting for a face, allowing rays to interact with both sides of the face, independent of the face normal 31

Eigenray Method

A method used to trace ray paths between transmitters and receivers which satisfies Fermat's Principle of least time, except that there is no refraction at transmissions 245

Faces

The components that make up a geometry, composed of planar polygons with three or more vertices 31

Free Space Propagation Model

A model which simulates propagation assuming no interactions with geometry and an electric field which decreases as $1/r$ with distance in all directions 263

Full 3D Propagation Model

A model which uses the SBR method and allows the user to set the ray spacing and the maximum number of reflections, transmissions, and diffractions 240

GUI (Graphical User Interface)

The Wireless InSite display screen 5

Geometry View

The window which displays the data which is currently loaded into the project 8

Geometry

A geometric object with material properties that comprises all city, terrain, building, foliage or object data loaded from a file 31

Hata Propagation Model

A model which evaluates the electric field of a single ray at the receiver point rather than the path loss 265

ISI

Inter-Symbol Interference 214

Images

A tab in the Main Window which lists all image files included in the project 6

MIMO

Multiple-input and multiple-output antenna systems can be defined for transmitters, receivers or transceivers. They are a collection of independent antenna sub-elements that perform together with a single waveform. 167

MPE

See Maximum Permissible Exposure. 289

Main Window

The parent window of Wireless InSite, which contains the main tabs that control the elements of the project 5

Material

A set of display and electromagnetic properties applied to a geometry, in part or whole 99

Maximum Permissible Exposure

The highest rms or peak electric or magnetic field strengths, their squares, or the plane-wave equivalent power densities associated with these fields, or the induced and contact currents to which a person may be exposed without incurring an established adverse health effect and with an acceptable margin of safety 289

Monte Carlo (MC)

Wireless InSite Monte Carlo simulations select random input values from a user-defined distribution of possible input values, perform the requested X3D simulations on each iteration and return the aggregate results. 299

Movie Player

The window used to play back available simulation results 14

Multi-Material

Monte Carlo Multi-Materials provide a mechanism for the Monte Carlo module to uniformly select a material from a used-defined set of materials. A Multi-Material is comprised of a set of previously-defined materials. Free Space may not be included as a constituent material. 299

One-Sided Face

A FULL 3D, VERTICAL PLANE, and URBAN CANYON specific setting for a face, allowing rays to interact only when incident on the side with the outward normal 31

Project Tree

The window used to navigate through the input and results of a project 13

Real Time - OPAR

A deterministic model which uses building depth between the transmitting and receiving antennas to enhance attenuation predictions 274

Real Time - Triple Path

A deterministic model designed to enhance vertical plane urban simulations 273

Real Time - VPUP

A deterministic model which achieves short simulation times by examining the propagation in vertical plane containing the transmitter and receiver 270

Real Time - Walfisch-Ikegami

A deterministic model used when the dominant energy is contributed by over-the-rooftop diffractions 275

Receiver

A point that collects radiation 177

Result Filters

Filters applied to Wireless InSite results to isolate ray paths with specific interactions and identify objects contributing to received power 371

Result Tree

The tree displayed in the Results tab of the Main Window 20

Results

Propagation predictions generated by the simulation engine 315

SINR

Signal to Interference plus Noise Ratio 214

Scale Bar

A color-coded representation of result values displayed in the Geometry View 351

Shooting and Bouncing Ray (SBR) Method

A method used to trace ray paths through the geometry without regard for the location of specific field points 241

Simulation Log

The window which records all information generated while performing simulations 14

Soft Rotation

A way to rotate Tx/Rx sets in the XY plane without them being X- or Y-axis aligned 205

Study Area

A user-defined region of the project in which to perform a simulation 225

Transmitter

The location of a source of input radiation 177

Tx/Rx

Transmitter/Receiver 2

Uniform Theory of Diffraction (UTD)

A high-frequency method for solving electromagnetic scattering problems 239

Urban Canyon Propagation Model

A model which simulates ground-level propagation in a high-rise urban environment among building footprints using SBR 257

Variable Parameter Material

Variable Parameter Materials are used in Monte Carlo simulations. They are defined by a distribution of values for dielectric permittivity, conductivity, thickness or roughness. 301

Vertical Plane Propagation Model

A ray-based model which predicts propagation over irregular terrain intersecting a two-dimensional vertical plane at VHF and UHF 261

Waveform

A signal radiated from a transmitting antenna which is time and/or frequency dependent 127

Wireless InSite Real Time Models

A deterministic path-loss-prediction solver made for ultra-fast response times in urban environments 270

X3D Ray Propagation Model

A model which uses the SBR method and allows the user to set the ray spacing and the maximum number of reflections, transmissions, and diffractions 249

Wireless InSite

A general purpose ray-based electromagnetic analysis tool 1

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